# The construction of four-dimensional quantum Yang–Mills theory

*Under construction. Version of November 11, 2026: this manuscript is incomplete. Sections are added and revised as they are written and checked against the formal proof; see the status table at* https://github.com/scottnarmstrong/yang-mills-informalization/tree/main/manuscript .

**Abstract**

For every closed subgroup $G \subset U(n)$ with simple nonabelian Lie algebra $\mathfrak{g}$ and every positive block penalty, we construct gauge-invariant local fields from four-dimensional Wilson measures. Exact scale integration and tuning to a common reference coupling give the full ultraviolet limit at fixed volume. Reflection positivity and chessboard bounds give volume-uniform control; compactness gives a limit along a refinement of any prescribed strictly increasing volume sequence. The limit decomposes into phases, each reconstructing a Wightman theory with unique vacuum. The fields carry a conserved stress tensor, an operator product expansion with spectator-independent coefficients and phase-uniform remainders, and distributional extensions through collisions. In a selected phase the centered stress trace has two-point asymptotic $(3 \dim_{\mathbb{R}} \mathfrak{g}/\pi^4) r^{-8} \log^{-2}(1/r)$ which agrees with no finite Wick profile on any interval $(0, \delta)$. No mass gap, uniqueness of the volume limit or phase, or independence from the block penalty is proved. This paper was written entirely by AI agents, working from OpenAI's Lean code for the construction of four-dimensional quantum Yang–Mills theory (commit `3456e5e`) and checking every statement against it; Section 1.2 describes the source and the method.

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# Part I
# The theorem and the architecture of its proof

## 1 Introduction

We construct continuum correlation functions of gauge-invariant local fields from four-dimensional Wilson lattice gauge theory. The lattice spacing is removed at each fixed physical volume; along a refinement of any prescribed increasing sequence of volumes, a subsequent limit gives a mixture of phases. Each phase has a relativistic reconstruction with positive energy, a unique vacuum and local fields. The same construction gives a conserved stress tensor, a quantitative operator product expansion and a short-distance law for the centered stress trace that excludes the finite Wick profiles defined below. The precise existence and construction statements are Theorems A and B.

Let $G \subset U(n_G)$ be a closed matrix group whose real Lie algebra $\mathfrak{g} = \mathrm{Lie}(G)$ is simple and nonabelian; connectedness of $G$ is not required. On the periodic lattice $\Lambda_s = (\mathbb{Z}/s\mathbb{Z})^4$, a configuration assigns $U(y,a) \in G$ to the positive link from $y$ in coordinate direction $a$, with inverse on the reversed link. Here $a = 1, \dots, 4$ and $e_a$ is the corresponding coordinate vector. For $\beta > 0$, the plaquette holonomy and Wilson measure are

$$\begin{aligned} U_{y;ab} &= U(y,a)U(y+e_a,b)U(y+e_b,a)^{-1}U(y,b)^{-1}, \\ E_s^{\mathrm{W}}(U) &= \sum_{y\in\Lambda_s}\sum_{a<b}\bigl(1 - n_G^{-1}\,\mathrm{Re}\,\mathrm{tr}\,U_{y;ab}\bigr), \\ d\mu_{s,\beta}(U) &= Z_{s,\beta}^{-1} e^{-\beta E_s^{\mathrm{W}}(U)} \prod_{y,a} dh(U(y,a))\,. \end{aligned}$$

The measure $dh$ is probability Haar measure, and $Z_{s,\beta}$ is the integral normalizing the density. Thus $\beta$ penalizes curvature around plaquettes. Section 2 fixes the normalization and defines the lattice observables.

The construction chooses increasing integers $D_N, L_p$, spacings $a_N = D_N^{-1}$ and bare couplings $\beta_{N,p}$, and uses $\mu_{N,p} = \mu_{L_pD_N,\beta_{N,p}}$. The physical torus has side $L_p$. The bare inverse coupling $\beta_{N,p}$ is tuned separately for each volume so that blocking reaches one common, sufficiently large reference value $\bar\beta$ at level zero. This level has spacing $a_0 = 1$ and fixes the reference length unit; the theorem does not identify it with a measured mass scale. For the trace metric $\langle X, Y\rangle_{\mathfrak{g}} = -n_G^{-1}\,\mathrm{Re}\,\mathrm{tr}(XY)$, let $(e_\alpha^{\mathfrak{g}})$ be an orthonormal basis and put

$$d_{\mathfrak{g}} = \dim_{\mathbb{R}} \mathfrak{g}, \qquad C_{\mathfrak{g}} = -d_{\mathfrak{g}}^{-1}\sum_{\alpha} \mathrm{tr}_{\mathfrak{g}}\bigl(\mathrm{ad}(e_\alpha^{\mathfrak{g}})^2\bigr), \qquad b_0 = \frac{11C_{\mathfrak{g}}}{48\pi^2}\,.$$

Here $\mathrm{ad}(X)Y = [X, Y]$, and $\mathrm{tr}_{\mathfrak{g}}$ is the real operator trace. These constants specify the running law proved by the construction:

$$\frac{\beta_{N,p}}{\log(1/a_N)} \longrightarrow \kappa_G = 2b_0 \qquad (N \to \infty), \quad \text{uniformly in } p\,. \tag{1.1}$$

An insertion label is a pair $(d, \mathbf{c})$, where $\mathbf{c}$ specifies an invariant connection-jet polynomial of engineering weight at most $d$. These polynomials describe gauge-invariant local expressions in curvature and its covariant derivatives, such as curvature square. Their engineering degree counts curvature with weight two and each covariant derivative with weight one. Their normalization is fixed by prescribing the coefficient vector at a reference level and inverting its evolution under blocking; the Maxwell comparison identifies the resulting short-distance fields afterwards. Each has a fixed finite stencil, is continuous and gauge invariant, and has coefficients depending on the cutoff and volume. Definition 3.9 gives their formula; Propositions 5.2 and 5.5 identify their short-distance comparison with invariant polynomials in free Maxwell curvature.

The renormalization flow stops at level zero. Reflection positivity and the chessboard estimate give bounds uniform in the physical volume; compactness then supplies the volume refinement, without a mass gap (Sections 14.2 and 15).

In words, Theorem A asserts that one regular Wilson family, with continuous gauge-invariant insertions on fixed finite stencils and a positive volume-uniform logarithmic running slope, can be chosen before any increasing sequence of volume indices is prescribed. Every such sequence has a refinement on which the correlations converge to a phase average. A selected phase continues to a nonzero Wightman theory, and the same hierarchy has an operator product expansion and a centered degree-four field with the trace asymptotics below. Theorem A requires a positive logarithmic running slope; the particular value in (1.1) is supplied by Theorem B.

To spell out the order of limits, let $v = (v_1, \dots, v_n)$ be a finite list of lattice field labels, with insertions $I_{N,p}(v_i, y_i; U)$. Their smeared correlation is

$$C^{\mathrm{lat}}_{N,p}(v; f) = a_N^{4n} \sum_{y \in \Lambda^n_{L_p D_N}} f(\hat{y}) \int \prod_{i=1}^{n} I_{N,p}(v_i, y_i; U)\, d\mu_{N,p}(U)\,.$$

Here $\hat{y}_i = a_N r_{L_p D_N}(y_i)$, where $r_s$ takes the integer representative in $(-s/2, s/2]$ coordinatewise. Take a smooth compactly supported test $f$ whose support avoids every collision $x_i = x_j$. For every prescribed strictly increasing $\mathsf{p} : \mathbb{N} \to \mathbb{N}$, a strictly increasing refinement $\mathsf{q}$ gives retained indices $P = \mathsf{p} \circ \mathsf{q}$ and limiting kernels $S_n(v; x)$ with

$$\lim_{k \to \infty} \left( \lim_{N \to \infty} C^{\mathrm{lat}}_{N,P(k)}(v; f) \right) = \int f(x) S_n(v; x)\, dx\,. \tag{1.2}$$

The inner limit exists along the full cutoff sequence at every volume containing the support of $f$ in its open centered cell. One refinement serves all field lists and all such tests. Theorems 4.2 and 4.7 prove these two limits; Definition 4.3 states the convergence precisely.

Theorem B identifies more of this construction. The fields have finite-dimensional spaces in each degree and a conserved symmetric stress tensor normalized by its translation and rotation fluxes. A shrinking cluster has an expansion into local fields with coefficients independent of the spectators and of the phase. Theorem 7.6 gives the truncation error and its uniformity over separated positions, phases and the mixture. Contact distributions extend these correlations through collision diagonals without changing their values away from collisions; the field spaces and the separated correlation functions are those already constructed.

Write $S^{\xi}$ for the phase correlations. Their compact parameter space $\mathsf{X}$ carries a full-support probability measure $\nu$, and the mixed correlations are their $\nu$-average. Each phase reconstructs fields on common observable labels. These fields act on a cyclic domain in a Hilbert space, satisfy Poincaré covariance and spacelike commutativity, and have energy-momentum spectrum in the forward cone. The reconstruction is faithful: every nonzero observable label gives a nonzero smeared field for some test function. Theorem 6.15 states the reconstruction and its relation to the Euclidean functions.

The construction selects a phase $\xi_*$ with a one-dimensional time-axis ground space before proving reconstruction in every phase. Theorem 7.2 states the trace law only in this selected phase. Its comparison estimates are uniform in the phase during the proof, but we do not assert an almost-everywhere or every-phase trace theorem.

For the stress tensor $T_{ab}$ in that phase, define

$$\tau = \sum_{a=1}^{4} T_{aa}, \qquad \Theta = \tau - S^{\xi_*}(\tau)\mathbf{1}_4, \qquad G_\Theta(r) = S_2^{\xi_*}(\Theta, \Theta; -re_4/2, re_4/2)\,,$$

where $\mathbf{1}_4$ is the unit field. Centering removes the selected phase's vacuum expectation. This field expresses the trace anomaly: the free Maxwell stress tensor is traceless, whereas the comparison below has a nonzero curvature-square coefficient of order $1/\log(1/r)$. Its square gives the inverse logarithm squared in the two-point law. The short-distance conclusion is

$$\lim_{r\downarrow 0} r^8 \log^2(1/r) G_\Theta(r) = \frac{3d_{\mathfrak{g}}}{\pi^4} > 0\,. \tag{1.3}$$

The limit holds through all positive radii. Theorem 7.2 proves it, and Proposition 7.4 shows that $G_\Theta$ agrees with no finite Wick profile on any interval $(0, \delta)$. Such profiles are finite sums of finite products of mass-shell Laplace kernels weighted by energy powers and homogeneous momentum factors, as specified in Definition 7.3. This is the precise free-field comparison excluded here.

## 1.1 Scope

The Clay Yang–Mills existence and mass gap problem [14, §4] asks for both existence and a positive mass gap. The present construction gives Wightman existence in each phase; no mass gap is proved. Uniqueness of the vacuum in a reconstructed phase means that the zero-energy space is one-dimensional; it gives no positive lower bound on the remaining energy spectrum. Nor do we prove uniqueness of the infinite-volume limit: the limiting table may depend on $\mathsf{p}$ and $\mathsf{q}$. The phase space is not shown to be a singleton, and the selected phase is not shown to be unique. Nor is non-Gaussianity proved in general: the trace law excludes finite Wick profiles, but a generalized free field with the spectral measure of $\Theta$ has the same two-point function.

The construction also depends on a positive block penalty $\mathsf{b}$, which weights the discrepancy between fine transport and coarse links in the probability kernel of Definition 3.1. Theorem B holds for every fixed $\mathsf{b} > 0$; independence of the resulting theory from this parameter is not proved. Theorem A follows, for example, by fixing $\mathsf{b} = 1$.

Convergence in (1.2) is tested on separated configurations. The contact extensions are additional distributions, constructed after these limits; the theorem does not assert convergence of the unsubtracted lattice correlations against tests meeting collision diagonals. The product theorem gives an estimate at each finite truncation degree, rather than convergence as that degree tends to infinity.

### 1.2 Source of the result and of this text

The mathematical result is due to OpenAI's Lean construction at commit `3456e5e` of the repository `yang-mills-group/OpenAI-YM-20261028` [20]. Theorem A states its top-level existence theorem; Theorem B records further conclusions of that construction. Checking a statement against the formal proof means comparing its hypotheses, definitions, quantifier order, normalizations and conclusions with declarations actually used by that theorem, as determined from the compiled dependencies. The separate correspondence table matches the statements here to used declarations; the source repository supplies a declaration-usage table. Its kernel-check record reports no unproved placeholders in the proved theorem and only propositional extensionality, classical choice and quotient soundness as axioms. This comparison does not make the prose a machine-checked proof: the mathematical deductions written here remain subject to mathematical reading and correction. This text was written with AI models under human direction, with separate writing and checking of statements and arguments, by the method described in *How the manuscript is written*. The work is carried out in collaboration with Martin Hairer's *humanise-yang-mills* project.

### 1.3 Earlier work and the method

Wilson's lattice formulation [32] makes the finite-cutoff theory a probability law on compact group-valued links. Bałaban's programme supplies much of the constructive gauge-theory setting: nonlinear group averages, gauge fixing, stationary backgrounds, local propagators and marginal curvature-square extraction. Analytic stationary backgrounds are treated in [2, Theorem 1 and Proposition 6]; symmetric block kernels and the marginal analysis occur in [3, pp. 252–254 and §§4–5]. The large-field result [4, Theorem 1, p. 355] assumes sufficiently small effective couplings, and its discussion on p. 356 leaves physical observables for further analysis. The present construction controls running couplings and local composite fields together.

Magnen, Rivasseau and Sénéor [17] give a four-dimensional ultraviolet construction for pure $SU(2)$ theory in the trivial topological sector, in a regularized axial gauge with a fixed infrared cutoff. They control background-dependent normalization and formulate a gauge-fixed Schwinger-function limit with Slavnov identities. Their account explicitly defers a fully detailed convergence proof (p. 327), retains the infrared cutoff (p. 327), and sets aside renormalized physical composite observables (p. 329). The endpoint here concerns an infinite-volume hierarchy of gauge-invariant local fields, together with its phases, stress tensor and local products.

Several operations used below have antecedents beyond gauge theory. Gawędzki and Kupiainen's critical lattice $\varphi^4_4$ construction [12, p. 212, (4.8), and pp. 218–221] uses disjoint polymers covering the large-field region and selective exponentiation of the remaining terms; its asymptotic freedom is an infrared statement for a scalar model. Brydges and Yau [6, §§1–2 and pp. 378–384] give exact polymer extraction and reblocking identities, analytic norms and Gaussian integration estimates. Covering the bad region, extracting marginal terms and restoring the interaction class are established operations.

The particular effective-density representation used here is closely related to OpenAI's 2026 sigma-model papers, especially the exact map in [19, §5, pp. 38–55] and its adaptation in [18, §3.3, pp. 23–28]. The former combines a clipped kinetic term, localized quadratic terms, regular errors and a compulsory covering sum, with an exact coupling reset. Those features have gauge-theoretic counterparts below. The comparison concerns their combination; it does not assign these operations a new origin: compulsory covering and selective exponentiation already occur in [12, p. 212]. The sigma-model papers have different infrared conclusions, which are not inputs here.

Reflection positivity of the Wilson measure and its transfer matrix go back to Osterwalder–

Seiler [23]; see [27, §2a, Theorem 2.1]. Chessboard estimates are due to Fröhlich–Israel–Lieb–Simon [11, §4, Theorems 4.1–4.2]; [27, Theorem 2.2] gives a lattice version for local matter functions. Analytic continuation follows the Osterwalder–Schrader programme [21, 22], including the growth and boundary-value issues addressed in the second paper. Section 17 supplies the slab and distribution estimates for these fields. Reconstruction does not follow from reflection positivity alone.

The logarithmic running in (1.1) has the one-loop slope of the asymptotic freedom found perturbatively by Gross–Wilczek and Politzer [13, 24]; the background-field coefficient is reviewed in [1, §3.3]. What requires proof here is its compatibility with the exact gauge block map, coupling reset and terminal-coupling tuning, followed by continuum identifications on one family. The substantive conclusions are simultaneous control of composite fields, restoration of rotations through stress-tensor fluxes, the all-radii trace law and product bounds uniform over phases. Operator product expansions with state-independent coefficients go back to Wilson [31, §II], the trace anomaly to Collins–Duncan–Joglekar [7, (3.19)–(3.21)], and Taylor subtraction at collision diagonals to [5, §5.2]. We make no priority claim for these renormalization-group principles or physical predictions.

### 1.4 Outline of the proof

The deduction in Section 8 fixes one lattice family, takes its limits, and identifies the resulting fields. The estimates apply to that same family throughout. We describe their roles using the tags in Table 1.

**Exact integration and the running coupling.** Section 10 realizes invariant connection-jet polynomials as local lattice functions, supplying the observables' finite stencils. Section 11 constructs a normalized probability kernel relating fine and coarse links, with symmetric gauge choices and alternatives where minimizing branches fail. Normalization preserves the partition integral exactly. The resulting density is described by a curvature action, a vacuum scalar, regular local terms and terms covering every large-field site.

On the small-field region, integration about a stationary background gives a Gaussian integral with determinant and Haar-coordinate factors. A large-field lower bound pays for the terms forced to cover the exceptional region. Section 12 matches the Wilson energy through fourth order, extracts the one-loop curvature-square coefficient and resets the kinetic coupling. Section 13 assembles the local terms and proves RG: exact integration returns a density in the next controlled class. For running coupling $\beta_{N,p,m}$ and odd block ratio $\ell_m$, the drift is

$$\beta_{N,p,m} = \beta_{N,p,m+1} - \kappa_G \log \ell_m + e_{N,p,m} \,,$$

with the error bounded as in RG. Blocking lowers the level from $m+1$ to $m$.

Section 14.1 starts from the Wilson density and varies the bare coupling continuously. Successive restrictions to parameter intervals keep the running coupling in its allowed bands and reach one common terminal value $\bar{\beta}$ at level zero, proving MT. Summing the drift gives (1.1). Analytic endpoint tuning then permits comparison of different cutoffs. These choices precede the prescribed volume sequence.

**Local fields and the two limits.** Section 14.2 follows insertions through the integration. An invertible map to their coefficients at a fixed reference level determines their normalization. Estimates for densities carrying the insertions follow their size through blocking; combined with chessboard bounds, they give UB, the volume-uniform bounds on correlations and physical lattice differences.

In Section 14.3, endpoint tuning compares normalized integrals at a common intermediate level. Increasing that level makes the comparison error small; taking both cutoffs large afterwards makes the original correlations Cauchy. UB extends convergence to all separated positions and derivatives. This proves UV along the full sequence at fixed volume.

Section 15 uses these bounds for a diagonal extraction along the prescribed volumes, followed by a further refinement fixing the corrected coefficient maps; together they give $\mathsf{q}$ and prove IV. Reflection positivity passes to the limit, and corrected coefficient maps identify inclusions and derivatives between degrees. For invariant polynomial labels $V_d$ of weight at most $d$, the observable fields are

$$F_d = V_d/\mathcal{N}_d(S)\,,$$

where $\mathcal{N}_d(S)$ consists of labels whose every separated correlation with arbitrary spectators vanishes. The quotient retains the fields detectable by correlations. FC gives inclusions, positional derivatives and dimensions.

**Stress, rotations and phases.** Section 16 compares short-distance correlations with free Maxwell fields, proving MC. The comparison identifies a line of conserved degree-four tensors. Their shell fluxes around one insertion act on fields; identifying these actions with translations and rotations fixes the stress normalization. The angular flux through a distant enclosing surface tends to zero; the remaining insertion fluxes give the rotation identities and recover full rotation covariance. This proves ST; SC adds differentiated comparison with the normalized stress insertions.

Section 17 bounds insertion operators between positive-time slabs and projects their action to the zero-energy subspace of the time-translation generator. The resulting commutative algebra has compact character space $\mathsf{X}$; its vacuum state gives $\nu$, and its characters give the phase correlations. This is PH, on the same quotient fields. Select $\xi_*$ from the full-measure set with one-dimensional time-axis ground space. Continuation and boundary estimates prove RC. Uniform low-energy estimates and continuity in the phase extend vacuum uniqueness to every character. The selected theory supplies the relativistic part of Theorem A.

**The trace and local products.** Section 18 centers the stress trace in the selected phase. After multiplying each trace insertion by $r^4$, its leading coefficient in the free curvature-square comparison is asymptotic to $-1/\log(1/r)$. The free two-point amplitude is $3d_{\mathfrak{g}}/\pi^4$; squaring the coefficient gives (1.3). The comparison can be stopped at levels chosen for an arbitrary sequence of radii. Positivity of the spectral measure makes the time-axis two-point function monotone, allowing nearby comparison radii to bound it from both sides and prove TR through all radii. Finite Wick profiles have finite-order asymptotic expansions in integer powers of the radius and nonnegative logarithmic powers. Their incompatibility with the nonzero inverse-logarithmic limit proves NF.

Section 19 recovers fields from finitely many correlations against fixed probe words and uses these projections to define product coefficients. Bounds in the compression algebra pass to every character and the mixture, proving OP with spectator-independent coefficients. The section treats nested clusters and constructs contact extensions by subtracting Taylor jets along collision diagonals. The contact projections and their clearance are separate choices; they use the existing field spaces and agree with the existing correlations away from collisions. This proves CT and completes Theorem B.

To obtain the hierarchy data of Theorem A, the deduction maps a lattice label to its quotient field by $\pi_{\mathrm{quo}}(d,\mathbf{c}) = (d,[\mathbf{c}])$ and assigns arbitrary correlation values on collision diagonals. Representative independence identifies the lattice limit with $S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}}$. The phase decomposition and

reconstruction give Q2–Q4, the product theorem gives Q5, and TR with NF gives Q6. Fixing $\mathsf{b} = 1$ gives Theorem A.

### 1.5 Reading guide

Part I introduces the objects and states the principal results with their proof ideas; Section 8 assembles them into Theorems A and B. Part II gives the proofs in Sections 10–19. Table 1 identifies the tags used throughout. The order-of-choice table in Section 9 records when the block penalty, scale data, couplings, insertions, volume refinement and continuum constructions are fixed. Later choices can depend on earlier ones.

Table 1: Principal results and their proofs.

| Tag | Statement | Section | Proof in Part II |
|---|---|---|---|
| RG | Exact integration of one scale and coupling drift | 3 | Sections 11–13 |
| MT | Common terminal coupling and running slope | 3 | Section 14.1 |
| UB | Volume-uniform bounds on lattice differences | 4 | Section 14.2 |
| UV | Full fixed-volume ultraviolet limit | 4 | Section 14.3 |
| IV | Volume refinement and Euclidean properties | 4 | Section 15 |
| FC | Inclusions, derivatives and Maxwell dimensions | 4 | Lemma 15.4 |
| MC | Maxwell comparison at short distances | 5 | Lemma 16.6 |
| ST | Conserved stress and normalized shell fluxes | 5 | Lemmas 16.10–16.11 |
| SC | Differentiated comparison with stress insertions | 5 | Lemma 16.12 |
| PH | Phase decomposition on the same fields | 6 | Section 17.1 |
| RC | Wightman reconstruction in every phase | 6 | Section 17.2 |
| TR | All-radii law for the centered stress trace | 7 | Section 18.1 |
| NF | Exclusion of finite Wick profiles | 7 | Lemma 18.7 |
| OP | Local product expansion with error bounds | 7 | Section 19.1 |
| CT | Extensions through collision diagonals | 7 | Section 19.2 |

### 1.6 Notation and conventions

Throughout, $\mathbb{N} = \{0, 1, \dots\}$, coordinate indices run from 1 to 4, and $e_a$ is the $a$th coordinate vector. Time is the fourth coordinate. For the later relativistic statements write

$$\mathsf{m}(x) = (x^4)^2 - |\vec{x}|^2, \qquad \overline{V}_+ = \{\mathbf{p} : |\vec{\mathbf{p}}| \le \mathbf{p}^4\}.$$

Inner products are antilinear in the first variable, and spacetime integrals use Lebesgue measure. Arity zero is included; empty sums are zero and empty products are one. The group $H_4$ consists of the signed permutation matrices; $\mathsf{s}_a^{\mathrm{refl}}$ denotes reflection in coordinate $a$.

## 2 Wilson theory

The starting measure assigns a group element to each lattice link and weights the curvature around each plaquette. We first fix the matrix normalization: it determines both the Wilson energy and the coefficient of the logarithmic coupling flow.

**Definition 2.1** (Gauge normalization)**.** Fix $n_G \geq 1$ and a closed subgroup $G \subset U(n_G)$ with simple nonabelian real Lie algebra

$$\mathfrak{g} = \{X \in M_{n_G}(\mathbb{C}) : \exp(tX) \in G \text{ for every } t \in \mathbb{R}\}, \qquad \langle X, Y\rangle_{\mathfrak{g}} = -n_G^{-1} \operatorname{Re}\operatorname{tr}(XY)\,.$$

For an orthonormal basis $(e_\alpha^{\mathfrak{g}})$, put

$$d_{\mathfrak{g}} = \dim_{\mathbb{R}} \mathfrak{g}, \qquad C_{\mathfrak{g}} = -d_{\mathfrak{g}}^{-1} \sum_\alpha \operatorname{tr}_{\mathfrak{g}}\big(\operatorname{ad}(e_\alpha^{\mathfrak{g}})^2\big),$$
$$b_0 = \frac{11 C_{\mathfrak{g}}}{48\pi^2}, \qquad \kappa_G = 2b_0\,.$$

This fixes the normalization of the running coupling.

Here $\operatorname{ad}(X)Y = [X, Y]$ and $\operatorname{tr}_{\mathfrak{g}}$ is the real operator trace on $\mathfrak{g}$. Simplicity means $[\mathfrak{g}, \mathfrak{g}] \neq 0$ and no proper nonzero real ideal. Connectedness of $G$ is not assumed. The adjoint maps are skew-adjoint for this metric; nonabelianity therefore gives $C_{\mathfrak{g}} > 0$ and $\kappa_G > 0$. For $u \in G$, set

$$\operatorname{pr}(u) = 1 - n_G^{-1} \operatorname{Re}\operatorname{tr} u, \qquad \operatorname{ch}(u) = n_G^{-1/2}\|u - 1\|_{\mathrm{F}}, \qquad \operatorname{ch}(u)^2 = 2\operatorname{pr}(u)\,. \tag{2.1}$$

The first quantity is the plaquette energy; the second measures its distance from the identity. Both are invariant under conjugation and inversion.

**Definition 2.2** (Wilson measure)**.** For an integer $s \geq 1$ and $\beta \in \mathbb{R}$, put $\Lambda_s = (\mathbb{Z}/s\mathbb{Z})^4$. A configuration assigns $U(y, a) \in G$ to every positive link, with inverse on the reversed link. Define

$$U_{y;ab} = U(y, a)U(y + e_a, b)U(y + e_b, a)^{-1}U(y, b)^{-1},$$
$$E_s^{\mathrm{W}}(U) = \sum_{y\in\Lambda_s} \sum_{a<b} \operatorname{pr}(U_{y;ab}),$$
$$d\mu_{s,\beta}(U) = Z_{s,\beta}^{-1} e^{-\beta E_s^{\mathrm{W}}(U)} \prod_{y,a} dh(U(y, a))\,.$$

Here $dh$ is probability Haar measure and $Z_{s,\beta}$ normalizes the measure. This is the lattice theory to be scaled.

The normalization is the positive finite integral

$$Z_{s,\beta} = \int e^{-\beta E_s^{\mathrm{W}}(U)} \prod_{y,a} dh(U(y, a))\,.$$

Indeed, the integrand is continuous and strictly positive on the compact configuration space. A site gauge transformation $\gamma_{\mathrm{g}} : \Lambda_s \to G$ acts by

$$U^{\gamma_{\mathrm{g}}}(y, a) = \gamma_{\mathrm{g}}(y)U(y, a)\gamma_{\mathrm{g}}(y + e_a)^{-1}\,.$$

It changes each plaquette by conjugation at its base point, so the energy is gauge invariant. Invariance of probability Haar measure under left and right multiplication then gives gauge invariance of the Wilson measure. Affine lattice isometries $x \mapsto \rho_{\mathrm{lat}} x + b$, with $\rho_{\mathrm{lat}} \in H_4$ and $b \in \Lambda_s$, transport links and invert reversed ones. The measure is invariant under these transformations as well; the finite-lattice identities are proved in Section 10.

A general Wilson family, indexed by $N, p \in \mathbb{N}$, consists of strictly increasing positive integer sequences $(L_p)$ and $(D_N)$, real couplings $\beta_{N,p}$, a label set $\mathcal{T}$, and complex insertions $I_{N,p}(v, y; U)$. The index $N$ controls the spacing, and $p$ controls the physical volume. Set

$$a_N = D_N^{-1}, \qquad \mu_{N,p} = \mu_{L_p D_N, \beta_{N,p}}, \qquad \hat{y} = a_N \big(r_{L_p D_N}(y^a)\big)_{a=1}^4 , \tag{2.2}$$

where $r_s(j)$ is the integer representative of $j \in \mathbb{Z}/s\mathbb{Z}$ in $(-s/2, s/2]$. Thus the physical torus has side $L_p$.

**Definition 2.3** (Lattice moments)**.** For a label list $v \in \mathcal{T}^n$, sites $y \in \Lambda_{L_p D_N}^n$, and a test $f : (\mathbb{R}^4)^n \to \mathbb{C}$, define

$$M_{N,p}^{\mathrm{lat}}(v; y) = \int \prod_{i=1}^{n} I_{N,p}(v_i, y_i; U)\, d\mu_{N,p}(U),$$
$$C_{N,p}^{\mathrm{lat}}(v; f) = a_N^{4n} \sum_{y \in \Lambda_{L_p D_N}^n} f(\hat{y}) M_{N,p}^{\mathrm{lat}}(v; y)\,.$$

These are the point correlations and their physical Riemann sums.

The construction below chooses the spacings, couplings and local insertions together. Its observables are defined by renormalization: for each token $v$, the insertion $I_{N,p}(v, y; \cdot)$ is a continuous gauge-invariant function of the links in a fixed finite stencil $y + \Sigma_v$, with coefficients depending on $N, p$ as specified by Definition 3.9. At and above the reference level, its terminal coefficient map is fixed by the block-transfer normalization. Propositions 5.2 and 5.5 compare the resulting continuum fields with free curvature polynomials at short distances. Its bare inverse coupling has logarithmic slope $\kappa_G$:

$$\frac{\beta_{N,p}}{\log(1/a_N)} \longrightarrow \kappa_G \qquad (N \to \infty), \quad \text{uniformly in } p\,. \tag{2.3}$$

This running law is proved in Section 3. At each fixed physical volume the ultraviolet limit will use the full sequence $N \to \infty$; only the subsequent infinite-volume limit requires a subsequence.

## 3 Renormalization

Blocking integrates one scale against a probability kernel. A curvature action, a vacuum scalar and two activity families describe the new density. The inductive step controls these coordinates and the coupling; the multiscale theorem chooses the bare coupling to reach one prescribed endpoint.

Fix $\mathsf{b} > 0$. Throughout we use the common choices of Sections 10–14, made once for this group realization and penalty in the order of Section 9. Their existence is proved with Theorems 3.5 and 3.6. Constants may depend on these choices; further dependence is stated where it occurs.

### 3.1 Scales and the block map

With odd ratios $\ell_m \geq 3$, set

$$D_0 = 1, \qquad D_{m+1} = \ell_m D_m, \qquad a_m = D_m^{-1}, \qquad s_m = \log D_m \,. \tag{3.1}$$

Level $m$ has spacing $a_m$; blocking passes from level $m+1$ to level $m$. The level-$m$ torus has side $L_p D_m$ in lattice units and side $L_p$ in physical units. Starting at cutoff $N$, blocking successively reaches levels $N-1, \dots, 0$.

The stage $J_m$ records the available derivative order and bounds. The stages satisfy $4 \leq J_m \leq J_{m+1} \leq J_m + 1$ and $J_m \to \infty$. Choose the reference endpoint $\bar{\beta}$ sufficiently large, as in row 7 of Section 9; in particular $\bar{\beta} > 1$. Define

$$\widehat{\beta}_m = \bar{\beta} + \kappa_G s_m, \qquad M_m^{\mathrm{cut}} = \exp\bigl((\log\log \widehat{\beta}_m)^2\bigr), \qquad t_m = \widehat{\beta}_m^{-1/2} M_m^{\mathrm{cut}}, \tag{3.2}$$

$$B_m = \lceil (\log \widehat{\beta}_m)^2 \rceil, \qquad \epsilon_m^{\mathrm{reg}} = \widehat{\beta}_m^{-11/10}, \qquad \epsilon_m^{\mathrm{for}} = e^{-(M_m^{\mathrm{cut}})^{1/5}} \,. \tag{3.3}$$

The reference coupling $\widehat{\beta}_m$ tracks the leading running slope. The factor $M_m^{\mathrm{cut}}$ enlarges the Gaussian field scale; $t_m$ is the threshold separating small and large plaquette fields. The radius $B_m$ sets the scale of the support weights. The quantities $\epsilon_m^{\mathrm{reg}}$ and $\epsilon_m^{\mathrm{for}}$ bound the regular and forced activity sums, respectively. These thresholds depend on the reference coupling, so they remain fixed as the actual coupling varies.

The allowed coupling interval is

$$\mathfrak{B}_m = [\widehat{\beta}_m/2, 2\widehat{\beta}_m] \,.$$

At stage $j$, write $\mathfrak{q}_j$ for the derivative order, $\varpi_j$ for the spatial power, $\sigma_j$ for the support exponent and $\omega_j^{\mathrm{RG}} \geq 0$ for the coupling drift bound. Their choices and growth conditions are in Section 9.

At a step $m+1 \to m$, the fine links $U$ and coarse links $V$ lie on $\Lambda_{L_p D_{m+1}}$ and $\Lambda_{L_p D_m}$. Put $j = J_{m+1}$ and $\ell = \ell_m$. Write $\mathbb{B}_R = \{-R, \dots, R\}^4$. In a patch of radius $(j+3)\ell$ about the fine site $\ell y$, let $h_{y,z}$ be the chosen frame at offset $z$. For a coarse edge $e = (y, a)$ the framed straight connector is

$$Y_{e,z} = h_{y,z} \left( \prod_{r=0}^{\ell-1} U(\ell y + z + r e_a, a) \right) h_{y+e_a,z}^{-1} \,. \tag{3.4}$$

This compares the fine transport between the two coarse endpoints in their chosen frames. The averaging weights and the predictor on its minimizing branch are

$$w_z^{\mathrm{av}} = \bigl(\mathrm{Unif}(\mathbb{B}_{(\ell-1)/2})\bigr)^{*j}(z), \qquad Q_e^{\mathrm{pred}} = \operatorname*{argmin}_{u \in G} \sum_z w_z^{\mathrm{av}} \, \mathrm{pr}(Y_{e,z} u^{-1}) \,. \tag{3.5}$$

Here Unif is uniform probability mass and $*j$ denotes $j$-fold convolution. This branch is used when the displayed minimum is below the fixed predictor cutoff; otherwise a tag selects $Y_{e,z}$ with mass $w_z^{\mathrm{av}}$. Symmetric minimizing frames and the alternative axial frames are specified in Section 11, Definitions 11.1–11.3. The finite tag list $\alpha_{\mathrm{tag}}$ specifies all frame and predictor choices. Their product mass satisfies

$$\lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) \geq 0, \qquad \sum_{\alpha_{\mathrm{tag}}} \lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) = 1 \,. \tag{3.6}$$

**Definition 3.1** (Exact block map)**.** For fine links $U$ and coarse links $V$, define

$$k_\beta^{(m)}(U,V) = \sum_{\alpha_{\mathrm{tag}}} \lambda^{\mathrm{tag}}(U,\alpha_{\mathrm{tag}}) \prod_{e\ \mathrm{coarse}} \frac{e^{-\mathsf{b}\beta\,\mathrm{pr}(V_e Q_e^{\mathrm{pred}}(U,\alpha_{\mathrm{tag}})^{-1})}}{\int_G e^{-\mathsf{b}\beta\,\mathrm{pr}(u)}\,dh(u)},$$
$$(\mathcal{B}_\beta^{(m)} f)(V) = \int k_\beta^{(m)}(U,V) f(U)\,dU\,.$$

Both product measures are probability Haar measures, and $\int k_\beta^{(m)}(U,V)\,dV = 1$. The map integrates one scale without changing the partition integral.

For fixed $U$ and tags, right invariance of Haar measure makes every coarse-link factor integrate to one. Summing the tag masses proves the normalization. Consequently, for integrable $f$,

$$\int (\mathcal{B}_\beta^{(m)} f)(V)\,dV = \int f(U)\,dU\,. \tag{3.7}$$

The branch choices, measurability and symmetry properties of this kernel are proved in Section 11.

## 3.2 The density and its norms

At level $m$, write $\Lambda = \Lambda_{L_p D_m}$. Let $\mathrm{Flag}_m(U)$ be the sites incident to a plaquette whose chord exceeds $t_m$. A support is healthy if it avoids this set. A regular activity $E_i^{\mathrm{reg}}$ has a site support $X_i$ and an anchor $o_i \in X_i$. A forced activity $K_i^{\mathrm{for}}$ has a support $Y_i$ and a nonempty inventory $I_i \subset Y_i$ of flagged sites. Its eligibility event is

$$\mathcal{E}_i^{\mathrm{elig}} = \{U : Y_i \cap \mathrm{Flag}_m(U) = I_i\} \cap \mathcal{E}_i^{\mathrm{extra}}, \qquad K_i^{\mathrm{for}}(U) = 0 \quad (U \notin \mathcal{E}_i^{\mathrm{elig}})\,. \tag{3.8}$$

The first test specifies exactly which sites of $Y_i$ are flagged. The extra event $\mathcal{E}_i^{\mathrm{extra}}$ is a fixed Borel gauge-invariant test supplied with that activity; it reads only links with both endpoints in $Y_i$ and the flags in $Y_i$. It restricts which local configurations can contribute that forced term. The activity families are countable.

The path action separates the kinetic coefficient from the quadratic corrections generated by integration. For the fixed level data, write $c_{\mathrm{dir}} > 16 c_{\mathrm{lin}} > 0$ for the clipping constants, chosen with $c_{\mathrm{dir}} \le \min(\gamma_{J_m}, 1)$ as in row 4 of Section 9, and $R_{J_m}^{\mathrm{rel}} > 0$ for the relaxation length. Distances use the periodic maximum-coordinate metric in lattice units: on a torus of integer side $n$, $\mathrm{dist}(x,y) = \min_{z\in n\mathbb{Z}^4} |\tilde{x} - \tilde{y} + z|_\infty$ for any lifts $\tilde{x}, \tilde{y} \in \mathbb{Z}^4$. Set

$$\mathrm{clip}_t(r) = \begin{cases} c_{\mathrm{dir}} r^2/2, & r \le t, \\ c_{\mathrm{lin}} t r, & r > t, \end{cases}$$
$$\chi_{m,x}^{\mathrm{hlth}}(U) = \mathbf{1}\Big\{ \mathrm{ch}(U_{y;ab}) \le t_m e^{\mathrm{dist}(x,y)/R_{J_m}^{\mathrm{rel}}} \text{ for every } y, a<b \Big\}\,. \tag{3.9}$$

The clip keeps a positive linear penalty beyond the small-field threshold, while $\chi_{m,x}^{\mathrm{hlth}}$ switches off a quadratic row when a plaquette violates the threshold prescribed by its distance from $x$. For each plaquette orientation $\mathfrak{p} = (a,b)$, let $\mathcal{F}_{\mathfrak{p}i}(U,x)$ be a curvature $\mathrm{proj}_{\mathfrak{g}}(U_\square - U_\square^{-1})/2$, transported along a specified finite path to $x$, and put

$$\mathcal{L}_{\mathfrak{p}}^{\mathrm{path}}(U,x) = \sum_i r_{\mathfrak{p}i} \mathcal{F}_{\mathfrak{p}i}(U,x)\,.$$

The projection is orthogonal for the trace metric of Definition 2.1. Let $\mathfrak{a}_i^{(q)}(U,x)$, $q=2,3,4$, denote the fixed real invariant $q$-linear expressions in transported curvatures, multiplied by the health indicator of their combined site supports. Then

$$\begin{aligned}(P^{\mathrm{path}})^{\mathrm{raw}}_{N,m}(\beta,U) &= \beta\sum_{x,\mathfrak{p}}\Big[\mathrm{clip}_{t_m}(\mathrm{ch}(U_{x;\mathfrak{p}})) + \tfrac{1}{2}\chi^{\mathrm{hlth}}_{m,x}(U)|\mathcal{L}^{\mathrm{path}}_{\mathfrak{p}}(U,x)|^2\Big]\\ &\quad+\beta\sum_{x,i}(\mathfrak{a}_i^{(3)}+\mathfrak{a}_i^{(4)})(U,x)+\sum_{x,i}\mathfrak{a}_i^{(2)}(U,x),\\ P^{\mathrm{path}}_{N,m}(\beta,U) &= |H_4|^{-1}\sum_{h\in H_4}(P^{\mathrm{path}})^{\mathrm{raw}}_{N,m}(\beta,h^*U)\,.\end{aligned}\tag{3.10}$$

Here $x\in\Lambda$ and $i\in\mathbb{N}$; $h^*U$ is the link transport by the lattice isometry $h$. The path coefficients and multilinear expressions are independent of $p$. Their absolutely convergent series and the clipping restrictions are given in Section 12, Definitions 12.1–12.2 and Lemmas 12.3–12.4. In particular the action is affine in $\beta$; its last quadratic sum is independent of $\beta$.

For these fixed path actions and thresholds, a *representation* at level $m$ is the datum

$$\rho=\big(\beta,v_{\mathrm{vac}},(X_i,o_i,E_i^{\mathrm{reg}})_i,(Y_i,I_i,\mathcal{E}_i^{\mathrm{extra}},K_i^{\mathrm{for}})_i\big)\,.\tag{3.11}$$

The scalar $v_{\mathrm{vac}}$ is its vacuum term, and $\beta(\rho)=\beta$ its running kinetic coefficient. These coordinates describe the integrated density below.

**Definition 3.2** (Effective density)**.** For regular activities $(X_i,o_i,E_i^{\mathrm{reg}})$ and forced activities $(Y_i,I_i,K_i^{\mathrm{for}})$, define

$$\mathrm{dens}\,\rho(U)=e^{v_{\mathrm{vac}}|\Lambda|-P^{\mathrm{path}}_{N,m}(\beta,U)-\sum_i \mathbf{1}_{X_i\ \mathrm{healthy}}E_i^{\mathrm{reg}}(U)}\cdot\sum_{\substack{\mathcal{A}_{\mathrm{for}}\subset\mathbb{N}\ \text{finite}\\ Y_i\ (i\in\mathcal{A}_{\mathrm{for}})\ \text{pairwise disjoint}\\ \bigcup_{i\in\mathcal{A}_{\mathrm{for}}}I_i=\mathrm{Flag}_m(U)}}\ \prod_{i\in\mathcal{A}_{\mathrm{for}}}K_i^{\mathrm{for}}(U)\,.$$

The sum covers every flagged site; each inventory is a nonempty subset of its support. This represents the density after integration.

If there are no flagged sites, only the empty family occurs and the forced sum equals one. When flags are present, their exact coverage and the disjointness of the supports assign each flagged site to precisely one selected forced term. Regular activities enter only on healthy supports. Their local regularity and cancellations, and the restrictions on forced activities, are specified in Section 13, Definitions 13.1–13.2.

For a link $e$, let $e^-$ be its initial site. The left derivative $D^L_{\dot{\mathcal{A}}}$ differentiates the increments $U_e\mapsto e^{t\dot{\mathcal{A}}_e}U_e$, with $\dot{\mathcal{A}}_e\in\mathfrak{g}$ and norm from Definition 2.1.

**Definition 3.3** (Local derivative seminorm)**.** For a regular activity anchored at $o_i$, set

$$\|E_i^{\mathrm{reg}}\|_m=\sup_{0\le q\le \mathfrak{q}_{J_m}}\ \sup_{U,\dot{\mathcal{A}}_1,\dots,\dot{\mathcal{A}}_q}\big|D^L_{\dot{\mathcal{A}}_1}\cdots D^L_{\dot{\mathcal{A}}_q}E_i^{\mathrm{reg}}(U)\big|\,.$$

The second supremum is over configurations healthy on $X_i$ and directions satisfying

$$|(\dot{\mathcal{A}}_h)_e|\le t_m\big(1+\mathrm{dist}(e^-,o_i)\big)^{\varpi_{J_m}}\,.$$

This measures ordered local derivatives in units of the permitted field size.

For a support $X$, let $\mathrm{cov}_{B_m}(X)$ be the minimum number of radius-$B_m$ lattice balls with connected union covering $X$. The exponential weight below penalizes supports that require many such balls.

**Definition 3.4** (Weighted activity sums)**.** For nonnegative majorants $e_i^{\mathrm{maj}} \geq \|E_i^{\mathrm{reg}}\|_m$ and $k_i^{\mathrm{maj}} \geq \sup_U |K_i^{\mathrm{for}}(U)|$, define

$$\mathcal{E}_m = \sup_x \sum_{o_i = x} e^{\sigma_{J_m} \mathrm{cov}_{B_m}(X_i)} e_i^{\mathrm{maj}},$$
$$\mathcal{K}_m = \sup_x \sum_{x \in Y_i} e^{\sigma_{J_m} \mathrm{cov}_{B_m}(Y_i)} k_i^{\mathrm{maj}}.$$

These sums measure regular and forced terms separately, with a penalty for large supports.

The chosen admissible classes $\mathcal{A}_{N,m,p}$, initial classes $\mathcal{I}_{N,p} \subset \mathcal{A}_{N,N,p}$, and adjacent-level relations

$$\mathcal{R}_{N,m,p} \subset \mathcal{A}_{N,m,p} \times \mathcal{A}_{N,m+1,p}$$

are constructed in Section 13, 13.14. They retain the Wilson initial representations, the coarse representations produced by the inductive assembly, and the estimates needed to continue that assembly and perform endpoint tuning. For every member, 13.11 supplies majorants with convergent weighted sums satisfying

$$\mathcal{E}_m \leq \epsilon_m^{\mathrm{reg}}, \qquad \mathcal{K}_m \leq \epsilon_m^{\mathrm{for}}. \tag{3.12}$$

These inequalities are properties of the chosen classes; the classes also retain the restrictions imposed by the inductive assembly. The relation $\mathcal{R}_{N,m,p}$ records its permitted adjacent pairs. The density identity and drift bound below hold for the pairs produced by RG and for the trajectories chosen in MT.

Along a continuous curve, supports, anchors, inventories and extra tests are fixed, and the coupling and vacuum scalar are continuous. Differences of regular derivatives on healthy configurations and of forced values admit majorants whose weighted sums over all addresses tend to zero as the parameter approaches any point. The weights are those of Definition 3.4 at the level of the representation, including $t_m$ in the derivative seminorm. Thus the input and output curves below use levels $m+1$ and $m$, respectively. This is the continuity of Definition 13.3 in Section 13: regular derivatives are measured on the healthy domain, and forced terms are compared in supremum norm on all configurations. The derivatives in this continuity condition act on regular activities on their healthy domains.

## 3.3 The inductive step

Exact integration removes one scale. Extracting its marginal curvature-square term into the coupling leaves a bounded drift error and returns a density in the next admissible class.

**Theorem 3.5** (RG. One step)**.** *For every $N, p$, $m < N$, reals $u_0 \leq u_1$ and continuous $\rho'_u \in \mathcal{A}_{N,m+1,p}$ with $\beta'_u = \beta(\rho'_u) \in \mathfrak{B}_{m+1}$ for all $u \in [u_0, u_1]$, there is a continuous $\rho_u \in \mathcal{A}_{N,m,p}$ with $(\rho_u, \rho'_u) \in \mathcal{R}_{N,m,p}$ and, for every $u \in [u_0, u_1]$ and coarse configuration $V$,*

$$\mathrm{dens}\, \rho_u(V) = (\mathcal{B}^{(m)}_{\beta'_u} \mathrm{dens}\, \rho'_u)(V),$$
$$\beta(\rho_u) = \beta'_u - \kappa_G \log \ell_m + e_u, \qquad |e_u| \leq \omega^{\mathrm{RG}}_{J_{m+1}}.$$

The proof occupies Sections 11–13, ending with Lemma 13.21. The density identity is exact; approximation enters the coupling drift and the activity estimates. Symmetric gauge fixing and the stationary background give the small-field integration; the large-field lower bound controls the forced terms. The one-loop action combines the inherited quadratic loop term, the fluctuation determinant and the Haar-coordinate Jacobian. Lemma 12.9 extracts its background curvature-square coefficient, giving the logarithmic change $-\kappa_G \log \ell_m$ up to a stage-dependent bounded error. Fourth-order Wilson matching fixes the remaining local terms, and the scalar coupling reset absorbs the residual kinetic contribution in Section 12. Section 13 assembles the activities with the bounds (3.12).

For the fixed construction, $\omega_j^{\mathrm{RG}}$ depends only on the stage; it is independent of $N, p$, the interval and the curve. Since $\widehat{\beta}_m = \widehat{\beta}_{m+1} - \kappa_G \log \ell_m$, a step changes $\beta - \widehat{\beta}$ by $e_u$ only. The input coupling lies in $\mathfrak{B}_{m+1}$; the trajectories of MT below keep both input and output couplings in their respective bands. Section 13, 13.21 supplies the continuous initial curve

$$\beta \longmapsto \rho_\beta^{\mathrm{W}} \in \mathcal{I}_{N,p}, \qquad \beta(\rho_\beta^{\mathrm{W}}) = \beta, \qquad \operatorname{dens} \rho_\beta^{\mathrm{W}} = e^{-\beta E^{\mathrm{W}}_{L_p D_N}} \quad (\beta \in \mathfrak{B}_N)\,, \tag{3.13}$$

whose vacuum scalar is zero. This is the base case for the induction.

### 3.4 Multiscale trajectories

A trajectory is a tuple $\rho_{N,p,m} \in \mathcal{A}_{N,m,p}$, $0 \le m \le N$, satisfying

$$\rho_{N,p,N} \in \mathcal{I}_{N,p}, \qquad (\rho_{N,p,m}, \rho_{N,p,m+1}) \in \mathcal{R}_{N,m,p} \quad (m < N)\,.$$

Write $\beta_{N,p,m} = \beta(\rho_{N,p,m})$. To select the bare coupling, start with the continuous Wilson curve across $\mathfrak{B}_N$ and apply RG. At the step $m+1 \to m$, the drift bound makes the endpoint output couplings bracket $[\bar\beta + (\kappa_G/2)s_m, \bar\beta + 2\kappa_G s_m] \subset \mathfrak{B}_m$. The intermediate value theorem gives a closed parameter subinterval whose endpoint couplings are the ends of this interval and whose couplings remain within it. Repeating this restriction reaches $\bar\beta$ at level zero. This is the shooting argument. It chooses the couplings for all $N, p$ before an increasing sequence of volume indices $\mathsf{p} : \mathbb{N} \to \mathbb{N}$ is prescribed.

**Theorem 3.6** (MT. Multiscale trajectory)**.** *There are bare couplings $\beta_{N,p}$ and trajectories $(\rho_{N,p,m})_{m=0}^N$, for all $N, p \in \mathbb{N}$, with $\rho_{N,p,N} = \rho^{\mathrm{W}}_{\beta_{N,p}}$ and $\beta_{N,p,N} = \beta_{N,p}$, satisfying*

$$\beta_{N,p,0} = \bar\beta, \qquad \beta_{N,p,m} \in \mathfrak{B}_m \quad (0 \le m \le N),$$
$$\operatorname{dens} \rho_{N,p,m} = \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} \operatorname{dens} \rho_{N,p,m+1} \quad (m < N)\,.$$

*For every $\varepsilon > 0$, some $m_0(\varepsilon)$ satisfies*

$$|\beta_{N,p,m} - \kappa_G s_m| \le \varepsilon s_m \qquad (N \ge m \ge m_0,\ p \in \mathbb{N})\,.$$

The proof is the shooting argument of Section 14, 14.1, with 14.1–14.2. Fix this family of trajectories for all subsequent constructions. The exact drift bound of RG holds on every adjacent pair of these trajectories, and all their integrals equal $Z_{L_p D_N, \beta_{N,p}}$ by (3.7) and the Wilson start. The path arrays remain independent of $p$.

The running law follows by summing the drift from the fixed endpoint:

$$|\beta_{N,p,m} - \widehat{\beta}_m| \le \sum_{i=0}^{m-1} \omega^{\mathrm{RG}}_{J_{i+1}}\,. \tag{3.14}$$

Row 5 of Section 9 bounds the sum by $(\kappa_G/4)s_m$, which also places the trajectory couplings in $\mathfrak{B}_m$. Its stronger bound $\omega^{\mathrm{RG}}_{J_{i+1}}/\log \ell_i \le J_{i+1}^{-2}$ makes the sum $o(s_m)$: split at a fixed level, then bound the tail by $\sup_{i\ge i_0} J_{i+1}^{-2}$ times $s_m$. Since $J_i \to \infty$ and $s_m \to \infty$, the finite initial sum and $\bar\beta$ disappear after division by $s_m$, uniformly in $N \ge m$ and $p$.

The same level choices give $\log \ell_m \to \infty$ and $\log \ell_m/s_m \to 0$. The physical periods $L_p$ are strictly increasing multiples of four, and every positive integer divides $L_p$ for infinitely many $p$.

Endpoint tuning varies the common level-zero coupling analytically, as needed for the full ultraviolet limit.

**Proposition 3.7** (Endpoint tuning)**.** *There is a width $w_{\mathrm{tun}} > 0$, independent of $N, p$, such that, for every $N, p$ and $s \in [-1, 1]$, a unique bare coupling $\beta_{N,p}(s)$ within the specified recursion reaches $\bar\beta + w_{\mathrm{tun}} s$ at level zero. It satisfies $\beta_{N,p}(0) = \beta_{N,p}$ and has a holomorphic extension $b_{N,p}$ to $|z| < 1$ with*

$$b_{N,p}(s) = \beta_{N,p}(s) \qquad (-1 < s < 1)\,. \tag{3.15}$$

The proof is in Section 14.1, Lemma 14.3.

## 3.5 Local fields

Local fields are normalized along the same trajectories. Their polynomial labels specify the continuum Taylor coefficients, while transfer to a fixed reference level determines the insertion at the fine cutoff.

The labels use radial, or Fock–Schwinger, gauge: the condition in the next definition sets the radial component of the connection to zero.

**Definition 3.8** (Polynomial labels)**.** Let $\mathcal{J}$ consist of formal $\mathfrak{g}$-valued arrays $\mathcal{A}_a(x) = \sum_{k\ge 0} \mathcal{A}_{k,a}(x)$, with $\mathcal{A}_{k,a}$ homogeneous of degree $k+1$ and

$$\sum_a x^a \mathcal{A}_{k,a}(x) = 0, \qquad V_d = \mathbb{C}[\mathcal{J}]^{\mathrm{Ad}(G)}_{\mathrm{weight}\le d}\,.$$

Each coordinate of $\mathcal{A}_k$ has bidegree $(k+2, 1)$: engineering weight and number of polynomial factors. Fix bihomogeneous bases, the supremum coefficient norm, conjugation and the induced $H_4$ action. These labels describe invariant curvature polynomials; curvature has weight two and a covariant derivative adds one.

Write $\mathbf{1}$ for the constant polynomial one. Section 10, 10.2 constructs a local section $P_d^{\mathrm{loc}}[\mathbf{c}](y; U)$, linear in $\mathbf{c} \in V_d$ and polynomial in the real and imaginary parts of link matrix entries on a fixed finite stencil $\Sigma_d \subset \mathbb{Z}^4$. It is gauge invariant, commutes with complex conjugation, is $H_4$ covariant, and sends the polynomial unit to one.

Its Taylor normalization compares lattice holonomies with a smooth connection jet. For $\mathsf{j} = (\mathcal{A}_k)_k \in \mathcal{J}$, put $\mathcal{A}^{\le d} = \sum_{k+2\le d} \mathcal{A}_k$ and let $U(h_{\mathrm{lat}}, \mathsf{j})$ be its test holonomies at spacing $h_{\mathrm{lat}}$. On a positive link $(y, a)$, with $y \in \mathbb{Z}^4$, these are defined by

$$U(h_{\mathrm{lat}}, \mathsf{j})(y, a) = u(1), \qquad u'(t) = u(t) h_{\mathrm{lat}} \mathcal{A}_a^{\le d}(h_{\mathrm{lat}}(y + te_a)), \qquad u(0) = 1_G\,. \tag{3.16}$$

The jet dilation is $\mathrm{sc}_{h_{\mathrm{lat}}} \mathsf{j} = (h_{\mathrm{lat}}^{k+2} \mathcal{A}_k)_k$. For every $\mathbf{c} \in V_d$ and $0 \le q \le d$, 10.2 proves

$$\frac{d^q}{dh_{\mathrm{lat}}^q} P_d^{\mathrm{loc}}[\mathbf{c}](0; U(h_{\mathrm{lat}}, \mathsf{j}))\Big|_{h_{\mathrm{lat}}=0} = \frac{d^q}{dh_{\mathrm{lat}}^q} \mathbf{c}(\mathrm{sc}_{h_{\mathrm{lat}}} \mathsf{j})\Big|_{h_{\mathrm{lat}}=0}\,. \tag{3.17}$$

Thus a homogeneous label of engineering weight $w$ specifies the coefficient at order $h_{\mathrm{lat}}^{w}$ in this expansion.

Choose the reference level $m_{\mathrm{ref}}(d)$ with $J_{m_{\mathrm{ref}}(d)} \geq d+3$ and $\mathfrak{q}_{J_m} \geq d+1$, $\varpi_{J_m} \geq d$ for $m \geq m_{\mathrm{ref}}(d)$, as in Section 14, 14.6. For a basis vector $b_{d,i}$ of bidegree $(w_i, k_i)$ define the factor-count scaling by

$$\mathsf{D}_t^{\mathrm{fac}} b_{d,i} = t^{k_i} b_{d,i}\,.$$

This rescales each polynomial factor in units of the small-field threshold. For $N \geq m_{\mathrm{ref}}(d)$ and terminal input $\mathbf{v} \in V_d$, the marked density at the origin is initialized and evolved by

$$H^{\mathrm{ins}}_{N,p,N;d}(\mathbf{v};U) = \mathrm{dens}\,\rho_{N,p,N}(U)\, P_d^{\mathrm{loc}}[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}}\mathbf{v}](0;U), \tag{3.18}$$

$$H^{\mathrm{ins}}_{N,p,m;d}(\mathbf{v}) = \ell_m^d \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} H^{\mathrm{ins}}_{N,p,m+1;d}(\mathbf{v}) \quad (m_{\mathrm{ref}}(d) \leq m < N)\,. \tag{3.19}$$

For fixed $N, p, d$, the transfer construction chooses coefficient vectors $\mathbf{x}_m(\mathbf{v})$ and regular and forced remainders $R^{\mathrm{reg}}_{m,i}(\mathbf{v})$ and $Q^{\mathrm{for}}_{m,i}(\mathbf{v})$. All are linear in $\mathbf{v}$. Write $\mathbf{x}_m = x_{m,0}\mathbf{1} + \mathbf{x}^{\circ}_m$, and abbreviate $\rho_{N,p,m}$ by $\rho_m$. The regular remainders have supports $Z_{m,i}$ containing the origin. Define their sum and the forced remainder by

$$\begin{aligned}
\mathcal{R}^{\mathrm{reg}}_m(\mathbf{v};U) &= \sum_i \mathbf{1}_{Z_{m,i}\ \mathrm{healthy}} R^{\mathrm{reg}}_{m,i}(\mathbf{v};U),\\
\mathcal{R}^{\mathrm{for}}_m(\mathbf{v};U) &= e^{v_{\mathrm{vac}}|\Lambda| - P^{\mathrm{path}}_{N,m}(\beta(\rho_m),U) - \sum_i \mathbf{1}_{X_i\ \mathrm{healthy}} E^{\mathrm{reg}}_i(U)}\\
&\quad \cdot \sum_{\mathcal{A}_{\mathrm{for}}\ \mathrm{eligible}} \sum_{i \in \mathcal{A}_{\mathrm{for}}} Q^{\mathrm{for}}_{m,i}(\mathbf{v};U) \prod_{j \in \mathcal{A}_{\mathrm{for}}\setminus\{i\}} K^{\mathrm{for}}_j(U)\,.
\end{aligned} \tag{3.20}$$

All unmarked quantities in this formula are taken from $\rho_m$, and eligibility is the disjointness and exact flag coverage in Definition 3.2. The marked density has the exact decomposition

$$\begin{aligned}
H^{\mathrm{ins}}_{N,p,m;d}(\mathbf{v};U) = \mathrm{dens}\,\rho_m(U) \Big[x_{m,0}(\mathbf{v}) + \mathbf{1}_{\Sigma_d\ \mathrm{healthy}} P_d^{\mathrm{loc}}[\mathsf{D}^{\mathrm{fac}}_{t_m^{-1}} \mathbf{x}^{\circ}_m(\mathbf{v})](0;U)\Big]\\
+ \mathrm{dens}\,\rho_m(U) \mathcal{R}^{\mathrm{reg}}_m(\mathbf{v};U) + \mathcal{R}^{\mathrm{for}}_m(\mathbf{v};U)\,.
\end{aligned} \tag{3.21}$$

Here the stencil $\Sigma_d$ is placed at the origin on the level-$m$ torus. If $10\,\mathrm{dist}(x,0) < L_p D_m$ for every $x \in Z_{m,i}$, the regular remainder $R^{\mathrm{reg}}_{m,i}$ has zero radial-jet Taylor coefficients through order $d$; the forced remainders vanish outside their unmarked eligibility events. Their format and bounds are proved in Section 14, 14.8–14.9. We use the fixed decomposition selected there. It supplies the invertible terminal map of 14.6:

$$\mathbf{x}_N(\mathbf{v}) = \mathbf{v}, \qquad \mathbf{x}_{m_{\mathrm{ref}}(d)}(\mathbf{v}) = \mathsf{Z}_{N,p,d}\mathbf{v}\,.$$

For $N < m_{\mathrm{ref}}(d)$ choose invertible maps to define the insertions; transfer normalization is required only at and above the reference level.

**Definition 3.9** (Renormalized insertions)**.** Put $\mathcal{T} = \bigsqcup_{d\geq 0} V_d$ and define, at every cutoff,

$$I_{N,p}((d,\mathbf{c}),y;U) = O^{(d)}_{N,p}(\mathbf{c},y;U) = a_N^{-d} P_d^{\mathrm{loc}}[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}} \mathsf{Z}^{-1}_{N,p,d}\mathbf{c}](y;U)\,.$$

This fixes the insertion along the trajectory. Its distinguished unit coefficient is $\mathbf{u}_d = a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}$, whose insertion equals one for $N \geq m_{\mathrm{ref}}(d)$. The scaling $\mathsf{D}_t^{\mathrm{fac}}$ counts factors, not engineering weight.

The inverse terminal map makes the transferred coefficient equal to the prescribed $\mathbf{c}$ at the reference level. Thus $\mathbf{c}$ is specified at the fixed physical scale $a_{m_{\mathrm{ref}}(d)}$, in that level's lattice units. The factor $\ell_m^d = (a_m/a_{m+1})^d$ uses the degree label of the whole space $V_d$; the terminal map allows mixing between its engineering weights. The factor $a_N^{-d}$ converts the normalized lattice insertion to physical units.

The unit identity follows from $\mathbf{x}_m(a_N^d\mathbf{1}) = a_m^d\mathbf{1}$, proved with the transfer symmetries in Section 14, 14.7. This explains why the coefficient giving the constant insertion is $a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}$, even though $V_d$ also contains polynomials of lower engineering weight.

The bare couplings of MT and these insertions now specify one Wilson family. Each insertion is continuous, gauge invariant and local on its fixed stencil. MT at $m = N$ gives the volume-uniform asymptotic slope $\kappa_G > 0$. These are the lattice inputs to the continuum limits.

## 4 The continuum limit

We first remove the lattice spacing at each fixed volume, then let the volume increase along one refinement. Throughout, the Wilson family and insertions are those of Section 3; all couplings $\beta_{N,p}$ and insertion maps are chosen before the increasing sequence of volume indices is prescribed. Constants may depend on the fixed group, its realization, the penalty and the construction choices of Section 9. Additional dependence is stated below.

For positions in $(\mathbb{R}^4)^n$, set

$$\mathrm{Conf}_n = \{x : x_i \neq x_j \text{ for } i \neq j\}, \qquad \mathrm{Gen}_n = \{x : x_i^a \neq x_j^a \text{ for } i \neq j,\ a = 1, \dots, 4\},$$
$$\mathrm{NoHit}_{L,n} = \{x : x_i - x_j \notin L\mathbb{Z}^4 \text{ for } i \neq j\}.$$

The last domain excludes collisions on a torus of physical side $L$. A separated test is a smooth function on $(\mathbb{R}^4)^n$ with compact closed support in $\mathrm{Conf}_n$. It is $p$-admissible if its support lies in $((-L_p/2, L_p/2)^4)^n$. Write $\mathrm{sep}_\infty(x) = \min_{i\neq j} |x_i - x_j|_\infty$, with value $+\infty$ for $n \leq 1$.

For a finite list $\boldsymbol{l}$ of pairs $(i, a) \in \{1, \dots, n\} \times \{1, \dots, 4\}$, define

$$\Delta_{\boldsymbol{l}} = \prod_{(i,a)\in\boldsymbol{l}} a_N^{-1}(\tau_{i,a} - 1), \qquad \partial_{\boldsymbol{l}} = \prod_{(i,a)\in\boldsymbol{l}} \partial_{x_i^a}.$$

Here $\tau_{i,a}$ shifts slot $i$ forward by one lattice link. These differences measure derivatives in physical units. Torus separation is the minimum, over distinct slots, of $\min_{z\in L_p\mathbb{Z}^4} |\hat{y}_i - \hat{y}_j + z|_\infty$, with value $+\infty$ for $n \leq 1$. An element $v_i = (d_i, c_i) \in \mathcal{T}$ retains both the polynomial label and its degree; we call it a token.

Marked transfers along the trajectories of Theorem 3.6 give bounds independent of the volume. They control compactness for each fixed list of fields and derivatives.

**Proposition 4.1** (UB. Uniform difference bounds)**.** *For every $v \in \mathcal{T}^n$, finite difference list $\boldsymbol{l}$ and $r > 0$, there are $B(v, \boldsymbol{l}, r) \geq 0$ and $N_0(v, \boldsymbol{l}, r)$ such that, for every $N \geq N_0$, every $p$ and every lattice tuple of torus separation at least $r$,*

$$|\Delta_{\boldsymbol{l}} M^{\mathrm{lat}}_{N,p}(v; y)| \leq B.$$

The proof is in Section 14, 14.8–14.25. Marked transfers control insertion coefficients and remainders. The chessboard estimate reflects an insertion into neighbouring blocks and bounds a mixed correlation by the product of the resulting reflected moments. Choosing an intermediate

level whose physical blocks fit below the prescribed separation permits its use for the differentiated insertions. The reflected-moment constant in 14.12 is chosen before that level, although its cutoff threshold may depend on it.

By Proposition 3.7, the common endpoint can be varied analytically.

This tuning gives convergence along the full ultraviolet sequence. The limiting kernel is periodic in each position and smooth away from periodic collisions, including configurations where positions share some coordinates.

**Theorem 4.2** (UV. Fixed-volume limit)**.** *For every $p, n$ and $v \in \mathcal{T}^n$, there is $M^{\mathrm{UV}}_{p,v} \in C^\infty(\mathrm{NoHit}_{L_p,n})$ such that, for every finite $\boldsymbol{l}$ and every $p$-admissible separated test $f$,*

$$\lim_{N\to\infty} a_N^{4n} \sum_{y\in\Lambda^n_{L_pD_N}} f(\hat{y})\Delta_{\boldsymbol{l}} M^{\mathrm{lat}}_{N,p}(v;y) = \int f(x)\partial_{\boldsymbol{l}} M^{\mathrm{UV}}_{p,v}(x)\,dx\,.$$

For $\boldsymbol{l} = \varnothing$, the left side is $\lim_N C^{\mathrm{lat}}_{N,p}(v;f)$. Thus the inner limits in Definition 4.3 below exist at every admissible volume. Admissibility places the test support inside $\mathrm{NoHit}_{L_p,n}$, so the displayed integrand is compactly supported and integrable.

The proof is in Section 14, 14.30–14.31. At a fixed intermediate level, the vacuum scalar cancels between the correlation numerator and its partition function. Analytic tuning and the fixed-level transfer estimates make the normalized integrals of the transferred insertions Cauchy as $N$ increases. For insertion sites obtained by repeated blocking from common finer sites, comparison with the original correlations has error bounded by a constant times $e^{-B_m/64}$. First choose the intermediate level $m$, then take both cutoffs large; the triangle inequality gives the Cauchy property for the original correlations. UB extends convergence from the dense set of these sites and identifies every derivative. 14.31 also treats multilinear interpolants evaluated at any sequence $x_N \to x \in \mathrm{NoHit}_{L_p,n}$, including when the points cross a boundary of the chosen fundamental cell. Thresholds may depend on $p$.

The following definition keeps the order of limits explicit. Write $C^{\mathrm{UV}}_p(v;f)$ for a fixed-volume smeared limit.

**Definition 4.3** (Ordered convergence)**.** A Wilson family converges to $K$ along $P : \mathbb{N} \to \mathbb{N}$ if, for every $n, v$ and separated test $f$, the test is eventually $P(k)$-admissible and there are numbers $C^{\mathrm{UV}}_p(v;f)$ with

$$C^{\mathrm{lat}}_{N,p}(v;f) \longrightarrow C^{\mathrm{UV}}_p(v;f) \quad (N\to\infty) \qquad \text{at every admissible } p,$$
$$C^{\mathrm{UV}}_{P(k)}(v;f) \longrightarrow \int f(x)K_n(v;x)\,dx\,.$$

A table assigns to each degree list $(d_i)_{i=1}^n$ and $x \in \mathrm{Conf}_n$ a complex multilinear function $S_n(c;x)$ of $c_i \in V_{d_i}$. For $v_i = (d_i, c_i)$ write also $S_n(v;x)$, and for the word $w = (v_i(x_i))_{i=1}^n$ write $S(w)$. Conjugation and the signed permutation action are those of the polynomial labels; a unit slot carries the distinguished coefficient $\mathbf{u}_d = a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}$ of Definition 3.9. The following conditions specify the Euclidean properties of the table to be constructed.

**Definition 4.4** (E1–E3. Euclidean symmetries)**.** E1 is simultaneous permutation symmetry, $S_0 = 1$ and deletion of a slot carrying $\mathbf{u}_d$. E2 is

$$S_n(hc;hx) = S_n(c;x) \quad (h\in H_4), \qquad S_n(\bar{c};x) = \overline{S_n(c;x)}\,.$$

E3 is translation invariance.

A word is positive-side for axis $a$ when all its positions have positive $a$th coordinate. The product of words concatenates their lists, so reflection positivity tests a correlation with one reflected group of insertions.

**Definition 4.5** (E4–E5. Reflection positivity)**.** E4 requires, for every axis $a$, positive-side separated words $w_i$ and complex coefficients $z_i$,

$$\sum_{i,j} \bar{z}_i z_j S(\mathsf{s}_a^{\mathrm{refl}} w_i^* w_j) \geq 0 \,.$$

The star reverses the word and applies conjugation to its labels; $\mathsf{s}_a^{\mathrm{refl}}$ reflects positions and acts on labels through $H_4$. E5 requires the same positivity for finite sums of words smeared against joint smooth compactly supported tests on positive-side separated configurations, and absolute integrability of every mixed pairing.

For a fixed label list, $D_x^k$ denotes the $k$th Fréchet derivative in the positions, with the product Euclidean norm. The next bound controls approach to collision sets uniformly over translations and over configurations with large diameter.

**Definition 4.6** (E6–E7. Euclidean regularity)**.** E6 is smoothness on $\mathrm{Conf}_n$. E7 requires, for each $n, d_i, c_i, k$, constants $C(n,d,c,k) \geq 0$ and an integer $A(n,d,c,k) \geq 0$ such that

$$\|D_x^k S_n(c;x)\| \leq C r^{-A} \qquad (0 < r \leq 1,\ \mathrm{sep}_\infty(x) \geq r) \,.$$

One volume refinement will serve every field and all subsequent constructions.

**Theorem 4.7** (IV. Infinite-volume limit)**.** *For every strictly increasing* $\mathsf{p} : \mathbb{N} \to \mathbb{N}$*, there are a strictly increasing* $\mathsf{q} : \mathbb{N} \to \mathbb{N}$ *and a multilinear table* $S$ *such that the Wilson family converges to* $S$ *along* $P = \mathsf{p} \circ \mathsf{q}$ *in the sense of Definition 4.3. The table satisfies E1–E7.*

*Remark* 4.8 (Dependence on the volume sequence). The limiting table $S$ may depend on the prescribed sequence $\mathsf{p}$ and the refinement $\mathsf{q}$, hence on the retained volume indices $P = \mathsf{p} \circ \mathsf{q}$.

Fix $\mathsf{p}$, $\mathsf{q}$, $P$ and $S$ from now on.

The proof is in Section 15, Lemmas 15.1–15.5, using the ultraviolet estimates of Lemmas 14.16–14.24. The extraction gives local smooth volume limits on $\mathrm{Conf}_n$ from Proposition 4.1 and Lemma 14.31, as recorded in (15.1) (E6). On the periodic generic domain the fixed-volume kernels are the restrictions of $M_{p,v}^{\mathrm{UV}}$, retaining UV's full inner limit at every admissible volume. The chessboard root bounds give E7 at order zero; the corrected derivative maps of Lemma 15.4 give it at positive orders. Lattice reflection positivity for finite sums of smeared words passes to the limit, giving E5; E4 follows from E5 together with E1 and E6. A further volume refinement in Lemma 15.4 fixes the corrected coefficient maps, which adjust labels of different degrees so their correlations agree or differ by a positional derivative; ordered convergence persists.

Write $v(x) = (v_1(x_1), \ldots, v_n(x_n))$ for a spectator word. Two polynomial labels represent the same observable when no separated correlation distinguishes them.

**Definition 4.9** (Observable fields)**.** Set

$$\begin{aligned} \mathcal{N}_d(S) &= \big\{ c \in V_d : S_{n+1}(c(x_0), v(x)) = 0 \\ &\qquad\qquad \text{for every } n, v \text{ and } x \in \mathrm{Conf}_{n+1} \big\}, \\ F_d &= V_d / \mathcal{N}_d(S), \qquad \overline{[c]} = [\bar{c}], \qquad h[c] = [hc] \quad (h \in H_4), \\ \iota_d[c](v;x) &= S_{n+1}(c(x_0), v(x)), \qquad \mathcal{L} = \mathrm{span} \bigcup_d \iota_d F_d \,. \end{aligned}$$

The maps $\iota_d$ give the responses to all spectators. Choose conjugation-invariant quotient norms.

E2 makes $\mathcal{N}_d(S)$ invariant under conjugation and $H_4$, so both displayed actions on $F_d$ are well defined. Permutation symmetry makes correlations independent of representatives in every slot: for $\boldsymbol{A} = ([c_1], \dots, [c_n])$, put $S_n(\boldsymbol{A}; x) = S_n(c; x)$. The unit field is $\mathbf{1}_d = [\mathbf{u}_d]$, with the normalization of Definition 3.9.

To state the dimension of $F_d$, introduce the free comparison space.

**Definition 4.10** (Free Maxwell comparison)**.** For $d_{\mathfrak{g}}$ independent colours, take

$$\mathbb{E}A_a^\alpha(x)A_b^\beta(y) = \delta_{\alpha\beta}\delta_{ab}\Gamma(x-y), \qquad \Gamma(x) = \frac{1}{4\pi^2|x|^2}, \qquad F_{ab} = \partial_a A_b - \partial_b A_a\,.$$

Let $M_d^{\mathrm{Max}}$ consist of complex $\mathrm{Ad}(G)$-invariant Wick polynomials in curvature jets of weight at most $d$, modulo antisymmetry, the linear Bianchi identities and $\sum_a \partial_a F_{ab} = 0$. Their separated Wick correlations are $S^{\mathrm{Max}}$.

The constant one is the Maxwell unit, coefficient conjugation gives its real structure, and inclusions retain the same polynomial at a larger weight bound. Free field differentiation is $\partial_a^{\mathrm{Max}}(:P:) =: \partial_a P :$ and the natural rotation action is $R_d^{\mathrm{Max}}(h)(:P:) =: hP :$, where the jet action is determined by

$$hF_{ab}^\alpha = \sum_{i,j} h_{ia}h_{jb}F_{ij}^\alpha, \qquad h\partial_a = \sum_i h_{ia}\partial_i h \quad (h \in O(4))\,.$$

These operations preserve the defining relations and give the structures used in the comparison below.

The quotient fields admit inclusions that preserve correlations and derivatives that act as positional derivatives. Their dimensions equal those of the Maxwell spaces.

**Proposition 4.11** (FC. Field calculus)**.** *For the table $S$ of Theorem 4.7, there are complex linear maps $\jmath_{d,e} : F_d \to F_e$ for $d \le e$ and $\partial_a : F_d \to F_{d+1}$ for every $d, a$ such that, for every $A \in F_d$,*

$$\iota_e \circ \jmath_{d,e} = \iota_d, \qquad \iota_{d+1}(\partial_a A) = \partial_{x_0^a}\iota_d(A),$$
$$\dim_{\mathbb{C}} F_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}}\,.$$

The proof is in Section 15, 15.4, using the corrected coefficient maps chosen with $S$. Each $\iota_d$ is injective by Definition 4.9, so the inclusions are unique and injective, compose as $\jmath_{e,f}\jmath_{d,e} = \jmath_{d,f}$ and satisfy $\jmath_{d,d} = \mathrm{id}$. By E1 they carry $\mathbf{1}_d$ to $\mathbf{1}_e$. Their images are nested, hence $\mathcal{L} = \bigcup_d \iota_d F_d$. These inclusions are characterized by equality of correlations across degrees, whose tokens $(d, c)$ and $(e, c)$ have different insertion normalizations.

Permutation symmetry gives, for every arity $n$, degree list $(d_i)$, $\boldsymbol{A} \in \prod_{i=1}^n F_{d_i}$, $x \in \mathrm{Conf}_n$, slot $i$ and axis $a$,

$$\partial_{x_i^a} S_n(\boldsymbol{A}; x) = S_n(A_1, \dots, \partial_a A_i, \dots, A_n; x)\,.$$

E2 and the injectivity of the response maps imply that inclusions and derivatives commute with conjugation, that inclusions commute with $H_4$, and that

$$h\partial_a = \sum_b h_{ba}\partial_b h \qquad (h \in H_4)\,.$$

## 5 Stress and rotations

We compare the fields of Section 4 with Maxwell fields at small distances, then normalize a conserved tensor by its shell fluxes. The volume refinement and the table $S$ remain fixed throughout. Subsequences taken below concern only comparison radii.

The comparison permits the basis of local fields to vary with the radius. This accommodates mixing among fields in a fixed finite-dimensional space.

**Definition 5.1** (Scaling frames)**.** A scaling sequence consists of $r_k > 0$, $r_k \to 0$, and complex linear isomorphisms

$$\mathsf{U}^{\mathrm{sc}}_{k,d} : M_d^{\mathrm{Max}} \longrightarrow F_d \,.$$

It is stable if, for every $D \in \mathbb{N}$, these maps eventually preserve conjugation, $H_4$, units and inclusions exactly in all degrees $d \le e \le D$. Scaling dimensions are included in the maps: an ordinary comparison insertion is $\mathsf{U}^{\mathrm{sc}}_{k,d} u$ at $r_k x$, with no further factor $r_k^d$.

The frames compare fields without identifying their null ideals.

After rescaling positions and choosing these frames, the interacting correlations approach their Maxwell counterparts. The first comparison measures the error by its integral on separated compact sets; one sequence works for every fixed list of fields.

**Proposition 5.2** (MC. Maxwell comparison)**.** *There is one stable scaling sequence such that, for every arity $n$, degree list $(d_i)$ and $u_i \in M_{d_i}^{\mathrm{Max}}$,*

$$S_n(\mathsf{U}^{\mathrm{sc}}_{k,d_i} u_i; r_k \cdot\,) \longrightarrow S_n^{\mathrm{Max}}(u; \cdot) \quad \textit{in } L^1_{\mathrm{loc}}(\mathrm{Conf}_n) \,.$$

The proof is in Section 16, 16.6. The lattice comparison stops near the requested physical scale and compares the resulting correlations with the free Gaussian ones. Finitely many separating correlations make the comparison maps injective; the upper bound $\dim_{\mathbb{C}} F_d \le \dim_{\mathbb{C}} M_d^{\mathrm{Max}}$ from Proposition 4.11 makes them isomorphisms. Finite-dimensional corrections impose the eventual equalities of Definition 5.1.

The integral norm retains uniformity over tests. Indeed, write $g_k(x)$ for the difference of the two smooth kernels in Proposition 5.2. For compact $\mathcal{K} \subset \operatorname{int} \mathcal{K}' \subset \mathcal{K}' \subset \mathrm{Conf}_n$, smooth approximation to $\overline{g_k}/|g_k|$ with a cutoff equal to one on $\mathcal{K}$ gives

$$\int_{\mathcal{K}} |g_k(x)|\, dx \le \sup_{\substack{f \in C_c^\infty,\ \operatorname{supp} f \subset \mathcal{K}' \\ \|f\|_\infty \le 1}} \left| \int f(x) g_k(x)\, dx \right| .$$

Conversely, the tested error on $\mathcal{K}$ is bounded by $\|f\|_\infty \int_{\mathcal{K}} |g_k|$. Thus local integral convergence is equivalent to convergence uniform over the sup-norm unit ball of tests on each fixed compact set. This comparison is used to construct and normalize the conserved current before the stress tensor is available.

To read translation and angular charge, integrate a conserved symmetric tensor through a shell surrounding just one insertion. Let $A \in F_d$, let $\boldsymbol{B}$ be a list of $n$ spectator fields at $x$, and take $(x_0, x) \in \mathrm{Conf}_{n+1}$.

**Definition 5.3** (Shell fluxes)**.** For $J_{ab} = J_{ba} \in F_4$ with $\sum_a \partial_a J_{ab} = 0$ and $K_{\mathrm{rot}} \in \mathfrak{so}(4)$, choose a real shell cutoff $\chi \in C_c^\infty(\mathbb{R}^4)$ equal to one near $x_0$ and supported away from spectators. Define

$$\mathcal{P}_b^J = -\sum_a \int \partial_a \chi(y) S(J_{ab}(y), A(x_0), \boldsymbol{B}(x))\, dy,$$
$$\mathcal{Y}_{K_{\mathrm{rot}}}^J = -\sum_{a,b} \int \partial_a \chi(y)(K_{\mathrm{rot}}(y - x_0))_b S(J_{ab}(y), A(x_0), \boldsymbol{B}(x))\, dy\,.$$

These integrals measure the flux without meeting an insertion point.

The fluxes are independent of the shell cutoff. For two choices, their difference vanishes near every insertion; integration by parts and Proposition 4.11 remove the divergence term. In the angular flux the additional term is $\sum_{a,b}(K_{\mathrm{rot}})_{ba} J_{ab} = 0$, since $K_{\mathrm{rot}}$ is skew-symmetric and $J$ is symmetric.

A field is real when it is fixed by conjugation. A family $J_{ab} \in F_4$ is $H_4$-tensorial when

$$hJ_{ab} = \sum_{i,j} h_{ia} h_{jb} J_{ij} \qquad (h \in H_4)\,.$$

It is conserved when $\sum_a \partial_a J_{ab} = 0$ in $F_5$ for every $b$, and centered when every component has zero one-point function. Section 16, 16.7 proves that the centered real symmetric $H_4$-tensorial conserved tensors of degree four form a real line. Along a refinement of the sequence in Proposition 5.2, positive multiples of the rescaled nonzero current approach the free stress line in the degree-four frames. This identifies the current needed for the shell laws.

For a nonzero member $J$ of this line, 16.8 constructs complex linear charge maps $\mathsf{p}_b : F_d \to F_{d+1}$ and $\mathsf{Y}_{K_{\mathrm{rot}}} : F_d \to F_d$ satisfying, for every degree, field, spectator list, separated configuration and shell cutoff $\chi$,

$$\mathcal{P}_b^J = S((\mathsf{p}_b A)(x_0), \boldsymbol{B}(x)), \qquad \mathcal{Y}_{K_{\mathrm{rot}}}^J = S((\mathsf{Y}_{K_{\mathrm{rot}}} A)(x_0), \boldsymbol{B}(x))\,.$$

The angular map is real linear in $K_{\mathrm{rot}}$.

Section 16, 16.10 identifies these maps with differentiation and rotations. It constructs continuous complex linear representations $R_d$ of $O(4)$, extending $H_4$, fixing units and commuting with conjugation and inclusions. For a real $c_\star \neq 0$ common to every degree, the generators and charge maps satisfy

$$\mathsf{L}_d(K_{\mathrm{rot}}) = \frac{d}{dt} R_d(e^{tK_{\mathrm{rot}}})\Big|_{t=0}, \qquad \mathsf{p}_b = c_\star \partial_b, \qquad \mathsf{Y}_{K_{\mathrm{rot}}} = c_\star \mathsf{L}_d(K_{\mathrm{rot}})\,.$$

The proof controls exterior angular flux before extending the rotation identity to all separated configurations. The resulting table has full covariance: for every arity $n$, degree list $(d_i)$, $\boldsymbol{A} \in \prod_i F_{d_i}$, $x \in \mathrm{Conf}_n$ and $h \in O(4)$, 16.11 gives

$$S_n(R_{d_i}(h) A_i; hx) = S_n(\boldsymbol{A}; x)\,.$$

For these representations, $O(4)$-tensorial means $R_4(h) J_{ab} = \sum_{i,j} h_{ia} h_{jb} J_{ij}$ for every $h \in O(4)$, with the same index convention as the discrete action above.

The common constant $c_\star$ fixes the normalization of the stress tensor by its action on fields.

**Theorem 5.4** (ST. Stress tensor). *There is a real symmetric $O(4)$-tensorial family $T_{ab} \in F_4$ with $\sum_a \partial_a T_{ab} = 0$ such that, for every degree, field, spectator list, separated configuration and shell cutoff of Definition 5.3, every $b$ and every $K_{\mathrm{rot}} \in \mathfrak{so}(4)$,*

$$\mathcal{P}_b^T = \partial_{x_0^b} S(A(x_0), \boldsymbol{B}(x)), \qquad \mathcal{Y}_{K_{\mathrm{rot}}}^T = S((\mathsf{L}_d(K_{\mathrm{rot}})A)(x_0), \boldsymbol{B}(x)) .$$

The proof is in Section 16, 16.10–16.11. Divide the current $J$ by $c_\star$ and average its tensor transform over $O(4)$ with probability Haar measure. Covariance and the transformation laws for derivatives and generators give the same shell actions for each transformed current; integrating these identities preserves both actions. Symmetry, reality, centering and conservation also pass through this finite-dimensional integral. Fix this averaged normalized tensor for all subsequent uses; it is centered, since $J$ is centered and $R_4$ preserves zero one-point values.

Write $\mathsf{L}_d^{\mathrm{Max}}(K_{\mathrm{rot}}) = \frac{d}{dt} R_d^{\mathrm{Max}}(e^{tK_{\mathrm{rot}}})\big|_{t=0}$ for the free rotation generators. The free stress $T^{\mathrm{Max}}$ is normalized by the same shell laws, with $\partial_a^{\mathrm{Max}}$ and $\mathsf{L}_d^{\mathrm{Max}}$, in 16.1. Their sign gives

$$T_{ab}^{\mathrm{Max}} = -\sum_{c,\alpha} : F_{ac}^\alpha F_{bc}^\alpha : + \frac{\delta_{ab}}{4} \sum_{i,j,\alpha} : F_{ij}^\alpha F_{ij}^\alpha : .$$

The final comparison includes any chosen stress insertions. For a free list $u_1, \dots, u_n$ with $u_i \in M_{d_i}^{\mathrm{Max}}$, a stress-slot choice is a subset $I \subset \{1, \dots, n\}$ with component pairs $(a_i, b_i)$ for $i \in I$. Put

$$A_i^0 = \begin{cases} T_{a_i b_i}^{\mathrm{Max}}, & i \in I, \\ u_i, & i \notin I. \end{cases}$$

The shell laws fix the stress normalization, so its comparison insertion needs only the engineering factor $r_k^4$. In the next limit $D_x^j$ differentiates the whole function $x \mapsto S_n(\boldsymbol{A}_k; r_k x)$, including the rescaled positions.

**Proposition 5.5** (SC. Differentiated short-distance comparison). *There is one stable scaling sequence such that, for every fixed free list and stress-slot choice, every derivative order $j$ and compact $\mathcal{K} \subset \mathrm{Conf}_n$, with $A_{k,i} = r_k^4 T_{a_i b_i}$ on $I$ for the fixed tensor of Theorem 5.4 and $A_{k,i} = \mathsf{U}_{k,d_i}^{\mathrm{sc}} u_i$ elsewhere,*

$$D_x^j S_n(\boldsymbol{A}_k; r_k x) \longrightarrow D_x^j S_n^{\mathrm{Max}}(\boldsymbol{A}^0; x) \quad \text{uniformly on } \mathcal{K} .$$

The proof is in Section 16, 16.12. With $I = \varnothing$ and $j = 0$, this gives Proposition 5.2 in uniform form along the same sequence: an integral against $f$ is bounded by $\|f\|_\infty$ times the volume of its compact support times the uniform error. The earlier comparison is needed first to construct the tensor. Along the sequence of Proposition 5.5, for every fixed $d, a, K_{\mathrm{rot}}$, the transported infinitesimal actions satisfy the operator-norm limits

$$r_k (\mathsf{U}_{k,d+1}^{\mathrm{sc}})^{-1} \partial_a \mathsf{U}_{k,d}^{\mathrm{sc}} \longrightarrow \partial_a^{\mathrm{Max}},$$
$$(\mathsf{U}_{k,d}^{\mathrm{sc}})^{-1} \mathsf{L}_d(K_{\mathrm{rot}}) \mathsf{U}_{k,d}^{\mathrm{sc}} \longrightarrow \mathsf{L}_d^{\mathrm{Max}}(K_{\mathrm{rot}}) .$$

## 6 Phases and reconstruction

Keep the table and quotient fields of Sections 4–5. Compression to zero energy separates the mixed correlations into phases. The slab estimate of 17.1 makes word insertion bounded after a positive time translation. The coordinate reflection $\mathsf{s}_a^{\mathrm{refl}}$ acts on labels through $H_4$.

**Definition 6.1** (OS space and compression)**.** For words at distinct points in $x^a > 0$, let $\mathcal{H}_a^{\mathrm{OS}}$ be the completion of their complex span modulo the null space of

$$\langle w, v\rangle_a = S(\mathsf{s}_a^{\mathrm{refl}} w^* v), \qquad e^{-tH_a^{\mathrm{OS}}}[v] = [\tau_{te_a} v] \quad (t \geq 0)\,.$$

Let $\Omega_a^{\mathrm{OS}}$ be the empty-word vector and $P_a^0$ the projection onto $\ker H_a^{\mathrm{OS}}$. For $h_{\mathrm{sl}} > 0$ and $w$ with times in $(0, h_{\mathrm{sl}})$, define, on positive-time words $v$,

$$A_w v = w\, \tau_{h_{\mathrm{sl}} e_4} v, \qquad M^0(w) = P_4^0 \mathsf{B}_{h_{\mathrm{sl}}}(w)|_{\ker H_4^{\mathrm{OS}}}\,,$$

where $\mathsf{B}_{h_{\mathrm{sl}}}(w)$ extends $A_w$ boundedly on OS classes, by 17.1.

Translate any separated word into a slab to define its compression; 17.4 proves independence of width and placement. The compressed operators generate the compression algebra

$$\mathfrak{A}_0 = C^*(M^0(w) : w \text{ separated})\,.$$

Section 17, 17.4 proves that this algebra is separable, commutative and unital, and that $\Omega_4^{\mathrm{OS}}$ is cyclic and separating. Its vacuum state is

$$a \longmapsto \langle \Omega_4^{\mathrm{OS}}, a\Omega_4^{\mathrm{OS}}\rangle, \qquad S(w) = \langle \Omega_4^{\mathrm{OS}}, M^0(w)\Omega_4^{\mathrm{OS}}\rangle\,.$$

The second identity uses translation invariance. The measure below represents this state on the character space; the separating property gives full support. A one-dimensional mixed ground space would make $M^0(w) = S(w)\mathrm{Id}$ and leave a single character.

Characters separate the mixed correlations without changing which fields are detectable. Call a character $\xi$ a phase and $S^\xi$ its correlation table.

**Theorem 6.2** (PH. Phase decomposition)**.** *The algebra $\mathfrak{A}_0$ has a compact metric character space $\mathsf{X}$ with a full-support probability measure $\nu$ such that, for every separated word $w$,*

$$S^\xi(w) = \xi(M^0(w)), \qquad S(w) = \int_{\mathsf{X}} S^\xi(w)\, d\nu(\xi)\,.$$

*Each phase satisfies E1–E7 and has exactly the null spaces $\mathcal{N}_d(S^\xi) = \mathcal{N}_d(S)$. The identities of FC, $O(4)$ covariance and ST hold with $S^\xi$ in place of $S$, for the same maps and tensor $T$. For fixed $w$, $\xi \mapsto S^\xi(w)$ is continuous.*

The proof is in Section 17, 17.1–17.5. Two positive coordinate directions give opposite orders to the middle two of four widely separated words. Rotation covariance identifies the limits, and the positive translation semigroups converge to the ground projections; the resulting identity makes the compressed operators commute.

For each phase apply Definition 6.1 to $S^\xi$, obtaining $\mathcal{H}_{4,\xi}^{\mathrm{OS}}$, $H_{4,\xi}^{\mathrm{OS}}$ and $\Omega_\xi^{\mathrm{OS}}$. Fix $\xi_*$ in the full-measure set with one-dimensional time-axis OS ground space supplied by 17.5. The selection uses the one-dimensional ground-space property. Theorem 6.15 will extend ground-space uniqueness to every phase.

Write $\mathcal{S}_n = \mathcal{S}(\mathbb{R}^{4n}; \mathbb{C})$ and $\mathcal{S} = \mathcal{S}_1$. The following definitions specify the relativistic theory.

**Definition 6.3** (Relativistic fields)**.** Take a complex Hilbert space $\mathcal{H}$, a complex vector space $\mathcal{L}$ of labels, a linear subspace $\mathcal{D} \subset \mathcal{H}$ and $\Omega \in \mathcal{D}$. Fields $\Phi(\ell, f) : \mathcal{D} \to \mathcal{D}$ are linear operators, complex bilinear in $(\ell, f) \in \mathcal{L} \times \mathcal{S}$. Include an antilinear map $\ell \mapsto \ell^*$, strongly continuous unitary translations $\mathsf{T}(z)$, $z \in \mathbb{R}^4$, and a Borel projection-valued measure $\mathsf{E}$, with finite diagonal measures $\mu_\psi(B) = \langle \psi, \mathsf{E}(B)\psi\rangle$ for Borel $B \subset \mathbb{R}^4$.

**Definition 6.4** (Poincaré action)**.** Take a topological group $\mathcal{P}$ with topological group isomorphism $\kappa_{\mathrm{P}} : \mathcal{P} \to SO^{+}(1,3) \ltimes \mathbb{R}^4$, writing $\kappa_{\mathrm{P}}(\gamma) = (\Lambda_\gamma, z_\gamma)$. Here $SO^{+}(1,3)$ preserves $\mathsf{m}$, has determinant one and preserves $\overline{V}_{+}$. Include a strongly continuous unitary representation $\mathsf{U}$ with $\mathsf{U}(\kappa_{\mathrm{P}}^{-1}(1,z)) = \mathsf{T}(z)$ and linear bijections $\rho_{\mathrm{lab}}(\gamma)$ of $\mathcal{L}$. Set $(\gamma \cdot f)(x) = f(\Lambda_\gamma^{-1}(x - z_\gamma))$.

In group arguments we identify $\mathcal{P}$ with $SO^{+}(1,3) \ltimes \mathbb{R}^4$ through $\kappa_{\mathrm{P}}$. The antilinear label map and label action are subject to W1 and W2 below. The spectral relation is imposed next.

**Definition 6.5** (W0. Spectrum and vacuum)**.** For every $\psi \in \mathcal{H}$ and $z \in \mathbb{R}^4$, require

$$\langle \psi, \mathsf{T}(z)\psi \rangle = \int e^{i\mathbf{p}\cdot z}\, d\mu_\psi(\mathbf{p}), \qquad \mathsf{E}(\mathbb{R}^4 \setminus \overline{V}_{+}) = 0\,,$$

with Euclidean pairing. Every vector fixed by all translations belongs to $\mathbb{C}\Omega$.

**Definition 6.6** (W1. Word domain and adjoints)**.** Require $\|\Omega\| = 1$, density of $\mathcal{D}$, and

$$\mathcal{D} = \operatorname{span}_{\mathbb{C}}\{\Phi(\ell_1, f_1)\cdots\Phi(\ell_k, f_k)\Omega : k \geq 0\}\,.$$

For each $\ell \in \mathcal{L}$, $\psi \in \mathcal{D}$, the map $f \mapsto \Phi(\ell, f)\psi$ is continuous from $\mathcal{S}$ to $\mathcal{H}$ in norm. For $\psi, \chi \in \mathcal{D}$, require $\langle \Phi(\ell, f)\psi, \chi \rangle = \langle \psi, \Phi(\ell^*, \bar{f})\chi \rangle$.

**Definition 6.7** (W2. Covariance)**.** Require $\rho_{\mathrm{lab}}(1) = 1$ and $\rho_{\mathrm{lab}}(\gamma\gamma') = \rho_{\mathrm{lab}}(\gamma)\rho_{\mathrm{lab}}(\gamma')$. For every $\gamma \in \mathcal{P}$, $\ell \in \mathcal{L}$, $f \in \mathcal{S}$ and $\psi \in \mathcal{D}$, require $\mathsf{U}(\gamma)\mathcal{D} \subset \mathcal{D}$, $\mathsf{U}(\gamma)\Omega = \Omega$, and

$$\mathsf{U}(\gamma)\Phi(\ell, f)\psi = \Phi(\rho_{\mathrm{lab}}(\gamma)\ell, \gamma \cdot f)\mathsf{U}(\gamma)\psi\,.$$

**Definition 6.8** (W3. Locality)**.** For $f, h \in \mathcal{S}$ with $\mathsf{m}(x - y) < 0$ for every $x \in \operatorname{supp} f$, $y \in \operatorname{supp} h$, require $[\Phi(\ell, f), \Phi(\ell', h)]\psi = 0$ for all $\ell, \ell' \in \mathcal{L}$ and $\psi \in \mathcal{D}$.

**Definition 6.9** (Wightman axioms)**.** A relativistic datum of Definitions 6.3–6.4 is a Wightman theory if it satisfies W0–W3.

**Definition 6.10** (Faithful labels)**.** The labels are faithful if, for every nonzero $\ell \in \mathcal{L}$, some $f \in \mathcal{S}$ has $\Phi(\ell, f) \neq 0$.

**Definition 6.11** (Bounded weak commutant)**.** A bounded operator $B$ belongs to the weak commutant if, for every smeared field $\Phi$ and all $u, v \in \mathcal{D}$, one has $u \in \operatorname{Dom}\Phi^*$ and $\langle \Phi^* u, Bv \rangle = \langle u, B\Phi v \rangle$. Here $\Phi^*$ is the unbounded Hilbert-space adjoint.

**Definition 6.12** (Hermitian scalar labels)**.** A label is Hermitian if $\ell^* = \ell$, and Lorentz scalar if $\rho_{\mathrm{lab}}(\Lambda, 0)\ell = \ell$ for every $\Lambda \in SO^{+}(1,3)$. These conditions give the adjoint and scalar covariance identities through W1 and W2.

For reconstruction use the response space $\mathcal{L}$ of Definition 4.9. Put $\mathbb{F} = \bigsqcup_d F_d$ and $\iota(d, A) = \iota_d A$.

**Definition 6.13** (Boundary condition)**.** For a phase $\xi$, a relativistic datum $(\mathcal{H}_\xi, \mathcal{L}, \mathcal{D}_\xi, \Omega_\xi, \Phi_\xi, \ldots)$ of Definitions 6.3–6.4, linear maps $\iota_d : F_d \to \mathcal{L}$, and functions $\mathsf{F}_n^\xi$ on $\mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n$, require, for every $n \in \mathbb{N}$, $\boldsymbol{A} \in \mathbb{F}^n$, real $h_1 < \cdots < h_n$ and $f_i \in \mathcal{S}$,

$$\lim_{r \downarrow 0} \int \mathsf{F}_n^\xi(\boldsymbol{A}; (x_i^4 + irh_i)_i, (\vec{x}_i)_i) \prod_i f_i(x_i)\, dx \\ = \langle \Omega_\xi, \Phi_\xi(\iota(A_1), f_1)\cdots\Phi_\xi(\iota(A_n), f_n)\Omega_\xi \rangle\,.$$

For this formula alone a nonintegrable function has integral zero.

**Definition 6.14** (Ordered-time continuation)**.** For a phase $\xi$, put $\mathcal{Z}_n = \{z \in \mathbb{C}^n : \operatorname{Im} z_1 < \cdots < \operatorname{Im} z_n\}$. A continuation $\mathsf{F}_n^\xi$ is holomorphic there at each fixed real spatial configuration and satisfies

$$\mathsf{F}_n^\xi(\boldsymbol{A}; (it_i)_i, \vec{x}) = S_n^\xi(\boldsymbol{A}; (\vec{x}_i, t_i)_i) \qquad (t_1 < \cdots < t_n)\,.$$

Together with Definition 6.13, it connects Euclidean and real-time correlations.

For a theory at phase $\xi$ in the next theorem, write $\mathcal{H}_\xi$, $\mathcal{D}_\xi$, $\Omega_\xi$, $\Phi_\xi$ and $\mathsf{T}_\xi$ for its Hilbert space, word domain, vacuum, fields and translations. Define its energy generator by $\mathsf{T}_\xi(te_4) = e^{itH_\xi}$. Its OS comparison map means the bounded linear map

$$\mathsf{J}_\xi^{\mathrm{OS}} : \mathcal{H}_\xi \longrightarrow \mathcal{H}_{4,\xi}^{\mathrm{OS}}, \qquad \mathsf{J}_\xi^{\mathrm{OS}}\Omega_\xi = \Omega_\xi^{\mathrm{OS}}\,,$$

sending each smeared field-word vector to the norm limit of its heat-regularized positive-time OS vectors as the ordered heights tend to zero. Section 17, 17.7 constructs this map and proves its uniqueness.

Each character yields relativistic fields on the common response labels. The continuation and boundary condition specify their relation to the phase correlations.

**Theorem 6.15** (RC. Reconstruction)**.** *For every $\xi \in \mathsf{X}$, there are a faithful Wightman theory (Definition 6.9) on $\mathcal{L}$ and functions $\mathsf{F}_n^\xi$ satisfying Definitions 6.14 and 6.13 with the response maps $\iota_d$ of Definition 4.9. Its OS comparison map $\mathsf{J}_\xi^{\mathrm{OS}}$ is onto, and*

$$\ker H_\xi = \mathbb{C}\Omega_\xi, \qquad \ker H_{4,\xi}^{\mathrm{OS}} = \mathbb{C}\Omega_\xi^{\mathrm{OS}}\,.$$

*Every bounded operator in the weak commutant of its fields is scalar.*

Faithfulness and the bounded weak commutant have the meanings of Definitions 6.10 and 6.11. The proof is in Section 17, 17.6–17.9. Slab bounds control the analytic boundary values of these continuations. A low-energy bound uniform in $\xi$, continuity of the vacuum-subtracted heat pairings and full support of $\nu$ extend almost-everywhere ground-space uniqueness to every phase. Thus the earlier choice of $\xi_*$ remains valid. 17.6–17.9 also give joint Schwartz distributions, phase averages and stress continuation.

The reconstructed label maps are common to all phases. In 17.6 the adjoint is induced by time reflection followed by conjugation:

$$(\iota_d A)^* = \iota_d\big(R_d(\mathsf{s}_4^{\mathrm{refl}})\overline{A}\big)\,.$$

The Lorentz label action is the continuation of $R_d$: spatial rotations retain their Euclidean action. For $i = 1, 2, 3$, the boost along $e_i$ has generator $-i\mathsf{L}_d(K_{4i})$, where $K_{4i} = e_4 e_i^{\mathsf{T}} - e_i e_4^{\mathsf{T}}$ generates rotation in the $(e_4, e_i)$ plane (17.6). In particular, a real $O(4)$-invariant field gives a Hermitian Lorentz scalar label. Retain the theory at $\xi_*$ and suppress its phase subscript in $\mathcal{H}, \mathcal{D}, \Omega, \Phi$ and the other relativistic data.

# 7 Trace and products

The trace is centered in the phase already selected in Section 6. Its short-distance logarithm will distinguish the resulting theory from finite Wick comparisons.

**Definition 7.1** (Centered trace)**.** For the tensor $T$ fixed after Theorem 5.4 and $r > 0$, put

$$\tau = \sum_{a=1}^{4} T_{aa}, \qquad e_{\mathrm{tr}} = S^{\xi_*}(\tau), \qquad \Theta = \tau - e_{\mathrm{tr}}\mathbf{1}_4,$$
$$G_\Theta(r) = S_2^{\xi_*}(\Theta, \Theta; -re_4/2, re_4/2), \qquad A_G = \frac{3d_{\mathfrak{g}}}{\pi^4}\,.$$

The one-point value is position independent. Centering removes the vacuum contribution in the selected phase.

Adding a real multiple of $\delta_{ab}\mathbf{1}_4$ to $T$ preserves its shell laws, the limit in Proposition 5.5 and its centered trace $\Theta$. The fixed stress tensor is centered in the mixture; the selected-phase centering subtracts its own one-point value. The next limit detects a nonzero centered trace and determines its power and logarithm.

**Theorem 7.2** (TR. Trace law)**.** *The selected field $\Theta$ is nonzero, real and $O(4)$ invariant; its label $\iota_4\Theta$ is Hermitian and Lorentz scalar (Definition 6.12). For every $f \in \mathcal{S}$,*

$$\langle \Omega, \Phi(\iota_4\Theta, f)\Omega\rangle = 0\,.$$

*As $r \downarrow 0$ through all positive radii,*

$$\lim_{r\downarrow 0} r^8 \log^2(1/r) G_\Theta(r) = A_G > 0\,.$$

The proof is in Section 18: Lemma 18.1 gives vacuum centering, Lemma 18.2 gives monotonicity, and Lemmas 18.3 and 18.5 give the comparison and Ward coefficient. These are combined in the proof of Theorem 7.2 there. To identify the amplitude, use the free scalar curvature square

$$\mathcal{O}_{F^2} = \frac{1}{4}\sum_{a,b,\alpha} :F^\alpha_{ab}F^\alpha_{ab}:, \qquad S_2^{\mathrm{Max}}(\mathcal{O}_{F^2}, \mathcal{O}_{F^2}; -e_4/2, e_4/2) = \frac{3d_{\mathfrak{g}}}{\pi^4}\,.$$

Here $\alpha = 1, \dots, d_{\mathfrak{g}}$ indexes an orthonormal colour basis. At running coupling $\beta$, the scalar trace coefficient is $4d_2/\beta$, with $d_2 = -b_0/2$. Since $\kappa_G = 2b_0$, its leading coefficient at radius $r$ is $4d_2/(\kappa_G \log(1/r)) = -1/\log(1/r)$. Squaring it and using the free correlation gives $A_G$. The comparison uses adjustable stopping levels and fourth-order information beyond Proposition 5.2; its estimates are uniform in the phase, although TR is stated only at the already selected $\xi_*$. For every prescribed sequence of positive radii tending to zero, the comparison holds along a subsequence. The positive spectral representation below makes $G_\Theta$ real and nonincreasing; a sandwich argument gives the limit on that subsequence. The subsequence criterion then gives the full-radius limit.

The same proof identifies centering on the word domain:

$$\Phi(\iota_4\Theta, f) = \Phi(\iota_4\tau, f) - e_{\mathrm{tr}}\left(\int f\right)\mathrm{Id}_{\mathcal{H}}\,.$$

It also constructs the unique positive tempered-growth spectral measure $\mu_\Theta$, supported in $\overline{V}_+$, for which

$$d\mu_{\Phi(\iota_4\Theta, f)\Omega} = |\widehat{f}_+|^2\, d\mu_\Theta, \qquad G_\Theta(r) = \int e^{-r\mathbf{p}^4}\, d\mu_\Theta(\mathbf{p}) \quad (r > 0)\,.$$

Here $\widehat{f}_+(\mathbf{p}) = \int e^{i\mathbf{p}\cdot x} f(x)\, dx$. The measure's Lorentz invariance and its high-energy limits against compactly supported continuous tests and Schwartz tests are stated in Lemmas 18.2 and 18.6.

To define the comparison class, fix $m_{\mathrm{sh}} \geq 0$, integers $j_h, k_E \geq 0$ and continuous real $h_{\mathrm{mom}}$ on $\mathbb{R}^3$ with $h_{\mathrm{mom}}(t\mathbf{k}) = t^{j_h} h_{\mathrm{mom}}(\mathbf{k})$ for $t > 0$.

**Definition 7.3** (Finite Wick profiles)**.** Put $\omega_{m_{\mathrm{sh}}}(\mathbf{k}) = \sqrt{m_{\mathrm{sh}}^2 + |\mathbf{k}|^2}$ and, for $r > 0$,

$$K_{m_{\mathrm{sh}},k_E,h_{\mathrm{mom}}}(r) = \int_{\mathbb{R}^3} \frac{\omega_{m_{\mathrm{sh}}}(\mathbf{k})^{k_E} h_{\mathrm{mom}}(\mathbf{k})}{2\omega_{m_{\mathrm{sh}}}(\mathbf{k})} e^{-r\omega_{m_{\mathrm{sh}}}(\mathbf{k})}\, d^3\mathbf{k}\,.$$

A finite Wick profile is a function

$$W_{\mathrm{Wick}}(r) = \sum_{i=1}^{N} c_i \prod_{j=1}^{n_i} K_{m_{ij},k_{ij},h_{ij}}(r) \quad (r > 0), \qquad c_i \in \mathbb{C}\,,$$

with finitely many terms and factors of the preceding kind; empty products are one. A zero denominator at zero momentum gives integrand zero.

The momentum integrals converge absolutely for $r > 0$. For a free scalar field of mass $m_{\mathrm{sh}}$, the time-axis two-point function is $(2\pi)^{-3} K_{m_{\mathrm{sh}},0,1}(r)$. Derivatives give momentum factors, and Wick contractions give finite sums of products at the same radius.

Such profiles cannot reproduce the factor $r^{-8}\log^{-2}(1/r)$ in Theorem 7.2.

**Proposition 7.4** (NF. Finite-profile exclusion)**.** *For every finite Wick profile $W_{\mathrm{Wick}}$ and every $\delta > 0$, there is $0 < r < \delta$ such that*

$$G_\Theta(r) \neq W_{\mathrm{Wick}}(r)\,.$$

The proof is in Section 18, 18.7. The mass-shell kernels have expansions to every finite order in integer powers of $r$ and nonnegative powers of $\log r$. Finite products and sums therefore give, to bounded error, a finite sum of monomials $r^a(\log r)^k$ with $a \in \mathbb{Z}$ and $k \in \mathbb{N}$. Multiplication by $r^8 \log^2(1/r)$ makes the terms with $a > -8$ vanish. The remaining nonzero leading term diverges, so any finite limit is zero. The positive limit in Theorem 7.2 therefore excludes equality on every interval $(0, \delta)$. Only that limit and its positivity enter this exclusion. 18.9 identifies polynomial mass-shell comparisons with their Laplace profiles and applies 18.7; 18.12–18.13 give the Fock countertests. Their comparison classes are defined there.

The product expansion is stated for the Euclidean tables on their common quotient fields. Write $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$ for a state index (a phase or the mixture), with $S^{\mathrm{mix}} = S$, and retain degree tags by $\mathbb{F} = \bigsqcup_d F_d$, $\deg(d, A) = d$. The underlying prescription consists of the original fields and their correlation table. From it, Section 19, 19.6 constructs finite channel sets $\mathrm{Ch}_j$, real fields $O_{j,\lambda} \in F_j$ and, for every finite $I$, functions

$$c_{j,\lambda} : \mathbb{F}^I \times (\mathbb{R}^4)^I \times \mathbb{R}^4 \longrightarrow \mathbb{C}\,.$$

Fix these choices once, before spectators, state indices and truncations. The fields form complex bases of conjugation-invariant complements of $F_{j-1}$ in $F_j$, with $F_{-1} = 0$. The degree-zero vector is a nonzero multiple of the unit. For finite cluster and spectator index sets $I_{\mathrm{cl}}, I_{\mathrm{sp}}$, take positions $x \in (\mathbb{R}^4)^{I_{\mathrm{cl}}}$, $y \in (\mathbb{R}^4)^{I_{\mathrm{sp}}}$ and expansion center $o \in \mathbb{R}^4$.

**Definition 7.5** (Product remainder)**.** For finite index sets $I_{\mathrm{cl}}, I_{\mathrm{sp}}$ and $D \in \mathbb{N}$, cluster and spectator lists $\boldsymbol{A} \in \mathbb{F}^{I_{\mathrm{cl}}}$, $\boldsymbol{B} \in \mathbb{F}^{I_{\mathrm{sp}}}$, put $|\boldsymbol{A}|_{\deg} = \sum_i \deg A_i$ and

$$\mathcal{R}_D^\zeta = S^\zeta(\boldsymbol{A}, x; \boldsymbol{B}, y) - \sum_{j=0}^{D} \sum_{\lambda \in \mathrm{Ch}_j} c_{j,\lambda}(\boldsymbol{A}; x, o) S^\zeta(O_{j,\lambda}(o), \boldsymbol{B}(y))\,.$$

Use a fixed enumeration of $I_{\mathrm{cl}} \sqcup I_{\mathrm{sp}}$ for the concatenated correlations. A pair $(x, o)$ is an $(r, M_{\mathrm{geo}}, \eta)$-cluster if $|x_i - o| \le M_{\mathrm{geo}} r$ and $|x_i - x_{i'}| \ge \eta r$ for distinct slots. Spectators are $\delta_{\mathrm{sp}}$-clear if $|o - y_j| \ge \delta_{\mathrm{sp}}$ and $|y_j - y_{j'}| \ge \delta_{\mathrm{sp}}$ for distinct slots. The remainder measures the error through degree $D$.

If $M_{\mathrm{geo}} r < \delta_{\mathrm{sp}}$, the concatenated positions are separated: $|x_i - y_j| \ge \delta_{\mathrm{sp}} - M_{\mathrm{geo}} r > 0$. Thus the remainder is defined in the small-radius range used below.

One field expansion approximates a cluster in every phase and in the mixture. Its coefficients are independent of all spectators.

**Theorem 7.6** (OP. Operator product expansion)**.** *For every finite list $\boldsymbol{A}$, $M_{\mathrm{geo}} \ge 0$, $\eta, \varepsilon > 0$, $j \in \mathbb{N}$ and $\lambda \in \mathrm{Ch}_j$, some $r_0 > 0$ gives, for $0 < r < r_0$ and every $(r, M_{\mathrm{geo}}, \eta)$-cluster,*

$$|c_{j,\lambda}(\boldsymbol{A}; x, o)| \le r^{j - |\boldsymbol{A}|_{\deg} - \varepsilon} .$$

*For every pair of finite lists $\boldsymbol{A}, \boldsymbol{B}$, $M_{\mathrm{geo}} \ge 0$, $\eta, \delta_{\mathrm{sp}}, \varepsilon > 0$ and $D \in \mathbb{N}$, some $r_1 > 0$ gives, for $0 < r < r_1$, such clusters, $\delta_{\mathrm{sp}}$-clear spectators and all state indices $\zeta$,*

$$|\mathcal{R}_D^{\zeta}| \le r^{D+1-|\boldsymbol{A}|_{\deg} - \varepsilon} .$$

The proof is in Section 19, Lemmas 19.6–19.10. The finite correlation decoder is chosen biorthogonally before averaging under rotations and conjugation and removing lower degrees. Lemma 19.1 gives the local continuation used in the lattice replacement of Lemma 19.8; Lemma 19.2 gives the physical quotient estimate used in Lemma 19.9, which implies Lemma 19.10. The discarded part is bounded in the compression algebra, so the bound holds in every phase and in the mixture. The lattice bounds have the required powers up to factors $e^{o(\log(1/r))}$. Multiplication by $r^{\varepsilon/2}$ makes these factors tend to zero, giving coefficient one for smaller radii. The coefficient and remainder thresholds have the separate dependencies in the statement.

Lemmas 19.11–19.16 give integrated and nested estimates; the nested error is a sum of gains from proper clusters.

Contact subtractions use the original fields and correlations underlying 19.6, with separate projections extracting fields through finite correlation probes. Their clearance is a radius keeping the collapsed cluster separated from these probes, independent of spectator clearance $\delta_{\mathrm{sp}}$.

A collision forest is a family of subsets of slots, any two disjoint or nested. The contact construction subtracts normal Taylor jets along their collision diagonals. The next result concerns this fixed subtraction; its local order bounds are stated in 19.23.

**Theorem 7.7** (CT. Contact extension)**.** *For every state index $\zeta$, arity $n$ and list $\boldsymbol{A} \in \mathbb{F}^n$, the contact construction of Section 19, 19.2, fixed on the underlying prescription of 19.6, gives a distribution $\mathfrak{D}_{\boldsymbol{A}}^{\zeta}$ on $\mathbb{R}^{4n}$ such that, for every separated test $f$,*

$$\mathfrak{D}_{\boldsymbol{A}}^{\zeta}(f) = \int f(x) S_n^{\zeta}(\boldsymbol{A}; x)\, dx .$$

The proof is in Section 19, Lemmas 19.20–19.23. A separated test is an $f \in C_c^{\infty}(\mathrm{Conf}_n)$, extended by zero to $\mathbb{R}^{4n}$. Every normal jet on a collision diagonal then vanishes, leaving the correlation integral. For a general test, Lemma 19.21 controls the contact rows and signed residuals after child subtractions. The shell induction of Lemma 19.22 gives a continuous functional. Lemma 19.20 supplies the positive subtraction scales for these contact extensions; the pairing uses the single contact length specified in Section 19.2. CT enters Theorem B; reconstruction and Theorem A use separated correlations.

# 8 Main theorems

The existence theorem concerns one Wilson family $\mathcal{W}$, chosen before any prescribed volume sequence. Its tokens $v \in \mathcal{T}$ label the lattice insertions. We first specify its continuum conclusion; the field calculus and stress tensor established above are additional properties of the construction.

**Definition 8.1** (Q0. Hierarchy data)**.** Hierarchy data over $\mathcal{W}$ consist of:

- complex Hilbert and vector spaces $\mathcal{H}, \mathcal{L}$, and the relativistic data of Definitions 6.3–6.4;
- finite-dimensional complex normed spaces $F_d$, linear maps $\iota_d : F_d \to \mathcal{L}$, and $\pi_{\mathrm{quo}} : \mathcal{T} \to \mathbb{F}$, where $\mathbb{F} = \bigsqcup_d F_d$;
- a compact metric space $\mathsf{X}$, a Borel measure $\nu$ and $\xi_* \in \mathsf{X}$;
- functions $S_n^\zeta : \mathbb{F}^n \times (\mathbb{R}^4)^n \to \mathbb{C}$ for $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$, and $\mathsf{F}_n : \mathbb{F}^n \times \mathbb{C}^n \times (\mathbb{R}^3)^n \to \mathbb{C}$;
- finite sets $\mathrm{Ch}_j$, fields $O_{j,\lambda} \in F_j$, and coefficients $c_{j,\lambda} : \mathbb{F}^{I_{\mathrm{cl}}} \times (\mathbb{R}^4)^{I_{\mathrm{cl}}} \times \mathbb{R}^4 \to \mathbb{C}$ for every finite index set $I_{\mathrm{cl}}$;
- an element $\Theta \in F_4$.

The spaces, maps and functions of Q0 are arbitrary on the displayed domains. Put $\deg(d, A) = d$ and $\iota(d, A) = \iota_d A$. The axioms below constrain the correlations at separated positions.

In Q2–Q6, all cited definitions and formulas are read with these hierarchy data. In definitions indexed by a phase, read $\Omega_{\xi_*}$, $\Phi_{\xi_*}$ and $\mathsf{F}_n^{\xi_*}$ as the $\Omega$, $\Phi$ and $\mathsf{F}_n$ of these data.

**Definition 8.2** (Q1. Regular Wilson family)**.** A Wilson family is regular if its insertions are continuous and gauge invariant, each token $v$ has a finite stencil $\Sigma_v \subset \mathbb{Z}^4$ valid for every $N, p, y$, and some $b_* > 0$ satisfies

$$\forall \varepsilon > 0 \ \exists N_0 \ \forall N \geq N_0 \ \forall p : \quad \left| \frac{\beta_{N,p}}{\log(1/a_N)} - b_* \right| < \varepsilon \,.$$

Only links with both ends in $y + \Sigma_v$ modulo the torus are read.

**Definition 8.3** (Q2. Wightman axioms)**.** The relativistic data of Q0 form a Wightman theory in the sense of Definition 6.9.

The following conditions specify the labels, phase average and selected continuation.

**Definition 8.4** (Q3. Labels and phases)**.** Every nonzero $\ell \in \mathcal{L}$ has $\Phi(\ell, f) \neq 0$ for some $f \in \mathcal{S}$. Each $\iota_d$ is injective, their images cover $\mathcal{L}$, $\pi_{\mathrm{quo}}$ is onto, and $\Phi(\iota(A), f) \neq 0$ for some $A \in \mathbb{F}$, $f \in \mathcal{S}$. The measure $\nu$ is a probability measure of full support. Each $S_n^\zeta$ is simultaneously permutation symmetric and smooth on $\mathrm{Conf}_n$. For every $n$, $\boldsymbol{A} \in \mathbb{F}^n$ and $x \in \mathrm{Conf}_n$,

$$\xi \longmapsto S_n^\xi(\boldsymbol{A}; x) \text{ is continuous}, \qquad S_n^{\mathrm{mix}}(\boldsymbol{A}; x) = \int_{\mathsf{X}} S_n^\xi(\boldsymbol{A}; x)\, d\nu(\xi) \,.$$

**Definition 8.5** (Q4. Selected continuation)**.** The functions $\mathsf{F}_n$ satisfy the ordered-time continuation condition of Definition 6.14 and the boundary condition of Definition 6.13 at $\xi_*$, using the relativistic datum and the maps $\iota_d$ of the hierarchy data.

**Definition 8.6** (Q5. Products)**.** For the channels, fields, coefficients and tables $S^\zeta$ of Q0, form $\mathcal{R}_D^\zeta$ by Definition 7.5. These coefficients and remainders satisfy the two inequalities of Theorem 7.6: the coefficient threshold depends on $I_{\mathrm{cl}}, \boldsymbol{A}, M_{\mathrm{geo}}, \eta, \varepsilon, j, \lambda$; the remainder threshold depends on $I_{\mathrm{cl}}, I_{\mathrm{sp}}, \boldsymbol{A}, \boldsymbol{B}, M_{\mathrm{geo}}, \eta, \delta_{\mathrm{sp}}, \varepsilon, D$. Each threshold is uniform in positions and state indices. The coefficients are functions of the cluster fields, their positions and the expansion center.

**Definition 8.7** (Q6. Trace test)**.** For the chosen $\Theta \in F_4$, put

$$G_\Theta(r) = S_2^{\xi_*}(\Theta, \Theta; -re_4/2, re_4/2), \qquad A_G = \frac{3d_{\mathfrak{g}}}{\pi^4}\,.$$

Require

$$\langle \Omega, \Phi(\iota_4\Theta, f)\Omega\rangle = 0 \qquad (f \in \mathcal{S})\,,$$

and

$$\lim_{r\downarrow 0} r^8 \log^2(1/r) G_\Theta(r) = A_G > 0 \quad \text{in } \mathbb{C}\,,$$

and require the finite-profile exclusion of Proposition 7.4.

For the lattice limit, $S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}}$ denotes the kernels

$$K_n(v; x) = S_n^{\mathrm{mix}}(\pi_{\mathrm{quo}}(v_1), \dots, \pi_{\mathrm{quo}}(v_n); x)\,.$$

The next theorem constructs a lattice approximation and one hierarchy satisfying these conditions. Its degree-four field $\Theta$ is constrained by Q6; the construction theorem below identifies it with a centered stress trace.

**Theorem A** (Existence)**.** *For every gauge group of Definition 2.1, there is a regular* (Q1) *Wilson family $\mathcal{W}$ such that, for every strictly increasing $\mathsf{p} : \mathbb{N} \to \mathbb{N}$, there are a strictly increasing $\mathsf{q}$ and hierarchy data over $\mathcal{W}$* (Q0) *satisfying* Q2–Q6*, with*

$$\mathcal{W} \longrightarrow S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}} \quad \textit{along } P = \mathsf{p} \circ \mathsf{q} \quad \textit{in the sense of Definition 4.3}\,.$$

Theorem A fixes the lattice theory before the volume sequence is chosen, removes the spacing at each fixed volume and then takes a volume refinement. Q2 gives relativistic fields with positive energy, a unique translation-invariant vacuum, Poincaré covariance, locality and continuous Schwartz smearing on a common cyclic domain with adjoints. Q3 and Q4 express the limiting Euclidean correlations as a phase average and continue the selected phase to those fields in real time. Q5 approximates local products uniformly in the state and in separated spectator positions, and Q6 fixes the vacuum centering and the leading short-distance asymptotics of the centered degree-four two-point function.

*Remark* 8.8 (Nontriviality). Q3 requires a nonzero smeared field, excluding the zero field theory. Q6 gives a positive limit $\lim_{r\downarrow 0} r^8 \log^2(1/r) G_\Theta(r) = 3d_{\mathfrak{g}}/\pi^4$ and excludes equality with every finite Wick profile on any interval $(0, \delta)$. Thus every Gaussian or finite free-field realization whose two-point profile for $\Theta$ belongs to the class of Definition 7.3 is excluded.

Theorem A, like the existence condition it expresses, leaves $b_* > 0$ unspecified; Theorem 3.6 supplies $b_* = \kappa_G = 2b_0 = 11C_{\mathfrak{g}}/(24\pi^2)$ for the construction.

The construction realizes the fields as $F_d = V_d/\mathcal{N}_d(S)$ and uses the token projection $\pi_{\mathrm{quo}}(d, \mathbf{c}) = (d, [\mathbf{c}])$. Its phase correlations are those of the same quotient fields; extending them arbitrarily to collision diagonals gives the domains required by Q0. The next theorem records the limits, reconstruction and local product structure of this family.

**Theorem B** (Construction)**.** *For every gauge group $G$ of Definition 2.1 and $\mathsf{b} > 0$, the renormalized insertions (Definition 3.9) on the penalty-$\mathsf{b}$ trajectories of Theorem 3.6 form a regular Wilson family $\mathcal{W}$ with $a_N \downarrow 0$, $L_p \uparrow \infty$ and $\beta_{N,p}/\log(1/a_N) \to \kappa_G$ uniformly in $p$. Moreover:*

(i) *For every $p$, token list $v$ and $p$-admissible separated $f$, $C_{N,p}^{\mathrm{lat}}(v; f) \to \int f M_{p,v}^{\mathrm{UV}}$ as $N \to \infty$.*

(ii) *For every strictly increasing* $\mathsf{p}$ *there are a strictly increasing* $\mathsf{q}$*, a table* $S$ *satisfying E1–E7 and hierarchy data over* $\mathcal{W}$ *satisfying Q2–Q6 with* $\mathcal{W} \to S$ *and* $\mathcal{W} \to S^{\mathrm{mix}} \circ \pi_{\mathrm{quo}}$ *along* $P = \mathsf{p} \circ \mathsf{q}$*. In (iii)–(v),* $S$ *is this table,* $F_d = V_d/\mathcal{N}_d(S)$*, and* $\nu$*,* $\xi_*$*,* $\Theta$ *are those of these hierarchy data.*

(iii) $S = \int_{\mathsf{X}} S^{\xi}\, d\nu(\xi)$ *with* $\nu$ *a fully supported probability measure on a compact metric space; each* $S^{\xi}$ *continues a faithful Wightman theory with unique vacuum.*

(iv) *A conserved stress tensor is normalized by its shell fluxes (Theorem 5.4); in the selected phase* $\xi_*$ *its centered trace* $\Theta$ *has* $r^8 \log^2(1/r) G_\Theta(r) \to 3d_{\mathfrak{g}}/\pi^4$*, and* $G_\Theta$ *agrees with no finite Wick profile on any interval* $(0, \delta)$*.*

(v) *The fields* $F_d$ *have an operator product expansion with spectator-independent coefficients and* $|\mathcal{R}_D^{\zeta}| \le r^{D+1-|\boldsymbol{A}|_{\deg}-\varepsilon}$*, with the quantifiers and uniformity of Theorem 7.6, and contact extensions from the same prescription.*

Theorem 3.6 gives the running law; Theorem 4.2 gives (i); Theorem 4.7 gives the refinement and table in (ii), and Steps 5–9 assemble its hierarchy data from Definition 4.9, Proposition 4.11 and Theorems 6.2, 6.15, 7.2, 7.4 and 7.6; Theorems 6.2 and 6.15 give (iii); Theorems 5.4 and 7.2 with Proposition 7.4 give (iv); and Theorems 7.6 and 7.7 give (v). Further conclusions about these same objects are proved in Part II: Lemma 14.3 gives endpoint tuning, Lemma 14.31 gives convergence at moving points, and Lemma 17.8 gives low-energy bounds. The spectral and nested product results are cited in Steps 8 and 9. The position-space two-point scaling consequence is given after Lemma 18.6. Theorem A follows from Theorem B with any $\mathsf{b} > 0$: clause (ii) supplies its hierarchy data. Steps 1–9 below prove Theorem B, and Step 10 deduces Theorem A.

*Deduction of Theorems A and B.*

1. Fix $G$ and $\mathsf{b} > 0$, and make the common choices in the order of Section 9. The Wilson initialization (3.13) starts the induction. Theorem 3.5 advances it, and Theorem 3.6 selects trajectories reaching the common endpoint. These choices precede any prescribed volume sequence.

2. Fix the local sections, reference levels and terminal maps on those trajectories. Definition 3.9 defines every insertion, including below its reference level. The local sections are polynomials in the links on their fixed stencils, hence continuous and local. Their gauge invariance is preserved by the terminal maps, which change only their coefficients. Taking $m = N$ in Theorem 3.6 gives the volume-uniform running law with $b_* = \kappa_G > 0$, completing Q1.

3. Theorem 4.2 gives the full fixed-volume limit for every fixed field and difference list. In particular, every admissible separated test has a full inner limit before volume extraction, proving (i).

4. Given any strictly increasing $\mathsf{p}$, Theorem 4.7 chooses $\mathsf{q}, S$ with ordered convergence along $P = \mathsf{p} \circ \mathsf{q}$ and E1–E7, proving (ii). The same extraction fixes the corrected coefficient maps.

5. Define $F_d = V_d/\mathcal{N}_d(S)$ and $\mathcal{L}$ by Definition 4.9. Proposition 4.11 and Lemma 15.4 give the inclusions and derivatives, with injective response maps whose images cover $\mathcal{L}$. Theorem 5.4, proved in Lemmas 16.10–16.11, gives the conserved tensor, normalized by its shell fluxes, on these fields.

6. Theorem 6.2 gives $\mathsf{X}, \nu, S^{\xi}$ on these quotient spaces. Choose $\xi_*$ from the full-measure set with one-dimensional time-axis OS ground space supplied by Lemma 17.5. Theorem 6.15 reconstructs every phase with a unique vacuum, proving (iii); retain its theory and continuation at $\xi_*$.

7. Set $\pi_{\mathrm{quo}}(d, \mathbf{c}) = (d, [\mathbf{c}])$; representatives make it onto. The unit response is nonzero since its one-point function is one. Faithfulness in Theorem 6.15 therefore gives a nonzero acting field and, with Step 5, the label clauses of Q3. Theorem 6.2 gives its phase clauses, and reconstruction gives Q2 and Q4. Extend the quotient mixture and phase correlations arbitrarily to collision diagonals. Separated tests vanish there, so the extensions preserve ordered convergence. Representative independence gives $S_n^{\mathrm{mix}}(\pi_{\mathrm{quo}} v; x) = S_n(v; x)$ on $\mathrm{Conf}_n$, identifying the lattice limit required in Theorem A.

8. Center the trace of the fixed tensor in the selected phase by Definition 7.1. Theorem 7.2, proved using Lemmas 18.3 and 18.5, gives vacuum centering, the positive amplitude and the full-radius law; Proposition 7.4 excludes every finite Wick profile. These give Q6 and (iv). For this same field, Lemma 18.1 gives the operator identity, Lemma 18.2 gives the unique spectral measure, and Lemma 18.6 gives its high-energy limits. Lemmas 18.9, 18.12 and 18.13 give further polynomial and Fock countertests in the same selected phase.

9. Choose the underlying prescription of Lemma 19.6 once, with its original fields and correlation table. Theorem 7.6 gives Q5 and its error bounds. Theorem 7.7 and Lemmas 19.20–19.23 use the same prescription, with separate contact projections, separating functionals and clearance, to give extensions, real bases and subtraction scales. The contact proof uses the lattice-derived estimates of Lemma 19.9 and the signed transfer of Lemma 19.21. This proves (v). Lemmas 19.11–19.16 give further integrated and nested estimates for this prescription.

10. Steps 1–9 give Theorem B and hierarchy data satisfying Q2–Q6 on the same family and volume refinement. Fix any $\mathsf{b} > 0$, for instance $\mathsf{b} = 1$. The regular family precedes $\mathsf{p}$; its refinement and hierarchy follow it, with the ordered convergence just identified. This proves Theorem A.

□

## 9 Constants and order of choice

Table 2 gives the choices for the construction. At each stage there are finitely many inequalities on constants chosen in the indicated order. Leading-map bounds are fixed before column bounds, columns before ratios, and these before the coupling entries. Later ratios and entries may be increased without changing earlier choices. The displayed inequalities are properties of the chosen construction.

Table 2: Choices in dependence order. Stage choices are made at each fixed $j \geq 4$ before entry levels.

| | Choice | Dependence, restrictions and purpose |
|---|---|---|
| 1 | Gauge normalization; $\mathsf{b} > 0$ | Fix $G$, its realization, probability Haar measure and metric, then $d_{\mathfrak{g}}, C_{\mathfrak{g}}, b_0, \kappa_G$ as in Definition 2.1; $\kappa_G > 0$. All later choices use these data. |
| 2 | Frame and predictor cutoffs; axial factor | Depend on $G$. The frame and predictor cutoffs select the unique-minimizer branches of the block kernel. On a radius-$R$ patch, the total link chord in an eligible axial frame is at most $a_{\mathrm{ax}}(2R+1)$ times the total plaquette chord, with $a_{\mathrm{ax}} \geq 16$. Choose the cutoffs small enough for branch validity on every patch (Lemma 11.4). |

Table 2 (continued)

| | Choice | Dependence, restrictions and purpose |
|---|---|---|
| 3 | Homogeneous bases, norms, sections, stencils | Depend on $G, d$. Impose Taylor matching, reality and $H_4$ covariance (Lemma 10.2 and Definitions 3.8–3.9). |
| 4 | Stage parameters | Depend on rows 1–3 and $j$. Choose derivative orders $\mathfrak{q}_j \geq 6$, spatial powers $\varpi_j \geq 2$ and support exponents $\sigma_j = 100$. In Section 12, 12.3, $w_j^{\mathrm{path}} > 0$ multiplies path length in the exponential weight, and $A_j^{\mathrm{path}} > 0$ bounds the weighted row, cubic and quartic sums. Fix the entire nondecreasing sequence $K_j \geq 1$ bounding the leading coefficient maps and their inverses, including the finite symmetry averages, as specified before Lemma 14.1. Then choose nondecreasing column bounds $C_j \geq 64K_{j+1}^2$, including the derivative factorials and finite basis-recombination constants, as in item 1 preceding that lemma. These columns bound marked coefficient transport (Lemma 14.8). Let $\gamma_j > 0$ be half a uniform positive lower bound for the free effective plaquette forms in Lemma 11.5, chosen for the top form and both possible incoming stages. Choose clipping coefficients with $0 < 16c_{\mathrm{lin},j} < c_{\mathrm{dir},j} \leq \min(\gamma_j, 1)$; the remaining free form then has the positive row factor used in Definition 12.2. Its health cutoff relaxes with distance through $e^{\mathrm{dist}/R_j^{\mathrm{rel}}}$, where $R_j^{\mathrm{rel}} > 0$ and $R_j^{\mathrm{rel}} w_j^{\mathrm{path}} \geq 12$. Let $d_j \geq 0$ be the block-ratio-independent one-loop bound of Lemma 12.9. Fix the reset half-width $r_{\mathrm{res}} = 1$, then choose $\omega_j^{\mathrm{RG}} \geq d_j + 1$ for Lemma 12.10. Orders and powers tend to infinity and, along levels, are nondecreasing, as are support exponents. All ratio costs are now fixed. |
| 5 | Odd ratios $\ell_j^{\mathrm{st}}$ | Depend on rows 1–4; choose $\ell_j^{\mathrm{st}} \geq 3$ nondecreasing and above every ratio threshold. At $j = J_{m+1}$ and $\ell_m = \ell_j^{\mathrm{st}}$ require $$\frac{\omega_j^{\mathrm{RG}}}{\log \ell_m} \leq \min(\kappa_G/4, j^{-2}),$$ $$\frac{\log(1 + A_j^{\mathrm{path}} + (w_j^{\mathrm{path}})^{-1} + C_j + R_j^{\mathrm{rel}} + \sigma_j)}{\log \ell_m} \leq j^{-2}.$$ These bounds control the drift and logarithmic stage costs relative to $\log \ell_m$. Impose also the graph and matching restrictions (14.5)–(14.6), the local marked error restriction in item 2 preceding Lemma 14.1, and the ratio inequalities of (19.2). Include the source comparison constants through (14.8). |

Table 2 (continued)

| | Choice | Dependence, restrictions and purpose |
|---|---|---|
| 6 | Exponents and coupling entry bounds | Depend on rows 1–5. The positive antitone exponents $\theta_j$ control the quadratic loop bound in row 9. The bounds $\beta_j^{\mathrm{ent}}$ are lower bounds on the reference coupling when stage $j$ is used. Choose them to meet the finite maximum of the remaining stage requirements, including those of Sections 16.3, 18.1 and 19. Impose the comparison entry bounds (14.7), for each arity admitted at that stage. For Section 18.1, raise the entry bounds to ensure $\ell_m \le \log \widehat{\beta}_m$ and $\max(\ell^{\mathrm{st}}_{J_m}, \ell^{\mathrm{st}}_{J_m-1}) \le M_m^{\mathrm{cut}}$ at every sufficiently large level, including stage transitions. Include both adjacent stage ratios and the finite coefficient-error thresholds in the proof of Lemma 18.5. Entry bounds may be raised. |
| 7 | $\bar{\beta}$, entry levels $m_j^{\mathrm{ent}}$, minimum periods $n_j^{\min}$ | Depend on rows 1–6. The levels $m_j^{\mathrm{ent}}$ delay the first use of stage $j$; $n_j^{\min}$ are its minimum lattice periods. Choose $\bar{\beta} > 1$ large enough for the initial coupling requirements and for $t_m \le 1$ at every level. Choose entry levels and periods to satisfy the remaining level and geometric requirements, including the scale restrictions (14.3) and comparison intervals (14.4). The marked construction uses Lemma 19.7, with weight restrictions (19.6) and entry restrictions (19.7). Impose also the support-enlargement and occupancy entry requirements in the proof of Lemma 19.8, and the regular marked weight inequalities stated before Lemma 19.7. Delaying entry preserves earlier requirements. |
| 8 | $J_m, \ell_m, D_m, a_m, L_p$, $m_{\mathrm{ref}}(d)$ | Depend on rows 1–7. Set $D_0 = 1$, $D_{m+1} = \ell_m D_m$, $a_m = D_m^{-1}$, $s_m = \log D_m$, with $\ell_m = \ell^{\mathrm{st}}_{J_{m+1}}$ and $4 \le J_m \le J_{m+1} \le J_m + 1$, $J_m \to \infty$. Require $0 < \theta_{J_m} \le J_m^{-1}$, $\log \ell_m \to \infty$, $\log \ell_m / s_m \to 0$. Choose strictly increasing $L_p$ with $4 \mid L_p$ and every positive integer dividing infinitely many $L_p$. Require $L_p D_m \ge 100(B_m + 1)$ and $L_p D_{m+1} \ge 100((J_{m+1} + 3)\ell_m + 1)$. Choose $J_{m_{\mathrm{ref}}(d)} \ge d + 3$, with $\mathfrak{q}_{J_m} \ge d + 1$, $\varpi_{J_m} \ge d$ for $m \ge m_{\mathrm{ref}}(d)$. |
| 9 | Derived level scales | The scales are determined by earlier choices. Set $$\widehat{\beta}_m = \bar{\beta} + \kappa_G s_m > 1, \qquad M_m^{\mathrm{cut}} = \exp((\log\log \widehat{\beta}_m)^2),$$ $$t_m = \widehat{\beta}_m^{-1/2} M_m^{\mathrm{cut}}, \qquad B_m = \lceil (\log \widehat{\beta}_m)^2 \rceil,$$ $$\epsilon_m^{\mathrm{reg}} = \widehat{\beta}_m^{-11/10}, \qquad \epsilon_m^{\mathrm{for}} = e^{-(M_m^{\mathrm{cut}})^{1/5}},$$ $$\Lambda_m^{\mathrm{loop}} = 2(M_m^{\mathrm{cut}})^{\theta_{J_m}}, \qquad \mathfrak{B}_m = [\widehat{\beta}_m/2, 2\widehat{\beta}_m].$$ These give the field thresholds, activity bounds and coupling bands. The quantity $\Lambda_m^{\mathrm{loop}}$ bounds the weighted quadratic loop sum in Section 12, 12.3. |
| 10 | Actions, classes, recursion, trajectories, couplings; width $w_{\mathrm{tun}}$ | Depend on rows 1–9. Choose the path actions and admissible subsets, the specified recursion and its trajectories for every finite $N, p$, all with the common level-zero endpoint (Theorem 3.6). Proposition 3.7 gives the analytic variation used in the ultraviolet limit. |

Table 2 (continued)

| | Choice | Dependence, restrictions and purpose |
|---|---|---|
| 11 | Marked transfers, terminal maps, insertions | Depend on rows 1–10; impose reference normalization (Definition 3.9). Fix them before any prescribed volume sequence. |
| 12 | $\mathsf{p}$, then $\mathsf{q}, P, S, F_d, \partial_a, \mathcal{L}$ | Prescribe arbitrary increasing $\mathsf{p}$ first. Theorem 4.7 and Proposition 4.11 give the retained volume indices $P = \mathsf{p} \circ \mathsf{q}$ and continuum fields. Theorem 4.2 retains the full inner limit. |
| 13 | Comparison radii, frames, $J$, charges, $c_\star, R, T$ | Depend on row 12 and the fixed lattice family. Use Proposition 5.2, then Theorem 5.4 and Proposition 5.5. Refine only comparison radii. |
| 14 | Slab operators, $\mathsf{X}, \nu, S^\xi$, then $\xi_*$ | Depend on earlier data. Theorem 6.2 gives the phase decomposition. Choose $\xi_*$ from the full-measure set with one-dimensional time-axis OS ground space (17.5). This choice precedes reconstruction. |
| 15 | Every-phase theories and continuations; $\Theta, \mu_\Theta$ | Depend on earlier data. Theorem 6.15 reconstructs the theories. Definition 7.1 centers the trace in the already selected phase; Theorem 7.2 gives its short-distance law; 18.2 supplies its spectral measure. |
| 16 | OPE prescription; contact projections, distributions, bases, scales | Depend on rows 12–14 and the fixed family. Theorems 7.6 and 7.7 use the same original fields and correlation table. The contact projections, separating functionals and clearance are chosen separately from those used for the product coefficients. |
| 17 | Estimate constants and thresholds | Depend only on the fixed data preceding them in each statement. Their dependence is uniform in positions, volumes and states wherever asserted. |

# Part II
# Proofs

## 10 Local polynomial fields

The polynomial labels of Definition 3.8 specify continuum Taylor coefficients, whereas the Wilson measure averages functions of group-valued links. We relate them by local sections: polynomials in the real and imaginary parts of finitely many link entries that reproduce each label on radial test holonomies through its engineering degree. They initialize the renormalized insertions in Definition 3.9, hence the Wilson family of Theorem B, independently of the lattice volume and later block ratios.

We must recover the connection coefficients visible to a label from finitely many measurements. Transported plaquette curvatures have linearizations given by ordinary square circulations. Vanishing of all these circulations forces the highest homogeneous curvature to vanish; radial gauge then forces the corresponding potential to vanish. Descending in degree proves injectivity. Finite dimensionality reduces this to finitely many plaquettes. A local inverse of their joint map, followed by Taylor expansion, gives the required polynomials. Averaging over signed permutations imposes covariance without changing the Taylor coefficients.

Each curvature factor also contributes one small-field threshold factor to the bounds in

Lemma 10.3. For the equation projections in Section 14, Lemma 10.4 constructs auxiliary sections with exact exterior zero moments. Their labels undergo a fixed invertible triangular change. We use Definitions 2.1, 2.2, 3.3 and 3.8, the chord (2.1), and the order of choices in Section 9; no earlier Part II result is used.

## Labels and test holonomies

Write $\mathcal{J}_j$ for the real space of homogeneous radial potentials of degree $j+1$ in Definition 3.8. Every label in $V_d$ is a polynomial on the finite-dimensional space

$$\mathcal{J}_{\le d} = \bigoplus_{j+2\le d} \mathcal{J}_j \,.$$

A coordinate of $\mathcal{A}_j$ has weight $j+2$, so only finitely many monomials have weight at most $d$. Adjoint action and conjugation preserve both gradings. Real and imaginary parts span each invariant component over $\mathbb{C}$, giving bihomogeneous bases as in Definition 3.8. Conjugation in this real basis is coefficientwise and agrees with conjugation of polynomial values on real jets. Write $k_i$ for the number of jet-coordinate factors in $b_{d,i}$. We identify a coefficient vector with its polynomial:

$$\mathbf{c}(\mathrm{j}) = \sum_i c_i b_{d,i}(\mathrm{j}), \qquad b_{d,0} = \mathbf{1} \,.$$

We use the test holonomies (3.16), on the lattice $\mathbb{Z}^4$ covering the torus. With $\mathcal{A}^{\le d} = \sum_{j+2\le d} \mathcal{A}_j$, their integral equation is

$$U(h_{\mathrm{lat}}, \mathrm{j})(x,a) = u(1), \qquad u(\tau) = 1 + \int_0^\tau u(\eta) h_{\mathrm{lat}} \mathcal{A}_a^{\le d}(h_{\mathrm{lat}}(x+\eta e_a))\, d\eta \,. \tag{10.1}$$

The defining linear matrix equation has a unique global solution for continuous coefficients [28, Theorem 3.9]. Piecewise constant Lie-algebra coefficients give products of exponentials in $G$; uniform convergence on compact intervals, by Gronwall's inequality [28, Lemma 2.7], and closedness of $G$ put the solution in $G$. Homogeneity and uniqueness give

$$U(h_{\mathrm{lat}}, \mathrm{j}) = U(1, \mathrm{sc}_{h_{\mathrm{lat}}} \mathrm{j}), \qquad \mathrm{sc}_{h_{\mathrm{lat}}} \mathrm{j} = (h_{\mathrm{lat}}^{j+2} \mathcal{A}_j)_j, \qquad U(0, \mathrm{j}) = 1 \,. \tag{10.2}$$

For a cover plaquette $\square$ based at $z$ and a path $\gamma$ from 0 to $z$, write $U_\gamma$ for the ordered product of successive oriented links, read from the initial point to the endpoint. Define its transported curvature by

$$\sigma_{\gamma,\square}(U) = \mathrm{Ad}(U_\gamma) F_\square(U), \qquad F_\square(U) = \mathrm{proj}_{\mathfrak{g}} \frac{U_\square - U_\square^{-1}}{2} \,.$$

The projection is orthogonal for the normalized Frobenius metric of Definition 2.1. These measurements transform by the adjoint action at 0. They are real polynomial functions of the real and imaginary parts of link entries, since $U_e^{-1} = U_e^*$.

**Definition 10.1** (Local sections)**.** For a finite site stencil $\Sigma_d \subset \mathbb{Z}^4$, a local section has the form

$$P_d^{\mathrm{loc}}[\mathbf{c}](U) = \sum_{\nu,\mu} p_{\nu\mu}^{(d)}(\mathbf{c}) \prod_{e,\alpha,\delta} U_e(\alpha,\delta)^{\nu_{e\alpha\delta}} \overline{U_e(\alpha,\delta)}^{\mu_{e\alpha\delta}} \,.$$

The product uses positive links with both endpoints in $\Sigma_d$ and matrix indices $1 \le \alpha, \delta \le n_G$. The sum is finite over nonnegative multi-indices; each coefficient is complex linear in $\mathbf{c}$. The construction below chooses such a map to represent each continuum label.

For a torus of side $n_{\mathrm{tor}}$ (the side $s$ of Definition 2.2), write $[x]$ for reduction of $x \in \mathbb{Z}^4$ modulo $n_{\mathrm{tor}}$. Placement at a torus site $y$ means evaluation on the periodic extension $U^{[y]}(x,a) = U(y+[x],a)$; denote it by $P_d^{\mathrm{loc}}[\mathbf{c}](y;U)$. On the cover, $P_d^{\mathrm{loc}}[\mathbf{c}](0;U) = P_d^{\mathrm{loc}}[\mathbf{c}](U)$. For $\rho \in H_4$, define the configuration and jet actions by

$$(\rho^* U)(x,a) = \begin{cases} U(\rho x, b), & \rho e_a = e_b, \\ U(\rho x - e_b, b)^{-1}, & \rho e_a = -e_b, \end{cases}$$
$$(\rho \mathsf{j})_{j,a}(z) = \sum_b \rho_{ab} \mathcal{A}_{j,b}(\rho^{-1} z), \qquad (\rho \mathbf{c})(\mathsf{j}) = \mathbf{c}(\rho^{-1}\mathsf{j})\,.$$

This action preserves homogeneity and radial gauge. Reality and covariance mean

$$P_d^{\mathrm{loc}}[\overline{\mathbf{c}}](U) = \overline{P_d^{\mathrm{loc}}[\mathbf{c}](U)}, \qquad P_d^{\mathrm{loc}}[\mathbf{c}](\rho^* U) = P_d^{\mathrm{loc}}[\rho \mathbf{c}](U) \quad (\rho \in H_4),$$

with conjugation as fixed above.

The following result supplies the sections specified in Section 3. The unit normalization is exact at every configuration; the Taylor assertion concerns the test configurations near zero spacing.

**Lemma 10.2** (Taylor-normalized sections)**.** *For every $d$, a box stencil $\Sigma_d = [-\varrho_d, \varrho_d]^4 \cap \mathbb{Z}^4$ and sections can be chosen gauge invariant, real and $H_4$ covariant, with $P_d^{\mathrm{loc}}[\mathbf{1}] = 1$, so that, for every $\mathbf{c} \in V_d$, radial jet $\mathsf{j}$ and $0 \le q \le d$,*

$$\frac{d^q}{dh_{\mathrm{lat}}^q} P_d^{\mathrm{loc}}[\mathbf{c}](U(h_{\mathrm{lat}}, \mathsf{j}))\Big|_{h_{\mathrm{lat}}=0} = \frac{d^q}{dh_{\mathrm{lat}}^q} \mathbf{c}(\mathrm{sc}_{h_{\mathrm{lat}}} \mathsf{j})\Big|_{h_{\mathrm{lat}}=0}\,. \tag{10.3}$$

Fix the sections constructed in the following proof.

*Proof.* We use the metric and lattice transports of Definitions 2.1–2.2 and the labels of Definition 3.8. If $d < 2$, the labels are constant and their constant sections suffice. Fix $d \ge 2$.

*Step 1: Curvatures determine the truncated jet.*

Use the $\mathrm{Ad}(G)$-invariant coefficient norm on $\mathcal{J}_{\le d}$ obtained from the metric of Definition 2.1. The Volterra term of order $l$ is bounded by $(C_{x,a,d}|\mathsf{j}|)^l/l!$, giving locally uniform convergence with all parameter derivatives and smooth measurements. The first variation of a link solution is its edge integral; that of a plaquette measurement is their circulation. Transport along $\gamma$ contributes nothing at first order, because $F_\square(1) = 0$.

Suppose all these linearized measurements vanish on $\mathcal{A} = \sum_{j=0}^m \mathcal{A}_j$. For $a < b$ set

$$N_{j;ab}(x,\tau) = \int_0^1 \int_0^1 (\partial_a \mathcal{A}_{j,b} - \partial_b \mathcal{A}_{j,a})(x + \tau(\xi e_a + \eta e_b))\, d\xi\, d\eta\,.$$

The fundamental theorem of calculus on each edge identifies the unit-square circulation with $\sum_{j \le m} N_{j;ab}(x,1)$. It is zero for every $x \in \mathbb{Z}^4$. Homogeneity gives $N_{j;ab}(\tau x, \tau) = \tau^j N_{j;ab}(x,1)$ and hence

$$\sum_{j=0}^m \tau^{m-j} N_{j;ab}(\tau x, \tau) = 0 \qquad (x \in \mathbb{Z}^4,\ \tau > 0)\,.$$

For $z \in \mathbb{R}^4$, set $x_\tau = (\lfloor z^a/\tau \rfloor)_a$, so $|\tau x_\tau - z|_\infty \le \tau$. Continuity as $\tau \downarrow 0$ leaves only the term $j = m$, giving $(\partial_a \mathcal{A}_{m,b} - \partial_b \mathcal{A}_{m,a})(z) = 0$. Antisymmetry covers the other ordered pairs. Differentiating the radial condition and using Euler's identity gives

$$(m+2)\mathcal{A}_{m,b}(z) = \sum_a z^a (\partial_a \mathcal{A}_{m,b} - \partial_b \mathcal{A}_{m,a})(z) = 0\,.$$

Remove this component and descend in $m$. The common kernel is zero. Adding a measurement whenever the kernel is nonzero decreases its dimension, so finitely many measurements $\sigma_1, \dots, \sigma_{M_{\mathrm{meas}}}$ give a joint map

$$\mathsf{S} : \mathcal{J}_{\le d} \longrightarrow \mathfrak{g}^{M_{\mathrm{meas}}}, \qquad \mathsf{S}(\mathsf{j}) = (\sigma_l(U(1,\mathsf{j})))_{l=1}^{M_{\mathrm{meas}}}$$

with injective derivative $D\mathsf{S}(0)$.

*Step 2: A polynomial inverse to the required order.* Uniqueness of the link equation shows that a constant gauge transformation takes the test links of $\mathsf{j}$ to those of $\mathrm{Ad}(v)\mathsf{j}$. Thus $\mathsf{S}$ and its derivative at zero are adjoint equivariant. The range of $D\mathsf{S}(0)$ and its orthogonal complement in $\mathfrak{g}^{M_{\mathrm{meas}}}$ are adjoint invariant. Compose its inverse on the range with orthogonal projection to obtain $C : \mathfrak{g}^{M_{\mathrm{meas}}} \to \mathcal{J}_{\le d}$. Then

$$C D\mathsf{S}(0) = I, \qquad C\,\mathrm{Ad}(v) = \mathrm{Ad}(v) C \quad (v \in G)\,.$$

Apply the inverse function theorem [25, Theorem 9.24] to $C\mathsf{S}$ on this finite-dimensional real space: it is smooth near zero, sends zero to zero and has derivative $I$. Its local inverse $\varphi$ is smooth, by differentiating $D\varphi = (D(C\mathsf{S}) \circ \varphi)^{-1}$. Choose an invariant source ball on which $C\mathsf{S}$ is injective, and then an invariant target ball whose inverse image lies in that source ball. Both exist because the adjoint actions are isometric. Put $f = \varphi \circ C$. Equivariance of $C\mathsf{S}$ and injectivity on the source ball give, near zero,

$$f\mathsf{S}(\mathsf{j}) = \mathsf{j}, \qquad f(0) = 0, \qquad f(\mathrm{Ad}(v)z) = \mathrm{Ad}(v) f(z) \quad (v \in G)\,.$$

Consequently the Taylor polynomial

$$T_d f(z) = \sum_{k=1}^{d} \frac{1}{k!} D^k f(0)[z, \dots, z]$$

is adjoint equivariant and has no constant term. Define

$$Q_i(z) = b_{d,i}(T_d f(z)), \qquad \Psi_{\mathbf{c}}(U) = \sum_i c_i Q_i(\sigma_1(U), \dots, \sigma_{M_{\mathrm{meas}}}(U))\,.$$

The polynomials $Q_i$ are invariant and real on real arguments, and $Q_0 = 1$. If $b_{d,i}$ has factor count $k_i$, every monomial of $Q_i$ has at least $k_i$ factors.

Taylor's formula gives $T_d f(\mathsf{S}(X)) = X + O(|X|^{d+1})$ near 0. Substitute $X = \mathrm{sc}_{h_{\mathrm{lat}}} \mathsf{j}$ truncated to $\mathcal{J}_{\le d}$, and use (10.2). Both sides are smooth in $h_{\mathrm{lat}}$, and

$$\Psi_{\mathbf{c}}(U(h_{\mathrm{lat}}, \mathsf{j})) = \mathbf{c}(\mathrm{sc}_{h_{\mathrm{lat}}} \mathsf{j}) + O_{\mathbf{c},\mathsf{j}}(|h_{\mathrm{lat}}|^{d+1})\,.$$

Their derivatives through order $d$ at zero agree. The polynomial $Q_i$ itself need not have degree at most $d$ in the curvature measurements.

*Step 3: Covariance and a common stencil.* Average the section by

$$P_d^{\mathrm{loc}}[\mathbf{c}](U) = \frac{1}{|H_4|} \sum_{\rho \in H_4} \Psi_{\rho^{-1}\mathbf{c}}(\rho^* U)\,. \tag{10.4}$$

Changing the summation variable to $\omega\rho$ proves $P_d^{\mathrm{loc}}[\mathbf{c}](\omega^* U) = P_d^{\mathrm{loc}}[\omega\mathbf{c}](U)$. Simultaneous adjoint covariance of the measurements gives gauge invariance. The real basis and signed-permutation action give reality; the constant column remains one. The action preserves factor count, so averaging retains the lower bound on the number of curvature factors.

For a positive image direction, the pulled-back link equation has coefficient

$$h_{\mathrm{lat}} \sum_{b} \rho_{ba} \mathcal{A}_{b}^{\leq d}(h_{\mathrm{lat}} \rho(x + \tau e_a)),$$

the coefficient for $\rho^{-1}\mathsf{j}$. Reversing the parameter and inverting the transport covers a negative image direction. Uniqueness gives $\rho^{*} U(h_{\mathrm{lat}}, \mathsf{j}) = U(h_{\mathrm{lat}}, \rho^{-1}\mathsf{j})$. Each term of (10.4) therefore has Taylor polynomial $\mathbf{c}(\mathrm{sc}_{h_{\mathrm{lat}}}\, \mathsf{j})$ through order $d$. A signed image of a transported curvature is again a transported curvature, up to sign: reversal changes $F_{\square}$ to $-F_{\square}$, and a change of base point gives conjugation by the intervening boundary transport. Enclose the finitely many resulting paths and plaquettes in a box $\Sigma_d = [-\varrho_d, \varrho_d]^4 \cap \mathbb{Z}^4$. This stencil contains zero and is signed-permutation invariant. Expanding the finite polynomial in matrix entries gives Definition 10.1. This proves (3.17). □

## Bounds and lattice symmetries

We bound normalized initial insertions using the ordered left derivatives of Definition 3.3. At a placement $y$, set $Y = y + [\Sigma_d]$; being healthy on $Y$ means that every plaquette meeting $Y$ has chord at most $t$. A direction is permitted when

$$|\dot{\mathcal{A}}_e| \leq t(1 + \mathrm{dist}(e^{-}, y))^{P}\,.$$

Here $[\Sigma_d]$ is the torus image of the stencil, and $\mathrm{dist}(x, y) = \min_{z \in n_{\mathrm{tor}}\mathbb{Z}^4} |\tilde{x} - \tilde{y} + z|_{\infty}$ is the periodic maximum-coordinate distance, for any lifts $\tilde{x}, \tilde{y}$.

**Lemma 10.3** (Section bounds)**.** *Each basis column is a polynomial in transported curvatures with at least $k_i$ factors per monomial. For all $d, k, P \in \mathbb{N}$ there are finite constants $C_{d,k,P}$ and $C_d$ such that, on every torus, for every placement $y$, $0 < t \leq 1$, configuration healthy on $Y$, and permitted direction list of length $q \leq k$,*

$$\left| D^{L}_{\dot{\mathcal{A}}_1} \cdots D^{L}_{\dot{\mathcal{A}}_q} P_d^{\mathrm{loc}}[b_{d,i}](y; U) \right| \leq C_{d,k,P} t^{k_i}\,. \tag{10.5}$$

*For arbitrary configurations,*

$$\sup_{i,y,U} |P_d^{\mathrm{loc}}[b_{d,i}](y; U)| \leq C_d\,. \tag{10.6}$$

The proof uses the curvature polynomials constructed in Lemma 10.2.

*Proof.* First note that $|F_{\square}(U)| \leq \mathrm{ch}(U_{\square})$: orthogonal projection is contractive and $\|U_{\square} - U_{\square}^{*}\|_{\mathrm{F}} \leq 2\|U_{\square} - 1\|_{\mathrm{F}}$. Adjoint transport is isometric, so an undifferentiated curvature measurement is bounded by $t$. An order-$q$ left derivative of a fixed path product is a finite sum of products of unitary matrices and exactly $q$ direction matrices. On the fixed stencil their norms are at most a constant times $t$, since periodic distance is bounded by cover distance. Thus each differentiated measurement is $O(t^q)$, and hence $O(t)$ when $q \geq 1$. This argument also applies when cover links coincide after placement: one first differentiates the cover expression and then assigns the same direction to coincident links.

Apply the product rule to the curvature polynomials of (10.4). Every differentiated monomial still has at least $k_i$ factors, each bounded by a constant times $t$. There are finitely many terms; since $t \leq 1$, their sum gives (10.5), including order zero. Each entry of a unitary matrix has absolute value at most one, so the finite matrix-entry polynomial gives (10.6). The formula also gives smoothness, measurability and locality. The constants depend on the sections, $G, d, k, P$ as applicable, and finite maxima make them nondecreasing in the integer parameters. They are fixed before the stage block ratios in Section 9. □

By linearity, Lemma 10.3 bounds permitted derivatives of $P_d^{\mathrm{loc}}[\mathsf{D}_{t^{-1}}^{\mathrm{fac}}\mathbf{c}]$ by $C_{d,k,P}\sum_i|c_i|$ on the healthy set: the factor $t^{-k_i}$ cancels $t^{k_i}$. Taking $t=t_N$, $k=\mathfrak{q}_{J_N}$ and $P=\varpi_{J_N}$ bounds the initial marked density of (3.18) on the healthy set, in the sense of Definition 3.3. On unhealthy configurations the global bound gives $C_d\sum_i|c_i|t^{-k_i}$; Lemmas 14.8–14.9 place this contribution in the forced remainders of (3.21).

**Finite-lattice invariance.** For the identities deferred from Definition 2.2, telescoping the path products shows that gauge transformations act on plaquette holonomies by conjugation. An affine isometry $h_{\mathrm{iso}}(x)=\rho x+b$, with $\rho\in H_4$, changes an image plaquette holonomy by conjugation, possibly composed with inversion. Both operations preserve pr, hence the Wilson energy. Fixed left and right multiplication, inversion and permutation preserve product probability Haar measure [9, Corollary 2.28 and Proposition 2.31]. Consequently, for every integer side $n_{\mathrm{tor}}\geq 1$, real $\beta$, site gauge transformation $\gamma_{\mathrm{g}}$ and bounded measurable $F$,

$$\int F(U^{\gamma_{\mathrm{g}}})\,d\mu_{n_{\mathrm{tor}},\beta}(U)=\int F(h_{\mathrm{iso}}^{*}U)\,d\mu_{n_{\mathrm{tor}},\beta}(U)=\int F(U)\,d\mu_{n_{\mathrm{tor}},\beta}(U)\,. \tag{10.7}$$

All integrals exist: $0\leq\mathrm{pr}\leq 2$ gives $0\leq E_{n_{\mathrm{tor}}}^{\mathrm{W}}\leq 12n_{\mathrm{tor}}^4$ and $0<Z_{n_{\mathrm{tor}},\beta}<\infty$. The terminal transfer normalization in Definition 3.9 is proved in Lemmas 14.6–14.7.

## Auxiliary equation sections

Exact lattice equation identities require different sections, still polynomial and local with the same small-field bounds. On $\mathcal{J}_j$ define the Maxwell subspace and its polynomial ideal by

$$K_j=\left\{\mathcal{A}_j:\ \sum_a\partial_a(\partial_a\mathcal{A}_{j,b}-\partial_b\mathcal{A}_{j,a})=0\text{ for every }b\right\},\qquad \mathcal{I}_d=\{p\in V_d:p|_{\prod_j K_j}=0\}\,.$$

Here restriction uses only $j+2\leq d$. Substitution into $\prod_j K_j$ preserves bidegree, so $\mathcal{I}_d$ is the direct sum of its bihomogeneous parts. Order bidegrees $(w,k)$ by the lexicographic order of $(k,w)$. A regrading is a complex-linear map whose difference from the identity sends each bihomogeneous polynomial into strictly later bidegrees. Such a map preserves the filtration by minimum factor count. For a torus site $y$, put

$$\Omega_r(y)=\{x:\mathrm{dist}(x,y)>r\}\,.$$

An exterior spectator is a bounded measurable function of the positive links having both endpoints in $\Omega_r(y)$; it need not be gauge invariant.

The following identity supplies the auxiliary equation observables used in Section 14. Its proof constructs them from derivatives of the Wilson energy with multipliers independent of the varied link.

**Lemma 10.4** (Auxiliary equation sections)**.** *For every $d$ there are a local section $V_d^{\mathrm{aux}}$ satisfying Lemmas 10.2 and 10.3 with its own stencil and constants, an invertible regrading $\mathsf{R}_d$ fixing $\mathbf{1}$ and commuting with conjugation and $H_4$, and integers $r_d,M_d$ such that every $n_{\mathrm{tor}}\geq M_d$, $\beta\neq 0$, placement $y$, $p\in\mathcal{I}_d$ and exterior spectator $S$ on $\Omega_{r_d}(y)$ satisfy*

$$\int V_d^{\mathrm{aux}}[\mathsf{R}_d p](y;U)S(U)\,d\mu_{n_{\mathrm{tor}},\beta}(U)=0\,. \tag{10.8}$$

The exact-equation subspace for these sections is $\mathsf{R}_d(\mathcal{I}_d)$. Its comparison with the selected primary family is deferred to Section 13. All choices in the following proof depend only on $G, d$ and the label basis.

*Proof.* We use Definitions 2.1, 2.2, 3.8 and Lemmas 10.2–10.3.

*Step 1: One-link equations.* Extend the curvature $F_\square$ defined above to $F_{ab}(U;x)$ for ordered axes by antisymmetry, with $F_{aa}=0$. For a positive link $e=(x,a)$ set

$$\mathcal{E}_e(U)=\sum_{b\neq a}\big(F_{ab}(U;x)-\mathrm{Ad}(U(x-e_b,b)^{-1})F_{ab}(U;x-e_b)\big)\ .$$

On a torus of side at least two, direct differentiation gives

$$D^L_{e,A}E^{\mathrm{W}}(U)=\langle A,\mathcal{E}_e(U)\rangle_{\mathfrak{g}}\,. \tag{10.9}$$

Indeed, the forward plaquette contributes $-n_G^{-1}\operatorname{Re}\operatorname{tr}(AU_\square)=\langle A,F_\square\rangle_{\mathfrak{g}}$. The backward plaquette, rebased at $x$, contributes the negative of the transported curvature in the display. These are the two incidences for each transverse axis; antisymmetry gives the signs for the opposite ordering.

If a continuous multiplier $g$ is independent of $U_e$, and a bounded measurable $S$ is also independent of $U_e$, left invariance of Haar measure gives

$$0=\frac{d}{d\epsilon}\int g(U)S(U)e^{-\beta E^{\mathrm{W}}(U^{e,A,\epsilon})}\,dU\Big|_{\epsilon=0}=-\beta Z_{n_{\mathrm{tor}},\beta}\int\langle A,\mathcal{E}_e\rangle_{\mathfrak{g}}gS\,d\mu_{n_{\mathrm{tor}},\beta}\,.$$

Here $U_e^{e,A,\epsilon}=e^{\epsilon A}U_e$ and all other links are fixed. Compactness bounds the derivative of the smooth density uniformly, justifying differentiation even for measurable $S$.

Choose a path $\gamma_e$ from $0$ to $e^-$ avoiding $e$. If $g$ also avoids $e$, then

$$\Theta(U)=\langle\xi,\mathrm{Ad}(U_{\gamma_e})\mathcal{E}_e(U)\rangle_{\mathfrak{g}}g(U)$$

has zero moment for every $\xi\in\mathfrak{g}$ against every exterior spectator. To see this, expand $\mathrm{Ad}(U_{\gamma_e}^{-1})\xi$ in a fixed orthonormal basis of $\mathfrak{g}$ and apply the preceding identity to each coefficient times $gS$. These multipliers are independent of $U_e$. There is one radius and one minimum torus side for any finite collection of such expressions: enlarge the radius to contain both endpoints of every varied link, and the side until distinct links in their finite cover supports remain distinct after wrapping. The same choices work at every placement.

*Step 2: Enough measurements with avoidance.* Any path can avoid one specified link: replace a traversal of that link by three sides of a square in a transverse direction. For each plaquette orientation only finitely many base points give a plaquette containing that link. Removing these points does not affect the shrinking-square argument in Lemma 10.2: for $z\neq 0$, the integer points $x_\tau=\lfloor z/\tau\rfloor$ eventually avoid any finite set, and continuity gives the conclusion at $z=0$. Thus curvature measurements whose paths and plaquettes avoid $e$ still have injective joint linearization. Finite-dimensional duality therefore realizes every linear functional on $\mathcal{J}_{\le d}$ by finitely many components.

For every linear functional $\ell$ on $\mathcal{J}_j$ vanishing on $K_j$, a finite combination of force linearizations equals $\ell\circ\mathrm{pr}_j$ on $\mathcal{J}_{\le j+2}=\bigoplus_{i\le j}\mathcal{J}_i$. Write $N_{j;ab}$ for the square averages in the proof of Lemma 10.2, extended antisymmetrically. The force linearization at $e=(x,a)$ is

$$L_a(\mathsf{j};x)=\sum_{i,b}\big(N_{i;ab}(x,1)-N_{i;ab}(x-e_b,1)\big)\,.$$

For a homogeneous component define

$$H_{j;ab}(z,\tau) = \int_{[0,1]^3} \partial_b(\partial_a \mathcal{A}_{j,b} - \partial_b \mathcal{A}_{j,a})\big(z + \tau(\xi e_a + (\eta + \zeta - 1)e_b)\big)\, d\xi\, d\eta\, d\zeta\,.$$

The fundamental theorem of calculus in the backward direction gives

$$N_{j;ab}(z,\tau) - N_{j;ab}(z - \tau e_b, \tau) = \tau H_{j;ab}(z,\tau)\,.$$

If all $L_a(\mathrm{j}; x)$ vanish for jets through degree $j = m$, homogeneity gives $\sum_{j \le m,b} \tau^{m-j} H_{j;ab}(\tau x, \tau) = 0$ for $\tau > 0$. Taking $x = x_\tau$ and $\tau \downarrow 0$ shows $\sum_b \partial_b(\partial_a \mathcal{A}_{m,b} - \partial_b \mathcal{A}_{m,a}) = 0$, hence $\mathcal{A}_m \in K_m$. A functional of this top component annihilating $K_m$ therefore annihilates the common kernel of the force measurements. Finite-dimensional duality puts it in their linear span. Transport paths do not enter the linearization and can be chosen to avoid their varied links. On $\mathcal{J}_{\le d}$ the same combination can have additional linear terms only at engineering weights above $m + 2$.

*Step 3: Polynomial lifts and the triangular map.* Choose linear projections $\pi_j : \mathcal{J}_j \to K_j$ and put $\pi = \prod_j \pi_j$. A polynomial belongs to $\mathcal{I}_d$ precisely when $p \circ \pi = 0$. For a product of linear coordinates,

$$\prod_{i=1}^{k_{\mathrm{tot}}} \ell_i - \prod_{i=1}^{k_{\mathrm{tot}}} (\ell_i \circ \pi) = \sum_{i=1}^{k_{\mathrm{tot}}} \left[\prod_{l<i} (\ell_l \circ \pi)\right] (\ell_i - \ell_i \circ \pi) \prod_{l>i} \ell_l\,.$$

Consequently every bihomogeneous equation label is a finite sum of products with one linear factor vanishing on some $K_j$, with the same engineering weight and factor count. The individual factors need not be invariant. For an observable $O$, its Taylor polynomial through engineering weight $d$ means the degree-$d$ Taylor polynomial at zero of $\mathrm{j} \mapsto O(U(1,\mathrm{j}))$ on $\mathcal{J}_{\le d}$, with monomials of weight above $d$ discarded. Every coordinate has weight at least two. By (10.2), matching this polynomial is equivalent to matching the spacing derivatives through order $d$ in (10.3).

Lift that distinguished factor by a finite combination of transported forces from Step 2. For each summand, lift the other factors by curvature measurements avoiding its varied link, matching their full linear parts through engineering weight $d$. The resulting expression has the zero-moment property of Step 1 and at least $k_{\mathrm{tot}}$ curvature factors. Indeed, every force is a difference of two transported curvatures per transverse axis. Its Taylor polynomial through engineering weight $d$ equals the prescribed product plus terms later in $(k, w)$: the additional linear terms of the force have greater engineering weight, and every nonlinear correction increases factor count. The square-measurement lifts have their prescribed full linear parts, so introduce only the latter kind of correction. Taylor polynomials exist by the smooth Volterra dependence used in Lemma 10.2.

Average over constant gauge transformations. All factors depend on a gauge transformation only at the origin, where they transform by adjoint matrices. Their orbit therefore lies in the finite-dimensional span of finitely many fixed-degree colour-coordinate monomials on the same measurements. Each orbit function retains zero moments, with colour vectors rotated by $\mathrm{Ad}(v^{-1})$. The Haar average is a finite linear combination of orbit functions, so preserves zero moments, minimum factor count and triangular Taylor identity, and is gauge invariant. The test-holonomy identity makes its Taylor polynomial invariant too.

In each bidegree extend a basis of $\mathcal{I}_d$ to a basis of the invariant label space in that bidegree, lifting the complementary vectors by Lemma 10.2. Lift $\mathbf{1}$ to 1, since $\mathcal{I}_d$ contains no nonzero

constant. Summing gives a linear lift $p \mapsto \mathcal{Y}_p$. Replace this lift by the two successive averages

$$\mathcal{Y}'_p(U) = \frac{1}{|H_4|} \sum_{\rho \in H_4} \mathcal{Y}_{\rho^{-1}p}(\rho^* U),$$
$$\mathcal{Y}''_p = \frac{1}{2}\left(\mathcal{Y}'_p + \overline{\mathcal{Y}'_{\bar{p}}}\right).$$

The Taylor map commutes with both operations; write $\mathcal{Y}_p$ for the final lift. The real Maxwell equations and the chain rule make $K_j$, and hence $\mathcal{I}_d$, stable under signed permutations, adjoint transformations and conjugation. The zero-moment property survives the signed average by (10.7), changing the spectator by the inverse isometry; the exterior set is preserved. Conjugation uses $\bar{S}$. A reversed varied link uses a right derivative of the Wilson energy, whose Haar identity is identical. Thus the averaged equation representatives remain finite sums of one-link action derivatives with link-independent multipliers and avoiding transport paths.

Let $\mathsf{R}_d p$ be the Taylor polynomial of the resulting $\mathcal{Y}_p$ through engineering weight $d$. Then $\mathsf{R}_d - I$ strictly raises $(k, w)$. There are finitely many bidegrees of weight at most $d$, so $(\mathsf{R}_d - I)^{h_0} = 0$ for some integer $h_0$, and

$$\mathsf{R}_d^{-1} = \sum_{l=0}^{h_0-1} (-1)^l (\mathsf{R}_d - I)^l, \qquad V_d^{\mathrm{aux}}[c] = \mathcal{Y}_{\mathsf{R}_d^{-1} c}.$$

This gives invertibility and Taylor normalization; the averages give the symmetry and unit identities. The inverse preserves minimum factor count, so Lemma 10.3 applies to these sections with their own constants. A signed-invariant box enclosing all paths and plaquettes gives a common stencil. Take $r_d, M_d$ as maxima for an ideal basis; linearity gives (10.8). □

## 11 Gauges and block integration

The block map of Definition 3.1 averages fine parallel transports in local frames. A fixed axial frame would distinguish an ordering of the axes. We instead minimize a symmetric average of transport energies when the field is sufficiently small, and average over curvature-controlled axial frames elsewhere. Lemma 11.4 and the paragraph following its proof show that these choices give a measurable, covariant probability kernel. The normalization preserves the partition function under the block integration in Theorem 3.5.

The remaining lemmas prepare the local integration used in Sections 12 and 13. The free block problem determines the quadratic action and the linear interpolation. An axial coercivity estimate then gives the nonlinear stationary branch. Its Gaussian integral contributes a determinant and the Haar-coordinate density; together with the inherited loop action they give the one-loop action. The third and fourth derivatives have bounds that improve the input-dependent part by one power of the block ratio. A stationary gauge identity removes the predictor contact term from the tested quadratic coupling coefficient. Lemmas 11.11 and 11.13 give the local large-field energy and the simultaneous differentiated integral used in Section 13.

Three uniformities require care. The frame cutoffs must work for arbitrarily many sites, and the free kernels must remain controlled after arbitrarily many integrations. The first follows from a contraction after centering the frame minimizers. For the second we separate the principal Fourier pole before inversion. In the determinant comparison, the fine cell contains $\ell^4$ sites. Weighted inverses and the summable cube of a regulator prevent this number from entering the comparison constant.

Throughout, use the group and metric of Definition 2.1, the Wilson term of Definition 2.1, and the level choices (3.1). At a step $m+1 \to m$, put $j = J_{m+1}$, $\ell = \ell_m$, and $R = (j+3)\ell$. Periods satisfy the minimum-period requirement in Section 9. All finite operators use counting inner products and are periodizations of the kernels described below. An adjoint at complex momentum means the analytic continuation of the real-momentum adjoint. Bounds for an input path action refer to Definition 12.2 and Lemmas 12.3–12.4 at the finer level. They are used in descending induction, starting with Wilson at the top level.

## Frames and predictors

Put $\mathbb{B}_R = \{-R, \dots, R\}^4$. For $o \in S_4$, let $T_U(y \to z; o)$ be the ordered product along the path that changes the four coordinates in the order $o$, using inverse links for negative steps.

**Definition 11.1** (Axial frames)**.** The frame tagged by $(z, o)$, its normalization and its link coordinates are

$$g_y^{z,o} = T_U(z \to y; o), \qquad h_y^{z,o} = (g_0^{z,o})^{-1} g_y^{z,o}, \qquad U_{(y,a)}^{z,o} = g_y^{z,o} U(y,a) (g_{y+e_a}^{z,o})^{-1}.$$

These frames provide the finite alternatives for the symmetric gauge choice.

For an origin-based axial tag $o_0$, the route matrix tested below is

$$\mathcal{T}_{xy;o}^{o_0} = g_x^{0,o_0} T_U(x \to y; o) (g_y^{0,o_0})^{-1}.$$

The common tag set on both branches is $\mathbb{B}_R \times S_4$; on the minimizing branch the frame is independent of its tag. For the patch centred at $\ell y$, the notation of (3.4) is $h_{y,r} = h_r$. Choose $\delta_{\mathrm{fr}} > 0$ small enough as in the proof of Lemma 11.4, and $a_{\mathrm{ax}} \geq 16$. Patch sums below count internal positive links and plaquettes.

**Definition 11.2** (Symmetric frame choice)**.** When $\mathrm{ch}(\mathcal{T}_{xy;o}^{o_0}) < \delta_{\mathrm{fr}}$ for every $x, y, o, o_0$, use the minimizer, with $h_0 = 1$, of

$$\frac{1}{24|\mathbb{B}_R|^2} \sum_{y,z \in \mathbb{B}_R} \sum_{o \in S_4} \mathrm{pr}(h_y T_U(y \to z; o) h_z^{-1}).$$

Otherwise use normalized axial frames with

$$\sum_{e \text{ patch}} \mathrm{ch}(U_e^{z,o}) \leq a_{\mathrm{ax}}(2R+1) \sum_{\square \text{ patch}} \mathrm{ch}(U_\square).$$

Tags have equal mass over all tags on the first branch and eligible tags on the second. This makes the choice symmetric.

For a coarse edge use the framed connector $Y_{e,z}$ of (3.4). The offset weight is supported in $\mathbb{B}_{j(\ell-1)/2} \subset \mathbb{B}_R$.

**Definition 11.3** (Coarse predictors)**.** Let $w^{\mathrm{av}} = (\mathrm{Unif}(\mathbb{B}_{(\ell-1)/2}))^{*j}$. Define

$$Q_e^{\mathrm{pred}} = \operatorname*{argmin}_{u \in G} \sum_z w_z^{\mathrm{av}} \mathrm{pr}(Y_{e,z} u^{-1}) \quad \text{if the minimum is below } \varepsilon_{\mathrm{pred}}.$$

Give predictor tags uniform mass on $\mathbb{B}_R$ on this branch. Otherwise choose $Q_e^{\mathrm{pred}} = Y_{e,z}$ with tag mass $w_z^{\mathrm{av}}$. The product of frame and predictor masses is $\lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}})$. These choices define every summand of Definition 3.1.

The branch choices require constants independent of the patch radius. Their normalization then follows from finite probability sums and Haar invariance.

**Lemma 11.4** (Branches and normalization)**.** *For the chosen frame and predictor cutoffs and axial factor $a_{\mathrm{ax}} \geq 16$, every stated minimizer is unique on its branch, eligible axial tags are nonempty, and*

$$\sum_{\alpha_{\mathrm{tag}}} \lambda^{\mathrm{tag}}(U, \alpha_{\mathrm{tag}}) = 1, \qquad \int k_\beta^{(m)}(U, V)\, dV = 1\,.$$

The proof below establishes the cutoffs first, then the two normalizations.

*Proof.* We use (2.1) and Definitions 11.1–11.3. Work in the real Euclidean space of complex matrices with Frobenius norm. The tubular projection lemma of [10, p. 153, Lemma], applied to the compact smooth embedded submanifold $G$, gives a neighbourhood with unique smooth nearest-point projection $\pi$. At the identity its derivative is the orthogonal projection $P_{\mathfrak{g}}$ onto $\mathfrak{g}$. Consequently, for any $\epsilon > 0$, shrinking this neighbourhood gives

$$\|\pi(A) - \pi(B) - P_{\mathfrak{g}}(A - B)\|_{\mathrm{F}} \leq \epsilon \|A - B\|_{\mathrm{F}} \tag{11.1}$$

when $A, B$ lie in a sufficiently small ball about 1. Integrate $D\pi - P_{\mathfrak{g}}$ along their joining segment.

For the predictor put $\bar{Y} = \sum_z w_z^{\mathrm{av}} Y_{e,z}$. Expansion of the square gives

$$2n_G \sum_z w_z^{\mathrm{av}}\, \mathrm{pr}(Y_{e,z} u^{-1}) = \sum_z w_z^{\mathrm{av}} \|Y_{e,z} - \bar{Y}\|_{\mathrm{F}}^2 + \|\bar{Y} - u\|_{\mathrm{F}}^2\,.$$

Thus the minimizers are precisely the nearest group points to $\bar{Y}$. If the minimum is below $\varepsilon_{\mathrm{pred}}$, then $\mathrm{dist}(\bar{Y}, G) < \sqrt{2n_G \varepsilon_{\mathrm{pred}}}$. Choose this smaller than a tubular radius. This choice uses neither the weights nor their number.

For the frame, first make an origin-based axial change of gauge. Its route matrices, denoted $\mathcal{T}_{xy,o}$, satisfy $\mathcal{T}_{xx,o} = 1$ and $\mathcal{T}_{yx,o'} = \mathcal{T}_{xy,o}^{-1}$ for the reversed order $o'$. On the minimizing branch choose $\delta_{\mathrm{fr}}$ so that $\|\mathcal{T}_{xy,o} - 1\|_{\mathrm{F}} \leq d$, where $d \leq 1/4$ will be fixed below. Write av for the uniform averages over the indicated indices. A global minimizer $h$ exists by compactness. Varying its coordinate $h_x$ and pairing the two orientations shows that $h_x$ is a nearest group point to

$$F_x(h) = \mathrm{av}_{y,o}\, h_y \mathcal{T}_{xy,o}^{-1}\,.$$

The self term is harmless: replacing it by its value at a proposed competitor adds a nonnegative square. Comparison with the constant frame and the triangle inequality give

$$\mathrm{av}_{x,y}\, \|h_x - h_y\|_{\mathrm{F}}^2 \leq 4d^2\,.$$

Left-multiply $h$ so that 1 is a nearest point to its centroid $\bar{h} = \mathrm{av}_y\, h_y$. Jensen's inequality then gives $\|\bar{h} - 1\|_{\mathrm{F}} \leq 2d$ and $\|F_x(h) - 1\|_{\mathrm{F}} \leq 3d$.

Center two minimizers $h, k$ in this way, and put $u = \max_y \|h_y - k_y\|_{\mathrm{F}}$. Their centroids are normal to $G$ at 1, so $P_{\mathfrak{g}}(\bar{h} - \bar{k}) = 0$. Moreover,

$$\|F_x(h) - F_x(k)\|_{\mathrm{F}} \leq u, \qquad \|F_x(h) - F_x(k) - (\bar{h} - \bar{k})\|_{\mathrm{F}} \leq du\,.$$

Apply (11.1) with $\epsilon = 1/4$, choosing $3d$ inside its ball. At a coordinate attaining $u$ it yields $u \leq (d + 1/4)u \leq u/2$. Hence the centered minimizers coincide. All minimizers differ by a common left multiple, which $h_0 = 1$ fixes uniquely.

For the axial alternatives let $E(X,U)$ and $F(X,U)$ be the sums of link and plaquette chords internal to a rectangular slice $X$. Induct on the number $r$ of varying axes in that slice. Choose a layer perpendicular to the last axis whose plaquette sum is at most the slice average, and apply the induction hypothesis there. Extend its frame along the last axis, making those links equal to 1. A link in another layer differs from its transported link in the chosen layer by a product of intervening plaquettes. The chord triangle inequality, summed over the links, gives

$$E(X,U^{\mathrm{ax}}) \le (2R+1)\{E(X_0,U^{\mathrm{ax}}) + 4F(X,U)\} \le 4r(2R+1)F(X,U)\,.$$

For the second inequality use $(2R+1)F(X_0,U) \le F(X,U)$ in the induction bound. At $r=4$ this proves eligibility, for each axis ordering, with factor 16.

Each frame mass therefore sums to one; each predictor mass sums to one since $w^{\mathrm{av}} \ge 0$ and $\sum_z w_z^{\mathrm{av}} = 1$. Successively summing the finite tag choices proves the first identity. For fixed tags, every coarse-link factor in Definition 3.1 integrates to one by right invariance of Haar measure. Their product and the tag sum prove the second identity. □

**The kernel assertions in Part I.** The branch tests are Borel; unique compact minimizers depend continuously on their data. Axial eligibility is a closed inequality. Thus the finite tag sum is Borel, and it is bounded for fixed $\beta$ since $0 \le \mathrm{pr} \le 2$ and the Haar denominator is positive. Gauge transformations act by conjugation on the rooted transports and transform both ends of a connector; lattice isometries permute the roots, orders and offsets, including inversion of reversed transports. The objective functions, eligibility inequalities and tag masses are preserved. Uniqueness on the minimizing branches and reindexing on the other branches therefore give

$$k_\beta^{(m)}(U^\gamma, V^{\gamma\circ\iota}) = k_\beta^{(m)}(U,V), \qquad k_\beta^{(m)}(s_{\ell b}U, s_b V) = k_\beta^{(m)}(U,V), \qquad \iota(y) = \ell y\,.$$

Here $s_b$ is the pullback by $x \mapsto Ox + b$, with $O \in H_4$; $s_{\ell b}$ uses the same $O$ on fine sites. Fubini's theorem and Lemma 11.4 give (3.7). Thus the block kernel is measurable, covariant and normalized.

## The free action

Write $d_r$ for the forward lattice exterior derivative on $r$-forms. Thus $(d_1 a)_{ab}(x) = a_b(x+e_a) - a_b(x) - a_a(x+e_b) + a_a(x)$. A history at level $m$ is the finite sequence $(\ell_i, J_{i+1})_{m \le i < N}$, read from fine to coarse. Its stages decrease by at most one. Let $\mathcal{Q}_m$ be the derivative of the predictor at the flat field. To specify its free metric, set

$$q_a(\theta) = \mathrm{e}^{i\theta_a} - 1, \quad \lambda(\theta) = \sum_a q_a(\theta) q_a(-\theta),$$

$$r_L(t) = L^{-1} \sum_{s=0}^{L-1} \mathrm{e}^{ist}, \quad b_L(\theta) = \prod_a L^{-1} \sum_{|s| \le (L-1)/2} \mathrm{e}^{is\theta_a}.$$

Starting from $c_a = \lambda^{-1}$, update these auxiliary diagonal covariances by

$$c_a'(p) = \mathsf{b}^{-1} + L^{-2} \sum_{z \bmod L} b_L(\theta_z)^{2j} r_L((\theta_z)_a) r_L(-(\theta_z)_a) c_a(\theta_z), \qquad \theta_z = (p + 2\pi z)/L. \tag{11.2}$$

For $p \ne 0$, define the unmodified two-form metric and the chosen metric by

$$\widehat{\mathsf{H}}^0_{ab,cd}(p) = \frac{\mathbf{1}_{(a,b)=(c,d)}}{c_a(p)c_b(p)\sum_h q_h(p)q_h(-p)/c_h(p)}, \qquad \widehat{\mathsf{H}}^0(0) = I,$$
$$\mathsf{H}^{-1} = (\mathsf{H}^0)^{-1} + \zeta_{J_m}\big(d_2(\mathsf{H}^0)^{-1}\big)^* d_2 (\mathsf{H}^0)^{-1}.$$

This modification permits positive square completion on all two-form coordinates, including those outside the curl range. Fix $\zeta_k = \max(\zeta^0_{k,k}, \zeta^0_{k+1,k})$, where the positive thresholds for the two incoming stage pairs are chosen in the proof below. These constants are fixed before $\gamma_k$ and the clipping coefficients. The ratio threshold also depends on the incoming $\zeta$. The free action is

$$\mathsf{S}^{\mathrm{free}}_{N,m}(a) = \tfrac{1}{2}\langle d_1 a, \mathsf{H}_{N,m} d_1 a\rangle, \qquad \mathsf{K}_{N,m} = d_1^* \mathsf{H}_{N,m} d_1.$$

Let $\mathcal{M}_m a(e) = \sum_z w_z^{\mathrm{av}} \sum_{r=0}^{\ell-1} a(\ell e^- + z + r e_{\mathrm{dir}(e)}, \mathrm{dir}(e))$ be straight-link averaging. The smooth interpolation $W_0$ minimizes the fine free action plus $\mathsf{b}\|\mathcal{Q}_m a - v\|^2/2$, with the remaining gradient fixed to match the predictor; the tree representative is obtained afterwards.

Let $T$ be the free two-form transfer (the lift of the straight-link block average), fixed on curls by

$$T^* d_1 = d_1 \mathcal{M}_m, \qquad T \mathsf{H}^0_c d_1 = \mathsf{H}^0_f d_1 W_0.$$

Its extension to all forms is obtained by the same alias formula, with a straight geometric-sum factor for each of the two indices. Define the adjacent three-form defect by

$$\mathsf{H}_c(\mathsf{H}_c^{-1} - T^* \mathsf{H}_f^{-1} T - \mathsf{b}^{-1} d_1 d_1^*)\mathsf{H}_c = d_2^* \mathcal{W}_3 d_2. \tag{11.3}$$

Write $k = J_m$, and let $C^{\mathrm{tr}}_{j,k}$ be the stage-pair transfer bound obtained below; row 5 of Section 9 imposes $\ell^2 \geq C^{\mathrm{tr}}_{j,k}\zeta_j$. At $m = N$ the free form is the identity on curls. This free integration fixes the normalization of the kinetic coefficient in Section 12.

**Lemma 11.5** (Free quadratic forms)**.** *For the chosen histories with $\ell^2 \geq C^{\mathrm{tr}}_{j,k}\zeta_j$,*

$$W_0^* \mathsf{K}_{N,m+1} W_0 + \mathsf{b}(\mathcal{Q}_m W_0 - I)^*(\mathcal{Q}_m W_0 - I) = \mathsf{K}_{N,m},$$
$$\widehat{\mathsf{H}}_{N,m}(0) = I, \qquad \mathsf{H}_{N,m} \geq 2\gamma_{J_m} I, \qquad \mathsf{H}_{N,m} - c_{\mathrm{dir},J_m} I = \rho^*_{N,m}\rho_{N,m}\,.$$

*Here $\gamma_k > 0$ is uniform over histories. The adjacent defect (11.3) has $\mathcal{W}_3 \geq c_k I$, $c_k > 0$. For $c_{\mathrm{dir},k} \leq \gamma_k$, the row factor is analytic on a fixed strip with exponentially decaying coefficients, uniformly in $N, m, p$.*

We prove the history bounds before the energy identity, so that inversion at zero momentum is justified.

*Proof.* Use the averaging weights of Definition 11.3. The straight-link multiplier is $L b_L(\theta)^j r_L(\theta_a)$. The frame adds a gradient and $\mathcal{Q}_m d_0 \phi = d_0(\phi \circ \iota)$. Gauge reduction therefore commutes with the block step. Before reduction the Gaussian marginal is

$$v = \mathcal{M}_m a + \mathsf{b}^{-1/2}\xi, \qquad \mathrm{Cov}(v) = \mathcal{M}_m \,\mathrm{diag}(c_a)\mathcal{M}_m^* + \mathsf{b}^{-1} I,$$

where $\xi$ is a standard Gaussian and $\mathcal{M}_m$ is straight-link averaging. Taking aliases gives (11.2); $L^{-2} = L^{-4}L^2$ combines counting adjoints with the two straight connectors. For $w_a = 1/c_a$, minimization of the diagonal precision over gradients gives

$$\min_\phi \sum_a w_a |a_a + q_a \phi|^2 = \frac{\sum_{a<b} w_a w_b |q_a a_b - q_b a_a|^2}{\sum_d w_d |q_d|^2}.$$

Indeed the minimizer is $\phi = -\sum_a w_a \bar{q}_a a_a / \sum_a w_a |q_a|^2$; expanding its square proves the identity. This is precisely the two-form metric defined above. For total history dilation $M$ at stage $j$, the cumulative averaging filter is

$$f(\theta) = b_M(\theta)^j \prod_{\text{stage increases at dilation } M'} b_{M'}(\theta)\,.$$

Successive dilations $M'$ have ratio at least 3. On the enlarged coarse cube $|\operatorname{Re} p_a| < 5\pi/4$, $|\operatorname{Im} p_a| < \eta$, the geometric-sum formula gives

$$|f((p+2\pi z)/M)| \le C_j \prod_a (1+|z_a|)^{-j}, \qquad |f(p/M)| \ge c_j > 0 \quad (p \text{ real}). \tag{11.4}$$

For the extra product, use $|b_{M'}(p/M)| \le \mathrm{e}^{4\eta M'/M}$ and the corresponding real lower bound $\mathrm{e}^{-CM'/M}$. The sum of $M'/M$ is bounded by a geometric series. The identities $b_L(\theta)b_{L'}(L\theta) = b_{LL'}(\theta)$ and $r_L(t)r_{L'}(Lt) = r_{LL'}(t)$ follow by writing each integer in its quotient and residue classes. They give the cumulative filter. The basic $b_M^j$ factor supplies the displayed alias decay. It also bounds $r_M((p+2\pi z)_a/M)$ by $C/(1+|z_a|)$, with a positive principal lower bound.

Separate the principal alias in the iterated (11.2). With $\lambda_M(p) = M^2\lambda(p/M)$ this reads

$$c_a(p) = \frac{A_a(p)}{\lambda_M(p)} + r_a(p), \qquad A_a(p) = f(p/M)f(-p/M)r_M(p_a/M)r_M(-p_a/M).$$

The remainder $r_a$ contains nonprincipal aliases, bounded by (11.4) and $|M^2\lambda((p+2\pi z)/M)| \ge c|z|^2$ for $z \ne 0$ on a smaller strip. Each earlier noise contributes $C_j(M')^{-2}$ by the squared filter bound. Summing these geometric contributions makes $r_a$ analytic and uniformly bounded; it is nonnegative on real momenta.

Put $B_a = A_a + \lambda_M r_a$ and $F = f(p/M)f(-p/M)$, so $c_a = B_a/\lambda_M$. The identity for a straight geometric sum implies

$$q_a(p)q_a(-p)F = M^2 q_a(p/M)q_a(-p/M)A_a.$$

It removes the apparent singularity in

$$\mathfrak{h}(p) = F^{-1} - \sum_a \frac{q_a(p)q_a(-p)r_a}{A_a B_a}, \qquad \lambda_M \mathfrak{h} = \sum_a \frac{q_a(p)q_a(-p)}{B_a}.$$

On the real enlarged cube $B_a$ is bounded above and below by positive constants. Since $\lambda(p)$ and $\lambda_M(p)$ are uniformly comparable there, the last identity bounds $\mathfrak{h}$ above and below, including $\mathfrak{h}(0) = 1$. Thin the strip using the uniform analytic bounds. The unmodified plaquette metric is then

$$\widehat{\mathsf{H}}^0_{ab,cd}(p) = \frac{\mathbf{1}_{(a,b)=(c,d)}}{\mathfrak{h}(p)B_a(p)B_b(p)} \qquad (a<b,\ c<d).$$

Substitution of $w_a = \lambda_M/B_a$ in the preceding minimization gives exactly $1/(\mathfrak{h}B_aB_b)$. Thus this formula extends the defined metric analytically to zero. It and its inverse are uniformly positive on real momenta, with $\widehat{\mathsf{H}}^0(0) = I$.

The inverse modification defining $\mathsf{H}$ has fixed stage coefficient $\zeta_j$. This positive addition has uniform analytic bounds. Since $d_2d_1 = 0$, it gives $\mathsf{H}d_1 = \mathsf{H}^0d_1$; it also leaves $\widehat{\mathsf{H}}(0) = I$.

For the unmodified metrics, divide the closed two-form defect by the two exterior derivatives, coordinate by coordinate, using $(f(p) - f(p|_{p_a=0}))/q_a(p) = \sum_x f_x \mathrm{e}^{ip\cdot x}(1-\mathrm{e}^{-ip_ax_a})/q_a(p)$. The quotient is a finite geometric sum for each Fourier coefficient. The two divisions lose two powers of path length; the exponential bounds leave a smaller strip. Symmetrize the resulting three-form kernel and call it $F_0$. Thus $F_0$ factors the defect formed with $\mathsf{H}^0_c, \mathsf{H}^0_f$. If $S_0$ is the corresponding lift of $d_2(\mathsf{H}^0_f)^{-1}T\mathsf{H}^0_c$, Woodbury inversion gives $d_2(\mathsf{H}^0_c)^{-1}\mathsf{H}_c = D_cd_2$. Substituting the two inverse modifications into the defect, the outgoing addition is $\zeta_k D_c^*D_c$ and the incoming subtraction is $-\zeta_j D_c^*S_0^*S_0D_c$. Hence

$$\mathcal{W}_3 = D_c^*(F_0 + \zeta_{J_m} I - \zeta_j S_0^*S_0)D_c, \qquad D_c = (I + \zeta_{J_m} d_2(\mathsf{H}^0_c)^{-1}d_2^*)^{-1}.$$

The pole-free alias estimates give $\|F_0\| \le C$ and $\|S_0\|^2 \le C^{\mathrm{tr}}_{j,k}/\ell^2$ on real momenta: the lift has per-residue order $\ell^{-3}$, whose square is summed over $\ell^4$ residues. Choose $\zeta^0_{j,k} = C + 2$ and then $\ell^2 \ge \zeta_j C^{\mathrm{tr}}_{j,k}$ for the stage pair $(j,k)$. The parenthesis is at least $I$. Both $D_c$ and its inverse have uniform analytic bounds at the fixed stages. Consequently $cI \le \mathcal{W}_3 \le CI$ on real momenta, uniformly in the history. This proves the required positive defect; it is the input for the square rows in Lemma 11.11.

Choose $2\gamma_k$ below the lower bounds for the top and the two incoming possibilities at stage $k$. Then $\mathsf{H} - c_{\mathrm{dir},k} I \ge \gamma_k I$. Its positive square root $\rho$ is analytic on a smaller common strip: expand the square root about a scalar exceeding the common upper bound. The resulting convergent matrix power series agrees with the positive root on the real torus. Contour displacement in Fourier inversion gives exponential coefficient decay and the same bound for all periodizations.

Finally, the normal equations for (11.2), with the gradient chosen to match the rooted predictor, give

$$\mathsf{K}_f W_0 = \mathcal{Q}_m^* \mathsf{K}_c, \qquad \mathcal{Q}_m W_0 = I - \mathsf{b}^{-1}\mathsf{K}_c\,.$$

At nonzero momentum write $\mathsf{C}_f = \mathrm{diag}(c_a)$ and $\mathsf{C}_c = \mathcal{M}_m \mathsf{C}_f \mathcal{M}_m^* + \mathsf{b}^{-1} I$. Gaussian conditioning gives $W_0 = \mathsf{C}_f \mathcal{M}_m^* \mathsf{C}_c^{-1}$ modulo gradients and $\mathcal{M}_m W_0 = I - \mathsf{b}^{-1}\mathsf{C}_c^{-1}$. Reduce these identities over gradients and select the gradient matching $\mathcal{Q}_m$. This gives the two normal equations. The pole-free formulas above extend them to zero; the constant fine potential is the coarse potential divided by $\ell$. Expanding the square gives

$$\langle a, \mathsf{K}_f a\rangle + \mathsf{b}\|\mathcal{Q}_m a - v\|^2 = \langle v, \mathsf{K}_c v\rangle + \langle a - W_0 v, (\mathsf{K}_f + \mathsf{b}\mathcal{Q}_m^*\mathcal{Q}_m)(a - W_0 v)\rangle\,.$$

Set $a = W_0 v$ and polarize. This is the asserted adjacent relation. The empty history gives $\mathsf{H}^0 = I$ and hence the Wilson form on curls, also after the inverse modification. □

### The stationary configuration

Root the axial tree in the block $\ell y + \mathbb{B}_{(\ell-1)/2}$ at $\ell y$, and set its tree links to 1. The remaining logarithmic link coordinates are $z$. For a fine edge $e = (x,a)$ put

$$(\mathrm{rep}\, V)_e = \begin{cases} V_{(y,a)}, & y = \mathrm{parent}(x),\ \mathrm{parent}(x + e_a) = y + e_a, \\ 1, & \mathrm{parent}(x + e_a) = \mathrm{parent}(x), \end{cases}$$
$$U(v,z)_e = \exp((\iota_{\mathrm{ax}} z)_e)(\mathrm{rep}\exp v)_e.$$

Here $\mathrm{dir}(e) = a$ for $e = (x,a)$, $\mathrm{parent}(e) = \mathrm{parent}(x)$, and parent takes the centred block containing a site, and $\iota_{\mathrm{ax}}$ extends by zero on tree edges. The copied field is one on every tree edge. The fine leading action is the hypercubic average of

$$\mathcal{A}_f(U) = c_{\mathrm{dir},j} \sum_{\square} \mathrm{pr}(U_\square) + \tfrac{1}{2} \sum_{x,\mathfrak{p}} |\mathcal{L}^{\mathrm{path}}_{\mathfrak{p}}(U,x)|^2 + \sum_{x,i} (\mathfrak{a}^{(3)}_i + \mathfrak{a}^{(4)}_i)(U,x).$$

The slots and coefficients are those of Definition 12.2, with masks and clipping removed. It excludes the inherited quadratic loop action and the common inverse coupling. Let $S(v,z)$ be this action at $U(v,z)$ plus

$$\mathsf{b} \sum_{e\ \mathrm{coarse}} \mathrm{pr}\big((\exp v)_e Q^{\mathrm{pred}}_e (U(v,z))^{-1}\big)\,.$$

Omit the common inverse coupling and put $\mathsf{Q} = D_z^2 S(0,0)$. The path bounds make these unclipped sums analytic near zero.

We specify the norms used for Fourier estimates. Put $\mu = \ell^{-1}$, $\theta_z(\xi) = (\xi + 2\pi z)/\ell$, with $x, z \in \{0, \dots, \ell - 1\}^4$, and

$$\omega_z(\xi) = \bigl(\mu^2 + \sum_a |q_a(\operatorname{Re}\theta_z(\xi))|^2\bigr)^{1/2},$$
$$\mathcal{U}_\ell(\xi)_{xz} = \ell^{-2} \mathrm{e}^{i\theta_z(\xi)\cdot x}, \quad T_\ell(\xi) = \mathcal{U}_\ell(\xi)\operatorname{diag}(\omega_z(\xi)^{-1})\mathcal{U}_\ell(0)^{-1}.$$

Use $T_\ell$ on fine links or sites, the identity on coarse coordinates, direct sums on products, and $T_\ell(-\xi)^{-T}$ on dual fields. Axial variables inherit the norm of their slice embedding. For kernels set

$$\|K\|_\eta^{\mathrm{bl}} = \sup \|T_{\mathrm{out}}(\xi_{\mathrm{out}})^{-1} K(\xi_{\mathrm{out}}, \xi_{\mathrm{in}}) T_{\mathrm{in}}(\xi_{\mathrm{in}})\|_{\mathrm{op}}.$$

The momenta and all their indicated shifts lie in $\mathcal{S}_\eta = \{\xi \in \mathbb{C}^4 : |\operatorname{Im}\xi_a| < \eta\}$. Use tensor products for several inputs; field insertions are normalized coarse directions. The weights themselves need not be analytic. A fine Bloch field obeys $u(x + \ell n) = \mathrm{e}^{i\xi\cdot n} u(x)$; $\mathcal{U}_\ell$ identifies its aliases with the fixed cell at $\xi = 0$. The last factor in $T_\ell$ fixes that reference cell. Write $W_{\mathrm{ax}} v = (v, Dz(0)v)$ for the axial tangent, and retain $W_0$ for the smooth free interpolation before axial gauge. They are distinct coordinate representatives.

The next lemma supplies the stationary configuration and identifies its first derivative with the free solution.

**Lemma 11.6** (Stationary interpolation)**.** *Near zero there is an analytic stationary branch with*

$$D_z S(v, z(v)) = 0, \quad z(0) = 0, \quad \mathsf{Q} \geq \nu_{j,\ell} I.$$

*Here $\nu_{j,\ell} > 0$ depends on the stage and ratio but not the volume. The axial representative of the free interpolation is $Dz(0)$. The averaging symbol in (3.5) vanishes to second order at every nonzero alias. At the zero alias, $\widehat{w}^{\mathrm{av}}(0) = 1$. For normalized coarse plane waves on a fixed strip,*

$$\|\nabla_f^r W_0 v\|_\infty \leq C_j \ell^{-1-r} |v| \qquad (0 \leq r \leq 2).$$

The proof uses a local axial estimate for coercivity and a finite-dimensional implicit equation for the branch.

*Proof.* Use Definitions 11.1–11.3 and Lemmas 11.4–11.5. At the flat field the frame objective has quadratic part $\operatorname{av}_{x,y} |h_x - h_y|^2/2$. On $h_0 = 0$ this is positive definite. The predictor objective has Hessian $I$ in its group variable. Their chart equations are real analytic. Their local complexifications and the analytic implicit function theorem make both minimizing choices real analytic near zero. Apply [16, Theorem 1.3.8] to the complexified chart equations, which vanish at zero and have the displayed invertible Hessians. The neighbourhoods may depend on the period.

For a fine infinitesimal field $a$, let $I_a(x \to y; o)$ be its signed sum along the route. Differentiating the frame equation gives

$$\tau_x(y) = -\operatorname{av}_{z,o} I_a(\ell x + y \to \ell x + z; o), \quad h_x'(y) = \tau_x(y) - \tau_x(0), \quad \mathcal{Q}_m a = \mathcal{M}_m a + d_0 \mathcal{F}_m a.$$

Here $\mathcal{M}_m a$ is the $w^{\mathrm{av}}$-average of straight length-$\ell$ sums, and $\mathcal{F}_m a(x) = \tau_x(0) - \sum_y w_y^{\mathrm{av}} \tau_x(y)$. In particular $\mathcal{Q}_m d_0 \phi = d_0(\phi \circ \iota)$.

We next bound a field on the axial slice. Telescoping the boundary of each coordinate rectangle expresses a free link as a copied coarse-link value plus curls in its enlarged block. Averaging the

straight connectors replaces the copied value by $\mathcal{Q}_m a$; the frame differences are route sums of the same curls. For example, interchanging two positive steps gives

$$a_a(x) + a_b(x + e_a) - a_b(x) - a_a(x + e_b) = (d_1 a)_{ab}(x).$$

Reverse steps change the sign and translate this identity. Repeated interchanges compare the tree route, a straight connector and its averaged translates. Thus, at each free edge $e$,

$$|a_e| \le C_{j,\ell} \sum_{\square \in \ell \ \mathrm{parent}(e)+\mathbb{B}_{(j+7)\ell}} |(d_1 a)_\square| + |(\mathcal{Q}_m a)_{\mathrm{parent}(e),\mathrm{dir}(e)}| \,.$$

This is an identity followed by a triangle inequality: tree-edge coordinates are zero, and every interchange of two successive route steps contributes exactly one signed curl. Cauchy–Schwarz and the bounded number of enlarged blocks containing a given plaquette give constants $A_{j,\ell}, B_{j,\ell} < \infty$, independent of the period, with

$$\|a\|_2^2 \le A_{j,\ell} \|d_1 a\|_2^2 + B_{j,\ell} \|\mathcal{Q}_m a\|_2^2 \,.$$

The kinetic cancellation of Lemma 12.4, at the finer input level, identifies the leading Hessian with $\mathsf{K}_f$; cubic and quartic curvature atoms have zero quadratic derivative. Consequently, for the slice embedding $\iota_{\mathrm{ax}}$,

$$\langle z, \mathsf{Q} z \rangle = \langle d_1 \iota_{\mathrm{ax}} z, \mathsf{H}_f d_1 \iota_{\mathrm{ax}} z \rangle + \mathsf{b} \|\mathcal{Q}_m \iota_{\mathrm{ax}} z\|^2 \ge \nu_{j,\ell} \|z\|^2 \,.$$

One can take $\nu_{j,\ell} = \min(2\gamma_j, \mathsf{b})/(1 + A_{j,\ell} + B_{j,\ell})$.

The map $\mathcal{E}(v, z) = D_z S(v, z)$ is analytic, $\mathcal{E}(0, 0) = 0$, and $D_z \mathcal{E}(0, 0) = \mathsf{Q}$. The same implicit function theorem gives $z(v)$. Differentiation yields

$$Dz(0) = -\mathsf{Q}^{-1} D_v D_z S(0, 0) \,.$$

The normal equations in Lemma 11.5 identify this with the free interpolation expressed relative to the copied field and the tree slice. The implicit neighbourhood may depend on the volume; the preceding coercivity constant does not.

Finally $\widehat{w}^{\mathrm{av}}(\theta) = b_\ell(\theta)^j$. At a nonzero alias $2\pi z/\ell$, at least one centered geometric sum has a simple zero, so the product has a zero of order at least $j \ge 4$. At zero it equals one. The fine-difference bound follows from (11.4): a difference multiplies alias $z$ by $q_a(\theta_z)$, of size at most $C\ell^{-1}(1 + |z|)$. The remaining covariance denominators and the $j$ averaging factors make these sums absolutely convergent through order two, including the pole-free principal term. □

## Gaussian coordinates

Write $U_v = U(v, z(v))$ and $\mathsf{Q}(v) = D_z^2 S(v, z(v))$. Let $s$ change axial logarithmic increments to left increments at $U_v$, and let $J_{\mathrm{ax}}$ be the product Haar density in these coordinates divided by its value at zero. For $x = \ell y + r$ in its centered block, set $\Psi(U, h_{\mathrm{fr}})(x) = h_{\mathrm{fr}}(y) h_{y,r}(U)$, using the patch frame of Definition 11.2. This extends a coarse frame $h_{\mathrm{fr}}$ over each fine block. The joint variable is $(U, h_{\mathrm{fr}})$, with gauge action $(U, h_{\mathrm{fr}}) \mapsto (U^\gamma, h_{\mathrm{fr}}(\gamma \circ \iota)^{-1})$. Let $\mathcal{R}$ insert fine site gauge parameters, and let $\mathcal{I}$ embed free-edge increments, with zero coarse-frame component. Thus $(\mathcal{R}, \mathcal{I}) : \mathfrak{g}^{\text{fine sites}} \oplus \mathfrak{g}^{\text{free edges}} \to \mathfrak{g}^{\text{fine edges}} \oplus \mathfrak{g}^{\text{coarse sites}}$ is square: the number of tree edges is the number of fine sites minus the number of block roots. Write $\mathsf{J}(U, h_{\mathrm{fr}})$ for the joint left Hessian of the leading action plus block penalty, with the coarse field fixed. We evaluate it along the stationary background; its value at the flat field has fine kinetic part $\mathsf{K}_f$. Put

$$X = D_{w=0}[-\log(\Psi(U, h_{\mathrm{fr}})^{-1} \Psi(\mathrm{e}^w(U, h_{\mathrm{fr}})))], \qquad \mathcal{L}_{m+1} = \sum_{x,i} \mathfrak{a}_i^{(2)}(\,\cdot\,, x) \,.$$

The last sum is the inherited quadratic loop action of Definition 12.2. Define its stationary Gaussian correction by

$$\mathsf{T}(v) = \mathcal{L}_{m+1}(U_v) + \tfrac{1}{2}\log\det \mathsf{Q}(v) - \log J_{\mathrm{ax}}(v).$$

It is the negative logarithm of the field-dependent Gaussian factor.

The determinant and coordinate density must be compared in the same counting coordinates.

**Lemma 11.7** (Gaussian coordinates)**.** *The stationary Gaussian integral, including the inherited loop action and Haar density, has field-dependent factor* $\mathrm{e}^{-\mathsf{T}(v)}$*. Moreover,*

$$J_{\mathrm{ax}} = |\det Ds| > 0, \qquad J_{\mathrm{ax}}(0) = 1, \qquad X\mathcal{R} = I, \qquad \det(\mathcal{R},\mathcal{I})^2 = 1.$$

*The Hessian* $\mathsf{Q}(v)$ *is positive near zero.*

The proof identifies the determinant in the coordinates used in Lemma 12.7. The sector version is given in Lemma 11.13.

*Proof.* Use Definitions 2.1, 3.1 and Lemma 11.6. Tree gauge is an exact Haar change of variables. In tree order, each tree link is a Haar translate of its child gauge variable, whose integral is one. For fixed $V$, gauge transformations fixing every block root leave the kernel invariant. Integration of these gauge variables reduces its integral to product Haar measure on free links; root transformations are not integrated here. In exponential coordinates write this measure as $J_*^{n_e}\prod_e J(z_e)\,dz$, with $J(0) = 1$ and $n_e$ the number of free links.

At a fixed stationary axial point $y = z(v)$ the coordinate change on each free link is

$$s_y(z)_e = \log(\exp z_e \exp(-y_e)), \qquad s_y(y) = 0\,.$$

Right invariance of Haar measure gives $\prod_e J(z_e) = |\det Ds_y(z)|\prod_e J(s_y(z)_e)$. Evaluate at $z = y$ to obtain $J_{\mathrm{ax}}(v) = |\det Ds_y(y)|$. Since $Ds_0(0) = I$, its value at zero is one, and it stays positive nearby. Stationarity removes the second-derivative term in the Hessian chain rule, giving $\mathsf{Q}(v) = (Ds)^*(\mathcal{I}^*\mathsf{J}\mathcal{I})Ds$ for the joint left Hessian $\mathsf{J}$.

Gauge covariance of $\Psi$ gives $X\mathcal{R} = I$. In tree-height order, each tree-edge row of $(\mathcal{R},\mathcal{I})$ has its child gauge coordinate $\pm I$ and earlier gauge coordinates; each coarse-frame row has its root gauge coordinate $-I$, and every free-edge slice column has its identity entry. Thus $\det(\mathcal{R},\mathcal{I})^2 = 1$. Positivity of $\mathsf{Q}(0)$ from Lemma 11.6 persists near zero.

Integrate the positive stationary quadratic form with covariance $\mathsf{Q}(v)^{-1}$. Its field-dependent factor is $\mathrm{e}^{-\mathcal{L}_{m+1}(U_v)}J_{\mathrm{ax}}(\det \mathsf{Q}(v))^{-1/2}$. Its negative logarithm proves the asserted action; the other factors enter the vacuum scalar. □

## Weighted inverses and insertions

For a fine site field put

$$(d_U\phi)_e = \phi(e^-) - \mathrm{Ad}(U_e)\phi(e^+), \quad \Delta_U = d_U^* d_U, \quad P_\mu = \mu^2(\Delta_U + \mu^2 I)^{-1}, \quad \mathfrak{m}_\mu = \mu^2 P_\mu^9\,.$$

The joint gauge completion is $\mathsf{J} + F_\mu^* F_\mu$, where $F_\mu = d_U^* P_f + \mathfrak{m}_\mu X$ and $P_f$ projects to fine links. Remove its predictor contact term to obtain one choice of $A_\mu$. The other is the folded fine operator $\mathsf{A} + d_U d_U^* + \mathsf{M}_\mu$, where $\mathsf{A}(U)$ is the fine leading Hessian and $\mathsf{M}_\mu$ applies $\mathfrak{m}_\mu$ to each link component. For weighted bounds use the nearest projection $\pi$ from Lemma 11.4 and centered coarse frames: $c_y(U) = \pi(\sum_r w_r^{\mathrm{av}} h_{y,r}(U))$, $\widetilde{h}_{\mathrm{fr}} = h_{\mathrm{fr}} c(U)$. Pull back the joint quadratic forms by this change. Its tangent is triangular with identity diagonal, so it preserves determinants. These are the operators

compared in Lemma 12.8; derivatives are at zero in free interpolated coarse directions. The constants in the following lemma depend on the free metric, frame and penalty bounds. After that lemma fix $E = \max(1, C_j^{\mathrm{ins}})$ for Lemma 11.9.

For inserted inverse bounds, let $M_r$ be the supremum of $\|D^r A_\mu\|$ from the primal block space to its dual, in normalized smooth coarse directions, for $r = 1, 2$. Use any common smaller strip for these inserted kernels. Define

$$Q_0(C) = C, \qquad Q_1(C) = C^2 M_1, \qquad Q_2(C) = 2C^3 M_1^2 + C^2 M_2.$$

The norms $M_1, M_2$ retain the actual cubic and quartic path contributions. On a smaller common strip, the bounds for the joint Hessians through two insertions give

$$\sup_{\ell} M_r \le \mathcal{P}_{j,r}(A_j^{\mathrm{path}}), \qquad r = 1, 2.$$

Here $A_j^{\mathrm{path}}$ is the stage path bound fixed in row 4 of Section 9; the fixed polynomials also depend on the interpolation, free metric, frame and penalty bounds. The derivation is included below.

These estimates control the determinant comparison without a factor equal to the number of fine sites.

**Lemma 11.8** (Scaled inverse insertions)**.** *There are $\eta_j^{\mathrm{inv}} > 0$ and $C_j^{\mathrm{ins}} < \infty$, fixed before the ratio and path bounds, such that, for $0 \le r \le 2$ and every admitted odd $\ell$,*

$$\sup_{\xi \in \mathcal{S}_{\eta_j^{\mathrm{inv}}}} \|P_\mu(\xi)^3\|_1 \le C_j^{\mathrm{ins}}, \qquad \|D^r P_\mu\|_{\eta_j^{\mathrm{inv}}}^{\mathrm{bl}} \le C_j^{\mathrm{ins}},$$
$$\|\mu^2 D^r (A_\mu^{-1})\|^{\mathrm{bl}} \le Q_r(C_j^{\mathrm{ins}}) \qquad (0 \le r \le 2).$$

*The inverse insertion bounds use the common smaller strip. The $r$th block-penalty and frame insertions have norms at most $C_j^{\mathrm{ins}}$.*

We first prove real coercivity in block units, then extend it to a strip and differentiate the inverse.

*Proof.* Use Definition 2.1, Lemmas 11.4–11.6. Constants depend on the free bounds, stage and $\mathsf{b}$.

At zero background, $\Delta_0$ has symbol $\lambda(\theta)$. With $x = \lambda(\theta) \ge 0$,

$$2^{-10}(x + \mu^2) \le x + \mu^2 \left(\frac{\mu^2}{x + \mu^2}\right)^9 \le x + \mu^2 . \tag{11.5}$$

For $x \ge \mu^2$ the first term supplies the lower bound; for $x < \mu^2$ the parenthesis is at least $1/2$. On links, the exterior identity $\|d_1 u\|^2 + \|d_0^* u\|^2 = \lambda(\theta)\|u\|^2$ and Lemma 11.5 give the same lower bound, with a fixed positive factor, for the auxiliary fine operator. Changing its quadratic form by $T_\ell$ therefore gives a positive lower bound independent of $\ell$.

In centered coordinates fine gauge insertion has coarse component $-\sum_r w_r^{\mathrm{av}} \phi(\ell y + r)$: differentiate $h_{y,r}(U^\gamma) = \gamma(\ell y) h_{y,r}(U) \gamma(\ell y + r)^{-1}$ and the centroid. The uncompleted joint form is

$$\langle d_1 u, \mathsf{H}_f d_1 u \rangle + \mathsf{b}\|\mathcal{M}_m u - d_0 \vartheta\|^2 .$$

Modulo a joint gauge transformation choose $\vartheta = 0$ and make every nonprincipal fine alias transverse. This is possible by dividing its longitudinal part by $\lambda(\theta_z)$; the remaining principal gauge parameter fixes $\vartheta$, since the principal averaging multiplier is bounded below by (11.4). Write $u_z$ for the

resulting normalized alias coordinates and $H_> = \sum_{z\neq 0} \omega_z^2 |u_z|^2$. The fine curvature energy bounds $cH_>$. The averaging multiplier on the principal alias is at least $c_j\mu$ in unitary cell coordinates; the weighted norm of all other alias columns is at most $C_j$. Therefore

$$(c_j\mu|u_0|)^2 \le 2\|\mathcal{M}_m u\|^2 + 2C_j^2 H_>, \qquad \omega_0 \le C\mu\,.$$

Combining these inequalities gives a uniform positive bound on the distance, in the weighted norm, to the joint gauge subspace.

Let $\mathcal{R}\phi$ be joint gauge insertion. Frame covariance gives $F_\mu \mathcal{R} = \Delta_0 + \mathfrak{m}_\mu$. By (11.5),

$$\|\mathcal{R}\phi\|_{\mathrm{bl}} \le C\|(\Delta_0 + \mathfrak{m}_\mu)\phi\|_2, \qquad \|F_\mu w\|_2 \le C\|w\|_{\mathrm{bl}}\,.$$

For the second bound one must also estimate $\mathfrak{m}_\mu X$ on arbitrary fine increments. In the centred frame, $X$ is a sum of normalized averages of routes of length at most $C_j\ell$. Its row and column sums are at most $C_j\ell$, so

$$\|Xu\|_2 \le C_j\ell\|u\|_2 \le C_j\mu^{-2}\|u\|_{\mathrm{bl}}, \qquad \|\mathfrak{m}_\mu X u\|_2 \le C_j\|u\|_{\mathrm{bl}}.$$

The coarse-frame part satisfies the same inequality since its embedding has $\ell^2$ as counting-norm factor and $\|\mathfrak{m}_\mu\| \le \mu^2$. Also $\|d_0^* u\|_2 \le C\|u\|_{\mathrm{bl}}$. For the first bound a fine gauge component contributes $\omega_z|q(\theta_z)\phi_z| \le \omega_z^2|\phi_z|$, and its averaged coarse component is bounded by the summable alias filter times $\omega_z^2|\phi_z|$. For $w = w_\perp + \mathcal{R}\phi$, the completed energy bounds $c\|w_\perp\|_{\mathrm{bl}}^2 + \|F_\mu w\|_2^2$. The last two inequalities give

$$\|w\|_{\mathrm{bl}} \le \|w_\perp\|_{\mathrm{bl}} + C\big(\|F_\mu w\|_2 + C\|w_\perp\|_{\mathrm{bl}}\big)\,.$$

Squaring proves real coercivity of the completed form with a constant independent of $\ell$.

The uninserted quadratic symbols and the geometric frame and penalty kernels have bounded weighted norms on a fixed coarse strip. To verify this, each fine derivative is bounded by $C\omega_z$, a free interpolated field by $C\mu$, and normalized frame or predictor averages by the summable filter (11.4). At the flat field these operators depend only on the free quadratic action, the interpolation, the frame and the penalty. Their defining inserted kernels use the smooth interpolation; its fine differences of order $s \le 2$ are $O(\ell^{-1-s})$ by the alias estimate (11.4). For complex momenta, freeze the weights at a real momentum: if the coarse displacement has norm at most one, the geometric-sum formula and $|q(\theta_z(\xi)) - q(\theta_z(\xi'))| \le C\mu|\xi - \xi'|$ make the two weights comparable by a fixed factor. Cauchy's estimate [16, Proposition 1.2.2] on a smaller disc bounds the change of the conjugated symbol by $C\eta$. Choose $\eta_j^{\mathrm{inv}}$ so this is less than half the real lower bound. A Neumann series gives the weighted inverse on that strip.

For inserted directions $v, w$, write $A_v = DA[v]$ and $\mathsf{G} = A^{-1}$. Direct differentiation gives

$$D\mathsf{G}[v] = -\mathsf{G}A_v\mathsf{G}, \qquad D^2\mathsf{G}[v,w] = \mathsf{G}A_v\mathsf{G}A_w\mathsf{G} + \mathsf{G}A_w\mathsf{G}A_v\mathsf{G} - \mathsf{G}D^2A[v,w]\mathsf{G}\,.$$

The geometric insertions have norm at most $C$: each smooth field is $O(\mu)$, each derivative on it is $O(\mu^2)$, and $\mu \le \omega_z$. With two arbitrary fine legs, Cauchy–Schwarz accounts for their two factors $\omega_z^{-1}$; a difference on either leg supplies its corresponding $\omega_z$. This bounds the block penalty, frame and $\Delta_U$ insertions independently of the nonlinear path coefficients. For the path contribution, expand each transported curvature through total degree four. Its third derivative has one curl and its fourth has zero or more differences after two smooth insertions. In either case two arbitrary fine legs contribute at most $\mu^{-2}$, while the two smooth insertions in the fourth derivative contribute $\mu^2$. For the third derivative a smooth insertion and the remaining curl give the same bound. Summation

by parts moves a difference off a smooth leg when necessary; its path-length loss is summable against the exponential path bound. Each coefficient is a product of at most two path-row coefficients, or one atom coefficient. The row and atom sums therefore give a fixed polynomial $\mathcal{P}_{j,r}(A_j^{\mathrm{path}})$, uniformly in $\ell$. Adding the geometric insertions proves the displayed bounds for $M_1, M_2$. The inverse identities therefore give dual-to-primal norms $C^2M_1$ and $2C^3M_1^2 + C^2M_2$. The inclusion of a primal source in the dual space has norm at most $C\mu^{-2}$; include its fixed factor in $C_j^{\mathrm{ins}}$. Multiplication by $\mu^2$ gives precisely $Q_1, Q_2$. Applying the same identities to $\Delta_U + \mu^2 I$, with the geometric insertion bounds, gives the stated bounds for $D^r P_\mu$. No derivative of a weight is taken.

Finally, in centered alias representatives,

$$\left| \frac{\mu^2}{\lambda(\theta_z(\xi)) + \mu^2} \right| \le C(1 + |z|^2)^{-1} .$$

The complex denominator differs from its real value by a small multiple of $\omega_z^2$, and the Bloch Fourier matrices have bounded norms. Hence

$$\|P_\mu^3\|_1 \le C d_{\mathfrak{g}} \sum_{z \in \mathbb{Z}^4} (1 + |z|^2)^{-3} < \infty .$$

In $P_\mu^9 = (P_\mu^3)^3$, two insertions leave one undifferentiated cube, whose trace norm bounds the regulator products; the remaining factors have bounded operator norms. □

## Vertices and the stationary gauge identity

For a translation-invariant Taylor kernel, momentum order at least $s$ means that every coefficient of total momentum degree less than $s$ vanishes at zero. Let $V_r$ be the degree-$r$ Taylor kernel of the fine leading action minus Wilson, for $r = 2, 3, 4$. Use smooth free directions $\mathcal{W}v = (v, W_0 v, 0)$ in the coarse-link, fine-link and centred coarse-frame variables. The joint action in these variables is

$$\mathcal{S}(v, u, \vartheta) = \mathcal{A}_f(\mathrm{e}^u) + \mathsf{b} \sum_e \mathrm{pr} \left( (\exp v)_e \left[ Q^{\mathrm{pred}}(\mathrm{e}^u)^{h_{\mathrm{fr}}} \right]_e^{-1} \right), \qquad h_{\mathrm{fr}} = \exp \vartheta\, c(\mathrm{e}^u)^{-1}.$$

Here the superscript is the coarse gauge action; $c$ is the centring map defined before Lemma 11.8. Denote its third and fourth derivatives at zero by $\mathcal{V}_3, \mathcal{V}_4$. On a smaller strip $\mathcal{S}_\eta$, define $C_r$ as the supremum of their absolute scalar Fourier entries on momentum-conserving tuples, with either all legs $\mathcal{W}v_i$, $|v_i| \le 1$, or one fine leg of block norm at most one. The remaining coarse legs are normalized. This definition includes the predictor and frame derivatives. Put $S_{\mathrm{stat}}(v) = S(v, z(v))$.

The extra momentum order of the difference makes the classical matching bounds independent of the input amplitude after enlarging $\ell$.

**Lemma 11.9** (Vertex bounds)**.** *The kernels $V_2, V_3, V_4$ have momentum orders at least $3, 2, 1$. For $\ell$ above a threshold depending on the input path data,*

$$C_3 \le a_3 + \ell^{-1} a_3', \qquad C_4 \le c_4 + \ell^{-1} c_4' .$$

*Here $a_3, c_4$ depend only on preceding stage data; $a_3', c_4'$ may depend on the input path bound. Moreover, in normalized coarse directions,*

$$|D^4 S_{\mathrm{stat}}| \le C_4 + 3C_3^2 E, \qquad E = \max(1, C_j^{\mathrm{ins}}).$$

We first prove the low-momentum cancellations, then count powers under interpolation.

*Proof.* Use Definition 12.2, its input bounds in Lemmas 12.3–12.4, and Lemmas 11.5–11.6. The quadratic difference is

$$\widehat{V}_2(p) = \tfrac{1}{2}\widehat{d}_1(-p)^T(\widehat{\mathsf{H}}_f(p) - I)\widehat{d}_1(p)\,.$$

Its middle factor vanishes at zero and its two curls each supply one momentum, proving order three.

To include the nonlinear transport terms, expand a connection with connection components of weight one and coordinate derivatives of weight one. A plaquette curvature starts at weight two, with leading term $\partial_a A_b - \partial_b A_a + [A_a, A_b]$. Changing its transport path changes this term only at higher weight. The row part at weight four is therefore its curvature square with coefficient $c_{\mathrm{dir}}\delta_{\gamma\delta} + \sum_\alpha s_{\alpha\gamma}s_{\alpha\delta} = \delta_{\gamma\delta}$ by Lemma 12.4 at the finer input level; $s_{\alpha\gamma}$ are the path-row coefficients of Definition 12.1. Cubic and quartic curvature atoms start at weights six and eight. The Wilson action has the same weight-four term. All contributions of weight at most four consequently cancel in their difference. Polynomial connection jets vary independently. Expand their finite parallel transports through weight four and sum using the exponential path bound. A degree-$r$ field term of weight at least five has at least $5 - r$ coordinate derivatives, equivalently that many momentum powers. This proves the three claimed orders, including cubic and quartic terms arising inside a quadratic curvature row.

For the quantitative bound, an analytic coefficient $f$ vanishing at zero factors as

$$f(p) = \sum_a p_a \int_0^1 \partial_a f(tp)\,dt\,.$$

Apply this to the vanishing Taylor coefficients, successively, on a smaller strip. Equivalently in position space, write $\mathrm{e}^{ip\cdot x} - 1$ as a sum of coordinate differences; the losses are fixed powers of the true path length. The exponential path norm bounds their sums. Wilson's degrees $r = 2, 3, 4$ have orders $4 - r$; the differences have orders $5 - r$.

Use the fine-difference bound for $W_0$ of Lemma 11.6. For one arbitrary fine leg use Cauchy–Schwarz over the block. If it carries $s = 0$ or 1 differences, its $\ell^1$ norm on the block is at most $C\ell^{3-s}\|u\|_{\mathrm{bl}}$: there are $\ell^4$ sites, $\|u\|_2 \le \mu^{-1}\|u\|_{\mathrm{bl}}$, and $\|\nabla_f u\|_2 \le C\|u\|_{\mathrm{bl}}$. Pair this with the other bounded legs. Summation by parts places at most one difference on the arbitrary leg. This gives the same bound in the dual weighted norm for a mixed vertex.

There are $\ell^4$ sites per block. A term with $r$ fields and $s$ fine differences thus has scale $\ell^{4-r-s}$. Wilson gives scale 1 at its leading orders, and the differences give $\ell^{-1}$. The penalty has a fixed bound: normalized route and offset averages have mass one, straight sums cancel the factor $\ell^{-1}$ in a smooth input, and implicit differentiation of the centered frame and predictor equations bounds their derivatives through order four. In centroid-normalized frame variables the linear equation is identity minus the constant projection; every remaining term is a normalized average, so these bounds have no factor counting patch sites. These contributions define $a_3, c_4$; the path-dependent sums define $a_3', c_4'$. Choose the stated ratio threshold so that $\ell^{-1}\max(a_3', c_4') \le 1$, giving $C_3 \le a_3 + 1$ and $C_4 \le c_4 + 1$ independently of the input path bound.

For the mixed pair in the fourth stationary derivative, use the joint vertices just defined. In the joint coordinates, the exact nonlinear tree gauge change and the stationary chain rule give

$$D^4 S_{\mathrm{stat}} = \mathcal{V}_4[\mathcal{W}^4] - \sum_{\text{three pairings}} \mathcal{V}_3[\mathcal{W}, \mathcal{W}, \cdot]\,\mathsf{G}\,\mathcal{V}_3[\cdot, \mathcal{W}, \mathcal{W}].$$

Here $\mathsf{G}$ is the inverse after gauge completion. Lemma 11.6 identifies the axial tangent with the tree representative of $W_0$: their difference is an infinitesimal joint gauge direction, including its coarse-frame component. A joint gauge transformation of the axial stationary curve remains stationary and

can be chosen with tangent $\mathcal{W}v$. Differentiating this curve gives the displayed minimum formula. Each paired functional annihilates the infinitesimal gauge directions by differentiated stationarity and gauge invariance. It may therefore be paired through the completed inverse of Lemma 11.8. If $b\mathcal{R} = 0$, the shift $x \mapsto x - \mathcal{R}(F_\mu \mathcal{R})^{-1} F_\mu x$ sets $F_\mu x$ to zero and preserves both the uncompleted energy and $b(x)$. Thus its pairing through the completed inverse equals its pairing through the slice inverse. That lemma gives a constant $E$ independent of $\ell$ and the input path bound. Duality then bounds the pairing by $C_3^2 E$. The three pairings in a symmetric fourth derivative have identical bounds. This proves both estimates and the mixed-insertion assertion. □

Denote the contact-removed completed Hessian by $A^{\mathrm{rem}}(v)$ and its inverse by $\mathsf{G}_\mu(v)$ on the smooth interpolated field. In joint left coordinates let $p_\lambda(w)$ be the $\lambda$ component of the logarithm of the framed predictor. The index $\lambda$ comprises a coarse link and a colour index; let $C_\lambda(v)$ be its second derivative at that field. Thus the full Hessian differs by $\sum_\lambda d^\lambda(v) C_\lambda(v)$, where $d(v)$ is the penalty discrepancy and

$$d(0) = 0, \qquad Dd(0) = -\mathsf{K}_c.$$

To keep the derivatives of the contact insertion as well as those of the inverse, introduce an independent coarse link source $u$ and put

$$\mathsf{R}(v,u) = \mathsf{G}_\mu(v) \sum_\lambda u^\lambda C_\lambda(v), \qquad \mathsf{F}(v,u) = \operatorname{tr} \mathsf{R}(v,u), \qquad \mathsf{F}_2(v,u) = \operatorname{tr} \mathsf{R}(v,u)^2.$$

The traces sum colour and cell coordinates. Their translation-invariant kernels, with internal momentum integrated using $d\theta/(2\pi)^4$, are

$$\mathsf{b}_{i\lambda} = \partial_{v_i} \partial_{u_\lambda} \mathsf{F}(0,0), \qquad \mathsf{c}_{\lambda\rho} = \partial_{u_\lambda} \partial_{u_\rho} \mathsf{F}_2(0,0).$$

In particular $\mathsf{c}_{\lambda\rho}$ includes the two product-rule terms. Put $\mathsf{a}_i(\xi) = -\mathsf{K}_c(\xi) e_i$. The contact difference $Y^{\mathrm{con}}$ is the Hessian kernel of one half of the full minus contact-removed logarithmic determinant, evaluated on the stationary branch. Use the simple Lie algebra of Definition 2.1 and the same stationary and analytic data as above.

The cancellation below is exactly the coefficient needed in Lemma 12.8.

**Lemma 11.10** (Stationary Ward estimates)**.** *The first penalty derivative has momentum order two;* $\mathsf{b}$ *vanishes at zero and* $\mathsf{c}$ *is analytic there. For a fixed colour and transverse link index* $i = (2,\alpha)$*, the tested diagonal entry satisfies*

$$Y^{\mathrm{con}}_{ii}(te_1) = O(t^3), \qquad \alpha_b(Y^{\mathrm{con}}) = 0.$$

*Here* $\alpha_b$ *takes twice the real coefficient of* $t^2$ *in this entry.*

The proof derives the trace contraction before extracting this coefficient.

*Proof.* Use Definitions 3.1, 12.2 and Lemmas 11.5–11.7, 11.8. The free normal equation gives $\mathsf{b}(I - \mathcal{Q}_m W_0) = \mathsf{K}_c$. With the chosen discrepancy sign its first derivative is $\mathsf{a} = -\mathsf{K}_c$, of order two because $\mathsf{K}_c = d_1^* \mathsf{H}_c d_1$.

The determinant difference along the stationary curve is

$$\tfrac{1}{2} \log\det(I + \mathsf{R}(v, d(v))) = \tfrac{1}{2} \mathsf{F}(v, d(v)) - \tfrac{1}{4} \mathsf{F}_2(v, d(v)) + O(|v|^3).$$

The first derivatives of each invariant scalar determinant vanish at the flat field: each is an adjoint-invariant linear functional on $\mathfrak{g}$, hence zero because $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$. Therefore its Hessian is

unchanged on replacing the stationary curve by its tangent, by the second-derivative chain rule. Also $\partial_u \mathsf{F}(0,0) = 0$ by the same argument. The potentially remaining chain-rule term consequently vanishes exactly:

$$\partial_u \mathsf{F}(0,0)\, D^2 d(0)[v,w] = 0.$$

Since $\mathsf{F}$ is linear in $u$, its pure $v$ and pure $u$ Hessians vanish at $(0,0)$. The mixed derivative retains both contributions

$$\partial_{v_i} \operatorname{tr}(\mathsf{G}_\mu C_\lambda) = \operatorname{tr}(-\mathsf{G}_\mu(\partial_{v_i} A^{\mathrm{rem}})\mathsf{G}_\mu C_\lambda + \mathsf{G}_\mu \partial_{v_i} C_\lambda).$$

The two mixed chain-rule terms of $\mathsf{F}$ and the pure $u$ Hessian of $\mathsf{F}_2$ now give

$$Y_{ii}^{\mathrm{con}}(\xi) = \tfrac{1}{2}\{W_i(\xi) + W_i(-\xi) - \tfrac{1}{2}V_i(\xi)\},$$
$$W_i(\xi) = \sum_\lambda \mathsf{a}_i^\lambda(\xi)\mathsf{b}_{i\lambda}(\xi), \qquad V_i(\xi) = \sum_{\lambda,\rho} \mathsf{a}_i^\lambda(-\xi)\mathsf{c}_{\lambda\rho}(\xi)\mathsf{a}_i^\rho(\xi).$$

These identities hold first for finite momentum sums. Lemma 11.8 bounds their traces on a common strip, so period limits give the indicated integrals and their derivatives.

Write $y_\lambda(v) = \operatorname{tr}(\mathsf{G}_\mu(v) C_\lambda(v))$. The stationary gauge identity is $Dy(0)d_0\phi = 0$: gauge covariance of the stationary field gives $y((\exp 0)^\gamma) = \mathrm{Ad}(\gamma)y(0) = 0$, and differentiation gives this identity since $y(0) = 0$. Lemma 11.6 identifies the stationary tangent with the smooth one, so it applies to the kernels just defined. In Fourier coordinates it gives

$$\sum_a q_a(-\xi)\mathsf{b}_{(a,\alpha),\lambda}(\xi) = 0$$

for each colour $\alpha$ and source index $\lambda$. Restrict to $\xi = te_a$. Only $q_a(-te_a)$ remains, hence the corresponding component is zero for $t \neq 0$. Continuity gives $\mathsf{b}(0) = 0$ for every component. Analyticity implies $\mathsf{b} = O(|\xi|)$ and $\mathsf{c} = O(1)$. On the tested ray $W_i = O(t^3)$ and $V_i = O(t^4)$. The displayed diagonal identity therefore has zero second Taylor coefficient, proving the assertion. □

## Local quadratic energy

The Gaussian integral is performed only on part of each field configuration. Its complement must retain a positive energy per selected large-field test. We define this energy as the local sum of the completed squares and their signed error rows, on a consistent selection of tests.

Fix one hypercubic orientation. Set $x = \widehat{\beta}_m$, $s = t_m$, $t = t_{m+1}$, $B = \lceil (\log x)^2 \rceil$, $R_* = \lfloor B/100000 \rfloor$, and

$$t_{\mathrm{hi}} = \big(x + \kappa_G \log \ell\big)^{-49/100}.$$

The threshold $t_{\mathrm{hi}}$ distinguishes very large tests from tests at scale $t$. Here $\widehat{\beta}_{m+1} = x + \kappa_G \log \ell$. By the formula for $t_m$ in Section 3 and $M_m^{\mathrm{cut}} = \exp((\log\log \widehat{\beta}_m)^2)$ in Section 9, $t/s \to 1$ and $B^n t/t_{\mathrm{hi}} \to 0$ as $x \to \infty$, with $\ell$ fixed, for every fixed $n$. The input and output row maps on two-forms are $\mathcal{B}_f = (\sqrt{c_{\mathrm{dir},j}}I, \rho_f)^T$ and $\mathcal{B}_c = (\sqrt{c_{\mathrm{dir},J_m}}I, \rho_c)^T$. They satisfy $\mathcal{B}_f^*\mathcal{B}_f = \mathsf{H}_f$ and $\mathcal{B}_c^*\mathcal{B}_c = \mathsf{H}_c$. The free square-completion maps are

$$M_f = \mathcal{B}_f \mathsf{H}_f^{-1} T \mathcal{B}_c^*, \qquad M_n = \mathsf{b}^{-1/2} d_1^* \mathcal{B}_c^*,$$
$$N = \begin{pmatrix} I - \mathcal{B}_c \mathsf{H}_c^{-1} \mathcal{B}_c^* \\ \mathcal{W}_3^{1/2} d_2 \mathsf{H}_c^{-1} \mathcal{B}_c^* \end{pmatrix}.$$

Equation (11.3) gives $M_f^*M_f + M_n^*M_n + N^*N = I$. For an infinitesimal pair of fine and coarse fields, let $X_f^{\mathrm{lin}}, X_c^{\mathrm{lin}}$ be their curls followed by $\mathcal{B}_f, \mathcal{B}_c$, and let $Z^{\mathrm{lin}}$ be $\sqrt{\mathsf{b}}$ times their linear predictor discrepancy. The free interpolation identities give

$$M_f^* X_f^{\mathrm{lin}} + M_n^* Z^{\mathrm{lin}} = X_c^{\mathrm{lin}}. \tag{11.6}$$

These identities apply to the untruncated flat kernels. We retain them when estimating the errors below.

For the actual rows, transport every kernel coefficient along the fixed coordinate route joining its endpoints, using the replicated coarse configuration $V$. Retain a coefficient only when that route, measured in fine units, has length at most $R_*$. For a fine-to-coarse coefficient also retain its residue and all the endpoints of that route. Denote the resulting maps, respectively truncating $M_f, M_n, N$, by $\Phi, \Phi_n, \mathcal{N}$. Write $L_{f,R_*}$ and $L_{c,R_*}$ for the path rows of Definition 12.2 truncated by their true lengths in fine units. At a plaquette $p$ put

$$\begin{gathered} X_f(p) = \big(\sqrt{c_{\mathrm{dir},j}}F_p(U), L_{f,R_*}(U,p)\big), \quad X_c(p) = \big(\sqrt{c_{\mathrm{dir},J_m}}F_p(V), L_{c,R_*}(V,p)\big), \\ v_c(p) = \big(\min(1, s/\operatorname{ch}(V_p))\sqrt{c_{\mathrm{dir},J_m}}F_p(V), \chi_p(V)L_{c,R_*}(V,p)\big), \\ Z(e) = \sqrt{\mathsf{b}}\,F(V_eQ_e^{-1}), \qquad Q = Q^{\mathrm{pred}}(U). \end{gathered}$$

Here $F(g) = P_{\mathfrak{g}}(g - g^{-1})/2$, the ratio in $v_c$ is one at zero chord, and $\chi_p$ is the output health test at the plaquette base. These thinned coordinates are bounded by $Ct$ everywhere. Put

$$K_f(p) = \operatorname{clip}_t(\operatorname{ch}(U_p)) + \tfrac{1}{2}\chi_p(U)|L_{f,R_*}(U,p)|^2, \quad K_c(p) = \operatorname{clip}_s(\operatorname{ch}(V_p)) + \tfrac{1}{2}\chi_p(V)|L_{c,R_*}(V,p)|^2.$$

The clips use the respective input and output coefficients. The six local rows are

$$\begin{aligned} E_f(p) &= K_f(p) + \tfrac{1}{2}|\Phi v_c(p)|^2 - \langle \Phi v_c(p), X_f(p)\rangle, \\ E_n(e) &= \tfrac{\mathsf{b}}{2}\operatorname{ch}(V_eQ_e^{-1})^2 + \tfrac{1}{2}|\Phi_n v_c(e)|^2 - \langle \Phi_n v_c(e), Z(e)\rangle, \\ E_c(p) &= \langle v_c(p), X_c(p)\rangle - \tfrac{1}{2}|v_c(p)|^2 - K_c(p), \\ E_N(q) &= \tfrac{1}{2}|\mathcal{N} v_c(q)|^2, \\ E_L(p) &= \langle v_c(p), \Phi^* X_f(p) + \Phi_n^* Z(p) - X_c(p)\rangle, \\ E_D(p) &= \tfrac{1}{2}\langle v_c(p), (I - \Phi^*\Phi - \Phi_n^*\Phi_n - \mathcal{N}^*\mathcal{N})v_c(p)\rangle. \end{aligned}$$

A doubled plaquette row is summed over its two components. The index $q$ runs over the two-form complement and three-form components of $N$. Summing over every anchor cancels all cross terms and squares, exactly:

$$\sum E_f + \sum E_n + \sum E_c + \sum E_N + \sum (E_L + E_D) = \sum K_f + \tfrac{\mathsf{b}}{2}\sum_e \operatorname{ch}(V_eQ_e^{-1})^2 - \sum K_c. \tag{11.7}$$

The adjoints in this identity are those of the truncated transported matrices, not the flat matrices. The right side is the truncated clipped kinetic energy difference; the higher atoms and omitted tails are separate factors in Section 13.

Fix a mismatch multiplier $W > 0$, chosen after $\ell$ below. In this subsection $c_{\mathrm{dir},f} = c_{\mathrm{dir},j}$ and $c_{\mathrm{dir},c} = c_{\mathrm{dir},J_m}$, with the same convention for $c_{\mathrm{lin}}$. A selection $A = (A_f, A_n, A_c)$ consists of fine plaquettes, coarse mismatch edges and coarse flagged sites; set $|A| = |A_f| + |A_n| + |A_c|$. Its location

includes every projected plaquette corner and both mismatch endpoints. Write $A^{[r]}$ for its radius-$r$ coarse neighbourhood. Consistency means that all tests meeting $A^{[5B]}$ obey

$$\begin{aligned} p \in A_f &\implies \operatorname{ch}(U_p) \geq t/4, & p \notin A_f &\implies \operatorname{ch}(U_p) \leq t/2, \\ e \in A_n &\implies \operatorname{ch}(V_e Q_e^{-1}) \geq Wt, & e \notin A_n &\implies \operatorname{ch}(V_e Q_e^{-1}) \leq 2Wt. \end{aligned}$$

In this neighbourhood $A_c$ is the set of coarse sites touched by an output plaquette with $\operatorname{ch}(V_p) > s$, and every chosen frame tag has positive mass. The fine field is in the actual tree slice. An allocation sector fixes these selected tests, their assignment to connected components of their neighbourhoods, a block tree and the frame and predictor tags. Its consistency conditions are exactly the inequalities just displayed and the coarse flag and positive-mass requirements. Define $\mathcal{E}_A$ by summing the six rows above over anchors in $A^{[2B]}$, projecting fine anchors to coarse sites. This fixes the energy whose lower bound is needed.

Let $D$ select the fine or mismatch tests exceeding $t_{\mathrm{hi}}$, and let $H$ be the output anchors within $4B/5$ of $D$. Put $r_p = \operatorname{ch}(U_p)$ or $\operatorname{ch}(V_p)$ according to the lattice and $n_e = \operatorname{ch}(V_e Q_e^{-1})$. All sums below are restricted to the local anchors defining $\mathcal{E}_A$.

The positive fine and mismatch rows absorb the signed-row losses estimated below. The order of choices in the next lemma leaves its energy constant independent of the later coupling threshold.

**Lemma 11.11** (Local large-field energy)**.** *Fix the stage path bounds, free interpolation and penalty. For sufficiently large odd $\ell$, and then sufficiently large $W$, there is $\kappa > 0$ such that, for all sufficiently large reference couplings and every period $L_p D_m$ admitted by Section 9,*

$$\mathcal{E}_A(U, V) \geq \kappa t_{m+1}^2 |A|$$

*for every rooted block-tree gauge, allocation sector and consistent selection $A$. The coupling threshold and $\kappa$ are uniform in the period and local sector.*

We first bound the fine, mismatch and coarse rows, then the affected and unaffected signed rows, and finally combine their contributions.

*Proof.* Use the path and cancellation bounds of Definition 12.2, Lemmas 12.3–12.4, and the free kernels of Lemma 11.5, all at the input and output levels already fixed in the descending construction. Exponential row bounds, together with unitary parallel transport, give row and column sums for $\Phi$ bounded by $C\ell^{-2}$ and $C\ell^2$ respectively. Both sums for $\Phi_n, \mathcal{N}$ are at most $C$. Truncation restricts every coarse displacement to $B/50$. The thinned coordinates give $\|\Phi v_c\|_\infty \leq C\ell^{-2} t$ and $\|\Phi_n v_c\|_\infty \leq Ct$.

Write $R_f = \sum_{A_f : r_p > t} r_p$ and $R_n = \sum_{A_n} n_e$. For the direct component, $|F(g)| \leq \operatorname{ch}(g)$ gives

$$E_f^{\mathrm{dir}}(p) \geq \tfrac{1}{2}(\sqrt{c_{\mathrm{dir},f}}\, r_p - |\Phi v_c(p, 0)|)^2 \quad (r_p \leq t).$$

Choose $\ell$ so $|\Phi v_c| \leq \epsilon_\ell t$, with $\epsilon_\ell \leq \sqrt{c_{\mathrm{dir},f}}/8$. Each selected $t/4 \leq r_p \leq t$ contributes $c_{\mathrm{dir},f} t^2/128$. For $r_p > t$, the linear clip and $\epsilon_\ell \sqrt{c_{\mathrm{dir},f}} \leq c_{\mathrm{lin},f}/4$ leave $3c_{\mathrm{lin},f} t r_p/4$.

The second component is a square when its health test holds. When it fails it has a cross term of either sign, which must be summed. If a row based at $p$ is dropped, some plaquette $q$ obeys $r_q > t \exp(\operatorname{dist}(p, q)/R_j^{\mathrm{rel}})$. For a fixed $q$, the number of possible $p$ is at most $C_{R_j^{\mathrm{rel}}}(r_q/t)$: its radius is $R_j^{\mathrm{rel}} \log(r_q/t)$, and $(1 + R_j^{\mathrm{rel}} \log y)^4 \leq C_{R_j^{\mathrm{rel}}} y$ for $y \geq 1$. Since $r_q \leq 2$, each witness is within $R_j^{\mathrm{rel}} \log(2/t) \leq B/50$ for large $x$. It therefore lies in the neighbourhood where consistency applies and belongs to $A_f$. Split a path row into endpoints with chord at most $t$ and those above $t$. The

row bound controls the first part by $Ct$ times the number of dropped rows; the column bound controls the second by $CR_f$. Hence

$$\sum_{\text{dropped } p} |L_{f,R_*}(U,p)| \le C(C_{R_j^{\mathrm{rel}}}+1)R_f.$$

Multiplication by $|\Phi v_c| \le \epsilon_\ell t$ loses at most $c_{\mathrm{lin},f}tR_f/4$ after increasing $\ell$. Therefore, for $d = c_{\mathrm{dir},f}/128$,

$$\sum E_f \ge \tfrac{1}{2}c_{\mathrm{lin},f}tR_f + dt^2|A_f \setminus \{r_p > t\}|.$$

For a selected mismatch $n_e \ge Wt$, $|\Phi_n v_c| \le Ct$. Choose $W$ so this is at most $\sqrt{\mathsf{b}}\,n_e/2$. The same square inequality gives

$$\sum E_n \ge \tfrac{\mathsf{b}}{8}\sum_{A_n} n_e^2.$$

For $E_c$ the path-row component cancels exactly, including its health test. Put $\alpha_p = \min(1, s/r_p)$. Its direct component is

$$E_c(p) = c_{\mathrm{dir},c}(\alpha_p - \alpha_p^2/2)|F(V_p)|^2 - \mathrm{clip}_s(r_p).$$

The nearest-point Taylor expansion gives $|F(g)|^2 \ge \mathrm{ch}(g)^2 - C\,\mathrm{ch}(g)^4$ for small chord: the deficit is the squared normal displacement of $g-1$, whose normal part is $O(\mathrm{ch}(g)^2)$. Fix $\delta$ small enough that also $|F(g)|^2 \ge 3\,\mathrm{ch}(g)^2/4$ below $\delta$. Since $c_{\mathrm{dir},c} > 16c_{\mathrm{lin},c}$, the three ranges yield

$$E_c(p) \ge \begin{cases} -Cs^4, & r_p \le s, \\ c_{\mathrm{dir},c}sr_p/4, & s < r_p \le \delta, \\ -c_{\mathrm{lin},c}sr_p, & r_p > \delta. \end{cases}$$

The last range uses only nonnegativity of the first term.

Fix

$$0 < a \le \min(c_{\mathrm{lin},f}/8, d, \mathsf{b}/32, c_{\mathrm{dir},c}/64).$$

Set $\epsilon = a/4$.

Here is the comparison for large output plaquettes. Let $\mathcal{P}_y(U) = \sum_q m_y(q)\,\mathrm{ch}(U_q)$ be the fine plaquette sum in the lifted macro-patch centred at $\ell y$, with $m_y(q)$ its actual multiplicity there. The macro radius is $(j+7)\ell$, so $\sum_y m_y(q) \le K_j$ independently of $\ell$ and the period. On its axial frame, denote a straight connector by $Y_z$ and its two root adjustments by $p_0, p_1$. The averaging weight satisfies $\ell^4 w_z^{\mathrm{av}} \le 1$. Telescoping each length-$\ell$ connector and the two end routes, then summing their crossings, gives

$$\ell^4 \sum_z w_z^{\mathrm{av}}\,\mathrm{ch}(p_0^{-1}Y_z p_1) \le C_j\ell^2\mathcal{P}_y(U).$$

For the straight segment, a fine edge occurs in at most $\ell$ translated segments, and the eligible axial link sum is at most $C_j\ell\mathcal{P}_y(U)$. The two end-frame sums obey the same $C_j\ell^2\mathcal{P}_y(U)$ bound: write their coordinate routes, count each link at most $C_j\ell$ times after the normalized transverse average, and use the nearest-point comparison for the minimizing frame. The axial alternatives use their defining link-sum bound instead. The predictor branch comparison bounds its adjusted chord by $C_{\mathrm{pred}}$ times this weighted chord sum, on either branch. Around the four sides of an output plaquette the root adjustments cancel. Comparing its four links with $Q$ gives

$$\mathrm{ch}(V_p) \le \sum_{e\in\partial p} n_e + C_j\ell^{-2}\mathcal{P}_y(U).$$

This proves both the inverse-square factor and its summable multiplicity. In an eligible axial frame a fine link differs from its replicated coarse link by at most $C$ times the macro-patch plaquette sum, plus the mismatch on a boundary edge. Compare the four sides of a coarse plaquette, average the straight paths, and use the predictor equation. Each fine plaquette occurs with total coefficient at most $C\ell^{-2}$; mismatch chords enter linearly. Split each fine chord into its part at most $t$ and its selected excess, and each mismatch into its part at most $2Wt$ and its selected part. The resulting bound has the form

$$r_p \le C_{j,\ell,W} t + H_p, \qquad \sum_p H_p \le C \left( \ell^{-2} \sum_{A_f : r_p > t} r_p + \sum_{A_n} n_e \right).$$

The excess constant depends on the fixed stage, predictor and penalty bounds and is fixed before $\ell, W$; only the bounded-field term may depend on them. This follows by summing the actual route multiplicities before enlarging a macro-patch. Applying the same bound to $\Phi^* X_f$ and $\Phi_n^* Z$, with the column estimates above, bounds their sums on $H$ by $C$ times that excess sum plus $C_{j,\ell,W} t|H|$. Since $|v_c| \le Ct$, these estimates give

$$\sum_H (E_L + E_D) \ge -Ct(\ell^{-2} R_f + R_n) - C_{j,\ell,W} t^2 |H|.$$

For $E_D$, expand its four operator terms. The bounds on $v_c, \Phi v_c, \Phi_n v_c, \mathcal{N} v_c$, together with both row and column sums of the three maps, bound each term by $Ct^2$ per anchor. The constants also depend on the predictor and penalty bounds.

Every output plaquette with $r_p > \delta$ lies in $H$ for large $x$: outside $H$ the macro-patch comparison has no fine or mismatch test above $t_{\mathrm{hi}}$, so $r_p \le C_{j,\ell,W} t_{\mathrm{hi}} < \delta$. On $H$ the same comparison and $s \le 2t$ give

$$c_{\mathrm{lin},c} s \sum_{r_p > \delta} r_p \le C_{j,\ell,W} t^2 |H| + Ct(\ell^{-2} R_f + R_n).$$

Add this inequality to the previous lower bound. There is a geometric constant $K$, fixed before the coupling threshold, with $|H| \le KB^4 |D|$ and at most $KB^4 |A|$ local output anchors. Increasing the constants defines $Z_{\mathrm{aff}} = C_{j,\ell,W} K$ and gives

$$\sum_H (E_L + E_D) \ge c_{\mathrm{lin},c} s \sum_{r_p > \delta} r_p - Ct(\ell^{-2} R_f + R_n) - Z_{\mathrm{aff}} t^2 B^4 |D|.$$

This is the positive term that cancels the third coarse-plaquette range.

For an anchor outside $H$, no fine chord or mismatch above $t_{\mathrm{hi}}$ occurs in the surrounding macro-patches. The preceding plaquette comparison gives coarse plaquette chords at most $C_{j,\ell,W} t_{\mathrm{hi}}$ on this region. Choose an axial gauge in a radius-$O(B)$ coarse cube. Lemma 11.4 bounds its link chords by $CB$ times the sum of its $O(B^4)$ plaquette chords, hence by $C_{j,\ell,W} B^5 t_{\mathrm{hi}}$. Its block-constant extension acts on the fine field. Comparing a fine link with its replicated coarse link uses the local block-tree route; its fine plaquette sum is $O_{j,\ell}(t_{\mathrm{hi}})$ and its boundary mismatch is at most $t_{\mathrm{hi}}$. Both fields are consequently small in this common gauge, with the convenient bound

$$\delta_{\mathrm{loc}} = C_{j,\ell,W} B^{10} t_{\mathrm{hi}}$$

through distance $2B/5$. In this region compare each transported kernel with its flat uncut kernel. A path of length $n$ has transport error at most $Cn\delta_{\mathrm{loc}}$, by telescoping its product. Let $\tau_0 > 0$ be a

common exponential decay rate of the fixed-stage kernels. Their tails are at most $Ce^{-\tau_0 R_*}$. Each curvature or predictor differs from its linearization by $C\delta_{\mathrm{loc}}^2$. With $\epsilon_{\mathrm{loc}} = (1+R_*)\delta_{\mathrm{loc}} + \mathrm{e}^{-\tau_0 R_*}$, subtract (11.6) and $M_f^* M_f + M_n^* M_n + N^* N = I$ before applying the triangle inequality. This gives

$$\|\Phi^* X_f + \Phi_n^* Z - X_c\| \le C\delta_{\mathrm{loc}}\epsilon_{\mathrm{loc}}, \qquad \|(I - \Phi^*\Phi - \Phi_n^*\Phi_n - \mathcal{N}^*\mathcal{N})v_c\| \le Ct\epsilon_{\mathrm{loc}}.$$

The assertion is local: extend the small logarithms by zero off the neighbourhood; the cut rows do not change there. Their range fits inside $2B/5$ for sufficiently large $x$. Thus each signed row is bounded below by $-C(t\delta_{\mathrm{loc}} + t^2)\epsilon_{\mathrm{loc}}$. Since $B^n t_{\mathrm{hi}}^2/t \to 0$ and $B^n e^{-\tau_0 R_*}/t \to 0$ for every fixed $n$, their total is at least $-\epsilon t^2|A|$.

The fine and mismatch bounds leave $at^2$ per selected fine or mismatch test, $att_{\mathrm{hi}}$ per test in $D$, and still retain the losses $Ct(\ell^{-2}R_f + R_n)$ from the signed rows. First increase $\ell$ to handle the fine loss, then $W$ to handle the mismatch loss. Coarse flags can be assigned to incident output plaquettes with chord above $s$, with at most four flags per plaquette. Plaquettes above $\delta$ number at most $K|D|$ by the macro-patch comparison; all local output anchors number at most $KB^4|A|$. Using $t \le 2s$, the middle range above leaves $4at^2$ per remaining such plaquette and so covers every other coarse flag.

The large output plaquette loss cancels against the positive term in the affected-row estimate; $E_N \ge 0$. The residual losses are at most $(4aK + Z_{\mathrm{aff}}B^4)t^2|D| + Cs^4KB^4|A| + at^2|A|/4$. Increase the coupling threshold, with $\ell, W, a$ fixed, until

$$(4aK + Z_{\mathrm{aff}}B^4)t^2 \le att_{\mathrm{hi}}, \qquad Cs^4KB^4 \le at^2/4.$$

The limits stated before the row construction justify both choices. At least $at^2|A|/2$ remains. Take $\kappa = a/2$. Every multiplicity used a fixed-radius local count or a row or column sum; none counts the period. This proves the asserted uniformity and the stated order of choices. □

## Sector covariances

We now specify the Gaussian law for simultaneous local integration. Fix the allocation, its actual block trees, the selected tests, all tags, and the frozen fine links $Y$. Active coordinates are the remaining free-edge logarithms in the complement of the selected radius-$10B$ neighbourhoods. The copied coarse field is the replica defined before Lemma 11.6. Linearize the retained fine and mismatch square rows at this field, applying their actual cutoffs and stencils. Their stacked affine map is $a \mapsto \mathsf{D}a + b_Y$, so their quadratic energy is

$$Q_Y(a) = \tfrac{1}{2}|\mathsf{D}a + b_Y|^2, \qquad \mathsf{A} = \mathsf{D}^*\mathsf{D}, \qquad \mathsf{b} = \mathsf{D}^* b_Y.$$

The stack includes the hypercubic average with its square-root normalization. A stencil for a row anchored at $z$ contains every fine edge with projected distance at most $R_* + 5 + r_{\mathrm{pred}}$ from $z$; $r_{\mathrm{pred}}$ is the maximum coarse distance read by a patch predictor. Thus no stencil is chosen by testing whether its coefficient vanishes.

For an edge-coordinate $i$ write $E_i$ for the two endpoints projected to the coarse lattice. The change of sector and the differentiated local integral use, respectively, the symmetric windows

$$\Omega_i^{\mathrm{sec}} = \{j : \max_{u\in E_i, v\in E_j} \mathrm{dist}(u,v) \le B/20\},$$
$$\Omega_i^{\mathrm{loc}} = \{j : \min_{u\in E_i, v\in E_j} \mathrm{dist}(u,v) \le B\}.$$

Each window is restricted to the coordinates of its sector; unused coordinates may be completed by identity modes. For either prescription put

$$P_{ji} = \mathbf{1}_{j\in\Omega_i}(\mathsf{A}|_{\Omega_i}^{-1})_{ji}, \qquad \Gamma = \tfrac{1}{2}(P + P^{*}), \qquad m_Y = -P^{*}\mathsf{b}. \tag{11.8}$$

These are compressed inverse columns, symmetrized for the covariance. Both windows contain a radius-$B/60$ ball for large $B$, since the two projected endpoints are at distance at most one. The following argument applies to both prescriptions, with their own retained affine rows and mean. The Schur norm of a kernel is the maximum of its absolute row and column sums.

Fix a coarse root $o$ for the derivative directions. Here the coarse flag tests remain the correct ones at the base field; derivatives hold their selected index sets, $Y$ and tags fixed, in coarse directions of size at most $s(1+\operatorname{dist}(o,e))^{q}$; $\upsilon_i = (1+\operatorname{dist}(o,i))^{qk}$. The proof gives the Schur bound needed below as well.

The following auxiliary estimate is needed to justify differentiating the common Gaussian law.

**Lemma 11.12** (Window covariance bounds)**.** *For each fixed ordered derivative order $k$ and direction power $q$, the actual sector matrices satisfy*

$$\mathsf{A} > 0, \quad \Gamma > 0, \quad |D^{r}\mathsf{A}_{ij}| \le C_k e^{-c\operatorname{dist}(i,j)}\upsilon_i^{r}, \quad |D^{r}(\mathsf{A}\Gamma - I)_{ij}| \le C_k e^{-cB}\mathrm{e}^{-c\operatorname{dist}(i,j)}\upsilon_i^{r},$$

*for $r \le k$. Moreover $|D^{r}m_Y(i)| \le C_k t\upsilon_i^{r}$ and $|D^{r}[g^{-1}(\mathsf{A}m_Y + \mathsf{b})]_i| \le C_k e^{-cB}\upsilon_i^{r}$, where $g = \beta^{-1/2}$. Constants are uniform in the period and sector for sufficiently large $x$, with $0 < \beta \le 2\widehat{\beta}_{m+1}$ and correct coarse flags.*

*Proof.* Use Definitions 3.1, 12.2, the axial estimate in Lemma 11.6, and the square rows above. The linearized square rows and an eligible predictor control a fine link by those row values in a fixed macro-patch plus $C\delta_x$ times the surrounding fine increments, where $\delta_x = C_{j,\ell}s$ tends to zero. Summing squares, using the bounded number of macro-patches meeting an edge, gives

$$\|a\|_2^2 \le C_{j,\ell}\|\mathsf{D}a\|_2^2 + C_{j,\ell}\delta_x^2\|a\|_2^2.$$

For large $x$ the last coefficient is at most $1/2$, proving $\mathsf{A} \ge cI$. All-frozen sectors have zero-dimensional active space and the assertion is vacuous there. Adding identity coordinates for inactive links preserves this lower bound.

Exponential path coefficients bound the weighted row and column sums of $\mathsf{D}$. Products give the same Schur bound for $\mathsf{A}$. More explicitly, for sufficiently small fixed $\theta > 0$, multiply coordinates by $\mathrm{e}^{\theta\operatorname{dist}(i,\cdot)}$ on the left and its inverse on the right. The resulting matrix differs from $\mathsf{A}$ by at most $C\theta$ in operator norm: $|\mathrm{e}^{\theta d} - 1| \le \theta d\mathrm{e}^{\theta d}$ is summable against the exponential coefficient bound. Choose $C\theta < c/2$. The inverse Neumann series now gives exponential row and column bounds for $\mathsf{A}^{-1}$ and every principal compression, with constants independent of the number of coordinates.

A compressed inverse column solves $\mathsf{A}Pe_i = e_i$ inside its window. Outside that window, its residual is a sum of products whose combined length crosses the boundary. Every window contains a radius-$B/60$ ball and there are only $C_{j,\ell}$ coordinates per coarse site. The exponential Schur bounds therefore give

$$\|P - \mathsf{A}^{-1}\|_{\mathrm{Schur}} \le Ce^{-cB}, \qquad \|\Gamma - \mathsf{A}^{-1}\|_{\mathrm{Schur}} \le Ce^{-cB}.$$

For sufficiently large $B$ this makes $\Gamma$ positive, since $\mathsf{A}^{-1} \ge \|\mathsf{A}\|^{-1}I$. Its row and column sums are bounded by a fixed constant. The same estimate gives $\mathsf{A}\Gamma - I = O(\mathrm{e}^{-cB})$.

Differentiate $\mathsf{A}_\Omega^{-1}$ by $D(\mathsf{A}_\Omega^{-1}) = -\mathsf{A}_\Omega^{-1}(D\mathsf{A}_\Omega)\mathsf{A}_\Omega^{-1}$ and iterate through order $k$. The index sets $\Omega$ stay fixed. Each derivative of a path coefficient is bounded by a fixed power of the path length and by the direction weight; $1 + \operatorname{dist}(o,j) \le (1 + \operatorname{dist}(o,i))(1 + \operatorname{dist}(i,j))$ moves it to $\upsilon_i^r$. A smaller exponential rate sums all remaining powers. In every differentiated residual, at least one chain still crosses the window boundary. This proves the displayed derivative bounds for $\mathsf{A}$ and $\mathsf{A}\Gamma - I$.

The affine source is formed from the same retained rows and from a bounded smooth extension of the frozen logarithms. Its derivatives are $O(t)$ with the same spatial weights. Multiplication by $P^*$ proves the mean bound. Subtracting the exact inverse solution gives $\mathsf{A}m_Y + \mathsf{b} = (I - \mathsf{A}P^*)\mathsf{b}$, whose derivatives are $O(te^{-cB})$. Finally $g^{-1} \le C\widehat{\beta}_{m+1}^{1/2}$ and $B = \lceil (\log x)^2 \rceil$; this fixed power is absorbed by decreasing $c$. The estimates refer to the actual $\mathsf{A}, \Gamma, m_Y$ of (11.8) throughout. □

## One differentiated local integral

We state the local estimate for the finite products used in the expansion. In a fixed allocation sector, a selected large-field test is called a seed. Join seeds when their fixed enlarged neighbourhoods intersect; their connected components are denoted $A_\eta$. Write $A(M)$ for the union of the fine-plaquette, mismatch and coarse-site selections belonging to the components chosen in a product $M$. Fix the support enlargement $h_{\mathrm{loc}}$ once for the local factors. Compatibility means that the radius-$h_{\mathrm{loc}}B$ neighbourhoods of the distinct chosen components are disjoint.

A factor has a complete read support $S_i$ containing the projected endpoints of every link it uses, every path and health test in its formula, every tag and frozen link, and the entire window $\Omega_a^{\mathrm{loc}}$ for each Gaussian coordinate $a$ it reads. Thus a change outside $S_i$ leaves its formula and its Gaussian marginal unchanged. A finite product $M$ is called a molecule when the intersection graph of its supports is connected. Choose $o$ in their union. Its size $s_M$ is the sum of the numbers of radius-$B$ blocks covering these supports, including retained path lengths in the same units.

For an old forced activity retain its support $Y_i$ and inventory $\mathcal{I}_i$ from Section 3. The admitted products satisfy

$$Y_i \cap Y_{i'} = \varnothing \quad (i \neq i'), \qquad \mathcal{I}_i \subset \bigcup_{p \in A_f(M)} \operatorname{corners}(p),$$
$$\operatorname{Flag}_m(V) \cap \bigcup_{i \in M} S_i = \varnothing.$$

They also satisfy the component compatibility just specified. These conditions retain exactly the incidence and separation used in local integration.

Here are the factors and their envelopes. The retained square corrections are the exponential differences between the exact fine or mismatch rows and their affine squares $|\mathsf{D}a + b_Y|^2/2$. The other smooth factors are inherited regular activities, chart densities, and determinant and Gaussian-density corrections. Old forced activities are bounded by their measurable supremum envelopes. For these factors let $a_i$ include their absolute coefficient bound, the selected seed or path suppression, and the Gaussian exponential bound described below.

For the chart cutoffs choose a smooth $\chi : [0,\infty) \to [0,1]$ equal to one on $[0,1]$ and zero on $[2,\infty)$. On an active logarithmic coordinate put $\chi_{a_*}(u_e) = \chi(|u_e|/(a_* t))$. The fixed mean bound is $|m_Y(e)| \le c_* t$, and choose $a_* > c_* + 1$. Fix a logarithm radius $r_{\log} > 0$ and take $x$ large enough that $4a_* t < r_{\log}/2$. The smaller chart is $|u_e| < a_* t$; the broader cutoff equals one for $|u_e| \le 2a_* t$. A protected link is an active link accompanied by the broader cutoff $\chi_{2a_*}$ whenever an old measurable factor reads it. Nonactive free links are the frozen links $Y$; their physical values are held fixed.

The nesting $\chi_{a_*} \neq 0 \Rightarrow \chi_{2a_*} = 1$ is what allows the physical-link change below on all derivative supports.

Warning factors are the departures from one of the fine-plaquette and mismatch cutoffs, the frame and predictor tag likelihoods, the inventory tests, or the active chart cutoffs. The addresses are fine plaquettes for fine cutoffs, coarse edges for mismatches and predictor tags, coarse sites for frame tags, fine sites for inventory tests, and fine edges for chart cutoffs. A site's location is its coarse projection; an edge or plaquette is located at its projected initial vertex. Near a compulsory component the corresponding departure is included in its component factor. Away from those components, each remaining warning and all its derivatives are supported on

$$E_w = \{z : \exists i,\ \operatorname{dist}(i, \operatorname{loc}(w)) \le 2B, \quad |z_i| \ge \xi t/(2g)\}.$$

Indeed, if all these coordinates are below that threshold, then $|m_Y(e) + gz_e| \le (c_* + \xi/2)t < a_* t$ after reducing $\xi$, so both chart cutoffs equal one. The local predictor bound then makes the fine chord less than $t/4$ and the mismatch less than $Wt$, and both tag likelihoods and the inventory test equal one. The strict margin persists on derivative supports. Choose the fixed $\xi > 0$ small enough for these finitely many local implications. Let $\mathcal{J}(M)$ be the set of distinct selected warning addresses and $E_{\mathcal{J}(M)} = \bigcap_{w \in \mathcal{J}(M)} E_w$ their simultaneous event. Their amplitudes are bounded by the fixed stage-and-ratio constant

$$M_w = 2 + |\mathbb{B}_R \times S_4| + |\mathbb{B}_R|.$$

This includes the likelihood relative to uniform tags. With $\varepsilon_w = \widehat{\beta}_{m+1}^{-\tau_w}$ for a fixed $\tau_w > 0$, the weights used below are

$$f_i = a_i \quad (i \text{ nonwarning}), \qquad f_w = M_w \varepsilon_w \quad (w \in \mathcal{J}(M)).$$

Each distinct address contributes once.

Assume $\widehat{\beta}_{m+1}/2 \le \beta \le 2\widehat{\beta}_{m+1}$. Use the local prescription $\Omega^{\mathrm{loc}}$ in (11.8), throughout this local integral. Conditional on the fixed sector and tags the reference law is

$$d\nu = dH_F(Y)\, d\gamma_\Gamma(z).$$

Here $H_F$ is product Haar measure on the frozen tree-slice links and $\gamma_\Gamma$ is the centered Gaussian. Reference tags are independent uniform tags; their actual likelihoods are included among the factors. All estimates below are uniform in those tags. A finite set of Gaussian reads $I$ has marginal covariance $\Gamma_I = \Gamma[I, I]$.

For a finite selection write its product as

$$F_M(V, Y, z) = \prod_{i \in M \setminus \mathcal{J}(M)} F_i(V, Y, z) \prod_{w \in \mathcal{J}(M)} W_w(V, Y, z).$$

Here $F_i$ are the nonwarning factors just specified and $W_w$ the warning factors. The transported product will use one change of Gaussian coordinates on their common set of reads.

To differentiate while holding physical fine links fixed, fix a base coarse field $V_0$ and write $u = m_Y(V_0) + gz$. For a nearby $V$, set

$$u_V(e) = \log\left(\exp(u(e))(\operatorname{rep} V_0)_e (\operatorname{rep} V)_e^{-1}\right), \qquad z_V = g^{-1}(u_V - m_Y(V)).$$

Use a fixed smooth extension of this formula outside $|u(e)| < r_{\log}/2$, equal to the identity plus a bounded correction there. Its derivative is invertible for $V$ sufficiently near $V_0$, uniformly over the

chart argument; the extension is used only with the stated protected cutoff. On active protected links,

$$\exp(u_V(e))(\operatorname{rep} V)_e = \exp(u(e))(\operatorname{rep} V_0)_e.$$

Every old measurable activity evaluated on those links is therefore constant under this differentiation. It is bounded by its supremum majorant and no field derivative is applied to it.

The density of the changed marginal, relative to the base one on the union $I$ of all reads, is

$$\mathcal{R}_I(V,z) = |\det D_z z_{V,I}| \, \frac{(\det \Gamma_I(V))^{-1/2} \exp[-\frac{1}{2} z_{V,I}^T \Gamma_I(V)^{-1} z_{V,I}]}{(\det \Gamma_I(V_0))^{-1/2} \exp[-\frac{1}{2} z_I^T \Gamma_I(V_0)^{-1} z_I]}.$$

Define the transported integrand by

$$\widetilde{F}_M(V,Y,z) = \mathcal{R}_I(V,z) F_M(V,Y,z_V).$$

The next lemma bounds its ordered derivatives at $V_0$ under the same base law, including products containing old measurable forced activities. It is the second local input to Section 13. Its bounds may depend on the fixed mismatch multiplier $W$ and energy constant $\kappa$ of Lemma 11.11.

**Lemma 11.13** (Simultaneous local integration)**.** *For odd $\ell$ above a threshold depending on the stage data, every fixed derivative order $k$ and spatial direction power $q$, and every admitted molecule $M$, the transported sector integrand has one integrable majorant $g_M$ through order $k$, with*

$$\int g_M \, d\nu \le P_{j,\ell,k,q}(x)(1+s_M)^{d_{j,\ell,k,q}} \prod_{i \in M} f_i.$$

*The function $P_{j,\ell,k,q}$ has scale rate zero. Simultaneous events at $r$ distinct warning addresses have probability at most $\varepsilon_w^{2r}$. Disjoint complete read supports factorize, including their Haar and tag coordinates.*

Scale rate zero means $P_{j,\ell,k,q}(x) \le C_\epsilon \widehat{\beta}_{m+1}^{\epsilon}$ for every $\epsilon > 0$ once $x$ is large. Directions and sector parameters have the restrictions of Lemma 11.12.

*Proof.* Use Definitions 3.1, 12.2, Lemmas 11.4, 11.7, 11.11, and 11.12. First retain the exact affine Gaussian substitution. For any chart-supported $F_Y$,

$$\begin{aligned}\int \mathrm{e}^{-\beta Q_Y(a)} F_Y(a)\, da &= (2\pi g^2)^{n/2} (\det \Gamma)^{1/2} \mathrm{e}^{-\beta Q_Y(m_Y)} \\ &\quad \times \int F_Y(m_Y + gz) \exp\left\{ \langle \eta_Y, z\rangle + \tfrac{1}{2}\langle z, (\Gamma^{-1} - \mathsf{A}) z\rangle \right\} d\gamma_\Gamma(z), \\ \eta_Y &= -g^{-1}(\mathsf{A} m_Y + \mathsf{b}).\end{aligned}$$

Here $n$ counts active real coordinates; the Haar density and chart cutoffs are included in $F_Y$. Expansion of $Q_Y(m_Y + gz)$ proves the identity, including its determinant and residual linear term. The identity holds for either window prescription. The local expansion in Section 13 uses $\Omega^{\mathrm{loc}}$.

For the common change of coordinates defined above, the first logarithmic derivative of the Gaussian density is

$$-\tfrac{1}{2}\operatorname{tr}(\Gamma_I^{-1}\dot{\Gamma}_I) + \tfrac{1}{2} z_V^T \Gamma_I^{-1} \dot{\Gamma}_I \Gamma_I^{-1} z_V - z_V^T \Gamma_I^{-1} \dot{z}_V.$$

Add the logarithmic Jacobian derivative. Repeated differentiation bounds every score through order $k$ by a polynomial in the absolute values of the marginal Gaussian coordinates, with coefficients

controlled by Lemma 11.12 and the fixed smooth chart extension. The number of placements is a fixed polynomial in $1 + s_M$, since $k$ is fixed and $|I| \le C_{j,\ell} B^4 s_M$. Connectivity and the chosen root bound all distances from $o$ by $C_{j,\ell} B(1 + s_M)$, so the spatial direction weights give another fixed polynomial. This explicitly includes covariance changes before any factors are multiplied.

The normalized coarse directions have size $O(s)$ with their stated spatial weights. On a fixed chart, derivatives of its multiplication, logarithm and bounded extension are bounded. The mean derivatives are $O(t)$, while $s/t$ stays bounded. In $z_V$ the additional normalization is $g^{-1}$. Thus the score coefficients are bounded by a fixed power of $g^{-1}s$, $B$ and $1 + s_M$. Since

$$g^{-1}s \le C \exp\{(\log\log \widehat{\beta}_{m+1})^2\},$$

all fixed powers have scale rate zero. The small inverse and stationarity residuals in Lemma 11.12 have the same property after this normalization. The chart-supported square corrections have deterministic envelopes: on their support all logarithms are $O(t)$, the affine remainder is $O(t^2)$, and $|\frac{1}{2}|r + e|^2 - \frac{1}{2}|r|^2| \le |r||e| + |e|^2/2$ gives an energy correction $O(\beta t^3)$. The exponentially summable path coefficients bound the same quantities with fixed path-length powers after differentiation. These bounds are included in $a_i$.

The unbounded exponentials come from the residual Gaussian density and its local covariance comparisons. Write their upper exponents as $\ell_i \cdot z + z^T K_i z/2$, with $K_i \ge 0$. At the base field the Gaussian likelihood is one; its derivatives are the polynomial scores already estimated. Its exponent tends to zero with $V - V_0$, so on a sufficiently small neighbourhood its weighted row sums are at most $\delta_B = C_{j,k} B^d \mathrm{e}^{-cB}$. The residual Gaussian density has this bound by Lemma 11.12. The resolvent identity for each compressed inverse gives $C\delta_B$ for its local quadratic corrections. Each factor's coefficients decay exponentially from its own address, and each address has bounded multiplicity. Thus the sum at a coordinate is bounded by $C\delta_B \sum_y \mathrm{e}^{-c\,\mathrm{dist}(y,a)} \le C'\delta_B$, uniformly in the period and number of factors. Here $y$ ranges over address locations and $a$ is the coordinate location. In the linear terms, weighted Cauchy–Schwarz at each coordinate gives

$$\left|\Gamma^{1/2} \sum_i \ell_i\right|^2 \le C \sum_i \Lambda_i, \qquad \left\|\Gamma^{1/2} \sum_i K_i \Gamma^{1/2}\right\| \le C\delta_B.$$

To specify the linear bound, let $w_{ia}$ be the sum of the absolute path weights contributing to coordinate $a$ of $\ell_i$; bounded endpoint multiplicity gives $\sup_a \sum_i w_{ia} \le C$. Put

$$\Lambda_i = \sum_a \frac{|\ell_i(a)|^2}{w_{ia}} + \mathrm{tr}(\Gamma^{1/2} K_i \Gamma^{1/2}),$$

with zero summand if both numerator and denominator vanish. This quantity assigns the common linear exponential bound to the individual factors. To see the first bound, use $|\sum_r x_r|^2 \le (\sum_r w_r) \sum_r |x_r|^2/w_r$ with the absolute path weights $w_r$; their sums are the preceding bounded row sums. The Schur bound for $\Gamma$ completes the estimate. In particular, this bounds $|\Gamma^{1/2}\ell_0|^2$ with $\ell_0 = \sum_i \ell_i$, including all terms $\ell_i^T \Gamma \ell_{i'}$ with $i \ne i'$. Include $\exp(C\Lambda_i)$ in $a_i$. Put $K = \sum_i K_i$. Since $0 \le K_i \le K$, each $\|\Gamma^{1/2} K_i \Gamma^{1/2}\|$ is at most $\|\Gamma^{1/2} K \Gamma^{1/2}\| < 1/8$ for large $B$, also after the fixed enlargement needed for two moments. Diagonalization and square completion, with $-\log(1-u) \le 2u$ for $0 \le u \le 1/2$, give

$$\mathbb{E}_\Gamma \mathrm{e}^{2\ell_0 \cdot z + z^T K z} \le \exp\left(C \sum_i \Lambda_i\right).$$

Under this tilt the covariance is at most $2\Gamma$ and its mean is $(I-2\Gamma K)^{-1}(2\Gamma\ell_0)$. Polynomial moments are bounded by the preceding linear estimate; increasing the fixed constant in $\exp(C\Lambda_i)$ includes their mean contributions. Thus the differentiated factors and likelihood have, after removing the warning amplitudes, one $L^2$ majorant bounded by $P_{j,\ell,k,q}(x)(1+s_M)^d\prod a_i$.

We next bound the simultaneous warning event defined above, including all derivative supports. Let $p_0$ bound the number of Gaussian coordinates per coarse site and $m_0$ the number of distinct warning addresses per site. These are fixed once the stage and ratio are fixed. For $r$ distinct warnings, choose one violating coordinate for each. There are at most $(p_0|\mathbb{B}_{2B}|)^r$ choices. A single coordinate can serve at most $m_0|\mathbb{B}_{2B}|$ warning addresses. The Schur bound gives covariance operator norm at most $M_0$. For a set $J$ of chosen distinct coordinates, Gaussian exponential moments and summation over signs give

$$\Pr\{|z_i|\ge T\ (i\in J)\}\le 2^{|J|}\exp\{(M_0/2-T)|J|\}.$$

Take $T=\xi t/(2g)$ and choose $H>0$ with $M_0/2+Hm_0|\mathbb{B}_{2B}|\le T$. The multiplicity bound gives

$$\Pr(E_{\mathcal{J}(M)})\le\bigl(2p_0|\mathbb{B}_{2B}|\mathrm{e}^{-H}\bigr)^r.$$

The factor $t/g$ grows faster than every fixed power of $B$ and $\log\widehat{\beta}_{m+1}$. Consequently $H$ can exceed $2\tau_w\log\widehat{\beta}_{m+1}+\log(2p_0|\mathbb{B}_{2B}|)$, proving the claimed bound $\varepsilon_w^{2r}$.

For the final majorant, take the finite sum of the absolute differentiated smooth expressions and scores, multiplied by the fixed old measurable envelopes, the warning amplitudes $\prod_w M_w$, and the simultaneous warning indicator. The product rule bounds every derivative through order $k$ by this same nonnegative function. Its integral is bounded by its polynomial $L^2$ bound times $\Pr(E_{\mathcal{J}(M)})^{1/2}$. This retains precisely $\varepsilon_w^r$ and gives the displayed estimate. The seed suppressions in $a_i$ follow from Lemma 11.11; the determinant and covariance corrections use the exponential bounds above. Integrating the independent frozen Haar variables and summing the finite tag probabilities preserve the bound.

Finally the Gaussian marginal on $I$ has covariance $\Gamma_I$. Every complete read support contains $\Omega_i^{\mathrm{loc}}$ for each coordinate $i$ it reads. The windows are symmetric. Hence if $i,j$ belong to disjoint complete read supports, $j\notin\Omega_i^{\mathrm{loc}}$ and $i\notin\Omega_j^{\mathrm{loc}}$, so $\Gamma_{ij}=0$ by (11.8). The Gaussian characteristic function therefore factorizes over these supports. The frozen Haar and reference tag coordinates are disjoint as well, proving the asserted exact factorization. □

## 12 Path actions and the coupling reset

The path action retains curvature terms through fourth order after each integration. We choose its quadratic part so that $\beta$ is the coefficient of the squared Wilson curvature at zero momentum at every level. Quadratic terms independent of $\beta$ have two vanishing displacement moments. The marginal part generated by the determinant is removed from these terms and absorbed into the coupling.

The path bounds enter Theorem 3.6, and the coupling formula enters Theorem 3.5. We evaluate the leading action at its stationary background and express its cubic and quartic terms in transported curvatures. We compare the coupling coefficient of the blocked determinant with that of a regulated fine-lattice determinant, with an error uniform in the block ratio. The vector and ghost bubbles give the logarithmic coefficient; an affine equation bounds the remaining reset. Lemmas 12.5–12.10 carry out these steps. The inputs are the branches, free forms, stationary interpolation, Gaussian coordinates, vertex, Ward and insertion estimates of Lemmas 11.4–11.8.

Lemmas 13.11, 13.8–13.13 supply the activity bounds, pre-reset expansion, exact density identity and continuity. Their proofs belong to Sections 11 and 13; the reset here bounds and identifies the correction in that identity.

Throughout, the group and metric are those of Definition 2.1, and the scales are those of Section 3. At a step $m+1 \to m$ put $\ell = \ell_m$ and $j = J_{m+1}$. Arrays depend on $N, m$ and the intervening block ratios, and are chosen on $\mathbb{Z}^4$ before periodic evaluation. They are independent of the volume index $p$ and of the running coupling. Constants declared uniform in $\ell$ are chosen before $\ell$.

The arrays are constructed by descending induction on $m$. Write $(I_m)$ for the path bounds and cancellations of Lemmas 12.3–12.4 at level $m$. The top-level Wilson matching in the proof of Lemma 12.6 starts the induction. Under $(I_{m+1})$, Lemmas 12.6, 12.8–12.9 treat the step $m+1 \to m$; the joint proof after Lemma 12.9 constructs the output arrays and proves $(I_m)$. Curl division and the gauge determinant identity are independent of this induction.

## Curvature coordinates

A path transports curvature to an anchor, where invariant colour contractions can be taken. Write $U_\gamma$ for the ordered link product along $\gamma$, $\mathfrak{p} = (a, b)$, $a < b$, for an orientation, and $U_{x;\mathfrak{p}}$ for a plaquette holonomy. The footprint of a slot is the set of vertices visited by its path and plaquette boundary. The footprint of an atom is the union of its slot footprints and the origin, translated to its anchor. Health means disjointness from $\mathrm{Flag}_m(U)$: no plaquette with a vertex in that footprint has chord exceeding $t_m$.

**Definition 12.1** (Curvature slots)**.** A slot $s = (\gamma, \mathfrak{p})$, with $\gamma : 0 \to z_s$, has length $|\gamma| + 4$ and value

$$\mathcal{F}_s(U, x) = \mathrm{Ad}(U_{x+\gamma})\,\mathrm{proj}_{\mathfrak{g}}\,\frac{U_{x+z_s;\mathfrak{p}} - U_{x+z_s;\mathfrak{p}}^{-1}}{2}\,.$$

Transport is from $x + z_s$ to $x$. Set $\mathcal{L}_{\mathfrak{p}}^{\mathrm{path}} = \sum_i r_{\mathfrak{p}i}\mathcal{F}_{\mathfrak{p}i}$. For $k = 2, 3, 4$, the atoms $\mathfrak{a}_i^{(k)}$ are real invariant $k$-linear expressions in slots, masked by health on their footprints. Atom length sums slot lengths; $|\mathfrak{a}_i^{(k)}|$ sums absolute colour coefficients. These arrays retain local terms through fourth order.

Here $r_{\mathfrak{p}i} \in \mathbb{R}$ and $\mathcal{F}_{\mathfrak{p}i}$ denotes the slot assigned to row $\mathfrak{p}$ and index $i \in \mathbb{N}$. More explicitly, in the orthonormal colour basis $(e_b^{\mathfrak{g}})_b$ of Definition 2.1, an atom has the form

$$\mathfrak{a}_i^{(k)}(U, x) = \mathbf{1}_{X_i + x \text{ healthy}} \sum_{b_1, \dots, b_k} c_i(b_1, \dots, b_k) \prod_{\nu=1}^{k} \langle e_{b_\nu}^{\mathfrak{g}}, \mathcal{F}_{s_{i,\nu}}(U, x)\rangle_{\mathfrak{g}}\,.$$

The coefficients are invariant under the simultaneous adjoint action on all slots. The finite set $X_i$ is the footprint just defined.

For each stage $k$, take $\gamma_k > 0$ from Lemma 11.5, so the free plaquette forms for the top and both incoming stages are at least $2\gamma_k I$. Write $c_{\mathrm{dir}} = c_{\mathrm{dir},J_m}$ and $c_{\mathrm{lin}} = c_{\mathrm{lin},J_m}$. Whenever $c_{\mathrm{dir}} \le \gamma_{J_m}$, the form $\mathsf{H}_{N,m} - c_{\mathrm{dir}} I$ is at least $\gamma_{J_m} I$ and has the positive row factor used below. In the relaxed health test, dist is the unit-lattice maximum distance on the torus. This test permits a larger chord farther from the anchor.

**Definition 12.2** (Path action)**.** Require $0 < 16c_{\mathrm{lin}} < c_{\mathrm{dir}} \le \min(\gamma_{J_m}, 1)$, and put

$$\mathrm{clip}_t(u) = \begin{cases} c_{\mathrm{dir}}u^2/2, & u \le t, \\ c_{\mathrm{lin}}tu, & u > t, \end{cases}$$
$$\chi^{\mathrm{hlth}}_{m,x}(U) = \mathbf{1}\{\mathrm{ch}(U_{y;\mathfrak{p}}) \le t_m e^{\mathrm{dist}(x,y)/R^{\mathrm{rel}}_{J_m}} \text{ for every } y, \mathfrak{p}\}\,.$$

Define

$$(P^{\mathrm{path}})^{\mathrm{raw}}_{N,m}(\beta, U) = \beta \sum_{x,\mathfrak{p}} \Big[\mathrm{clip}_{t_m}(\mathrm{ch}(U_{x;\mathfrak{p}})) + \tfrac{1}{2}\chi^{\mathrm{hlth}}_{m,x}|\mathcal{L}^{\mathrm{path}}_{\mathfrak{p}}(U,x)|^2\Big] + \beta \sum_{x,i}(\mathfrak{a}^{(3)}_i + \mathfrak{a}^{(4)}_i)(U,x) + \sum_{x,i}\mathfrak{a}^{(2)}_i(U,x),$$
$$P^{\mathrm{path}}_{N,m}(\beta, U) = \frac{1}{|H_4|}\sum_{\varsigma\in H_4}(P^{\mathrm{path}})^{\mathrm{raw}}_{N,m}(\beta, \varsigma^* U)\,.$$

This separates the kinetic coefficient from loop corrections.

The sums over $x$ are over the torus at level $m$, and $i \in \mathbb{N}$. The clip has a downward jump immediately to the right of $t$: its value at $t$ is $c_{\mathrm{dir}}t^2/2$, whereas its right limit is $c_{\mathrm{lin}}t^2$. For $u > t$ its linear branch gives $\mathrm{clip}_t(u) = c_{\mathrm{lin}}tu \ge c_{\mathrm{lin}}t^2$, the large-field lower bound used in Lemma 11.11. The bounds below control actual path lengths on every period.

**Lemma 12.3** (Path bounds)**.** *For every $N$ and $m \le N$, the construction below chooses arrays independent of $p$ and $\beta$, with*

$$\sum_i |r_{\mathfrak{p}i}| e^{w^{\mathrm{path}}_{J_m}\,\mathrm{length}(\mathcal{F}_{\mathfrak{p}i})} \le A^{\mathrm{path}}_{J_m},$$
$$\sum_i |\mathfrak{a}^{(k)}_i| e^{w^{\mathrm{path}}_{J_m}\,\mathrm{length}(\mathfrak{a}^{(k)}_i)} \le \begin{cases} \Lambda^{\mathrm{loop}}_m, & k = 2, \\ A^{\mathrm{path}}_{J_m}, & k = 3, 4. \end{cases}$$

*All these series converge. The constants $A^{\mathrm{path}}_k, w^{\mathrm{path}}_k > 0$ depend only on the fixed stage data.*

The joint proof of Lemmas 12.3 and 12.4 follows Lemma 12.9. To state the cancellations, put

$$s_{\mathfrak{p}\mathfrak{q}} = \sum_{i:\,\mathrm{orientation}(\mathcal{F}_{\mathfrak{p}i})=\mathfrak{q}} r_{\mathfrak{p}i}\,.$$

For fixed ordered orientations $\mathfrak{q}, \mathfrak{r}$ and colours $b_1, b_2$, let $c_i^{\mathfrak{q}\mathfrak{r};b_1b_2}$ be the corresponding coefficient of the quadratic atom, zero unless its slots have those orientations. Write $z_{i,1}, z_{i,2}$ for their offsets. The cancellations remove the kinetic component from the loop atoms.

**Lemma 12.4** (Kinetic and quadratic cancellations)**.** *For each pair of orientations $\mathfrak{q}, \mathfrak{r}$,*

$$c_{\mathrm{dir},J_m}\delta_{\mathfrak{q}\mathfrak{r}} + \sum_{\mathfrak{p}} s_{\mathfrak{p}\mathfrak{q}}s_{\mathfrak{p}\mathfrak{r}} = \delta_{\mathfrak{q}\mathfrak{r}}\,.$$

*For each ordered orientation-colour pair,*

$$\sum_i c_i^{\mathfrak{q}\mathfrak{r};b_1b_2} = 0, \qquad \sum_i c_i^{\mathfrak{q}\mathfrak{r};b_1b_2}(z_{i,2} - z_{i,1}) = 0\,.$$

*All these moment series converge absolutely.*

The joint proof is given after Lemma 12.9.

### Fourth-order matching

Near the flat field all health indicators equal one. By $\mathrm{ch}^2 = 2\,\mathrm{pr}$ from Definition 2.1, the leading analytic action is obtained from Definition 12.2 by replacing the clipped term by $c_{\mathrm{dir}}\,\mathrm{pr}$ and omitting the indicators and loop atoms. Denote this action, without its factor $\beta$, by $\mathsf{S}_{N,m}$. Its prescribed quadratic part is

$$\frac{1}{2}\langle d_1 a, \mathsf{H}_{N,m} d_1 a\rangle, \qquad \mathsf{H}_{N,m} = c_{\mathrm{dir},J_m} I + \rho^*_{N,m}\rho_{N,m}, \qquad \widehat{\mathsf{H}}_{N,m}(0) = I\,.$$

Here $d_1$ is the lattice curl. Lemma 11.5 gives the exact adjacent free relation and the row factor, with $\mathsf{H}_{N,m} - c_{\mathrm{dir},J_m} I \geq \gamma_{J_m} I$. Placing its coefficients on coordinate paths gives $\mathcal{L}^{\mathrm{path}}_{\mathfrak{p}}$. The cell volume contributes $\ell^4$ and each curvature contributes $\ell^{-2}$; hence $\ell^4(\ell^{-2})^2 = 1$, preserving the coefficient $\beta$.

We need a division that expresses a gauge-invariant Taylor difference in curvatures. For a momentum $\xi$ put $q_a(\xi) = e^{i\xi_a} - 1$; its exterior action $d_1(\xi)a = q(\xi) \wedge a$ is the curl symbol. Let $\xi_1, \dots, \xi_{k-1}$ be independent momenta and $\xi_0 = -\sum_{\nu=1}^{k-1} \xi_\nu$. Write $\delta_\nu = \iota_{q(\xi_\nu)}$ for bilinear contraction in leg $\nu$. Consider periodic analytic kernels $f$ valued in $\bigotimes_{\nu=0}^{k-1} \Lambda^1\mathbb{C}^4$, with $\delta_\nu f = 0$ in every leg. On the strip $|\operatorname{Im}\xi^a_\nu| < \sigma$, assume each coordinate of $f$ is bounded by $A \geq 0$. Momentum order $s$ means vanishing of all Taylor coefficients of total momentum degree less than $s$.

**Lemma 12.5** (Curl division)**.** *For $k \geq 3$ and $\sigma > 0$, there are $0 < \eta \leq \sigma$ and $K \geq 0$, depending only on $k, \sigma$, such that every such $f$ admits a periodic analytic kernel $g$ valued in $\bigotimes_{\nu=0}^{k-1} \Lambda^2\mathbb{C}^4$, with*

$$f = \delta_0 \cdots \delta_{k-1} g, \qquad \sup_{\eta\text{-strip}} |g| \leq KA\,.$$

*Its Fourier coefficients decay exponentially. For $k = 2$ the same conclusion holds if $f$ has momentum order $s + 2$, with $g$ of order $s$; $\eta, K$ are independent of $s \geq 0$.*

*Proof.* We give the division for a general transverse kernel. Scalar division requires vanishing when the divided leg has zero momentum; positive exterior degree requires only closedness. For one leg let $\delta = \iota_q$ be contraction on exterior tensors. Split a $\delta$-closed tensor as $f = e_a \wedge u + v$, with $u, v$ containing no $e_a$. Define

$$D_a f(\xi) = \frac{f(\xi) - f(\xi)|_{\xi_a=0}}{q_a(\xi)}, \qquad h_a f = e_a \wedge D_a v, \qquad R_a f = v|_{\xi_a=0}\,.$$

The identity $\delta f = 0$ gives $\delta' v = -q_a u$, where $\delta'$ contracts the other axes. Since $q_b$ is independent of $\xi_a$ for $b \neq a$, $D_a$ commutes with $\delta'$, and $D_a(q_a u) = u$. Consequently

$$\delta h_a f = q_a D_a v - e_a \wedge D_a \delta' v = v - v|_{\xi_a=0} + e_a \wedge u = f - R_a f\,.$$

The remainder $R_a f$ is again closed and has no component containing $e_a$. Applying this in the four axes gives the homotopy

$$f = \delta\left(\sum_{a=1}^{4} h_a R_{a-1} \cdots R_1 f\right) + R_4 R_3 R_2 R_1 f\,.$$

The final remainder vanishes in positive exterior degree because it contains none of the four axes. In degree zero it is the restriction to zero momentum of this leg, which must vanish for scalar division. In that case the formula reads $f = \sum_a q_a D_a(R_{a-1} \cdots R_1 f)$.

The quotients are analytic even at the zeros of $q_a$. On a Fourier monomial the identity

$$\frac{e^{in\xi_a}-1}{e^{i\xi_a}-1}=\begin{cases}\sum_{v=0}^{n-1}e^{iv\xi_a}, & n>0,\\ -\sum_{v=n}^{-1}e^{iv\xi_a}, & n<0,\\ 0, & n=0\end{cases}$$

bounds the quotient on a smaller strip by a constant times $|n|$ times its original exponential weight. The strip loss absorbs $|n|$, so the Fourier series converges absolutely. Each division loses at most one momentum order.

For independent legs, their remaining transverse subspaces do not depend on the divided momentum, so each restriction and division preserves them. The dependent-leg induction has the following claim: with $r$ independent momenta and dependent exterior degree $d\ge 1$, a tensor closed in the dependent leg and transverse in all independent legs has a dependent-leg primitive preserving those transverse conditions, provided $d+2r>4$. Analytic bounds hold after a strip reduction depending only on $r,d$ and the original strip. For $r=0$ the condition gives $d>4$, so the tensor is zero. The induction step below reduces $(r,d)$ to $(r-1,d+2)$, preserving the condition. Initially $r=k-1, d=1$, so it applies for $k\ge 3$.

For the dependent momentum $\xi_0=-\sum_{i=1}^{k-1}\xi_i$, first shear the coordinates to replace $\xi_1$ by $\xi_0$. The other independent contractions remain unchanged. The one-leg formula gives $\delta_0 X=f$, retaining the contractions in legs $2,\dots,k-1$. Write $c_1$ for contraction in leg 1. Its defect obeys $\delta_0(c_1X)=c_1f=0$. Solve $\delta_0 Y=c_1X$ by the same formula. After $c_1$, leg 1 has exterior degree zero, so restoring that leg will require only scalar division.

On $\xi_1=0$, the restriction $Y_0$ is $\delta_0$-closed, since $c_1X$ vanishes there. The induction hypothesis with one fewer independent momentum and dependent degree two higher gives $\delta_0 Z=Y_0$, preserving the other transverse conditions. Extend $Z$ independently of $\xi_1$. Then $Y'=Y-\delta_0 Z$ vanishes on $\xi_1=0$, and scalar division gives $c_1D=Y'$. Thus

$$V=X-\delta_0 D,\qquad \delta_0 V=f,\qquad c_1V=c_1X-\delta_0 Y'=0\,.$$

All corrections preserve the other independent contractions. The two successive solves $f\mapsto X$ and $c_1X\mapsto Y$ each raise the dependent exterior degree by one; contraction $c_1$ changes only leg 1. Thus the restricted problem for $Y_0$ has one fewer independent momentum and dependent degree two higher. For three or more legs it terminates above exterior degree four, where the tensor is zero. The independent-leg divisions then finish the factorization.

For two legs, the restriction $Y_0$ is at zero external momentum. Starting from order $s+2$, the first division loses one order, contraction restores one, and the solve for $Y$ leaves order $s+1\ge 1$. Hence $Y_0=0$ directly. The correction and final division leave order $s$. This verifies the scalar vanishing hypothesis where it is needed. For the Taylor differences of Definition 12.2, gauge invariance supplies transversality because all lower Taylor differences vanish. □

For $z=(z_1,\dots,z_{k-1})$, contour displacement gives $|K(z)|\le Ce^{-\zeta\sum_\nu|z_\nu|_1}$. Choose the anchor at zero and coordinate paths to $z_\nu$. Their total slot length is at most $4k+\sum_\nu|z_\nu|_1$. Every path width below $\zeta$ therefore gives a convergent weighted sum. Reality and simultaneous adjoint invariance pass to the coefficients by conjugation and invariant projection.

At a step, let $v$ be the coarse chart variable and $z$ the fine axial variable. Let $S(v,z)$ be the fine leading action plus $\mathsf{b}\sum_e \mathrm{pr}$ of the predictor discrepancy, without $\beta'$. It excludes the inherited loop action. Lemma 11.6 supplies

$$D_zS(v,z(v))=0,\qquad z(0)=0,\qquad \mathsf{Q}=D_z^2S(0,0)\ge\nu_jI,\qquad \nu_j>0\,.$$

Put $Wv = (v, Dz(0)v)$ and $\mathsf{B}(v,v)[u] = D^3S(0)[Wv, Wv, (0,u)]$. For the bounds use the joint fine-link and centered-frame vertices of Lemma 11.9, before choosing an axial representative. Let $C_3, C_4$ bound their scalar Fourier entries on the fixed smaller strip, both on normalized smooth coarse legs and with one arbitrary fine leg of block norm at most one. In particular, the mixed cubic functional has dual block norm at most $C_3$. Take $E = \max(1, C_j^{\mathrm{ins}})$ from Lemma 11.8, bounding the completed inverse from the dual block norm to the primal block norm. The mixed functionals annihilate joint gauge directions, so their pairing through this inverse equals their axial pairing through $\mathsf{Q}^{-1}$, by Lemma 11.9. Thus $E$ is independent of $\ell$ and of the input path amplitude. The constants $a_3, c_4$ depend only on preceding stage data; $a_3', c_4'$ may also depend on the input path bound.

The next formula constructs the higher arrays while keeping their bound independent of the input amplitude after the block ratio has been enlarged. The stationary identities hold for every admitted ratio; only the uniform array bound requires a larger ratio. Lemma 11.9 gives $C_3 \le a_3 + \ell^{-1}a_3'$ and $C_4 \le c_4 + \ell^{-1}c_4'$.

**Lemma 12.6** (Classical matching)**.** *The stationary action* $\mathsf{S}_{\mathrm{cl}}(v) = S(v, z(v))$ *satisfies*

$$D^3\mathsf{S}_{\mathrm{cl}}(0)[v^3] = D^3S(0)[(Wv)^3],$$
$$D^4\mathsf{S}_{\mathrm{cl}}(0)[v^4] = D^4S(0)[(Wv)^4] - 3\mathsf{B}(v,v)\mathsf{Q}^{-1}\mathsf{B}(v,v)\,.$$

*Its factorial-normalized kernels have bounds* $C_3/6$ *and* $(C_4 + 3C_3^2E)/24$*. For* $\ell$ *above a threshold depending on the input path data, cubic and quartic arrays satisfying the bounds of Lemma 12.3 make* $\mathsf{S}_{N,m}$ *agree with* $\mathsf{S}_{\mathrm{cl}}$ *through degree four. Their bound depends only on fixed stage data. The degree-*$r$ *fine-minus-Wilson vertex has momentum order* $5 - r$*, for* $r = 2, 3, 4$*.*

*Proof.* Use the block map of Definition 3.1 and Lemmas 11.6, 11.9. Differentiating $D_zS = 0$ once gives $D^2S(0)[Wv, (0,u)] = 0$; differentiating twice gives $D^2z(0)[v,v] = -\mathsf{Q}^{-1}\mathsf{B}(v,v)$. In the third derivative of $S(v, z(v))$, the terms containing $D^2z$ or $D^3z$ vanish by these two stationary identities. In the fourth derivative the surviving terms involving $w = D^2z(0)[v,v]$ are

$$6\mathsf{B}(v,v)[w] + 3\langle w, \mathsf{Q}w\rangle = -3\mathsf{B}(v,v)\mathsf{Q}^{-1}\mathsf{B}(v,v)\,.$$

The other term is $D^4S(0)[(Wv)^4]$. In the joint coordinates of Lemma 11.9, the cubic and quartic stationary kernels have the same identities with smooth interpolated legs. Each mixed cubic functional has dual block norm at most $C_3$, and the completed inverse has dual-to-primal norm at most $E$. Their pairing is therefore bounded by $C_3^2E$. There are three pairings in the symmetric fourth derivative. This gives the kernel estimates after division by 3! and 4!.

The fine-minus-Wilson vertices have orders $3, 2, 1$ by Lemma 11.9. The quadratic order uses $\widehat{\mathsf{H}}(0) = I$ and two curls. Differentiated gauge invariance then gives the higher orders, including nonlinear curvature transport. Wilson has orders $2, 1, 0$: the extra power becomes $\ell^{-1}$ under free interpolation and rescaling. The Wilson vertices and penalty give $a_3, c_4$, independently of the input atom bound; the differences give $a_3', c_4'$.

Increase $\ell$ until $\ell^{-1}a_3', \ell^{-1}c_4' \le 1$. Apply Lemma 12.5 to the cubic difference between $\mathsf{S}_{\mathrm{cl}}$ and the output row action. Subtract the resulting full nonlinear cubic atoms before treating degree four. The remaining quartic difference is transverse, and a second application gives quartic atoms with a fixed bound. This makes the two actions agree through degree four. At $m = N$, the free form acts as the identity on curls, so the same construction matches the Wilson energy through degree four and starts with zero loop atoms. Lemma 13.21 places the remaining top-level difference, including clipping and unhealthy configurations, in the initial activities. □

The common factor $\beta'$ in the penalty and fine leading action cancels from the stationary equation; hence the arrays are independent of $\beta'$. In $\log\det(\beta' M)$, the term $(\dim M)\log\beta'$ is independent of the field and contributes only to the vacuum scalar.

### The determinant comparison

The Gaussian change of coordinates in Lemma 11.7 gives the one-loop action

$$\mathsf{T}(v) = \mathcal{L}_{m+1}(U_v) + \tfrac{1}{2}\log\det M(v) - \log J_{\mathrm{ax}}(v)\,.$$

Here $U_v$ is the stationary fine configuration, $\mathcal{L}_{m+1}$ is the inherited quadratic loop action, $M$ is the axial Hessian without $\beta'$, and $J_{\mathrm{ax}}$ is the Haar chart ratio, normalized to one at zero.

For a quadratic Taylor kernel $Y$, including the Taylor factor $1/2!$, define

$$[t^2]g(t) = \tfrac{1}{2}g''(0), \qquad \alpha_b(Y) = 2\,\mathrm{Re}[t^2]Y_{(2,b),(2,b)}(te_1), \qquad \alpha_{N,m} = \alpha_b(Y^{\mathrm{blk}}_{N,m})\,.$$

The kernel $Y^{\mathrm{blk}}_{N,m}$ is the infinite-volume quadratic kernel of $\mathsf{T}$. Its periodic evaluation has a small finite-period error retained in the activities. The prescribed kinetic kernel has $\alpha_b = 1$. Simplicity makes $\alpha_{N,m}$ independent of $b$, as shown in the proof of Lemma 12.4; by construction it is independent of $p$ and $\beta'$.

Let $\mathsf{J}$ be the left-coordinate Hessian of the joint fine action and block penalty, including the coarse frame. Let $R$ insert infinitesimal gauge transformations and $Q$ the axial slice. For $\mu > 0$, put

$$(d_U\phi)_e = \phi(e^-) - \mathrm{Ad}(U_e)\phi(e^+), \qquad \Delta_U = d_U^* d_U, \qquad \mathfrak{m}_\mu(U) = \mu^2[\mu^2(\Delta_U + \mu^2 I)^{-1}]^9\,.$$

For the rooted frame $\Psi$ of Lemma 11.7, define

$$X = D_{w=0}\big[-\log\big(\Psi(U,h_{\mathrm{fr}})^{-1}\Psi(e^w(U,h_{\mathrm{fr}}))\big)\big]\,.$$

Frame covariance gives $XR = I$. With $P_f$ the projection to fine links, set

$$F_\mu = d_U^* P_f + \mathfrak{m}_\mu X, \qquad \mathsf{C}_\mu = \mathsf{J} + F_\mu^* F_\mu, \qquad \mathsf{G}_\mu = \Delta_U + \mathfrak{m}_\mu\,.$$

The completed Hessian has the following local determinant identity.

**Lemma 12.7** (Gauge completion)**.** *For $v$ in a neighbourhood of zero,*

$$\mathsf{T}(v) = \mathcal{L}_{m+1}(U_v) + \tfrac{1}{2}\log\det\mathsf{C}_\mu(v) - \log\det\mathsf{G}_\mu(v)\,. \tag{12.1}$$

*Proof.* Use Definitions 2.1, 3.1 and Lemmas 11.6–11.7. Gauge invariance and stationarity give $\mathsf{J}R = 0$. Ordering the axial tree by height makes the gauge block triangular with diagonal entries $\pm 1$, while the slice block is the identity; hence $\det(R,Q)^2 = 1$. Since $F_\mu R = \mathsf{G}_\mu$, in these columns the completed form is

$$\begin{pmatrix} \mathsf{G}_\mu & F_\mu Q \\ 0 & I \end{pmatrix}^{*} \begin{pmatrix} I & 0 \\ 0 & Q^*\mathsf{J}Q \end{pmatrix} \begin{pmatrix} \mathsf{G}_\mu & F_\mu Q \\ 0 & I \end{pmatrix}\,.$$

Thus $\det\mathsf{C}_\mu = (\det\mathsf{G}_\mu)^2\det(Q^*\mathsf{J}Q)$. If $s$ changes axial coordinates to left increments at the stationary point, then

$$\det M = (\det Ds)^2\det(Q^*\mathsf{J}Q), \qquad J_{\mathrm{ax}} = |\det Ds|\,.$$

The second equality follows by comparing the two Haar densities. Positivity of $M$, $J_{\mathrm{ax}}$ and $\mathsf{G}_\mu$ near zero permits logarithms and proves (12.1). □

The auxiliary fine determinant used for comparison is

$$\Pi_\mu(U) = \tfrac{1}{2}\log\det(\mathsf{A}(U) + d_U d_U^* + \mathsf{M}_\mu(U)) - \log\det(\Delta_U + \mathfrak{m}_\mu(U))\,.$$

Here $\mathsf{A}$ is the left-coordinate Hessian of the fine leading action and $\mathsf{M}_\mu$ applies $\mathfrak{m}_\mu$ to each link component. Let $Y_\mu$ be its infinite-volume quadratic Taylor kernel. The following comparison controls the coupling coefficient. Enlarge the threshold $u_j$ of Lemma 11.8 to include that of Lemma 11.9; rows 5 and 8 of Section 9 choose odd ratios above it.

**Lemma 12.8** (Block-to-fine coefficient comparison)**.** *For every odd $\ell \geq u_j$ as in rows 5 and 8 of Section 9, put $\mu = \ell^{-1}$. There is a constant $d_{\mathrm{cmp}} \geq 0$, depending only on the fixed stage bounds, such that*

$$|\alpha_{N,m} - \alpha_b(Y_\mu)| \leq d_{\mathrm{cmp}}\,.$$

*The constant is independent of $N, p, \ell$.*

*Proof.* Use Lemmas 12.4, 12.7 and Lemmas 11.6, 11.9–11.8. The input arrays satisfy Lemma 12.4 by $(I_{m+1})$. The full completed Hessian contains a contact term: the second derivative of the predictor multiplied by the first derivative of the penalty. Along the free interpolation, write $\mathsf{a}(\xi)$ for this penalty derivative and $\mathsf{b}(\xi)$ for the mixed fine/frame trace. Lemma 11.10 gives $\mathsf{a}(\xi) = O(|\xi|^2)$ and $\mathsf{b}(\xi) = O(|\xi|)$. The change in the tested entry on removing the contact term consists of $\mathsf{ab}$, its reflected counterpart, and a term quadratic in $\mathsf{a}$, with bounded analytic factors. It is $O(|\xi|^3) + O(|\xi|^4)$ and has zero $\alpha_b$.

For the remaining comparison, let $D$ denote external field differentiation. Differentiate the logarithm and use the resolvent identity:

$$D^2 \log\det A_0 = \mathrm{tr}(A_0^{-1}D^2A_0 - A_0^{-1}DA_0\,A_0^{-1}DA_0), \qquad A_0^{-1} - A_1^{-1} = A_0^{-1}(A_1 - A_0)A_1^{-1}\,.$$

The difference consists of the block penalty, coarse-frame terms and regulator terms. The block penalty factors through the coarse links and the frame terms through the coarse frame variables: their dimensions per block are $4d_{\mathfrak{g}}$ and $d_{\mathfrak{g}}$. Lemma 11.8 bounds the resulting products in the block-scaled operator norms. Its trace is therefore bounded by that dimension times the norm, independently of $\ell$.

For the regulator put $P_\mu = \mu^2(\Delta_U + \mu^2 I)^{-1}$. At zero background and coarse momentum $\xi \in [-\pi, \pi]^4$, its fibre is diagonal in the $\ell^4$ aliases $k_z = (\xi + 2\pi z)/\ell$, where $z$ ranges over centered representatives of $(\mathbb{Z}/\ell\mathbb{Z})^4$. These lie in the fine momentum cube. Since $|q(k)| \geq 2|k|_\infty/\pi$, $|\xi + 2\pi z|_\infty \geq \pi|z|_\infty$ and $\mu = \ell^{-1}$,

$$\frac{\mu^2}{|q(k_z)|^2 + \mu^2} \leq \frac{C}{1 + |z|^2}, \qquad \mathrm{tr}_{\mathrm{fibre}}\,|P_\mu^3| \leq C\sum_{z\in\mathbb{Z}^4}(1 + |z|^2)^{-3} < \infty\,.$$

Here $C$ includes the component count, at most $4d_{\mathfrak{g}}$; the trace is at one coarse momentum, with no sum over coarse sites. The same bounds hold on a fixed smaller complex strip in coarse momentum. In $P_\mu^9 = (P_\mu^3)^3$, two derivatives leave at least one undifferentiated cube. A regulator insertion has the prefactor $\mu^2$, paired with an inverse bounded in block units by $C\mu^{-2}$. Lemma 11.8 supplies these bounds through two field insertions for the inverse and $P_\mu$. Thus the two powers cancel before taking the trace. Put the undifferentiated cube in trace norm and all other factors in operator norm; $|\mathrm{tr}\,XY| \leq \mathrm{tr}\,|X|\,\|Y\|$ gives a uniform bound $C_*$ for the analytic difference kernel on a strip of width $\zeta_* > 0$.

To extract the coefficient, restrict its tested entry to $\xi = te_1$. On a circle $|t| = \zeta_*/2$, the analytic bound gives $|[t^2]g| \le 4C_*/\zeta_*^2$; hence $|\alpha_b(g)| \le 8C_*/\zeta_*^2$. This gives $d_{\mathrm{cmp}}$. The estimate applies after the zero-coefficient contact term has been removed.

Finally, folding the auxiliary operator into coarse momentum does not alter $\alpha_b$. The symbol of the weights $w^{\mathrm{av}}$ in (3.5) vanishes at every nonzero alias to order at least two in each external leg. These aliases contribute $O(|\xi|^4)$. The zero alias in the tested entry is

$$\ell^4 \ell^{-2} Y_\mu(\xi/\ell) + O(|\xi|^3)\,.$$

The factors are the cell volume and $\ell^{-1}$ per interpolated field. Extracting the quadratic coefficient supplies another $\ell^{-2}$, so $\ell^4\ell^{-2}\ell^{-2} = 1$. This proves the comparison. For a period of side $n$, Fourier coefficients at nonzero multiples of $n$ give an error in each quadratic coefficient tested on Fourier basis modes bounded by $C_\ell|\Lambda|e^{-c_\ell n}$. The Fourier basis modes here have unit amplitude at every site, so the bound includes the volume factor; the preparatory expansion of Lemma 13.8 retains it in the activities. □

## The logarithmic coefficient

The vertex comparison in the proof of Lemma 12.6 gives momentum orders $3, 2, 1$ for the quadratic, cubic and quartic differences from Wilson. The input arrays satisfy its hypotheses by $(I_{m+1})$. The resulting logarithmic term is the part of the one-step drift which remains after the block ratio has been chosen.

**Lemma 12.9** (One-loop coefficient)**.** *There is a constant $d_j \ge 0$, depending only on the fixed stage data, such that, for every step $m+1 \to m$ at stage $j$ with odd $\ell_m \ge u_j$ as in Section 9,*

$$|\alpha_{N,m} + \kappa_G \log \ell_m| \le d_j\,.$$

*It is independent of $N, p, \ell_m$ and the running coupling.*

*Proof.* Use Definition 2.1, Lemmas 12.4, 12.8 and the vertex orders just described. Choose a colour basis index $b$, external polarization $e_2$ and external momentum $te_1$. Signed permutations reduce other transverse axis pairs to this one. For nonzero loop momentum $k$, write $\varkappa = |k|$. The leading ghost vertex is $-2ik_2\,\mathrm{ad}(e_b^{\mathfrak{g}})$. The vector vertex has the same orbital term on its four components and the spin term

$$2it(\delta_{i2}\delta_{j1} - \delta_{i1}\delta_{j2})\,\mathrm{ad}(e_b^{\mathfrak{g}})\,.$$

Simplicity and the trace metric give $\mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}(e_b^{\mathfrak{g}})^2) = -C_{\mathfrak{g}}$. The bubble pairs vertices at $te_1$ and $-te_1$. The orbital factor $-2ik_2$ is unchanged when $k \mapsto k + te_1$ and $te_1 \mapsto -te_1$. Put $S_{\mathrm{sp}} = E_{21} - E_{12}$. Since $\mathrm{tr}\, S_{\mathrm{sp}} = 0$, the orbital–spin cross terms vanish; the spin product is $(2itS_{\mathrm{sp}})(-2itS_{\mathrm{sp}}) = 4t^2S_{\mathrm{sp}}^2$, with trace $-8t^2$. The directional traces of the paired vertices are therefore $4(-2ik_2)^2 - 8t^2$ for the vector and $(-2ik_2)^2$ for the ghost. Include the colour trace, divide by $\varkappa^2|k+te_1|^2$ and take the coefficient of $t^2$. In $\frac{1}{2}\log\det \mathsf{C}_\mu - \log\det \mathsf{G}_\mu$, the bubble terms have weights $-1/4$ for the vector and $+1/2$ for the ghost. The extraction of $\alpha_b$ supplies the factor 2. Before taking the coefficient, ghost minus half vector is

$$-\frac{4C_{\mathfrak{g}}(k_2^2 + t^2)}{\varkappa^2|k+te_1|^2}\,.$$

Its quadratic coefficient is

$$h(k) = 4C_{\mathfrak{g}}\left(\frac{k_2^2}{\varkappa^6} - \frac{4k_2^2k_1^2}{\varkappa^8} - \frac{1}{\varkappa^4}\right)\,. \tag{12.2}$$

Expanding the denominator gives $\varkappa^{-4} - 2tk_1\varkappa^{-6} + t^2(4k_1^2\varkappa^{-8} - \varkappa^{-6}) + O(t^3)$; substitution yields (12.2) with its sign.

Let $k_{\mu,b}(k)$ be the integrand of $\alpha_b(Y_\mu)$, obtained with these weights and Taylor factors. Then

$$\alpha_b(Y_\mu) = (2\pi)^{-4} \int_{[-\pi,\pi]^4} \operatorname{Re} k_{\mu,b}(k)\, dk\,.$$

For constants $A_{\mathrm{int}}, D_{\mathrm{int}} \geq 0$ depending only on the fixed fine-stage bounds, uniformly in $\mu$,

$$|k_{\mu,b}(k)| \leq \frac{A_{\mathrm{int}}}{(\varkappa+\mu)^4}, \qquad |k_{\mu,b}(k) - h(k)| \leq D_{\mathrm{int}} \left(\varkappa^{-3} + \mu^2\varkappa^{-6}\right) \quad (\varkappa \geq \mu)\,. \tag{12.3}$$

To obtain them, use $q_a(k) = ik_a + O(\varkappa^2)$ and the vertex orders above. For the free inverse $R_\mu(k)$, differentiation of $R_\mu A_\mu = I$ gives

$$\|R_\mu\| \leq C(\varkappa+\mu)^{-2}, \quad \|\partial R_\mu\| \leq C(\varkappa+\mu)^{-3}, \quad \|\partial^2 R_\mu\| \leq C(\varkappa+\mu)^{-4}\,.$$

Each first-order vertex has order $\varkappa$, with bounded first external derivative. In a bubble, two momentum derivatives on one inverse give $\varkappa^2(\varkappa+\mu)^{-2}(\varkappa+\mu)^{-4} \leq (\varkappa+\mu)^{-4}$; moving derivatives to vertices gives the same bound. On $\varkappa \geq \mu$, a lattice-to-leading replacement gains one power of $\varkappa$, giving $C\varkappa^{-3}$. A regulator insertion contributes at most $C\mu^2\varkappa^{-6}$.

The terms with one second-order vertex are absent from $h$. Their quadratic coefficients have bounds

$$C\left((\varkappa+\mu)^{-2} + \mu^2(\varkappa+\mu)^{-6}\right).$$

The first term comes from one inverse and a bounded second-order vertex derivative; the second retains the regulator factor $\mu^2$ with the differentiated inverse bounds. On the cube, $\varkappa \leq 2\pi$ and $\mu \leq 1$, so this is at most $C'(\varkappa+\mu)^{-4}$. On $\varkappa \geq \mu$, it is at most $C(2\pi\varkappa^{-3} + \mu^2\varkappa^{-6})$. Thus these terms satisfy both inequalities of (12.3). The product rule gives the same bounds for the ninth-power regulator. Summing proves (12.3).

On the spherical shell $\mu < \varkappa \leq 1$, rotational symmetry gives angular means $k_2^2/\varkappa^2 = 1/4$ and $k_2^2k_1^2/\varkappa^4 = 1/24$. For the latter, write the fourth moments as $a = \mathbb{E}\theta_1^4$ and $b = \mathbb{E}\theta_1^2\theta_2^2$ on the unit sphere: rotation by $\pi/4$ gives $a = 3b$, and $(\sum_i \theta_i^2)^2 = 1$ gives $4a + 12b = 1$. Since the sphere has area $2\pi^2$,

$$\begin{aligned}(2\pi)^{-4} \int_{\mu<\varkappa\leq 1} h(k)\, dk &= \frac{8\pi^2 C_{\mathfrak{g}}}{(2\pi)^4}\left(\frac{1}{4} - \frac{1}{6} - 1\right)\log\mu^{-1} \\ &= -\frac{11C_{\mathfrak{g}}}{24\pi^2}\log\mu^{-1} = -\kappa_G \log\mu^{-1}\,.\end{aligned} \tag{12.4}$$

The errors in (12.3) have shell integrals at most $2\pi^2 D_{\mathrm{int}}$ and $\pi^2 D_{\mathrm{int}}$. On $\varkappa \leq \mu$ the bound is $A_{\mathrm{int}}\mu^{-4}$ and the volume is $\pi^2\mu^4/2$; outside $\varkappa \leq 1$ the bound is $A_{\mathrm{int}}$. Hence

$$|\alpha_b(Y_\mu) + \kappa_G \log\mu^{-1}| \leq (2\pi)^{-4}\{A_{\mathrm{int}}\pi^2/2 + A_{\mathrm{int}}(2\pi)^4 + 3\pi^2 D_{\mathrm{int}}\}\,.$$

Combine this bound with Lemma 12.8 and put $\mu = \ell^{-1}$. Their sum gives $d_j$. The computed slope is $\kappa_G = 2b_0$ of Definition 2.1, with its stated trace metric. □

*Proof of Lemmas 12.3 and 12.4.* Use Definitions 2.1, 3.1, 12.1–12.2, the free forms of Lemma 11.5, and Lemmas 12.5, 12.6, 12.9. Start at $m = N$ and descend. The condition $c_{\mathrm{dir},k} \leq \gamma_k$ supplies the positive row factor at every level. Its zero-momentum entries give

$$\widehat{\mathsf{H}}(0)_{\mathfrak{q}\mathfrak{r}} = c_{\mathrm{dir},J_m}\delta_{\mathfrak{q}\mathfrak{r}} + \sum_{\mathfrak{p}} s_{\mathfrak{p}\mathfrak{q}}s_{\mathfrak{p}\mathfrak{r}} = \delta_{\mathfrak{q}\mathfrak{r}}\,.$$

This proves the first cancellation and also forces $c_{\mathrm{dir},J_m} \le 1$ by taking a diagonal entry.

Subtract $\alpha_{N,m}$ times the kinetic kernel from the quadratic kernel of $\mathsf{T}$. The residual is transverse. Its colour bilinear form is scalar: its self-adjoint representing operator commutes with the adjoint action, and its eigenspaces are ideals, so simplicity leaves one eigenvalue. Signed permutations identify the diagonal orientation coefficients. Reflection in an axis belonging to exactly one of two distinct orientations cancels the off-diagonal coefficient. Thus there is one degree-two coefficient, removed by the subtraction. Reflection through the origin removes degree three. The residual link kernel consequently vanishes to order four.

Apply the two-leg part of Lemma 12.5. Removing its two curl symbols leaves a plaquette kernel of order two. If its Fourier coefficients for a fixed orientation-colour pair are $a_{\mathrm{loop}}(z)$, then

$$\sum_z a_{\mathrm{loop}}(z) = 0, \qquad \sum_z z_a a_{\mathrm{loop}}(z) = 0 \quad (a = 1, \dots, 4)\,.$$

Exponential decay permits termwise differentiation. Put the coefficients on coordinate paths to obtain the quadratic atoms and their absolutely convergent moments. Inherited loop atoms already have these moments, hence a link kernel of order four. Pullback by the free interpolation preserves this order. Only the quadratic Taylor kernel of $\mathsf{T}$ is assigned to these atoms; its higher terms enter the preparatory expansion of Lemma 13.8.

For the bounds, contour displacement and the path-length estimate after Lemma 12.5 give a sufficiently small width $w_k^{\mathrm{path}} > 0$. Choose $A_k^{\mathrm{path}}$ to bound the top construction and the two possible incoming stages $k, k+1$. Lemma 12.6 permits this choice before the corresponding ratios, by making the input-dependent terms at most one. Choose the widths small enough for these same three constructions.

The new loop contribution has a weighted bound $a_j^{\mathrm{lp}}\ell^4 + b_j^{\mathrm{lp}} \log \ell + c_j^{\mathrm{lp}}$. Here $\ell^4$ bounds the trace dimension per coarse cell, and the logarithm comes from the subtracted coefficient. For an inherited order-four link kernel, rescaling gives

$$\ell^4 \ell^{-2} O(|\xi/\ell|^4) = O(\ell^{-2}|\xi|^4)\,.$$

Its weighted pullback is therefore bounded by $C_j^{\mathrm{pb}}\ell^{-2}\Lambda_{m+1}^{\mathrm{loop}}$. Choose $\ell$ so that $C_j^{\mathrm{pb}}\ell^{-2} \le 1/4$, then raise the coupling entry threshold until

$$2(a_j^{\mathrm{lp}}\ell^4 + b_j^{\mathrm{lp}} \log \ell + c_j^{\mathrm{lp}}) \le \Lambda_m^{\mathrm{loop}}, \qquad \Lambda_{m+1}^{\mathrm{loop}} \le 2\Lambda_m^{\mathrm{loop}}\,.$$

The sum is at most $\frac{1}{2}\Lambda_m^{\mathrm{loop}} + \frac{1}{4}\cdot 2\Lambda_m^{\mathrm{loop}} = \Lambda_m^{\mathrm{loop}}$. These choices are in the order of Section 9: widths and amplitudes, ratios, then entry thresholds. The first inequality is possible because $2(M_m^{\mathrm{cut}})^{\theta_{J_m}} \to \infty$ as $\widehat{\beta}_m \to \infty$ with $\ell_m, J_m$ fixed. For the second, antitonicity of $\theta_k$, $J_m \le J_{m+1} \le J_m + 1$, and $\theta_k \le 1$ give, once $M_m^{\mathrm{cut}} > 1$,

$$\frac{\Lambda_{m+1}^{\mathrm{loop}}}{\Lambda_m^{\mathrm{loop}}} \le \left(\frac{M_{m+1}^{\mathrm{cut}}}{M_m^{\mathrm{cut}}}\right)^{\theta_{J_m}} \le \frac{M_{m+1}^{\mathrm{cut}}}{M_m^{\mathrm{cut}}} \longrightarrow 1$$

as $\widehat{\beta}_m \to \infty$ with $\ell_m$ fixed. The descent starts with zero loops and the fourth-order Wilson matching of Lemma 12.6. The kernels and paths are chosen once for all periods. Finally $|\mathcal{F}_s| \le 2$ and $|\mathfrak{a}_i^{(k)}(U,x)| \le 2^k |\mathfrak{a}_i^{(k)}|$ give absolute convergence of the action. □

## The scalar reset

Use Definitions 3.2, 3.3–3.4 and Lemmas 12.3–12.4. Write $P_{N,m}^{\mathrm{path}}(\beta, U) = \beta \mathsf{K}_m^{\mathrm{kin}}(U) + \mathcal{L}_m(U)$. Here $\mathsf{K}_m^{\mathrm{kin}}$ contains the clipped term, row squares, and cubic and quartic atoms; $\mathcal{L}_m$ contains the

quadratic atoms, with the hypercubic average in both. For an input curve $\rho'_u$ of Theorem 3.5, put $\mathcal{H}_u = \mathcal{B}^{(m)}_{\beta'_u} \operatorname{dens} \rho'_u$ and $b_u = \beta'_u + \alpha_{N,m}$. Lemma 13.12 supplies regular and forced sums $E^{\mathrm{sum}}_u, \mathcal{F}_u$, independent of $d$, with trial density

$$q_{u,d}(U) = e^{-(b_u+d)\mathsf{K}^{\mathrm{kin}}_m(U) - \mathcal{L}_m(U) - E^{\mathrm{sum}}_u(U)} \mathcal{F}_u(U)\,.$$

The flat-field normalizations give $q_{u,d}(1) = 1$. Fix $r_{\mathrm{res}} = 1$ before $\omega^{\mathrm{RG}}_j \geq d_j + 1$ as in Section 9.

Lemma 13.8 gives a preparatory sum $E^{\mathrm{pre}}_u$ with bound $c_{\mathrm{pre}}\epsilon^{\mathrm{reg}}_m/4$, for a stage constant $c_{\mathrm{pre}} \geq 4$. Lemmas 13.11, 13.12 give the final bound (3.12) and $q_{u,\Delta_u} = \mathcal{H}_u/\mathcal{H}_u(1)$ for some real $\Delta_u$. The estimate below proves $|\Delta_u| \leq 1$; that bound is no input to the identity.

The remaining scalar correction can be read from a single configuration with a definite kinetic energy.

**Lemma 12.10** (Coupling reset)**.** *For each input curve of Theorem 3.5, take the families and real $\Delta_u$ supplied by Lemma 13.12, with no bound on $\Delta_u$ assumed. Then $|\Delta_u| \leq 1$, and $\beta_u = \beta'_u + \alpha_{N,m} + \Delta_u$ is continuous and satisfies*

$$\beta_u = \beta'_u - \kappa_G \log \ell_m + e_u, \qquad |e_u| \leq d_j + 1 \leq \omega^{\mathrm{RG}}_j\,.$$

*The bound is independent of $N, p, u$ and the input curve.*

*Proof.* Use Definitions 2.1, 3.2, Lemmas 12.3, 12.9, and Lemmas 13.11, 13.8–13.13. Recall $t = t_m \leq 1$ from row 7 of Section 9. Put a group element of chord $h_{\mathrm{pr}} = \tau t$ on direction-1 links whose second coordinate is even, and the identity elsewhere. Continuity along a nonconstant one-parameter subgroup gives $\varepsilon_G > 0$ such that every chord in $[0, \varepsilon_G]$ occurs. Choose

$$0 < \tau \leq \min\left(\varepsilon_G, \frac{1}{4}, \frac{c_{\mathrm{dir},J_m}}{512(1 + A^{\mathrm{path}}_{J_m})}\right)\,.$$

Every $(1,2)$ plaquette has chord $h_{\mathrm{pr}}$ and all other plaquettes have chord zero. Thus this configuration $V$ is everywhere healthy. All periods here are even because $4 \mid L_p$. Signed permutations preserve the sum of direct plaquette charges, so the direct contribution to $\mathsf{K}^{\mathrm{kin}}_m(V)$ is at least $|\Lambda| c_{\mathrm{dir},J_m} h^2_{\mathrm{pr}}/2$. Row squares are nonnegative. Since $|\operatorname{proj}_{\mathfrak{g}}(U - U^{-1})/2| \leq \operatorname{ch}(U)$ and adjoint transport is isometric, every slot has norm at most $h_{\mathrm{pr}} \leq 1/4$. Lemma 12.3 bounds the absolute sum of cubic and quartic terms by

$$2|\Lambda| A^{\mathrm{path}}_{J_m} h^3_{\mathrm{pr}} \leq |\Lambda| c_{\mathrm{dir},J_m} h^2_{\mathrm{pr}}/4\,.$$

Each signed image of $V$ has the same plaquette chords with a single nonzero orientation. Hence the bounds apply to every term of the hypercubic average. Consequently

$$K_{\mathrm{pr}} = \mathsf{K}^{\mathrm{kin}}_m(V) \geq |\Lambda| \chi_{\mathrm{pr}} t^2, \qquad \chi_{\mathrm{pr}} = c_{\mathrm{dir},J_m} \tau^2/4 > 0\,.$$

Lemma 13.8 gives $\mathcal{H}_u(1), \mathcal{H}_u(V) > 0$. Put $v_u = |\Lambda|^{-1} \log \mathcal{H}_u(1)$ and $\varepsilon = \epsilon^{\mathrm{reg}}_m$. Here $\mathsf{K}^{\mathrm{kin}}_{2,m}$ is the nonlinear direct and row functional of Lemma 13.8, omitting cubic and quartic atoms. Taking the logarithmic contrast gives

$$\log \frac{\mathcal{H}_u(1)}{\mathcal{H}_u(V)} = \beta'_u \mathsf{K}^{\mathrm{kin}}_m(V) + \alpha_{N,m} \mathsf{K}^{\mathrm{kin}}_{2,m}(V) + \mathcal{L}_m(V) + E^{\mathrm{pre}}_u(V) - E^{\mathrm{pre}}_u(1)\,. \tag{12.5}$$

The pre-reset bound of Lemma 13.8, summed over anchors as in Definition 3.4, bounds each of the last two values by $|\Lambda| c_{\mathrm{pre}} \varepsilon/4$. No vanishing condition at 1 is needed for this preparatory sum.

On everywhere-healthy configurations every curvature slot has norm at most $t$. Lemma 12.3 and $t \le 1$ therefore give

$$|\mathsf{K}_m^{\mathrm{kin}} - \mathsf{K}_{2,m}^{\mathrm{kin}}| \le 2|\Lambda| A_{J_m}^{\mathrm{path}} t^3 .$$

After $\ell$ is fixed, Lemma 12.9 bounds $|\alpha_{N,m}|$ by a stage constant. Raise the coupling entry threshold until $2|\alpha_{N,m}| A_{J_m}^{\mathrm{path}} t^3 \le c_{\mathrm{pre}}\varepsilon/2$. This is possible since $t^3/\varepsilon = \widehat{\beta}_m^{-2/5}(M_m^{\mathrm{cut}})^3 \to 0$. Together with (12.5) this yields

$$\left| b_u K_{\mathrm{pr}} + \mathcal{L}_m(V) - \log \frac{\mathcal{H}_u(1)}{\mathcal{H}_u(V)} \right| \le |\Lambda| c_{\mathrm{pre}} \varepsilon .$$

For the final regular sum, Lemma 13.11 and (3.12) give

$$|E_u^{\mathrm{sum}}(V)| \le \sum_x \sum_{o_i = x} e_i^{\mathrm{maj}} \le |\Lambda| \mathcal{E}_m \le |\Lambda| \varepsilon .$$

Since the activity families are fixed as $d$ varies, the scalar discrepancy is affine:

$$\begin{aligned} f(u,d) &= (b_u + d) K_{\mathrm{pr}} + \mathcal{L}_m(V) + E_u^{\mathrm{sum}}(V) - \log \frac{\mathcal{H}_u(1)}{\mathcal{H}_u(V)} \\ &= f(u,0) + d K_{\mathrm{pr}} . \end{aligned}$$

Since $V$ has no flagged sites, $\mathcal{F}_u(V) = 1$. Evaluating the exact identity of Lemma 13.12 at $V$ gives $f(u, \Delta_u) = 0$. As $K_{\mathrm{pr}} > 0$, this determines

$$\Delta_u = -f(u,0)/K_{\mathrm{pr}}, \qquad |\Delta_u| \le \frac{(c_{\mathrm{pre}} + 1)\epsilon_m^{\mathrm{reg}}}{\chi_{\mathrm{pr}} t_m^2} \le 1 .$$

Increase the entry threshold to obtain the final inequality, using $\epsilon_m^{\mathrm{reg}}/t_m^2 = \widehat{\beta}_m^{-1/10}(M_m^{\mathrm{cut}})^{-2} \to 0$. The block integral and activity sums are continuous in $u$ by Lemma 13.13, so $\Delta_u$ is continuous.

Multiplication by $\mathcal{H}_u(1) = e^{v_u|\Lambda|}$ gives the density identity at $\Delta_u$. Its coupling is $\beta_u = \beta'_u + \alpha_{N,m} + \Delta_u$. Lemma 12.9 and $|\Delta_u| \le 1$ prove the claimed drift. □

## 13 Activities and the inductive step

An exact block integral contains interactions on arbitrarily large supports. We express them as a sum of regular activities and a disjoint-support sum of forced activities. Regular terms are smooth where their supports are healthy; forced terms retain configurations with flagged plaquettes. Their norms serve different purposes: derivatives are required of the first family, whereas the second is estimated uniformly over configurations. The estimates and exact density identity enter Theorem 3.5. The Wilson representation satisfies Theorem 3.5 with the same classes and supplies the initial curve of Theorem 3.6.

The first difficulty is to estimate products under one Gaussian law. Lemmas 11.11 and 11.13 supply the local energy and simultaneous integration bounds. Connected components then separate the ordinary interactions from those attached to flags. Their initial regular bound is too large for iteration. We improve it by subtracting the flat value of a finite connected sum and evaluating that sum on five reflected tori. A fourth difference cancels the boundary terms exactly.

We next remove the affine Hessian of each regular term. Lattice symmetry reduces the sum of these Hessians to one curvature-square coefficient. Changing its health masks creates further local terms, which are included in the forced sum. This gives the exact density before the scalar

estimate of Lemma 12.10. The last part constructs one recursion on continuous curves and proves its weighted analytic estimates. Section 14 uses these estimates for endpoint tuning and the full fixed-volume limit. The Wilson initial condition is proved at the end.

Throughout, the group and trace metric are those of Definition 2.1. Scales, flags, path actions and densities are those of Section 3 and Definitions 3.2, 12.1–12.2. At a step $m+1 \to m$ put $j = J_{m+1}$, $\Lambda = \Lambda_{L_p D_m}$, $t = t_m$, $B = B_m$ and $\sigma = \sigma_{J_m}$. Set $h_X(U) = \mathbf{1}\{X \cap \mathrm{Flag}_m(U) = \varnothing\}$. Constants may depend on the fixed stage data and, when indicated, on the chosen ratio, but are independent of the volume.

## Local terms and their norms

Locality on $X$ means dependence only on links with both endpoints in $X$. Write $\xi_o(y)$ for the centered lift of $y - o$ to $\mathbb{Z}^4$. All lattice distances in this section use the maximum norm. Set $N_R(X) = \{y : \mathrm{dist}(y, X) \le R\}$ and $N_R(o) = N_R(\{o\})$. A link internal to $X$ has both endpoints in $X$. The nonwrapping condition is $10\,\mathrm{dist}(y, o) < L_p D_m$ for every $y \in X$; it permits local affine tests without crossing a choice of lift.

**Definition 13.1** (Regular activities)**.** A regular activity is a Borel real gauge-invariant function $E_i$, local on $X_i$ and anchored at $o_i \in X_i$. It satisfies $E_i(1) = 0$ and is $C^{\mathfrak{q}_{J_m}}$ in exponential left-link increments on healthy configurations, with all ordered derivatives. On nonwrapping supports the Hessian of $A \mapsto E_i(\exp A)$ at zero vanishes on every pair of affine directions

$$A_{(y,a)} = u_a + \sum_c \xi_{o_i}(y)^c v_{ac}, \qquad u_a, v_{ac} \in \mathfrak{g}\,.$$

Here $(\exp A)_e = \exp(A_e)$. This removes the affine quadratic contribution from the error.

For forced terms the inventory specifies exactly the flags inside the support. Additional tests distinguish local alternatives with the same support and inventory.

**Definition 13.2** (Forced activities)**.** For $\varnothing \ne I_i \subset Y_i$, set

$$\mathcal{E}_i = \{U : Y_i \cap \mathrm{Flag}_m(U) = I_i\} \cap \mathcal{E}_i^{\mathrm{extra}}\,.$$

A forced activity is a Borel real gauge-invariant function $K_i$ vanishing off $\mathcal{E}_i$. Both $K_i$ and its Borel gauge-invariant extra test depend only on internal links and flags in $Y_i$. These conditions restrict the forced sum to the prescribed large-field inventory.

We use the ordered derivative seminorm and weighted sums of Definitions 3.3–3.4. Regular sums are bounded at their anchors; forced sums are bounded at every site of their supports. For a function $f$ on $Z$ anchored at $o$, $\|f\|_{m,Z}$ denotes the seminorm of Definition 3.3 with that support and anchor.

Continuity concerns the coefficients at fixed addresses. Health sets remain fixed as the curve parameter varies.

**Definition 13.3** (Continuity of representations)**.** A continuous curve has fixed activity addresses and tests, and continuous coupling and vacuum scalar. At each $u_0$, differences of regular derivatives and forced values admit nonnegative majorants $e_i(u), k_i(u)$ in the sense of Definition 3.3, with

$$\sum_i e^{\sigma\,\mathrm{cov}_B(X_i)} e_i(u) + \sum_i e^{\sigma\,\mathrm{cov}_B(Y_i)} k_i(u) \longrightarrow 0 \quad (u \to u_0)\,.$$

Both sums run over all addresses. This is continuity for coupling tuning, with no derivative of a flag indicator.

At fixed volume these sums control the density. Summing regular bounds over anchors gives $\sum_i \|E_i\|_m \le |\Lambda|\mathcal{E}_m$. Each forced support is nonempty, so summing forced bounds over sites gives $\sum_i \sup|K_i| \le |\Lambda|\mathcal{K}_m$. The absolute sum over all finite selections is at most $\prod_i(1+\sup|K_i|) \le \exp(\sum_i \sup|K_i|)$. Restricting to disjoint supports and exact coverage preserves this bound. Thus the density sums of Definition 3.2 converge. No volume-independent bound on the full density follows.

## 13.1 The local expansion

Fix a compact parameter interval and a continuous input curve $\rho'_u$ in the sense of Definition 13.3. Its prescribed path satisfies Lemmas 12.3–12.4; its regular and forced families satisfy Definitions 13.1–13.2 and the caps of Definitions 3.3–3.4; and $\beta'_u \in \mathfrak{B}_{m+1}$. Write

$$\mathcal{H}_u = \mathcal{B}^{(m)}_{\beta'_u} \operatorname{dens} \rho'_u, \qquad P^{\mathrm{path}}_{N,m}(\beta, U) = \beta \mathsf{K}^{\mathrm{kin}}_m(U) + \mathcal{L}_m(U)\,.$$

The direct and row part of $\mathsf{K}^{\mathrm{kin}}_m$ is denoted by $\mathsf{K}^{\mathrm{kin}}_{2,m}$; it remains nonlinear in the links. The coefficient $\alpha_{N,m}$ is the quadratic coefficient of the determinant comparison in Section 12. It is independent of the curve parameter and the period.

For $a < b$ define the clover curvature at $o$ by

$$C_{ab}(U, o) = \frac{1}{4} \sum_{\varepsilon,\eta=\pm1} \varepsilon\eta \operatorname{proj}_{\mathfrak{g}} \frac{U_{o;\varepsilon a,\eta b} - U^{-1}_{o;\varepsilon a,\eta b}}{2}\,.$$

The loop starts at $o$, takes steps $\varepsilon e_a, \eta e_b, -\varepsilon e_a, -\eta e_b$, and returns to $o$. Thus each term transforms by the adjoint action at $o$ and uses links within distance two. For an affine field as in Definition 13.1, differentiating the four link factors gives

$$\left.\frac{d}{dz}\right|_{z=0} C_{ab}(e^{zA}, o) = v_{ba} - v_{ab}\,. \tag{13.1}$$

Each inverse contributes the negative link field; the four orientations have the same signed curl.

Expand each row square into pairs of its slots. Together with the direct plaquette terms these give a countable family $\lambda$. Let $r_\lambda$ be its unclipped quadratic curvature function, including its coefficient, and $c_\lambda$ the same function with each curvature replaced by the clover curvature of its orientation at the anchor. Let $L_\lambda$ be its actual clipped or health-tested contribution. A direct term has $r_\lambda = c_{\mathrm{dir}} \operatorname{pr}(U_\square)$ and $c_\lambda = c_{\mathrm{dir}}|C_q|^2/2$. A row pair has coefficient $r_{\alpha i} r_{\alpha k}/2$ multiplying its two curvature factors. Include translations and the $H_4$ average, with weight $|H_4|^{-1}$, in the addresses.

Choose $Z_\lambda$ to contain the slot footprints and the radius-$B$ neighborhood of the anchor needed to decide the relaxed health test. Queries outside that neighborhood are automatic once $t\exp((B-2)/R^{\mathrm{rel}}_{J_m}) \ge 2$, since chords are at most two. Lemmas 12.3–12.4 give absolute convergence and

$$\mathsf{K}^{\mathrm{kin}}_m = \sum_\lambda L_\lambda + \mathcal{A}_m, \qquad \mathcal{Q}_m = \sum_\lambda h_{Z_\lambda} r_\lambda,$$
$$\sum_\lambda h_{\{o_\lambda\}} c_\lambda = \frac{1}{2} \sum_o h_{\{o\}} \sum_q |C_q(U, o)|^2\,. \tag{13.2}$$

Here $\mathcal{A}_m$ is the cubic and quartic sum. A signed image in these formulas uses the pulled-back anchor, support and clover; the final equality follows from the coefficient identity of Lemma 12.4 and reindexing orientations. On $Z_\lambda$ healthy, $L_\lambda = r_\lambda$; without flags, $\mathcal{Q}_m = \mathsf{K}^{\mathrm{kin}}_{2,m}$.

The local integral is expanded before any regular term is normalized. Its elementary factors are the remainders of the stationary action, the determinant and covariance ratios, the Haar Jacobian, the incoming activities, and the failures of the fluctuation and chart tests. For a factor $F_a$, write $W_a = F_a - 1$ when its unperturbed value is one. For an incoming forced factor use $W_a = K_a$ directly. The support includes every field, flag and covariance window read by the factor. We join two factors when these complete supports intersect. A molecule is a finite connected selection in this graph, with pairwise disjoint incoming forced supports. It retains the union of those supports and inventories.

For an ordinary molecule $M$ on a healthy output support, its integrated weight and an absolute derivative majorant are

$$z_M(U) = \int \prod_{a\in M} W_a(U,\zeta)\, d\nu(\zeta), \qquad \int g_M\, d\nu \le P_j(\widehat{\beta}_m)(1+s_M)^{d_j} \prod_{a\in M} f_a\,. \tag{13.3}$$

Here $\nu$ is the single sector Gaussian/Haar law of Lemma 11.13, $g_M$ majorizes the product and its required ordered derivatives, and $f_a$ are that lemma's factor bounds. The integer $s_M$ counts the retained local positions and paths. The function $P_j$ has subpower growth after the ratio has been fixed. The formula is used only on compatible sectors; their sum retains the exact allocation indicators. Terms whose reads meet an output flag component remain attached to that component and have a nonempty forced inventory. The ordinary connected integrals are taken on the complement, where their read supports are disjoint from the output flags. Components with empty inventory are used in the regular logarithm on their healthy supports.

We first give the connected identity with its initial estimates. The stronger regular estimate needed for subtraction is obtained in Lemma 13.7.

**Lemma 13.4** (Connected local expansion)**.** *For every input curve above, the block integral has fixed countable local addresses and an absolutely convergent connected expansion on all configurations. Its regular terms have an anchored derivative bound of order $\widehat{\beta}_{m+1}^{-7/16}$ at exponent $\sigma+3$. Its forced terms have a site bound*

$$C_j \exp\{-r_j \beta'_u t_{m+1}^2\}$$

*at exponent $\sigma+2$, with $r_j > 0$. The families are equivariant and continuous at fixed addresses in the corresponding weighted norms.*

*Proof.* Use the kernel and density of Definitions 3.1, 3.2, the path bounds of Lemma 12.3, the local energy of Lemma 11.11, and the simultaneous majorant and disjoint-support factorization of Lemma 11.13. Expand finite products first. Fubini's theorem applies to each term by (13.3). Disjoint complete supports use disjoint Gaussian marginal blocks and Haar variables, so their integrals multiply. This gives a gas of the connected weights $z_M$. Overlapping factors are always integrated together.

Put $\mathfrak{b} = \widehat{\beta}_{m+1}$ and $g = (\beta'_u)^{-1/2}$. For a healthy local chart write its link logarithm as $m_Y + gz$; the mean is $O(t_{m+1})$ by Lemma 11.13. The following bounds include the prescribed finite number of ordered derivatives. A factor $P_j(\mathfrak{b})$ below absorbs fixed powers of $B$, $M_{m+1}^{\mathrm{cut}}$ and the read count

after the ratio is fixed:

| factor | integrated majorant |
|---|---|
| stationary-action remainder | $P_j \beta'_u g^3 \le P'_j \mathfrak{b}^{-1/2}$ |
| centered determinant remainder | $P_j (t_{m+1} + g) \le P'_j \mathfrak{b}^{-1/2}$ |
| Gaussian-density and stationarity residuals | $P_j \mathfrak{b}^{c_j^{\mathrm{res}}} e^{-c_j^{\mathrm{tail}} B}$ |
| Haar-Jacobian remainder | $P_j (t_{m+1} + g)^2$ |
| incoming regular sum | $P_j \epsilon_{m+1}^{\mathrm{reg}}$ |
| incoming forced component | $P_j \epsilon_{m+1}^{\mathrm{for}}$ |
| warning | $M_w \mathfrak{b}^{-1/2}$ |

(13.4)

For the first row, Taylor's formula after the stationary quadratic part starts with $\beta'_u (m_Y + gz)^3$; substituting $m_Y = O(g M_{m+1}^{\mathrm{cut}})$ gives the displayed bound. For the determinant, subtract its flat value and the quadratic term already assigned to $\alpha_{N,m} \mathcal{Q}_m$. The inverse identity bounds the remaining variation by $P_j (t_{m+1} + g)$. The covariance and stationary residuals are exponentially small by Lemma 11.13; their rescaling costs only the displayed fixed power. The chart density has value one and zero first derivative at zero, giving its quadratic bound. Incoming sums retain their own allowances. Chart-test failures, fine-plaquette and mismatch failures, and tag or inventory departures are precisely the warning factors of Lemma 11.13.

Choose its warning exponent $\tau_w = 1/2$, so $\varepsilon_w = \mathfrak{b}^{-1/2}$. This choice is allowed by the simultaneous-event proof there: with at most $C_j B^4$ warnings sharing a coordinate, its Gaussian tail exponent per warning is bounded below by $c_j^{\mathrm{res}} (t_{m+1}/g)^2 / B^4$. Thus

$$\frac{(t_{m+1}/g)^2}{B^4 \log \mathfrak{b}} \longrightarrow \infty .$$

It eventually exceeds the fixed exponent required for $\Pr(E_{w_1} \cap \cdots \cap E_{w_r}) \le \mathfrak{b}^{-r}$, including the coordinate-choice multiplicity. The fixed amplitudes $M_w$ enter $P_j$. All rows of (13.4) are consequently at most a subpower factor times $\mathfrak{b}^{-1/2}$. Choose that factor at most $C_j \mathfrak{b}^{1/32}$, then raise the threshold until $C_j \mathfrak{b}^{-1/32} \le 1$. This gives $\mathfrak{b}^{-7/16}$, with $1/32$ of the power available before the last threshold choice. Path lengths are summed with the exponential bounds of Lemma 12.3. Every fixed polynomial in the number of retained positions is bounded by $C_{j,\eta} e^{\eta s_M}$; choose $\eta$ from the unused support exponent. These choices precede the coupling threshold.

A compulsory component is a connected component of the selected seed neighbourhoods in the allocation sector of Lemma 11.11. Its seed selection $A = (A_f, A_n, A_c)$ consists of the selected fine plaquettes, mismatch edges and coarse flags defined there. Write $\kappa_B$ for that lemma's constant. It retains $\exp(-\kappa_B \beta'_u t_{m+1}^2 |A|)$ before integration. Choose the exponent in the present lemma with $0 < r_j < \kappa_B$. Keep all reads meeting this component in its local factor, estimated in supremum norm with this energy bound. Its frozen Haar integrals are probability integrals. The surrounding ordinary factors have supports disjoint from the output flags, so Lemma 11.13 applies to their common Gaussian integral. The polynomial factor in (13.3), the local sector multiplicity, and a fixed number of derivatives use a fixed part of the retained exponent. A simultaneous set of $r$ warning addresses contributes $\varepsilon_{\mathrm{w}}^{2r}$ to the Gaussian probability in Lemma 11.13; Cauchy–Schwarz leaves $\varepsilon_{\mathrm{w}}^{r}$ after the other factors have been estimated under that same law. Thus a warning still has its own factor in $\prod f_a$. This is necessary when distinct warnings share Gaussian coordinates.

To sum connected molecules, root at a specified site and order the children of a spanning tree by their first support intersection. A support requiring $n$ radius-$B$ balls has at most $(2B+1)^4 n$ sites. If $v_a$ is the resulting factor majorant at a site, the attachment sum is bounded by $n(2B +$

$1)^4 \sup_x \sum_{a:x\in Z_a} v_a$. The coupling threshold makes this quantity at most $\eta n$, with $\eta$ smaller than the reserved exponent. Summing $k$ children contributes the exponential series $1/k!$; iteration gives the rooted bound by $v_a e^{\eta n_a}$. This proves absolute summability, including all sectors and all differentiated terms. The same argument with a prescribed forced seed retains one factor $\exp(-r_j \beta'_u t_{m+1}^2)$ and gives the claimed site bound. Increasing the fine support exponent before choosing the ratio leaves the output exponents $\sigma+3$ and $\sigma+2$.

Write $\Xi(z) = \sum_{\mathcal{M}} \prod_{M\in\mathcal{M}} z_M$, where $\mathcal{M}$ runs over finite selections with disjoint supports, including the empty selection. This is the ordinary hard-core gas. Its logarithm has the connected coefficients

$$\log \Xi(z) = \sum_{n\geq 1} \frac{1}{n!} \sum_{M_1,\dots,M_n} \phi^T(M_1,\dots,M_n) \prod_{h=1}^{n} z_{M_h}, \qquad \phi^T = \sum_{\Gamma \text{ connected}} \prod_{\{h,k\}\in\Gamma} [-\mathbf{1}_{Z_{M_h}\cap Z_{M_k}\neq\varnothing}]\,. \tag{13.5}$$

The graph in this formula has vertex set $\{1,\dots,n\}$, and the one-vertex coefficient is one. Apply [30, Theorem 1, equations (3)–(4)] to the counting measure with masses $z_M$ and interaction $\zeta(M,M') = -\mathbf{1}_{Z_M\cap Z_{M'}\neq\varnothing}$. Its hypotheses are $|1+\zeta|\leq 1$, finite total variation, finite exponential mass, and

$$\sum_{M':Z_{M'}\cap Z_M\neq\varnothing} |z_{M'}| e^{\mathfrak{w}(M')} \leq \mathfrak{w}(M), \qquad \mathfrak{w}(M) = \eta \operatorname{cov}_B(Z_M)\,.$$

The rooted estimate just proved verifies the last condition. At fixed volume the site sum gives finite total exponential mass. Equation (13.5) and its rooted estimate therefore apply. Keep the components joined to forced molecules as coefficients of the exact-cover sum; take the logarithm only of the remaining ordinary gas. Expanding and regrouping finite selections proves this separation. Absolute convergence then proves the identity for the full integral. The deterministic part is exactly $\beta'_u \mathsf{K}_m^{\mathrm{kin}} + \mathcal{L}_m + \alpha_{N,m}\mathcal{Q}_m$: the stationary terms and one-loop quadratic term are those of Lemmas 12.6–12.9, with the masks in (13.2) retained.

For two inputs at fixed addresses, telescope each product with one difference factor. Lemma 11.13 bounds that marked factor by its own difference norm, including changes of the mean and covariance. The tree estimate uses the same strict attachment margin for the unmarked factors. It bounds the weighted sum of the differences by a constant times the input weighted discrepancy plus the coupling discrepancy. The constant may depend on this fixed volume. For a merely continuous curve this proves continuity locally; compactness and a finite covering give fixed addresses over the whole interval. Translation and signed permutations transport complete read supports, factors and sectors. Their finite average and the induced reindexing of connected selections prove the claimed equivariance. □

The terms linear in an incoming regular activity must be treated separately from products of small factors. Their supports can be large, so they are not replaced by auxiliary tori. The function $E_i$ and its derivative norm in this paragraph are at level $m+1$. For such an address $i$, let $T_i$ be its contribution to the negative connected logarithm. The local Gaussian substitution of Lemma 11.13, restricted to the windows incident to $X_i$, reconstructs a fine configuration $U_i^0(V,z)$. With the same inner chart protector $\Gamma_i$, define

$$P_i(V) = \frac{1}{|\mathcal{T}_i|} \sum_{\theta\in\mathcal{T}_i} \operatorname{Ext}_{Y_i} \int \Gamma_{i,\theta}(z) E_i\big(U_{i,\theta}^0(V,z)\big)\, d\gamma_{C_{i,\theta}(V)}(z), \qquad R_i = T_i - P_i\,.$$

Here $\mathcal{T}_i$ is the finite set of rooted frames and tags in that local substitution. The operator $\operatorname{Ext}_{Y_i}$ extends the function from the healthy domain by its local Borel representative; on that domain it

changes no value. The coarse support $Y_i$ contains the projected fine support and every incident window. This principal Gaussian average retains the incoming function itself.

**Lemma 13.5** (Inherited regular terms)**.** *The functions $P_i, R_i$ are regular before flat-value and Hessian subtraction, on the same fixed coarse addresses as $T_i$. At exponent $\sigma + 3$ their anchored derivative sums satisfy*

$$\mathcal{E}(P) \le \frac{V_j}{\ell_m} \epsilon^{\mathrm{reg}}_{m+1}, \qquad \mathcal{E}(R) \le \omega^{\mathrm{inh}}_{j,\ell_m}(\widehat{\beta}_m)\epsilon^{\mathrm{reg}}_{m+1},$$

*where $V_j$ is fixed before the ratio and $\omega^{\mathrm{inh}}_{j,\ell}$ has scale rate $-1/2$. The identity $T_i = P_i + R_i$ and the corresponding weighted comparisons hold on the entire input curve.*

*Proof.* Use Definitions 3.3, 3.4, 3.2, Lemmas 11.13, 12.3–12.4 and 13.4. For the principal average use the same normalized directions as in the incoming norm. The linear background lifts a unit coarse curl to fine curvature of size $O(\ell_m^{-2})$. Replace the background on $X_i$ by its affine part at the anchor. In the Taylor formula for $E_i$ the constant and first derivative vanish; its Hessian on two such affine fields vanishes by Definition 13.1. Every surviving background term therefore has either three curvature factors or one spatial remainder. The third-order integral remainder has the product of its three normalized direction sizes. The spatial remainder is bounded by the first difference of the prediction kernel and supplies one further inverse ratio. In the normalized curvature directions, the linear prediction $\mathsf{P}$ and its affine approximation $\mathsf{P}_{\mathrm{aff}}$ satisfy

$$\|\mathsf{P}\| \le C_j \ell_m^{-2}, \qquad \|\mathsf{P} - \mathsf{P}_{\mathrm{aff}}\| \le C_j \ell_m^{-3}\,.$$

Indeed each discrete derivative of the rescaled prediction kernel contributes $\ell_m^{-1}$, and its exponential row sum is bounded by Lemma 11.13. For the Hessian remainder use one difference and one undifferenced direction; for the cubic remainder use three directions. Thus their factors before the anchor sum are

$$(\ell_m^{-2})^3 = \ell_m^{-6}, \qquad \ell_m^{-3}\ell_m^{-2} = \ell_m^{-5}\,.$$

If the fine support wraps, or is too large for the affine replacement, its connected cover contains at least $c\ell_m$ fine radius-$B_{m+1}$ balls: its diameter is at least a fixed fraction of the fine period or of the enlarged coarse neighborhood. Reserve half of its incoming support exponent. The resulting $e^{-c\sigma_{J_{m+1}}\ell_m}$ bounds the polynomial anchor count and yields the same inverse-ratio estimate without an affine test. For the remaining supports there are $\ell_m^4$ fine anchors over a coarse anchor. Three curvature factors give $\ell_m^{-6}$, and the spatial-remainder terms give at least $\ell_m^{-5}$ after the kernel sums. Multiplying by the anchor count gives at most $C_j\ell_m^{-1}$. Powers of the fine support diameter from the Taylor formula and derivative placements are bounded by its covering weight. The support compression gives

$$(\sigma+3)\operatorname{cov}_{B_m}(Y_i) \le C_j + \tfrac{1}{2}\sigma_{J_{m+1}} \operatorname{cov}_{B_{m+1}}(X_i)\,;$$

the remaining half of the incoming exponent bounds those diameter powers and the possible coarse anchors. This is why the fine support exponent and its compression condition are chosen before the ratio. Gaussian terms in the average have additional factors $g/t_{m+1}$; their fixed moments and the chart tails tend to zero uniformly after the ratio is fixed. Increase the coupling threshold until their total is bounded by another $C_j/\ell_m$ times the incoming sum. This proves the principal estimate, with $V_j$ independent of the eventual ratio.

For $\operatorname{cov}_{B_m}(Y_i) \le B_m$, compare the actual local integral with this principal Gaussian average before integrating. Subtract their means, covariances and transported kernels using the inverse

difference identity, and telescope the remaining factor product with one marked difference. The inverse difference identity $C^{-1} - \widetilde{C}^{-1} = C^{-1}(\widetilde{C} - C)\widetilde{C}^{-1}$ puts a reconstruction difference in one factor. Taylor's formula for the link exponential makes that factor at most $P_j(\mathfrak{b})(t_{m+1}+g)$ in each of the requested normalized derivatives; stationarity and window residuals contribute $P_j(\mathfrak{b})g^{-c_j}e^{-c'_jB}$. Consequently their common bound is

$$P_j(\mathfrak{b})\{t_{m+1} + g + g^{-c_j}e^{-c'_jB}\} = \mathfrak{b}^{-1/2+o(1)}, \qquad \mathfrak{b} = \widehat{\beta}_{m+1}\,.$$

A second occurrence of the incoming activity has bound $C_j(\epsilon^{\mathrm{reg}}_{m+1})^2$, which is also bounded by a scale-rate-$-1/2$ factor times $\epsilon^{\mathrm{reg}}_{m+1}$. In every term of the difference, one of these factors is retained before the common Gaussian majorant is applied. Lemma 11.13 bounds its remaining moments by a fixed polynomial in the read count and $B$. Summing with the unused incoming support exponent gives a function $\omega^{\mathrm{inh}}_{j,\ell}$ of rate $-1/2$. This proves the remainder estimate in its weighted norm, rather than only a pointwise comparison of integrals. For the remaining supports use the unrefined bound for $T_i$ from Lemma 13.4 and the same Gaussian majorant for $P_i$. Support compression implies that the incoming covering number is at least $cB_m - C_j$. Retaining a further fixed part of its exponent gives an extra $e^{-cB_m}$ in the weighted sum, for some $c > 0$ fixed before the coupling threshold. Since $B_m \geq (\log \widehat{\beta}_m)^2$, this factor, times any fixed polynomial in $B_m$, is bounded by $\widehat{\beta}_m^{-1/2}$ after the threshold is raised. This supplies the stated remainder bound on large supports without extending the local comparison beyond its domain.

Local Gaussian marginals read only $Y_i$. A gauge transformation acts orthogonally on their colour coordinates and leaves $\Gamma_i$ invariant; changing Gaussian variables proves gauge invariance of $P_i$. The frame average transports the support and anchor under lattice symmetries. The local extension preserves these identities, measurability and the healthy derivatives. The same properties hold for $R_i$ by subtraction. For a parameter difference use that incoming difference in the displayed integral and in the marked comparison. The covariance score and protected-coordinate change are those of Lemma 11.13, so the same weighted estimates give continuity at fixed addresses. Finally choose the ratio so $V_j/\ell_m$ is less than the prescribed pre-ratio fraction, and then the threshold so $\omega^{\mathrm{inh}}_{j,\ell_m}$ is less than that fraction. □

The initial regular estimate does not fit $\epsilon^{\mathrm{reg}}_m = \widehat{\beta}_m^{-11/10}$. The following localization gives the additional power. We describe the finite-period cancellation because an estimate of each separate local term would miss it.

Write $f^0_a$ for a regular summand of the negative logarithm in (13.5), including its coefficient. Keep the inherited terms of Lemma 13.5 separately. Select finitely many of the remaining terms not estimated by (13.8) below, and all short path terms selected at this truncation. Give an ordinary term the shares $p_{xa} = \mathbf{1}_{X_a}(x)/|X_a|$, and a path term the share $\mathbf{1}_{\{o_a\}}(x)$. Put $F_x = \sum_{a \text{ selected}} p_{xa} f^0_a$. The remaining terms form $f^{\mathrm{tail}}$. Let $R$ contain every complete read support of this finite sum relative to a site receiving a nonzero share. Its radius is a stage multiple of $B$. Set $w = 4R + 100(1 + B)$, $e = w + 5$, and let $\Lambda_i$ have side $2(w + i)$, $0 \leq i \leq 4$. A configuration on $\Lambda$ gives a configuration $V_{i,x}$ on $\Lambda_i$ by repeated reflection of its cube about $x$. The direction of a reflected link is reversed when necessary. Put

$$\lambda_i = \frac{(-1)^i \binom{4}{i}}{2^4\, 4!}, \qquad G_x(V) = \sum_{i=0}^{4} \lambda_i \sum_{y \in \Lambda_i} F^{(i)}_y(V_{i,x})\,. \tag{13.6}$$

Here $F^{(i)}$ is the same finite local formula on the auxiliary torus, with its value at 1 subtracted at each site. This operation recovers a local function from bounds on the total auxiliary action.

**Lemma 13.6** (Finite-period localization)**.** *For a finite local connected sum with complete read radius $R$, the function $G_x$ in (13.6) is local on $N_e(x)$. Its redistribution has the exact form*

$$\sum_a h_{X_a} f_a^0 = c|\Lambda| + \sum_a h_{X_a} f_a^{\mathrm{tail}} + \sum_x h_{N_e(x)} G_x + \sum_x L_x \,.$$

*Each $L_x$ reads links and flags in $N_{2e}(x)$ and vanishes when that set is healthy. The estimates in the proof give $G_x$ the additional regular smallness.*

*Proof.* Use Definitions 3.1–3.2, Lemmas 12.3–12.9, and 13.4. Give each connected summand equal shares at the sites of its complete support; give each path term to its anchor. These shares sum to one. Let $F_x$ be the resulting finite sum at $x$, and let $f^{\mathrm{tail}}$ be the terms not selected. Define

$$L_x = \sum_{a \text{ selected}} p_{xa} h_{X_a} f_a^0 - F_x(1) - \sum_{i=0}^{4} \lambda_i \sum_{y \in \Lambda_i} h_{N_e(x-\widetilde{y})} F_y^{(i)}(V_{i,x-\widetilde{y}}), \qquad c = F_0(1)\,. \tag{13.7}$$

The coefficients $p_{xa}$ are the shares, and $\widetilde{y}$ is the integer representative used in the reflection. Summing over $x$ and changing $x$ to $x - \widetilde{y}$ gives exactly the asserted identity. Translation symmetry makes $F_x(1)$ independent of $x$.

We verify the vanishing of $L_x$ on a healthy $N_{2e}(x)$. The complete read supports then lie in that set, every mask in (13.7) is one, and every reflected configuration occurring there is healthy. Classify a point of $\Lambda_i$ by its distance, through $R$, from each reflection face. In the $y$-summand of (13.7) the reflection cube is centred at $x - \widetilde{y}$, so its site $y$ corresponds to $x$. For a fixed class the local formula is therefore independent of $i$. In one coordinate the number of points in an interior class is $2(w - 2R + 1) + 2i$; for a specified boundary class it is a fixed integer. Consequently each class has a multiplicity polynomial in $i$ of degree at most four. Its degree is four only when all four coordinates are interior. The identity

$$\sum_{i=0}^{4} (-1)^i \binom{4}{i} i^k = 0 \quad (0 \le k < 4), \qquad \sum_{i=0}^{4} (-1)^i \binom{4}{i} i^4 = 4!$$

cancels every boundary class and retains the interior class with coefficient $2^4 4!/(2^4 4!) = 1$. Locality and gauge invariance identify its value with $F_x(V) - F_x(1)$. This proves $L_x = 0$. It also proves its stated read support: the largest displacement is at most $w + 4 < e$, and each mask reads a further radius $e$.

Here are the quantitative estimates on the sum before this localization. Put $\mathfrak{b} = \widehat{\beta}_{m+1}$ and $\mathfrak{h} = \mathfrak{b}^{3/8}$. In the convergent connected sums multiply every ordinary molecule weight by $\mathfrak{h}$. The rooted attachment parameter still tends to zero, since $\mathfrak{h}\mathfrak{b}^{-7/16} = \mathfrak{b}^{-1/16}$. A term with at least three ordinary factors gains $\mathfrak{h}^{-3}$ upon restoring its original weights. A term with a retained large-field overlap has the same gain because its suppression is at least $\mathfrak{h}^{-3}$. The sum after multiplication has size at most a subpower factor times $\mathfrak{h}\mathfrak{b}^{-7/16}$. Thus the terms in these two classes have bound

$$P_j(\mathfrak{b})\mathfrak{h}^{-3}(\mathfrak{h}\mathfrak{b}^{-7/16}) = P_j(\mathfrak{b})\mathfrak{b}^{-19/16} \le \mathfrak{b}^{-23/20} \tag{13.8}$$

after the coupling threshold is raised. The reserve is $19/16 - 23/20 = 3/80$. This estimate also bounds fixed ordered derivatives: their placement polynomial was included before the rooted sum.

First retain the inherited terms using Lemma 13.5; their principal sum is made smaller than a prescribed multiple of $H_j$ by choosing the ratio, and their comparison remainder by raising the threshold. Among all other connected terms retain also those whose total support size reaches the

truncation radius: their support decay is at most $\mathfrak{h}^{-3}$ after the threshold is raised, so the estimate in (13.8) applies to them. The remaining terms have at most two elementary factors, no seed component and bounded complete read radius. They are the finite connected contributions and path terms of length at most $B$ to which the reflected-period construction is applied, together with a summable truncation tail. Truncation on the countable connected addresses makes the latter arbitrarily small in the weighted norm at fixed volume. Long paths have an additional factor $e^{-w_{J_m}^{\mathrm{path}} B/4}$ by Lemma 12.3; it converts their $\mathfrak{b}^{-7/16}$ bound to $\mathfrak{b}^{-23/20}$. For the finite sum use the stationary integral on each $\Lambda_i$. Subtract its flat value and the classical cubic, quartic and quadratic one-loop terms of Lemmas 12.6–12.9. Differentiation of its logarithm gives connected expectations under the common Gaussian law. For the terms not involving incoming errors, write the normalized local integral as $\mathbb{E}\exp(V_3 + V_4 + V_{\geq 5})$, after the stationary action and its Gaussian determinant have been extracted. The subscripts refer to the Taylor degree in the link chart. Gaussian parity removes a lone odd cubic term. The fourth-order stationary terms are already in the classical path action, and the field-dependent quadratic part of the determinant is already in $\alpha\mathcal{Q}_m$ and the quadratic loop atoms. Subtracting the value at 1 removes the remaining vacuum terms. Expanding $\log \mathbb{E}e^V$ through two factors gives $\mathbb{E}V + \frac{1}{2}(\mathbb{E}V^2 - (\mathbb{E}V)^2)$; its remaining terms have at least three factors. In normalized fluctuation coordinates each nonquadratic factor starts at order $\mathfrak{b}^{-1/2}$. The common Gaussian majorant bounds the last terms by $P_j(\mathfrak{b})\mathfrak{b}^{-3/2}$. In the one- and two-factor terms that remain after the stated cancellations, Taylor's integral remainder and the healthy field bound give

$$P_j(\mathfrak{b})\{\mathfrak{b}^{-3/2} + \mathfrak{b}\, t_m^5 + \mathfrak{b}^{-1} t_m + t_m^3 + e^{-c_j B}\} \leq \mathfrak{b}^{-23/20}\,.$$

Every radius and chart loss is included in the subpower factor. Each of the four power terms has exponent at most $-3/2$ before that factor; the path exponential decreases faster than any fixed power. Terms involving incoming errors are divided into the finite sum and its tail above; those with at least three ordinary factors or a large-field overlap obey (13.8). Lemma 11.13 bounds these terms simultaneously, so the derivative bound for the centered total auxiliary sum is

$$C_j \mathfrak{b}^{-23/20} + 2|\Lambda_i|\{C_j \mathfrak{b}^{-23/20} + \varepsilon_{\mathrm{tr}}\}\,. \tag{13.9}$$

Here $\varepsilon_{\mathrm{tr}} > 0$ is the chosen truncation error. The first term is the centered local integral; the two volume terms bound the omitted regular sum at the field and at 1. Approaching a boundary point of the healthy domain by radial contraction preserves these estimates, so they hold on the closed domain used in Definition 3.3.

A direction of that definition, restricted to $N_e(x)$ and reflected, is bounded by $2t(1+e)^{\varpi_{J_m}}$. Rescaling each of at most $\mathfrak{q}_{J_m}$ directions in (13.9) therefore gives

$$\|G_x\|_{m,N_e(x)} \leq 2^{\mathfrak{q}_{J_m}} (1+e)^{\varpi_{J_m}\mathfrak{q}_{J_m}} \sum_{i=0}^{4} |\lambda_i|[C_j \mathfrak{b}^{-23/20} + 2|\Lambda_i|(C_j \mathfrak{b}^{-23/20} + \varepsilon_{\mathrm{tr}})]\,.$$

All the factors outside the negative powers are fixed powers of $1+B$. Also $\mathrm{cov}_B(N_e(x))$ is bounded by a stage constant. The regular bound divided by $\epsilon_m^{\mathrm{reg}}$ tends to zero, since $23/20 > 11/10$. Finally, (13.7) and the uncentered $\mathfrak{b}^{-7/16}$ bound give $\sup|L_x| \leq P_j(\mathfrak{b})\mathfrak{b}^{-7/16}$, with the same polynomial read radius. These logarithmic corrections tend to zero in the site norm at exponent $\sigma+2$ and vanish on healthy supports. This proves all the estimates claimed for the localization. □

To specify the small coefficient, let $C_j^{\mathrm{sub}}$, $C_j^{\mathrm{low}}$ and $C_j^{\mathrm{high}}$ be the operator bounds for local subtraction, low path replacement and the part of high path replacement proportional to $H_j$. These

are the three linear operations in the following subsections; they are fixed from the output derivative norm and path bounds. Put $C_j^0 = C_j^{\mathrm{sub}} + C_j^{\mathrm{low}} + C_j^{\mathrm{high}}$. These constants use only derivative orders, direction powers, support exponents, and the output path bounds. For a prescribed $0 < \theta \le 1$, choose before the ratios

$$c_j = \frac{1}{64(1 + C_j^0)}, \qquad H_j = \theta c_j \epsilon_m^{\mathrm{reg}}.$$

The next assertion concerns the very same block integral and the masks of (13.2).

**Lemma 13.7** (Preparatory activities)**.** *For every input curve above, fixed local families $f_{i,u}, k_{i,u}$ and a continuous scalar $\tau_u$ satisfy (13.10)–(13.12). They are equivariant and continuous in the weighted norms with exponents $\sigma + 3$ and $\sigma + 1$, respectively. The regular functions have the regularity of Definition 13.1 before flat-value and Hessian subtraction; the forced functions satisfy Definition 13.2. Restricting the interval preserves the identities and estimates.*

$$\mathcal{H}_u(U) = e^{\tau_u|\Lambda| - \beta_u' \mathsf{K}_m^{\mathrm{kin}}(U) - \mathcal{L}_m(U) - \alpha_{N,m}\mathcal{Q}_m(U) - \sum_i h_{X_i}(U) f_{i,u}(U)} \mathcal{F}_u^0(U), \tag{13.10}$$

$$\sup_x \sum_{o_i = x} e^{(\sigma+3)\operatorname{cov}_B(X_i)} \mathfrak{e}_{i,u} \le H_j, \qquad \mathfrak{e}_{i,u} \ge \|f_{i,u}\|_{m,X_i}, \tag{13.11}$$

$$\sup_x \sum_{i : x \in Y_i} e^{(\sigma+1)\operatorname{cov}_B(Y_i)} \mathfrak{k}_{i,u} \le e^{-\kappa_j^{\mathrm{pre}} (M_{m+1}^{\mathrm{cut}})^2}, \qquad \mathfrak{k}_{i,u} \ge \sup_U |k_{i,u}(U)|. \tag{13.12}$$

Here $\kappa_j^{\mathrm{pre}} > 0$ is fixed after the ratio, and $\mathcal{F}_u^0$ is the disjoint-support exact-cover sum of Definition 3.2, formed from $k_{i,u}$.

*Proof.* Use Definitions 3.1–3.2, Lemmas 11.11, 11.13, 12.3–12.9, 13.4 and 13.6. First choose the ratio so that the inherited principal terms of Lemma 13.5 use at most $H_j/32$, and then raise the threshold so that their comparison remainders use at most $H_j/32$. Bound the terms retained in (13.8) and the long paths together by $H_j/16$. The localization bound is $o(H_j)$ with the stronger support exponent. This fixes a coupling threshold independent of the period.

Next choose one finite truncation on the whole parameter interval. For each parameter, summability gives a tail smaller than a prescribed $\varepsilon_{\mathrm{tr}}$. Weighted continuity gives a neighborhood on which that same tail is smaller than $2\varepsilon_{\mathrm{tr}}$. Finitely many such neighborhoods cover the interval; take the union of their finite address sets and include their symmetry images. Increasing the truncation preserves the tail bound. Choose the tail bound to be at most $H_j/16$. The regular terms outside the finite sum then have total at most $(1/16 + 2/16)H_j$. Choose $\varepsilon_{\mathrm{tr}}$ so that the two localization errors in (13.9), including the direction and support losses, have total at most $12H_j/16$. The resulting bound is $15H_j/16 < H_j$.

The change in the scalar is $\tau_u^{\mathrm{old}} - F_0(1)$, with the sign fixed by (13.7). Multiply the preliminary forced sum by $\exp(-\sum_x L_x)$. Expand each $e^{-L_x}$ about one and collect intersecting supports, as in the first paragraph of the proof of Lemma 13.10. The logarithmic factors vanish unless their supports contain flags, so each resulting connected component still contains a forced term. Their site bound at exponent $\sigma + 2$ is a subpower factor times $\mathfrak{b}^{-7/16}$, whereas each forced component retains $C_j e^{-r_j \beta_u' t_{m+1}^2}$. Choose a smaller positive $r_j' < r_j$. The site attachment parameter tends to zero and the remaining exponential gives

$$C_j' e^{-r_j' \beta_u' t_{m+1}^2} \le e^{-\kappa_j^{\mathrm{pre}} (M_{m+1}^{\mathrm{cut}})^2}$$

for some $\kappa_j^{\mathrm{pre}} > 0$, since $\beta'_u \geq \widehat{\beta}_{m+1}/2$. One extra support power is used in this collection, leaving $\sigma + 1$. The rooted-tree induction written in Lemma 13.10 proves the asserted site bound and absolute convergence. No scalar depends on the output configuration, and all identities hold also when a chord is exactly at a threshold.

For continuity, the truncation and all reflected tori are now fixed. Finite linear combinations preserve the weighted comparison from Lemma 13.4. A difference of forced products has one marked difference factor; the rooted sum has the same strict attachment margin. Dominated convergence proves the weighted comparison after the collection. Flat evaluation gives continuity of the scalar. The support enlargement, inventories and conjunctions of tests are fixed. The same constructions commute with translations and signed permutations by reindexing. Every assertion is pointwise in the interval parameter or a convergence assertion within the interval, so restriction to a closed subinterval preserves it. □

The preparatory regular sum is $E_u^{\mathrm{pre}} = \sum_i f_{i,u}$ on the healthy domain. Its weighted bound is the anchored sum of Definition 3.4, before flat-value and Hessian subtraction. The next identity supplies the scalar reset.

**Lemma 13.8** (Pre-reset expansion)**.** *For every input curve above and every $u$, there is a preparatory regular family with sum $E_u^{\mathrm{pre}}$ and weighted bound $\epsilon_m^{\mathrm{reg}}$ such that, when $\mathrm{Flag}_m(U) = \varnothing$,*

$$\mathcal{H}_u(U) = \exp\{\tau_u|\Lambda| - \beta'_u \mathsf{K}_m^{\mathrm{kin}}(U) - \alpha_{N,m}\mathsf{K}_{2,m}^{\mathrm{kin}}(U) - \mathcal{L}_m(U) - E_u^{\mathrm{pre}}(U)\}\,.$$

*In particular $\mathcal{H}_u(1), \mathcal{H}_u(U) > 0$ on this set.*

*Proof.* Use Definition 3.2, Lemmas 12.3–12.4, and (13.10)–(13.11). Set $E_u^{\mathrm{pre}} = \sum_i f_{i,u}$ on configurations without flags. Every health mask is one, and nonempty forced inventories give $\mathcal{F}_u^0 = 1$. Equation (13.2) gives $\mathcal{Q}_m = \mathsf{K}_{2,m}^{\mathrm{kin}}$. Dropping the extra three support powers gives the stated bound, since $\theta c_j \leq 1/64$. Absolute convergence makes the exponent finite and real, proving positivity, including at 1. □

## 13.2 Local quadratic subtraction

The subtraction is local and uses no bound on the eventual coupling correction. We use the clover curvature of (13.1) for the local quadratic subtraction.

For a preliminary $f$ on $X$ anchored at $o$, put $Z = X \cup N_2(o)$. For $q = (a, b)$ and $T \in \mathfrak{g}$, set $A_{(y,c)}^{q,T} = \mathbf{1}_{c=b}\xi_o(y)^a T$, and fix a unit vector $T_* \in \mathfrak{g}$. Define

$$\begin{aligned} q_f(q, r) &= D^2[f(\exp\cdot)](0)[A^{q,T_*}, A^{r,T_*}], \\ Q_f(U) &= \frac{1}{2}\sum_{q,r} q_f(q, r)\langle C_q(U, o), C_r(U, o)\rangle_{\mathfrak{g}}\,. \end{aligned}$$

Set $q_f = 0$ when $Z$ wraps. The matrix extracts the response to constant curls; its color independence is proved below. Define the regular remainder and the discrepancy of masks by

$$\begin{aligned} R_f &= f - f(1) - Q_f, \\ \zeta_f &= \mathbf{1}_{Z \text{ unhealthy}}\{f(1) + h_{\{o\}}Q_f - h_X f\}\,. \end{aligned} \tag{13.13}$$

The following identity is checked before any activities are summed.

**Lemma 13.9** (Local subtraction)**.** *For every preliminary regular $f$,*

$$h_X f = f(1) + h_{\{o\}} Q_f - \zeta_f + h_Z R_f \,.$$

*The function $R_f$ is regular on $Z$, vanishes at 1, and has zero affine Hessian when $Z$ is nonwrapping. The Borel gauge-invariant function $\zeta_f$ depends only on internal links and flags in $Z$ and vanishes when $Z$ is healthy.*

*Proof.* Use Definitions 2.1, 3.3, 13.1 and the clover formula. The first derivative of $f$ at the flat field vanishes: constant gauge transformations make its coefficient on each internal link an adjoint-invariant vector of $\mathfrak{g}$, hence a central vector, which is zero. Differentiating local gauge invariance again makes its Hessian annihilate an infinitesimal gradient in either argument. On a nonwrapping support an affine field with zero curl is a local gradient. Its constant and symmetric linear parts are differences of a quadratic site field, with a constant adjustment for the discrete difference. Multiply this site field by a periodic cutoff equal to one on $Z$. The internal links read by $f$ are unchanged. The second differentiation of gauge invariance uses $Df(1) = 0$, as proved above, so no second derivative of the gauge orbit remains. Hence the affine Hessian depends only on the constant curls.

For each pair of orientations its color form is an invariant bilinear form. Represent it by an operator on $\mathfrak{g}$ using the trace inner product. The operator commutes with the adjoint action. Its self-adjoint part has ideal eigenspaces, so simplicity makes it scalar. Its skew-adjoint part $J$ commutes with infinitesimal adjoint actions, giving

$$[JX, Y] = J[X, Y] = [X, JY] \,.$$

Thus $\operatorname{ad}(JX) = J \operatorname{ad}(X)$ is both self-adjoint, as a product of commuting skew-adjoint operators, and skew-adjoint. It is zero, and $JX$ is central. Hence $J = 0$. The color coefficient is therefore the one obtained by testing $T_*$.

The side satisfies $L_p D_m \geq 100(B_m + 1) > 20$, which permits the clover loops and affine tests without degeneracies. Equation (13.1) identifies the affine Hessian of $Q_f$ with that of $f$. Subtracting gives zero diagonal affine Hessian for $R_f$; polarization gives zero on pairs. The value and first derivative of $R_f$ vanish at 1. If $Z$ wraps, no Hessian cancellation is required. The clover polynomial is smooth, Borel, gauge invariant and local on $N_2(o)$, which gives the other regularity assertions on $Z$.

All indicators in $\zeta_f$ read flags in $Z$. If $Z$ is healthy, every mask in the identity is one and $\zeta_f = 0$; the identity reduces to the definition of $R_f$. If $Z$ is unhealthy, $h_Z R_f = 0$ and the definition of $\zeta_f$ gives the identity directly. This proves the equality on every configuration and the stated locality and cancellations. □

The same subtraction has a quantitative bound. If $a_f$ majorizes $f$, the diagonal tests $tA^{q,T_*}$ and their half-sum are permitted directions in Definition 3.3. They give

$$t^2 |q_f(q, q)| \leq a_f, \qquad t^2 |q_f(q, r)| \leq 3a_f \,. \tag{13.14}$$

Indeed four times the half-sum's diagonal Hessian is $t^2(q_f(q, q) + q_f(r, r) + 2q_f(q, r))$. Its absolute value and the two diagonal bounds give the second estimate. Product differentiation of the four-link formula bounds every permitted derivative of $C_q$ by $C_j t$ at a healthy anchor. For order zero this follows from the plaquette chord bound; at positive order the direction factors supply $t$, and all undifferentiated group matrices are bounded. Hence

$$\|R_f\|_{m,Z} + \sup_U |\zeta_f(U)| \leq C_j a_f \,. \tag{13.15}$$

For $\zeta_f$, the terms $Q_f$ and $f$ occur only under their own health indicators, so the estimates still apply when $Z$ is unhealthy.

Enlargement costs at most $\operatorname{cov}_B(Z) \le \operatorname{cov}_B(X) + 1$: one radius-$B$ ball centered at $o \in X$ contains $N_2(o)$ and meets the original connected cover. Also $\operatorname{dist}(x, o) \le CB \operatorname{cov}_B(Z)$ for $x \in Z$, by following the connected cover. For a support of covering number $n$ through $x$, its anchor lies within distance $CBn$ of $x$, giving at most $C(1+B)^4(1+n)^4$ possible anchors. The inequality $(1+n)^4 \le Ce^n$ converts exponent $\sigma+3$ at the anchor to exponent $\sigma+1$ at a site: one support power bounds this polynomial and one remains unused. At exponent $\sigma$ the enlargement itself costs only a stage constant. Without these extra powers the conversion would leave a polynomial in $n$ at the original exponent.

Sum $q_{f_{i,u}}$ at each anchor. Translation covariance makes the result independent of the anchor. Reflection in an axis belonging to exactly one of two distinct orientations changes the sign of the off-diagonal entry, so that entry is zero. Permuting axes identifies the diagonal entries. Thus a single real $\Delta_u$ satisfies

$$\begin{aligned} \sum_{o_i = o} q_{f_{i,u}}(q, r) &= \Delta_u \delta_{qr}, \\ \sum_i h_{\{o_i\}} Q_{f_{i,u}} &= \frac{\Delta_u}{2} \sum_o h_{\{o\}} \sum_q |C_q(U, o)|^2 . \end{aligned} \tag{13.16}$$

The coefficient sums converge absolutely by (13.14), and $t^2|\Delta_u| \le H_j$. This estimates the subtractions; the bound $|\Delta_u| \le 1$ is obtained in the proof of Lemma 12.10.

## 13.3 Forced fusion and the activity bounds

Put $\alpha = \alpha_{N,m}$. The additional regular functions and logarithmic corrections are

$$\begin{aligned} e_{\lambda,u} &= \Delta_u (c_\lambda - r_\lambda), \\ \ell_{\lambda,u} &= (\alpha + \Delta_u)(L_\lambda - h_{Z_\lambda} r_\lambda) + \Delta_u (h_{Z_\lambda} - h_{\{o_\lambda\}}) c_\lambda . \end{aligned} \tag{13.17}$$

The regular function is used with mask $h_{Z_\lambda}$. Its affine Hessian vanishes because its two terms have the same constant-curl response. Add also the regular cubic and quartic functions $-(\alpha + \Delta_u)\mathfrak{a}_i^{(k)}$, with their prescribed masks; they have zero Hessian because $k \ge 3$. Every $\ell_{\lambda,u}$ vanishes on healthy $Z_\lambda$.

Write the logarithmic corrections $\zeta_{f_{i,u}}$ and $\ell_{\lambda,u}$ as $\ell_a$ on nonempty supports $Z_a$. A connected selection $P$ consists of preliminary forced labels and logarithmic labels, contains at least one forced label, and has pairwise disjoint preliminary forced supports. Connected means connected in the graph joining intersecting supports. Its support $Y_P$ is the union of all supports, including logarithmic supports; its inventory $I_P$ is the union of the preliminary forced inventories. Define

$$K_P(U) = \mathbf{1}_{\{Y_P \cap \operatorname{Flag}_m(U) = I_P\}} \prod_{i \in P_{\mathrm{for}}} k_{i,u}(U) \prod_{a \in P_{\log}} (e^{\ell_a(U)} - 1) . \tag{13.18}$$

The extra test is the conjunction of the preliminary forced tests. Selections and tests are fixed as $u$ varies.

**Lemma 13.10** (Forced fusion)**.** *Suppose* $\sum_a \sup_U |\ell_a(U)| < \infty$. *For* $\kappa_{\mathrm{fus}}, \gamma \ge 0$ *assume nonnegative* $v_a$ *for all forced and logarithmic labels with*

$$|k_i| \le e^{-\kappa_{\mathrm{fus}}} v_i, \quad |e^{\ell_a} - 1| \le v_a, \quad \sup_x \sum_{a : x \in Z_a} e^{(\sigma+1)\operatorname{cov}_B(Z_a)} v_a \le \gamma, \quad (2B+1)^4 \gamma \le 1 .$$

*Then the exact-cover gas of* (13.18) *equals* $\mathcal{F}_u^0 \exp(\sum_a \ell_a)$ *and its forced norm of Definition 3.4 is at most* $e^{-\kappa_{\mathrm{fus}}}(e^{\gamma}-1)$.

*Proof.* Use Definition 3.2 and the inventories of Definition 13.2. Expand $\exp(\sum_a \ell_a) = \prod_a (1 + (e^{\ell_a}-1))$. Decompose a finite selection, together with a selection from $\mathcal{F}_u^0$, into support-overlap components. A nonzero logarithmic factor meets a flag. Exact coverage assigns that flag to a preliminary forced label, whose support meets the logarithmic support. Every component therefore contains a forced label. Their union supports are disjoint; their union inventories cover all flags. Conversely the components recover the original selection uniquely. The eligibility indicator in (13.18) equals one on any such nonzero product: another component cannot supply a flag in $Y_P$, since its support is disjoint. This proves the identity for finite expansions.

For the absolute bound retain $e^{-\kappa_{\mathrm{fus}}}$ from one forced label per connected selection. Apply the rooted-tree bound to the remaining $v_a$. The total weight of labels meeting a support $Z$ is at most $|Z|\gamma \le \mathrm{cov}_B(Z)$, using the extra exponent in the hypothesis. Successive attachments are therefore paid by that exponent. More explicitly, put $\mathfrak{v}_i = v_i e^{\sigma\,\mathrm{cov}_B(Z_i)}$ and let $T_i^{(n)}$ sum rooted trees of height at most $n$, permitting repeated labels. The exponential formula for their unordered branches is

$$T_i^{(0)} = \mathfrak{v}_i, \qquad T_i^{(n+1)} = \mathfrak{v}_i \exp\left( \sum_{k : Z_k \cap Z_i \neq \varnothing} T_k^{(n)} \right).$$

The hypothesis gives $T_i^{(n)} \le \mathfrak{v}_i e^{\mathrm{cov}_B(Z_i)}$ by induction. The sum of these upper bounds over roots whose supports contain a fixed site is at most $\gamma$. Every connected selection has a spanning tree and an eligible root, so its sum is at most $\gamma \le e^{\gamma}-1$. The covering number is subadditive on intersecting unions, which retains $e^{\sigma\,\mathrm{cov}_B(Y_P)}$ in this comparison. Restoring $e^{-\kappa_{\mathrm{fus}}}$ gives the claimed bound. The resulting summable envelope justifies passage from finite expansions to the infinite identity. □

Let $E_u^{\mathrm{sum}}$ be the masked sum of the $R_{f_{i,u}}$, the $e_{\lambda,u}$, and the cubic and quartic corrections. Let $\mathcal{F}_u$ be the fused gas. Hold these functions fixed while a trial parameter $\vartheta$ varies, and put

$$q_{u,\vartheta} = e^{-(\beta'_u + \alpha_{N,m} + \vartheta)\mathsf{K}_m^{\mathrm{kin}} - \mathcal{L}_m - E_u^{\mathrm{sum}}} \mathcal{F}_u .$$

The output representation is the following single list: regular addresses $(Z_i, o_i)$ with functions $R_{f_{i,u}}$, addresses $(Z_\lambda, o_\lambda)$ with functions $e_{\lambda,u}$, the cubic and quartic addresses of Definition 12.1 with functions $-(\alpha + \Delta_u)\mathfrak{a}_i^{(k)}$, and forced addresses $(Y_P, I_P, \mathcal{E}_P^{\mathrm{extra}})$ with functions $K_P$. Each regular function carries its own health mask. We prove the inequalities

$$e^{-\kappa_{\mathrm{fus}}}(e^{\gamma}-1) \le \epsilon_m^{\mathrm{for}}, \qquad \mathcal{E}_m \le \epsilon_m^{\mathrm{reg}} . \tag{13.19}$$

in Lemma 13.11 before using them for the density identity. They impose no condition on the trial coupling.

**Lemma 13.11** (Activity bounds and covariance)**.** *For every coarse output constructed above and every Wilson representation of Lemma 13.21, both activity families are countable and equivariant under translations and* $H_4$*, including supports, anchors, inventories and tests. There exist nonnegative majorants as in Definitions 3.3–3.4 whose weighted series converge and which satisfy*

$$\mathcal{E}_m \le \epsilon_m^{\mathrm{reg}}, \qquad \mathcal{K}_m \le \epsilon_m^{\mathrm{for}} .$$

The proof follows; the Wilson case is treated in the proof of Lemma 13.21. Lemma 13.14 extends these conclusions to all the chosen classes.

*Proof for a coarse output.* Use Definitions 3.3–3.4, Lemmas 12.3–12.4, 13.7, 13.9 and 13.10. Equation (13.15) controls each subtracted function. The low regular terms cost $C_j H_j$, since $t^2|\Delta_u| \le H_j$. The high terms cost $C_j(|\alpha|t^3+H_j)$: each has at least three curvature factors, and $|\Delta_u|t^3 \le H_j t \le H_j$. A derivative of a word of length $l$ costs a fixed power of $1+l$, by the product rule and the allowed spatial growth of the directions. The support obeys $\operatorname{cov}_B(Z_\lambda) \le C + l/B$. Raise the threshold until $\sigma/B \le w_{J_m}^{\mathrm{path}}/4$. A quarter of the path exponent then bounds the support weight, and another quarter bounds this polynomial, since $\sup_{l\ge 0}(1+l)^k e^{-wl/2} < \infty$. Summing Lemma 12.3 gives the previously defined constant $C_j^0$ and a constant $C_j^1$ with

$$\mathcal{E}_m \le C_j^0 H_j + C_j^1 |\alpha| t^3 .$$

The constant $c_j = [64(1+C_j^0)]^{-1}$ was chosen before the ratios in Lemma 13.7. After the ratio is fixed, Lemma 12.9 bounds $|\alpha|$; raise the entry threshold until $C_j^1|\alpha|t^3 \le \epsilon_m^{\mathrm{reg}}/2048$. This is possible because $t^3/\epsilon_m^{\mathrm{reg}} = \widehat{\beta}_m^{-2/5}(M_m^{\mathrm{cut}})^3 \to 0$. Take $\theta \le 1/32$ before the ratio. Since $C_j^0 c_j \le 1/64$, the sum is at most $\epsilon_m^{\mathrm{reg}}/1024$. The same argument allows any prescribed smaller coefficient by choosing $\theta$ first and raising the threshold for the second term.

For the logarithmic corrections, conversion from anchored sums to site sums costs at most $C_j(1+B)^4$, after the extra support powers pay the powers of the covering number. The local subtraction costs $C_j H_j$. The low terms in (13.17) cost

$$C_j(1+B)^4\{(|\alpha| + H_j/t^2)t + H_j\} =: m_j .$$

The linear clipped tail is bounded by $2c_{\mathrm{lin}}t$ times its coefficient, since chords are at most two. For a failed relaxed health test, the remote threshold supplies $e^{l/R_{J_m}^{\mathrm{rel}}}$ in a path of length $l$. It is paid by the path decay because $R_{J_m}^{\mathrm{rel}} w_{J_m}^{\mathrm{path}} \ge 12$. The common-anchor clover term costs $|\Delta_u|t^2 \le H_j$. These estimates give exactly the displayed bound, including unhealthy configurations. Since $H_j/t = O_j(\widehat{\beta}_m^{-3/5}/M_m^{\mathrm{cut}})$ and $|\alpha|$ is fixed after the ratio, $(1+B)^4 m_j \to 0$.

Let $\kappa_{\mathrm{fus}} = \kappa_j^{\mathrm{pre}}(M_{m+1}^{\mathrm{cut}})^2/2$. The majorants in (13.12), multiplied by $e^{\kappa_{\mathrm{fus}}}$, have rooted total at most $e^{-\kappa_{\mathrm{fus}}}$. For each logarithm use $|e^{\ell_a} - 1| \le 2|\ell_a|$ once its individual majorant is at most $\log 2$; this follows by increasing the entry threshold until $m_j \le \log 2$. Their rooted total is at most $2m_j$. Increase the entry threshold so that $(2B+1)^4\gamma \le 1/2$ for $\gamma = e^{-\kappa_{\mathrm{fus}}} + 2m_j$. Lemma 13.10 applies. Increase the threshold further so that $\gamma \le 1$ and $e^{-\kappa_{\mathrm{fus}}}(e^{\gamma}-1) \le \epsilon_m^{\mathrm{for}}/1024$. Indeed $e^{\gamma} - 1 \le 2\gamma$ and $\kappa_{\mathrm{fus}} \gg (M_m^{\mathrm{cut}})^{1/5}$. This proves the forced inequality and summability in (13.19).

Every address is a finite combination of countable preliminary and path addresses, so the families are countable. Subtraction and path replacement transport the entire address under translations. Signed permutations transform the curvature orientations with their signs; the finite group average includes every transformed term. These permutations act on supports, anchors and functions. On a connected selection they also transform the union inventory and conjunction of tests. Equation (13.18) therefore has the required covariance. For an inverse link, $(e^{zA}U)^{-1} = e^{-z\,\mathrm{Ad}_{U^{-1}}A}U^{-1}$. Differentiating this identity repeatedly gives only finitely many product-rule terms through order $\mathfrak{q}_{J_m}$; the adjoint-invariant norm bounds them by a constant depending on that order. This proves the complete address covariance and the convergent weighted bounds for a coarse output. The Wilson start is established below; induction in Lemma 13.14 then gives the class assertion. □

The identity below fixes the whole density before the scalar estimate of Lemma 12.10. Its proof uses the bounds just established.

**Lemma 13.12** (Exact density representation)**.** *The final activities satisfy Lemma 13.11, vanish at the flat field in their regular part, and are independent of $\vartheta$. For every $u$ there is $\Delta_u \in \mathbb{R}$ such that*

$$q_{u,\Delta_u}(U) = \frac{\mathcal{H}_u(U)}{\mathcal{H}_u(1)} \qquad \text{for every } U\,.$$

*Moreover $\mathcal{F}_u(U) = 1$ when $\mathrm{Flag}_m(U) = \varnothing$.*

*Proof.* Use Definition 3.2, Lemmas 12.3–12.4, 13.9, 13.10, and (13.10)–(13.11), including (13.19). Choose $\Delta_u$ by (13.16) and put $c_u = \sum_i f_{i,u}(1)$. Local subtraction gives

$$\sum_i h_{X_i} f_{i,u} = c_u + \frac{\Delta_u}{2} \sum_o h_{\{o\}} \sum_q |C_q|^2 - \sum_i \zeta_{f_{i,u}} + \sum_i h_{Z_i} R_{f_{i,u}}\,.$$

All sums converge absolutely. For each low term (13.17) gives

$$\begin{aligned}(\alpha + \Delta_u) L_\lambda + \Delta_u h_{Z_\lambda}(c_\lambda - r_\lambda) - \ell_{\lambda,u} \\ = \alpha h_{Z_\lambda} r_\lambda + \Delta_u h_{\{o_\lambda\}} c_\lambda\,.\end{aligned}$$

Sum and apply (13.2). The high corrections cancel $(\alpha + \Delta_u)\mathcal{A}_m$. Thus the exponent in (13.10) becomes

$$\begin{aligned}(\tau_u - c_u/|\Lambda|)|\Lambda| - (\beta'_u + \alpha + \Delta_u)\mathsf{K}_m^{\mathrm{kin}} - \mathcal{L}_m - E_u^{\mathrm{sum}} \\ + \sum_i \zeta_{f_{i,u}} + \sum_\lambda \ell_{\lambda,u}\,.\end{aligned}$$

Lemma 13.10 incorporates the last two sums into $\mathcal{F}_u$. Hence, on every configuration,

$$\mathcal{H}_u(U) = e^{(\tau_u - c_u/|\Lambda|)|\Lambda|} q_{u,\Delta_u}(U)\,. \tag{13.20}$$

At 1 all path and final regular terms vanish. The only exact-cover forced selection is empty, so $q_{u,\vartheta}(1) = 1$ for every $\vartheta$. Equation (13.20) proves the quotient identity and positivity of its denominator. The same empty-selection argument gives $\mathcal{F}_u = 1$ whenever there are no flags.

The regular cancellations were proved in Lemma 13.9 and after (13.17); the weighted bounds are those of Lemma 13.11. The value $\Delta_u$ used in the activity construction is already fixed by the preliminary functions. The trial variable $\vartheta$ occurs only in the exponential defining $q_{u,\vartheta}$. This proves the exact identity without using Lemma 12.10 or its reset estimate. □

## 13.4 Continuity at fixed volume

The scalar reset needs continuity of the integral and both final sums. The next deduction uses the weighted comparison in Lemma 13.7.

**Lemma 13.13** (Continuity of the activity sums)**.** *For input curves of Theorem 3.5, the final activity families have fixed addresses and are continuous in the weighted sense of Definition 13.3. For every fixed configuration $U$,*

$$u \longmapsto \mathcal{H}_u(U), \qquad u \longmapsto E_u^{\mathrm{sum}}(U), \qquad u \longmapsto \mathcal{F}_u(U)$$

*are continuous.*

*Proof.* Use Definitions 3.1, 3.2, 13.3, Lemmas 12.3, 13.9, 13.10, and Lemma 13.7. The maps $f \mapsto f(1), q_f, R_f, \zeta_f$ are linear on functions with fixed address. Equation (13.14) applied to a difference bounds its projection coefficients by its derivative majorant divided by $t^2$. Weighted preliminary continuity therefore gives continuity of $c_u$, $\Delta_u$, and the subtracted regular and logarithmic series. The low and high corrections are linear in $\Delta_u$, with fixed summable path coefficients, so they retain this continuity. Every mask is a fixed function of $U$.

For fusion each coefficient is a finite product and hence continuous in supremum norm. To justify summing weighted differences, work first along subsequences with one summable majorant. Given $u_n \to u_0$, pass to a subsequence whose total weighted preliminary discrepancies are at most $2^{-n}\varepsilon$. Add their nonnegative majorants to those at $u_0$; the extra mass is at most $\varepsilon$. The margin $(2B+1)^4\gamma \le 1/2$ arranged in the proof of Lemma 13.11 permits this addition for small $\varepsilon$. Lemma 13.10 gives one summable envelope for the fused coefficients along this subsequence. Dominated convergence gives convergence of their weighted differences. If the original sequence failed to converge, a subsequence with discrepancies bounded away from zero would admit such a further subsequence, a contradiction. This proves weighted continuity without assuming a common envelope on an arbitrary neighborhood. Constants here may depend on the fixed volume.

For the pointwise gas, put $M_i = \max(\sup|K_{i,u}|, \sup|K_{i,u_0}|)$. Telescoping finite products and summing selections gives

$$|\mathcal{F}_u(U) - \mathcal{F}_{u_0}(U)| \le e^{\sum_i M_i} \sum_i \sup|K_{i,u} - K_{i,u_0}|\,.$$

The restrictions on exact coverage use fixed inventories. Weighted continuity makes the right side tend to zero. The regular sum is continuous by the analogous sum of order-zero differences. The same bounds give continuity in $u$, uniform in $U$, for the input density at fixed volume, since its path action is affine in the continuous coupling.

Finally the kernel of Definition 3.1 is uniformly bounded on a compact coupling interval at this volume. The tag masses sum to one; its single-link normalizing integrals are positive and continuous in the coupling, and its link factors are bounded on the compact group. Dominated convergence against product probability Haar measure gives continuity of $\mathcal{H}_u(U)$. This proves all three pointwise continuities and weighted continuity, by Lemma 13.7. □

## 13.5 The specified recursion

A choice of output for one representation does not determine a step on every continuous input curve. We choose the local construction for an entire curve, including its truncation, auxiliary tori, tree gauge and address identifications, and then allow restrictions of that choice. The estimates used to repeat the step are the following explicit ones. At level $m$ the path is the fixed $P^{\mathrm{path}}_{N,m}$ of Definition 12.1, with the bounds of Lemmas 12.3–12.4. The activity functions satisfy Definitions 13.1–13.2 and Lemma 13.11. At an adjacent pair the output is exactly the functions in Lemma 13.12, its scalar is $|\Lambda|^{-1}\log\mathcal{H}(1)$, and its correction obeys

$$t_m^2|\Delta| \le H_j, \qquad \mathcal{E}_m \le \epsilon_m^{\mathrm{reg}}/1024, \qquad \mathcal{K}_m \le \epsilon_m^{\mathrm{for}}/1024, \qquad |\Delta| \le 1\,. \tag{13.21}$$

The first three bounds precede the scalar reset; the last is obtained in the proof of Lemma 12.10. The stronger preparatory bounds, fixed tests, and whole-interval comparison of Lemma 13.7 are retained with the step. The input coupling belongs to $\mathfrak{B}_{m+1}$; no output-band condition is imposed. These are the continuation conditions used here.

Define $\mathcal{I}_{N,p} = \{\rho_\beta^{\mathrm{W}} : \beta \in \mathfrak{B}_N\}$ by the Wilson construction of Lemma 13.21, and put $\mathcal{A}_{N,N,p} = \mathcal{I}_{N,p}$. Inductively downwards, let $\mathcal{A}_{N,m,p}$ contain all outputs of the construction on continuous curves in $\mathcal{A}_{N,m+1,p}$ with coupling in $\mathfrak{B}_{m+1}$, retaining their restrictions to closed subintervals and their fixed address identifications. Define $\mathcal{R}_{N,m,p}$ to consist of the output–input pairs from these same curves and restrictions. The analytic construction below fixes these choices simultaneously on the common address spaces; its single step map and its restrictions are used for the selected recursion and endpoint tuning. Thus an adjacent pair always uses the actual input density and its own coupling in the kernel.

The following statement fixes these choices for the later trajectory argument. Its proof also shows that it applies to every admitted continuous curve, including curves assembled from previously retained outputs.

**Lemma 13.14** (Specified recursion)**.** *For fixed $G, \mathsf{b}$, the classes just defined consist of representations satisfying Lemmas 12.3–13.11, with*

$$\mathcal{I}_{N,p} \subset \mathcal{A}_{N,N,p}, \qquad \mathcal{R}_{N,m,p} \subset \mathcal{A}_{N,m,p} \times \mathcal{A}_{N,m+1,p}\,.$$

*Its trajectories are exactly the tuples satisfying*

$$\rho_m \in \mathcal{A}_{N,m,p}\ (0 \le m \le N), \quad \rho_N \in \mathcal{I}_{N,p}, \quad (\rho_m, \rho_{m+1}) \in \mathcal{R}_{N,m,p}\ (m < N)\,.$$

*With these classes, Theorem 3.5 holds and every member satisfies the activity bounds in* (13.21)*.*

*Proof.* Use Definition 3.2, Lemmas 12.3–12.4, 12.10, 13.7, and 13.11, 13.8–13.13. The Wilson construction at the end supplies the initial curve and both caps. At a step, Lemma 13.7 uses only the prescribed paths, the local regularity and forced eligibility, the bounds and the weighted continuity of its input. The proof does not require that the input curve coincide with an earlier chosen curve. In particular its common truncation is chosen afresh on the entire compact interval from these hypotheses. Local subtraction, path replacement and fusion preserve the read supports, gauge invariance, cancellations and countability. Lemma 13.11 proves their numerical bounds, and Lemma 13.13 proves weighted continuity on the interval.

The normalized identity of Lemma 13.12 and the reset of Lemma 12.10 identify the same $\Delta$. Its proof identifies $\Delta$ with the continuous scalar root and gives $|\Delta| \le 1$; its statement gives the drift in Theorem 3.5. Defining the vacuum scalar by flat evaluation gives the exact integral, since the normalized representation is one at 1. Hence (13.21) is inherited by the outputs and the prescribed paths are unchanged. Finite induction gives the classes just described.

Restriction preserves every estimate, all supports and tests, and each curve choice. Its weighted continuity is continuity in the relative topology of the smaller interval, so it is preserved as well. Declare a step selected on a larger interval and its restriction to be the same choice. Other input curves are treated by the preceding construction, rather than by pointwise ancestry. This proves readiness on every continuous input curve in the classes. The displayed trajectory conditions are exactly the resulting finite recursion. Section 14 uses its closure under restriction for shooting; Lemma 13.19 below supplies the weighted analytic estimates for the selected recursion. The Wilson identity and the final one-step assembly are proved after those estimates. □

## 13.6 Weighted analytic estimates

We compare cutoffs using activity functions on fixed domains. At level $m$ and period $p$, put all regular addresses into one countable list and all forced addresses into a second list. A missing

term has coefficient zero. The lists contain every finite support, anchor and inventory used above, and every finite conjunction of their eligibility tests. The output of a fusion is indexed by its finite selection, so the lists are closed under the construction. Keep the same permutations for translations and signed permutations. This identifies different cutoffs without changing their health domains.

For a difference $A = (E, K)$ of activity families set

$$\|A\|_{m,p} = \max\left\{ \frac{\sup_x \sum_{o_i = x} e^{\sigma_{J_m} \operatorname{cov}_{B_m}(X_i)} \|E_i\|_m}{\epsilon_m^{\mathrm{reg}}}, \frac{\sup_x \sum_{x \in Y_i} e^{\sigma_{J_m} \operatorname{cov}_{B_m}(Y_i)} \sup_U |K_i(U)|}{\epsilon_m^{\mathrm{for}}} \right\}. \tag{13.22}$$

The seminorms in this formula are the least majorants in Definitions 3.3–3.4. Quotient by zero seminorm and complete the space with the prescribed local cancellations, eligibility and permutations. Denote the resulting real Banach space by $\mathcal{X}_{m,p}$. This retains regular derivatives on healthy configurations and forced values on every configuration. The vacuum scalar is excluded.

The coupling coordinate must allow relative changes of $\beta$ while keeping a uniform analytic radius. Define

$$\chi_m = \widehat{\beta}_m^{-1/32}, \quad b_m = \chi_m \widehat{\beta}_m, \quad x_m = \chi_m^{-1} \sqrt{\beta_m / \widehat{\beta}_m}, \quad \beta_m = \widehat{\beta}_m (\chi_m x_m)^2 . \tag{13.23}$$

Thus $b_m$ converts a change of $x_m$ into a change of $\beta_m$, up to the factor $2\chi_m x_m$. Monotonicity gives $b_m \geq \chi_0 \widehat{\beta}_0 > 0$. Raise the entry threshold so $\bar{\beta} \geq 2^{224}$; then $\chi_m \leq 1/128$. In the following paragraphs write $b_* = \chi_0 \widehat{\beta}_0$, $g_* = 1/128$ and $\lambda_* = 1/16$.

Let $F_{N,m}$ denote the selected activity map and $\mathcal{D}_{N,m}$ its scalar correction, so that

$$A_m = F_{N,m}(x_{m+1}, A_{m+1}), \qquad \beta_m = \beta_{m+1} + \alpha_{N,m} + \mathcal{D}_{N,m}(x_{m+1}, A_{m+1}) .$$

The top map is denoted by $F_N^{\mathrm{W}}(x_N)$. The maps are constructed for $\|A\| \leq 1$ and $33/40 \leq \chi_m x \leq 47/40$. We use expansions centred in the smaller domain

$$\|A\| \leq 1/8, \qquad 17/20 \leq \chi_m x \leq 23/20 . \tag{13.24}$$

A ball of radius $1/4$ about such a centre stays in the larger domain: its activity norm is at most $3/8$, and the change of $\chi_m x$ is at most $1/512 < 1/40$. For a retained history with $|\beta_0 - \bar{\beta}| \leq \bar{\beta}/4$, summing the drift and the error bound in Section 9 gives $|\beta_m - \widehat{\beta}_m| \leq \widehat{\beta}_m/4$. Its square root therefore lies in $[17/20, 23/20]$. Every activity in the history, including the Wilson start, has norm at most $1/1024$ by the proofs of Lemmas 13.11 and 13.21. Thus the entire history lies in (13.24).

For a finite stencil with real coefficients $c_\nu$ and nodes $s_\nu \in [-1, 1]$, define

$$\mathfrak{s}(c, s; \rho) = \sup_{k \geq 0} \left| \sum_\nu c_\nu (s_\nu / \rho)^k \right| , \qquad \rho > 1 . \tag{13.25}$$

We apply it along $(x(s), A(s)) = (x + sv, A_0 + sA_1)$, with $|v| \leq 1$ and $\|A_0 + sA_1\| \leq 1$ for $|s| \leq 1$. Supports, health tests and local function domains are fixed along the line. Write $F(s) = F_{N,m}(x(s), A(s))$ and $\mathcal{D}(s) = \mathcal{D}_{N,m}(x(s), A(s))$. Let $\mathcal{N}_{\mathrm{pre}}$ be the anchored regular norm at exponent $\sigma + 3$, before division by $\epsilon_m^{\mathrm{reg}}$ and before subtraction. Let $\mathcal{N}_{\mathrm{for}}$ be the corresponding forced site norm at exponent $\sigma + 1$.

The coupling varies through a square of an affine function. The next estimate controls that dependence at a radius growing with the reference coupling; a bound at one fixed field-derivative order would not suffice.

**Lemma 13.15** (Coupling-line stencils)**.** *Put* $\mathfrak{b} = \widehat{\beta}_{m+1}$ *and* $\mathfrak{R} = \mathfrak{b}^{1/256}$*. For the lines above centred in* $33/40 \le \chi_{m+1}x \le 47/40$*, the preparatory families satisfy*

$$\mathcal{N}_{\mathrm{pre}}\left(\sum_{\nu} c_\nu f(s_\nu)\right) \le \mathfrak{b}^{-1/4}\mathfrak{s}(c,s;\mathfrak{R}/2),$$
$$\mathcal{N}_{\mathrm{for}}\left(\sum_{\nu} c_\nu k(s_\nu)\right) \le C_j e^{-\kappa_j^{\mathrm{pre}}(M_{m+1}^{\mathrm{cut}})^2}\mathfrak{s}(c,s;\mathfrak{R}/4).$$

*For any prescribed* $\theta_{\mathrm{W}} > 0$*, the Wilson map satisfies the analogous normalized estimate with radius* $\mathfrak{R}$ *and bound* $4\theta_{\mathrm{W}}$ *after the entry threshold is raised.*

The proof keeps physical fields fixed and then applies the connected and localization operations.

*Proof.* Use Definitions 3.1–3.2, Lemmas 11.11, 11.13, 12.3–12.9, 13.4 and 13.6. Put $\upsilon = \mathfrak{b}^{-1/128}$ and raise the threshold until

$$128\mathfrak{R}\chi_{m+1} \le \upsilon \le 1\,. \tag{13.26}$$

This is possible since $1/256 - 1/32 + 1/128 = -5/256 < 0$. Set $\zeta = \chi_{m+1}x$ and $L = \zeta - 6\mathfrak{R}\chi_{m+1}$. For $|s| \le 6\mathfrak{R}$,

$$\beta(s) = \mathfrak{b}L^2\left(\frac{\zeta}{L} + \frac{\chi_{m+1}\upsilon}{L}s\right)^2, \qquad \frac{3}{4} \le \chi_{m+1}(x+s\upsilon) \le \frac{4}{3}\,.$$

The displayed margins follow from (13.26) and $33/40 \le \zeta \le 47/40$. In particular the real couplings lie in $\mathfrak{B}_{m+1}$. The relative scale change is at most a fixed multiple of $\upsilon$.

To estimate parameter coefficients, use physical link variables in each sector integral. All flag, allocation, tag and chart tests are then independent of $s$: their thresholds are $t_{m+1}$ and the prescribed stage constants, not $\beta(s)^{-1/2}$. Old measurable forced functions occur as $K_0 + sK_1$ at the same physical links. Gaussian rescaling is used to estimate the coefficients, after these functions have been held fixed. Thus parameter coefficients do not differentiate their field arguments or any test.

Here is the coefficient bound for this change of Gaussian scale. For a positive local precision matrix $\mathsf{A}$ and affine source $\mathsf{b}_0$, put $m_0 = -\mathsf{A}^{-1}\mathsf{b}_0$ and $Q(a) = \langle a - m_0, \mathsf{A}(a-m_0)\rangle/2$. This is the centred quadratic energy in the sector Gaussian density. The parameter occurs through $\beta(s)Q$, its normalizing determinant, and the local stationary corrections. Expand the inverse by

$$(\mathsf{A}+H)^{-1} = \sum_{n\ge 0}(-\mathsf{A}^{-1}H)^n\mathsf{A}^{-1}, \qquad \log\det(I+H) = \sum_{n\ge 1}\frac{(-1)^{n+1}}{n}\operatorname{Tr} H^n\,.$$

In these formulas $H$ denotes, respectively, the precision perturbation and its dimensionless version. The inverse bound and exponential Schur sums of Lemma 11.13, together with (13.26), make their radius-$\mathfrak{R}$ coefficient sums less than a fixed constant. For the Gaussian density, expansion of its exponential is dominated by $e^{c|z|^2}$ with $c$ smaller than half its precision lower bound. Indeed the coefficient of $|z|^2$ changes by $O(\mathfrak{R}\chi_{m+1})$. Integrating this majorant bounds every parameter degree at once. The local number of coordinates is a fixed power of $1+B$; hence its additional loss is bounded because $(1+B)^C\mathfrak{R}\chi_{m+1} \to 0$. The same Neumann series solves each stationary correction with the uniformly bounded inverse. Ordered field derivatives only add the fixed read polynomials already present in Lemma 11.13.

Each incoming factor is affine in $s$. For a regular occurrence its exponential has radius-$\mathfrak{R}$ majorant $\exp(\mathfrak{R}\|E_1\|)$; for a forced occurrence the majorant is $|K_0| + \mathfrak{R}|K_1|$. Products use Cauchy

products of coefficients. Reserve the support powers before summing these expressions. The initial small factor in (13.4) loses at most one factor $\mathfrak{R}$ for an incoming affine occurrence; the change of Gaussian scale has the bounded sums just proved. Thus the weighted attachment parameter is bounded by

$$P_j(\mathfrak{b})\mathfrak{R}\mathfrak{b}^{-7/16} = P_j(\mathfrak{b})\mathfrak{b}^{-111/256} \longrightarrow 0\,.$$

The connected logarithm therefore has an absolutely summable parameter series by the same rooted estimate as in Lemma 13.4. A forced component retains a smaller fixed portion of the energy in Lemma 11.11; coefficient multiplication and sector counting use the rest. Its coefficient majorant is $C_j e^{-\kappa_j^{\mathrm{pre}}(M_{m+1}^{\mathrm{cut}})^2}$.

Apply the five reflected tori of Lemma 13.6 to these same parameter coefficients. A stencil commutes with their finite linear combinations. The direction rescaling, auxiliary volumes, support conversion and flat subtraction cost a subpower factor. The resulting regular coefficient sum has bound $P_j(\mathfrak{b})\mathfrak{b}^{-3/8}$: the preceding attachment bound is smaller, since $111/256 > 3/8$. Raise the threshold until this is at most $\mathfrak{b}^{-1/4}$. The regular logarithm and the forced collection may use radii $\mathfrak{R}/2$ and $\mathfrak{R}/4$, respectively; their Cauchy products then have geometric reserve. Multiplying degree $n$ by $\sum_\nu c_\nu s_\nu^n$ proves the asserted stencil bounds.

For the Wilson map use the formulas in the proof of Lemma 13.21. Its regular term is $\beta(s)$ times a fixed fifth-order remainder. Its radius-$\mathfrak{R}$ coefficient sum is at most $C_j\mathfrak{b}t_N^5/\epsilon_N^{\mathrm{reg}}$, which tends to zero. For a forced charge $d_I \geq 0$, write $\beta(s) = \beta(0) + b_1 s + b_2 s^2$. Its exponential has coefficient sum at most

$$\exp\{-\beta(0)d_I + (|b_1|\mathfrak{R} + |b_2|\mathfrak{R}^2)d_I\} \leq e^{-\beta(0)d_I/2}$$

by (13.26). The inventory count in that proof is still summable with this half exponent. Both normalized coefficient sums can be made at most $\theta_{\mathrm{W}}$; enlarging the bound to $4\theta_{\mathrm{W}}$ gives the stated estimate. All bounds were taken before the period was counted, so they are uniform in the period. □

We now reduce the growing radius to one fixed radius chosen before the ratios. Choose $\rho > 1$, and then positive $\delta_{\mathrm{map}} \leq g_*/2$ and $\eta_{\mathrm{par}} \leq \delta_{\mathrm{map}}$. Put $\varphi_m = \eta_{\mathrm{par}} H_{J_{m+1}}/t_m^2$ and $\varrho_m = 2\varphi_m$; here $H_{J_{m+1}} = \theta c_{J_{m+1}}\epsilon_m^{\mathrm{reg}}$ is the coefficient for this step. The small fraction $\theta$ is chosen below before the ratios.

**Lemma 13.16** (Real parameter estimates)**.** *For every line specified above, the selected local maps satisfy*

$$\left\|\sum_\nu c_\nu F(s_\nu)\right\| \leq \delta_{\mathrm{map}}\mathfrak{s}(c,s;\rho/2), \qquad \left|\sum_\nu c_\nu \mathcal{D}(s_\nu)\right| \leq \varphi_m \mathfrak{s}(c,s;\rho), \tag{13.27}$$

*with $\sum_{i<m}\varrho_i \leq g_* b_m$. The radius $\rho$ can be arbitrarily large. The Wilson map satisfies the first estimate with the same $\delta_{\mathrm{map}}$.*

*Proof.* Use Definitions 3.3, 3.4, 3.2 and Lemmas 13.7, 13.11 and 13.15. Recover the regular line coefficients from the latter's stencil bound by finite differences at zero, as in Lemma 13.17 below. Put $\mathfrak{Q} = \mathfrak{b}^{1/256}/2$. Split after degree 300 and interpolate the finite part at 301 fixed distinct points of $[-1,1]$. The norm of this finite interpolation matrix gives a constant $L_\rho$ independent of the coupling and volume. The radius-$\rho$ coefficient sum is bounded by

$$L_\rho H_j + (1+L_\rho)(\rho/\mathfrak{Q})^{301}\mathfrak{b}^{-1/4}\,.$$

For the tail each degree $n \geq 301$ contributes $(\rho/\mathfrak{Q})^n$; its geometric denominator is bounded once $\mathfrak{Q} \geq 2\rho$. The tail has rate

$$-301/256 - 1/4 = -365/256 < -11/10\,.$$

Apply Lemma 13.7 with the smaller real-interval target $H_j^{\mathrm{aux}} = \eta_{\mathrm{par}} H_j/[4(1+L_\rho)]$, while retaining $H_j$ as its stated upper bound. This changes no identity. Choose this fraction before the ratio, and raise the threshold until the tail is at most $\eta_{\mathrm{par}} H_j/4$. The radius-$\rho$ bound is then at most $\eta_{\mathrm{par}} H_j/2$. Projection by (13.14) gives a scalar bound at most $\eta_{\mathrm{par}} H_j/t_m^2$. The scalar reset in the proof of Lemma 12.10 is exactly this extracted coefficient.

For the regular output, apply its three linear conversions with total bound $C_j^0$. Choose $\theta$ so the radius-$\rho$ preparatory contribution is at most $\epsilon_m^{\mathrm{reg}}/1024$ after this conversion, and raise the threshold so the high path term contributes at most another $\epsilon_m^{\mathrm{reg}}/1024$. Explicitly it suffices to choose the real-interval coefficient so that $C_j^0 L_\rho \theta c_j \leq 1/2048$ and then to use the tail reserve just proved. For the forced output, retain its coefficient majorant from Lemma 13.15; subtraction adds logarithmic factors with the bounds in the proof of Lemma 13.11. At fixed $\rho$, their radius-$\rho$ sums have the same small attachment parameter, with a constant depending on $\rho$. Fusion therefore gives

$$\mathcal{K}_m(K)_{\rho/2} \leq C_{j,\rho} \exp\{-\kappa_j^{\mathrm{out}} (M_{m+1}^{\mathrm{cut}})^2\} \leq \epsilon_m^{\mathrm{for}}/1024$$

for a smaller $\kappa_j^{\mathrm{out}} > 0$ and a larger threshold. Here $\mathcal{K}_m$ is the forced norm at the output exponent $\sigma$, and the subscript denotes the parameter coefficient sum at that radius. The exponent $\sigma + 1$ of the preparatory bound supplies the extra power used in fusion. The maximum norm (13.22) is thus at most $1/512$, and can be made smaller than the prescribed $\delta_{\mathrm{map}}$. This derives both the order-zero bound and the full stencil bound. For Wilson choose $4\theta_{\mathrm{W}} \leq \delta_{\mathrm{map}}$ in Lemma 13.15 and reduce its radius to $\rho/2$.

Since $\theta \leq 1$ and $c_j \leq 1/64$, the scalar normalization gives

$$\varrho_m \leq (\eta_{\mathrm{par}}/32)\widehat{\beta}_m^{-1/10}, \qquad \sum_{i<m} \varrho_i \leq C_G \eta_{\mathrm{par}} \widehat{\beta}_m^{9/10}\,.$$

The second inequality follows by comparison with an integral, since the reference-coupling increments have a fixed positive lower bound. Divide by $b_m = \widehat{\beta}_m^{31/32}$ and raise the entry threshold until $C_G \eta_{\mathrm{par}} \bar{\beta}^{-11/160} \leq g_*$. This proves the weighted scalar bound. □

Uniform line estimates at nearby centres control mixed finite differences. This supplies the multilinear coefficients for the endpoint equations. Let $X, Y$ be real normed spaces, with $Y$ complete, and let $f$ be defined near $x \in X$. Suppose that for all centres $x'$ in a neighbourhood of $x$, all $\|w\| \leq r$ and all stencils in (13.25),

$$\left\| \sum_\nu c_\nu f(x' + s_\nu w) \right\| \leq C\mathfrak{s}(c, s; R), \qquad r > 0, \quad R > 1\,.$$

The domain contains all the indicated real segments.

**Lemma 13.17** (Recovery from real stencils)**.** *Under these hypotheses there are continuous symmetric multilinear maps $T_k : X^k \to Y$ such that*

$$f(x+h) = \sum_{k \geq 0} T_k(h, \dots, h), \qquad \|T_k\| \leq 2C \left(\frac{4e}{rR}\right)^k$$

*whenever $\|h\| \leq r$ and $4e\|h\| < rR$. The coefficients extend with the same bounds to the completion of $X$.*

*Proof.* Associate to a stencil the sequence $(\sum_\nu c_\nu (s_\nu/R)^k)_{k\ge 0}$ with supremum norm. The assumed inequality defines a bounded linear map, of norm at most $C$, from their span to $Y$. Extend it to the closure. The $k$th coordinate vector lies in that closure: the $k$th finite difference at 0, divided by $k!(a/R)^k$, has zero moments below $k$, moment one at $k$, and higher moments tending uniformly to zero as its mesh $a \to 0$. To see the uniformity, split the higher moments at any fixed order and use $|ja/R|^n \le (ka/R)^n$ for the remaining tail. Evaluation on a single point then gives the exact line series. Its coefficient sum at radius $R/2$ is at most $2C$. Uniqueness follows by successively dividing a vanishing series by its first possible power.

Write $P_k(h)$ for the coefficient of $s^k$ in $f(x+sh)$. Rescaling $s$ gives $P_k(ah) = a^k P_k(h)$ and

$$\|P_k(h)\| \le 2C \left(\frac{2\|h\|}{rR}\right)^k .$$

We must prove the polynomial identity in the direction $h$. Put $\delta_h g(a) = g(a+h) - g(a)$. Uniform line remainders at nearby centres imply

$$\delta_{sh}^n f(x+sa) = O(s^n) \quad (s \to 0).$$

Indeed insert the line series at $x+sa$ into the $n$th forward difference. The coefficients below degree $n$ cancel by $\sum_{j=0}^n (-1)^{n-j}\binom{n}{j} j^k = 0$ for $k < n$; the uniformly bounded tail is $O(s^n)$.

For $h_0, \dots, h_{n-1}$ and $T \subset \{0, \dots, n-1\}$ set

$$v_T = \sum_{i\in T} \frac{h_i}{n-i}, \qquad w_T = -\sum_{i\in T} \frac{i h_i}{n-i}.$$

There is the exact mixed-difference identity

$$\delta_{sh_0} \cdots \delta_{sh_{n-1}} f(x+sa) = \sum_T (-1)^{n-|T|} \delta_{sv_T}^n f(x + s(a+w_T)) \,. \tag{13.28}$$

Expand the right side as a sum over its $n+1$ forward-difference points. At the point indexed by $j < n$, pairing sets $T$ differing by the index $i = j$ cancels the terms, since that index has coefficient $(j-i)/(n-i) = 0$. At $j = n$ the coefficient of each $h_i$ is one, leaving precisely the left side. Hence every mixed $n$th difference above is $O(s^n)$.

Expand each of its finitely many values in its line series at $x$ and compare powers of $s$. It follows that

$$\delta_{h_0} \cdots \delta_{h_{n-1}} P_k(a) = 0 \qquad (k < n).$$

For $n = k+1$ this makes the $k$th difference of $P_k$ independent of its base point. Define

$$T_k(h_1, \dots, h_k) = \frac{1}{k!} \delta_{h_1} \cdots \delta_{h_k} P_k(0).$$

The identity $\delta_{u+v} = \delta_u + \delta_v + \delta_u\delta_v$ proves additivity in each slot, since the extra $(k+1)$st difference vanishes. The displayed bound on $P_k$ makes these additive maps locally bounded, hence continuous and real linear in each slot. The operators commute, giving symmetry. Homogeneity gives $\delta_h^k P_k(0) = k! P_k(h)$, so their diagonal is the original line coefficient.

There are $2^k$ terms in this polarization sum, and the argument in each has norm at most $k$ on unit directions. Thus $k^k/k! \le e^k$ gives the stated operator bound. The diagonal series equals $f(x+h)$ by the line recovery; the norm bound proves its convergence on the stated domain. Continuous multilinear maps extend uniquely through the dense inclusion into the completion, with unchanged norm. □

The real parameter estimate yields norm-convergent multilinear expansions in the common completed spaces. The strict path reserve also permits their comparison across cutoffs.

**Lemma 13.18** (Local analytic estimates)**.** *At every point of the real domain above, the centered maps $F_{N,m}$, $\mathcal{D}_{N,m}$ and $F_N^{\mathrm{W}}$ have convergent real multilinear expansions on the ball of radius $1/4$. Their degree-$k$ coefficients, $k \geq 1$, have norms at most $g_*$, $\varrho_m$ and $g_*$, respectively, where $\varrho_m \geq 0$ is independent of $N, p$ and*

$$\sum_{i<m} \varrho_i \leq g_* b_m \,.$$

*At fixed $m, p$ these maps are compatible with cutoff limits in the common spaces, including the prescribed paths with a strictly larger path exponent.*

*Proof.* Use Definitions 3.3, 3.4, 3.2 and Lemmas 13.16 and 13.17. The larger real domain preceding (13.24) supplies the neighbouring centres required in the recovery lemma. Apply it with $r = 1/4$ and $R = \rho/2$ for the activity map, and $R = \rho$ for the scalar correction. Their coefficient bounds are respectively

$$2\delta_{\mathrm{map}}(32e/\rho)^k, \qquad 2\varphi_m(16e/\rho)^k \,.$$

Choose $\rho \geq 128e$ before the ratios. Since $2\delta_{\mathrm{map}} \leq g_*$ and $2\varphi_m = \varrho_m$, these are the asserted bounds. The same argument applies to Wilson. Completion preserves each coefficient, its norm and its diagonal identity. The fixed permutation subspace preserves equivariance.

For cutoff compatibility, compare first the quadratic rows and then the cubic, quartic and quadratic loop kernels in that order. The free block-averaged forms converge by Lemma 11.6; its uniform inverse bound and the resolvent identity give convergence of the rows. In each subsequent comparison fix these rows and all preceding kernels. Let $\mathcal{V}_a$ be the affine coefficient space for the kernel of the current degree, with its prescribed symmetries and cancellation identities. Divide its weighted coefficient norm by the stage allowance. The corresponding coefficient map, denoted $\mathcal{T}_m^a$, obeys

$$\mathcal{T}_m^a(K + \lambda(H - K)) = \mathcal{T}_m^a(K) + \lambda(\mathcal{T}_m^a(H) - \mathcal{T}_m^a(K)) \,. \tag{13.29}$$

For the cubic coefficient this follows by retaining degree three in the stationary substitution: it is linear in the incoming cubic kernel. At degree four, terms quadratic in the cubic kernel are fixed, and the incoming quartic kernel occurs once. The quadratic loop coefficient is affine in its incoming loop kernel, with the preceding classical kernels fixed. These are the degree extractions in Lemmas 12.6–12.9.

Take the ratio threshold of Lemma 12.6 for the enlarged input path bound $64A^{\mathrm{path}}_{J_{m+1}}$, and, for the enlarged quadratic loop bound $64\Lambda^{\mathrm{loop}}_{m+1} \leq 128\Lambda^{\mathrm{loop}}_m$, the threshold in the proof of Lemmas 12.3–12.4 with $C_j^{\mathrm{pb}}\ell^{-2} \leq 1/256$, so that $\frac{1}{2}\Lambda^{\mathrm{loop}}_m + \frac{1}{256} \cdot 128\Lambda^{\mathrm{loop}}_m = \Lambda^{\mathrm{loop}}_m$. These choices precede the coupling threshold. With the preceding kernels fixed, the coefficient constructions then accept normalized input size 64 and give normalized output size at most one. The enlarged input must still belong to $\mathcal{V}_a$ and satisfy these bounds. For admitted inputs $K, H$ of normalized size at most one put $\delta_K = \|H - K\|$. If $\delta_K > 0$, the extrapolated input $K_* = K + (32/\delta_K)(H - K)$ belongs to the same affine space and has norm at most $33 < 64$. Its linear cancellation and symmetry conditions hold because their homogeneous differences form a linear subspace. Hence its stage bound applies. Equation (13.29) gives

$$\frac{32}{\delta_K}\|\mathcal{T}_m^a(H) - \mathcal{T}_m^a(K)\| \leq \|\mathcal{T}_m^a(K_*)\| + \|\mathcal{T}_m^a(K)\| \leq 2 \,.$$

The output difference is at most $\delta_K/16 \le \delta_K/4$; at $\delta_K = 0$ it is zero by the same affine identity. This calculation concerns one kernel, with all preceding kernels fixed.

For two cutoffs, the change of those fixed preceding kernels contributes an error $e_{N,N',i}$ tending to zero at each fixed level $i$, once their preceding comparisons have been made. The normalized difference of the current kernels therefore satisfies $\delta_i \le \delta_{i+1}/4 + e_{N,N',i}$. Truncate this coefficient recursion at level $k$, retaining any admitted terminal kernel there. Its terminal difference is at most two, and iteration gives

$$\delta_m \le 2(1/4)^{k-m} + \sum_{i=m}^{k-1} (1/4)^{i-m} e_{N,N',i}\,.$$

At fixed $k$ the sum tends to zero; then $k \to \infty$ proves the Cauchy property, successively for each degree.

Retain now finitely many addresses and finite path lengths. Window covariances and means on these sets converge with their derivatives by the uniform inverse bound and the resolvent identity. Stationary corrections use this same inverse, and determinant differences are integrals of traces of resolvent differences. Lemmas 12.6–12.9 express the path coefficients by these kernels. Their exponential estimates hold at an exponent $w'$ strictly larger than the output exponent $w$. A tail of length at least $l$ at exponent $w$ is bounded by its uniform $w'$ norm times $e^{-(w'-w)l}$. Thus finite-coefficient convergence and uniformly small tails prove path convergence with a strict reserve.

For the connected weights, subtract the two local integrals using the same coordinates and covariance interpolation as in Lemma 11.13. A finite collection converges by dominated convergence. The rooted estimates bound the discarded collections uniformly. Localization uses only five fixed auxiliary tori, finite linear combinations and such uniformly bounded tails. Subtraction is linear, and fusion has a strict attachment margin. Consequently input coordinates converging in (13.22) and converging normalized couplings give outputs and corrections converging in the same common spaces. This proves the compatibility assertion. □

For two histories at the same $N, p$, let $x_m, A_m$ and $x'_m, A'_m$ be these coordinates and set

$$\mathfrak{d} = \max_{0\le m\le N} \max\{|x'_m - x_m|, \|A'_m - A_m\|_{m,p}\}, \qquad \delta = \beta'_0 - \beta_0\,. \tag{13.30}$$

The parameter series below belong to $\mathcal{S}(X) = \{\sum_{k\ge 0} z^k X_k : \sum_k \|X_k\| < \infty\}$, with norm $\sum_k \|X_k\|$. Evaluation has norm at most one for real $|z| \le 1$. This space is complete, and its Cauchy product is bounded by the product of the norms.

For the series equations fix a history reaching $\bar\beta$ and set $z_m = \chi_m x_m$. Write $u_m$ for the perturbation of $x_m$, $W_m$ for that of $A_m$, and $\mathfrak{e}$ for the endpoint series. For any map $F$ let $\widehat F$ be its centered series. The simultaneous equations are

$$\begin{aligned} W_N &= \widehat F^{\mathrm W}_N(u_N), \\ W_m &= \widehat F_{N,m}(u_{m+1}, W_{m+1}) \quad (m < N), \\ u_m &= (2z_m)^{-1}\left[ b_m^{-1}\left( \mathfrak{e} - \sum_{i<m} \widehat{\mathcal{D}}_{N,i}(u_{i+1}, W_{i+1}) \right) - \chi_m u_m^2 \right]. \end{aligned} \tag{13.31}$$

The last line is obtained by expanding the square in (13.23); it retains its quadratic term.

These right sides define a map on the product of the spaces $\mathcal{S}(\mathbb{R})$ and $\mathcal{S}(\mathcal{X}_{m,p})$, with the maximum of their series norms. The real endpoint interval is

$$\mathfrak{J} = [\bar\beta - \lambda_* b_*/4, \bar\beta + \lambda_* b_*/4]\,.$$

The following is the estimate used by Section 14. It applies to the selected step maps, their actual Wilson initial family and the same fixed address spaces.

**Lemma 13.19** (Analytic endpoint equations)**.** *The histories above satisfy*

$$\mathfrak{d} \le |\delta|/b_* + 8g_*\mathfrak{d}\,.$$

*For* $\|\mathfrak{e}\|_{\mathcal{S}(\mathbb{R})} \le \lambda_* b_*/4$*, equations* (13.31) *preserve the closed radius-*$\lambda_*$ *ball in the product of the spaces* $\mathcal{S}(\mathbb{R})$ *and* $\mathcal{S}(\mathcal{X}_{m,p})$ *and contract by at most* $1/2$*. Their unique solution evaluates to the selected real recursion. At fixed* $m, p$*, its scalar couplings and activity coordinates converge along the full cutoff sequence for every convergent real endpoint sequence in* $\mathfrak{J}$*.*

The maximum estimate and the series contraction are proved next; the cutoff conclusion is proved in Lemma 13.20.

*Proof of the comparison and series assertions.* Use Definitions 3.3, 3.4, 3.2, Lemmas 12.10, 13.14 and 13.18. The latter's centered expansions give

$$\begin{aligned} \|A'_m - A_m\| &\le 8g_* \max\{|x'_{m+1} - x_{m+1}|, \|A'_{m+1} - A_{m+1}\|\}, \\ |\mathcal{D}'_{N,m} - \mathcal{D}_{N,m}| &\le 4\varrho_m \max\{|x'_{m+1} - x_{m+1}|, \|A'_{m+1} - A_{m+1}\|\}\,. \end{aligned} \tag{13.32}$$

For distances at most $1/4$ these follow by subtracting the series and summing $k(1/4)^{k-1}$. For larger distances the separate bounds $\|A_m\| \le g_*/2$ and $|\mathcal{D}_{N,m}| \le \varrho_m/2$ give the displayed, weaker constants. The Wilson top map has the first bound as well.

Write $z_m = \chi_m x_m$ and $z'_m = \chi_m x'_m$ in this proof. Both are in $[17/20, 23/20]$. Equation (13.23) gives the exact identity

$$\beta'_m - \beta_m = b_m(x'_m - x_m)(z'_m + z_m)\,.$$

Summing the scalar recursion from the endpoint cancels $\alpha_{N,i}$ between the two histories, and hence

$$b_m|x'_m - x_m| \le |\delta| + \sum_{i<m} |\mathcal{D}'_{N,i} - \mathcal{D}_{N,i}| \le |\delta| + 4g_* b_m \mathfrak{d}\,.$$

Take the maximum together with the activity estimates and use $b_m \ge b_*$. This proves the asserted inequality before absorbing its last term. In particular $\delta = 0$ gives $\mathfrak{d} = 0$, and otherwise $\mathfrak{d} \le 2|\delta|/b_*$. Thus real evaluation can be identified by uniqueness.

On the product ball of radius $\lambda_*$ the series pair $(u_m, W_m)$ has norm at most $2\lambda_* < 1/4$. A centered series with coefficients bounded by $c$ consequently has norm at most $c/2$, and the difference of its two evaluations is bounded by $2c$ times the difference of its arguments. For the scalar prefix the latter gives $4\sum_{i<m} \varrho_i$ times the product distance, because the two coordinates contribute at most twice that distance. Also

$$\|u_m^2 - v_m^2\| \le 2\lambda_* \|u_m - v_m\|\,.$$

Since $2z_m \ge 1$ and $\chi_m \le 1$, the scalar part of (13.31) has norm at most

$$\lambda_*/4 + g_* + \lambda_*^2 < \lambda_*$$

and Lipschitz constant at most $4g_* + 2\lambda_* = 5/32 < 1/2$. The activity part has norm at most $g_*/2 < \lambda_*$ and Lipschitz constant at most $4g_* < 1/2$. Thus the map preserves the closed ball and contracts there. Starting at zero, its successive differences are bounded by a geometric series with ratio $1/2$; completeness gives its unique fixed point.

Evaluation commutes with sums, products and each absolutely convergent multilinear series. It turns the fixed point into the real recursion and Wilson start. The comparison already proved identifies it with the selected history. The endpoint width $\lambda_* b_*/4$ is smaller than the real shooting width: $b_* = \chi_0 \bar{\beta} \le \bar{\beta}/128$. Hence the whole closed real interval stays in that shooting interval. □

Only the endpoint coupling is compact in the next argument; no compactness of a ball in the activity Banach space is used.

**Lemma 13.20** (Scalar-free fixed-level limits)**.** *For fixed $p, m$, the selected scalar couplings and activity coordinates in $\mathcal{X}_{m,p}$ converge along every full cutoff sequence with convergent endpoints in the interval of Lemma 13.19. The prescribed paths converge with the strict reserve of Lemma 13.18. The limits are independent of subsequences. No additive vacuum scalar is included.*

*Proof.* Use Definitions 3.3, 3.4, 3.2, Lemmas 12.3–12.10, 13.18 and 13.19. First extract a subsequence on which every scalar coordinate $x_i$ converges. Each fixed coordinate is in a compact real interval by $17/20 \le \chi_i x_i \le 23/20$; a diagonal selection gives the simultaneous subsequence.

Fix a stopping index $k > m$. At level $k$ replace the activity coordinate by zero and run the same selected maps downward using the unchanged scalar coordinates. Write $A_m^{[k]}$ for the resulting coordinate. The first inequality in (13.32), applied successively at equal scalar inputs, gives

$$\|A_m - A_m^{[k]}\|_{m,p} \le (8g_*)^{k-m}\,. \tag{13.33}$$

The terminal difference is at most one. For fixed $k$ the truncated coordinates converge by a finite downward induction, using cutoff compatibility in Lemma 13.18. The geometric bound is uniform in the cutoff and tends to zero as $k \to \infty$. Given $\varepsilon > 0$, choose $k$ to make both truncation errors smaller than $\varepsilon/3$, then choose the cutoffs so that the truncated difference is smaller than $\varepsilon/3$. Thus the actual activity coordinates are Cauchy in their common completed spaces on the extracted subsequence. The scalar corrections converge by the second inequality in (13.32) and cutoff compatibility.

It remains to identify two possible subsequential limits. Denote them by $x_i, A_i, \mathcal{D}_i$ and $x_i', A_i', \mathcal{D}_i'$. Their common endpoint is $\beta_0$. Passing to the limit in the finite scalar recursion gives

$$\beta_i - \beta_i' = -\sum_{j<i} (\mathcal{D}_j - \mathcal{D}_j')\,. \tag{13.34}$$

The limits of $\alpha_{N,j}$ cancel because the prescribed paths have one limit. Put

$$\mathfrak{d}_\infty = \sup_{i \ge 0} \max\{|x_i - x_i'|, \|A_i - A_i'\|_{i,p}\}\,.$$

This supremum is finite with bound $1/4$. Indeed the activity difference is at most $1/4$, and (13.34) together with $|\mathcal{D}_j|, |\mathcal{D}_j'| \le \varrho_j/2$ gives $|x_i - x_i'| \le g_*$, after division by $b_i$ as in the previous proof. Passing the one-step estimates to the limit gives $\|A_i - A_i'\| \le 8g_*\mathfrak{d}_\infty$. Equations (13.34) and (13.32) give $|x_i - x_i'| \le 4g_*\mathfrak{d}_\infty$. Hence $\mathfrak{d}_\infty \le 8g_*\mathfrak{d}_\infty$, so $\mathfrak{d}_\infty = 0$.

Every subsequence therefore has a further subsequence with these same scalar and activity limits. If the full sequence failed to converge at some fixed level, a subsequence staying a positive distance from the limit would contradict the preceding construction. This proves full convergence. The continuous map $x \mapsto \widehat{\beta}_m(\chi_m x)^2$ gives convergence of $\beta_m$. The path assertion is already part of the cutoff comparison, with a strictly larger exponent than the one defining the local charts. None of these arguments compares the vacuum scalars, which disappear from the normalized activity and coupling equations. □

## 13.7 The Wilson start

At the finest level the identity free form permits a local choice. Put $\delta_{\mathrm{dir}} = c_{\mathrm{dir},J_N}$ and $F(u) = \mathrm{proj}_{\mathfrak{g}}(u - u^{-1})/2$. To define the quartic term, let $\psi$ be the local inverse of $X \mapsto F(e^X)$ at zero and set

$$T_4(Y) = \frac{1}{4!} D^4[\mathrm{pr}(e^{\psi(\cdot)})](0)[Y,Y,Y,Y]\,.$$

The inverse exists since $DF(e^{\cdot})(0) = I$. This invariant quartic polynomial is chosen so that

$$R(u) = \mathrm{pr}(u) - \tfrac{1}{2}|F(u)|^2 - T_4(F(u)) \tag{13.35}$$

has zero Taylor terms through degree four at 1. This is the case $m = N$ in the proof of Lemma 12.6. The path action of Definition 12.1 at $m = N$ has row $\sqrt{1-\delta_{\mathrm{dir}}}\,F$, no cubic or loop terms, and quartic term $(1-\delta_{\mathrm{dir}})T_4(F)$ with its prescribed mask. These finite path sums satisfy Lemma 12.3.

The initial activities retain the difference between Wilson energy and this action also on flagged configurations. The following proof constructs the full curve. The same construction of the classes is used throughout.

**Lemma 13.21** (Wilson initialization)**.** *For every $N, p$ there is a continuous curve $\beta \mapsto \rho_\beta^{\mathrm{W}} \in \mathcal{I}_{N,p}$ on $\mathfrak{B}_N$, with coupling $\beta$, vacuum scalar zero, and*

$$\mathrm{dens}\,\rho_\beta^{\mathrm{W}} = e^{-\beta E^{\mathrm{W}}_{L_p D_N}}\,.$$

*The curve has the activity bounds of Lemma 13.11 and the continuity of Definition 13.3.*

*Proof of the initialization identity, bounds and continuity.* Use Definitions 2.1, 3.2, 13.1–13.3, and Lemmas 12.3, 12.4, 12.6. Everything here is at level $N$: $j = J_N$, $t = t_N$, $B = B_N$ and $\sigma = \sigma_{J_N}$. For a plaquette $q$, write $\rho_q = \mathrm{ch}(U_q)$, $k_q^{\mathrm{kin}} = (1-\delta_{\mathrm{dir}})|F(U_q)|^2/2$, $h_q = h_{\mathrm{vertices}(q)}$, and $\chi_q$ for its relaxed health test. Define its unassigned action by

$$\mathcal{W}_q(U) = \begin{cases} \mathrm{pr}(U_q) - c_{\mathrm{lin}} t \rho_q, & \rho_q > t, \\ h_q k_q^{\mathrm{kin}} + (1-h_q)(1-\delta_{\mathrm{dir}})\,\mathrm{pr}(U_q) - \chi_q k_q^{\mathrm{kin}}, & \rho_q \le t. \end{cases}$$

The relaxed health test $\chi_q = 1$ implies $\rho_q \le t$. The identity $(u-u^{-1})/2 = (u-1)(1+u^{-1})/2$ and contraction of the projection give $|F(u)| \le \mathrm{ch}(u)$. Together with $\mathrm{ch}(u)^2 = 2\,\mathrm{pr}(u)$ this gives $\mathcal{W}_q \ge 0$. For $\rho_q > t$, $c_{\mathrm{lin}} < \delta_{\mathrm{dir}}/16 \le 1/16$ gives $\mathcal{W}_q \ge \rho_q^2/4 \ge t^2/4$. Substitution in the two cases gives

$$\mathrm{pr}(U_q) = L_q(U) + (1-\delta_{\mathrm{dir}})h_q T_4(F(U_q)) + (1-\delta_{\mathrm{dir}})h_q R(U_q) + \mathcal{W}_q(U)\,. \tag{13.36}$$

Here $L_q$ is the direct clipped term plus the health-tested row square. The identity includes small plaquettes next to a flag and both sides of the clipping threshold.

Take the regular activities $\beta(1-\delta_{\mathrm{dir}})R(U_q)$ on their plaquette supports, anchored at the starting vertex, with the signed average of the path action. They vanish at 1 with their Hessians. Through the prescribed order their normalized healthy derivatives are bounded by $C_j\beta t^5$. To verify this without assuming small links, use the plaquette holonomy as one coordinate and the other three links as compact parameters. The remainder vanishes to order five in the holonomy chart. At derivative order $k \le 5$, Taylor's formula leaves order $5-k$ and the direction factors supply $t^k$. For $k > 5$, bounded derivatives and $t^k \le t^5$ give the same estimate. The four-link formula controls the

remaining compact factors. There are finitely many plaquettes at each anchor, so the regular sum is at most $C_j \beta t^5$. On the top band,

$$\frac{\widehat{\beta}_N t_N^5}{\epsilon_N^{\mathrm{reg}}} = \widehat{\beta}_N^{-2/5} (M_N^{\mathrm{cut}})^5 \longrightarrow 0 \,.$$

Increasing the entry threshold gives the regular bound $\epsilon_N^{\mathrm{reg}}/1024$ uniformly in volume.

Let $\mathfrak{F} = \mathrm{Flag}_N(U)$. Join two flags when their radius-$2B$ neighborhoods overlap. For each nonempty connected inventory $I$ use $Y_I = N_{2B}(I)$ and

$$K_I(U) = \mathbf{1}_{\{Y_I \cap \mathfrak{F} = I\}} e^{-\beta d_I(U)} \,.$$

The charge $d_I$ is the signed average of those $\mathcal{W}_q$ whose anchors lie within distance $B$ of $I$, in the coordinates of that signed image. Every nonzero $\mathcal{W}_q$ is assigned: a failure of the relaxed health test has a witnessing flag within this distance, since $t \exp((B-2)/R_{J_N}^{\mathrm{rel}}) \geq 2$. The other cases already meet a flag at a plaquette vertex. The anchors lie within $B$ of $I$, and each nonautomatic relaxed health test reads within $B-2$ of its anchor. Thus $N_{2B}(I)$ contains every link and flag used in the charge. Distinct actual components have disjoint supports, so each nonzero term is assigned once. Hence the sum of the component charges is the signed average of $\sum_q \mathcal{W}_q$.

Every flag touches a plaquette with $\rho_q > t$, and a plaquette has at most four vertices. Retaining the identity orientation in the nonnegative signed average gives

$$d_I(U) \geq \frac{t^2 |I|}{16 |H_4|} \qquad (I \subset \mathfrak{F}) \,.$$

A nonzero exact-cover product of $K_I$ selects precisely the overlap components of $\mathfrak{F}$. Indeed any two flags in one selected connected inventory are connected in $\mathfrak{F}$. Conversely a flag joined to a selected inventory cannot lie in another selected inventory, since their radius-$2B$ supports would intersect. Exact coverage then forces every component to be selected. The actual components satisfy eligibility. Thus the forced sum has exactly one nonzero term, equal to the exponential of minus $\beta$ times the unassigned action. Equation (13.36), summed and averaged, proves the Wilson density identity for all configurations with scalar zero.

For the forced norm, a connected inventory of $r$ sites is encoded by a rooted spanning walk of length at most $2(r-1)$ in the overlap graph. Each step has at most $C(1+B)^4$ choices, and $\mathrm{cov}_B(N_{2B}(I)) \leq 5^4 r$. Counting possible roots for a support through a specified site costs another $C(1+B)^4$. With $c = 1/(16|H_4|)$, the weighted sum is therefore bounded by

$$C_j (1+B)^4 \sum_{r \geq 1} \left[ C_j (1+B)^8 e^{5^4 \sigma - c\beta t^2} \right]^r \,.$$

The top band gives $\beta t^2 \geq (M_N^{\mathrm{cut}})^2/2$. This dominates both $\log(1+B)$ and the exponent in $\epsilon_N^{\mathrm{for}}$, so the geometric sum is eventually at most $\epsilon_N^{\mathrm{for}}/1024$, uniformly in volume. Both normalized bounds can be made arbitrarily smaller by increasing the entry threshold.

Addresses and tests are independent of $\beta$. Regular differences have the same majorants with $\beta$ replaced by $|\beta - \beta_0|$. For the forced functions $d_I \geq 0$ is bounded at fixed volume and

$$|e^{-\beta d_I} - e^{-\beta_0 d_I}| \leq |\beta - \beta_0| \sup_U d_I(U) \qquad (\beta, \beta_0 \geq 0) \,.$$

There are finitely many subsets of this torus, so the weighted difference sums tend to zero. Translations and signed permutations transport complete plaquette terms, component supports,

inventories and tests. This gives covariance as well as continuity. We have proved the Wilson representation, both bounds, and its full continuous curve with scalar zero. The common one-step assertion follows from the assembly below. □

*Proof of Theorem 3.5.* Use Definitions 3.1, 3.2, 13.3, and the classes of Lemma 13.14. Lemmas 13.11, 13.8–13.13 give the final functions and their exact normalized identity. Put

$$v_u = |\Lambda|^{-1} \log \mathcal{H}_u(1), \qquad \beta_u = \beta'_u + \alpha_{N,m} + \Delta_u .$$

The denominator is positive by Lemma 13.8. Lemma 13.13 gives continuity of $v_u$ and of the activity families. The proof of Lemma 12.10 identifies this same $\Delta_u$ with its continuous scalar root; its drift estimate gives

$$\beta_u = \beta'_u - \kappa_G \log \ell_m + e_u, \qquad |e_u| \le \omega^{\mathrm{RG}}_{J_{m+1}} .$$

Define $\rho_u$ by these scalars and the activities in Definition 3.2. Multiply the identity of Lemma 13.12 by $\mathcal{H}_u(1)$ to obtain $\operatorname{dens} \rho_u(U) = \mathcal{H}_u(U)$ for every configuration. Lemma 13.11 gives the activity caps and Lemma 13.13 gives continuity. The retained conditions and the curve construction of Lemma 13.14 give $\rho_u \in \mathcal{A}_{N,m,p}$ and $(\rho_u, \rho'_u) \in \mathcal{R}_{N,m,p}$. The Wilson construction above supplies the common initial curve with scalar zero. □

## 14 Trajectories, marked fields and ultraviolet convergence

This section constructs the trajectories and the renormalized insertions used in Part I, and proves their volume-uniform bounds and full fixed-volume limits. The conclusions are Theorem 3.6, Proposition 4.1 and Theorem 4.2. We also establish the cutoff equation identities, coefficient corrections and finite rank comparisons used to construct the quotient fields in Section 15. All these results concern the same Wilson measures of Definition 2.2 and the insertions of Definition 3.9.

There are three different arguments. A continuous shooting argument first chooses the bare coupling to reach one prescribed coarse coupling. Weighted estimates for the specified recursion then give uniqueness and analytic dependence near that endpoint. Finally, at fixed volume, convergence of the effective coordinates at each fixed level is combined with a bound on the error made by stopping the insertion transfer there. The additive vacuum scalar cancels from normalized expectations; its convergence is unnecessary.

The main difficulty in the bounds is differentiation in position. Each lattice difference introduces an inverse spacing, so a bound for undifferentiated insertions does not suffice. Local comparison replaces a differentiated insertion by one of higher degree, with a small chessboard error. Exact equation identities remove its equation component. A finite matrix of free Maxwell correlation tests bounds the remaining coefficient. This argument takes place at finite cutoff, before any continuum field space has been constructed.

Throughout, the group, metric, penalty, scales and admissible classes are those of Sections 3 and 9. We use the paths of Lemmas 12.3–12.4 and the Wilson initialization of Lemma 13.21 and the inductive step of Theorem 3.5. Additional dependence of constants is indicated. Write $r(d) = m_{\mathrm{ref}}(d)$, $M_{N,p} = M^{\mathrm{lat}}_{N,p}$ and $\langle F \rangle_{N,p} = \int F \, d\mu_{N,p}$.

For a token $v = (d, c)$ put $O_{N,p}(v, y) = O^{(d)}_{N,p}(c, y)$. Physical points $x$ in a moment are read as lattice sites $x/a_N$, modulo $L_p D_N$. We use the torus separation

$$\mathrm{sep}_{\mathrm{tor}}(y) = \min_{i \ne j} \min_{z \in L_p \mathbb{Z}^4} |a_N(y_i - y_j) + z|_\infty ,$$

with value $+\infty$ for at most one slot. Multilinear interpolation on the physical grid means the affine interpolation in each coordinate of each grid cube, using its vertex values.

## 14.1 Trajectories and tuning

The entry levels allow the ratios and stage constants to increase sufficiently slowly. Besides the conditions of Section 9, we use the following quantitative choices. Let $c_{\mathrm{dir},j}$ be the direct clipping coefficient at stage $j$, and let $\nu_G^{\mathrm{mat}}$ be the real dimension of the ambient matrix space. Put

$$Q_j(b) = 2(2+4A_j^{\mathrm{path}})+1+6(c_{\mathrm{dir},j}+(A_j^{\mathrm{path}})^2)+4\nu_G^{\mathrm{mat}}(\log 16+\tfrac{1}{2}\log b), \qquad q_m = Q_{J_m}(\widehat{\beta}_m). \tag{14.1}$$

Here $A_j^{\mathrm{path}}$ is the path bound of Lemma 12.3. For the unmarked density on $V$ sites write $\Phi_m = v_{\mathrm{vac},m}V - P_{N,m}^{\mathrm{path}} - \sum_i \mathbf{1}_{X_i \text{ healthy}} E_i^{\mathrm{reg}}$. To verify the normalization cost, put $\widehat{g}_m = \widehat{\beta}_m^{-1/2}$ at this level. The scalar-free unmarked exponent is at most $VE_m$, and its absolute forced sum is at most $e^{V\epsilon_m^{\mathrm{for}}}$, where the path bounds give $E_m \le 2 + 4A_{J_m}^{\mathrm{path}}$. Restrict the denominator to links of chord at most $\widehat{g}_m/4$. There are no flags there, the forced sum is one, and its scalar-free exponent is at least $-V(E_m + 6(c_{\mathrm{dir},J_m} + (A_{J_m}^{\mathrm{path}})^2))$. The elementary Haar-ball bound gives this set measure at least $(\widehat{g}_m/16)^{4V\nu_G^{\mathrm{mat}}}$. Thus

$$\begin{aligned} &\frac{\int e^{\Phi_m(U)} \sum_{\mathcal{A}} \prod_{Y\in\mathcal{A}} |K_{m,Y}(U)|\, dU}{\int \mathrm{dens}\, \rho_m\, dU} \\ &\quad \le \exp\Big\{ V\Big[2E_m + \epsilon_m^{\mathrm{for}} + 6(c_{\mathrm{dir},J_m} + (A_{J_m}^{\mathrm{path}})^2) + 4\nu_G^{\mathrm{mat}} \log(16/\widehat{g}_m)\Big]\Big\} \le e^{Vq_m}. \end{aligned} \tag{14.2}$$

The common vacuum scalar in numerator and denominator cancels. This estimate uses all four links per site without gauge fixing; the sharper gauge-fixed calculation in Lemma 14.23 has a different purpose. Enlarge $Q_j$ for any finite initial levels, so that $q_m \ge 1$ at every level. The choices include

$$B_{i+1}^2 \le 2B_i^2, \qquad (\ell_i B_i^2)^4 q_{i+1} = o(s_{i+1}), \qquad \varpi_{J_i} = o(s_i). \tag{14.3}$$

Define $k(m)$ to be the first $k \ge m$ with $B_k^2 \le a_m/a_k$.

The short-distance comparison uses the following intervals of levels. For $b > 0$ and a level $t$ in stage $j$, write $L = \ell_j^{\mathrm{st}}$ and

$$h_t(b) = \left\lceil \frac{(b+2)\log M_t^{\mathrm{cut}}}{\log L} \right\rceil, \quad v_t = \left\lceil \frac{10 \log B_t}{\log L} \right\rceil, \quad k_t = t - h_t(b) - v_t. \tag{14.4}$$

We collect the numerical requirements used in the marked proofs. For each stage $j$, let $K_j \ge 1$ be a nondecreasing bound for the leading coefficient maps and their inverses through that stage's finitely many degrees and symmetry averages. These bounds are fixed before the column bounds $C_j$. Let $A_d^{\mathrm{sym}}$ bound the finite basis recombination in degree $d$. Let $C_j^{\mathrm{loc}}$ bound the local marked error before choosing the ratio. The constant $\mathcal{C}_j \ge 2$ includes these bounds and the finite source comparison constants. Each step below has $j = J_{m+1}$, $K = K_j$, $S = C_j$, $L = \ell_m$ and $\varepsilon_L = L^{-3/8}$.

1. Choose nondecreasing columns with $C_j \ge 64K_{j+1}^2$, above the derivative factorials and finite basis-recombination constants at stage $j$. The support exponent is $\sigma_j = 100$; we also retain the estimates at exponent 98.

2. Choose $L^{3/5} > 2C_j^{\mathrm{loc}}$. The local marked error $\eta_m$ satisfies $\eta_m \le L^{-2/5}$, and

$$\eta_m(1+S)A_d^{\mathrm{sym}} \le \varepsilon_L, \quad 2K\varepsilon_L(1+S) \le 1, \quad 4S\varepsilon_L \le 1, \quad 2K \le S, \quad 4K^2 \le C_{J_m}. \tag{14.5}$$

3. For local matching require

$$4K\, 32(1+S)^2\varepsilon_L \le 1, \qquad (4K^2+1)L^{1/8}\varepsilon_L \le 1. \tag{14.6}$$

For each auxiliary section through the stage's degrees, require also $2(1+C_j)H_{j,d}A_d^{\mathrm{sym}}L^{-1/8} \le 1/4$, where $H_{j,d}$ bounds its normalized healthy derivatives.

4. Choose ratios to satisfy (14.8), including every constant in $\mathcal{C}_j$.

5. On a comparison interval, put $\alpha_t = (M_t^{\mathrm{cut}})^{1/16}$. Let $\mathcal{E}_t^{\mathrm{G}}$ be the sum of the covariance, rounding, Taylor and localization errors in the fixed finite source comparison. Require

$$\alpha_t^2 \mathcal{C}_j^{h_t(b)+v_t+2}\mathcal{E}_t^{\mathrm{G}} \le 1, \qquad Q_j^{\mathrm{cmp}} \le \log M_t^{\mathrm{cut}}/(16n), \qquad \widehat{\beta}_i \in [\widehat{\beta}_t/2, \widehat{\beta}_t] \quad (k_t \le i \le t). \tag{14.7}$$

Here $Q_j^{\mathrm{cmp}}$ is the stage normalization constant in (14.42), enlarged to cover the source modes and amplitudes of the finitely many arity bounds $n \ge 2$ admitted at stage $j$, and the second bound is required for each of these $n$. These constants are fixed first; the entry is then raised until $\log M_t^{\mathrm{cut}} \ge 16n\, Q_j^{\mathrm{cmp}}$. The first bound is imposed with the stronger smallness used in (14.40). The entry also meets the local marked bound in item 2 and the forced-source bounds at every exponent between 98 and 100.

The finite comparisons in item 5 exhaust bounded separated configurations and finite derivative orders as $j$ increases. Their errors and constants are calculated in Lemma 14.22; none depends on the period.

The following choice gives both the sparse insertion grids and the intervals used for free comparison.

**Lemma 14.1** (Scale and period choices)**.** *All requirements 1–5 and* (14.3) *can be met with*

$$\ell_m \ge 3 \text{ odd}, \qquad \log \ell_m \to \infty, \qquad \log \ell_m/s_m \to 0, \qquad 4 \mid L_p,$$
$$(a_m/a_{k(m)})^4 q_{k(m)} = o(s_{k(m)}).$$

*The sides $L_p$ increase strictly; every positive integer divides infinitely many. For every fixed $b \ge 1$ and finite comparison request, arbitrarily late intervals* (14.4) *meet the comparison requirements, with $k_t \to \infty$ and $J_i = J_t$ throughout.*

*Proof.* Use the scale definitions of Section 3 and the order of choices in Section 9. First fix the entire sequence $K_j$ and the finite section bounds. Choose $C_j \ge 64K_{j+1}^2$, and then fix $\mathcal{C}_j$ to include these columns and all source constants. Since $J_{m+1} \le J_m + 1$, this gives $4K_{J_{m+1}}^2 \le C_{J_m}$ even at a stage transition. Next choose nondecreasing odd stage ratios $\ell_j^{\mathrm{st}} \ge \max(3, j)$ large enough that

$$\frac{\omega_j^{\mathrm{RG}}}{\log \ell_j^{\mathrm{st}}} \le \min(\kappa_G/4, j^{-2}), \qquad \frac{\log \mathcal{C}_j}{\log \ell_j^{\mathrm{st}}} \le j^{-2}. \tag{14.8}$$

Each denominator can be increased after its numerator is fixed. The ratio requirements in items 2 and 3 are lower bounds on this same later choice: $L^{-2/5}/L^{-3/8} = L^{-1/40} \to 0$ and $L^{1/8}L^{-3/8} = L^{-1/4} \to 0$. Choose also $L^{3/5} > 2C_j^{\mathrm{loc}}$, so $C_j^{\mathrm{loc}}/L < \frac{1}{2}L^{-2/5}$. For this fixed $L$, raise the coupling entry until the absolute value of the $o_{\widehat{\beta}_m}(1)$ term is at most $\frac{1}{2}L^{-2/5}$. The local error $C_j^{\mathrm{loc}}/L + o_{\widehat{\beta}_m}(1)$ is then at most $L^{-2/5}$. The column and ratio choices also give the fixed auxiliary-section inequality. Thus no column is chosen in response to a ratio or to a later-stage column.

Choose consecutive constant-stage intervals, postponing each new stage until all its fixed coupling requirements hold. On a fixed stage $s_i$ grows linearly, while $B_i = O((\log(2+s_i))^2)$ and $q_i = O(1+\log(2+s_i))$. Thus

$$(\ell_i B_i^2)^4 q_{i+1}/s_{i+1} \longrightarrow 0 \quad \text{on that stage as its entry is postponed.}$$

For the initial stage, choose $\bar\beta$ after its fixed ratio $L_0$. For $B(b) = \lceil(\log b)^2\rceil$, the quotient $B(b+\kappa_G \log L_0)^2/B(b)^2$ tends to one as $b\to\infty$. Choose $\bar\beta$ beyond a threshold where it is at most two for every $b\ge\bar\beta$. This covers every step of the first stage, including $i=0$. At a transition to a later ratio, impose the same bound at its entry and thereafter, with that ratio fixed. Impose the second condition of (14.3) with bound $1/j$ after entry to stage $j$. Impose also $\varpi_j \le s_i/j$ and $\log \ell_j^{\mathrm{st}} \le s_i/j$ at its entry. These are lower bounds on that entry. They give both little-oh assertions and $\log \ell_i/s_i \to 0$; $s_i \ge i\log 3$ at every level.

For each stage, the length $h_t(b)+v_t$ in (14.4) is $O_j((\log\log(2+s_t))^2 + \log\log(2+s_t)) = o(t)$. Lengthen that constant-stage interval to contain a terminal point with the whole interval (14.4) in the same stage and with its left endpoint beyond all earlier requested levels. Do this for $b\le j$ and the finitely many comparison orders at stage $j$, on the compact sets of row configurations with coordinate separations at least $1/j$ and coordinates of absolute value at most $j$. Their finitely many derivative bounds are uniform on each such compact set. The estimates (14.38)–(14.40) show that the first requirement in (14.7) holds once that entry is late enough. The second is a lower bound on $\log M_t^{\mathrm{cut}}$ after $Q_j^{\mathrm{cmp}}$ is fixed. The total interval length is $o(t)$, so its coupling variation is $o(\widehat\beta_t)$ and gives the third requirement. These are lower bounds on a later entry or interval length. Each fixed request is then met at arbitrarily large stages. These choices do not alter earlier intervals. Taking $\ell_i = \ell_{J_{i+1}}^{\mathrm{st}}$ gives odd ratios, and $\ell_j^{\mathrm{st}} \ge j$ gives $\log \ell_i \to \infty$.

The level $k(m)$ exists: $a_i B_i^2$ decreases by at least the factor $2/3$ at each step. If $k(m) = u+1 > m$, minimality gives

$$a_m/a_{k(m)} \le \ell_u B_u^2. \tag{14.9}$$

Consequently (14.3) gives $(a_m/a_{k(m)})^4 q_{k(m)} = o(s_{k(m)})$. The case $k(m)=m$ satisfies the same conclusion directly from $q_m = o(s_m)$.

Choose $L_p = 4(M+p+1)$. The two required geometric inequalities are

$$L_p e^{s_m} \ge 100(B_m+1), \qquad L_p e^{s_{m+1}} \ge 100((J_{m+1}+3)\ell_m + 1).$$

The first follows by choosing $M$ large, since $(B_m+1)e^{-s_m}$ is bounded. After canceling $\ell_m$ in the second, postpone each stage until $e^{s_m} \ge 100(J_{m+1}+4)$; enlarge $M$ for the finite initial interval. This proves both inequalities. The periods are strictly increasing, divisible by four, and every integer divides infinitely many of them. □

The next identities retain the normalization of the original Wilson integral through every integration.

**Lemma 14.2** (Exact trajectory identities)**.** *For the trajectories constructed below with endpoint $\bar\beta$ and every $m<N$,*

$$\rho_{N,p,N} = \rho^{\mathrm{W}}_{\beta_{N,p}}, \qquad |\beta_{N,p,m+1} - \beta_{N,p,m} - \kappa_G \log \ell_m| \le \omega^{\mathrm{RG}}_{J_{m+1}},$$
$$\int \mathrm{dens}\, \rho_{N,p,m} = Z_{L_p D_N, \beta_{N,p}}.$$

*All representations are admissible, and the terminal integral has the same value.*

The following proof also proves Theorem 3.6.

*Proof of Lemma 14.2 and Theorem 3.6.* Use the bands and block map of Definitions 3.1 and 3.2, Lemma 14.1, the continuous Wilson curve of Lemma 13.21, and the continuous step of the specified recursion in Lemma 13.14. Set

$$b_m^- = \bar\beta + \tfrac12 \kappa_G s_m, \qquad b_m^+ = \bar\beta + 2\kappa_G s_m.$$

These intervals lie in $\mathfrak{B}_m$. Suppose the fine endpoint couplings are $b^-_{m+1}$ and $b^+_{m+1}$. The drift estimate and (14.8) give coarse endpoint bounds

$$\beta^-_m \le b^-_m - \tfrac{1}{4}\kappa_G \log \ell_m < b^-_m, \qquad \beta^+_m \ge b^+_m + \tfrac{3}{4}\kappa_G \log \ell_m > b^+_m.$$

For a continuous function whose endpoints bracket $[b^-_m, b^+_m]$, restrict between its last lower hit before its first upper hit. Its values then remain in this interval. Restrict the parameter interval accordingly. Every preceding representation is restricted along with the new one, so all already obtained identities persist.

Start at level $N$ with the Wilson curve restricted to $[b^-_N, b^+_N]$ and repeat this operation down to zero. Since $b^-_0 = b^+_0 = \bar\beta$, it produces a trajectory with the required endpoint. The case $N = 0$ is the Wilson representation at $\bar\beta$. All retained representations and adjacent pairs belong to the chosen classes and relation because Theorem 3.5 returns admissible representations whose adjacent pairs lie in the chosen relation.

The exact density identity at each step is the one in Theorem 3.5, now applied only on the retained interval. By Lemma 11.4, integration of the block kernel in its coarse argument is one. Fubini therefore preserves the integral, and the Wilson start gives its displayed value at every level. The path arrays were fixed before $p$ in Lemma 12.3; restricting the parameter interval does not change them.

Summing the drift from zero gives

$$|\beta_{N,p,m} - \bar\beta - \kappa_G s_m| \le \sum_{i<m} \omega^{\mathrm{RG}}_{J_{i+1}}. \tag{14.10}$$

For a fixed splitting level $k$, the tail is at most $\sup_{i\ge k} J^{-2}_{i+1} s_m$ by (14.8). The initial sum is fixed. Divide by $s_m$, let $m \to \infty$ and then $k \to \infty$. Since $\bar\beta/s_m \to 0$, this proves the uniform running law in Theorem 3.6; neither bound depends on $N \ge m$ or $p$. The construction, exact identities and running asymptotic asserted there are all proved. □

The uniqueness estimate concerns the specified recursion, rather than arbitrary representations of a density. Lemma 13.19 supplies its weighted analytic coordinates and their real power-series extension. Section 14.3 uses only its fixed-level convergence assertion.

**Lemma 14.3** (Analytic endpoint tuning)**.** *There is $w_{\mathrm{tun}} > 0$, depending only on the construction, such that, for every $N, p$ and $s \in [-1, 1]$, exactly one bare coupling $\beta_{N,p}(s)$ starts a trajectory of the specified recursion ending at $\bar\beta + w_{\mathrm{tun}} s$. Moreover,*

$$\beta_{N,p}(0) = \beta_{N,p}, \qquad \beta_{N,p}(s) = b_{N,p}(s) \quad (-1 < s < 1),$$

*where $b_{N,p}$ is holomorphic on $|z| < 1$.*

*Proof.* Use Definitions 3.2, 13.3 and Lemma 13.19. Write $\mathcal{D}$ for the maximum distance between the normalized coordinates of two admitted histories, including the coupling coordinate, and $\delta$ for their coarse endpoint difference. That lemma gives constants $b_*, \lambda_* > 0$, independent of $N, p$, and

$$\mathcal{D} \le |\delta|/b_* + 8 g_* \mathcal{D}, \qquad g_* = 1/128. \tag{14.11}$$

For equal endpoints, $\mathcal{D} \le \mathcal{D}/16$ implies equality of the histories and hence of their bare couplings.

Take $w_{\mathrm{tun}} = \lambda_* b_*/4$, where $\lambda_* = 1/16$ is fixed in Lemma 13.19. The local action is affine in the varying coupling; the masks and sector domains are fixed in its analytic coordinates. Its

power-series map preserves the radius-$\lambda_*$ ball and contracts by $1/2$ for endpoint series $w_{\mathrm{tun}}z$. Successive substitution converges in the norm $\sum_j \|u_j\|$ of absolutely summable series. The map preserves real coefficients, so evaluation gives an admitted real trajectory for every $s \in [-1,1]$, pointwise in $s$. This supplies existence on the whole stated interval, including its endpoints. The selected coefficient at $s = 0$ is the trajectory of Lemma 14.2, by uniqueness. The bare-coupling series is holomorphic in $|z| < 1$ and has the claimed real values. All clipping and health conditions are covered by the admitted analytic neighbourhood in Lemma 13.19; they are not differentiated across their boundaries. □

## 14.2 Marked fields and volume-uniform estimates

Fix a degree $d$. The local section $P_d = P_d^{\mathrm{loc}}$ is that of Lemma 10.2, and $\mathsf{D}_t^{\mathrm{fac}}$ is the factor-count scaling of Definition 3.9. Suppress $N, p, d$ while describing one transfer. Write $\rho_m$ for its unmarked representation and $v_{\mathrm{vac},m}$ for its vacuum scalar. Its site torus is $\Lambda_{n_{p,m}}$. An anchor is the lattice site at which the insertion is placed. Denote the marked regular supports by $Z_i$, to distinguish them from the unmarked supports $X_i$. For the moment $v \in V_d$ denotes a bare polynomial coordinate. Prescribe

$$H_N(v) = \operatorname{dens} \rho_N \, P_d[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}} v], \qquad H_m(v) = \ell_m^d \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} H_{m+1}(v). \tag{14.12}$$

These densities keep track of one insertion under exact integration. The formula below describes the coordinates in which they will be represented. Write $x_0$ for the unit coefficient of $x_m(v)$ and $x_m(v)^\circ = x_m(v) - x_0\mathbf{1}$.

**Definition 14.4** (Marked density)**.** A representation of the one-insertion density has the form

$$\begin{aligned} H_m(v) = \operatorname{dens} \rho_m \Bigg[ x_0 + \mathbf{1}_{\Sigma_d \text{ healthy}} P_d[\mathsf{D}^{\mathrm{fac}}_{t_m^{-1}} x_m(v)^\circ] + \sum_i \mathbf{1}_{Z_i \text{ healthy}} R_i \Bigg] \\ + e^{v_{\mathrm{vac},m}|\Lambda_{n_{p,m}}| - P^{\mathrm{path}}_{N,m} - \sum_i \mathbf{1}_{X_i \text{ healthy}} E_i^{\mathrm{reg}}} \sum_{\mathcal{A} \text{ eligible}} \sum_{i \in \mathcal{A}} Q_i \prod_{j \in \mathcal{A}\setminus\{i\}} K_j^{\mathrm{for}}. \end{aligned}$$

Here $R_i, Q_i$ are the marked regular and forced remainders; eligibility is that of Definition 3.2. The section and footprint are translated to the anchor. The remainders restore the exact density in (14.12).

The marked regular norm below is anchored at the insertion site, taken to be zero; it is not anchored separately at each support. For selected majorants $e_i \ge \|R_i\|_m$, $k_i \ge \sup|Q_i|$, set

$$\mathcal{E}_m = \sum_i e^{\sigma_{J_m} \operatorname{cov}_{B_m}(Z_i)} e_i, \qquad \mathcal{K}_m = \sum_i e^{\sigma_{J_m} \operatorname{cov}_{B_m}(Y_i)} k_i, \qquad \|g_m\|_{\mathrm{rem}} = \mathcal{E}_m + (\epsilon_m^{\mathrm{for}})^{-1/2} \mathcal{K}_m. \tag{14.13}$$

The derivative seminorm is Definition 3.3. A coefficient graph at level $m$ is a linear choice $x \mapsto g_m(x) = (R(x), Q(x))$ of such remainders, with the supports fixed and $\|g_m(x)\|_{\mathrm{rem}} \le C_{J_m}|x^\circ|_1$. Its purpose is to use the current polynomial coordinate to bound the entire inserted density.

The following step retains the complete inserted density while reexpressing its remainders in the new polynomial coordinate. The inverse estimate is essential: a bound in the original bare coordinate would accumulate at every step.

**Lemma 14.5** (One marked step)**.** *Above $r(d)$, exact integration sends the selected fine coefficient graph to a coarse coefficient graph. Its coefficient map $L_m$ is invertible and satisfies*

$$|L_m x|_1 \le C_{J_{m+1}} \ell_m^d |x|_1, \qquad |x|_1 \le C_{J_{m+1}} |L_m x|_1,$$
$$\|g_m(L_m x)\|_{\mathrm{rem}} \le C_{J_m} |(L_m x)^\circ|_1.$$

*The graphs are linear, vanish on the unit, and respect conjugation and signed permutations. Writing $u = L_m x$, fine graphs differing by at most $tC_{J_{m+1}}|x|_1$ give coarse graphs differing by at most $tC_{J_m}|u|_1/2$.*

The proof gives the construction and the bounds, including the contraction between two choices of incoming remainders.

*Proof.* Use the local sections of Definition 3.8, the exact block map of Definition 3.1, and the expansion and local subtraction of Lemmas 13.4, 13.8 and 13.12. Put $L = \ell_m$, $s = C_{J_{m+1}}$, $C = K_{J_{m+1}}$ and $\kappa_m = (\epsilon_m^{\mathrm{for}})^{1/2}$. All norms in this proof are coefficient sums or the anchored norm (14.13).

*Step 1: Marked terms in the local integration.* Introduce $z^2 = 0$. Replace the regular exponential factor by its product with $1 + z(x_0 + \mathbf{1}_{\Sigma_d \text{ healthy}} P_d[\mathsf{D}^{\mathrm{fac}}_{t_{m+1}^{-1}} x^\circ] + \sum_i \mathbf{1}_{Z_i \text{ healthy}} R_i)$ and replace a forced factor $K_i$ by $K_i + zQ_i$. The coefficient of $z$ in the eligible covering sum is exactly Definition 14.4. Each connected term in Lemma 13.4 now contains one distinguished factor. For a term with ordinary factors $a_1, \dots, a_h$, its marked coefficient is

$$[z] \prod_{i=1}^{h} (a_i + z b_i) = \sum_{i=1}^{h} b_i \prod_{j \ne i} a_j.$$

Root its connected support at this distinguished factor. The sums over all other factors are bounded by the rooted sums in Lemma 13.4; there is no sum over possible translates of the marked anchor. Absolute convergence justifies the displayed extraction and integration term by term. A regular mark has its derivative majorant $e_i$; a forced mark has its amplitude majorant $k_i$. The latter retains its original eligibility event, so the large-field estimate used in the forced expansion still applies.

The three contributions to a basis input are bounded as follows. Denote by $T_m$ the invertible leading coefficient change from the local Gaussian polynomial calculation, and by $\mathsf{S}_L$ the diagonal map

$$(\mathsf{S}_L x)_i = L^{d - w_i} x_i, \qquad 0 \le w_i \le d,$$

where $w_i$ is the weight of the homogeneous basis polynomial. The unit has $w_0 = 0$. The map $T_m$ fixes the unit and preserves the nonconstant subspace. Let $C_{\mathrm{loc}} \ge 1$ bound its two operator norms and the following local remainder estimates before symmetry averaging; the leading stage bound was chosen with $K_{J_{m+1}} \ge 2A_d^{\mathrm{sym}} C_{\mathrm{loc}}$. Taylor subtraction of a primary mark leaves coefficient error at most $\eta|x^\circ|_1$ and remainder at most $C_{\mathrm{loc}}|\mathsf{S}_L x^\circ|_1$. A regular remainder of mass $e^{\sigma \operatorname{cov}(Z)} e_i$ contributes at most $\eta e^{\sigma \operatorname{cov}(Z)} e_i$ to each of these two bounds. A forced remainder contributes at most $\eta e^{\sigma \operatorname{cov}(Y)} k_i / \kappa_{m+1}$.

The common local error bound is $\eta = \eta_m$ from item 2 of Lemma 14.1. The local polynomial is subtracted through degree $d$ before rescaling. Each term left from a nonwrapping regular mark has at least one additional scaled jet weight; the block rescaling gives the inverse block factor in its norm. The finite connected sums and the ordered derivative product rule introduce only

stage constants. For a forced mark, the support sum keeps half the original forced suppression: its normalization is $\kappa_{m+1}$, not $\epsilon^{\mathrm{for}}_{m+1}$. The remaining suppression absorbs the same derivative and support factors. Wrapping terms are assigned to the forced remainder. Thus the bounds before choosing parameters have the form $C_j^{\mathrm{loc}}/L + o_{\widehat{\beta}_m}(1)$, with $C_j^{\mathrm{loc}}$ fixed before $L$ and the second term tending to zero at fixed stage and block factor. Choose $L^{3/5} > 2C_j^{\mathrm{loc}}$ first. The first term is then less than $\frac{1}{2}L^{-2/5}$; raise the coupling entry until the absolute value of the second term is at most $\frac{1}{2}L^{-2/5}$. This is item 2 in the scale choices of Lemma 14.1. The degree-$d$ Taylor extraction is on healthy charts; the indicators themselves are never differentiated.

Let $u$ be the extracted coarse polynomial coordinate, $r$ the incoming anchored remainder norm and $r'$ the resulting coarse remainder norm, including its forced normalization. Summing the contributions gives

$$|u - T_m \mathsf{S}_L x|_1 \le \eta(|x^\circ|_1 + r), \qquad r' \le C_{\mathrm{loc}}|\mathsf{S}_L x^\circ|_1 + \eta r, \qquad r = \|g_{m+1}(x)\|_{\mathrm{rem}}. \tag{14.14}$$

The same sums with the primary mark canceled give the Lipschitz bound $\eta\|g - \widetilde{g}\|_{\mathrm{rem}}$ for the coefficient and remainder differences. This cancellation keeps the entire forced marked family; estimating the two families separately would lose this bound.

*Step 2: Linear choice and inverse bound.* Perform the construction on each homogeneous basis vector and on all its signed and conjugated images. Take their union of supports, with zero entries where necessary, and recombine using the basis coefficients. Average over the finite signed-permutation group and over conjugation. This gives a complex-linear family with the required symmetries and exact density identity. Let $A_d^{\mathrm{sym}} \ge 1$ bound the sum of absolute recombination coefficients in this finite averaging. After averaging, the error in (14.14) is at most $\eta(1+s)A_d^{\mathrm{sym}}|x^\circ|_1$, and the remainder is at most $(C_{\mathrm{loc}} + \eta s)A_d^{\mathrm{sym}}|\mathsf{S}_L x^\circ|_1$.

Use (14.5) with $s = C_{J_{m+1}}$, $C = K_{J_{m+1}}$ and $\varepsilon_L = L^{-3/8}$. The cross-stage inequality follows from item 1, fixed before the ratios. Since $\eta s \le 1 \le C_{\mathrm{loc}}$, the averaged remainder constant is at most $K_{J_{m+1}} = C$. In the notation $y = \mathsf{S}_L x$, the resulting inequalities are

$$|u - T_m y|_1 \le \varepsilon_L(|x^\circ|_1 + r), \qquad r' \le C|y^\circ|_1 + \varepsilon_L r, \qquad r \le s|x^\circ|_1.$$

Since $|x|_1 \le |\mathsf{S}_L x|_1 \le L^d|x|_1$, the first two conditions in (14.5) imply

$$|y|_1 \le 2C|u|_1, \qquad |u|_1 \le 2C|y|_1, \qquad |y^\circ|_1 \le 2C|u^\circ|_1.$$

For example, apply $T_m^{-1}$ and use $2C|u - T_m y|_1 \le |y^\circ|_1$ to absorb the error; the same argument on the nonconstant subspace gives the last bound. Moreover $\varepsilon_L r \le C|y^\circ|_1$, so

$$r' \le 2C|y^\circ|_1 \le 4C^2|u^\circ|_1 \le C_{J_m}|u^\circ|_1.$$

The linear map $x \mapsto u = L_m x$ is injective by $|x|_1 \le s|u|_1$, hence invertible in finite dimension. This defines the new graph by $g_m(u) = g_m(L_m x)$. The two coefficient estimates now follow. In particular the inverse estimate has no factor $L^d$: $\mathsf{S}_L^{-1}$ has norm at most one.

For the unit, the primary extraction is exact, $T_m$ fixes it and its nonconstant coordinate is zero. All the preceding remainder bounds are then zero. It is sent to $L^d\mathbf{1}$, with zero remainders. Finally, when two fine graphs differ by at most $tC_{J_{m+1}}|x|_1$ in the remainder norm, the Lipschitz estimate above and rebasing by the just-proved inverse map give a coarse difference at most $\frac{1}{2}tC_{J_m}|u|_1$: the two contributions are the changed remainder at fixed input and the changed input coordinate, each bounded by $t/4$ using (14.5). This is the graph contraction used in Lemma 14.28. □

Choose $r(d)$ with $J_{r(d)} \geq d+3$, $\mathfrak{q}_{J_m} \geq d+1$ and $\varpi_{J_m} \geq d$ for $m \geq r(d)$. Apply Lemma 14.5 to the Wilson start, and let $L_m$ be its coefficient map. Define, before normalizing the lattice fields,

$$\mathsf{Z}_{N,p,d} = L_{r(d)} \cdots L_{N-1} \quad (N \geq r(d)). \tag{14.15}$$

An empty product is the identity. For $N < r(d)$ choose an arbitrary invertible map, as allowed in Definition 3.9.

**Lemma 14.6** (Reference normalization)**.** *For every $d$, $N \geq r(d)$ and $p$, the invertible maps (14.15) satisfy*

$$x_N(v) = v, \qquad x_{r(d)}(v) = \mathsf{Z}_{N,p,d} v,$$
$$H_m(v) = \ell_m^d \mathcal{B}^{(m)}_{\beta_{N,p,m+1}} H_{m+1}(v) \quad (r(d) \leq m < N).$$

*At the cutoff, $H_N(v) = \operatorname{dens} \rho_N \, P_d[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}} v]$.*

*Proof.* Use Definitions 3.1, 3.9, 14.4 and Lemma 14.5. At level $N$, split the unmasked section into its healthy part and its complement. The latter is assigned to the marked forced addresses meeting the stencil, with the original eligibility tests. Their sum restores the complement exactly. The regular remainder is zero and $x_N = v$.

Iterate the marked step on the same unmarked trajectory. It gives $x_m = L_m x_{m+1}$ and the exact equality in (14.12). Each $L_m$ is invertible, so their product is invertible and $x_{r(d)} = \mathsf{Z}_{N,p,d} v$. No choice of a different unmarked representation is made. The factor $\ell_m^d$ agrees with $a_m^d = \ell_m^d a_{m+1}^d$. □

The normalization retains the symmetries of the local sections. Its unit includes the degree-dependent spacing factor.

**Lemma 14.7** (Symmetry and the unit)**.** *For every $d$, $N \geq r(d)$ and $p$, the maps $\mathsf{Z}_{N,p,d}$ commute with conjugation and $H_4$. For $r(d) \leq m \leq N$,*

$$x_m(a_N^d \mathbf{1}) = a_m^d \mathbf{1}.$$

*Consequently $\mathbf{u}_d = a_{r(d)}^d \mathbf{1}$ has insertion one.*

*Proof.* Use the section symmetries in Lemma 10.2, the symmetric block map of Definition 3.1, and the equivariant coefficient extraction in Lemma 14.5. Complex conjugation and signed permutations commute with integration, Taylor extraction and the chosen coefficient graph. Each operation therefore commutes with every $L_m$ and their product.

For a unit input, the marked density is $a_N^d \operatorname{dens} \rho_N$. Exact integration and multiplication by $\ell_m^d$ give $a_m^d \operatorname{dens} \rho_m$ at level $m$. The selected extraction has zero nonconstant coordinate and zero remainder for this density. Induction proves the displayed identity. In particular $\mathsf{Z}_{N,p,d}^{-1} \mathbf{u}_d = a_N^d \mathbf{1}$. Substitute into Definition 3.9; factor scaling fixes the polynomial unit, so its insertion equals $a_N^{-d} a_N^d = 1$. □

The coefficient graph bounds give the following uniform control through the trajectory.

**Lemma 14.8** (Marked remainders and transport)**.** *For $N \geq r(d)$ and $r(d) \leq m \leq N$, the coefficients and both remainder families are linear in $v$, with fixed supports, and*

$$\mathcal{E}_m(v) + (\epsilon_m^{\mathrm{for}})^{-1/2} \mathcal{K}_m(v) \leq C_{J_m} |x_m(v)^\circ|_1.$$

*For $m < N$,*

$$|x_{m+1}(v)|_1 \leq C_{J_{m+1}} |x_m(v)|_1, \qquad |x_m(v)|_1 \leq C_{J_{m+1}} \ell_m^d |x_{m+1}(v)|_1.$$

*Proof.* Use Definitions 3.3, 3.4, 3.9, 14.4 and Lemmas 14.5–14.7. The graph inequality of Lemma 14.5, in the norm (14.13), is the first display. Its two coefficient inequalities give the second display. Iterating retains the graph at every level. For fixed supports, take the union of the addresses selected for the finitely many basis inputs and fill unused addresses by zero. The basis construction in that lemma then makes the coefficients and both remainders linear in $v$. Lemma 14.5 states that the selected graph vanishes on the unit; the density decomposition alone would not imply this. □

These cancellations are the marked format (3.21) of Section 3.

**Lemma 14.9** (Marked local format)**.** *For $N \geq r(d)$ and $r(d) \leq m \leq N$, the complex regular remainders are Borel, gauge invariant, local and smooth as in Definition 13.1. Their supports contain the anchor. Their values at the identity, and their radial-jet coefficients through degree $d$ on nonwrapping supports, vanish. The forced remainders satisfy Definition 13.2, and*

$$Q_i \neq 0 \quad \Longrightarrow \quad \text{the anchor belongs to } Y_i.$$

*Proof.* Use Definitions 3.3, 3.2, 13.1–14.4 and the anchored extraction in Lemma 14.5. Subtracting the value at the identity and the radial Taylor polynomial gives exactly the two stated regular cancellations. On wrapping supports no radial Taylor cancellation is imposed. In the connected expansion every marked component contains the original insertion anchor. The support is the union of the supports in that component, so it still contains that anchor after enlargement. The same argument applies when the component is assigned to a forced address. It is assigned only on that address's eligibility event. Locality, gauge invariance and Borel measurability are preserved by local integration and by these tests. The regular derivatives are taken on healthy configurations, as in Definition 13.1; no derivative of a health indicator is used. The fixed basis construction and its support unions retain these properties under iteration. □

A bond tiling has cell boundaries through link midpoints. Its reflected copies place an observable in every cell, with conjugation on odd reflections. The copy $(F)_C$ is the pullback of $F$ under the composition of bond reflections taking the base cell to $C$, conjugated when that composition has odd parity.

**Definition 14.10** (Chessboard root)**.** For a tiling $\mathcal{C}^{\mathrm{tile}}$ into congruent reflected cells and an observable $F$ in its base cell, set

$$\|F\|_{\mathcal{C}^{\mathrm{tile}}} = \left| \left\langle \prod_{C \in \mathcal{C}^{\mathrm{tile}}} (F)_C \right\rangle_{N,p} \right|^{1/|\mathcal{C}^{\mathrm{tile}}|}.$$

Interior means dependence only on links with both endpoints strictly inside the base cell. The centered tiling $\mathcal{C}_m^{\mathrm{tile}}(y)$ has physical cell side $a_m$ and base cell centered at $y$. This quantity converts reflected-product estimates into bounds on one insertion.

For interior complex observables, these roots are seminorms and satisfy

$$\left| \left\langle \prod_{C \in \mathcal{C}^{\mathrm{tile}}} (F_C)_C \right\rangle_{N,p} \right| \leq \prod_{C \in \mathcal{C}^{\mathrm{tile}}} \|F_C\|_{\mathcal{C}^{\mathrm{tile}}}. \tag{14.16}$$

Apply the abstract chessboard theorem, [11, Theorem 4.1], successively in the four axes. The real vector space is the space of complex cell functions, and the reflection includes conjugation.

Translation invariance gives cyclicity; reflection positivity gives the required Schwarz inequality. For the Wilson measure this positivity is [27, Theorem 2.1, pp. 14–15], with compact matrix group, positive coupling and observables on one side of the bond planes. For $\mathcal{C}_m^{\mathrm{tile}}$, there are $L_pD_m$ cells per axis, an even integer. The cell side $a_m/a_N = D_N/D_m$ is odd, so the centered boundaries pass through link midpoints. Bounded Borel cell functions follow by approximation in the finite-dimensional configuration measure. We will also use the interior-root comparison of Lemma 15.2; its proof uses only (14.16), the fixed stencils and their symmetries, and so is independent of the continuum assertions of Section 15.

**Several-insertion expansion**

For a finite set of slots $I_0$, work in the commutative algebra $\mathbb{C}[z_i : i \in I_0]/(z_i^2 : i \in I_0)$. Write $z_I = \prod_{i\in I} z_i$. The exact source density starts at

$$\mathcal{H}_N(z;U) = \operatorname{dens}\rho_N(U)\prod_{i\in I_0}(1+z_iF_i(U)), \qquad \mathcal{H}_m = \mathcal{B}^{(m)}_{\beta_{N,p,m+1}}\mathcal{H}_{m+1}. \tag{14.17}$$

Thus its normalized $z_I$ integral is the cutoff moment of the corresponding physical insertions. At level $m$ write its local representation as

$$e^{\Phi_m(U)+\sum_i z_i f_{i,m}(U)+\sum_X H_{m,X}(z;U)} \sum_{\mathcal{A}\ \text{eligible}} \prod_{Y\in\mathcal{A}} \left( K_{m,Y}(U) + \sum_i z_i k_{i,m,Y}(U) + K^{\ge 2}_{m,Y}(z;U) \right). \tag{14.18}$$

Here $\Phi_m$ is the unmarked exponent, $f_{i,m}$ and $k_{i,m,Y}$ are the complete one-insertion exponential and forced terms, including their physical spacing factors. The polynomials $H_{m,X}$ and $K^{\ge 2}_{m,Y}$ contain only modes $z_I$ with $|I| \ge 2$. Their support contains every anchor indexed by a nonzero mode. The regular terms include their health masks, and each forced term retains its eligibility test. Formula (14.18) distinguishes the exact multiple-source density from the one obtained by keeping only its unmarked and singleton data.

The next estimate controls one insertion in every reflected cell. Its normalization cost is proportional to the number of marks and to the number of coarse sites, rather than to the number of fine sites.

**Lemma 14.11** (Replicated source bound)**.** *For every $d$ and bounded coefficient ball there are radii $\mathfrak{r}_k > 0$, independent of $p$, such that, for $m \ge r(d)$, $k \ge m$ and $B_k^2 \le a_m/a_k$, eventually in $N > k$, uniformly in $p$,*

$$\left| \left\langle \prod_{C\in\mathcal{C}_m^{\mathrm{tile}}} (O(c))_C \right\rangle_{N,p} \right| \mathfrak{r}_k^{|\mathcal{C}_m^{\mathrm{tile}}|} \le e^{|\Lambda_{n_{p,k}}|q_k + 6|\mathcal{C}_m^{\mathrm{tile}}|}.$$

*For every $\eta > 0$, eventually $6 - \log\mathfrak{r}_k \le (d+\eta)s_k$.*

*Proof.* Use Definitions 3.3–3.9, Lemmas 14.1, 14.5 and 14.9, and the expansion in Lemma 13.4. Enlarge the fixed coefficient radius to include its signed and conjugated images. Put

$$E_k = R'C_{J_k}\prod_{i<k} C_{J_{i+1}}, \qquad \mathfrak{r}_k = a_k^d\widehat{\beta}_k^{-w_d}/E_k,$$

where $R' > 0$ depends on the coefficient ball and $w_d$ is a fixed nonnegative integer large enough for its finite polynomial evaluations and forced counterparts. The transport inequality of Lemma 14.8,

iterated from $r(d)$, bounds the level-$k$ coefficients by the product in $E_k$. Threshold powers introduce only a fixed power of $\widehat{\beta}_k$; enlarging $w_d$ includes them. Hence multiplying every mark by $\mathfrak{r}_k$ bounds its full singleton size by the positive source allowance $\widehat{\beta}_k^{-w_d} \le 1$.

For the local expansion, give a mode $z_I$ the weight $\prod_{i\in I} w_i$, with $w_i > 0$. The resulting coefficient norm is submultiplicative: multiplication sums over disjoint splittings of $I$, exactly the splittings in the product of the two weighted norms. The expansion of Lemma 13.4 therefore applies coefficientwise. Its connected terms with at least two marks have the additional support estimate

$$B_k|I| \le 64 \operatorname{cov}_{B_k}(X), \qquad |I| \ge 2, \tag{14.19}$$

when distinct anchors are separated by $B_k^2$. To verify it, take a minimal cover by $M$ radius-$B_k$ balls with connected union. Join their centers by a tree with edges of length at most $2B_k + 1$. Its doubled traversal has length at most $6B_k(M-1)$. Assign each anchor to one covering center. Distinct anchors have distinct centers, and the distance between any two assigned centers is at least $B_k^2 - 2B_k$. Visit these centers in their cyclic order in the traversal. The triangle inequality gives $|I|(B_k^2 - 2B_k) \le 6B_k(M-1)$. For $B_k \ge 8$ this implies the stated bound, with room in the constant. The argument uses the periodic max-distance and needs no nonwrapping assumption. This is the marked covering estimate used in the support norm of Lemma 13.4.

It permits each source weight to be enlarged by $e^{\delta B_k/64}$ when $\delta$ units of the support exponent are removed. Singleton terms are bounded directly by their one-mark norms. Taylor's product rule introduces at most the factorial of the derivative order; the stage column constant was chosen above this factorial. With $j = J_{k+1}$, Lemma 13.4 supplies the rooted derivative bound at support exponent $\sigma_{J_k} + 3$ and the forced bound $C_j e^{-r_j \beta_{N,p,k+1} t_{k+1}^2}$ at exponent $\sigma_{J_k} + 2$. Multiplying by a mode weight changes only the marked letter in its tree sum. Item 1 pays the derivative factorial. The weighted sums therefore retain those bounds; decreasing the final exponent from 100 to any value in $[98, 100]$ only decreases the sums. The forced normalization uses half the suppression, leaving the other half for the marked derivative and attachment losses. Their spatial anchor is always one of the marks; a nonzero mode is assigned only to a support containing all its marked anchors. This proves the support assertion in (14.18), as well as its absolute convergence. Restricting the set of marks commutes with the construction, since every operation takes disjoint unions of their modes.

The reflected anchors at level $k$ are the embedded sites of the level-$m$ torus. They have spacing $a_m/a_k \ge B_k^2$. Their parents agree under each subsequent block map. The inequality persists at all finer levels because $B_{i+1}^2 \le 2B_i^2$ and $\ell_i \ge 3$. Occupancy is therefore one at every such level. These are precisely the conditions needed for the preceding weighted sums; the number of marks does not enter their local bounds.

At the stopping level, the weighted supremum of each complete singleton exponential term is at most 3, and that of its forced profile is at most 1. Indeed, its polynomial part is at most its coefficient bound, its marked regular sum is at most that bound, and the normalization of the forced part is at most one; Lemma 14.8 and the definition of $E_k$ bound the sum by 3. The forced profile follows from its $\kappa_k$ normalization. For (14.18), the local absolute sums give an additional two per mark: exponentiation of the connected regular terms and the eligible forced sum each contribute at most one. Integrating the remaining scalar-free unmarked exponent and dividing by the unmarked partition function gives at most $e^{|\Lambda_{n_{p,k}}| q_k}$, with $q_k$ as in (14.1). Thus coefficient extraction in the weighted source norm yields

$$\mathfrak{r}_k^{|I_0|} \left| \left\langle \prod_{i\in I_0} F_i \right\rangle_{N,p} \right| \le \exp\{|\Lambda_{n_{p,k}}| q_k + (3+1+2)|I_0|\}.$$

Exact integration preserves these normalized coefficients, proving the first assertion. The ratio of site and cell counts is $(a_m/a_k)^4$.

Finally (14.8), summed as in (14.10), gives

$$\sum_{i<k} \log C_{J_{i+1}} = o(s_k), \qquad \log \widehat{\beta}_k = o(s_k).$$

The separate final column is included in this sum for $k > 0$. Hence, for every $\eta > 0$, eventually $6 - \log \mathfrak{r}_k \le (d+\eta)s_k$. All choices precede the period, and all the estimates are on full cutoff tails. □

The centered estimate fixes its constant before the physical cell size. A sufficiently fine cutoff then places the fixed insertion stencil strictly inside that cell.

**Lemma 14.12** (Centered bound)**.** *For every $d$ and $R, \varepsilon > 0$, there is $A(d, R, \varepsilon) \ge 0$ such that, for every $m \ge r(d)$, some $N_0(d, R, \varepsilon, m) \ge m$ gives, for $N \ge N_0$, every $p, y$ and $\|c\| \le R$,*

$$O^{(d)}_{N,p}(c, y) \text{ is interior}, \qquad \|O^{(d)}_{N,p}(c, y)\|_{\mathcal{C}^{\mathrm{tile}}_m(y)} \le A a_m^{-d-\varepsilon}.$$

*Proof.* Use Definition 3.9 and Lemmas 14.1 and 14.11. The replicated integral and radius bounds, together with the stopping-scale estimate in Lemma 14.1, give

$$\begin{aligned} \|O(c,0)\|_{\mathcal{C}^{\mathrm{tile}}_m(0)} &\le \exp\{(a_m/a_k)^4 q_k + 6 - \log \mathfrak{r}_k\}, \\ 6 - \log \mathfrak{r}_k \le (d+\eta)s_k, &\qquad (a_m/a_{k(m)})^4 q_{k(m)} \le \eta s_{k(m)} \quad \text{eventually.} \end{aligned} \tag{14.20}$$

The cutoff threshold is uniform in the coefficient ball and in $p$; $k = k(m)$ is the first level specified in Lemma 14.1.

Put $\tau = s_{k(m)}$, $\upsilon = s_m$ and $W = (a_m/a_k)^4$. Since $q_k \ge 1$,

$$\tau - \upsilon = \log(a_m/a_k) \le W \le W q_k \le \eta\tau.$$

The exponent in the first line of (14.20) is therefore at most $(d+2\eta)\tau \le (d+2\eta)\upsilon/(1-\eta)$. Choose $0 < \eta < 1$ so that $(d+2\eta)/(1-\eta) \le d+\varepsilon$. This proves the required estimate with constant one for all sufficiently large $m$. For the finitely many remaining $m \ge r(d)$, use the first line of (14.20) at the selected fixed $k(m)$ and take the maximum of its ratio to $a_m^{-d-\varepsilon}$ and one. This fixes $A$ before $m$.

Finally the stencil of Definition 3.9 has fixed lattice radius. For fixed $m$ its physical radius tends to zero as $N \to \infty$, uniformly in $p$. Translation covariance moves the centered calculation to any $y$. This proves interiority and the bound, using precisely the expansion estimates (14.20). □

### Equation and difference corrections

The following estimates concern the insertions of Definition 3.9, with the terminal maps $\mathsf{Z}_{N,p,d}$ of Lemma 14.6. They will be used to identify the quotient fields in Section 15. A statement holds on full cutoff tails if it holds for all $N \ge N_0$ and every $p$. A nonnegative family $b_{N,p}$ has subpower growth when, for every $\varepsilon > 0$, $b_{N,p} \le a_N^{-\varepsilon}$ on such a tail. A coefficient family is admissible if

$$\forall \varepsilon > 0 \ \exists N_0 \ \forall N \ge N_0 \ \forall p : \|\mathsf{Z}^{-1}_{N,p,d} c_{N,p}\|_\infty \le a_N^{-\varepsilon}. \tag{14.21}$$

For later volume extraction we also use strictly increasing subsequences $N = \mathsf{n}(j)$, $p = P(k)$ and the ordered limit

$$\operatorname{olim} X_{N,p} = x \iff \forall \varepsilon > 0 \ \exists k_0 \ \forall k \ge k_0 \ \exists j_0(k) \ \forall j \ge j_0(k) : |X_{\mathsf{n}(j),P(k)} - x| < \varepsilon. \tag{14.22}$$

The ordered versions of admissibility and smallness replace full cutoff tails by these quantifiers. We prove the stronger versions for the bounds below. The same constructions work for ordered-admissible families, with their ordered thresholds; this is the form used in Section 15.

We use the chessboard root and interior condition of Definition 14.10. Translates of a bond tiling are by lattice vectors. An anchor is the lattice site at which an insertion is placed. In a base cell of side $a_k$, let $z^a$ be its integer coordinate measured from the first site in direction $a$, and put

$$M_{n_*,k,N} = \left\lfloor \frac{a_k}{8(n_*+1)a_N} \right\rfloor, \qquad M_{n_*,k,N} \le z^a, \quad z^a + M_{n_*,k,N} + 1 \le a_k/a_N\,. \tag{14.23}$$

These inequalities specify a relative interior margin.

**Definition 14.13** (Small errors)**.** An observable family $E_{N,p}(y)$ is small if some $\eta_{N,p} \ge 0$ with $\sup_p \eta_{N,p} \to 0$ as $N \to \infty$ satisfies the following condition. For every arity $n_*$ and lower level $m$, there is $k \ge m$ such that, on a full cutoff tail, $k \le N$ and every translated bond tiling $\mathcal{C}^{\mathrm{tile}}$ of side $a_k$ and every base-cell anchor satisfying (14.23) obey

$$E_{N,p}(y) \text{ is interior}, \qquad \|E_{N,p}(y)\|_{\mathcal{C}^{\mathrm{tile}}} \le \eta_{N,p}\,.$$

Here $n_*$ is the number of insertions in a later product test; the lower level $m$ requires the chosen cell scale to be arbitrarily fine. The tolerance family is chosen before $n_*, m, k$. All residuals below are translation covariant, and their signed and conjugated images obey the same estimates. To pass from centered to arbitrary interior roots, choose $k' \ge k$ with $24(n_*+1)a_{k'} \le a_k$ and apply the finite reflection argument of Lemma 15.2. It uses only fixed stencils and the chessboard inequality, so also applies to these residuals on full cutoff tails.

For a local observable $F_{N,p}(y)$, let $a_N^e F_{N,p}$ be its dimensionless representative. We require it to be continuous on its fixed stencil, regular on healthy configurations in the sense of Definition 3.3, with derivative and supremum bounds of subpower growth. The supremum bound is on all configurations. These two bounds have different uses: Taylor extraction uses the first, and the forced complement uses the second.

Local matching makes no assumption on the size of the physical coefficient it produces. It bounds its inverse terminal coordinate, which is the quantity in (14.21).

**Lemma 14.14** (Local comparison)**.** *For every such degree-$e$ family $F$, there are admissible coefficients $b_{N,p} \in V_e$ with $O^{(e)}(b) - F$ small. For the auxiliary section of Lemma 10.4, there are invertible coefficient changes $P_{N,p}$, with $\|P_{N,p}^{-1}\|_1 \le 2$ eventually uniformly in $p$, and maps*

$$B_{d,N,p} = \mathsf{Z}_{N,p,d} P_{N,p} \mathsf{D}^{\mathrm{fac}}_{t_N} \mathsf{R}_d$$

*from polynomial labels to $V_d$, such that $O^{(d)}(B_{d,N,p}\pi) - a_N^{-d} V_d^{\mathrm{aux}}[\mathsf{R}_d\pi]$ is small for equation labels with admissible $B_{d,N,p}\pi$.*

In particular, for $\pi_{N,p} \in \mathcal{I}_d$ with admissible $B_{d,N,p}\pi_{N,p}$, the comparison is

$$O^{(d)}_{N,p}(B_{d,N,p}\pi_{N,p}, y) - a_N^{-d} V_d^{\mathrm{aux}}[\mathsf{R}_d\pi_{N,p}](y) \quad \text{is small.} \tag{14.24}$$

We prove both assertions using the same backward coefficient correction.

*Proof.* Use Definitions 3.3, 3.8, 3.9, Lemmas 14.5, 14.11 and the local sections of Lemma 10.2. Write $h_N = a_N^e F_{N,p}(0)$. Enlarge its stencil by the section stencil. The radial Taylor readout through

degree $e$ gives a coefficient $v_N$ and a remainder. The readout is bounded by a fixed multiple of the derivative bound: it is a linear map on a finite jet space. Subtract the section with that coefficient on the healthy part. Its regular remainder has zero value and zero radial jet through degree $e$. Assign the complementary observable to the forced supports meeting the enlarged stencil. This gives an exact one-mark representation of $\operatorname{dens}\rho_N h_N$, with

$$|v_N|_1 + r_N \le E_{N,p}, \qquad \text{forced majorant} \le E'_{N,p}, \tag{14.25}$$

where $E, E'$ have subpower growth. The support weight of this fixed enlarged stencil is bounded by $e^{81\sigma_{J_N}}$ eventually; its $B_N$ covering number is at most 81. Threshold powers, the readout bound and this support weight are included in $E, E'$. The entry conditions make each of them subpower. No division by the forced allowance is made on the uncontrolled complement: it is retained with its original eligibility tests and the supremum bound $E'$.

To compare with the selected graph, carry this representation backwards and at the reference level choose the primary coefficient with the same polynomial coordinate. Subtract the corresponding selected inserted density. Its polynomial coordinate at the reference level is zero. The following calculation carries this matching condition towards the fine scale.

For one step, let $u, v$ be the coarse and fine coordinates, $R, T$ their remainder bounds, $Y = |\mathsf{S}_L v|_1$, $K$ the norm bound for the leading map and its inverse, $\varepsilon_L = L^{-3/8}$ and $\vartheta_L = L^{1/8}$. The marked-step inequalities give

$$Y \le K\big(|u|_1 + \varepsilon_L(Y+T)\big), \qquad R \le KY + \varepsilon_L T. \tag{14.26}$$

Suppose $a|u|_1 \le R$, with $a = 32(1+C_{J_i})^2 \ge 4K^2$. Then $4K|u|_1 \le Y + \varepsilon_L T$. Substitution in the first inequality, using $4K\varepsilon_L \le 1$, gives $Y \le 4K\varepsilon_L T$. Item 3 in Lemma 14.1 gives

$$4K\,32(1+C_{J_{i+1}})^2\varepsilon_L \le 1, \qquad (4K^2+1)\vartheta_L\varepsilon_L \le 1.$$

Consequently $32(1+C_{J_{i+1}})^2|v|_1 \le T$ and $\vartheta_L R \le T$. The matching condition starts this induction with $u = 0$ at the reference level. Multiplying the second inequality through the interval gives the inverse-scale gain, while the finest-level bound is closed by absorbing the coefficient contribution into its remainder bound. In particular, with subpower $E''$ absorbing $E, E'$,

$$|\Delta_N|_1 \le \frac{2(1+C_{J_N})E''_{N,p}}{\ell_{N-1}^{1/8}}, \qquad R_k \le 2(1+C_{J_N})E''_{N,p}e^{-(s_N-s_k)/8}. \tag{14.27}$$

Here $\Delta_N$ is the difference from the selected fine polynomial coordinate. The product $\prod_{i=k}^{N-1}\ell_i^{-1/8}$ is exactly $e^{-(s_N-s_k)/8}$. Thus this factor comes from the matched remainder inequalities, rather than from the physical diameter of the original stencil.

To estimate the replicated residual, for a source of size $E$ started at $N$, use at level $k$ the radius

$$w = \frac{\zeta_{e,k}/4}{a_k^{-e}E\exp\{-(s_N-s_k)/8\}},$$

where $\zeta_{e,k} > 0$ is a source allowance fixed at that level before $N, p$. At every finer step multiply this radius by the source contraction factor of that step. Its product with the dimensionless singleton size is at most one quarter of the available source allowance; the other three quarters bound the graph remainder, the forced complement and their sum. This follows inductively from the two inequalities in (14.14). For the first step the forced complement is bounded by $E'$ in (14.25); the large-field suppression absorbs its fixed support and derivative factors before division by the new

forced normalization. Thus every step is admitted. The singleton polynomial coordinate at the reference level is zero, so the representation constructed this way is exactly the difference with the selected primary coefficient, rather than a new approximation to it.

Here $m$ labels the tiling and $k$ the finer stopping level at which its replicated sources are integrated. Replicate the difference at an admissible pair $m \le k$, $B_k^2 \le a_m/a_k$, and use the weighted source representation (14.18). Coefficient extraction contributes $w^{-1}$ per mark; the complete singleton bounds and the normalized unmarked ratio (14.2) contribute a constant depending on $m, k$. Consequently, for every fixed such pair,

$$\|O^{(e)}(b) - F\|_{\mathcal{C}_m^{\mathrm{tile}}(0)} \le C_{m,k}\mathfrak{e}_{N,p}a_N^{1/8}, \qquad \|\mathsf{Z}_{N,p,e}^{-1}b_{N,p}\|_\infty \le 2E'''_{N,p}, \tag{14.28}$$

where $\mathfrak{e}, E'''$ have subpower growth. The factor $e^{s_k/8}$ and the fixed radius $\zeta_{e,k}^{-1}$ belong to $C_{m,k}$. The same argument applies after each signed permutation and conjugation, with translation covariance and the union of their fixed stencils. Absorbing two factors $a_N^{-1/32}$ bounds the first expression by $a_N^{1/16}$ on every fixed pair. The interior-root comparison after Definition 14.13 gives the common tolerance $a_N^{1/32}$, chosen before all cell requests. This proves smallness and admissibility on full cutoff tails.

For the auxiliary section, perform the construction on all its basis polynomials together, with the original Taylor normalization. Let $u_j$ be the primary bare coefficient matching its $j$th basis vector, and set $Px = \sum_j x_j u_j$, after the finite symmetry averaging. For this fixed section the regular derivative bound $H_{j,d}$ at stage $j$ is fixed before its ratio, independently of the later coupling. The sharper coefficient estimate in (14.27) therefore has numerator bounded by $2(1 + C_j)H_{j,d}$ on each basis input. The ratio condition, including the finite basis-recombination factor, gives

$$|x - Px|_1 \le \tfrac{1}{4}|x|_1.$$

Indeed, before summing over the fixed basis the right side is the last-step factor $\ell_{N-1}^{-1/8}$ times the stage coefficient bound; the stage ratio was chosen after that bound. The matrix is therefore invertible and $|P^{-1}x|_1 \le 2|x|_1$. The construction is common to all later cell requests. Multiplication by $\mathsf{D}_{i_N}^{\mathrm{fac}}\mathsf{R}_d$ changes a polynomial label into its auxiliary bare insertion coordinate; multiplication by $\mathsf{Z}$ changes that coordinate to the reference normalization. These are exactly the maps $B$ in the statement. The inverse bound preserves admissibility, and (14.28) compares the two sections on its transported equation ideal. Finite combinations use the same subpower factors. The argument only replaces eventual bounds on coefficients, so ordered-admissible families obey the corresponding ordered version as well. □

The next positional estimate is used for the rank test before any bounds on differentiated fields have been established. It concerns generic configurations, where every coordinate separates the insertion points.

**Lemma 14.15** (Displacement of correlation tests)**.** *Near each of the fixed generic row configurations, moments of bounded coefficient lists have a modulus of continuity independent of $N, p$ on full cutoff tails. More precisely, for sufficiently small $r > 0$ and all sufficiently fine fixed centered levels, if all relevant centered roots and reflected roots are at most $A$, then*

$$|M_{N,p}(v; x/a_N) - M_{N,p}(v; y/a_N)| \le \frac{4nA^n}{r}|x - y|$$

*for sufficiently close grid configurations in that neighbourhood.*

*Proof.* Use Definition 14.10, Lemma 14.12 and the reflection inequality (14.16). Choose the neighbourhood so that in each coordinate every directed torus distance between distinct points exceeds $4r$, uniformly in $p$. This is possible for the row configurations: they are generic and lie strictly inside the smallest physical period box. Choose a centered level with $a_m < r$. The fixed stencil and its reflected stencils fit in the centered cells on a full cutoff tail.

Fix an axis and all other coordinates. Reflection positivity and translation invariance give a positive transfer operator, rescaled to $0 \le T \le 1$. On the periodic axis the expectation is a normalized trace. If $X$ denotes the insertion being moved, its two adjacent empty slabs have lengths $n_1^{\mathrm{sl}}, n_2^{\mathrm{sl}}$ in fine sites. A move by $l \ge 0$ changes their factor by

$$\begin{aligned} T^{n_1^{\mathrm{sl}}+l} X T^{n_2^{\mathrm{sl}}-l} - T^{n_1^{\mathrm{sl}}} X T^{n_2^{\mathrm{sl}}} &= T^{n_1^{\mathrm{sl}}}(T^l X - X T^l) T^{n_2^{\mathrm{sl}}-l} \\ &= -T^{n_1^{\mathrm{sl}}}(1-T^l) X T^{n_2^{\mathrm{sl}}-l} + T^{n_1^{\mathrm{sl}}} X (1-T^l) T^{n_2^{\mathrm{sl}}-l}. \end{aligned} \tag{14.29}$$

Both expressions are inserted in the same remaining cyclic product and divided by the same partition trace. The stencil is contained in a centered cell of lattice side $d_{\mathrm{cell}} = a_m/a_N$. Choose the neighbourhood and level so the directed separations exceed $2d_{\mathrm{cell}} + 2r/a_N$ and $la_N \le r$. Splitting the two surrounding empty slabs at their reflection planes leaves $n_{\mathrm{sl}} + 1 \ge 2r/a_N$ fine sites next to each occurrence of $1 - T^l$. The spectral theorem gives

$$\|(1-T^l)T^{n_{\mathrm{sl}}}\| \le \sup_{0\le t\le 1} l(1-t)t^{n_{\mathrm{sl}}} \le l/(n_{\mathrm{sl}}+1) \le la_N/(2r).$$

For each normalized trace in (14.29), reflection Schwarz at the slab ends and the chessboard inequality bound the remaining factors by the product of their centered roots, at most $A^n$. One applies this inequality to the complete cyclic product, so no supremum norm of $X$ is required. The two terms together are bounded by $A^n la_N/r$. An arity-one moment is translation invariant and has change zero.

Replace the $4n$ coordinates one at a time. Choose the initial neighbourhood with a smaller concentric neighbourhood so that every intermediate configuration retains the same separations. Each one-coordinate displacement is at most $|x-y|$. Summing gives the display. The centered estimates apply to the fixed coefficient ball and its reflected images uniformly in $p$, proving the claimed uniform modulus without using Lemma 14.25. □

The equation projections separate the coefficient directions controlled by exact cutoff identities. The free equation space consists of polynomial labels that vanish on curvature jets satisfying the Maxwell equations. Its transported image is the kernel of the projections chosen below; the free quotient has dimension $\dim_{\mathbb{C}} M_d^{\mathrm{Max}}$. The projections and the equation observables in Lemmas 14.16–14.17 are chosen together, before the spacing and volume subsequences.

**Lemma 14.16** (Equation projections)**.** *For every $d$, there are complex linear idempotents $\Pi_{d,N,p} : V_d \to V_d$ and $C_d < \infty$ such that*

$$\|\Pi_{d,N,p}\| \le C_d, \qquad \dim V_d \le \dim \ker \Pi_{d,N,p} + \dim_{\mathbb{C}} M_d^{\mathrm{Max}}.$$

*More precisely,* $\ker \Pi_{d,N,p} = B_{d,N,p}\mathcal{I}_d$*, with the maps $B$ of Lemma 14.14.*

*Proof.* Use the labels of Definition 3.8, the free comparison of Definition 4.10 and Lemma 14.14. Let $\mathcal{I}_d \subset V_d$ be the polynomials belonging to the ideal generated by the linear Maxwell equations in radial jet coordinates. Radial coordinates already satisfy the linear Bianchi identities. Taking invariant polynomials commutes with the on-shell quotient: average any representative over the

compact adjoint group to obtain an invariant representative. Thus the on-shell map is onto $M_d^{\mathrm{Max}}$ and

$$\dim V_d - \dim \mathcal{I}_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}}. \tag{14.30}$$

This dimension calculation uses invariant averaging and the definition of $M_d^{\mathrm{Max}}$; it does not use positivity.

Lemma 14.14 constructs invertible coefficient maps $B_{d,N,p} : V_d \to V_d$. They transport $\mathcal{I}_d$ to the coefficient equations for auxiliary local representatives, including their change to the selected insertion coordinates. Set $U_{d,N,p} = B_{d,N,p}\mathcal{I}_d$. In the Euclidean norm on the fixed basis take the orthogonal projection $Q_{d,N,p}$ onto $U_{d,N,p}$ and set $\Pi_{d,N,p} = 1 - Q_{d,N,p}$. Then $\Pi^2 = \Pi$ and $\ker \Pi = U_{d,N,p}$. Equivalence with the fixed coefficient norm bounds $\|\Pi\|$ independently of $N, p$, even when $B$ is poorly conditioned. Equation (14.30) gives the claimed dimension inequality, in fact equality.

The same local comparison maps will be used in Lemma 14.17. They and these projections are chosen for all cutoffs before any subsequences. This proves the projection and rank assertions for the maps used in the equation comparison. □

The next estimate converts a coefficient equation into an observable with exact separated moment identities.

**Lemma 14.17** (Equation observables)**.** *For these same projections, every admissible family $c_{N,p} \in \ker \Pi_{d,N,p}$ admits a measurable observable family $\mathcal{Q}_{N,p}(y)$, bounded at each anchor and cutoff on a full cutoff tail, for which $O_{N,p}^{(d)}(c_{N,p}, y) - \mathcal{Q}_{N,p}(y)$ is small. For every fixed spectator list and $\delta > 0$, on a full cutoff tail,*

$$\int \mathcal{Q}_{N,p}(y_i) \prod_{h \neq i} O_{N,p}(v_h, y_h)\, d\mu_{N,p} = 0 \qquad (\mathrm{sep}_{\mathrm{tor}}(y) \geq \delta)\,.$$

*Proof.* Use Definitions 3.8–3.9, the coefficient maps of Lemma 14.14 and the kernel identity in Lemma 14.16, and Lemmas 10.4 and 14.14. Lemma 10.4 supplies $\widetilde{P}_d = V_d^{\mathrm{aux}}$ and $\mathsf{R}_d$. Write $c_{N,p} = B_{d,N,p}\pi_{N,p}$ with $\pi_{N,p} \in \mathcal{I}_d$, and set

$$\mathcal{Q}_{N,p}(y) = a_N^{-d} V_d^{\mathrm{aux}}[\mathsf{R}_d \pi_{N,p}](y). \tag{14.31}$$

Lemma 14.14, specifically (14.24), proves that $O^{(d)}(c) - \mathcal{Q}$ is small. Its inverse change has subpower growth for admissible $c$. The common smallness tolerance and interior-root comparison were proved there and apply to this family.

The exact zero identity is (10.8) in Lemma 10.4: every auxiliary equation term is a Wilson action derivative whose multiplier and transport avoid the varied link, so Haar invariance makes its exterior moment zero. The auxiliary section has one finite equation radius. At prescribed positive physical separation, every fixed spectator stencil is exterior to it for all sufficiently fine cutoffs, uniformly in $p$. Its product is bounded and measurable, as required by that lemma. Apply the identity with that product as spectator. Finally $\widetilde{P}_d$ is continuous on a finite product of compact groups. At each fixed cutoff its evaluation with the chosen finite coefficients is bounded. No bound uniform in those cutoffs is used. This proves the lemma from (14.31) and its local comparison estimate. □

Changing the degree changes the insertion normalization. The next estimate corrects that change before taking limits.

**Lemma 14.18** (Corrected inclusion)**.** *For every $d \le e$ and $c \in V_d$, there is an admissible family $b_{N,p} \in V_e$ such that*

$$O^{(e)}_{N,p}(b_{N,p}, y) - O^{(d)}_{N,p}(c, y) \quad \textit{is small}.$$

*Proof.* Use Definition 3.9, Lemmas 14.6–14.9, and Lemma 14.14. After multiplication by $a_N^e$, the target observable is

$$a_N^{e-d} P_d[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}} \mathsf{Z}^{-1}_{N,p,d} c](y).$$

It is a local continuous function on the fixed degree-$d$ stencil. Its ordered derivatives in the allowed link directions and its supremum have subpower bounds. Indeed, the inverse terminal coordinate has this growth by the first inequality of the second display in Lemma 14.8 and (14.8); threshold powers have subpower growth because $\log \widehat{\beta}_N = o(s_N)$. The factor $a_N^{e-d} \le 1$ can only decrease these bounds.

The local matching estimate extracts a degree-$e$ coefficient $b_{N,p}$ whose inverse terminal coordinate still has subpower growth. Its residual root is bounded by $C_{m,k} \mathfrak{e}_{N,p} a_N^{1/8}$ at every fixed admissible pair of levels, with $\mathfrak{e}$ of subpower growth. The absorption and interior-root argument in Lemma 14.14 gives one tolerance $a_N^{1/32}$ for all the later cell requests. The residual is therefore small. For $d = e$ one may simply take $b_{N,p} = c$, with zero residual. This proves the assertion, including that case. □

The physical difference quotient requires a correction in the next degree.

**Lemma 14.19** (Corrected difference)**.** *For every degree $d$, $c \in V_d$ and axis $a$, there is an admissible family $b_{N,p} \in V_{d+1}$ such that*

$$O^{(d+1)}_{N,p}(b_{N,p}, y) - \frac{O^{(d)}_{N,p}(c, y + e_a) - O^{(d)}_{N,p}(c, y)}{a_N} \quad \textit{is small}.$$

*Proof.* Use Definition 3.9 and Lemma 14.14. Multiplying the target difference by $a_N^{d+1}$ gives exactly

$$P_d[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}} \mathsf{Z}^{-1}_{N,p,d} c](y + e_a) - P_d[\mathsf{D}^{\mathrm{fac}}_{t_N^{-1}} \mathsf{Z}^{-1}_{N,p,d} c](y).$$

Its stencil is the union of one fixed stencil and its unit translate. It is continuous and local. Moving an ordered derivative anchor by one lattice site multiplies its spatial weight by at most a stage-dependent factor. The factor is at most $2^{\varpi_{J_N}}$ per derivative. The logarithmic condition $\varpi_{J_N} = o(s_N)$ in Lemma 14.1 makes each fixed product of these factors subpower. The other bounds are those in the preceding proof. Local matching in degree $d+1$ now supplies an admissible coefficient family and the same small residual. The inverse spacing has been removed before applying matching, so it causes no further loss in that residual. The four axes use translates of this one computation and their fixed spatial-weight bounds. This proves the corrected difference. □

### Short-distance comparison

Fix $d$ and a real basis $\mathfrak{m}_1 = 1, \mathfrak{m}_2, \dots, \mathfrak{m}_{q_d}$ of $M_d^{\mathrm{Max}}$, where $q_d = \dim_{\mathbb{C}} M_d^{\mathrm{Max}}$. Choose finitely many one- or two-point tests. In row $i$, the first label is the variable $\mathfrak{m}_j$; the optional second label is a fixed basis element. Let $y^i \in \mathrm{Gen}_{n_i}$, $n_i \in \{1, 2\}$, be the row's configuration. The free test matrix is

$$\mathcal{G}_{ij} = S^{\mathrm{Max}}_{n_i}(\mathfrak{m}_j,\ \text{row } i \text{ spectator}; y^i)\,. \tag{14.32}$$

The next lemma supplies a finite family that distinguishes every free field of degree at most $d$.

**Lemma 14.20** (Free test rows)**.** *For every $d$ and real basis $\mathfrak{m}_1 = 1, \mathfrak{m}_2, \dots, \mathfrak{m}_{q_d}$, there are finitely many one- or two-point rows at generic configurations, with every second label a basis element, for which the matrix* (14.32) *has rank $q_d$.*

The proof uses only the Maxwell correlations of Definition 4.10 and the free positivity in Lemma 16.1, independently of Section 15.

*Proof.* Let $c \neq 0$ in $M_d^{\mathrm{Max}}$. Lemma 16.1 gives a strictly positive reflected two-point value for $c$ and the reflection of $\bar{c}$. Expand that second label in the basis $(\mathfrak{m}_j)$. At least one of the resulting two-point functionals is nonzero on $c$. Each free Wick correlation is a finite sum of products of derivatives of $\Gamma(x) = (4\pi^2|x|^2)^{-1}$ between distinct sites, hence is smooth there. This permits a perturbation of the pair to $\mathrm{Gen}_2$ preserving this nonzero value. Thus the common kernel of all permitted row functionals is zero. Starting with $M_d^{\mathrm{Max}}$, choose a row nonzero on a nonzero vector in the current common kernel. Each choice strictly lowers its dimension. After finitely many choices the common kernel is zero, so $\mathcal{G}$ has rank $q_d$. □

Fix such a family for the rank comparison below.

For a trial $\nu$, a common coefficient list $b^{\nu}_{j,N,p} \in V_d$ and lattice-compatible configuration $x^{\nu,i} = r_\nu y^{\nu,i}$ define the row insertions $F_h^{\mathrm{row}} = O^{(d)}_{N,p}(b^{\nu}_{j_h,N,p}, x_h^{\nu,i}/a_N)$, with $j_1 = j$ and the spectator index fixed by row $i$. Physical points are read modulo the lattice period. Let $\mathcal{G}^{\nu}_J$ be the Maxwell moment of the labels $(\mathfrak{m}_{j_h})_{h\in J}$ at $(y_h^{\nu,i})_{h\in J}$, with $\mathcal{G}^{\nu}_{\varnothing} = 1$. Define the relative errors by

$$R_{\varnothing} = 1, \qquad R_I = \prod_{h\in I} F_h^{\mathrm{row}} - \sum_{\varnothing \neq J \subseteq I} \mathcal{G}^{\nu}_J R_{I\setminus J}\,. \tag{14.33}$$

Only singleton and pair errors occur. The lifts $b^{\nu}$ include the scale normalization, so $F_h^{\mathrm{row}}$ carries no additional power of $r_\nu$. A lift is the reference coefficient whose Wick-ordered leading polynomial at scale $r_\nu$ is the specified basis element $\mathfrak{m}_j$. The construction uses $r_\nu = a_{t_\nu}(M^{\mathrm{cut}}_{t_\nu})^b$ for one fixed $b > 0$ chosen from the degree and the finite tests, before $\nu$ varies.

For a pair of slots define

$$\mathsf{C}^{\mathrm{Max}}(z) = \mathcal{G}_{\{1\}} z_1 + \mathcal{G}_{\{2\}} z_2 + (\mathcal{G}_{\{1,2\}} - \mathcal{G}_{\{1\}}\mathcal{G}_{\{2\}}) z_1 z_2, \qquad z_1^2 = z_2^2 = 0;$$

for one slot retain its linear term. The coefficients to be estimated are

$$\mathfrak{c}_I = [z_I]\left( e^{-\mathsf{C}^{\mathrm{Max}}(z)} \prod_h (1 + z_h F_h^{\mathrm{row}}) \right). \tag{14.34}$$

For the source comparison, use the intervals (14.4). Fix $b > 0$ large enough for the degree and the finite tests, put $l = t - h_t(b)$, $k = l - v_t$ and $r = a_t(M_t^{\mathrm{cut}})^b$. The interval from $t$ to $l$ separates the reading points from the last fluctuation scale; the interval from $l$ to $k$ permits the corrected sources to increase in size while their remainders decrease. With $L = \ell^{\mathrm{st}}_{J_t}$, these intervals satisfy

$$L^{h_t(b)} \geq (M_t^{\mathrm{cut}})^{b+2}, \qquad L^{v_t} \geq B_t^{10}, \qquad \widehat{\beta}_i \in [\widehat{\beta}_t/2, \widehat{\beta}_t] \quad (k \leq i \leq t) \tag{14.35}$$

on sufficiently late intervals. The last assertion follows from $h_t(b) + v_t = o(t)$ at fixed stage and the entry choices. All rounded row points lie strictly inside the level-$k$ cell: their distance from the origin is $O(r)$, while $a_l/r \geq (M_t^{\mathrm{cut}})^2$. Rounding changes their rescaled coordinates by $O((M_t^{\mathrm{cut}})^{-b})$.

Put $\mathcal{M} = M_t^{\mathrm{cut}}$. Let $q_{\mathrm{der}}$ bound the required derivative orders, and let $c, \delta > 0$ be the localization and healthy-chart exponents in the estimates below. Define

$$\mathcal{E}_t^{\mathrm{G}} = \mathcal{M}^{-b} + C_j \mathcal{M}^{q_{\mathrm{der}}} (\widehat{\beta}_t^{-\delta} + e^{-cB_t}). \tag{14.36}$$

The polynomial comparison supplies the initial small error for the subsequent source iteration.

**Lemma 14.21** (Middle-level source comparison)**.** *For each fixed finite test family, sufficiently late intervals admit common real lifts, fixing the unit and bounded in $N, p$ for that interval. At radius one, the corrected singleton and connected pair terms at $l$ have total regular and normalized forced bound*

$$\mathcal{C}_j^{h_t(b)+2} \mathcal{E}_t^{\mathrm{G}}.$$

*The assertion holds on a full cutoff tail, uniformly in $p$.*

The proof gives the Wick correction and the inverse-separation gain.

*Proof.* Use Definitions 3.1, 3.8, 4.10, Lemmas 12.3–12.4, 13.4, 14.5, 14.11 and the covariance estimates of Lemmas 11.5 and 11.12. The sources have at most two marks in each cell; the number of cells is unrestricted.

On a fixed healthy chart, subtract the radial Taylor polynomial of each local insertion. The coefficient norm of the remainder is bounded by its next ordered derivative times the corresponding power of the coordinate radius. All derivatives needed here are among the finitely many stage orders. Express each polynomial in curvature and differentiated-curvature variables, carrying along their powers of $r$. For a polynomial $\pi$, the finite Gaussian contraction operator is

$$\mathsf{H}_C \pi = \sum_{j \geq 0} \frac{1}{2^j j!} (\Delta_C)^j \pi, \qquad \mathsf{H}_C^{-1} \pi = \mathsf{H}_{-C} \pi.$$

The sums are finite on the chosen polynomial space, since a contraction removes two factors. Use the inverse operator for contractions internal to a single insertion. This is the Wick-ordering correction; contractions between distinct insertions remain. The leading coefficient change $T_m$ of Lemma 14.5 and this finite contraction operator, in the jet normalization of Definition 3.8, are invertible and preserve the unit and real form. The error in replacing the selected coefficient map by this leading map has norm less than $1/4$ after its inverse, by the marked estimate (14.14) and the entry condition. Solving this finite linear system gives one lift for each free basis polynomial, common to every row. Transport only between the fixed levels $r(d)$ and $t$ bounds these reference coefficients independently of $N, p$ for that interval. All matrices are real on the real form, and the unit equation has the exact solution $\mathbf{u}_d$.

The covariance calculation involves finitely many derivatives of $\Gamma(x) = (4\pi^2 |x|^2)^{-1}$. For each pair of coloured plaquette stencils the rescaled Fourier chart $K(\xi)$ is supported in $|\xi| \leq 2$. Its leading symbol is the Maxwell curvature symbol $H(\xi)$. For every fixed required order $q_{\mathrm{der}}$, the quadratic-form calculation of Lemma 11.5, differentiated in frequency, gives

$$|\partial^j K(\xi)| \leq C |\xi|^{-j}, \qquad |\partial^j (K - H)(\xi)| \leq C |\xi|^{1-j} \quad (0 < |\xi| \leq 1,\ j \leq q_{\mathrm{der}}).$$

These bounds follow by differentiating the inverse quadratic symbol: its curl factors remove the inverse-square singularity; replacing each lattice difference by its linear term leaves one extra power of $|\xi|$. The regulator and its derivatives obey the same fixed bounds; different colours have zero cross covariance. To convert the second estimate to position space, decompose a cutoff of $K - H$

into annuli $|\xi| \asymp 2^{-v}$. A derivative of order $s$ in position inserts $s$ powers of $\xi$. After $q_{\mathrm{der}} > 5 + s$ integrations by parts on each annulus its transform at $Rx$, with $x$ in a compact separated set, is bounded by

$$C2^{-v(5+s)} \min\{1, (R2^{-v})^{-q_{\mathrm{der}}}\}.$$

Splitting the sum at $2^v \asymp R$ gives $CR^{-5-s}$. The smooth part away from zero decreases faster than this. The leading homogeneous symbol is the corresponding curvature derivative of $\Gamma$ away from the origin, with the Fourier normalization of Part I. Including the curvature scaling powers therefore gives relative error $O(R^{-1})$.

For the finite jet samples, terms of the matching homogeneity give this leading derivative exactly. Every other term has at least one extra inverse power of $R$; summing their finitely many absolute coefficients preserves the $O(R^{-1})$ bound. Here $R = (M_t^{\mathrm{cut}})^b$, and rounding in the rescaled variables gives an additional $O(R^{-1})$. Localizing the sector covariance introduces $C(M_t^{\mathrm{cut}})^{q_{\mathrm{der}}} e^{-cB_t}$ by Lemma 11.12. Its nonlinear healthy-chart error is $C(M_t^{\mathrm{cut}})^{q_{\mathrm{der}}} \widehat{\beta}_t^{-\delta}$ for some fixed $\delta > 0$. The derivative orders and powers $q_{\mathrm{der}}$ are fixed by the degree and the finite row family; their stage constants are fixed before entry. Since $\log M_t^{\mathrm{cut}} = o(\log \widehat{\beta}_t)$ and $B_t \asymp (\log \widehat{\beta}_t)^2$, both errors are smaller than any prescribed fixed negative power of $M_t^{\mathrm{cut}}$ after entry. Choose $b$ so that the inverse-separation error is smaller than the finite power required by the source radius below. The chart and inspected supports lie inside the nonwrapping comparison interval in (14.35), so these local estimates hold on every period, with the same constants.

The finite-polynomial estimate combining these errors is as follows. Let $U$ bound the Gaussian comparison polynomials, $T$ the one-factor error, and $Z$ the two-factor error in their coefficient norm. For the connected two-point term, subtract the product of the one-point terms. Expanding that product gives the bound

$$T + Z + (U + T)T + TU. \tag{14.37}$$

For example, if $a_i$ and $a_i^0$ are the two one-point terms, then $a_1 a_2 - a_1^0 a_2^0 = (a_1 - a_1^0)a_2 + a_1^0(a_2 - a_2^0)$, whose norm is at most $(U + T)T + UT$. The terms $T, Z$ are the errors in the remaining one- and two-mark polynomial terms. Gaussian contraction expands into pairings, so the bound for any fixed polynomial is a finite polynomial in the bounds for its contractions. After $b$ has been fixed sufficiently large, the preceding covariance and Taylor estimates make (14.37) smaller than every source bound required at $l$.

The free polynomial corrections are the singleton means and the connected pair moment in $\mathsf{C}^{\mathrm{Max}}$. Subtract them before carrying the sources to coarser levels. The coefficient of $z_1 z_2$ in a connected logarithm is the second moment minus the product of the means. Consequently (14.37) bounds exactly the residual pair term; the leading free term cancels, including the product of the means. When an admissible two-mark support is absent, both its local source coefficient and the corresponding restricted polynomial are zero. This preserves the same estimate without adding a source at an unsupported address.

For terms meeting a flagged region, keep that region's complete source support and its original eligibility test. At a step $i + 1 \to i$ in stage $j$, Lemma 13.4 gives $C_j e^{-r_j \beta_{N,p,i+1} t_{i+1}^2}$ before division by the forced source allowance. Half this suppression pays the derivative and support factors of the finite marks; the other half bounds their normalized forced remainder by the error above after entry. The finite polynomial contractions, Taylor subtractions and connected attachments cost at most $\mathcal{C}_j$ at each of the $h_t(b)$ steps. Their product is the factor in the statement. At each step the source norm is submultiplicative and the rooted sum is anchored at a mark, so the bound is independent of the number of replicated cells. □

The next estimate increases the source radius while retaining the full regular and forced

representation.

**Lemma 14.22** (Corrected Gaussian sources)**.** *For arbitrarily late intervals* (14.4)*, there are common real lifts of the free basis, bounded in $N, p$ for each fixed interval and fixing the unit, such that the flowed corrected source $e^{-\mathsf{C}^{\mathrm{Max}}(z)} \prod_h (1 + z_h F_h^{\mathrm{row}})$ has a representation* (14.18) *at $k$ with radius*

$$\alpha_t = \exp(\log M_t^{\mathrm{cut}}/16).$$

*At this radius its regular and forced source sums are each at most one, with scalar source bound at most one per cell. The assertion is eventual in $N$, uniformly in $p$.*

Its proof iterates the middle-level error over the second part of the interval.

*Proof.* Use Definitions 3.1, 3.3, Lemmas 13.4, 14.5, 14.11 and 14.21. Work in the submultiplicative weighted mode norm of Lemma 14.11, with the same support weights and forced eligibility tests.

We now bound the loss over all steps; their number grows with $t$. Use (14.36). Increasing the fixed power $q_{\mathrm{der}}$ includes the finite Taylor and rounding bounds above. The source radius contributes at most $\alpha_t^2 = \mathcal{M}^{1/8}$ for a singleton or pair in one cell. Enlarge $\mathcal{C}_j$ to include all per-step finite polynomial and support constants. From (14.8) and the two ceilings in (14.4), on sufficiently late intervals with fixed $b$,

$$\mathcal{C}_j^{h_t(b)+v_t+2} \le \mathcal{M}^{(b+3)/j^2} B_t^{10/j^2} \le \mathcal{M}^{1/64}. \tag{14.38}$$

Indeed $\log B_t = 2 \log\log \widehat{\beta}_t + o(1)$ and $\log \mathcal{M} = (\log\log \widehat{\beta}_t)^2$, so $B_t = \mathcal{M}^{o(1)}$. The constant factors from the ceilings fit in the extra $\mathcal{M}^{1/j^2}$; the final inequality holds at all sufficiently large stages for each fixed $b$.

We use the following one-step source estimate. For the fixed lifts of Lemma 14.21, let $S_i$ majorize the corrected singleton terms, and $P_i$ the connected pair terms, in the weighted regular norm at radius $\alpha_t$, including their scalar terms per cell. Use the same quantities for the forced norm after division by its positive source allowance, and take the larger values. For $k \le i < l$, the connected product expansion gives

$$\begin{aligned} S_i &\le K_j^{\mathrm{src}} S_{i+1} + \mathcal{M}^{1/2} S_{i+1}^2, \\ P_i &\le K_j^{\mathrm{src}} P_{i+1} + \mathcal{M}^{1/2} (S_{i+1} + P_{i+1})^2. \end{aligned} \tag{14.39}$$

Here $K_j^{\mathrm{src}} \ge 1$ is fixed at the stage, and $\mathcal{C}_j \ge K_j^{\mathrm{src}} + 2$. To obtain these bounds, separate connected terms linear in the incoming corrected sources from those with at least two source factors. The former use the one-mark integration bound. In the latter, the product rule leaves a product of two source norms; summing the additional attachments uses the rooted estimate of Lemma 13.4. Its finite derivative and localization losses are bounded by $\mathcal{M}^{1/2}$ after entry. Every higher product is absorbed in the same quadratic bound while the source norm is at most $\mathcal{M}^{-1/2}$. For the forced terms the retained eligibility event supplies the forced suppression before its allowance is divided out. The entry requirement in item 5 includes this normalized forced bound. Disjoint cell modes use the multiplicative weights of Lemma 14.11; their connecting supports keep the same estimate. No factor counts the number of cells.

Set $\mathfrak{a}_i = S_i + P_i$. Equations (14.39) imply $\mathfrak{a}_i \le K_j^{\mathrm{src}} \mathfrak{a}_{i+1} + 2\mathcal{M}^{1/2} \mathfrak{a}_{i+1}^2$. The first part of the interval gives $\mathfrak{a}_l \le \alpha_t^2 \mathcal{C}_j^{h_t(b)+2} \mathcal{E}_t^{\mathrm{G}}$. By (14.38), for $b \ge 1$ and sufficiently late entry,

$$\alpha_t^2 \mathcal{C}_j^{h_t(b)+v_t+2} \mathcal{E}_t^{\mathrm{G}} \le \mathcal{M}^{-3/4}, \qquad \mathfrak{a}_i \le \mathcal{C}_j^{l-i} \mathfrak{a}_l \le \mathcal{M}^{-3/4} < \mathcal{M}^{-1/2}. \tag{14.40}$$

The second inequality follows by backward induction: below $\mathcal{M}^{-1/2}$ the quadratic term in the recurrence is at most $2\mathfrak{a}_{i+1}$. For the first, the covariance contribution is at most $\mathcal{M}^{1/8+1/64-b}$, whose exponent is at most $-55/64$. The other contributions tend to zero faster than every fixed negative power of $\mathcal{M}$, since $\log \mathcal{M} = o(\log \widehat{\beta}_t)$ and $B_t \asymp (\log \widehat{\beta}_t)^2$. This proves compatibility of the initial-error condition with the whole growing interval, rather than treating its constants as a finite product independent of $t$.

The extra length $v_t$ also provides the support margin: a level-$l$ neighborhood of radius $B_t^2$ has level-$k$ radius at most $B_t^2/L^{v_t} \le B_t^{-8}$. It therefore fits inside the protected cell around its rounded reading points. The successive kernel neighborhoods add a bounded geometric sum in those units; their fixed margin was included in the choice of the comparison cell. Thus the inspection and attachment tests used in the iteration stay inside the stated interval. The unit has no remainder by Lemma 14.5.

The support margin in (14.35) makes every replicated inspected stencil fit its own cell. Terms meeting flagged regions keep their original forced eligibility and the suppression used in Lemma 13.4; their source amplitude replaces only the marked letter's amplitude. Thus the local source sums at the enlarged radius are at most one in both regular and forced norms. The constant source contribution is at most one per cell. These are estimates on the complete source representation, not on its Gaussian polynomial alone. They prove the lemma on arbitrarily late intervals, with all thresholds independent of $p$. □

The remaining normalization uses the complete positive unmarked density. In particular it cannot discard the forced sum or replace it by its healthy part.

**Lemma 14.23** (Normalized source integral)**.** *For the corrected sources of Lemma 14.22, put* $u_t = \log M_t^{\mathrm{cut}}/32$. *Eventually on its intervals, uniformly in p, for every nonempty mode* $I$,

$$\left| \left\langle \prod_{C \in \mathcal{C}_k^{\mathrm{tile}}} (\mathfrak{c}_I)_C \right\rangle_{N,p} \right| \le e^{|\mathcal{C}_k^{\mathrm{tile}}| q_t^{\mathrm{cmp}}} \alpha_t^{-|I||\mathcal{C}_k^{\mathrm{tile}}|}, \qquad q_t^{\mathrm{cmp}} \le u_t.$$

*Proof.* Use Definition 3.2 and positivity of $\operatorname{dens} P$, which follows from the density identity of Theorem 3.5, the positive Wilson start and the probability kernels of Lemma 11.4, and Lemma 14.22. Let $P$ be the unmarked representation at the width level $k$, $V = |\Lambda_{n_{p,k}}| = |\mathcal{C}_k^{\mathrm{tile}}|$, $g = \beta(P)^{-1/2}$, $H_{\mathrm{path}} = A_{J_k}^{\mathrm{path}}$, and let $\delta, \Lambda^{\mathrm{loop}}$ bound its regular and loop terms. The admissibility inequalities give $0 < g \le t_k$, $\delta \le 1$, $\Lambda^{\mathrm{loop}} \le \Lambda_k^{\mathrm{loop}}$ and $\beta(P) \le 4\widehat{\beta}_k$.

Choose constants $s_H \in (0,1]$, $b_H \ge 1$ such that small Haar balls have measure at least $s_H r^{d_{\mathfrak{g}}}$ and the clipped one-link integrals are at most $b_H (g/\sigma_*)^{d_{\mathfrak{g}}}$, where $\sigma_*$ is the smaller of one and the linear clip coefficient. These follow from the nonsingular exponential chart and compactness of the group, with constants chosen before all levels. Write

$$\mathfrak{h} = \log b_H - d_{\mathfrak{g}} \log \sigma_*, \qquad \mathcal{P}(P, X, g) = 2E_{\mathrm{ex}} + 6 + \beta(P) E_{\mathrm{base}} + X + 3\mathfrak{h} - 4 \log s_H.$$

Here $E_{\mathrm{ex}}$ bounds the loop and regular exponent per site, including the exponent discrepancy $\delta$ in the clipped comparison, and on links of size at most $g/4$, $\beta(P) E_{\mathrm{base}} = 3(c_{\mathrm{dir},J_k} + H_{\mathrm{path}}^2)$. For any integrable complex $F$ with $z|F(U)| \le e^{\Phi_P(U) + VX}$, $z \ge 0$, the ratio estimate is

$$\frac{z \left| \int F \, dU \right|}{\int \operatorname{dens} P \, dU} \le e^{V\mathcal{P}(P,X,g) - 4d_{\mathfrak{g}} \log g}. \tag{14.41}$$

For the denominator, fix an axial spanning tree with $V-1$ links. Gauge invariance removes those links, leaving $3V+1$ free links. Restrict each free link to the ball of chord $g/4$. All plaquettes are then healthy, so the forced covering sum equals one. The lower action bound and the Haar-ball bound give

$$\int \operatorname{dens} P \, dU \geq e^{v_{\mathrm{vac}} V - V(E_{\mathrm{ex}} + \beta(P) E_{\mathrm{base}})} (s_H g^{d_{\mathfrak{g}}})^{3V+1}.$$

For the numerator, the clipped plaquette inequality is

$$\beta(P) E_{\mathrm{plaq}}(U) \geq \frac{\operatorname{ch}(U_\square)}{g/\sigma_*} - 1.$$

It follows in the quadratic branch from $(a/g-1)^2 \geq 0$ and $\sigma_* \leq \min(1, c^{\mathrm{dir}})$, and in the linear branch from $\sigma_* \leq c^{\mathrm{lin}}$ and $g$ below the threshold. Summing over the six plaquettes per site retains the product of $e^{-\operatorname{ch}(U_\square)/(g/\sigma_*)}$ in the upper integrand.

Keep the $3V-3$ plaquette factors selected as follows and discard the others, whose values lie in $[0,1]$. The axial tree returns each site towards zero in its highest nonzero coordinate. Order free links by that coordinate, treating wrap links first in each coordinate, except for the four links at zero, and then transverse links in decreasing height. Associate to each treated link its plaquette towards the tree. That plaquette contains the link once; every other remaining selected plaquette is independent of that link in this order. Haar invariance therefore replaces its factor by the one-link integral. The four untreated links account for the difference $(3V+1)-4 = 3V-3$. All discarded plaquette factors lie in $[0,1]$, including at the periodic boundary. Thus

$$z \Big| \int F \, dU \Big| \leq e^{v_{\mathrm{vac}} V + V(E_{\mathrm{ex}}+6+X)} \big(b_H (g/\sigma_*)^{d_{\mathfrak{g}}}\big)^{3V-3}.$$

Divide these inequalities. The powers of $g$ differ by $-4d_{\mathfrak{g}}$. Using $3V-3 \leq 3V$, $3V+1 \leq 4V$, $b_H \geq 1$ and $s_H \leq 1$ gives (14.41). This proves the estimate for the complete positive unmarked denominator without removing its forced terms outside the restricted lower-bound set.

One may use the clipped comparison exponent in this calculation even if it is not the exponent used for the source bound. Their difference has absolute value at most $V\delta$. Multiplying the upper source bound by $e^{V\delta}$ and bounding the two positive partition integrals in the opposite direction by the same factor accounts for this change. These two factors are included in $2E_{\mathrm{ex}}$ because each occurrence of $E_{\mathrm{ex}}$ includes $\delta$. Thus (14.41) applies to the actual source integral on the original exact history.

The enlarged-radius bounds of Lemma 14.22 give $X \leq 1+2$: one is the scalar bound, and each of at most two marks contributes at most one. The local coefficient estimate before integration includes $z = \alpha_t^{|I|V}$. The entry inequalities imply

$$E_{\mathrm{ex}} \leq 8H_{\mathrm{path}} + 2, \qquad -4d_{\mathfrak{g}} \log g/V \leq 1.$$

For the second, $-4d_{\mathfrak{g}} \log g = 2d_{\mathfrak{g}} \log \beta(P)$ is $O(\log \widehat{\beta}_k)$, whereas $V \geq D_k = e^{s_k}$. The first follows from $\widehat{\beta}_k t_k^3 \leq 1$, $\widehat{\beta}_k t_k^4 \leq 1$ and $\Lambda_k^{\mathrm{loop}} t_k^2 \leq 1$ on sufficiently late levels. Hence a stage constant

$$Q_j^{\mathrm{cmp}} = 2(8A_j^{\mathrm{path}} + 2) + 6 + 3(c_{\mathrm{dir},j} + (A_j^{\mathrm{path}})^2) + 3 + 3\mathfrak{h}_j - 4 \log s_H + 1 \tag{14.42}$$

bounds the exponent per cell in (14.41). All its quantities are fixed before the entry to stage $j$. Item 5 in Lemma 14.1 requires $Q_j^{\mathrm{cmp}} \leq \log M_t^{\mathrm{cut}}/32$ on the chosen intervals. This proves $q_t^{\mathrm{cmp}} \leq u_t$ and the asserted coefficient estimate. In particular the volume appears only through the number of cells, so the cutoff threshold is independent of $p$. □

The next estimate uses the corrected source comparison in Lemma 14.22. Each relative error is interior whenever its insertions are interior, since (14.33) is a polynomial in those insertions.

**Lemma 14.24** (Relative rank comparison)**.** *For every such finite test family, there are $r_\nu \downarrow 0$, $y^{\nu,i} \to y^i$, levels $k_\nu \le t_\nu$ tending to infinity, $\gamma_\nu \ge 0$ tending to zero, and real lifts $b^\nu_{j,N,p}$ of the free basis $(\mathfrak{m}_j)$, bounded in $N,p$ for fixed $\nu$ and common to all rows. They satisfy $b^\nu_{1,N,p} = \mathbf{u}_d$. Each $x^{\nu,i}$ is generic on the spacing-$a_{t_\nu}$ grid. Eventually uniformly in $p$, all row insertions are interior to $\mathcal{C}^{\mathrm{tile}}_{k_\nu}(0)$, and, for every nonempty $I$,*

$$\|R_I\|_{\mathcal{C}^{\mathrm{tile}}_{k_\nu}(0)} \le \gamma_\nu\,.$$

*Proof of Lemma 14.24.* Use Definition 4.10 and Lemmas 14.22 and 14.23. Use the coefficients $\mathfrak{c}_I$ of (14.34). With $u_t = \log M_t^{\mathrm{cut}}/32$, $\alpha_t = e^{2u_t}$, the normalized source lemma gives

$$\left|\left\langle \prod_{C\in\mathcal{C}^{\mathrm{tile}}} (\mathfrak{c}_I)_C \right\rangle_{N,p}\right| \le \exp\{|\mathcal{C}^{\mathrm{tile}}| q_t^{\mathrm{cmp}}\}\alpha_t^{-|I||\mathcal{C}^{\mathrm{tile}}|}, \qquad q_t^{\mathrm{cmp}} \le u_t. \tag{14.43}$$

The cutoff threshold is independent of $p$.

Taking the $|\mathcal{C}^{\mathrm{tile}}|$-th root in (14.43) gives

$$\|\mathfrak{c}_I\|_{\mathcal{C}^{\mathrm{tile}}} \le e^{u_t - 2|I|u_t} \le e^{-u_t} = (M_t^{\mathrm{cut}})^{-1/32} \longrightarrow 0.$$

Since $e^{\mathsf{C}^{\mathrm{Max}}}$ has coefficient $\mathcal{G}_J$ at $z_J$, multiplication by $e^{\mathsf{C}^{\mathrm{Max}}}$ shows that $\mathfrak{c}_I$ is exactly $R_I$ in (14.33). Choose successive admissible windows with both levels increasing, reading radii strictly decreasing to zero, and $e^{-u_t}$ decreasing to zero. The scale-selection windows in Lemma 14.1 allow these finitely many requests at each step. Call the resulting levels $k_\nu \le t_\nu$, radii $r_\nu$ and errors $\gamma_\nu = e^{-u_{t_\nu}}$. Rounding on the finer grid preserves genericity for each fixed trial once that grid is fine enough. Interior insertion supports imply interiority of every $R_I$, by its polynomial formula. This proves the relative-error assertion and the common choice of lifts for all rows. □

### Volume-uniform difference bounds

All the preceding comparison lemmas hold on full cutoff tails, uniformly in $p$. The local matching estimate also applies to any fixed iterated difference: after multiplying by $a_N^{d+h}$, a difference of order $h$ is a finite sum of translates of the degree-$d$ section, with subpower derivative bounds. Matching in degree $d+h$ leaves a root error tending to zero at every fixed cell size, uniformly in $p$.

This finite-cutoff argument supplies the difference bounds required in Part I.

**Lemma 14.25** (Difference estimate)**.** *For every arity $n$, token list $v$, difference list $\boldsymbol{l}$ and $r > 0$, the constants in Proposition 4.1 can be chosen independently of $p$:*

$$\sup_p \sup_{\mathrm{sep}_{\mathrm{tor}}(y)\ge r} |\Delta_{\boldsymbol{l}} M_{N,p}(v;y)| \le B(n,v,\boldsymbol{l},r) \qquad (N \ge N_0(n,v,\boldsymbol{l},r)).$$

*Proof.* Use Definitions 3.9 and 14.10, Lemmas 14.8–14.12, the local matching and equation comparison of Lemmas 14.17–14.19 on full cutoff tails, and the free tests and rank comparison of Lemmas 14.20–14.24. We first bound the roots of a differentiated insertion at each sufficiently fine fixed cell size.

Let $F = O^{(d)}(c)$, and let $\Delta_{\boldsymbol{a}} F$ be a fixed iterated normalized difference in its anchor. Local matching gives admissible coefficients $c^{\boldsymbol{a}}$ in degree $e = d + |\boldsymbol{a}|$ with $O^{(e)}(c^{\boldsymbol{a}}) - \Delta_{\boldsymbol{a}} F$ of vanishing

root. Write $c^{\mathrm{pr}} = \Pi_{e,N,p} c^{\boldsymbol{a}}$. For an admissible coefficient in $\ker \Pi_e$, the auxiliary equation section has root zero. To see this, take all the other reflected copies as the spectator in the exact Haar identity in Lemma 10.4. For a fixed physical cell, its side in lattice units eventually exceeds twice the equation radius plus two; all those copies are then exterior. The section-comparison residual tends to zero in root. In particular it applies to $z = (c^{\boldsymbol{a}} - c^{\mathrm{pr}})/(1 + \|c^{\mathrm{pr}}\|)$. Indeed, $c^{\boldsymbol{a}}/(1 + \|c^{\mathrm{pr}}\|)$ is admissible, and $c^{\mathrm{pr}}/(1 + \|c^{\mathrm{pr}}\|)$ has norm at most one. Every bounded coefficient family is admissible, because Lemma 14.8 gives

$$\|\mathsf{Z}_{N,p,e}^{-1}\| \le C_e \prod_{i=r(e)}^{N-1} C_{J_{i+1}} = a_N^{-o(1)}$$

by (14.8). Subtracting these two admissible families proves admissibility of $z$ before $c^{\mathrm{pr}}$ is bounded.

It remains to bound $c^{\mathrm{pr}}$. Use a fixed sufficiently short-distance trial in Lemma 14.24 for degree $e$. Average the variable slot of every row over $Q_N = \lceil b/a_N \rceil$ sites in each direction occurring in $\boldsymbol{a}$, with $b > 0$ small enough that all translated tests remain separated. Denote this linear map on coefficients by $\mathcal{P}_{N,p}$. Telescoping in one direction gives

$$\frac{1}{Q_N} \sum_{j=0}^{Q_N - 1} \frac{f(z + (j+1)e_a) - f(z + je_a)}{a_N} = \frac{f(z + Q_N e_a) - f(z)}{Q_N a_N}.$$

Since $Q_N a_N \ge b$, iterating bounds the tests of $\Delta_{\boldsymbol{a}} F$ by $(2/b)^{|\boldsymbol{a}|}$ times the supremum of tests of $F$. The latter is bounded by Lemma 14.12, the interior-root comparison and (14.16). Matching errors tend to zero in these tests by the finite-cutoff argument of Lemma 15.3: put the error and the bounded spectator insertions in distinct cells with fixed margins, use Lemma 15.2 for their interior roots, and apply (14.16). All these bounds hold on full cutoff tails and use no positional difference estimate. Thus $\|\mathcal{P}_{N,p} c^{\boldsymbol{a}}\| \le C_{\mathrm{test}}$ eventually, uniformly in $p$.

The finite-dimensional lower bound is obtained before any cutoff limit. The free matrix has a positive lower bound on the unit sphere. The relative-error estimate and the equation-root bound put the tests of the projected trial lifts within $O(\gamma_\nu) + o_{N\to\infty}(1)$ of the free matrix, uniformly in $p$. Choose the fixed trial $\nu$ with this error smaller than its matrix lower bound. Averaging over sufficiently small $b$ preserves the lower bound: the undifferentiated correlation tests have uniform positional continuity, proved in Lemma 14.15. The projected lifts are therefore linearly independent. Their number equals $\dim M_e^{\mathrm{Max}}$, whereas Lemma 14.16 bounds $\dim \operatorname{ran} \Pi_e$ by that number. They consequently span the range. Their uniform coefficient bounds give

$$\|c^{\mathrm{pr}}\| \le C_{\mathrm{det}} \|\mathcal{P}_{N,p} c^{\mathrm{pr}}\| \tag{14.44}$$

with $C_{\mathrm{det}}$ independent of $N, p$ on the retained tails. Choose an invertible square minor $H_0$ of the free matrix and the corresponding interacting minor $H$. For $\|H - H_0\| \le (2\|H_0^{-1}\|)^{-1}$, the inverse bound $\|H^{-1}\| \le 2\|H_0^{-1}\|$ follows. This proves (14.44) without assuming a continuum rank identity.

The equation coefficient $z = (c^{\boldsymbol{a}} - c^{\mathrm{pr}})/(1 + \|c^{\mathrm{pr}}\|)$ is admissible. Its tests tend to zero, so eventually $C_{\mathrm{det}} \|\mathcal{P}_{N,p} z\| \le 1/2$. Equation (14.44) yields

$$\|c^{\mathrm{pr}}\| \le C_{\mathrm{det}} C_{\mathrm{test}} + \tfrac{1}{2}(1 + \|c^{\mathrm{pr}}\|), \qquad \|c^{\mathrm{pr}}\| \le 2 C_{\mathrm{det}} C_{\mathrm{test}} + 1.$$

Lemma 14.12 now bounds the roots of $O^{(e)}(c^{\mathrm{pr}})$ at each fixed sufficiently fine scale. Now $c^{\boldsymbol{a}} - c^{\mathrm{pr}}$ is admissible, so its equation residual has vanishing root. Together with the matching residual, the

triangle inequality for roots gives the same type of bound for $\Delta_{\boldsymbol{a}} F$. All thresholds and constants in this argument are independent of $p$.

Distribute the differences in $\boldsymbol{l}$ among their indicated slots. A difference in slot $i$ acts only on that insertion, so the resulting moment is exactly the expectation of the product of these differentiated insertions. Choose a fixed tiling side smaller than $r$ and translate its boundaries away from all anchors. In each axis the excluded boundary strips occupy less than one full cell length if their widths are chosen below $a_m/[4(n+1)]$. For sufficiently large $N$ a lattice translation with these margins exists. Distinct anchors occupy distinct cells, since their torus separation exceeds the cell side. The interior-root comparison, with a finer centered scale, bounds all the factors uniformly. Apply (14.16) and put one in every unused cell. Their product gives the asserted $B$. Empty difference lists and arities at most one obey the same argument. This proves the lemma and Proposition 4.1. □

### Bounded coarse insertions

The later phase argument needs bounded approximations with a bound independent of the cutoff. Compactness at each cutoff gives no such bound. We obtain it by stopping at one fixed coarse lattice and pulling bounded observables back through the probability block kernels.

In the next two lemmas, a statement holds eventually in the ordered limits when it holds for all sufficiently large indices of the chosen volume sequence and, at each such volume, all sufficiently fine cutoffs of the chosen spacing sequence. No common cutoff threshold in other subsequently fixed parameters is asserted. The chessboard roots used below are defined in Definition 14.10.

**Lemma 14.26** (Bounded coarse approximations)**.** *For every $d$, $k \geq r(d)$, real $c \in V_d$ and $\delta > 0$, there are measurable real observables $\mathcal{B}_{N,p}(c,y)$ and $M_{\mathrm{bnd}} < \infty$, independent of $N, p, y$, with $|\mathcal{B}_{N,p}(c,y)| \leq M_{\mathrm{bnd}}$ and*

$$\|\mathcal{B}_{N,p}(c,y) - O^{(d)}_{N,p}(c,y)\|_{\mathcal{C}_k^{\mathrm{tile}}(y)} \leq \delta$$

*eventually in the ordered limits. Both observables are interior to that cell, and the same holds for their signed and reflected images.*

*Proof.* Use Definitions 3.1, 3.9, Lemmas 14.5, 14.9, and the several-source estimate in Lemma 14.11. The construction in this proof is independent of the phase decomposition. Choose a fixed $m \geq k$ with $B_m^2 \leq a_k/a_m$, using Lemma 14.1. At a finer level $h \geq m$, the bounded coarse polynomial with coordinate $v$ is

$$\mathsf{V}_h(v;U) = a_h^{-d} \left( v_0 + \mathbf{1}_{\Sigma_d \text{ healthy}} P_d[\mathsf{D}^{\mathrm{fac}}_{t_h^{-1}} v^\circ](U) \right).$$

Its supremum for each basis input is bounded by a constant depending on $d, h$, uniformly in $N, p$: the healthy polynomial uses a fixed compact coordinate range, and the unit term is constant. Its inspected stencil has a fixed radius, including the links needed for the health test.

Let $\mathcal{K}_{N,h}(U, dV)$ be the probability kernel obtained by composing the block kernels from $N$ to $h$, and set

$$(\mathcal{P}_{N,h} F)(U) = \int F(V)\, \mathcal{K}_{N,h}(U, dV).$$

It preserves real values and does not increase the supremum norm. The locality of the block kernels confines its dependence to the fine preimage of the inspected stencil. For reflected cells these ancestor cones are disjoint once $h$ has the stated margin. At each intermediate level the block kernels are products in the coarse-link variables given the preceding field; integrating disjoint ancestor cones therefore gives conditional independence, and hence $\prod_C \mathcal{P}_{N,h} F_C = \mathcal{P}_{N,h}(\prod_C F_C)$.

Choose $h$ sufficiently large that this preimage lies strictly inside the centered level-$k$ cell, including its boundary-link margin.

To compare with the actual insertion, choose trial fine coefficients $u_j$ whose selected level-$h$ polynomial coordinates differ by at most $1/4$ from the basis vectors. They are obtained by carrying the pure coarse polynomial back through the inverse graph construction in Lemma 14.5. Apply (14.26) to the difference with this primary coarse polynomial. Put $S_h = C_{J_h}$ temporarily for the column bound, and let $\mathcal{J}_h$ be the matrix of the selected level-$h$ coordinates of these trials. For a basis input, the cone and the finest-level boundary estimate give

$$32(1+S_h)^2|e_j - \mathcal{J}_h e_j|_1 \le R_h, \qquad R_h \le 2(1+S_h), \qquad |e_j - \mathcal{J}_h e_j|_1 \le \frac{1}{16(1+S_h)} \le \frac{1}{4}. \tag{14.45}$$

Taking the maximum over columns gives $\|\mathcal{J}_h - 1\|_1 \le 1/4$, which implies $\|\mathcal{J}_h^{-1}\|_1 \le 2$. If $c$ is the required reference coefficient, linear recombination therefore gives

$$O(c) = \sum_j t_j O(\mathsf{Z}_{N,p,d} u_j), \qquad \sum_j |t_j| \le 2e^{\sum_{i<h} \log C_{J_{i+1}}} |c|_1.$$

Replace each term on the right by $\mathcal{P}_{N,h}\mathsf{V}_h(e_j)$. This defines a bounded complex observable $\mathcal{B}'_{N,p}$; its bound depends on $h, d, c$, and hence may depend on $\delta$, but is independent of $N, p$.

The discrepancy is controlled by replicating all its reflected copies before taking the marginal kernel. Formula (14.17) applies to the fine insertions; at level $h$ multiply the source density by the corresponding coarse factors. The conditional factorization above identifies the coefficient at each mark with $\mathcal{P}_{N,h}\mathsf{V}_h(e_j) - O(\mathsf{Z} u_j)$. The probability kernel supplies no additional supremum factor. Weighted coefficient extraction and (14.2) then give, after recombination,

$$\|\mathcal{B}'_{N,p} - O(c)\|_{\mathcal{C}_k^{\mathrm{tile}}} \le C_{k,m,d,c}\, e^{\sum_{i<h} \log C_{J_{i+1}}} (1 + C_{J_h}) e^{-(s_h - s_m)/8}. \tag{14.46}$$

The prefactor has subpower growth in $a_h^{-1}$ by (14.8); the last factor tends to zero. Choose $h$ so that this expression is at most $\delta$, before choosing the cutoff. This also meets the stencil-fit condition.

For real $c$, the original insertion is real by Lemmas 10.2 and 14.7. Set $\mathcal{B} = \operatorname{Re}\mathcal{B}'$. In the centered cell,

$$\mathcal{B} - O(c) = \tfrac{1}{2}(\mathcal{B}' - O(c)) + \tfrac{1}{2}\overline{\mathcal{B}' - O(c)}.$$

The root seminorm and its invariance under conjugation show that this does not increase (14.46); the supremum bound also does not increase. Translation, signed permutation and conjugation use the same finite collection of stencils and bounds. This proves the lemma on the stated ordered tails. □

Fix a degree bound $d$, a lower level $m$ and an ordered extraction. A real grid word is a list $w = ((d_i, c_i), x_i)_{i=1}^n$ with $d_i \le d$, $\bar{c}_i = c_i$ and distinct spacing-$a_k$ positions. We assume its configuration is generic. Write $\theta_{\mathrm{lin}}$ for linear time reflection on positions and labels, without scalar conjugation. Put

$$F_{N,p}(w) = \prod_i O_{N,p}^{(d_i)}(c_i, x_i/a_N), \qquad z_j(T) = T j a_k(1,1,1,1),$$

$$f_{j,N,p,T} = F_{N,p}(\tau_{z_j(T)} w) - F_{N,p}(\tau_{z_j(T)} \theta_{\mathrm{lin}} w).$$

These are the real word differences used in the off-diagonal variance estimate in Section 17.

**Lemma 14.27** (Bounded real word approximations)**.** *There is $k \geq m$, chosen before $w$, such that every such word has, for each $\varepsilon > 0$, measurable real approximations $g_{j,N,p,T}$ and $R_\varepsilon < \infty$ with $|g_{j,N,p,T}| \leq R_\varepsilon$ eventually for each fixed $T, j$. For each $i < j$ and every sufficiently large integer $T$, eventually in the ordered limits,*

$$|\langle g_{i,N,p,T} g_{j,N,p,T}\rangle_{N,p} - \mathrm{Re}\langle f_{i,N,p,T} f_{j,N,p,T}\rangle_{N,p}| \leq \varepsilon.$$

*Proof.* Use Definitions 3.9, 14.10, Lemmas 14.26, 14.12 and the finite reflection argument of Lemma 15.2. Choose $k \geq \max(m, \max_{e \leq d} r(e))$. It depends only on $d, m$, before the length, labels and positions of $w$. All its finer comparison levels remain available by Lemma 14.1.

Include every label in $w$ and its signed and conjugated images in a finite list. Lemma 14.12 gives a bound $C$ for their centered roots at any one fixed sufficiently fine level. Choose that level so that its cells and the margins required by Lemma 15.2 fit between distinct spacing-$a_k$ points. Apply Lemma 14.26 with error $\delta$ at a still finer centered level, preserving these margins. Denote its real bounded approximations by $\mathcal{B}_i$. The same family, translated to any anchor, is used for every copy of the word. Its roots are at most $C + \delta$, independently of its possibly much larger supremum bound.

For a product of $n_{\mathrm{word}}$ such insertions at distinct grid sites, telescope the products, replacing one factor at a time. Put all $n_{\mathrm{word}}$ factors into distinct cells with the chosen margins. The interior-root comparison and (14.16) give

$$\left| \left\langle \prod_{a=1}^{n_{\mathrm{word}}} \mathcal{B}_a - \prod_{a=1}^{n_{\mathrm{word}}} O_a \right\rangle_{N,p} \right| \leq n_{\mathrm{word}} \delta (C + \delta)^{n_{\mathrm{word}} - 1}. \tag{14.47}$$

Only the roots, not the supremum bounds on $\mathcal{B}_a$, occur on the right. For each fixed finite configuration, sufficiently large periods avoid wrapping identifications; its stencils then lie in the prescribed cells on the inner cutoff tail. This explains the order of the two eventualities.

Let $G_{j,N,p,T}$ be the product of the $\mathcal{B}_i$ for the translated word, and let $G^{\theta}_{j,N,p,T}$ be the corresponding product for its reflected word. Set $g_j = G_j - G^{\theta}_j$. Choose $M_i^{\mathrm{bnd}}$ to bound both the original and reflected approximations in slot $i$. Then

$$|g_j| \leq 2 \prod_i M_i^{\mathrm{bnd}} =: R_\varepsilon,$$

independently of $N, p, T, j$. Every $g_j$ is real and measurable. For fixed $i < j$, sufficiently large $T$ separates the two translated finite lists, for each of the four combinations of a word and a reflected word. Indeed their first coordinates differ by $T(j - i)a_k$ plus a fixed finite difference. Each list is itself distinct because the original word is generic and reflection is injective.

Expand $g_i g_j$ and $f_i f_j$ into their four product terms and apply (14.47) with $n_{\mathrm{word}} = 2n$. Choose $0 < \delta \leq 1$ so that $4(2n)\delta(C + 1)^{2n-1} \leq \varepsilon$. Taking real parts cannot increase the error. Intersecting the four eventual sets proves the assertion. The empty word has zero difference and may be assigned $g_j = 0$. There is no common cutoff threshold in $T, i, j$; a finite list of pairs is handled by intersecting its eventual sets. The choice gives such a grid level above every prescribed lower level $m$, which is the cofinality required in Section 17. □

## 14.3 Full ultraviolet convergence

Fix $p$. The normalized density at a fixed level is

$$\frac{\mathrm{dens}\, \rho_{N,p,m}}{\int \mathrm{dens}\, \rho_{N,p,m}} = \frac{e^{-P^{\mathrm{path}}_{N,m} - \sum_i \mathbf{1}_{X_i \text{ healthy}} E_i^{\mathrm{reg}}} \mathcal{F}_{N,p,m}}{\int e^{-P^{\mathrm{path}}_{N,m} - \sum_i \mathbf{1}_{X_i \text{ healthy}} E_i^{\mathrm{reg}}} \mathcal{F}_{N,p,m}}. \tag{14.48}$$

Here $\mathcal{F}_{N,p,m}$ is the forced sum of Definition 3.2. The factor $e^{v_{\mathrm{vac},m}|\Lambda_{n_{p,m}}|}$ has canceled. This identity is needed because no convergence of that factor is proved or used.

For a fixed finite token list and a separated common-grid configuration $x$, let $o_{i,m} = x_i/a_m$ be its sites at all sufficiently fine levels. They are read modulo the period; their parents agree at consecutive levels. In (14.18), form $\mathcal{H}^0_{N,p,m}$ by setting $H_{m,X} = K^{\geq 2}_{m,Y} = 0$, keeping the complete singleton functions $f_{i,m}$ and $k_{i,m,Y}$. Define

$$C_{N,p,m}(v;x) = \frac{\int [z_{\{1,\dots,n\}}] \mathcal{H}^0_{N,p,m}(z;U)\, dU}{\int \operatorname{dens} \rho_{N,p,m}\, dU}. \tag{14.49}$$

The numerator is therefore specified by the unmarked representation and the one-mark families, rather than by a new choice of a several-insertion density. The difference from the exact source is estimated separately.

The first lemma compares different cutoffs at fixed volume and fixed level. It uses the convergence assertion for the unmarked recursion, rather than applying a same-cutoff uniqueness inequality to histories of different lengths.

**Lemma 14.28** (Fixed-level marked convergence)**.** *For every fixed $p, v, x$ as above and every sufficiently fine fixed level $m$, the scalar-free unmarked density and the complete singleton functions in (14.18) are Cauchy in their common function spaces as $N \to \infty$. Their normalized integrals satisfy*

$$C_{N,p,m}(v;x) \text{ is Cauchy as } N \to \infty. \tag{14.50}$$

*The scalar-free partition integral is bounded below by a positive constant depending on $p, m$.*

*Proof.* Use Definitions 3.2, 3.9, 13.3 and Lemmas 13.19, 13.20, 14.5, 14.9. Lemma 13.19 gives full-sequence convergence of the scalar couplings and scalar-free activities in fixed weighted spaces at each fixed $p, m$. Lemma 13.20 includes the prescribed paths with the strict reserve. Thus the cutoff-dependent path arrays converge in the same charts. The vacuum scalar is excluded, as in (14.48).

Use $\kappa_i = (\epsilon_i^{\mathrm{for}})^{1/2}$ in the remainder norm, and normalize graph differences by $C_{J_i}$ and the coefficient sum. The graph bound of Lemma 14.5 bounds this distance between two admitted graphs by 2. The contraction in Lemma 14.5 reduces it by $1/2$ through a step on the same unmarked representation. For two different cutoffs, first compare both steps with a step on their limiting unmarked coordinates. The one-mark local integrals, their Taylor coefficients and the forced regrouping are continuous in these coordinates. To check this continuity, retain the common summable majorants in (14.14); each finite local integral is continuous in its coefficients and chart, and the remaining tails are bounded by these same summable majorants. The inverse coefficient maps are continuous because their inverses have the common bound already proved. Hence for each fixed $i$ the extra discrepancy $\delta_i(N, N')$ tends to zero as both cutoffs tend to infinity.

Iterating from a remote fixed level $h$ down to $m$ gives the genuine cross-cutoff estimate

$$d_m(N, N') \leq 2^{1-(h-m)} + \sum_{i=m}^{h-1} 2^{-(i-m)} \delta_i(N, N'). \tag{14.51}$$

This estimate compares graphs after the fixed-level unmarked identification, not histories of the same length. First make $h - m$ large. For that fixed $h$, the sum tends to zero, proving the Cauchy property at $m$. The finitely many coefficient maps between the reference level and $m$ converge as well. Solving their fixed-dimensional linear systems therefore gives Cauchy polynomial coordinates for each fixed reference label. Substituting in the graph gives the complete singleton functions.

On the fixed finite torus, weighted convergence bounds the sums of the regular terms uniformly and bounds the forced terms after grouping identical supports and inventories. There are only finitely many such supports and inventories. The resulting scalar-free density and the singleton functions are uniformly Cauchy on their fixed health domains. The health threshold at level $m$ is fixed; no moving indicator is being differentiated or replaced. Formula (14.18), with its higher modes set to zero, is a finite polynomial in the grouped functions and their exponentials. Its scalar-free coefficients are consequently uniformly Cauchy and integrable against Haar probability measure.

The denominator has a common positive lower bound. Restrict to an axial tree gauge with all free links of chord at most $g/4$, choosing $g > 0$ below every admitted $\beta_{N,p,m}^{-1/2}$ on the retained fixed-level tail. The forced sum on this set is one, and the scalar-free action is uniformly bounded above there. Its product Haar measure is positive. The same lower bound applies to all cutoffs on that tail. Dividing the Cauchy numerator integrals by these positive Cauchy denominators proves (14.50). □

The source density transfers exactly, but its singleton part does not: connected components with two or more marks remain. The next estimate bounds the effect of omitting all those components while keeping the bound independent of the stopping level.

**Lemma 14.29** (Stopping estimate)**.** *For fixed $p, v, x$, there are $F_0 < \infty$ and $m_0$ such that, for each $m \geq m_0$ and every sufficiently large $N$,*

$$|M_{N,p}(v; x/a_N) - C_{N,p,m}(v; x)| \leq 2F_0 e^{-B_m/64}. \tag{14.52}$$

*The constant $F_0$ is independent of $m, N$; the cutoff threshold may depend on $m$.*

*Proof.* Use Definitions 3.2, 3.9, Lemmas 14.11, 14.28, and the exact source evolution (14.17). Choose a base level $k_0$ above the reference levels and the grid containing $x$. At all sufficiently fine levels, distinct ancestral sites have separation at least $B_i^2$: their separation is a fixed positive physical number divided by $a_i$, while $B_i^2$ grows only as a power of $\log \widehat{\beta}_i$. Their occupancy is one. Enlarge $k_0$ so these assertions and all required positive source allowances hold thereafter.

Choose positive weights $w_i$ at this base level so the weighted complete singleton sizes lie below those allowances. This choice is independent of $N$: Lemma 14.28 bounds each singleton uniformly along the fixed-level cutoff tail. Let $s_0^{\mathrm{reg}}, s_0^{\mathrm{for}}$ bound the weighted regular singleton supremum and the sum of forced singleton suprema at $k_0$. Set

$$F_0 = \frac{\exp\{|\Lambda_{n_{p,k_0}}| q_{k_0} + n(s_0^{\mathrm{reg}} + s_0^{\mathrm{for}} + 2)\}}{\prod_i w_i}. \tag{14.53}$$

The numerator is the normalized-integral bound of the several-mark expansion, now with these fixed base weights. It depends on the base level, not on the later stopping level.

At level $m > k_0$, every nonzero mode of $H_{m,X}$ or $K_{m,Y}^{\geq 2}$ contains at least two marks and its support contains their anchors. The covering estimate (14.19) gives

$$\mathrm{cov}_{B_m}(X) \geq B_m/64.$$

Let $R_m = e^{B_m/64}$ and multiply every one of these higher-mode functions by a common complex variable $z$. For $|z| \leq R_m$, its amplitude increase is absorbed by decreasing the support exponent by two:

$$|z| e^{(\sigma-2)\,\mathrm{cov}_{B_m}(X)} \leq e^{\sigma\,\mathrm{cov}_{B_m}(X)}.$$

Singleton functions are not multiplied by $z$. The source representation therefore remains integrable on $|z| < R_m$, with its prescribed singleton functions, original forced eligibility tests and two fewer units of support exponent. Here $\sigma = \sigma_{J_m} = 100$. For $s \geq 0$ and $98 \leq \tau \leq 100$, $e^{\tau s} \leq e^{100s}$, so every regular or forced weighted sum decreases when its exponent decreases. Lemma 13.4 bounds the connected regular terms at exponent $\sigma + 3$ and the forced terms at $\sigma + 2$. The marked product factors are paid at the entry chosen in item 5 of Lemma 14.1. Applying these bounds in the weighted source norm of (14.18) leaves the two units used here, with final exponent 98.

Carry this modified density back to the same base level $k_0$, using the rooted sums of Lemma 13.4 with these reduced weights. Every nonzero mode contains only marks whose anchors lie in its support, so multiplying modes retains their positive weights without adding a mark outside that support. At the base level, the singleton exponential and forced factors contribute at most $n(s_0^{\mathrm{reg}} + s_0^{\mathrm{for}})$ to the logarithmic majorant; the remaining connected regular and forced sums contribute at most $2n$. Equation (14.2) contributes $|\Lambda_{n_{p,k_0}}|q_{k_0}$. Coefficient extraction divides by $\prod_i w_i$, giving exactly (14.53). Absolute convergence on $|z| < R_m$ makes the normalized full-mode integral $F_m(z)$ holomorphic there, with

$$|F_m(z)| \leq F_0.$$

At the base level this is precisely the bound (14.53). It avoids the level-$m$ volume factor that would arise from estimating the normalized integral only where it was stopped.

For $R_m > 2$, the Cauchy derivative estimate on circles of radius $R_m/2$ centered on $[0, 1]$ gives

$$|F_m(1) - F_m(0)| \leq 2F_0/R_m.$$

At $z = 1$ this is the exact multiple-source integral. Its equality to the original cutoff moment follows from Fubini and the probability normalization of every block kernel in (14.17). At $z = 0$ it is exactly (14.49). Substitution proves the estimate. For each fixed $m$, the required bound on its singleton functions holds on a cutoff tail by Lemma 14.28; it changes the threshold but leaves $F_0$ fixed. □

The fixed-level and stopping estimates, together with the positional bounds already proved, give the full limit.

**Lemma 14.30** (Lattice-point convergence)**.** *For every $p, n, v$, there is $f_{p,v} \in C^\infty(\mathrm{NoHit}_{L_p,n})$ such that, for every $\boldsymbol{l}$, compact $K \subset \mathrm{NoHit}_{L_p,n}$ and $\varepsilon > 0$, some $N_0(p, n, v, \boldsymbol{l}, K, \varepsilon)$ gives*

$$|\Delta_{\boldsymbol{l}} M_{N,p}(v; x/a_N) - \partial_{\boldsymbol{l}} f_{p,v}(x)| < \varepsilon$$

*at every spacing-$a_N$ point of $K$ in the periodic lift, for every $N \geq N_0$.*

*Proof.* Use Definitions 2.3 and 3.9, Lemma 14.25, and Lemmas 14.28–14.29. At a separated configuration on a fixed common grid, the triangle inequality gives

$$|M_{N,p} - M_{N',p}| \leq 4F_0 e^{-B_m/64} + |C_{N,p,m} - C_{N',p,m}|.$$

First take $m$ large, using $B_m \to \infty$, and then take $N, N'$ large. This proves the Cauchy property at every such configuration. Since every ratio is an integer, each coarser grid is contained in every finer grid. The union of these grids is dense in the periodic noncollision domain.

We next extend this limit and identify its derivatives. Choose a compact neighbourhood $K'$ of $K$ contained in that domain, and a positive lower bound for its torus separation. Lemma 14.25

bounds the lattice functions and all their fixed differences there. One more difference gives, by telescoping along coordinate lattice paths,

$$|G_N(z) - G_N(z')| \le B\,|z - z'|_1$$

for nearby physical grid points, where $G_N = \Delta_{\boldsymbol{l}} M_{N,p}$. For order zero, a finite dense grid in $K'$ and this estimate extend the pointwise Cauchy property to uniform convergence on $K$. The multilinear interpolants obey the same bound, with an additional $O(a_N)$ displacement to their vertices. The resulting limit $f_{p,v}$ is continuous and periodic.

Suppose $G_N$ already converges locally uniformly to $g$. Bounds for two more differences compare its first forward difference with a difference over a fixed physical interval. Put $h_N = Q_N a_N \to h > 0$, with $Q_N \in \mathbb{N}$. Summing successive first differences and bounding their variation by the second difference gives

$$\left| \frac{G_N(z + h_N e) - G_N(z)}{h_N} - \Delta_e G_N(z) \right| \le \tfrac{1}{2} B h_N. \tag{14.54}$$

Choose $h$ small and then let $N \to \infty$. The endpoint values converge uniformly, so (14.54) proves that $\Delta_e G_N$ is uniformly Cauchy on $K$. Call its continuous limit $g_e$. The exact telescoping identity on grid segments, followed by Riemann summation of the uniformly convergent first differences, gives

$$g(z + te) - g(z) = \int_0^t g_e(z + ue)\,du.$$

Thus $\partial_e g = g_e$. Induct on the length of the difference list. Every required additional order is available from Lemma 14.25, so all derivative limits exist and $f_{p,v}$ is smooth. Finite covers by such neighbourhoods give the asserted uniform statement on $K$. □

The same limit applies when observation points move across periodic representative boundaries. Smearing then follows from a compact-support Riemann sum.

**Lemma 14.31** (Interpolation and smearing)**.** *For fixed $p, n, v, \boldsymbol{l}$ and $x_N \to x \in \mathrm{NoHit}_{L_p,n}$, the multilinear interpolants of $\Delta_{\boldsymbol{l}} M_{N,p}$ at $x_N$ converge to $\partial_{\boldsymbol{l}} f_{p,v}(x)$. For every $p$-admissible separated test $t$,*

$$t\partial_{\boldsymbol{l}} f_{p,v} \in L^1, \qquad a_N^{4n} \sum_y t(\hat{y}) \Delta_{\boldsymbol{l}} M_{N,p}(v; y) \longrightarrow \int t\partial_{\boldsymbol{l}} f_{p,v}.$$

*Proof.* Use the placement and moment conventions of Definition 2.3, admissibility from Section 4, and Lemma 14.30. Choose a compact neighbourhood of $x$ inside the periodic noncollision domain. Every vertex used by the interpolant at $x_N$ lies there eventually and tends to $x$. The interpolant is a convex combination of the vertex values. Uniform convergence at these vertices and continuity of the limiting derivative give the moving-point assertion. Work on the periodic lift throughout; no continuity of the choice of centered representatives is needed.

The closed support of $t$ is a compact subset of that domain, strictly inside the chosen period box. The smooth limiting derivative is bounded there, so its product with $t$ is integrable. On a fixed compact neighbourhood, Lemma 14.30 makes the supremum of the difference between the lattice values and the limiting derivative tend to zero. Its contribution to the sum is at most that supremum times $\|t\|_\infty$ times the total volume of the grid cubes meeting the support. This volume is bounded independently of $N$. The remaining sum is the Riemann sum for the continuous compactly supported function $t\partial_{\boldsymbol{l}} f_{p,v}$ and converges to its integral. □

*Proof of Theorem 4.2.* Set $M^{\mathrm{UV}}_{p,v} = f_{p,v}$. Lemma 14.30 gives its smoothness on the full periodic noncollision domain, and Lemma 14.31 gives the asserted interpolation and smearing limits for every fixed derivative list. Thus every inner limit in Definition 4.3 exists at every admissible fixed volume along the full cutoff sequence. □

# 15 Infinite volume and field calculus

We pass from the fixed-volume kernels of Theorem 4.2 (UV) to correlations on $\mathbb{R}^4$. The order of limits matters: the lattice spacing tends to zero at every fixed admissible volume, and only then does the volume increase. Lemmas 15.1–15.5, using the ultraviolet estimates of Lemmas 14.16–14.24, prove Theorem 4.7 (IV). The same volume refinement supplies the coefficient maps needed for Proposition 4.11 (FC). Lemma 15.6 controls repeated insertions and will be used to construct the bounded slab operators in Lemma 17.1.

There are two compactness arguments. The bounds of Proposition 4.1 (UB), including one additional difference, make the ultraviolet kernels and their derivatives equicontinuous on compact separated sets. A countable diagonal extraction therefore chooses the volume limit for all polynomial fields. Finite-dimensional compactness also retains certain corrected coefficient maps. These maps are needed because a positional difference changes the degree and the insertion normalization; the polynomial embedding between two coefficient spaces need not preserve correlations.

The identification of null coefficients is the additional difficulty. An approximation to a derivative may have unbounded coefficients. We normalize its projected part and use correlation tests to exclude a nonzero null vector in the range of the limiting projection. This produces one fixed coefficient vector that represents the derivative against every spectator. Finally, chessboard estimates give bounds uniform in the positions, including configurations of arbitrarily large diameter. For repeated words, two cell sizes keep the cost of the exterior independent of the number of repetitions.

Throughout we use the Wilson family, tokens, differences and separated tests of Section 4. Write $f_{p,v} = M^{\mathrm{UV}}_{p,v}$ and $\langle F\rangle_{N,p} = \int F\, d\mu_{N,p}$. All constants below may depend on the fixed construction. Any further dependence is indicated.

## 15.1 Volume limits

Fix a strictly increasing $\mathsf{p} : \mathbb{N} \to \mathbb{N}$. For each $p, n, v$, let $S^{(p)}_v$ be the continuous restriction of $f_{p,v}$ to the periodic generic domain

$$\{x : x^a_i - x^a_j \notin L_p\mathbb{Z} \quad (i \neq j,\ a = 1, \dots, 4)\}.$$

These restrictions retain the fixed-volume kernels required in IV. The next extraction chooses the limiting table and volume sequence together, with convergence of all derivatives.

**Lemma 15.1** (Generic volume kernels)**.** *There are a strictly increasing* $\mathsf{q}$ *and a complex multilinear table* $S$*. Put* $P = \mathsf{p} \circ \mathsf{q}$*. For every* $n, v, \boldsymbol{l}$*,*

$$S_n(v;\cdot) \in C^\infty(\mathrm{Conf}_n), \qquad \partial_{\boldsymbol{l}} f_{P(k),v} \longrightarrow \partial_{\boldsymbol{l}} S_n(v;\cdot) \quad \textit{locally uniformly on } \mathrm{Conf}_n\,. \tag{15.1}$$

*Every compact* $K \subset \mathrm{Conf}_n$ *eventually lies in* $\mathrm{NoHit}_{L_{P(k)},n}$*. For every separated test* $f$*,*

$$C^{\mathrm{UV}}_{P(k)}(v; f) \longrightarrow \int f(x) S_n(v; x)\, dx. \tag{15.2}$$

*Both convergences persist under further refinement.*

The proof below also gives eventual inclusion of every compact generic set in the periodic generic domains and uniform convergence of $S_v^{(P(k))}$ there. After Lemma 15.4, $P$ denotes its final refinement.

*Proof.* Fix a compact $K \subset \mathrm{Conf}_n$ and a compact neighbourhood $K'$ contained in $\mathrm{Conf}_n$. For all sufficiently large $p$, $K'$ lies in $\mathrm{NoHit}_{L_p,n}$, and its torus separation has a positive lower bound $2r$ independent of $p$. UB bounds $\Delta_{\boldsymbol{l}} M^{\mathrm{lat}}_{N,p}$ by $B(v,\boldsymbol{l},r)$ on lattice tuples near $K'$, for all sufficiently large $N$, uniformly in $p$. By Lemma 14.31, the multilinear interpolants converge at each fixed $p$ to $\partial_{\boldsymbol{l}} f_{p,v}$. Consequently

$$\sup_{p\ge p_0}\sup_{x\in K'} |\partial_{\boldsymbol{l}} f_{p,v}(x)| \le B(v,\boldsymbol{l},r). \tag{15.3}$$

The ultraviolet threshold for taking this limit may depend on $p$. One more derivative bounds variation along short segments in $K'$, giving equicontinuity on $K$.

Apply Arzelà–Ascoli, in the form for a pointwise bounded equicontinuous sequence of continuous complex functions on a compact metric space [25, Theorem 7.25]. Diagonalize over compact exhaustions of $\mathrm{Conf}_n$, arities, degree lists, basis coefficient lists and derivative lists. These indices are countable. Starting from any prescribed strictly increasing $\mathsf{p}$, we obtain a strictly increasing $\mathsf{q}$ and $P = \mathsf{p}\circ\mathsf{q}$. Multilinearity extends the limits from basis lists to every token list. Integration by parts against a compactly supported test identifies the limit of each first derivative with the derivative of the limiting kernel. Repeating this for all derivative lists gives a smooth multilinear table $S$, with local convergence of every derivative.

A compact generic set has bounded diameter and a positive minimum coordinate separation, so it eventually lies in the displayed periodic generic domain. The asserted uniform convergence of $S_v^{(P(k))}$ follows from the extraction. The same bounded-diameter argument puts every compact separated set in $\mathrm{NoHit}_{L_{P(k)},n}$ eventually. For every separated test $f$, UV gives, at every admissible $p$, the full inner limit $C_p^{\mathrm{UV}}(v;f) = \int f f_{p,v}$. Its support is eventually admissible along $P$, and local uniform convergence proves (15.2). Thus Definition 4.3 holds. Both convergences persist on every further volume refinement. □

**Symmetries of the limit.** We first prove E1–E3 and E6, using Definition 3.9, Lemmas 14.7 and 14.31 and (15.1). Permutation symmetry, the empty correlation, deletion of the distinguished unit, conjugation and signed-permutation covariance hold on the lattice once $N$ exceeds the finitely many reference levels. Approximate any fixed separated configuration by lattice points and use Lemma 14.31, then (15.1). For translations, approximate the translation vector on the same lattice. Translation invariance and (15.3) pass the identity to arbitrary real translations. This proves E1–E3 independently of the coefficient corrections. Equation (15.1) gives E6; multilinearity was retained in the extraction.

## 15.2 Interior roots and errors

We use the chessboard root of Definition 14.10. A signed image means the image under $H_4$; conjugation is also allowed. Bond tilings, anchors, margins and small errors are specified before Definition 14.13. We consider tilings at the construction levels. Two such tilings are aligned when their cells have the same coordinate axes and every coarse boundary is a fine boundary. Their side ratio is an odd integer, since the block ratios are odd.

For interior functions in the base cell of a bond tiling $\mathcal{T}$, the chessboard inequality reads

$$\left| \left\langle \prod_{C \in \mathcal{T}} (F_C)_C \right\rangle_{N,p} \right| \le \prod_{C \in \mathcal{T}} \|F_C\|_{\mathcal{T}} \,. \tag{15.4}$$

Apply the abstract chessboard theorem [11, Theorem 4.1] axis by axis, on real vector spaces of complex observables, with reflection followed by conjugation. The functional has $L_p D_k$ factors in each axis, an even number since $4 \mid L_p$. Lattice translation invariance gives cyclicity. Reflection positivity across every translated band cut gives the required Schwarz inequality. For the Wilson action this is Osterwalder–Schrader positivity on the periodic lattice [27, Theorem 2.1, pp. 14–15], applied to a compact matrix gauge group, positive coupling, and observables supported on one side of the bond planes. Their boundaries pass through link midpoints, as required there.

The centered root estimate must also control anchors near arbitrary, separated configurations. Write $v = (d, c)$ for the insertion token.

**Lemma 15.2** (Interior root bound)**.** *Fix $m \le k \le N$, $N \ge m_{\mathrm{ref}}(d)$, and $B_{\mathrm{rt}} \ge 0$. Suppose, for every anchor $y'$, every signed image or image under conjugation of $O_{N,p}(v, y')$ has root at most $B_{\mathrm{rt}}$ in $\mathcal{T}_k(y')$ and stencil within $a_k$ of $y'$. In every translated bond tiling $\mathcal{T}$ of side $a_m$, an anchor at distance at least $3a_k$ from its cell boundary satisfies*

$$O_{N,p}(v, y) \text{ is interior}, \qquad \|O_{N,p}(v, y)\|_{\mathcal{T}} \le B_{\mathrm{rt}} \,. \tag{15.5}$$

The proof transfers the centered bound first at aligned cell centers, then between them.

*Proof.* Use Definition 3.9, its symmetry in Lemma 14.7, the cell interior condition of Definition 14.10, and (15.4). Since $a_k/a_N$ is odd, the fine-cell centers are lattice sites. At an aligned fine-cell center, reflect all coarse copies into distinct fine cells. Their signs and conjugations are among the assumed images. Apply (15.4) with 1 in every unoccupied fine cell. Taking the coarse root bounds it by $B_{\mathrm{rt}}$.

For intermediate anchors, let $H(z)$ be the coarse root after $z$ lattice translations in one axis, with the other coordinates fixed. Translation invariance and reflection give

$$H(z)^2 \le H(z-1)H(z+1) \,. \tag{15.6}$$

Indeed, translate the reflected product by one lattice step. Alternate cells then contain shifts $z-1$ and $z+1$; apply (15.4). On a finite interval a nonnegative sequence satisfying this inequality cannot exceed its endpoint maximum: an interior positive maximum extends to both neighbours and hence to an endpoint. This includes zero endpoints.

Bracket each coordinate of $y$ by its two neighbouring fine-center coordinates. All anchors in the resulting rectangle have margin at least $2a_k$, so their stencils are interior. Induct on $i = 0, \dots, 4$: the root is at most $B_{\mathrm{rt}}$ when the first $i$ coordinates are arbitrary lattice coordinates in these intervals and the remaining coordinates are endpoints. The case $i = 0$ is the aligned-center bound. In the next axis, the endpoint bounds hold uniformly in the coordinates already moved; (15.6) supplies the intervening bounds. The case $i = 4$ proves (15.5). □

Small errors are defined by a root tolerance chosen before the cell scale. This order makes them negligible against each fixed spectator test.

**Lemma 15.3** (Vanishing errors)**.** *For every small error $E$ of Definition 14.13, fixed token list $v$, distinguished slot $i$, and $\delta > 0$, there is $C(v,\delta) < \infty$ such that, eventually in (14.22),*

$$\left| \left\langle E_{N,p}(y_i) \prod_{h \neq i} O_{N,p}(v_h, y_h) \right\rangle_{N,p} \right| \leq C(v,\delta)\eta_{N,p} \qquad (\mathrm{sep}_{\mathrm{tor}}(y) \geq \delta)\,. \tag{15.7}$$

*Consequently its smear against any fixed separated test tends to zero in that order.*

The proof uses distinct cells for the error and each spectator.

*Proof.* Use Definition 3.9, the centered estimate of Lemma 14.12, Definition 14.13, Lemma 15.2 and (15.4). Let $n$ be the arity, and take the arity parameter in (14.23) equal to $n$. Take $m$ with $a_m < \delta$; the small-error condition gives $k \geq m$. Here $m$ only imposes a lower bound on $k$; the tiling has side $a_k$. Choose a finer spectator level $k_{\mathrm{sp}}$ with $3a_{k_{\mathrm{sp}}} \leq a_k/[8(n+1)]$. Lemma 14.12 bounds all signed spectator roots and their images under conjugation there by a fixed constant.

In each axis choose the translation of the level-$k$ tiling so that no boundary meets a strip of half-width $a_k/[4(n+1)]$ around any anchor. The excluded length is at most $na_k/[2(n+1)] < a_k$, so such a lattice translation exists for sufficiently large $N$. It supplies the integer margins (14.23), including their extra lattice unit, and margin $3a_{k_{\mathrm{sp}}}$ for spectators. Any pair of anchors differs in some coordinate by at least $\delta > a_k$, so they occupy distinct cells. Apply Lemma 15.2 to each spectator with its levels $(m,k)$ equal to $(k,k_{\mathrm{sp}})$ here; (15.4) then gives (15.7). On a compact test support, the number of lattice tuples multiplied by $a_N^{4n}$ is bounded independently of $N,p$. Multiplication by the test supremum therefore proves the smeared assertion. □

## 15.3 Corrected coefficients

We now use the equation projections and corrections of Lemmas 14.16–14.19, and the relative rank comparison of Lemma 14.24. The other inputs are Proposition 4.1, Definitions 3.9, 4.9 and 4.10, Lemmas 14.7, 14.8 and 14.12, Lemma 14.31, Lemmas 15.1, 15.2 and 15.3, and E1–E3 established above. Lemma 14.20 supplies the free test rows independently of the interacting arguments here. Auxiliary coefficient limits may use a common spacing subsequence $\mathsf{n}$ at every $p$; the full ultraviolet kernel limits remain those of UV.

These estimates distinguish coefficients that vanish in every separated correlation. The inclusions and derivatives in the next statement are the maps asserted in FC; the following proof constructs them and proves their defining correlation identities.

**Lemma 15.4** (Quotient compatibility)**.** *For every $d \leq e$, the inclusions of FC preserve correlations and units and compose compatibly. Inclusions and derivatives commute with conjugation, and*

$$h\partial_a = \sum_b h_{ba}\partial_b h \qquad (h \in H_4).$$

*Each response map is injective; every response lies in one $\iota_d F_d$; every field has a token representative. The unit is $\mathbf{1}_d = [a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}]$.*

The proof below constructs the inclusions and derivatives asserted in FC, proves its dimension equality, and then derives the compatibility properties. E1–E3 are already available.

*Proof.* We use the inputs listed above. Bounded coefficient families satisfy (14.21). In fact Lemma 14.8 bounds their inverse terminal coordinates by a fixed norm factor times $\prod_{m<N} C_{J_{m+1}}$,

and the constant choices in Section 9 give $\log C_{J_{m+1}} \le J_{m+1}^{-2} \log \ell_m$. Split the sum at a fixed level. Its finite initial part is $o(s_N)$, while its tail divided by $s_N = \sum_{m<N} \log \ell_m$ is bounded by the largest $J_{m+1}^{-2}$ in that tail. Sending the splitting level to infinity gives $\sum_{m<N} \log C_{J_{m+1}} = o(s_N)$. There are countably many projection matrices and comparison coefficient lists. The uniform bound in Lemma 14.16 gives convergent subsequences by finite-dimensional compactness. The same countable diagonal extraction retains every projection and each comparison lift, first in $N$ at every $p$, then in volume, retaining (15.1). Thus, in operator norm,

$$\Pi_{d,\mathrm{n}(j),p} \longrightarrow \Pi_d^{(p)}, \qquad \Pi_d^{(P(k))} \longrightarrow \Pi_d, \qquad \Pi_d^2 = \Pi_d\,. \tag{15.8}$$

Continuity of composition gives idempotence at both stages. For later use, the convergence also holds in the order (14.22).

Let $L_{N,p} : V_d \to \mathbb{C}$ be any fixed one-slot smeared correlation map, with arbitrary fixed spectators and test. Its operator-norm limit is $L$, by finite dimensionality and (15.2). Apply Lemma 14.17 and (15.7) to $c - \Pi_{d,N,p}c$. This gives $L((1-\Pi_d)c) = 0$ for every $L$. Since $S$ is smooth by (15.1),

$$\ker \Pi_d \subset \mathcal{N}_d(S). \tag{15.9}$$

For the reverse dimension inequality, use the rank-$q_d$ free matrix $\mathcal{G}$ of Lemma 14.20 and the common lifts of Lemma 14.24. Rearrange (14.33) to isolate $\mathcal{G}_I^\nu$. Every remaining term contains a nonempty $R_J$; its expectation is bounded by $\gamma_\nu$ using (15.4). Thus

$$\left| \left\langle \prod_{h\in I} I_h \right\rangle - \mathcal{G}_I^\nu \right| \le \gamma_\nu \sum_{J \subsetneq I} |\mathcal{G}_J^\nu|. \tag{15.10}$$

The finite sum stays bounded as $y^{\nu,i} \to y^i$. Pass through both coefficient and kernel limits. Each matrix entry is a fixed spectator functional on $V_d$, applied to the limiting lift of $m_j$. That functional vanishes on $\mathcal{N}_d(S)$, so each matrix factors through $F_d$ and has rank at most $\dim F_d$. The matrices tend to $\mathcal{G}$. A nonzero $q_d$-minor therefore gives $\dim F_d \ge q_d$. Ranks of converging idempotents are eventually constant: if $\|P-Q\| < 1$, $P$ is injective on $\operatorname{ran} Q$, and conversely. Passing the dimension inequality in Lemma 14.16 to $\Pi_d$ using (15.8) and using (15.9) now gives

$$\mathcal{N}_d(S) = \ker \Pi_d, \qquad \dim_{\mathbb{C}} F_d = q_d. \tag{15.11}$$

For inclusion, Lemma 14.18 gives a corrected family that has target smeared limit $\int f S(c,\dots)$. For the corrected difference of Lemma 14.19, torus summation by parts gives exactly

$$a_N^{4n} \sum_y f(\hat{y}) \Delta_{i,a} M_{N,p}^{\mathrm{lat}}(v;y) = a_N^{4n} \sum_y \frac{f(\hat{y} - a_N e_{i,a}) - f(\hat{y})}{a_N} M_{N,p}^{\mathrm{lat}}(v;y). \tag{15.12}$$

Here $e_{i,a}$ is the coordinate vector in slot $i$, axis $a$. The backward quotient tends uniformly to $-\partial_{i,a} f$; its residual is bounded by $a_N \|\partial_{i,a}^2 f\|_\infty / 2$. UB at order zero controls the residual sum on a fixed compact neighbourhood. Thus the derivative target is $-\int (\partial_{i,a} f) S(c,\dots)$.

For either corrected family $w_{N,p}$ in its representing degree $e$, write $\Pi = \Pi_{e,N,p}$. The normalized remainder $(w - \Pi w)/(1 + \|\Pi w\|)$ is admissible and belongs to $\ker \Pi$: its first term is a bounded scalar multiple of an admissible family, and its second term is bounded. Lemma 14.17 and Lemma 15.3 give

$$L_{N,p}\big((w - \Pi w)/(1 + \|\Pi w\|)\big) \longrightarrow 0. \tag{15.13}$$

The ordered tails

$$\{(k,j) : k \ge k_0,\ j \ge j_0(k)\}$$

form a filter base. An iterated limit $x_{k,j} \to x_k$ as $j \to \infty$, followed by $x_k \to x$, converges to $x$ on this filter: given $\varepsilon > 0$, first make $\|x_k - x\| < \varepsilon/2$ for $k \ge k_0$, then make $\|x_{k,j} - x_k\| < \varepsilon/2$ for $j \ge j_0(k)$. Consequently (15.8) and the fixed-volume kernel limits give $\Pi_{e,N,p} \to \Pi_e$ and $L_{N,p} \to L$ in norm on this filter, where $N = \mathsf{n}(j)$, $p = P(k)$. The corrected inclusion or difference and Lemma 15.3 also give $L_{N,p}(w) \to t_L$, its target value, for every spectator test.

The pairs $((1+\|\Pi w\|)^{-1}, \Pi w/(1+\|\Pi w\|))$ lie in the compact product of $[0,1]$ and the closed unit ball of $V_e$. Choose an ultrafilter $\mathcal{U}$ containing the ordered tails along which these pairs converge to $(\lambda, z)$. Set $b_{N,p} = (1 + \|\Pi w\|)^{-1}$. Passing the identity $\Pi(b_{N,p}\Pi w) = b_{N,p}\Pi w$ and the norm identity to the limit gives $\Pi_e z = z$ and $\lambda + \|z\| = 1$. The spectator limits and (15.13) hold along $\mathcal{U}$, so

$$\begin{aligned} L(z) &= \lim_{\mathcal{U}} b_{N,p} L_{N,p}(\Pi w) \\ &= \lim_{\mathcal{U}} \big[ b_{N,p} L_{N,p}(w) - L_{N,p}\big(b_{N,p}(w - \Pi w)\big) \big] = \lambda t_L. \end{aligned}$$

If $\lambda = 0$, then $L(z) = 0$ for every spectator test, hence $z \in \mathcal{N}_e(S) = \ker \Pi_e$ by (15.11). Together with $\Pi_e z = z$, this gives $z = 0$, contradicting $\|z\| = 1$. Thus $\lambda > 0$, and $z/\lambda$ represents the target for all spectators simultaneously. Choose these representatives on a basis and extend linearly, obtaining $\kappa_{d,e}$ and $\tau_{d,a}$ with

$$S(\kappa_{d,e} c, \dots) = S(c, \dots), \qquad S(\tau_{d,a} c, \dots) = \partial_{x_0^a} S(c, \dots). \tag{15.14}$$

Equation (15.1) converts the distributional identities to pointwise identities.

These maps preserve null coefficients, so they descend to $\jmath_{d,e}$ and $\partial_a$. A response vanishes precisely on $\mathcal{N}_d(S)$; hence $\iota_d$ is injective. Equality of responses gives injectivity, composition and uniqueness of inclusions. E1 and Lemma 14.7 give preservation of $[a^d_{m_{\mathrm{ref}}(d)}\mathbf{1}]$. Any finite sum of responses lies in the largest degree involved; quotient surjectivity gives token representatives. Finally E2, conjugation and the chain rule applied to $S(hc; hx) = S(c; x)$ give the asserted symmetry identities by injectivity of the response maps. The same comparison without differentiation gives $h\jmath_{d,e} = \jmath_{d,e} h$. Permuting the distinguished slot proves FC's positional identity in every slot. This proves FC and the lemma using Lemmas 14.16–14.24. □

## 15.4 Euclidean properties

The lattice symmetries and positivity pass through the ordered limits. The centered root estimate gives the additional control as separated points approach one another.

**Lemma 15.5** (Euclidean properties)**.** *The table supplied by IV is complex multilinear and satisfies E1–E7, including joint-test reflection positivity, absolute integrability of its pairings, and bounds for every derivative order:*

$$\|D_x^\nu S_n(c; x)\| \le C(n, d, c, \nu) r^{-A(n,d,c,\nu)} \quad (0 < r \le 1,\ \mathrm{sep}_\infty(x) \ge r).$$

The proof below passes reflection positivity to the limit and proves the power bound; its positive derivative orders use (15.14).

*Proof.* We use Definitions 3.9 and 4.9, Lemmas 14.7, 14.12 and 14.31, equations (15.1) and (15.4), Lemma 15.2, and the derivative maps (15.14) from Lemma 15.4. E1–E3 and E6 were proved after Lemma 15.1; only the positive derivative orders of E7 use the corrected maps.

We prove E5 and deduce E4. Fix finitely many joint smooth tests $f_i$ with compact support on positive-side separated configurations in one axis. Their lattice smears $F_i$ are supported strictly inside the positive half-torus for large $p, N$: the support has positive distance from the reflection plane, and the physical radius of each fixed stencil tends to zero. Use the bond reflection $x^a \mapsto a_N - x^a$, with planes $x^a = a_N/2$ and $x^a = (a_N + L_p)/2$. It is a lattice symmetry at every cutoff. For large $p$, the supports also avoid a neighbourhood of the second plane $x^a = L_p/2$; for large $N$ their stencils avoid both actual bond planes. The periodic Wilson reflection positivity cited before (15.4) gives

$$\sum_{i,j} \bar{z}_i z_j \langle \overline{F_i \circ \mathsf{r}_{a,N}} F_j \rangle_{N,p} \geq 0.$$

Here $\mathsf{r}_{a,N}$ maps a configuration to its reflected configuration. Insertion covariance reflects the positions and applies $\mathsf{s}_a^{\mathrm{refl}}$ to their labels; conjugation acts on the labels in the first word. The reflected joint test converges, with all derivatives, to $\bar{f}_i \otimes f_j$ with its first group reflected through $x^a = 0$. Its support is compact and separated, including across the reflection plane. All these supports lie in one compact separated set. UB at order zero bounds the difference between the moving tests and their limiting test by their uniform difference times a bounded weighted site count. It tends to zero. Apply Lemma 14.31 and (15.2) to the fixed limiting test in each entry. Smoothness bounds the kernel on the support, so every pairing is absolutely integrable. The limiting matrix is positive semidefinite. Concentrating normalized smooth bumps at prescribed positive-side words gives E4 by continuity. This works in every axis with its corresponding reflection.

For the power bound first consider order zero and a nonempty token list. Put $d_{\mathrm{pow}} = 1 + \max_i d_i$. Lemma 14.12 with $\varepsilon = 1$, for all the finitely many signed images and images under conjugation, bounds their centered roots by $C_{\mathrm{ctr}} a_k^{-d_{\mathrm{pow}}}$ for $k \geq k_0$. The constants $k_0, C_{\mathrm{ctr}}$ precede the choice of separation. The scale condition $\log \ell_u / s_u \to 0$ in Lemma 14.1 gives

$$a_{u+1} \geq c_{\mathrm{mesh}} a_u^2. \tag{15.15}$$

Indeed $\log \ell_u / \log a_u^{-1} \to 0$ implies $\ell_u \leq a_u^{-1}$ eventually; choose $c_{\mathrm{mesh}} > 0$ to include the remaining levels. The first $u \geq k_0$ with $a_u < \delta \leq 1$ consequently satisfies $a_u \geq c_{\mathrm{mesh},k_0} \delta^2$, including $u = k_0$.

Choose the first $m \geq k_0$ with $a_m < r/2$, then the first $k \geq k_0$ with $a_k < a_m/[6(n+1)]$. Thus $k \geq m$ and $a_k \geq c'(n) r^4$, with $c'(n) > 0$ also depending on $k_0$. In each coordinate the boundary strips of half-width $3a_k$ excluded by the $n$ anchors have total length less than $a_m$. Choose a translated tiling outside them. All anchors occupy distinct cells. Lemma 15.2, followed by (15.4), gives the lattice bound $(C_{\mathrm{ctr}} a_k^{-d_{\mathrm{pow}}})^n$. Passing to the limits by Lemma 14.31 and (15.1), with slightly smaller separation to accommodate lattice approximation, gives

$$|S_n(v; x)| \leq C(n, d, c) r^{-4 d_{\mathrm{pow}} n}. \tag{15.16}$$

All constants are independent of the positions and the volume.

For higher orders use (15.14): every coordinate derivative is a correlation with fixed, degree-raised tokens. At a fixed order there are finitely many such lists. Apply (15.16) to each, take the largest exponent, and bound the Fréchet operator norm by the sum of its coordinate coefficients in an orthonormal basis of $\mathbb{R}^{4n}$. This proves E7 using Lemma 15.4 and hence Lemmas 14.16–14.24. For arity zero, $S_0 = 1$ and all positive-order derivatives vanish. □

*Proof of Theorem 4.7.* Compose the volume refinements in Lemmas 15.1 and 15.4. Equation (15.2) proves ordered convergence, retaining UV's full inner limit at every admissible volume. Lemma 15.5 proves E1–E7. The coefficient corrections do not change any kernel limit already obtained. □

## 15.5 Repeated reflected words

Use the OS space of Definition 6.1. For any $s > 0$ and word $w$ with times in $(0, s)$, set

$$A_w v = w\, \tau_{se_4} v, \qquad w^\dagger = \tau_{se_4} \mathsf{s}_4^{\mathrm{refl}} w^*.$$

Here $A_w$ initially acts on words; boundedness on OS classes is not assumed. A buffer $r$ means pairwise sup-distance at least $r$ and times in $[r, h_{\mathrm{sl}} - r]$ for a slab of width $h_{\mathrm{sl}}$.

Fix a degree bound $d$ and coefficient bound $R \geq 0$, and put $Q = 4(d+1)$. The next estimate bounds the growth of repeated insertion before taking any operator extension.

**Lemma 15.6** (Repeated reflected words)**.** *There is $C_0 = C_0(d, R) > 0$ such that every nonempty length-$n$ word $w$ with these bounds and buffer $0 < r \leq 1$ in a slab of width $h_{\mathrm{sl}}$ admits $s = n_{\mathrm{lat}} a_k \in (0, h_{\mathrm{sl}}]$, $k, n_{\mathrm{lat}} \in \mathbb{N}$, for which $w$ has buffer $r/2$ in the slab of width $s$. For every separated exterior word $v$ with positive times,*

$$\|[(A_{w^\dagger} A_w)^j v]\|_{\mathrm{OS}}^2 \leq B_v [C_0^n (n/r)^{Qn}]^{4j} \qquad (j \geq 0).$$

*Here $B_v = B_v(w, r, s) < \infty$ is independent of $j$.*

The operators in the statement use the chosen width $s$. Its choice and the constants $C_0, Q$ precede $v$.

*Proof.* We use Definitions 3.9 and 6.1, Lemmas 14.12 and 14.31, Lemmas 15.2 and 15.5, and (15.4), (15.1) and (15.15). Enlarge the coefficient ball to contain every signed image and image under conjugation in degrees at most $d$. Lemma 14.12, with $\varepsilon = 1$, gives $k_* \in \mathbb{N}$ and $B_* > 0$ such that all these centered roots at levels $k_{\mathrm{w}} \geq k_*$ are bounded by $B_* a_{k_{\mathrm{w}}}^{-d_{\mathrm{pow}}}$, where $d_{\mathrm{pow}} = d + 1$. This choice precedes $n, r, w$.

Choose the first $m \geq k_*$ with $a_m < r/32$, and put $s = a_m \lfloor h_{\mathrm{sl}}/a_m \rfloor$. The word is nonempty with buffer $r$, so $s > 0$, $s \leq h_{\mathrm{sl}}$, and it still contains $w$ with buffer at least $r/2$. Equation (15.15) gives $a_m \geq c_1 r^2$. Fix $0 < \alpha \leq 1$ with $20\alpha < c_1$, and choose the first $k_{\mathrm{w}} \geq k_*$ with $a_{k_{\mathrm{w}}} < \alpha (r/n)^2$. Then

$$20 n a_{k_{\mathrm{w}}} < a_m, \qquad a_{k_{\mathrm{w}}} \geq c_2 (r/n)^4, \qquad (B_* a_{k_{\mathrm{w}}}^{-d_{\mathrm{pow}}})^n \leq (B_* c_2^{-d_{\mathrm{pow}}})^n (n/r)^{4 d_{\mathrm{pow}} n}. \tag{15.17}$$

These choices precede the exterior. In the statement take $k = m$ and $n_{\mathrm{lat}} = \lfloor h_{\mathrm{sl}}/a_m \rfloor$.

Expanding the OS norm gives a correlation of $4jn$ main insertions and two copies of $v$. The main insertions are mutually separated, and separated from the exterior, by at least $r/2$. Translations by multiples of $s$ do not change residues modulo $a_m \mathbb{Z}^4$. Thus the main anchors have at most $2n$ residue patterns, those of $w$ and its reflection with conjugation, independently of $j$.

For nonempty $v$, let $t_{\min} > 0$ be its minimum time and choose $m_{\mathrm{v}} \geq k_{\mathrm{w}}$ with $a_{m_{\mathrm{v}}} < 2t_{\min}$ and $a_{m_{\mathrm{v}}}$ smaller than every distance between distinct anchors of $v$. The two exterior copies have times $\pm(2js + t)$, where $t$ is a time of $v$. Their mutual distance is at least $2t_{\min} > a_{m_{\mathrm{v}}}$, for every $j$. Next choose $k_{\mathrm{v}} \geq m_{\mathrm{v}}$, beyond the exterior reference levels, with $12|v| a_{k_{\mathrm{v}}} < a_{m_{\mathrm{v}}}$. Lemma 14.12 gives a common finite bound $X \geq 0$ for all exterior roots and their images at level $k_{\mathrm{v}}$. Both $m_{\mathrm{v}}, k_{\mathrm{v}}$ are independent of $j$.

We apply the chessboard estimate at two cell sizes. In each axis, avoid strips of half-width $5a_{k_{\mathrm{w}}}$ around the $2n$ main residues modulo $a_m$. This is possible by (15.17). Then change the translation by less than $a_{m_{\mathrm{v}}}$ to place all $2|v|$ exterior anchors at distance at least $3a_{k_{\mathrm{v}}}$ from the finer boundaries; $12|v| a_{k_{\mathrm{v}}} < a_{m_{\mathrm{v}}}$ permits this. Since $a_{m_{\mathrm{v}}} \leq a_{k_{\mathrm{w}}}$, the main anchors retain margin at least $3a_{k_{\mathrm{w}}}$. Main insertions occupy distinct coarse cells, none containing an exterior anchor.

Several exterior factors may occupy one coarse cell. Reflect their product into every coarse cell, and apply (15.4) to the aligned tiling of side $a_{m_{\mathrm{v}}}$. The fine and coarse tilings share an origin and have an odd integer side ratio. Coarse reflections preserve the fine grid and its interior margins. Within each reflected coarse cell, exterior anchors are separated by more than $a_{m_{\mathrm{v}}}$ and so occupy distinct fine cells; across coarse boundaries their fine cells are disjoint. The uncentered estimate of Lemma 15.2 bounds every factor by $X$. If the coarse cell contains $n_C$ exterior factors, its root is therefore at most $X^{n_C}$. Applying the coarse chessboard estimate now gives

$$\|[(A_{w^\dagger}A_w)^j v]\|_{\mathrm{OS}}^2 \le X^{2|v|}(B_* a_{k_{\mathrm{w}}}^{-d_{\mathrm{pow}}})^{4jn}.$$

For each fixed $j$, take lattice approximations of all anchors, preserving these strict margins and separations, then apply Lemma 14.31 and (15.1). No cutoff threshold uniform in $j$ is needed. The constants on the right are already independent of $j$. Choose $B_v = \max(1, X^{2|v|})$, $C_0 = B_* c_2^{-d_{\mathrm{pow}}}$, and $Q = 4d_{\mathrm{pow}}$. Equation (15.17) proves the assertion. For empty $v$ omit its factors and take $B_v = 1$. □

# 16 Free fields, Ward identities and rotations

The stress tensor is determined by its action on local fields. Conservation alone leaves its scale and sign undetermined. We first compare the mixed correlations with the Maxwell correlations of Definition 4.10. This comparison identifies a one-dimensional space of conserved tensors. Integrating a member of that space over a small shell gives local translation and angular charges. We then identify these charges with the derivatives and rotation generators of the fields.

Lemmas 16.6, 16.11 and 16.12 prove, respectively, Proposition 5.2, Theorem 5.4 and Proposition 5.5. The free calculation in Lemma 16.1 also supplies the positivity used by Lemma 14.20. Its proof is independent of the interacting quotient and hence of Section 15.

Two issues require more than convergence at fixed separated points. The lattice comparison must control tests uniformly in their supremum norm, including near configurations with a shared coordinate. We remove narrow strips around these configurations, using two different scales for the bound on the kernel and its variation. Later, conservation removes the inner boundaries in the angular identity, but does not by itself remove the boundary at infinity. An anisotropic cutoff, reflection positivity and a low-energy estimate remove that boundary for collinear configurations. Analyticity of one-insertion vectors then extends the identity to arbitrary separated configurations.

Except in the free calculation, fix the table, volume refinement, quotient fields and calculus of Definitions 4.3, 4.9, 4.10 and Lemmas 15.1, 15.4, 15.5. Subsequences below change only comparison radii. Write $\jmath_{d,e}$ for the inclusions and $\partial_a$ for field differentiation. The coordinate Hilbert spaces used to remove the exterior boundary are those of Lemma 17.3; their construction uses coordinate reflections and precedes the use of continuous rotations.

## 16.1 Free fields

Put $z = x - y$ and $\mathsf{h}_{ab}(z) = \partial_a \partial_b \Gamma(z)$. The curvature covariance in Definition 4.10 is

$$C_{ab,ij}(z) = -\delta_{bj}\mathsf{h}_{ai} + \delta_{bi}\mathsf{h}_{aj} + \delta_{aj}\mathsf{h}_{bi} - \delta_{ai}\mathsf{h}_{bj}, \qquad \mathsf{h}_{ab} = \frac{8 z_a z_b - 2\delta_{ab}|z|^2}{4\pi^2 |z|^6}. \tag{16.1}$$

Different colours have zero covariance. Set

$$\begin{aligned} Q_{ab,ij} &= \sum_{\alpha} :F^{\alpha}_{ab}F^{\alpha}_{ij}:, \qquad P_{ab} = \sum_{c} Q_{ac,bc}, \qquad Q_{\mathrm{tot}} = \sum_{a} P_{aa}, \\ T^{\mathrm{Max}}_{ab} &= -P_{ab} + \tfrac{1}{4}\delta_{ab}Q_{\mathrm{tot}}, \qquad \mathcal{O}_{F^2} = \tfrac{1}{4}Q_{\mathrm{tot}}. \end{aligned} \tag{16.2}$$

The minus sign is fixed by the shell convention in Definition 5.3.

For the coordinate reflection $s_a$, strict reflected positivity means

$$S_2^{\mathrm{Max}}(R^{\mathrm{Max}}(s_a)\bar{u}, u; s_a x, x) > 0 \qquad (u \in M_d^{\mathrm{Max}} \setminus \{0\},\ x = te_a,\ t > 0).$$

Write $\mathsf{L}^{\mathrm{Max}}(K)$ for the derivative at zero of $R^{\mathrm{Max}}(e^{tK})$. The two free shell normalizations, with the fluxes of Definition 5.3, are

$$\mathcal{P}_b^{T^{\mathrm{Max}}} = S^{\mathrm{Max}}((\partial_b^{\mathrm{Max}}u)(x_0), \boldsymbol{v}(x)), \qquad \mathcal{Y}_K^{T^{\mathrm{Max}}} = S^{\mathrm{Max}}((\mathsf{L}^{\mathrm{Max}}(K)u)(x_0), \boldsymbol{v}(x)). \tag{16.3}$$

The next lemma gives both the nondegeneracy of free correlation tests and the normalization used in the interacting comparison. Its proof follows immediately below.

**Lemma 16.1** (Free positivity and stress)**.** *The Maxwell table satisfies E1–E7, the field calculus and $O(4)$ covariance. Every nonzero on-shell invariant Wick polynomial has a strictly positive reflected two-point Gram value at distinct reflected points. For simple $\mathfrak{g}$, its centered real symmetric $H_4$-tensorial conserved degree-four tensors form the line of $T^{\mathrm{Max}}$, normalized by both shell laws, and*

$$S_2^{\mathrm{Max}}(\mathcal{O}_{F^2}, \mathcal{O}_{F^2}; x, y) = \frac{3d_{\mathfrak{g}}}{\pi^4|x-y|^8} \qquad (x \neq y).$$

*Proof.* Use Definition 4.10; no interacting input is needed. Wick correlations sum pairings between different insertion sites. Formula (16.1) satisfies Bianchi and $\sum_a \partial_a C_{ab,ij} = 0$ off zero, so these correlations descend to the on-shell quotient. It also gives permutation symmetry, reality, units, translations and orthogonal covariance. Each differentiated contraction is homogeneous of a fixed negative degree and bounded on the unit sphere. A finite product of these bounds proves E6–E7. Differentiating the pairings proves the derivative law; inclusions retain the same polynomial.

Here is the nondegeneracy calculation, including the equations imposed on jets. Let $\mathcal{H}_j$ be the real space of $\mathfrak{g}$-valued homogeneous polynomial two-forms of degree $j$ satisfying $df = \delta f = 0$. Its Frobenius inner product identifies it with the dual of the order-$j$ on-shell curvature jets. Use Taylor coefficients $D^j/j!$ and the inner product summing both ordered form indices and ordered derivative indices; $p(\partial)$ denotes contraction with this normalized derivative. For $z \neq 0$, put $I(z) = z/|z|^2$ and $\mathcal{R}_z = I_4 - 2zz^{\mathsf{t}}/|z|^2$. Define $R_p(y)$ to be the covariance of $F(y)$ with the jet probe $p$ at zero. Then

$$R_p(y) = s_j|y|^{-4}(\mathcal{R}_y \otimes \mathcal{R}_y)p(I(y)), \qquad s_j = \frac{(j+1)(j+2)(-2)^j}{\pi^2}. \tag{16.4}$$

Indeed $p(\partial)\Gamma = (-2)^j|y|^{-2j}p(y)\Gamma$; induction proves this identity, since traces of the coefficient tensor vanish by harmonicity. In taking the two curls, terms containing $\sum_a \partial_a p_{ab}$ and $\partial_a p_{bc} + \partial_b p_{ca} + \partial_c p_{ab}$ vanish. The remaining radial derivatives use $\sum_a y_a \partial_a p = jp$ and $\partial_a|y|^{-2j-2} = -(2j+2)y_a|y|^{-2j-4}$. They give

$$s_0 = 2/\pi^2, \qquad (j+1)s_{j+1} = -2(j+3)s_j,$$

which yields the coefficient in (16.4).

For towers $p = (p_j)$ and $q = (q_j)$, let $B_z(p, q)$ be their two-site covariance at separation $z$, with $\mathcal{R}_z$ acting on all tensor indices of the second probe. This is the reflected jet pairing at $\pm z/2$.

Define the triangular map $\sigma_z$ by replacing a single order-$j$ probe $p$ with the tower whose order-$k$ component is $(s_j/s_k)T_k[p(\cdot - I(z))]$, $k \le j$, where $T_k$ denotes the homogeneous Taylor term at zero. Its generated field is the right side of (16.4) with $p(I(y) - I(z))$. Since $DI(z) = |z|^{-2}\mathcal{R}_z$, its Taylor terms at $z$ below $j$ vanish, and its top term is $s_j|z|^{-2j-4}\mathcal{R}_z p$. Symmetry removes the entries above the diagonal as well. Thus

$$B_z(\sigma_z p, \sigma_z q) = \sum_j \frac{(j+1)(j+2)2^j}{\pi^2|z|^{2j+4}}\langle p_j, q_j\rangle. \tag{16.5}$$

Evaluating the second order-$j$ probe supplies $(-1)^j$, since its positional derivatives are opposite to derivatives of the separation. This cancels the sign in $s_j$; reflection acts on all its tensor indices as specified above. All pivots are positive. In coordinates diagonalizing this form, Wick pairing between two sites gives $\langle X^\mu, X^\nu\rangle = \mathbf{1}_{\mu=\nu}\mu!\prod_i b_i^{\mu_i}$, with $b_i > 0$. Every polynomial uses a finite tower, proving strict positivity.

For several positive-side sites the reflected curvature covariance is positive semidefinite: spatial Fourier transformation of $\Gamma$ gives $e^{-|p|(t+s)}/(2|p|)$, and its two curls, with reflection of the form indices, give the sum of squares of the two transverse polarizations. Jet derivatives differentiate these same factors. Include the contractions within each positive-side word in its Fock vector. Its reflected pairing is then the symmetric Fock pairing of this positive semidefinite form; hence finite sums of words have nonnegative squared norm. On compact separated supports every pairing is continuous and bounded, so integration gives E5 and point evaluation gives E4.

For the stress line, simplicity makes every invariant linear colour form zero and every invariant bilinear form a multiple of the fixed metric. Through weight four, only the unit and the quadratics $Q$ remain. Sign changes and permutations therefore give the following basis of centered real symmetric tensors:

$$\mathcal{A}_{ab} = \delta_{ab}P_{aa}, \qquad \mathcal{B}_{ab} = \delta_{ab}(Q_{\mathrm{tot}} - 2P_{aa}), \qquad \mathcal{D}_{ab} = (1-\delta_{ab})P_{ab}.$$

Sign parity reduces the corresponding degree-five vectors to contractions of $F\partial F$ with the free index $b$. The equations remove $\sum_d \partial_d F_{dc}$; Bianchi gives $2\sum_{c,d} F_{cd}\partial_c F_{bd} = \sum_{c,d} F_{cd}\partial_b F_{cd}$. The two remaining sums are

$$\sum_c {:}\,F_{bc}\partial_b F_{bc}\,{:} = \tfrac{1}{2}\partial_b P_{bb}, \qquad \sum_{c<d,\ c,d\neq b} {:}\,F_{cd}\partial_b F_{cd}\,{:} = \tfrac{1}{4}\partial_b(Q_{\mathrm{tot}} - 2P_{bb}),$$

with colour sums understood. Test the first divergence component at $e_1$ against $Q_{12,12}(0)$ and $Q_{23,23}(0)$, dividing by $d_{\mathfrak{g}}/(4\pi^2)^2$. For example, $S_2^{\mathrm{Max}}(P_{11}(te_1), Q_{12,12}(0)) = 2d_{\mathfrak{g}}/(\pi^4 t^8)$, whose derivative gives the first entry below. Differentiating (16.1) in the other entries gives

$$\begin{pmatrix} -256 & 0 & 128 \\ 0 & -512 & -128 \end{pmatrix}. \tag{16.6}$$

A further test of $\mathcal{D}_{12}$ against $Q_{13,23}$ gives $-16$ and gives zero for $\mathcal{A}, \mathcal{B}$. The three tensors are independent, the vector space has dimension two, and divergence is onto. Its kernel is the line of $\frac{1}{2}\mathcal{A} - \frac{1}{4}\mathcal{B} + \mathcal{D} = -T^{\mathrm{Max}}$.

To compute its charge, polarize the tensor $T^{\mathrm{Max}}$ between a smooth Maxwell wave $f$ and the field generated by a probe at $x_0$. Replace $\Gamma$ by $\Gamma_\varepsilon$ with $-\Delta\Gamma_\varepsilon = \rho_\varepsilon$, where $\rho_\varepsilon$ is a smooth approximate identity. For a probe $F_{ij}$ the divergence of its generated field is $\delta_{bj}\partial_i\rho_\varepsilon - \delta_{bi}\partial_j\rho_\varepsilon$.

Write $g$ for the generated field and polarize $T^{\mathrm{Max}}$ as $t_{ab} = -\sum_c (f_{ac}g_{bc} + g_{ac}f_{bc}) + \frac{1}{2}\delta_{ab}\sum_{i,j} f_{ij}g_{ij}$. Bianchi cancels its derivative terms to give $\sum_a \partial_a t_{ab} = -\sum_c f_{bc}\sum_a \partial_a g_{ac}$. Hence the minus sign of the shell integral gives $\int \chi v_b \partial_a t_{ab} = -\int \chi v_b f_{bc}(\delta_{cj}\partial_i\rho_\varepsilon - \delta_{ci}\partial_j\rho_\varepsilon)$. A second integration by parts gives

$$\partial_i(v_b f_{bj})(x_0) - \partial_j(v_b f_{bi})(x_0) = (\mathcal{L}_v f)_{ij}(x_0).$$

There is no derivative of $\chi$ at $x_0$. The contraction with $Dv$ vanishes by symmetry. For $v = K(\cdot - x_0)$ the last expression is $\sum_c (K_{ci}f_{cj} + K_{cj}f_{ic})(x_0)$, the generator of the free action on curvature; for constant $v$ it is $v \cdot \nabla f$. Higher probes follow by differentiating in $x_0$ on the compact shell. In a Wick contraction, two stress legs ending at $x_0$ have zero shell: dilation by 2 multiplies it by $2^{3+p-8-j-l} \neq 1$, for probe orders $j, l$ and weight degree $p = 0, 1$. Two legs ending outside have zero shell by conservation. The remaining contractions give the polynomial derivation $\mathcal{L}_v$, proving (16.3).

Finally, at $z = re_1$ the six unordered curvature components have diagonal covariances $\pm(\pi^2 r^4)^{-1}$. Since $\mathcal{O}_{F^2} = \frac{1}{2}\sum_{a<b,\alpha} :(F^\alpha_{ab})^2:$, its two-point Wick sum is $\frac{1}{4}\cdot 2\cdot 6 d_{\mathfrak{g}}/(\pi^4 r^8)$. Orthogonal covariance gives the asserted formula for every $z \neq 0$. □

The next two lemmas concern only the invariant polynomial algebra of Definition 4.10, realized by the Taylor sections of Lemma 10.2. A curvature jet of order $j$ has weight $j + 2$. Let $\mathfrak{P}$ be the complex polynomial algebra on all on-shell jets, and $\mathfrak{P}_{\le j}$ its subalgebra using orders at most $j$. The invariant polynomials of weight at most $d$ identify with $M_d^{\mathrm{Max}}$. For a fixed spectator degree $d$, put

$$\mathfrak{N}_d = \{P \in \mathfrak{P} : S^{\mathrm{Max}}(P(0), u_1(x_1), \dots, u_n(x_n)) = 0 \text{ for all } n, u_i \in M_d^{\mathrm{Max}}, (0, x) \in \mathrm{Conf}_{n+1}\}.$$

All polynomials in this display are Wick ordered at their own site. The degree of $P$ is unrestricted. Bounded spectator degree will suffice to detect it.

Let $\mathcal{C}_d$ be the real space of complex linear tuples $P_a : M_d^{\mathrm{Max}} \to M_{d+1}^{\mathrm{Max}}$ which commute with conjugation, kill the unit, are vector-covariant under $H_4$, and send $M_e^{\mathrm{Max}}$ into $M_{e+1}^{\mathrm{Max}}$ for $e \le d$. For $P \in \mathcal{C}_d$, its Ward sum is

$$\mathcal{E}_b(P; \boldsymbol{u}, x) = \sum_i S_n^{\mathrm{Max}}(u_1, \dots, P_b u_i, \dots, u_n; x).$$

The following lemma identifies its kernel using only finitely many tests on lines. Its proof first obtains a bounded spectator degree that detects arbitrarily high output jets.

**Lemma 16.2** (Finite vector Ward tests)**.** *For all sufficiently large $d$, finitely many positional derivatives of $\mathcal{E}_b(P; \boldsymbol{u}, x)$, with $u_i \in M_d^{\mathrm{Max}}$ and $x$ on a coordinate line perpendicular to $e_b$, define a real linear map $\mathcal{T}_d$ such that*

$$\ker \mathcal{T}_d = \mathbb{R}\partial^{\mathrm{Max}} \quad \text{in } \mathcal{C}_d. \tag{16.7}$$

*Proof.* Use Definition 4.10 and Lemma 16.1.

We first establish detection with one bounded spectator degree:

$$\mathfrak{N}_{d_0} \cap \mathfrak{P}^G = \{0\} \qquad \text{for some } d_0. \tag{16.8}$$

Here $\mathfrak{P}^G$ is the invariant subalgebra. Fix a jet order $j$. The algebra $\mathfrak{P}_{\le j}$ is a polynomial algebra on finitely many variables, hence Noetherian by the Hilbert basis theorem [29, Lemma 10.31.1]. Its ideal generated by invariant polynomials of positive ordinary polynomial degree therefore has finitely many invariant, ordinary-homogeneous generators $f_\alpha$. Averaging coefficients over

the compact gauge group and taking homogeneous components shows, inductively in ordinary degree, that every invariant ordinary-homogeneous polynomial is a sum of products of invariant homogeneous polynomials of weight at most some $b_j$. Indeed write $P = \sum a_\alpha f_\alpha$, keep only terms of total degree $\deg P$, and replace $a_\alpha$ by its gauge average; their degrees are smaller. This proves the assertion by induction, including constants.

These products detect every invariant $P \in \mathfrak{P}_{\le j}$. Suppose otherwise and take its lowest nonzero ordinary-homogeneous part $P_r$. Split a product of homogeneous factors of total ordinary degree $r$ into distinct nearby spectator sites. Every leg of $P$ must pair with a spectator leg. Since its lowest ordinary degree is $r$, all terms with internal spectator contractions, or with higher-degree parts of $P$, vanish identically. Wick's rule, as the spectator sites coalesce, therefore gives the two-site pairing of $P_r$ with their ordinary product. Each factor has weight at most $b_j$, so the assumed annihilation makes the limit zero. Choose the product sum to be the reflection of the complex conjugation of $P_r$, using the preceding generation result. Lemma 16.1 makes this pairing strictly positive. Induction on $r$ proves $P = 0$.

We now use induction on the weight bound of $P$ to remove the restriction on jet order. Put $g = d_{\mathfrak{g}}$ and choose $d_0 \ge \max\{b_g, 2g+5\}$. In a real colour basis, write $X_{j,\alpha}(p)$ for the jet coordinate tested by a scalar order-$j$ probe $p$. For scalar probes $p, q$ of orders $k, l$, the invariant quadratic $\sum_\alpha : X_{k,\alpha}(p) X_{l,\alpha}(q) :$ has weight $k + l + 4$. Its collision with a polynomial using jets through order $j$ has, in the coefficient where just the order-$k$ leg contracts at that order, the polynomial

$$\sum_\alpha X_{l,\alpha}(q) \sum_\mu C_{k,j}(p, e_\mu; z) \frac{\partial P}{\partial X_{j,\alpha}(e_\mu)}. \tag{16.9}$$

Here $(e_\mu)$ is a basis of order-$j$ probes, and $C_{k,j}$ is their differentiated curvature covariance. The extraction distinguishes harmonic poles from products of poles, rather than comparing the weights of their remaining polynomials. Put $r = 4 + k + j$. After testing against fixed exterior spectators, the term (16.9) is harmonic and homogeneous of degree $-r$ in $z$; multiplication by $|z|^{2(r-1)}$ makes it real analytic at zero. Taylor expansion of the other terms at each singular order gives homogeneous functions which become real analytic after multiplication by $|z|^{2(r-2)}$ at order $r$. In particular, a double contraction is a product of poles of orders $r_1, r_2 \ge 2$: its product is cleared by $|z|^{2(r_1+r_2-2)}$. Taylor terms preserve this two-power improvement. The contraction of the order-$l$ leg alone has order at most $4 + l + j < r$.

A harmonic homogeneous function $K$ of degree $-r$, $r \ge 2$, cannot have the latter improvement unless it is zero. Indeed, in four dimensions,

$$\Delta(|z|^{2s} K) = 4s(s + 1 - r)|z|^{2s-2} K.$$

If $|z|^{2(r-2)} K$ extends analytically, successive Laplacians show that $K$ itself extends analytically; negative homogeneity then makes it zero. Expand the collision through any order exceeding $8 + k + l + 2j$. If $P \in \mathfrak{N}_{d_0}$, its correlation with the quadratic test and any further bounded-degree spectators is identically zero. Comparing successive radial homogeneities in this finite expansion makes its order-$r$ harmonic pole zero by the preceding argument. Thus (16.9) belongs to $\mathfrak{N}_{d_0}$. This conclusion uses only $l < k$ and $k + l + 4 \le d_0$, without a weight induction. The induction is used next to make this extracted polynomial zero.

Take $k = g+1$ and $l = 0, \dots, g-1$. For each $l$ choose a probe $q_l$ and a jet value $v_l$ with $q_l(v_l) = 1$. The matrix $(X_{l,\alpha}(q_l))_{l,\alpha}$ has nonzero polynomial determinant: assign $v_l$ to colour $l$ and zero to the other colours in order $l$, making this matrix the identity. For $j > g$ the polynomial (16.9) replaces an order-$j$ jet by an order-$l$ jet, so its weight is strictly smaller. The induction hypothesis therefore

makes it zero. Multiplication by the transposed cofactor matrix, in the polynomial domain, shows that every one-colour contraction in (16.9) vanishes. As $z, p$ vary, the forms $C_{k,j}(p, \cdot; z)$ span the dual of the order-$j$ jets. Their common annihilator generates a Maxwell wave whose order-$k$ jets vanish everywhere. That wave is a polynomial of degree below $k$; its negative homogeneity makes it zero. Formula (16.4) then makes the probe zero. Hence every order-$j$ partial derivative of $P$ vanishes. Starting above $g$ and descending removes all these jet orders. The case of orders at most $g$ was proved above with spectator bound $b_g$. This closes the induction on weight and proves (16.8).

Free correlations are real analytic off the diagonals by (16.1). If all the stated jets vanish, every Ward sum vanishes near each testing line. For any separated configuration choose distinct ordered line coordinates respecting the strict order of its coordinates in that axis. The straight segment to that line configuration stays separated: strict inequalities persist, and coordinate ties are separated by the line coordinates. Analytic continuation along this segment gives $\mathcal{E}_b = 0$ on all of $\mathrm{Conf}_n$.

We first determine $P$ on weight four. Let $V_j$ be the scalar on-shell order-$j$ jet space and set $\kappa_{ij}(p, q; z) = C_{i,j}(p, q; z)$ in one colour. For $p \in V_0^*$ put $U(p) = \sum_\alpha : X_{0,\alpha}(p)^2 :$. Simplicity, the filtration and inversion parity force $P_b U(p)$ to be a colour contraction of one order-zero and one order-one jet. Represent it by a linear map $A_p : V_0 \to V_1^*$. In the Ward identity of $U(p), U(q), U(w)$, let the first two points coalesce along $tu$ and multiply by $t^5$. The term with $P_b$ at the distant point tends to zero. The remaining two triangular Wick contractions give, for every $X \in V_0$,

$$\kappa_{10}(A_p X, q; u)\, q(X) = \kappa_{10}(A_q X, p; u)\, p(X).$$

To obtain arbitrary $X$, choose the distant probe $w$ using the invertible order-zero covariance (16.4). A common nonzero coefficient in the two surviving terms cancels. If $p(X) = 0$, choose $q(X) \neq 0$; varying $q, u$ makes $A_p X = 0$, again by (16.4). Thus $A_p X = p(X) f_b p$. The same displayed identity and nondegeneracy show that $f_b : V_0^* \to V_1^*$ is real linear and

$$\kappa_{10}(f_b p, q; u) = \kappa_{10}(f_b q, p; u).$$

Here is the signed-covariant calculation of $f$. Write $e^{ij}$ for the curvature coordinate and $\partial_h e^{ij}$ for its first jet. For $f_1 e^{12}$, sign parity permits only $\partial_1 e^{12}, \partial_3 e^{23}, \partial_4 e^{24}$; interchange of axes $3, 4$ and $\sum_i \partial_i e^{i2} = 0$ reduce their span to $\mathbb{R}\partial_1 e^{12}$. For $f_1 e^{23}$ the candidates are $\partial_1 e^{23}, \partial_2 e^{13}, \partial_3 e^{12}$; antisymmetrizing axes $2, 3$ and Bianchi reduce them to $\mathbb{R}\partial_1 e^{23}$. Permutations leave two constants. They agree because

$$\kappa_{10}(\partial_1 e^{12}, e^{23}; e_1 + e_3) = -\frac{1}{2\pi^2} \neq 0,$$

as follows by differentiating (16.1). Therefore $f_b = c_{\mathrm{jet}} \partial_b$ with one real $c_{\mathrm{jet}}$. Since $\partial_b U(p)$ has two identical terms, $P_b = c\partial_b$ on weight four, with $c = c_{\mathrm{jet}}/2$.

Set $Q_b = P_b - c\partial_b$. Translation invariance makes its Ward sums zero, and it kills every free stress component. Insert a free stress shell for the dilation field $v(y) = y - x_0$ into such a sum. The term where $Q_b$ acts on the stress is zero. Evaluate all other shell terms by the polarized calculation in Lemma 16.1, now with this $v$. The extra derivative of $v$ contracts with the trace of the free stress, which is zero by (16.2). Choose the shell to surround only the distinguished point, so its Ward sum has just that slot. The local Lie derivative of a two-form under dilation is $y \cdot \nabla + 2$; differentiating its jets gives the weight operator $\mathcal{H} X_{j,\alpha} = (j + 2) X_{j,\alpha}$; subtraction of the Ward sum with the distinguished label $\mathcal{H}u$ shows that $[\mathcal{H}, Q_b]u \in \mathfrak{N}_d$. For $d \geq d_0$ Equation (16.8) removes this polynomial. Thus $Q_b$ preserves each homogeneous weight. Inversion acts there by $(-1)^{\mathrm{weight}}$, whereas vector covariance makes $Q_b$ odd under inversion. It follows that $Q_b u = -Q_b u$, hence $Q_b = 0$.

Conversely translation invariance annihilates every test on $\partial^{\mathrm{Max}}$. Finally the real domain $\mathcal{C}_d$ is finite-dimensional. Successively choose a test nonzero on the current kernel modulo this line. Each choice lowers its dimension, so finitely many tests have precisely the asserted kernel. □

Let $\mathcal{B}_d$ consist of alternating real-bilinear families $\Delta : \mathfrak{so}(4)^2 \to \mathrm{End}_{\mathbb{C}}\, M_d^{\mathrm{Max}}$ which commute with conjugation, kill the unit, preserve every lower-degree subspace, and satisfy

$$R^{\mathrm{Max}}(h)\Delta(K,L) = \Delta(hKh^{-1}, hLh^{-1})R^{\mathrm{Max}}(h) \qquad (h \in H_4,\ \det h = 1).$$

The next result will remove the defect in the rotation commutator.

**Lemma 16.3** (Finite internal Ward tests)**.** *For every $d \geq 4$, finitely many sums*

$$\sum_i S_n^{\mathrm{Max}}(u_1, \dots, \Delta(K,L)u_i, \dots, u_n; x), \qquad u_i \in M_d^{\mathrm{Max}}, \quad x \in \mathrm{Conf}_n, \tag{16.10}$$

*separate $\mathcal{B}_d$.*

*Proof.* Use Definition 4.10 and Lemma 16.1. Suppose all these sums vanish and fix $\mathsf{Z} = \Delta(K,L)$. On the weight-four space let $b_x(u,v) = S_2^{\mathrm{Max}}(u, v; x, 0)$. The order-zero Kelvin formula and Wick's rule give, for $|x| = |y| \neq 0$,

$$b_x(u,v) = b_y(R^{\mathrm{Max}}(\mathcal{R}_y\mathcal{R}_x)u, v).$$

Both $\mathsf{Z}$ and the orthogonal action preserve this space. Applying $b_x(\mathsf{Z}u, v) + b_x(u, \mathsf{Z}v) = 0$ at $x$ and $y$, then using nondegeneracy of $b_y$, shows that $\mathsf{Z}$ commutes with $R^{\mathrm{Max}}(\mathcal{R}_y\mathcal{R}_x)$ there. In particular it commutes with every coordinate quarter turn.

Combine this with the signed covariance of $\Delta$. For each weight-four $u$, the alternating bilinear function $(K,L) \mapsto \Delta(K,L)u$ is invariant under the action of coordinate quarter turns on both arguments. Such a function is zero. Indeed in the plane basis $K_{ab} = E_{ba} - E_{ab}$, overlapping distinct planes have opposite sign under a determinant-one diagonal sign change, and disjoint planes are interchanged by an even permutation; invariance and alternation kill both cases. Thus $\mathsf{Z}$ vanishes on weight four, including the stress.

Insert a free angular stress shell into its Ward sum, as in the dilation argument of Lemma 16.2. The shell surrounds only the distinguished point. Subtracting the Ward sum with its angularly differentiated label gives $[\mathsf{L}^{\mathrm{Max}}(M), \mathsf{Z}]u \in \mathfrak{N}_d$. This polynomial has weight at most $d$, so strict two-point positivity in Lemma 16.1 already makes it zero. Hence $\mathsf{Z}$ commutes with all infinitesimal rotations. Differentiating $e^{t\mathsf{L}(M)}\mathsf{Z} - \mathsf{Z}e^{t\mathsf{L}(M)}$ proves commutation with coordinate quarter turns in every degree. Repeating the preceding alternating-plane argument gives $\Delta = 0$ in all degrees. Finite-dimensionality selects a finite separating subfamily, by successively reducing the common kernel. □

## 16.2 Lattice inputs

The insertions are those of Definition 3.9; their coefficient polynomials are constructed in Definition 10.1 and Lemma 10.2. We use the inverse coefficient estimate of Lemma 14.5, the corrected sources and normalized integral of Lemmas 14.22 and 14.23, and the backward matching estimate (14.27). The positional bound is Lemma 14.15. The precise normalizations used below are (16.14), (16.15) and (16.18); the marked local format is Lemma 14.9. Roots always mean Definition 14.10. Cutoff limits have the order (14.22), followed by the fixed extraction of Lemma 15.1.

Fix arity and degree bounds $n \ge 2, D \ge 1$ and $R \ge 1$. A basis word means a list of members of fixed real bases of $M_d^{\mathrm{Max}}$, $d \le D$. Put

$$K_{R,m} = \{x : |x| \le R,\ \mathrm{sep}_\infty(x) \ge R^{-1}\}, \qquad E_\sigma = \{x : |x_i^a - x_j^a| > \sigma\ (i \ne j,\ a = 1, \dots, 4)\}.$$

Choose the construction's exponent triple for a degree horizon containing $n, D$, and denote its exponents by $\alpha, \beta, \gamma > 0$ and $\zeta < 0$. They satisfy

$$\beta + 6 < \alpha, \quad \gamma + 5 < \alpha + \zeta, \quad (\alpha - \beta)Dn + \zeta < 0, \quad (\alpha - \gamma)Dn - \alpha - \zeta < 0. \tag{16.11}$$

The choice is explicit: with $h_{\max} = \max(n, D)$, take $v = 7h_{\max}(h_{\max} + 2) + 1$, $w = 6 + v$, $\alpha = 17 + h_{\max} + v + wh_{\max}(h_{\max} + 2) + w$, $\beta = \alpha - 7$, $\gamma = \alpha - w$ and $\zeta = -v$. The first two inequalities place the root cells; the last two will remove coordinate coincidence strips. Write $M_k = M_k^{\mathrm{cut}}$ and set

$$r_k = a_k M_k^{\alpha}, \qquad \mathfrak{s}_k = M_k^{-\alpha}, \qquad \sigma_k = M_k^{\zeta}, \qquad \mathfrak{a}_k \le C M_k^{(\alpha-\beta)D}, \quad \mathfrak{b}_k \le C M_k^{(\alpha-\gamma)D}. \tag{16.12}$$

In particular $r_k \mathfrak{s}_k = a_k$: the normalized grid is the physical construction grid. For an ordered coefficient cluster limit $c_{k,d} : M_d^{\mathrm{Max}} \to V_d$ of the lifts in Lemma 16.4, define

$$Q_k(\boldsymbol{u}; x) = S_m(\pi_{d_i} c_{k,d_i} u_i; r_k x),$$

where $\pi_d : V_d \to F_d$ is the quotient of Definition 4.9.

The following comparison uses one coefficient family for every word within the fixed bounds. Its constants may depend on these bounds, the word and the construction. Its estimates hold for every coefficient cluster limit, rather than for one specially chosen limit.

**Lemma 16.4** (Lattice comparison at small radii)**.** *There are arbitrarily large levels $k$, real $H_4$-equivariant lifts $\Lambda_{k,N,p,d} : M_d^{\mathrm{Max}} \to V_d$ preserving the degree-zero unit, and $\varepsilon_k \to 0$ such that every ordered coefficient cluster limit of these lifts satisfies, for every basis word with $m \le n$,*

$$\begin{aligned} |Q_k| &\le \mathfrak{a}_k^m && \text{on } K_{R,m}, \\ \|\nabla Q_k\| &\le 4m\mathfrak{b}_k^m/\sigma_k && \text{on } K_{2R,m} \cap E_{4\sigma_k}, \\ |Q_k - S_m^{\mathrm{Max}}| &\le \varepsilon_k && \text{on } (\mathfrak{s}_k\mathbb{Z})^{4m} \cap K_{2R,m} \cap E_0. \end{aligned} \tag{16.13}$$

*At fixed $k$ the lifts are bounded on ordered tails.*

*Proof.* Choose a real $H_4$-equivariant polynomial representative map $\iota_d^{\mathrm{free}}$ for the on-shell quotient, using the Taylor sections of Lemma 10.2 and averaging a real splitting over $H_4$. Let $\mathsf{C}_{k,N,p,d}$ be the marked coefficient readout after the finite Gaussian contraction correction of Lemma 14.21. Write $\delta_t^{\mathrm{wt}}$ for multiplication of a weight-$j$ monomial by $t^j$; the factor-count scaling $\mathsf{D}_t^{\mathrm{fac}}$ is Definition 3.9. Abbreviate the actual coupling by $b_{N,p,k} = \beta_{N,p,k}$. The Gaussian field divided by the small-field threshold has scale

$$\vartheta_{k,N,p} = (t_k \sqrt{b_{N,p,k}})^{-1}, \qquad 0 < \vartheta_{k,N,p}^{-1} \le 2M_k.$$

The inequality follows from $t_k = M_k/\sqrt{\widehat{\beta}_k}$ and $b_{N,p,k} \le 2\widehat{\beta}_k$. Here $t_k$ is the small-field threshold of Part I. Solve

$$\mathsf{C}_{k,N,p,d}\Lambda_{k,N,p,d}u = \mathsf{D}^{\mathrm{fac}}_{\vartheta_{k,N,p}^{-1}} \delta^{\mathrm{wt}}_{M_k^{\alpha}} \iota_d^{\mathrm{free}} u, \qquad |v|_1 \le K_{k,d} a_k^d |\mathsf{C}_{k,N,p,d} v|_1. \tag{16.14}$$

Here $K_{k,d}$ is independent of $N, p$ at fixed $k$. To obtain the second inequality, iterate the inverse estimate of Lemma 14.5 from $r(d)$ to $k$ and apply the coefficient comparison in the proof of Lemma 14.21; its relative readout error is less than $1/4$, so its inverse costs at most 2. One may take $K_{k,d}$ bounded by twice the product of the finitely many column and block factors up to $k$, enlarged by the fixed basis norms. The readout is injective, hence invertible on its finite coefficient space. Its unit, conjugation and signed laws give these laws for the inverse. The displayed upper bound on $\vartheta^{-1}$ bounds the right side at fixed $k$, proving boundedness of the lifts on cutoff tails.

Here is the all-arity estimate. Write $I_i = O_{N,p}^{(d_i)}(\Lambda_{k,N,p,d_i} u_i, r_k x_i / a_N)$ at compatible lattice points. For a subset $J$ put $G_J = S_{|J|}^{\mathrm{Max}}((u_i)_{i\in J}; (x_i)_{i\in J})$ and define

$$R_\varnothing = 1, \qquad R_I = \prod_{i\in I} I_i - \sum_{\varnothing \neq J \subseteq I} G_J R_{I\setminus J}.$$

This subtracts the full continuum Gaussian moment on every subset of slots. For exact continuum Gaussian variables its nonempty coefficients have expectation zero, by induction on $|I|$. The contraction correction and the $\vartheta$-normalized equation (16.14) identify the leading Gaussian correlations with these continuum moments at the normalized grid points, as calculated in Lemma 14.21. Covariance, Taylor and localization errors remain in the regular and forced source terms.

Here is the root estimate for the whole reflected array. Use the interval (14.4) with exponent $\alpha$ and terminal level $k$. Denote its width level by $w_\alpha(k) = k - h_k(\alpha) - v_k$ and its centered tiling by $\mathcal{C}^{\mathrm{tile}}_{w_\alpha(k)}(0)$, as in Definition 14.10. Put $q_k = \log M_k/(16n)$. Give each marked slot amplitude $e^{2q_k}$. This amplitude is at most $M_k^{1/16}$ because $n \geq 2$. The weighted mode norm in the proof of Lemma 14.11 multiplies amplitudes over subsets, so coefficient extraction for a nonempty $I$ contributes $e^{-2q_k|I|}$ in each reflected cell. The corrected-source iteration of Lemma 14.22 keeps this radius on the selected comparison intervals. For the arity bound $n$, work in $\mathbb{C}[z_1, \dots, z_m]/(z_i^2)$ and multiply by the reciprocal of the continuum free moment polynomial. Its coefficient of $z_I$ is exactly the recursion defining $R_I$. Compute the finite Gaussian contractions on every subset; the remaining source modes have zero leading Gaussian coefficient. In the source iteration, collect their absolute weighted norms. Terms linear in these modes cost the finite stage constant; every other connected term contains at least two residual modes and is bounded by their squared norm times a fixed power of $M_k$. This is the submultiplicative mode-norm calculation in Lemma 14.11. The covariance and localization errors in Lemma 14.21 are chosen below the corresponding finite source tolerance at entry, as in (14.40). The induction of Lemma 14.22 therefore applies to the full finite set of modes. Its sums are rooted at a mark and do not count the replicated cells.

The remaining cost is the normalized positive integral. Formula (14.41) bounds its logarithm per cell by the stage constant $Q_{J_k}^{\mathrm{cmp}}$ of (14.42), with the maximum number $n$ of marks in place of two. This replaces only the finite source-amplitude contribution by at most $n$; include it in that stage constant. For the arity bound $n$, we require in item 5 the stronger bound $Q_j^{\mathrm{cmp}} \leq \log M_t/(16n)$, hence $Q_{J_k}^{\mathrm{cmp}} \leq q_k$, in place of its two-mark bound $\log M_t/32$. This is possible by increasing its entry level, since $\log M_k \to \infty$ within a fixed stage. Taking the root over the reflected cells gives

$$\|R_I\|_{\mathcal{C}^{\mathrm{tile}}_{w_\alpha(k)}(0)} \leq e^{q_k - 2q_k|I|} \leq M_k^{-1/(16n)} \qquad (\varnothing \neq I \subseteq \{1, \dots, m\}). \tag{16.15}$$

The smallness here comes from the corrected density at large coupling: its source terms remain bounded at an increasing amplitude, while the unmarked normalization grows more slowly than that amplitude. The estimate applies to the complete reflected array, before absolute values are taken, and requires no independence of its insertions.

The source stencils are interior. Distinct coordinate differences at normalized grid points are at least $\mathfrak{s}_k$, so their physical differences are at least $a_k$. The reading map assigns the corresponding level-$k$ sites without rounding, and the interval margin (14.35) contains the inspected stencils and their reflected copies. Expanding the recursion, applying the chessboard inequality (14.16) to each $R_I$, and bounding the free submoments on $K_{2R,m}$ gives $C_{R,\boldsymbol{u}} M_k^{-1/(16n)}$. This proves the last line of (16.13).

Two further root levels bound the same insertions between grid points. At its own normalization, the singleton source comparison just proved gives a root at most $1 + |S_1^{\mathrm{Max}}(u)|$ by the triangle inequality with the unit. For exponent $\beta$, changing normalization to $M_k^{\alpha}$ multiplies a weight-$j$ coefficient by $M_k^{(\alpha-\beta)j}$. The finite coefficient sum therefore gives $\mathfrak{a}_k \le C M_k^{(\alpha-\beta)D}$. The same calculation at exponent $\gamma$ gives $\mathfrak{b}_k \le C M_k^{(\alpha-\gamma)D}$. Include every reflected basis label and its image under conjugation in these finite maxima.

The two interval lengths of (14.4) give, with the stage block ratio denoted by $L$,

$$\frac{a_{w_b(k)}}{a_k} = L^{h_k(b)+v_k} \le L^2 M_k^{b+2} B_k^{10} \le M_k^{b+5} \qquad (b = \beta, \gamma). \tag{16.16}$$

The last inequality holds after the stage entry: $L$ is fixed there and $B_k = M_k^{o(1)}$. In particular the $\beta$ cell is at most $a_k M_k^{\beta+6} = o(r_k)$, and the $\gamma$ cell is $o(r_k\sigma_k)$ by (16.11). The chessboard inequality with the first roots bounds separated continuum moments by $\mathfrak{a}_k^m$. The second cells fit between points of $E_{4\sigma_k}$; applying Lemma 14.15 with margin $r_k\sigma_k$ gives $4m\mathfrak{b}_k^m/\sigma_k$ after rescaling positions. These choices use the same coefficient readout and lifts; only their weight normalization changes. The finitely many comparisons can be imposed simultaneously in the choices of Lemma 14.1, and their cutoff tails can be intersected.

Finally take any convergent coefficient subfilter of these bounded lifts in the ordered limit. Lemma 15.1 passes the moment estimates to the fixed table; the difference bound passes by Proposition 4.11 and Lemma 15.4. This proves all three inequalities for every coefficient cluster limit. No extraction of the table or volume sequence is made here. □

For the shell estimate take $p \in \{0,1\}$, $A \in F_d$, $B \in F_4$ and $D = d + 1 - p$. A grid test is a separated configuration $(0, x_i)$ on $a_j\mathbb{Z}^4$ for some level $j$, generic in each coordinate, with a fixed spectator list $\boldsymbol{B}_i$. Define its response functional and marked response by

$$\mathcal{L}_i(v) = S(v(0), \boldsymbol{B}_i(x_i)), \qquad \mathfrak{h}_i(y) = S(B(y), A(0), \boldsymbol{B}_i(x_i)).$$

Finite collections may use different grids. The union of all these configurations is dense. The estimate below supplies the degree bound needed to represent a shrinking shell by a field.

**Lemma 16.5** (Annular annihilator estimate)**.** *For every fixed finite list of grid tests and coefficients $g_i \in \mathbb{C}$ with $\sum_i g_i \mathcal{L}_i = 0$ on $F_D$, there are $\rho_l \downarrow 0$ such that*

$$\sup_{1/4 \le |z| \le 2} \left| \sum_i g_i \rho_l^{3+p} \mathfrak{h}_i(\rho_l z) \right| \longrightarrow 0. \tag{16.17}$$

*Proof.* Use Definitions 3.9, 4.9, the local sections of Lemma 10.2, Lemmas 14.8, 14.9, 14.12, together with Proposition 4.11 and Lemma 15.4. Put $h = 3 + p$, so $D + h = d + 4$. Write $s_m = \log(a_m^{-1})$. Use the support radius $B_m = \lceil (\log \widehat{\beta}_m)^2 \rceil$ of Part I and take $\rho_m = 128 B_{m+1}^2 a_{m+1}$. Its logarithmic rate is $\log \rho_m = -s_m + o(s_m)$.

Fix a centered cell around the distinguished point, small enough to be disjoint from all spectator stencils, and a finer stopping level at which its replicated source estimate applies. Choose a slightly finer level $j(m)$ with

$$\frac{s_{j(m)}}{s_m} \longrightarrow t = 1 + \frac{1}{64(d+12)}.$$

Such levels exist since $s_{m+1}/s_m \to 1$, by Lemma 14.1. At a grid point $y \in a_{j(m)}\mathbb{Z}^4$ with $\rho_m/16 < |y| < 4\rho_m$, choose representatives $a \in V_d, b \in V_4$ for $A, B$, block their two marked insertions at $y, 0$ together, and retain the Taylor terms of weight at most $D$. Their coefficient vector $c_{N,p}(y) \in V_D$ is chosen by inverting the marked coefficient map. The exact difference is

$$\rho_m^h O_{N,p}^{(4)}(b, y/a_N) O_{N,p}^{(d)}(a, 0) - O_{N,p}^{(D)}(c_{N,p}(y), 0).$$

Estimate this difference in the root of the fixed cell. We give the quantitative marked calculation, with the powers that determine its smallness. Let $u_{m,1}, u_{m,2}$ be the available amplitudes of the degree-$d$ and rescaled degree-four marks at level $j(m)$. The one-mark bounds, including the factor $\rho_m^h$, give

$$\log u_{m,1} = (-dt + o(1))s_m, \qquad \log u_{m,2} = (-4t + h + o(1))s_m.$$

The support weights, coefficient columns and threshold powers have logarithm $o(s_m)$ by (14.3) and (14.8). Transporting each of the two sources to level $m+1$ adds the four-dimensional volume factor $(a_{j(m)}/a_{m+1})^4$ to its available amplitude. Thus the joint amplitude $v_m$, and the dimensionless coefficient bound $e_m$ of its Taylor remainder, satisfy

$$\begin{aligned} \log v_m &= [-D - (d+12)(t-1) + o(1)]s_m, \\ e_m &\le \exp\{[(d+12)(t-1) + o(1)]s_m\}. \end{aligned} \tag{16.18}$$

Indeed the two transported exponents sum to $-(d+4)t + h + 8(1-t) = -D - (d+12)(t-1)$. Taylor extraction costs the reciprocal joint amplitude times $a_{m+1}^D$, which cancels the $D$ term and gives the second line. This is the two-mark use of the source representation (14.18), retaining its forced terms and their eligibility tests.

The matched remainder gains $e^{(s_m - s_{k_0})/8}$ in its inverse bound, where $k_0$ is the fixed stopping level: it is the product of the factors $\ell_i^{1/8}$ in (14.27). Put $c_m = e^{s_m/32}$. We require the residual root to be $O(c_m^{-1})$, so multiply its source by $c_m$ before applying that bound. Its size is then at most

$$c_m e_m e^{-(s_m - s_{k_0})/8} \le \exp\{[-5/64 + o(1)]s_m\}, \qquad \frac{1}{32} + (d+12)(t-1) - \frac{1}{8} = -\frac{5}{64}.$$

It eventually fits every fixed source allowance at $k_0$. The normalized replicated-source estimate, Lemma 14.11, consequently bounds the unamplified residual root by $2C/c_m$. The inverse coefficient estimate (16.14), applied in degree $D$ at the fixed stopping level, also bounds $c_{N,p}(y)$ on cutoff tails for each fixed $m, y$. Interior support follows from $32\rho_m \to 0$ and the support diameter $a_{m+1}(B_{m+1}^2 + 1 + 2048B_{m+1}^2) \to 0$; both fit the fixed centered cell.

Pass to a coefficient cluster limit $c(y)$ and project it to $F_D$. Applying the chessboard inequality (14.16) to the residual and the fixed exterior stencils gives, simultaneously for the finite tests,

$$|\rho_m^h \mathfrak{h}_i(y) - \mathcal{L}_i(\pi_D c(y))| \le C_i c_m^{-1}.$$

The same coefficient vector is used for every spectator. Multiplication by $g_i$ and summation cancels the response term exactly. Consequently the sum in (16.17) is $O(c_m^{-1})$ at all grid points of this enlarged annulus whose four coordinates are nonzero.

We interpolate across the excluded coordinate planes. Let $f_m(y) = \sum_i g_i \rho_m^h \mathfrak{h}_i(y)$. For a derivative order $q$, Proposition 4.11 replaces the moving label by one of degree $4+q$. The centered and reflected root bounds of Lemmas 14.12 and 14.11, with (14.16), bound a field of fixed degree $e$ by $\exp[(e+2+o(1))s_m]$. This uses the one-insertion bound with exponent loss two; its constant depends on that field but has logarithm $o(s_m)$. The finitely many later levels required for the fixed spectator word have logarithmic scale $s_m + o(s_m)$, by Lemma 14.1. Choose degree bounds $e_{ij}$ for its spectator labels and put $C_0 = 8 + d + \max_i \sum_j (e_{ij}+2)$. Multiplying these root bounds, and using $\rho_m^h \le 1$, gives

$$|\partial_a^q f_m(y)| \le \exp[(q + C_0 + o(1))s_m] \qquad (\rho_m/16 < |y| < 4\rho_m), \tag{16.19}$$

where $C_0$ depends on the fields and spectators, independently of $q$. Choose $q$ so $q(t-1) > C_0$. Since $a_{j(m)} = \exp[-(t+o(1))s_m]$, all the $q$th derivatives along grid-sized coordinate segments tend to zero. Choose $2^q$ nonzero grid coordinates, separated by at least one normalized mesh, within distance $4(2^q+2)a_{j(m)}$ in each axis. On an interval of that fixed normalized length, repeated mean value estimates bound a function by a constant times its values at these nodes plus its $q$th derivative: pair adjacent nodes to bound the derivative at intermediate points, repeat $q$ times, and integrate back. Apply this to the real and imaginary parts in each of the four axes. The constant depends on $q$, not $m$.

These segments stay in the enlarged annulus because $4(2^q+2)a_{j(m)} = o(\rho_m)$. Values with all four coordinates on the nonzero grid tend to zero by the preceding cancellation; the derivative remainders tend to zero by the choice of $q$. Four successive interpolations therefore give $\sup_{\rho_m/4 \le |y| \le 2\rho_m} |f_m(y)| \to 0$. Select a decreasing subsequence of $\rho_m$. These choices depend on the fixed tests and $g$, as allowed. □

## 16.3 Free comparison and the conserved line

The next argument converts Lemma 16.4 into the stable comparison of Part I. Its proof also explains why no volume refinement is needed.

**Lemma 16.6** (Stable weak comparison)**.** *The fixed family has the scaling sequence and estimate of Proposition 5.2. At every fixed degree horizon its frames eventually preserve exactly conjugation, $H_4$ actions, units and inclusions.*

*Proof.* Use Definitions 4.3, 4.9, 4.10, 5.1, Lemmas 15.1, 15.4, 16.1 and 16.4. The set $K_{R,m} \setminus E_{8\sigma_k}$ has volume at most $C_{m,R}\sigma_k$: it lies in finitely many strips about coordinate coincidence hyperplanes. Its contribution to a tested difference is at most $C_{m,R}\|f\|_\infty(\mathfrak{a}_k^m + \|S_m^{\mathrm{Max}}\|_{K_{R,m}})\sigma_k$. Outside these strips, rounding to the grid moves a point by at most $4m\mathfrak{s}_k$. Eventually $4m\mathfrak{s}_k \le \sigma_k$, so the segment remains in $K_{2R,m} \cap E_{4\sigma_k}$. The mean value bound and the free modulus of continuity $\varpi$ give total tested error at most

$$C_{m,R}\|f\|_\infty\big[\varepsilon_k + (\mathfrak{a}_k^m + \|S_m^{\mathrm{Max}}\|_{K_{R,m}})\sigma_k + \mathfrak{b}_k^m \mathfrak{s}_k/\sigma_k + \varpi(4m\mathfrak{s}_k)\big]. \tag{16.20}$$

Every term tends to zero by the strict exponent inequalities. Expand arbitrary fields in the finitely many basis words; their coefficient sums are bounded by a constant times the product of their norms.

At a fixed comparison level the lifts are bounded in a finite-dimensional space. Take a convergent subfilter in the ordered cutoff limit and project to $F_d$. Ordered convergence holds also for convergent coefficients, by multilinearity. This proves (16.20) for linear maps $U_{k,d} : M_d^{\mathrm{Max}} \to F_d$.

Every subfilter uses the same table, by Lemma 15.1; the bounds apply to all its coefficient cluster limits by Lemma 16.4.

Choose successively arity, degree and radius bounds tending to infinity, tolerances tending to zero, and strictly decreasing radii tending to zero. The latter is possible because $\log M_k = o(\log a_k^{-1})$ by the scale definitions (3.2) and the running law (14.10), whence $a_k M_k^{\alpha} \to 0$ for every fixed $\alpha$. This selects one sequence before any fixed field list. To impose inclusions exactly, choose real $H_4$ equivariant left inverses $L_d$ of the free inclusions $i_d$, by averaging an inner product over the finite group. Replace the maps recursively by

$$\widetilde{U}_{k,0} = U_{k,0}, \qquad \widetilde{U}_{k,d+1} = \jmath_{d,d+1}\widetilde{U}_{k,d}L_d + U_{k,d+1}(I - i_d L_d). \tag{16.21}$$

Expansion into the two summands preserves the comparison, since free and interacting inclusions preserve correlations. It also preserves reality and signed covariance; the degree-zero unit now gives all units. Strict free positivity supplies finitely many separating tested correlations in each degree. Their matrices converge, so a nonzero full-rank minor makes $\widetilde{U}_{k,d}$ eventually injective. Proposition 4.11 makes it an isomorphism. Fill the finitely many earlier indices in each degree with any of these isomorphisms. This changes no eventual assertion.

Uniformity of (16.20) over the supremum-norm unit ball of tests gives local $L^1$ convergence by smooth approximation to the phase of the difference. This proves Proposition 5.2 for the single sequence of Definition 5.1. □

We record a consequence needed before identifying the current. For the stable frames of Lemma 16.6, put

$$\mathsf{d}_{k,a} = r_k(\mathsf{U}^{\mathrm{sc}}_{k,d+1})^{-1}\partial_a \mathsf{U}^{\mathrm{sc}}_{k,d}. \tag{16.22}$$

Then $\mathsf{d}_{k,a} \to \partial_a^{\mathrm{Max}}$ in operator norm. To prove this, choose finitely many tested free responses injective on $M_{d+1}^{\mathrm{Max}}$. Their interacting matrices converge to this injective matrix by Lemma 16.6. Testing $\mathsf{d}_{k,a}u$ gives, by integration by parts, minus the correlation tested against $\partial_{x_0^a} f$. These values converge to the tests of $\partial_a^{\mathrm{Max}}u$. An injective finite matrix has a bounded left inverse on all sufficiently small perturbations, proving the limit on a basis and hence in operator norm.

The chain rule and (16.22) make every positional derivative converge locally in $L^1$, also for convergent free label lists. For a compactly supported smooth difference $g$ and $M > 4n + j$, integration by parts gives

$$|\widehat{g}(p)| \le C_M(1 + |p|)^{-M} \sum_{|\alpha| \le M} \|\partial^{\alpha} g\|_{L^1}.$$

Fourier inversion [8, Theorem 8.26] applies since both $g$ and $\widehat{g}$ are integrable. Multiplying by $p$ through order $j$ and integrating proves uniform convergence of those derivatives on a smaller compact set. Consequently

$$S_n(\mathsf{U}^{\mathrm{sc}}_{k,d_i}u_{k,i}; r_k\,\cdot\,) \longrightarrow S_n^{\mathrm{Max}}(u_i;\cdot) \quad \text{in } C^{\infty}_{\mathrm{loc}}(\mathrm{Conf}_n) \quad (u_{k,i} \to u_i). \tag{16.23}$$

The rank calculation (16.6) now transfers to the mixed fields. The following proof also chooses the sign of one fixed current.

**Lemma 16.7** (Conserved line)**.** *The real space of centered real symmetric $H_4$-tensorial tensors $J_{ab} \in F_4$ with $\sum_a \partial_a J_{ab} = 0$ is one-dimensional. For some nonzero $J$, a refinement of the sequence of Proposition 5.2 and positive $\xi_k$,*

$$(\mathsf{U}^{\mathrm{sc}}_{k,4})^{-1}(\xi_k^{-1} r_k^4 J_{ab}) \longrightarrow T_{ab}^{\mathrm{Max}}$$

*componentwise.*

*Proof.* Use Definitions 4.9, 4.10, 5.1, Lemmas 15.4, 16.1, 16.6 and (16.22). Transport a free tensor by $\mathsf{U}^{\mathrm{sc}}_{k,4}$ and subtract its mixed one-point value times the unit. On centered tensors this is a real isomorphism $C_k$; the inverse transports back and subtracts the free expectation. Unit preservation makes the two composites identities. Let $V_k$ be transport by $\mathsf{U}^{\mathrm{sc}}_{k,5}$ on real covariant vectors. Then

$$V_k^{-1}(r_k\,\mathrm{div})C_k \longrightarrow \mathrm{div}^{\mathrm{Max}}.$$

The centering term disappears since derivatives kill the unit. Composition with a right inverse of free divergence tends to the identity, so mixed divergence is onto. The domain and target dimensions are three and two by these isomorphisms and (16.6); its kernel is a real line.

Choose a nonzero $B$ on this line and set $\xi_k = \|(\mathsf{U}^{\mathrm{sc}}_{k,4})^{-1}(r_k^4 B)\|/\|T^{\mathrm{Max}}\|$, with norms on whole tensors. The normalized preimages have a convergent subsequence on this sphere. Stability preserves symmetry, reality and signed tensoriality. Equation (16.23) at arity one gives centering of the limit, and (16.22) gives zero free divergence. Lemma 16.1 makes the limit $T^{\mathrm{Max}}$ or $-T^{\mathrm{Max}}$. Choose $J = B$ or $-B$ once to obtain the stated limit with positive $\xi_k$. □

The shell estimate in Lemma 16.5 now applies to each component of the current.

The resulting fields represent both fluxes of Definition 5.3. The proof combines that estimate with finite-dimensional duality, conservation, translation covariance and continuity of separated responses.

**Lemma 16.8** (Raw shell charges)**.** *For the chosen $J$ there are complex linear maps $\mathsf{p}_b : F_d \to F_{d+1}$ and $\mathsf{Y}_K : F_d \to F_d$, the latter real linear in $K$, such that, for every allowed cutoff, field, spectator list and separated configuration,*

$$\mathcal{P}_b^J = S((\mathsf{p}_b A)(x_0), \boldsymbol{B}(x)), \qquad \mathcal{Y}_K^J = S((\mathsf{Y}_K A)(x_0), \boldsymbol{B}(x)).$$

*Proof.* Use Definitions 4.9, 5.3, Lemma 15.4, Lemma 16.7 and Lemma 16.5. More generally use a constant Killing field $v$ ($p = 0$) or $v(y) = Ky$ ($p = 1$). Shell independence follows by integration by parts: the divergence vanishes, and $\sum_{a,b}(\partial_a v_b)J_{ab} = 0$ by symmetry.

Fix finitely many grid response tests $\mathcal{L}_i$ as in Lemma 16.5, and coefficients $g_i$ annihilating $F_D$, where $D = d+1-p$. Shrink a common window to $\chi(y/\rho_l)$, with $\chi = 1$ near zero and $\nabla\chi$ supported in $1/2 \le |y| \le 1$. Changing variables gives the sum of their fluxes as

$$-\sum_{a,b}\int (\partial_a\chi)(z)v_b(z)\sum_i g_i\rho_l^{3+p}S(J_{ab}(\rho_l z), A(0), \boldsymbol{B}_i(x_i))\,dz.$$

It is independent of $l$ and tends to zero by (16.17). Thus every linear relation between the response tests is also a relation between their flux values. Finite-dimensional duality gives a field in $F_D$ matching these finitely many values: the annihilator of the image of the response matrix annihilates the flux vector.

Choose a finite separating family of grid tests. Such a family exists because the common kernel of all separated responses is zero, grid configurations are dense, and responses are continuous. A field matching this family is unique. Adjoin any additional grid test and apply the preceding finite-dimensional argument; uniqueness makes the same field match it. Continuity of the responses and of shell integrals, with the window fixed locally, extends the identity to all separated configurations with distinguished point zero. Translation covariance then moves that point to arbitrary $x_0$. Linearity of the integral and injectivity of responses give complex linearity in $A$ and real linearity in $K$. Taking $v = e_b$ and $v = K(\cdot - x_0)$ proves exactly the two asserted raw shell laws. □

## 16.4 Ward identities and rotations

Write $\boldsymbol{q} = (A_i(x_i))_i$ and define the angular sum

$$\mathcal{W}_K[Y,p](\boldsymbol{q};x) = \sum_i \left[ S(\dots, Y_K A_i, \dots; x) + \sum_b (Kx_i)_b S(\dots, p_b A_i, \dots; x) \right]. \tag{16.24}$$

The coordinate spectral representation of Lemma 17.3 uses a reflection Hilbert space $\mathcal{H}_\mu^{\mathrm{OS}}$ and positive generator $H_\mu^{\mathrm{OS}}$ for each coordinate axis $\mu$. Here $\mathsf{T}_\perp(y)$ is the unitary translation by $y$, $y^\mu = 0$, on this space. For a point field or finite collinear word $U$, write $U_t$ for its vector after translating it into the positive half-space by $t$. Its positive spectral measure $\nu_\mu^U$ is specified by

$$\langle U_t, e^{-sH_\mu^{\mathrm{OS}}} \mathsf{T}_\perp(y) U_t \rangle = \int e^{-(2t+s)\epsilon} e^{ip\cdot y}\, d\nu_\mu^U(\epsilon, p).$$

Here $t$ is larger than the fixed extent of the word; the point-field measures are allowed to have infinite total mass before this damping. The damped moments are (17.5), and mixed spectral Cauchy–Schwarz is (17.4). Positive-time analyticity is supplied by Lemma 17.3. These results use coordinate reflection positivity, E1–E7, Proposition 4.11 and the reflected-word bound of Lemma 15.6; conservation enters the tensor estimates, and continuous rotations are not used.

Fix distinct axes $a, b$ and a line axis $l$. Choose even $\eta \in C_c^\infty(\mathbb{R})$ equal to one on $[-1,1]$ and put $\ell(t) = t\eta(t)$. Choose an even nonnegative $\psi \in C_c^\infty(\mathbb{R})$ equal to one on $[-1,1]$; it will dominate the spatial Fourier factors on the momentum box. Set $\omega_j = -100$ for $j = a, b$, $\omega_l = -150$ if $l \notin \{a,b\}$, and $\omega_j = -300$ otherwise. For $\lambda = e^k$, $k \in \mathbb{N}$, $k \to \infty$, put

$$\mathcal{K}_\lambda(p) = \prod_{j \neq \mu} w_j^{-1} \psi(p_j / w_j), \qquad I_\mu(U;\upsilon) = \int_{\epsilon \le 1} \frac{\mathcal{K}_\lambda(p)}{\epsilon + \lambda^\upsilon}\, d\nu_\mu^U, \quad w_j = \lambda^{1+\omega_j}.$$

For a word $B$ on $\mathbb{R}e_l$ projected to odd parity in axes $a, b$ and even parity in the other axes, Lemma 17.3 gives, for every $\delta > 0$,

$$I_\mu(B; -3000) = O(\lambda^u), \quad u = \begin{cases} 1, & \mu = l, \\ -\omega_l, & \mu \neq l, \end{cases} \qquad I_\mu(J_{\mu b}; \omega_\mu) = O_\delta(\lambda^\delta) \quad (\mu \neq b). \tag{16.25}$$

For the diagonal component it gives, with $W = \lambda^{-99}$,

$$\int_{\epsilon \le 1} \frac{\epsilon W^4}{(\epsilon + W)^4} \mathcal{K}_\lambda(p)\, d\nu_\mu^{J_{\mu\mu}} = O_\delta(\lambda^\delta W^4). \tag{16.26}$$

Constants may depend on the fields, configuration, axes, profiles and $\delta$, independently of $\lambda$. These bounds include $\epsilon = 0$. The zeroth damped moment and $\mathcal{K}_\lambda \le C\lambda^{900}$ imply $I_\mu(J_{\mu\mu}; 0) = O(\lambda^{900})$. For the Fourier–Laplace test integrals used below, Lemma 17.3 bounds the integral over $\{\epsilon > 1\} \cup \{\exists j \neq \mu : |p_j| > w_j\}$ by $C_M \lambda^{-M}$ for every fixed $M$, after any fixed polynomial prefactor.

The free vector and internal tests are those of Lemmas 16.2 and 16.3.

The coordinate bounds remove the angular boundary at infinity before continuous rotations are known. The proof below uses an anisotropic cutoff so that each reflected column gains an inverse power of its largest spatial extent.

**Lemma 16.9** (Collinear angular identity)**.** *For every separated configuration $x$ on a coordinate line, every field list $\boldsymbol{q}$ and every $K \in \mathfrak{so}(4)$, the raw charges satisfy*

$$\mathcal{W}_K[\mathsf{Y}, \mathsf{p}](\boldsymbol{q}; x) = 0. \tag{16.27}$$

*Proof.* Apply the raw shell laws of Lemma 16.8, with Definition 5.3 and the calculus of Proposition 4.11, and the coordinate estimates (16.25), (16.26) of Lemma 17.3. It suffices to treat $K = E_{ba} - E_{ab}$, $a \neq b$. With the profiles fixed above, set

$$f_{ab}(y) = \lambda^{100}\ell(\lambda^{-100}y_a)\prod_{j\neq a}\ell'(\lambda^{\omega_j}y_j), \qquad V_b = f_{ab}, \quad V_a = -f_{ba}, \quad V_j = 0\ (j \neq a, b).$$

The vector field $V$ equals $Ky$ near every insertion for large $\lambda$. Disjoint windows and conservation give $\mathcal{W}_K = -\int \sum_{\mu,b} \partial_\mu V_b\, S(J_{\mu b}(y), \boldsymbol{q})\, dy$. The two terms with $\mu = a, b$ cancel exactly, because $\partial_a f_{ab} = \partial_b f_{ba}$ and $J_{ab} = J_{ba}$.

Consider the $V_b$ terms with $\mu \neq a$. Reflecting the integration variable in each axis, including the tensor signs, replaces the spectator word by its odd projection in $a, b$ and its even projections in the other axes. If the line axis is reflected, positions on that line are reflected too; the projection is a finite linear combination of collinear words. Lemma 17.3 applies to each such combination. The differentiated profile is supported where $|y_\mu| \geq \lambda^{-\omega_\mu}$. Its Laplace transform therefore has the factor $e^{-\epsilon/\lambda^{\omega_\mu}}$. For each of the three spatial factors $f_j = \ell$ or $\ell'$, Fourier scaling gives, on $|p_j| \leq w_j$,

$$|\widehat{f_j(\lambda^{\omega_j}\cdot)}(p_j)| \leq \lambda^{-\omega_j}\|f_j\|_{L^1} \leq C\lambda w_j^{-1}\psi(p_j/w_j).$$

Their product gives $\lambda^3\mathcal{K}_\lambda$; the remaining time integral contributes the displayed $\lambda^{\omega_\mu}$ below. For the off-diagonal component, $e^{-t} \leq (1+t)^{-1}$ and spectral Cauchy–Schwarz bound the integral, before the prefactor $\lambda^{100}$, by

$$C\lambda^{3+\omega_\mu}\,[I_\mu(B; -3000)I_\mu(J_{\mu b}; \omega_\mu)]^{1/2}.$$

Take $\delta = 1$. For this off-diagonal term $\mu \notin \{a, b\}$. Thus if $\mu = l$ then $\omega_\mu = -150$ and $u = 1$; otherwise $\omega_\mu = -300$ and $u \leq 150$. The exponents are at most $-146$ and $-221.5$, respectively.

If $\mu = b$, split at $\epsilon = \lambda^{-3000}$. Above this value use $e^{-\epsilon/\lambda^{-100}} \leq 4[W/(\epsilon+W)]^2$ and (16.26); Cauchy–Schwarz gives exponent at most $3 + (150 + 1 - 396)/2 = -119.5$. Below it, $\epsilon + \lambda^{-3000} \leq 2\lambda^{-3000}$ supplies a factor $\lambda^{-1500}$ in Cauchy–Schwarz. The crude bound $I_\mu(J_{\mu\mu}; 0) = O(\lambda^{900})$ then gives exponent at most $3 - 1500 + (150 + 900)/2 = -972$. The complementary integration region is $O(\lambda^{-M})$ by Lemma 17.3, with $M > 201$. The $V_a$ terms are identical after exchanging $a, b$, since $\omega_a = \omega_b$; this includes their diagonal component $J_{aa}$. All remaining terms are consequently $O(\lambda^{-101})$ before the prefactor, and $O(\lambda^{-1})$ after it. The angular sum is independent of $\lambda$, so it is zero. Linearity in $K$ proves (16.27). □

For a finite-dimensional rotation action write $\mathsf{L}_d(K) = \frac{d}{dt}R_d(e^{tK})\big|_{t=0}$ for its generator. The shell charges can now be identified without assuming rotation covariance. The proof first fixes their common scale and then constructs an actual orthogonal-group action.

**Lemma 16.10** (Rotation and charge identification)**.** *There are $c_\star \in \mathbb{R} \setminus \{0\}$ and continuous complex linear $O(4)$ representations $R_d$ extending $H_4$, fixing units and commuting with conjugation and inclusions, with*

$$\mathsf{p}_b = c_\star\partial_b, \qquad \mathsf{Y}_K = c_\star\mathsf{L}_d(K), \qquad R_{d+1}(h)\partial_a = \sum_i h_{ia}\partial_i R_d(h).$$

*For every $d$, an isomorphism $\mathcal{V}_d : M_d^{\mathrm{Max}} \to F_d$ intertwines the free and interacting $O(4)$ actions.*

*Proof.* Use Definitions 4.9, 4.10, 5.1, 5.3, Lemma 15.4 and Lemmas 16.1, 16.2, 16.3, 16.6, 16.7, 16.8. Fix a window and move its distinguished point. Differentiating its two shell identities under the compact integral gives

$$\partial_b \mathsf{p}_a = \mathsf{p}_a \partial_b, \qquad \partial_b \mathsf{Y}_K + \sum_i K_{ib} \mathsf{p}_i = \mathsf{Y}_K \partial_b. \tag{16.28}$$

The derivative of $K(y - x_0)$ gives the second term. Testing the unit, applying conjugation or a signed permutation, and including a field show, respectively, that the charges kill units, are real, have vector or angular covariance, and respect inclusions. Injectivity of responses turns these tested identities into field identities.

Lemma 16.9 gives (16.27) for the raw charges.

Translate a collinear configuration by $e_l$ in (16.27), with $K = E_{bl} - E_{lb}$. Subtraction gives $\sum_i S(\dots, \mathsf{p}_b A_i, \dots; x) = 0$ for $b \neq l$. Equation (16.28) gives every positional jet of these identities. Translation invariance gives the same identities for $\partial$.

Apply the finite free vector tests (16.7) to the transported mixed vector maps. Their test matrices converge by (16.23). Adjoin a real functional nonzero on $\partial^{\mathrm{Max}}$; the resulting free matrix is injective and remains so under small perturbations. The mixed kernel therefore has dimension at most one. It contains both $\mathsf{p}$ and $\partial$, so $\mathsf{p} = c_d \partial$ in every sufficiently large degree. Inclusions and nonzero free differentiation make the constants equal and give the identity also in lower degrees. Rescale the two shell laws by $r_k/\xi_k$ and $1/\xi_k$, respectively. Changing variables $y = r_k z$ and using (16.23) yields

$$\frac{r_k}{\xi_k} (\mathsf{U}^{\mathrm{sc}}_{k,d+1})^{-1} \mathsf{p}_a \mathsf{U}^{\mathrm{sc}}_{k,d} \to \partial_a^{\mathrm{Max}}, \qquad \xi_k^{-1} (\mathsf{U}^{\mathrm{sc}}_{k,d})^{-1} \mathsf{Y}_K \mathsf{U}^{\mathrm{sc}}_{k,d} \to \mathsf{L}_d^{\mathrm{Max}}(K). \tag{16.29}$$

Finite separating tests recover these operator limits from the shell integrals. In particular $c_\star = c_d \neq 0$ and $c_\star/\xi_k \to 1$. Put $\mathsf{D}_K = c_\star^{-1} \mathsf{Y}_K$. Then (16.28) reads $\partial_b \mathsf{D}_K + \sum_i K_{ib} \partial_i = \mathsf{D}_K \partial_b$.

We next extend the collinear identity. Its angular sum has derivatives of the same form, by this commutator. Hence all its jets vanish on a line. In a coordinate plane write $\Phi_{u,v}(t,s) = S(u(te_a + se_b), v(0))$ and $(\varrho, \theta)$ for polar coordinates. Put $\Psi_{u,v}(t,s) = S((e^{-\theta \mathsf{D}_K} u)(\varrho e_a), (e^{-\theta \mathsf{D}_K} v)(0))$. For positive separation the axial function is a matrix coefficient of $e^{-tH_a^{\mathrm{OS}}}$ in $\mathcal{H}_a^{\mathrm{OS}}$, after fixed positive-time damping of its end vectors. Lemma 17.3 gives its analyticity, so $\Psi$ is analytic. The angular equation holds for $\Psi$, and $\mathcal{F} = \Phi - \Psi$ vanishes on the axis. Its angular jets give the recurrence

$$t \partial_s^{m+1} \mathcal{F}_{u,v}(t,0) = -\partial_s^m (\mathcal{F}_{\mathsf{D}_K u, v} + \mathcal{F}_{u, \mathsf{D}_K v})(t,0) + m \partial_t \partial_s^{m-1} \mathcal{F}_{u,v}(t,0).$$

For $m = 0$ omit the last term. Induction, for all labels at once, gives every transverse Taylor coefficient of $\mathcal{F}$ equal to zero. In a reflection Hilbert space let $v(s)$ be a one-insertion vector translated transversely. Its Gram kernel depends on $s' - s$ and has just these analytic Taylor coefficients. On a compact positive axial interval choose $\rho > 0$ inside a common complex analyticity neighbourhood of the finitely many matrix entries of $\Psi$. Cauchy estimates for its Gram kernel give $\|v^{(m)}(s)\|^2 \leq C \rho^{-2m} (2m)! \leq C (2/\rho)^{2m} (m!)^2$. Taylor's remainder proves transverse analyticity, uniformly in $s$. Pairing with a fixed exterior word proves analyticity whenever the moving insertion is earliest or latest in some coordinate.

Move one point at a time along coordinate segments while keeping it extreme in another coordinate. Every configuration with distinct first coordinates can be joined this way to a configuration on the first axis: move an extreme point beyond the others in the second coordinate, then beyond them in the first, then set its remaining coordinates to zero; repeat. Along each

segment the angular sum is analytic and all its jets vanish at the endpoint. It vanishes along the segment. Such configurations are dense, so continuity proves the global angular identity.

Its commutator gives $[\mathsf{D}_K, \mathsf{D}_L] = \mathsf{D}_{[K,L]}$. To justify this inference, its defect has vanishing correlation sums, is real, alternating in $K, L$, covariant under signed permutations of determinant one, kills the unit, and preserves the filtration. The finite free tests of (16.10) are injective on this defect space. Their injectivity transfers by (16.23); pulling the mixed defect back by the stable frames therefore makes it zero in sufficiently large degree. Injective inclusions give the same conclusion in all lower degrees.

We now construct the representation. Let $C_i = \mathsf{L}_d^{\mathrm{Max}}(K_i)$ and $\mathsf{B}_{k,i} = (\mathsf{U}_{k,d}^{\mathrm{sc}})^{-1}\mathsf{D}_{K_i}\mathsf{U}_{k,d}^{\mathrm{sc}}$ for the six plane generators. By (16.29), $\mathsf{B}_k \to C$. On endomorphisms put $\Omega(B, C) = \sum_i (B_i \cdot - \cdot C_i)^2$. An invariant inner product makes $\Omega(C, C)$ self-adjoint, with kernel the common commutant. A real polynomial $z$, with $z(0) = 1$, gives its kernel projection $P = z(\Omega(C, C))$. The operators $z(\Omega(\mathsf{B}_k, C))$ commute with the relative generators: expanding the commutator with the sum of squares cancels the antisymmetric structure constants in pairs. To correct them, iterate $\varphi(t) = 3t^2 - 2t^3$. The identities $\varphi(t) - t = (1-2t)(t^2-t)$ and $\varphi(t)^2 - \varphi(t) = (4(t^2-t)-3)(t^2-t)^2$ show that, near $P$, the errors square and the displacements have a convergent geometric sum. The limits are projections $P_k \to P$, each commuting with its relative generators $\mathsf{B}_{k,i} \cdot - \cdot C_i$. The bracket identity just proved makes the relative generators a Lie-algebra representation; the compressed operators therefore satisfy the same plane commutator relations and tend to zero. Their maximum norm $m_k$ satisfies $m_k \le 2m_k^2$, so they vanish eventually. Thus $f_k = P_k(I) \to I$ is invertible and $\mathsf{B}_{k,i} f_k = f_k C_i$. All operations preserve the real signed commutant. Set $\mathcal{V}_d = \mathsf{U}_{k,d}^{\mathrm{sc}} f_k$ for one sufficiently large $k$, and

$$R_d(h) = \mathcal{V}_d R_d^{\mathrm{Max}}(h) \mathcal{V}_d^{-1}.$$

This is continuous, real and extends the signed action, with generator $\mathsf{D}$. The global angular identity makes $S(R(e^{tK})A; e^{tK}x)$ constant. Plane rotations and coordinate reflections generate $O(4)$, proving full covariance. Covariance, unit deletion, inclusion and differentiation then imply the unit, inclusion and vector laws by injectivity of responses. This proves all assertions, with $\mathsf{L} = \mathsf{D}$ and the same $c_\star$ in every degree. □

Put $J_0 = c_\star^{-1} J$ and define, with probability Haar measure on $O(4)$,

$$T_{ab} = \int_{O(4)} \sum_{i,j} h_{ai} h_{bj} R_4(h)(J_0)_{ij} \, dh. \tag{16.30}$$

This average imposes exact tensor covariance while retaining the normalized shell actions. The following proof establishes Theorem 5.4 for this $T$.

**Lemma 16.11** (Covariance and normalized tensor)**.** *For every field list, $x \in \mathrm{Conf}_n$ and $h \in O(4)$, the table and the tensor $T$ defined in (16.30) satisfy*

$$S_n(R_{d_i}(h)A_i; hx) = S_n(\boldsymbol{A}; x), \qquad R_4(h)T_{ab} = \sum_{i,j} h_{ia} h_{jb} T_{ij}.$$

*The tensor is real, symmetric and conserved and has both normalized shell laws of Theorem 5.4.*

*Proof.* Use Definition 5.3 and Lemmas 16.8, 16.10. Covariance was obtained at the end of the preceding proof. This is a finite-dimensional integral of a continuous function on a compact group. Haar invariance gives the displayed tensor law. Reality and symmetry pass through the

integral, and the vector law for derivatives gives conservation. Changing variables $y = hz$ in each transformed shell, and using covariance, gives the translation and angular actions $\partial_b$ and $\mathsf{L}_d(K)$; for the latter use $R(h)\mathsf{L}(h^{-1}Kh) = \mathsf{L}(K)R(h)$. Integration preserves both identities. The positional derivative law in Proposition 4.11 identifies the translation action with $\partial_{x_0^b} S$. Thus (16.30) proves precisely Theorem 5.4, as well as the lemma. The tensor is centered: covariance gives $S_1(R_4(h)J_{ij}) = S_1(J_{ij}) = 0$ in every integrand. □

Finally the charge normalization removes the factors $\xi_k$ from stress insertions. No new choice depending on the arity or derivative order is needed.

**Lemma 16.12** (Transported infinitesimal actions)**.** *Along the single stable sequence of Proposition 5.5, for every fixed $d, a, K$,*

$$r_k(\mathsf{U}^{\mathrm{sc}}_{k,d+1})^{-1}\partial_a \mathsf{U}^{\mathrm{sc}}_{k,d} \longrightarrow \partial_a^{\mathrm{Max}}, \qquad (\mathsf{U}^{\mathrm{sc}}_{k,d})^{-1}\mathsf{L}_d(K)\mathsf{U}^{\mathrm{sc}}_{k,d} \longrightarrow \mathsf{L}_d^{\mathrm{Max}}(K)$$

*in operator norm. Proposition 5.5 holds for every fixed arity, free list, subset of stress slots, component choices, derivative order and separated compact set.*

*Proof.* Use Definitions 5.1, 5.3, Lemmas 16.6, 16.7, 16.10, 16.11 and (16.23). The derivative limit is (16.22); the rotation limit follows from (16.29) and $\xi_k/c_\star \to 1$.

The centered tensor $T$ is signed-tensorial and conserved, so Lemma 16.7 gives $J = cT$ for a real $c$. It is nonzero: otherwise its translation shell would force $\partial_a = 0$ in degree four, contradicting (16.22) and nonzero free differentiation of a quadratic curvature polynomial. Comparing translation shells gives $c = c_\star$. Consequently

$$(\mathsf{U}^{\mathrm{sc}}_{k,4})^{-1}(r_k^4 T_{ab}) = \frac{\xi_k}{c_\star}(\mathsf{U}^{\mathrm{sc}}_{k,4})^{-1}(\xi_k^{-1} r_k^4 J_{ab}) \longrightarrow T_{ab}^{\mathrm{Max}}.$$

Use these convergent free preimages in any chosen stress slots and use the fixed free labels in the others. Equation (16.23) gives convergence of every positional derivative, uniformly on every separated compact set. The derivative there differentiates $x \mapsto S(\boldsymbol{A}_k; r_k x)$, so all chain-rule factors are included. This is exactly Proposition 5.5, for the fixed tensor (16.30). The sequence is the refinement already chosen in Lemma 16.7; each fixed arity, degree and derivative order is covered on a tail of that single sequence. □

## 17 Slabs, phases and reconstruction

We prove the phase decomposition of Theorem 6.2 and the reconstruction of Theorem 6.15, starting from the correlations of Theorem 4.7 and quotient fields of Proposition 4.11. The phase construction uses the rotations and tensor of Theorem 5.4. The rotation proof in Section 16.4 needs coordinate spectral estimates before continuous rotations are available. Lemma 17.3 supplies them; they are not used later in this section.

Reflection positivity enters at several stages. Repeated reflected insertions bound the operator that inserts a word into a time slab. Compressing these operators to zero energy gives a commutative algebra; its characters define the phases. A lattice variance inequality then makes each character invariant under reflection, and small chessboard roots preserve the rank of a finite family of field tests in every character. Neither conclusion follows from commutativity alone.

Slab bounds control continuation to complex times and its boundary distributions. Vacuum uniqueness initially holds almost everywhere in the character space. Locality gives a phase-uniform

low-energy estimate; continuity of heat pairings and full support remove the exceptional set. The selected phase is chosen by its Euclidean ground space before reconstruction, and retained throughout. The energy estimate is polynomial; no positive lower bound on nonzero energies is used.

Throughout, a word is $w = \prod_{i=1}^{n} A_i(x_i)$ with distinct points, and $\tau_z w = \prod_i A_i(x_i + z)$. Fix a norm $\|\cdot\|_{V_d}$ on each coefficient space. Token bounds refer to representatives in the spaces $V_d$ of Definition 3.9; the resulting operators depend only on their classes in $F_d$. Write $\vartheta w$ for linear time reflection, acting on labels by $R_d(\mathsf{s}_4^{\mathrm{refl}})$, and put $\theta w = \vartheta \bar{w}$ with the order reversed. Thus $\theta$ is the antilinear reflection in the OS form of Definition 6.1.

## 17.1 Slabs and phases

Put $E_t = e^{-tH_4^{\mathrm{OS}}}$, $P_0 = P_4^0$ and $\Omega = \Omega_4^{\mathrm{OS}}$ in this subsection. Spatial translations on the OS space are denoted by $U_y$, $y^4 = 0$. The insertion rule and buffer are those preceding Lemma 15.6: $A_w v = w\, \tau_{h_{\mathrm{sl}} e_4} v$, and an $r$-buffered word in width $h_{\mathrm{sl}}$ has mutual sup-distance at least $r$ and times in $[r, h_{\mathrm{sl}} - r]$.

Write $\partial_{i,a} w$ for the word with $A_i$ replaced by $\partial_a A_i$. The reflected-word bound eliminates the exterior-dependent constant when one takes arbitrarily high powers. This gives the bounded operators needed for compression. Below words are $r$-buffered, $d$ bounds degrees, $R$ bounds coefficient norms, and $h, k$ are admissible slab widths.

**Lemma 17.1** (Slab operators)**.** *For $d \in \mathbb{N}$, $R \geq 0$, there are $C > 0$, $Q \in \mathbb{N}$ such that each admitted length-$n$ word has a bounded insertion operator satisfying*

$$\|\mathsf{B}_{h_{\mathrm{sl}}}(w)\| \leq C^n (n/r)^{Qn} \qquad (h_{\mathrm{sl}} > 0,\ 0 < r \leq 1)\,.$$

$$\mathsf{B}_h(w)^* = \mathsf{B}_h(\tau_{he_4}\theta w), \qquad \mathsf{B}_h(w)\mathsf{B}_k(v) = \mathsf{B}_{h+k}(w\,\tau_{he_4} v), \tag{17.1}$$

$$\partial_{x_i^a} \mathsf{B}_h(w) = \mathsf{B}_h(\partial_{i,a} w). \tag{17.2}$$

*Derivatives are in operator norm; empty words give $E_h$. Phase operators $\mathsf{B}_h^{\xi}(w)$, constructed after Lemma 17.5, satisfy this bound.*

The phase assertion is completed after Lemma 17.5.

*Proof.* Use Definition 6.1, Lemmas 15.5 and 15.6, and replace $R$ by $\max(1, R)$. The translation rule defines contractions: on the span of positive words,

$$\|E_t v\|^2 = \langle v, E_{2t} v\rangle \leq \|v\|\, \|E_{2t} v\|\,.$$

E7 bounds the separated orbit; iteration gives $\|E_t v\| \leq \|v\|$ and preserves null vectors. Translation invariance gives symmetry and commuting unitary spatial translations; Gram continuity gives strong continuity. Since $E_1 \psi = 0$ implies $E_{2^{-j}} \psi = 0$, strong continuity excludes zero spectral mass. Spectral calculus [15, Theorem 2.5.1 and Proposition 4.2.3] defines $H_4^{\mathrm{OS}} = \int_{(0,1]} -\log v \, dP(v)$ from the spectral measure of $E_1$. Positive roots and continuity give $E_t = e^{-tH_4^{\mathrm{OS}}} \to P_0$ strongly.

For a nonempty word, Lemma 15.6 supplies a width $s$ and $D = C_0^n (n/r)^{Qn}$. On the word span put $A = A_w$, $B = A_{w^\dagger}$, $T = BA$. The OS form gives $\langle Av, u\rangle = \langle v, Bu\rangle$; the reflected-word bound gives $\|T^j v\| \leq K_v D^{2j}$. Thus $d_j = D^{-2j}\|T^j v\|$ is bounded and $d_j^2 \leq \|v\| d_{2j}$. Taking suprema yields $\sup_j d_j \leq \|v\|$, hence $\|Av\|^2 = \langle v, Tv\rangle \leq D^2 \|v\|^2$. The bounded extension is unique by density. Right multiplication by $E_{h_{\mathrm{sl}} - s}$ gives the required width.

Positive-word matrix entries prove (17.1). FC bounds differentiated-label operators in a smaller buffer. Twice integrating their matrix entries along segments gives an operator norm Taylor remainder; induction proves (17.2) and norm smoothness. □

With $e_d$ as time the same construction gives $\mathcal{H}_d^{\mathrm{OS}}$, $E_t^{(d)} = e^{-tH_d^{\mathrm{OS}}}$ and commuting transverse translations $U_x^{(d)}$, $x^d = 0$, using only coordinate reflection positivity and signed permutations. Their joint projection-valued measure $P_d$ is on $[0,\infty) \times \mathbb{R}^3$, with variables $(E,p)$: apply Gelfand theory to the commutative unital C*-algebra generated by $E_1^{(d)}$ and the translation representation of $C_0(\mathbb{R}^3)$ [15, Theorems 2.4.1, 3.7.11, equation (3.89) and Proposition 4.2.3], then take the logarithm of the heat coordinate.

For a word combination $U$, let $v_{d,u}(U)$ be its OS vector translated by $ue_d$ into positive times. Undo its damping by

$$d\nu_d^U(E,p) = e^{2uE}\, d\langle v_{d,u}(U), P_d(E,p) v_{d,u}(U)\rangle. \tag{17.3}$$

This is independent of $u$ and finite on $E \le M$. Polarize to $\nu_d^{UV}$ on integrable tests. For nonnegative integrable $g_1, g_2$, spectral Cauchy–Schwarz gives

$$\left| \int f\, d\nu_d^{UV} \right|^2 \le \int g_1\, d\nu_d^U \int g_2\, d\nu_d^V \quad \text{if } |f|^2 \le g_1 g_2. \tag{17.4}$$

For $\vartheta_j U = \varepsilon_j U$, $\varepsilon_j \in \{1,-1\}$, ordinary conjugation gives, with $x^d = 0$ and $t$ separating the words,

$$S\big(\bar{U}^{\mathrm{rev}}\, \tau_{te_d + x} U\big) = \varepsilon_d \int e^{-tE + ip\cdot x}\, d\nu_d^U(E,p)\,.$$

Here the bar includes scalar conjugation. Both point fields and collinear words are represented.

Opposite reflection signs compare coordinate spectra. Fix a set of axes $Q$ and a finite word combination $U$ whose offsets have zero coordinates in $Q$, with $\vartheta_j U = \varepsilon_j U$ for $j \in Q$. Assume that both signs occur. For even nonnegative $f_j \in C_c^\infty(\mathbb{R})$, set $f_{j,v}(p) = v^{-1} f_j(p/v)$. The next estimate is used in Lemma 17.3.

**Lemma 17.2** (Opposite reflection signs)**.** *For every $i \in Q$, $q < 0$, $\beta_j \le 0$ and $\delta > 0$,*

$$\int_{E \le 1} \frac{\prod_{j \in Q\setminus\{i\}} f_{j,\lambda^{\beta_j}}(p_j)}{E + \lambda^q}\, d\nu_i^U = O_\delta(\lambda^\delta) \qquad (\lambda = e^s,\ s \in \mathbb{N})\,.$$

*Constants may depend on $U, Q, q, \beta, f, \delta$.*

*Proof.* Use Definitions 4.9, 6.1, E1–E7 and the coordinate derivative law of Proposition 4.11.

For a placement of margin $r > 0$, E7 bounds the $2m$th transverse derivative of the reflected-pair correlation by $Cr^{-D}$. Its centered difference has spectral integrand $[2\sin(tp_j/2)/t]^{2m} e^{-2uE}$. The integral remainder formula and Fatou give

$$\int e^{-2uE} |p_j|^{2m}\, d\nu_d^U \le C r^{-D}. \tag{17.5}$$

Extra heat supplies energy powers. Thus for every admissible $u$ and $m \in \mathbb{N}$,

$$\int e^{-2uE} (1 + E + |p|)^m\, d\nu_d^U < \infty\,. \tag{17.6}$$

For offsets perpendicular to $d$, take $u = t/2$, $r \asymp t$. Integrating $Ct^{-D}$ against $t^D\,dt$ on $(0,1)$ gives, enlarging $D$, $\int (1+E)^{-D}|p_j|^{2m}\,d\nu_d^U < \infty$.

Choose even $\sigma \geq 0$ of integral one supported in $1 \leq |v| \leq 2$. For $a_j = j+2$, let $L_j$ interpolate at $a_j^2$, $0 \leq j < m$. For radius $\varrho > 0$, define the annular correction by

$$\int \mathcal{C}_{m,\varrho} f(q) g(q)\,dq = \int \sigma(v) \int f(p) \sum_{j<m} L_j((p/(\varrho v))^2) g(\varrho v a_j)\,dp\,dv.$$

This gives smooth annular profiles with moment coefficients containing $\varrho^{-2r}$. For even $f$, the residual has $2m$ vanishing moments. Put $\operatorname{Lap} f(E) = \int e^{-E|t|}\widehat{f}_+(t)\,dt$. Interpolating $(p^2+E^2)^{-1}$ gives its residual response

$$\mathcal{R}_{m,\varrho} f(E) = 2E \int \frac{f(p)}{p^2+E^2} \int \sigma(v) \prod_{j<m} \frac{(\varrho v a_j)^2 - p^2}{(\varrho v a_j)^2 + E^2}\,dv\,dp. \tag{17.7}$$

At $E = 0$ its value is $2\pi f(0)$. For $f \geq 0$ supported in $[-\varrho, \varrho]$, all factors lie in $[0,1]$. For $\varrho \geq 1$ with $\operatorname{supp} f_v \subset [-\varrho, \varrho]$, $f_v(p) = v^{-1} f(p/v)$ and sufficiently large $m$, the moment bound gives

$$\int \mathcal{R}_{m,\varrho} f_v(E) |p_j|^{2r}\,d\nu_d^U \leq C(v^{-1} + \varrho^D). \tag{17.8}$$

Indeed the response is bounded by $2\pi\|f_v\|_\infty$ on $E \leq 1$ and $C_m(\varrho/E)^D$ on $E \geq 1$. Choose $m \geq D/2$ after the moment exponent $D$. For a bump equal to one on $[-1,1]$, the product is bounded below by $2^{-m}(1+E)^{-2m}$. Integrating over $0 < p < vE/(E+v)$ in (17.7) gives $\mathcal{R}_{m,\varrho} f_v(E) \geq C_m/(E+v)$ on $0 < E \leq 1$. The value at zero follows by continuity.

Thus $\mathcal{R}_{m,\varrho} f = \operatorname{Lap}(f - \mathcal{C}_{m,\varrho} f)$. For $i, j \in Q$, write $\operatorname{Resp}_i(g) = \int \operatorname{Lap} g_i(E) \prod_{j \in Q\setminus\{i\}} g_j(p_j)\,d\nu_i^U$. For $Q$ containing all axes, the same product test gives

$$(2\pi)^{-3} \int S(\bar{U}^{\mathrm{rev}} \tau_x U) \prod_k \widehat{g}_{k,+}(x_k)\,dx = \varepsilon_i \operatorname{Resp}_i(g) = \varepsilon_j \operatorname{Resp}_j(g)\,.$$

Indeed $\int e^{ipx}\widehat{g}_+(x)\,dx = 2\pi g(-p) = 2\pi g(p)$ for the even profiles, whereas the time integral is $\operatorname{Lap} g(E)$. Both temporal profiles have the zero moments required at collisions. For smaller $Q$, choose an even cutoff $h \in C_c^\infty(\mathbb{R})$ equal to one near zero and put $g_k(p) = h(p/T)$ in omitted axes, then let $T \to \infty$ by (17.8). Hence $\operatorname{Resp}_i(g) = -\operatorname{Resp}_j(g)$ when $\varepsilon_i = -\varepsilon_j$. Subtract annular corrections $a_i, a_j$ from the two head profiles. Positivity of the residual responses gives

$$\operatorname{Resp}_i(g_i - a_i; g_{\neq i}) \leq \operatorname{Resp}_i(g_i - a_i; g_j \mapsto a_j) + \operatorname{Resp}_j(g_j - a_j; g_i \mapsto a_i). \tag{17.9}$$

Expand the signed identity and discard $\operatorname{Resp}_j(g_j - a_j; g_{\neq j}) \geq 0$.

Iterate (17.9) along a finite sequence of opposite signs visiting every axis in $Q$. Insert an opposite sign between visits of the same sign. Replacing width $\lambda^\beta$, $\beta \leq 0$, by $\lambda^\alpha$, $\alpha > 0$, gives bounded coefficients $\lambda^{2r(\beta-\alpha)}$. A correction at $\lambda^{\alpha'}$ is at most $C_r \lambda^{-2r\alpha'} |p_j|^{2r}$. Other heights costing $\lambda^K$ and head width $\lambda^\beta$ give total cost $\lambda^{K - 2r\alpha' + \max(-\beta, \alpha D)}$. Choose $\alpha'$ from the remaining visits first, then $r$ with $2r\alpha' > K + \max(-\beta, \delta) + 1$, then its moment exponent $D$, and finally $0 < \alpha < \min(\alpha', \delta/(D+1))$. The reverse choices close the induction. End on an already visited axis; all widths in $Q$, including the head, are then at least one. The preceding upper and lower response bounds prove the claim.

□

Fix $a \neq b$, a line axis $l$, and $\lambda = e^{s}$, $s \in \mathbb{N}$. Set $c_a = c_b = -100$, $c_l = -150$ when $l \notin \{a, b\}$, and $c_j = -300$ for $j \notin \{a, b, l\}$. For fixed even nonnegative $h \in C_c^{\infty}(\mathbb{R})$ put

$$w_j = \lambda^{1+c_j}, \quad H_{\lambda}^{(d)}(p) = \prod_{j \neq d} w_j^{-1} h(p_j / w_j), \quad I_d(U; q) = \int_{E \leq 1} \frac{H_{\lambda}^{(d)}(p)}{E + \lambda^{q}} \, d\nu_d^{U} .$$

Set $B = 2^{-4} \prod_j (1 + \varepsilon_j \vartheta_j) U$ for a word combination $U$ with offsets on $\mathbb{R}e_l$, $\varepsilon_a = \varepsilon_b = -1$ and the other signs positive. Let $J_{ij} \in F_4$ be real, symmetric, $H_4$-tensorial and conserved: $\sum_j \partial_j J_{jb} = 0$. The index $J_{db}$ in a measure denotes the one-point word $J_{db}(0)$. No Ward normalization is assumed.

For the exterior estimate, consider tests $\Psi_{\lambda}(E, p)$ such that, for some fixed $K$ and every $L \geq 0$,

$$|\Psi_{\lambda}(E, p)| \leq C_L \lambda^{K} e^{-\lambda^{-c_d} E} \prod_{j \neq d} (1 + \lambda^{-c_j} |p_j|)^{-L} . \tag{17.10}$$

Write $X_{\lambda} = \{E > 1\} \cup \bigcup_{j \neq d} \{|p_j| > w_j\}$. The scaled Fourier–Laplace tests in Section 16.4, with fixed polynomial prefactors, have this bound. All constants below may depend on the fixed words, tensor, profiles and exponents.

These are the coordinate inputs to the rotation proof; their proof uses no result of Section 16.

**Lemma 17.3** (Coordinate spectral estimates)**.** *For every projected collinear $B$ and $\delta > 0$,*

$$I_d(B; -3000) = O(\lambda^{\kappa_d}), \qquad \kappa_d = \begin{cases} 1 & d = l, \\ -c_l & d \neq l, \end{cases}$$

$$I_d(J_{db}; c_d) = O_{\delta}(\lambda^{\delta}) \qquad (d \neq b),$$

$$\int_{E \leq 1} \frac{E W^4}{(E + W)^4} H_{\lambda}^{(d)} \, d\nu_d^{J_{dd}} = O_{\delta}(\lambda^{\delta} W^4), \qquad W = \lambda^{-99}.$$

*The pairings obey (17.4) and (17.6); one-insertion heat matrix elements are holomorphic for $\operatorname{Re} t > 0$. Moreover, for $N \geq 0$,*

$$I_d(J_{dd}; 0) = O(\lambda^{900}), \qquad \int_{X_{\lambda}} \Psi_{\lambda} \, d\nu_d^{UV} = O_N(\lambda^{-N}) .$$

We prove the sign bounds first, then the diagonal estimate by conservation.

*Proof.* In direction $l$, choose $2u$ separating the offsets and use $R_v(E) = \int e^{-E|t|} (1 - \chi(t)) \phi(vt) \, dt$, where $0 \leq \chi \leq 1$ is compactly supported and equals one on $[-2u, 2u]$, and $\phi = \widehat{\sigma * \sigma}_{+} \geq 0$, $\phi(0) = 1$. It satisfies $C/(E+v) \leq R_v(E) \leq C' v^{-1} e^{-2uE}$ for $E \leq 1$ and small $v$. For the lower bound integrate over $R < t < R + c/(E+v)$ beyond $\operatorname{supp} \chi$; for the upper bound drop $1 - \chi$ and scale. The removed profile has bounded supremum. Finite moments (17.5) permit the induction (17.9) through an opposite transverse sign. Taking $\delta = 1$ in this sign-transfer argument gives $O(\lambda)$ when $d = l$. For $d \neq l$, omit the $l$ profile, apply Lemma 17.2 on $Q = \{j : j \neq l\}$, and restore its supremum $O(\lambda^{-1-c_l})$. Both signs occur in $Q$ for the specified parity. Taking $\delta = 1$ gives $O(\lambda^{-c_l})$. The same lemma applied to the point word $J_{db}$, $d \neq b$, gives the second estimate.

It remains to treat the diagonal tensor. Write $C_{ab,cd}(x) = S(J_{ab}(0) J_{cd}(x))$, $x \neq 0$. Choose $D_{\mathrm{col}}$ bounding the collision powers of these pairs and their first derivatives. For even $\chi = 1$ near zero with compact support put $\Pi_{D_{\mathrm{col}}} f(t) = \chi(t) \sum_{r < D_{\mathrm{col}}} f^{(r)}(0) t^{r} / r!$, and define

$$R_{ab,cd}^{0}(f) = \int C_{ab,cd}(x) \left[ \prod_j f_j(x_j) - \prod_j \Pi_{D_{\mathrm{col}}} f_j(x_j) \right] dx.$$

Taylor vanishing gives integrability; a zero Taylor factor recovers the unsubtracted integral. Average under signed symmetries and pair exchange to obtain $R$. Conservation makes $\sum_i \partial_i R_{ib,cd}$ a jet at zero. Inversion $x \mapsto -x$ makes $R$ even, eliminating even divergence orders. To remove the linear orders, put $\chi_0(x) = \prod_j \chi(x_j)$ and $q_{ab,cd} = \sum_i R_{ib,cd}(\partial_i(x_a\chi_0))$, and replace the position extension by $R_{ab,cd} - q_{ab,cd}\delta_0$. Reflections kill coefficients unless indices occur in pairs; permuting them proves the pair symmetries. The subtraction preserves these and cancels linear divergence. The Fourier transform, still $R$, has divergence polynomials with no terms below degree three. Write $F_{b,cd} = \sum_i p_i R_{ib,cd}$ for the Fourier divergence polynomial. Symmetry gives

$$\sum_{j,k\neq d} p_j p_k R_{kd,jd} = \sum_{j\neq d} p_j (F_{d,jd} - p_d R_{dd,jd}) = p_d^2 R_{dd,dd} + \sum_{j\neq d} p_j F_{d,jd} - p_d F_{d,dd}\,.$$

Applying the same two contractions to the other index pair gives

$$\begin{aligned} \mathcal{Q}_0 &= R_{dd,dd}, \\ \mathcal{Q}_1 &= \sum_{j,k\neq d} p_j p_k R_{kd,jd} = p_d^2 \mathcal{Q}_0 + D_1, \\ \mathcal{Q}_2 &= \sum_{j,k,r,s\neq d} p_j p_k p_r p_s R_{sk,rj} = p_d^2 \mathcal{Q}_1 + D_2. \end{aligned} \tag{17.11}$$

Here multiplication by $p_j$ acts on the test, and $D_1, D_2$ are the polynomial remainders.

We also need profiles that do not annihilate the remainders. On a fixed temporal annulus, Lemma 17.2 and (17.4) bound each $R_{ab,cd}$ by $O_\delta(\lambda^\delta)$. For a diagonal entry, divide the annular profile by $p_d$ and move the derivative to the label, which then has both signs. Moment subtraction justifies this integration by parts at the collision. To remove the remaining zero-moment restrictions, expand one transverse profile into its residual and its fixed annular correction. For an odd profile $f$ supported in $[-r_0, r_0]$, use $p^{-1}\mathcal{C}_{m,\varrho}(pf)$ as correction; for $f_v(p) = v^{-1}f(p/v)$, splitting at $E = v$ gives $\int_{|p|\le r_0 v} |pf_v(p)|/(p^2+E^2)\,dp \le C/(E+v)$ from $|f(p)| \le C|p|$. The kernel factor $(1+E)^{-2m}$ makes the even lower bound dominate this residual, including zero. Equations (17.4) and (17.9) control it. Each correction fixes one more transverse annulus with bounded coefficients $\lambda^{2r\beta}$, $\beta \le 0$. Induction ends after at most three coordinates; finite Taylor terms are bounded on normalized shrinking profiles.

Since each transverse momentum costs at most $W$, the fixed-annulus bounds for $\mathcal{Q}_0, \mathcal{Q}_1, \mathcal{Q}_2$ are respectively $O_\delta(\lambda^\delta)$, $O_\delta(\lambda^\delta W^2)$ and $O_\delta(\lambda^\delta W^4)$. The errors cost $O(W^4)$ and $O(W^6)$: divergence has degree at least three, and the contractions add one and three powers. Choose one subtraction order $m$ above these degrees; its zero temporal moments annihilate the errors.

Put $g_{v,r}(p) = v^{-1}(p/v)^{-2r}\sigma(p/v)$, so $p^2 g_{v,r+1} = v^2 g_{v,r}$. Keep the transverse profile $H_\lambda^{(d)}$ fixed and set

$$c_r(t) = W^{-2r}\,\mathrm{Re}\,\mathcal{Q}_r(g_{tW,r}), \qquad d_r = W^{-2r}\,\mathrm{Re}\,\mathcal{Q}_r\big((1-\mathcal{C}_{m,8})(g_{W,r} - g_{2W,r})\big).$$

Conservation gives $c_{r+1}(t) = t^2 c_r(t) + O_\delta(\lambda^\delta W^2)$ for $t = 1, 2$, $r = 0, 1$. The correction of $g_{W,r} - g_{2W,r}$ has zero zeroth moment and other even moments $O(W^2)$, so the annular bounds give $d_r = c_r(1) - c_r(2) + O_\delta(\lambda^\delta W^2)$. Reflection positivity for $J_{dd}$ makes the spectral representation of $\mathcal{Q}_0$ positive. On corrected profiles it gives $d_0 \ge 0$, $d_1 \le 0$, $d_2 \ge 0$, since multiplication by $p_d^2$ becomes $-E^2$. Consequently $4d_0 = 5d_1 - d_2 + O_\delta(\lambda^\delta W^2)$ and $d_0 = O_\delta(\lambda^\delta W^2)$.

The response difference is nonnegative. Subtracting $(E^2 + 4W^2v^2)^{-1}$ from $(E^2 + W^2v^2)^{-1}$ in (17.7), with $1 \le |v| \le 2$, bounds it below by $c_m E W^2/(E+W)^4$ for $0 < E \le 1$; both vanish at zero. Multiply the $d_0$ bound by $W^2$.

On $E > 1$, (17.10) gives $C_N \lambda^{-N} e^{-t_0 E}$ for fixed admissible $t_0$. On $|p_j| > w_j$, the inequality $\lambda^{-c_j}|p_j| > \lambda$ gives arbitrary Schwartz gain. Integrate by (17.6) and use (17.4) for mixed terms. Finiteness of $\nu_d^{J_{dd}}(E \le 1)$ and $\|H_\lambda^{(d)}\|_\infty = O(\lambda^{897})$ give the crude bound. A positively buffered insertion has matrix elements $\langle v, A e^{-tH_d^{\mathrm{OS}}} u\rangle$ with bounded $A$, holomorphic for $\operatorname{Re} t > 0$ by spectral calculus. □

Rotations make spatial translations trivial at zero energy. Compression then removes slab width and placement.

**Lemma 17.4** (Compression algebra)**.** *The operators $M^0(w)$ are independent of admissible slab width and placement. They generate a separable commutative unital $C^*$-algebra $\mathfrak{A}_0$ on $\ker H_4^{\mathrm{OS}}$, with cyclic and separating vector $\Omega$. Its spectrum $\mathsf{X}$ and vacuum measure $\nu$ satisfy*

$$S^\xi(w) = \xi(M^0(w)), \qquad S(w) = \int_{\mathsf{X}} \xi(M^0(w))\, d\nu(\xi)\,.$$

*The space $\mathsf{X}$ is compact metric and $\nu$ is a full-support probability measure.*

*Proof.* Use Definition 6.1, Lemmas 17.1 and 16.11. For fixed words and large $z^4$, the correlation is $\langle v, U_{\vec z} E_{z^4 - c} v'\rangle$. Strong heat convergence makes fixed time shifts negligible, uniformly in $\vec z$.

Choose an orthogonal reflection $K$ taking $y + e_4$ to $|y + e_4| e_4$. Since $\langle K(y + e_4), K e_4\rangle = 1$, its ray $tKe_4$ has positive time component. Rotating the displacements $te_4 + y + e_4$ and $te_4$ makes their difference purely temporal, so their correlations become equal in the limit. Removing the extra $e_4$ as above gives $U_y P_0 = P_0$ by density.

The insertion identities $\mathsf{B}_{h+t}(w) = \mathsf{B}_h(w) E_t$ and $\mathsf{B}_{h+t}(\tau_{te_4} w) = E_t \mathsf{B}_h(w)$ give width and placement independence of $M^0(w) = P_0 \mathsf{B}_h(w)|_{\ker H_4^{\mathrm{OS}}}$, and $M^0(w)\Omega = P_0[w]$, $\langle \Omega, M^0(w)\Omega\rangle = S(w)$.

Successive use of (17.1) proves the cluster identity

$$\lim_{j\to\infty} S\big(\tau_{z_1(j)} w_1 \cdots \tau_{z_m(j)} w_m\big) = \langle \Omega, M^0(w_1) \cdots M^0(w_m)\Omega\rangle \tag{17.12}$$

whenever $(z_{i+1}(j) - z_i(j))^4 \to \infty$. In the induction, $\psi_j \to \psi$ and $t_j \to \infty$ imply $U_{y_j} E_{t_j} \psi_j \to P_0 \psi$ for arbitrary $y_j$, by contraction and spatial invariance on the ground space.

Place four words at 0, $j(e_4 + 2e_1)$, $j(2e_4 + e_1)$ and $3j(e_4 + e_1)$. Exchanging the first and fourth axes interchanges the two middle time orders. Covariance and (17.12) equate the resulting products after simultaneous rotation of their labels and points. A second application with all displacements along $e_4 + e_1$ removes that simultaneous rotation. The middle factors therefore commute on the dense span $P_0[w] = M^0(w)\Omega$. This also gives cyclicity; the empty word gives the unit. If $a\Omega = 0$, then $ab\Omega = 0$ on this dense span, so $a = 0$.

Basis labels and dense countable position sets generate $\mathfrak{A}_0$ by norm continuity. Gelfand theory and the vacuum state give its compact metric spectrum and probability measure [15, Theorems 2.4.1 and 2.8.2]. Separation gives full support: $\int |\xi(a)|^2 d\nu = \|a\Omega\|^2$ vanishes only for $a = 0$. □

A generic grid word has positions in $a_k \mathbb{Z}^4$ with pairwise distinct coordinates along each axis. For real coefficient representatives $(d_i, c_i)$ write $\langle\cdot\rangle_{N,p}$ for the Wilson expectation of Section 2, and put $\mathcal{O}_{N,p}(w) = \prod_i O_{N,p}^{(d_i)}(c_i, x_i/a_N)$. The word differences used below are

$$f_{j,N,p,T} = \mathcal{O}_{N,p}(\tau_{z_j(T)} w) - \mathcal{O}_{N,p}(\tau_{z_j(T)} \vartheta w), \qquad z_j(T) = T j a_k (1,1,1,1). \tag{17.13}$$

For each degree bound and lower grid level $m$, Lemma 14.27 chooses $k \geq m$ before $w$; these are the cofinal grids used below. For $\varepsilon > 0$ it supplies real $g_{j,N,p,T}$ bounded by $R_\varepsilon$ independently of $N, p, T, j$. For each $i < j$ and sufficiently large $T$, eventually in the ordered cutoffs,

$$|\langle g_i g_j \rangle_{N,p} - \langle f_i f_j \rangle_{N,p}| \leq \varepsilon \,.$$

The threshold may depend on $T, i, j$. Reality removes the real part from the cited lemma. Bounded $g_i$ control the diagonal variance terms, for which $f_i^2$ has no uniform bound.

Lattice variance gives characterwise reflection invariance; chessboard roots preserve the quotient fields.

**Lemma 17.5** (Phase fields and selection)**.** *For every $\xi \in \mathsf{X}$, its table satisfies E1–E7, retains the inclusions, derivatives, conjugation, rotations and stress laws of the mixture, and*

$$\mathcal{N}_d(S^\xi) = \mathcal{N}_d(S) \qquad (d \in \mathbb{N}) \,.$$

*Each separated value is continuous in $\xi$. For $\nu$-almost every $\xi$, the time-axis OS ground space is one-dimensional.*

*Proof.* We first prove the operator identities

$$M^0(\vartheta w) = M^0(w), \qquad M^0(hw) = M^0(w) \quad (h \in O(4)) \,. \tag{17.14}$$

Use Lemma 17.4, the point limits of Lemma 14.30 and (15.1). For a real grid word set $Z = M^0(w) - M^0(\vartheta w)$. Equation (17.1) gives $Z^* = -Z$, so $a = \langle \Omega, Z^2 \Omega \rangle = -\|Z\Omega\|^2 \leq 0$. Lattice variance gives

$$0 \leq \left\langle \left( \sum_{i=0}^{q} g_i \right)^2 \right\rangle \leq (q+1) R_\varepsilon^2 + 2 \sum_{i<j} \left( \mathrm{Re} \langle f_i f_j \rangle + \varepsilon \right) \,.$$

Take the ordered cutoffs, then $T \to \infty$. Expand each pair into four separated moments; their signed sum tends to $a$ by (17.12). Thus $0 \leq (q+1)[R_\varepsilon^2 + q(a+\varepsilon)]$. First $q \to \infty$, then $\varepsilon \downarrow 0$, gives $a \geq 0$. Separation of $\Omega$ gives $Z = 0$. Grid density, norm continuity and multilinearity give all words.

The cluster identity gives simultaneous orthogonal invariance of two-factor vacuum entries along rays with positive time before and after rotation. Such a ray exists unless time is reversed, when composition with $\vartheta$ suffices. Writing $v = M^0(u)\Omega$ and $x = M^0(w)\Omega$, simultaneous invariance and time-reflection invariance give $\langle v, x \rangle = \langle g^{-1} v, \vartheta g^{-1} x \rangle = \langle v, g \vartheta g^{-1} x \rangle$; transformations on vectors act on their words. Density gives every hyperplane reflection. Successively fixing basis images expresses an orthogonal map as at most four reflections, proving the second identity.

Use also Definitions 4.9, 6.1, FC and ST. Testing on positive words gives the permutation, unit and linearity identities and annihilates mixed null labels. Thus $\mathcal{N}_d(S) \subset \mathcal{N}_d(S^\xi)$. Character evaluation gives reality, covariance and smooth phase values, continuous in $\xi$.

For positive words $w_i$ in a common slab set $Y_i = \mathsf{B}_h(w_i)|_{\ker H_4^{\mathrm{OS}}}$. Direct evaluation of the OS form gives

$$M^0(\theta w_i w_j) = Y_i^* Y_j, \qquad \sum_{i,j} \bar{c}_i c_j M^0(\theta w_i w_j) = \left( \sum_i c_i Y_i \right)^* \left( \sum_j c_j Y_j \right) \geq 0 \,.$$

Spectral permanence and character evaluation preserve positivity [15, Theorems 2.5.1(1) and 2.6.4]; coordinate rotations give every axis. Complete this continuous positive Gram kernel. Integrating

its vectors against compact positive-time joint tests expresses the smeared OS form as a norm square.

Inclusions and derivatives pass through $T \mapsto \xi(P_0 T|_{\ker H_4^{\mathrm{OS}}})$. Put a stress shell and its insertions in one buffered slab. Its weights vanish near collisions, so the operator integral exists. Testing against exterior positive words gives ST; character evaluation gives both phase shell laws. The bound $|S^\xi(w)| \le \|\mathsf{B}_h(w)\|$, also for derivative labels, gives E7 and separated-support integrability.

For the reverse inclusion use the rows of Lemma 14.20, corrected lifts of Lemma 14.24 and coefficient limits of Lemma 15.4. Row $\ell$ consists of a singleton $u$ and optional spectator $v$. At the comparison positions $r_\ell y^{\ell,i}$ set $s = \mathcal{G}(u)$, $s' = \mathcal{G}(v)$ and $c = \mathcal{G}(uv) - ss'$, where $\mathcal{G}$ denotes the corresponding free correlation. In $\mathfrak{A}_0$ put

$$\Delta_1 = M^0(u) - s1, \quad \Delta_1' = M^0(v) - s'1, \quad \Delta_2 = M^0(uv) - sM^0(v) - s'M^0(u) + (ss' - c)1\,.$$

Apply (15.4) to $2j$ alternating conjugated copies of each polynomial in (14.33), translated by multiples of $a_{k_\ell}(1,1,1,1)$. Translating the periodic pattern and reality preserve the centered norm $\|\cdot\|_\mathcal{T}$ under signed images and conjugation. Large translations put the copies in distinct cells, with bound $(\gamma_\ell + \varepsilon)^{2j}$. Expand, take ordered coefficient and point limits, then (17.12). Reflections identify the alternating copies with adjoints. Let $\varepsilon \downarrow 0$ to obtain

$$\int_{\mathsf{X}} |\xi(\Delta)|^{2j}\, d\nu(\xi) \le \gamma_\ell^{2j} \qquad (j \in \mathbb{N})\,. \tag{17.15}$$

A value $|\xi_0(\Delta)| > b > \gamma_\ell$ would give an open set $O$ of positive measure with $\nu(O) b^{2j} \le \gamma_\ell^{2j}$ for all $j$, contradicting (17.15). Thus $|\xi(\Delta_1)|, |\xi(\Delta_1')|, |\xi(\Delta_2)| \le \gamma_\ell$ in every phase. Expanding $\Delta_2$ gives $|\xi(M^0(uv)) - ss' - c| \le \gamma_\ell(1 + |s| + |s'|)$.

Phase singleton and pair evaluations of the corrected lifts converge to the injective free row map. Its positive minimum on the unit sphere makes these maps eventually injective. They factor through $V_d/\mathcal{N}_d(S^\xi)$. The first null inclusion and (15.11) bound its dimension above by $\dim M_d^{\mathrm{Max}}$; injectivity bounds it below. Thus the null spaces coincide.

Reflection positivity and E7 construct the phase heat semigroup as in Lemma 17.1. Write $[b]_\xi$ for a positive-word class and $P_{0,\xi}^{\mathrm{OS}}$ for its ground projection. Strong heat convergence and the mixture identity give, by dominated convergence,

$$\int_{\mathsf{X}} \|P_{0,\xi}^{\mathrm{OS}}[b]_\xi\|^2\, d\nu(\xi) = \|P_0[b]\|^2 = \int_{\mathsf{X}} |S^\xi(b)|^2\, d\nu(\xi)\,.$$

The heat entries decrease from their zero-time bounds. The integrands differ by $\|P_{0,\xi}^{\mathrm{OS}}[b]_\xi - S^\xi(b)\Omega_\xi^{\mathrm{OS}}\|^2$. It vanishes almost everywhere. Rational configurations and basis coefficients give a countable dense family in every phase OS space, by Gram continuity. Outside the union of its null sets the ground range is $\mathbb{C}\Omega_\xi^{\mathrm{OS}}$. □

Choose $\xi_*$ from the full-measure set in Lemma 17.5. Only its ground-space dimension enters this choice.

*Completion of Lemma 17.1.* In the phase OS space put $v_j = (A_{w^\dagger} A_w)^j v$. The mixed-space map $Y(v_j) = \mathsf{B}(v_j)|_{\ker H_4^{\mathrm{OS}}}$ equals $(\mathsf{B}_h(w)^* \mathsf{B}_h(w))^j Y(v)$. Hence

$$\|[v_j]_\xi\|^2 = \xi\big(Y(v_j)^* Y(v_j)\big) \le \|Y(v)\|^2 \|\mathsf{B}_h(w)\|^{4j}\,.$$

The same power argument gives $\mathsf{B}_h^\xi(w)$ of norm at most $\|\mathsf{B}_h(w)\|$. The dense-vector argument gives its identities and norm derivatives, with the original constants. □

*Proof of Theorem 6.2.* Lemma 17.4 gives the spectrum, full-support probability measure and mixture formula. Lemma 17.5, using Lemma 14.27, gives E1–E7, equal null spaces, the common FC and ST identities and phase continuity. This proves Theorem 6.2. □

## 17.2 Reconstruction

Use the response labels $\mathcal{L}$ of Definition 4.9 and the star

$$(\iota_d A)^* = \iota_d\big(R_d(\mathsf{s}_4^{\mathrm{refl}})\bar{A}\big)\,. \tag{17.16}$$

The boost generators $-i\mathsf{L}_d(K_{4i})$ and spatial rotations integrate on $SL(2,\mathbb{C})$, the simply connected cover of $SO^+(1,3)$. A full spatial turn acts trivially, giving descent to $SO^+(1,3)$. Inclusions intertwine the actions; the time-reflection sign of $K_{4i}$ and antilinearity make (17.16) commute with boosts.

For $\mathcal{P} = SO^+(1,3) \ltimes \mathbb{R}^4$ take $\kappa_{\mathrm{P}} = \mathrm{id}$ and $\rho_{\mathrm{lab}}(\Lambda, z)$ the continued label action of $\Lambda$. In the field-word algebra, multiply label tensors and joint Schwartz tests by concatenation and tensor product. Its antilinear star reverses order and applies label and test conjugation. Allow finite sequences.

Fix a phase and write $\mathcal{K}_\xi = \mathcal{H}^{\mathrm{OS}}_{4,\xi}$, $E^\xi_t = e^{-tH^{\mathrm{OS}}_\xi}$ and $U^\xi_y$ for spatial translation. The next four proofs follow the statements: continuation, stress, then comparison and low energy.

Positive-height operators give positive boundary distributions without a Euclidean extension across collisions.

**Lemma 17.6** (Fields and boundary distributions)**.** *For every $\xi \in \mathsf{X}$, there is a faithful theory on $\mathcal{L}$ satisfying W0–W3 and Definitions 6.13–6.14. Its word-domain vacuum expectations extend to joint Schwartz distributions $W^\xi_n(A;\psi)$, and*

$$W^\xi(F^*F) \geq 0$$

*for every finite Schwartz test sequence $F$ in the field-word algebra. The adjoint and Lorentz label maps are the common maps specified above.*

Surjectivity transfers vacuum uniqueness between the two Hilbert spaces.

**Lemma 17.7** (OS comparison and irreducibility)**.** *For every $\xi$, the bounded OS comparison map $\mathsf{J}^{\mathrm{OS}}_\xi : \mathcal{H}_\xi \to \mathcal{K}_\xi$ is surjective, and both zero-energy spaces are vacuum lines. A bounded operator $B$ is scalar if and only if, for every smeared field $\Phi$ and $u, v \in \mathcal{D}_\xi$, one has $u \in \operatorname{Dom}\Phi^*$ and*

$$\langle \Phi^* u, Bv\rangle = \langle u, B\Phi v\rangle\,.$$

Locality controls low energies uniformly in the phase; full support then extends vacuum uniqueness to every character.

**Lemma 17.8** (Uniform low-energy bounds)**.** *For every fixed finite complex combination $A$ of field words smeared with compactly supported smooth tests, there are $M_A, M'_A \geq 0$, depending on its words, labels and tests, such that for every $\xi \in \mathsf{X}$,*

$$\begin{aligned} \mu_{A\Omega_\xi}(\{0 < \mathbf{p}^4 \leq u\}) &\leq M_A u^2 && (0 < u \leq 1),\\ 0 \leq \langle A\Omega_\xi, (e^{-sH_\xi} - P_{0,\xi})A\Omega_\xi\rangle &\leq M'_A s^{-2} && (s \geq 1). \end{aligned}$$

*Here $P_{0,\xi}$ projects onto $\mathbb{C}\Omega_\xi$.*

For the charge law, set $\eta = \operatorname{diag}(-1,-1,-1,1)$, $\zeta_i = 1$ for $i \le 3$, $\zeta_4 = -i$, and

$$T^{\mathrm{M}}_{ab} = -\zeta_a \zeta_b T_{ab}, \qquad T^{\mathrm{M}}_{ab}(f) = \Phi_\xi(\iota_4 T^{\mathrm{M}}_{ab}, f)\,.$$

Let $K(x) = k + Vx$, $V^{\mathsf{T}}\eta + \eta V = 0$, and let $\Sigma_K$ be its infinitesimal Lorentz label action. For real $f \in C^\infty_c(\mathbb{R})$ of integral one and $c \in C^\infty_c(\mathbb{R}^3)$, put $Q_K(f,c) = \sum_b T^{\mathrm{M}}_{4b}(K^b(c \otimes f))$. Suppose $m \in C^\infty_c(\mathbb{R}^4)$ and $c = 1$ on a ball such that all $(y,s)$ over its complement, $s \in \operatorname{supp} f$, are spacelike to $\operatorname{supp} m$. Then the charge law is

$$i[Q_K(f,c), \Phi_\xi(\iota_d A, m)] = \Phi_\xi(\Sigma_K \iota_d A, m) - \sum_a \Phi_\xi(\iota_d A, \partial_a(K^a m))\,. \tag{17.17}$$

A common polynomial bound permits boundary limits under the phase integral and continuation of local stress identities.

**Lemma 17.9** (Phase dependence and continued stress)**.** *For every $n$, label list $A$ and $\psi \in \mathcal{S}_n$, the map $\xi \mapsto W^\xi_n(A;\psi)$ is continuous and*

$$\int_{\mathsf{X}} W^\xi_n(A;\psi)\, d\nu(\xi) = W^{\mathrm{mix}}_n(A;\psi)\,,$$

*where $W^{\mathrm{mix}}_n$ is the boundary distribution of the mixed continuation. In every phase $T$ continues to a symmetric Hermitian conserved Lorentz tensor of operator-valued distributions, with the local charge law (17.17) under its stated support hypothesis. The selected theory and continuation are those at $\xi_*$.*

*Continuation and boundary distributions in Lemmas 17.6 and 17.9.* Use Definitions 6.3–6.14, Lemmas 17.1–17.5 and FC. Vacuum uniqueness follows in the comparison proof. For a label $a$ put $B_{\xi,\epsilon}(a) = \mathsf{B}^\xi_{2\epsilon}(a(\epsilon e_4))$. Lemma 17.1 gives $\|B_{\xi,\epsilon}(a)\| \le C_a \epsilon^{-q_a}$, $0 < \epsilon < 1$, uniformly in $\xi$. For $\operatorname{Im} z > 2\epsilon$ set $G_{\xi,\epsilon}(z,y) = e^{-(\operatorname{Im} z - 2\epsilon)H^{\mathrm{OS}}_\xi} e^{i \operatorname{Re} z H^{\mathrm{OS}}_\xi} U^\xi_y$. For $n \ge 1$ set

$$\mathsf{F}^\xi_n(a;z,\vec{x}) = \langle \Omega^{\mathrm{OS}}_\xi, B_{\xi,\epsilon}(a_1) \prod_{j=2}^{n} [G_{\xi,\epsilon}(z_j - z_{j-1}, \vec{x}_j - \vec{x}_{j-1}) B_{\xi,\epsilon}(a_j)] \Omega^{\mathrm{OS}}_\xi \rangle\,. \tag{17.18}$$

Choose $0 < \epsilon < 1$ with $4\epsilon < \min_j \operatorname{Im}(z_{j+1} - z_j)$; set $\mathsf{F}^\xi_0 = 1$. At imaginary times the insertion rule gives $S^\xi_n$ independently of $\epsilon$; the identity theorem extends this agreement. It is holomorphic in time and continuous in space; extend by zero outside ordered times. Taking $\epsilon = \delta/8$ gives

$$|\mathsf{F}^\xi_n(a; x^4 + iy, \vec{x})| \le C\delta^{-m} \quad (0 < \delta \le 1,\ y_{j+1} - y_j \ge \delta)\,, \tag{17.19}$$

uniformly in $x$ and $\xi$.

To construct the boundary, write $\psi_\xi(y;f) = \int \mathsf{F}^\xi_n(a; x^4 + iy, \vec{x}) f(x)\, dx$, let $\rho_j = j$, and put $\mathcal{A} = -i \sum_j \rho_j \partial_{x^4_j}$. By (17.19) and holomorphy, integration by parts gives $\partial_t \psi_\xi(y + t\rho; f) = \psi_\xi(y + t\rho; \mathcal{A} f)$. Repeated integration of this identity gives

$$\psi_\xi(y;f) = \sum_{k=0}^{m} \frac{(-1)^k}{k!} \psi_\xi(y + \rho; \mathcal{A}^k f) - \int_0^1 \frac{(-t)^m}{m!} \psi_\xi(y + t\rho; \mathcal{A}^{m+1} f)\, dt\,. \tag{17.20}$$

The integrand is bounded by $C\|\mathcal{A}^{m+1} f\|_{L^1}/m!$, uniformly in $y, \xi$. At $y = 0$ this defines $W^\xi_n$. Dominated convergence gives every ordered boundary limit and a common Schwartz seminorm bound.

The mixed operators of Lemma 17.1 define $\mathsf{F}^{\mathrm{mix}}$ by (17.18) and $W^{\mathrm{mix}}$ by (17.20), with the same bound. The mixture formula at imaginary times extends to strict times by the identity theorem and integration of Cauchy's formula. For continuity, set $z_j = iy_j + \varsigma x_j^4$. The common strip bound gives a normal family in $\varsigma$. If $\xi_\ell \to \xi$, each subsequential limit agrees on the imaginary segment by Lemma 17.5, hence everywhere. Fubini and the common integrable bound in (17.20) give the distributional continuity and barycenter.

Display the $j$th label in $\psi_\xi(y; f; \dots, a_j, \dots)$, keeping the others fixed. Differentiating Euclidean $(4, i)$ covariance and using the time Cauchy–Riemann identity gives

$$\sum_j \psi_\xi(y; f; \dots, -i\mathsf{L}(K_{4i})a_j, \dots) = \psi_\xi\left(y; \sum_j [x_j^i \partial_{x_j^4} + (x_j^4 + iy_j)\partial_{x_j^i}]f\right).$$

It holds first at imaginary times, then by holomorphy. At the boundary (17.20) removes $y$. Along a real boost the derivative of simultaneously transformed labels and tests is therefore zero. Spatial rotations and translations retain their Euclidean identities, proving Poincaré covariance.

Define $\widetilde{W}_n^\xi(b) = W_n^\xi(\widehat{b}_+(-\cdot))$. For compactly supported smooth frequency tests $b$, differentiating the height and the exponential multiplier gives cancellation, hence

$$\widetilde{W}_n^\xi(b) = \psi_\xi\big(y; \widehat{e^{\sum_j y_j p_j^4} b_+(-\cdot)}\big). \tag{17.21}$$

The boundary formula includes the zero endpoint. Sending ordered heights to infinity kills negative time-tail momenta. Translation and Lorentz covariance therefore give support in

$$\mathcal{V}_n = \left\{ p : \sum_j p_j = 0, \quad \sum_{k=j}^n p_k \in \overline{V}_+ \ (j \geq 2) \right\}.$$

Let $\chi = \mathbf{1}_{\{\mathrm{dist}(p,\mathcal{V}_n)<2\}} * \rho$, where $\rho$ is a nonnegative smooth unit-integral function supported in the unit ball. Then $\chi = 1$ near $\mathcal{V}_n$, has bounded derivatives, and vanishes at distance greater than three. For consecutive imaginary differences in the open forward cone define

$$\mathcal{W}_n^\xi(z) = (2\pi)^{-4n} \widetilde{W}_n^\xi\left(\chi(p) e^{i\sum_j p_j \cdot z_j}\right).$$

Put $q_j = \sum_{k=j}^n p_k$. Then $\sum_j p_j \cdot \mathrm{Im}\, z_j = \sum_{j\geq 2} q_j \cdot \mathrm{Im}(z_j - z_{j-1}) \geq c|p|$ on $\mathcal{V}_n$. On $\mathrm{supp}\,\chi$ this holds up to a bounded subtraction, giving a holomorphic Schwartz test. Equation (17.21) identifies this with (17.18). On the closed cone the damping multipliers act continuously on Schwartz tests. Thus joint tests admit continuous partial boundaries as any consecutive imaginary differences vanish. Set the arity-zero value to one.

Tie the heights of two adjacent spacelike points, retaining strict height differences elsewhere. For their difference $(v, t)$, $|t| < |v|$, an imaginary boost along $v/|v|$ gives imaginary difference $\sin b\,(t\, v/|v|, |v|)$ in the forward cone for small $b > 0$; all other differences stay in the cone. The opposite boost treats the exchanged order. Let $\rho_n^\vee$ act on label tensors, so $\mathcal{W}(\Lambda z) = \rho_n^\vee(\Lambda)\mathcal{W}(z)$. Define the two germs by

$$G_\beta(z) = \rho_n^\vee(\Lambda_{i\beta})^{-1}\mathcal{W}(\Lambda_{i\beta} z), \qquad N_\beta = \{z : \Lambda_{i\beta} z \text{ lies in that order's tube}\}.$$

Choose the sign of $\beta$ for that order. The cone inequalities make $N_\beta$ a neighbourhood of the tied point. Real covariance and the boost identity theorem identify $G_\beta$ with the tube function

on overlap. This works along the path scaling real times to zero: spacelikeness and the other strict height margins persist. At zero real times, Euclidean permutation symmetry identifies every derivative on the totally real Wick slice, hence the germs. The overlap identities carry this equality through a finite chain covering the compact path. Letting the tied heights vanish gives equality on Schwartz tests supported in the spacelike set, proving locality.

Apply the propagators and insertions in (17.18) to the vacuum, with an initial heat interval, and smear against a word test $b$. Reflection and the insertion rule identify mixed-height Gram entries with $\mathsf{F}^{\xi}(b^{*}c)$. Their limits as both height lists vanish are $W^{\xi}(b^{*}c)$ by (17.20). The vectors are therefore Cauchy; their limits $v_{\xi}(b) \in \mathcal{K}_{\xi}$ have this Gram form. The common seminorm bound and product-test density extend it to joint tests. Hence $W^{\xi}(F^{*}F) = \| \sum_{b} c_{b} v_{\xi}(b) \|^{2} \geq 0$.

Complete the word form modulo null vectors to obtain $\mathcal{H}_{\xi}$, with $\Phi_{\xi}(a,f)[b] = [(a,f)b]$. For null $x$, the inserted vector pairs with $c$ as $\langle x, (a^{*}, \bar{f})c \rangle = 0$. Taking $c$ to be that vector preserves nullity and gives the adjoint law. Test continuity gives strongly continuous covariance unitaries. Translations integrate to a nondegenerate representation of $C^{*}(\mathbb{R}^{4}) = C_{0}(\mathbb{R}^{4})$ [15, Theorem 3.7.11 and equation (3.89)]; its spectral measure comes from [15, Proposition 4.2.3]. The Fourier support puts its spectrum in $\overline{V}_{+}$. Locality passes to the word domain. A zero-acting label has zero Euclidean correlations by (17.21), symmetry and continuity; Lemma 17.5 and response injectivity give faithfulness. Only vacuum uniqueness remains. □

*Stress assertion of Lemma 17.9.* Use ST, Lemma 17.5 and the continuation. The factors give $T^{\mathrm{M}}_{44} = T_{44}$, $T^{\mathrm{M}}_{4i} = iT_{4i}$, $T^{\mathrm{M}}_{ij} = -T_{ij}$. Euclidean symmetry and (17.16) give symmetry and Hermiticity; differentiation and exponentiation of covariance give the Lorentz action. Continuing the derivative identity gives $\Phi_{\xi}(\iota_{d+1}\partial_{a}A, f) = -\zeta_{a}^{-1}\Phi_{\xi}(\iota_{d}A, \partial_{a}f)$. Hence Euclidean conservation gives $\sum_{a} \eta_{aa} T^{\mathrm{M}}_{ab}(\partial_{a}f) = 0$ on the common word domain.

For real $c$, insert $c(y)h(t-t_{0})$ into ST and integrate conservation on each side of $t_{0}$. The $\nabla c$ term stays away from the distinguished point and remains bounded, so $\sum_{b} \int c(y) K^{b}_{\mathrm{E}}(y,t) T_{4b}(y,t)\, dy$ is Cauchy on each side. Its jump is $\Sigma_{K}A + \sum_{b} K^{b}_{\mathrm{E}} \partial_{b} A$, where $K^{b}_{\mathrm{E}}(y) = \zeta_{b} K^{b}(\vec{y}, iy^{4})$.

Continue this jump as follows. Fix strictly ordered heights $y_{j}$ for all fields other than the current, including the distinguished field with index 0. For their real times $t_{j}$ put $z_{j}(\varsigma) = iy_{j} + \varsigma t_{j}$, and place the current at $z_{0}(\varsigma) \pm i\varepsilon$. The center follows the distinguished time, retaining Euclidean separation $\varepsilon > 0$. Write

$$I^{\pm}_{\varepsilon}(\varsigma) = \sum_{b} \int c(x) K^{b}_{\mathrm{E}}(x, -iz_{0}(\varsigma) \pm \varepsilon)\, \mathsf{F}^{\xi}_{\pm}(\dots, T_{4b}(x, z_{0}(\varsigma) \pm i\varepsilon), \dots)\, dx,$$

where $\mathsf{F}_{\pm}$ inserts the current immediately after or before the distinguished field. Choose a common complex neighbourhood $\mathcal{U}$ of $[0,1]$ preserving the other strict height differences. On $\operatorname{supp} \nabla c$ the spacelike germs bound the side integrals, uniformly for small $\varepsilon$ and $\varsigma$ in compact $L \subset \mathcal{U}$. Conservation gives

$$\sup_{\varsigma \in L} |I^{\pm}_{\varepsilon}(\varsigma) - I^{\pm}_{\varepsilon'}(\varsigma)| \leq C_{L} |\varepsilon - \varepsilon'|.$$

The limits are holomorphic. At small imaginary $\varsigma$ the Euclidean calculation identifies their jump; the identity theorem extends it to $\varsigma = 1$. Test against $m, f$ and take the continuous partial boundaries. Integration by parts differentiates $K^{a}m$, giving (17.17). Schwartz cutoffs and the joint bounds extend from compact spectators. For complex $c$, its imaginary part has charge commuting with the distinguished field by locality. Constant and rotational $K$ give the required actions, retaining $\xi_{*}$. □

*Comparison and low energy: Lemmas 17.7 and 17.8.* Use Definitions 6.5–6.8 and 6.11, with Lemma 17.5.

The isometry $J = \mathsf{J}_\xi^{\mathrm{OS}} : [b] \mapsto v_\xi(b)$ takes vacuum to vacuum and intertwines spatial translations and heat: $Je^{-sH_\xi} = E_s^\xi J$ by (17.21). Density makes the boundary comparison unique. After surjectivity, strong heat convergence identifies ground projections.

For compact frequency $b$, let $f_y = e^{-\widehat{\sum_j y_j p_j^4}} b_+(-\cdot)$. Equation (17.21) equates the norms and mixed pairing of its boundary-word image under $J$ and the corresponding positive-height OS vector. They therefore agree. Frequency-test density and closed range include every smeared positive-height vector; approximate identities give point vectors. Permutation and perturbation of equal times give all positive words, proving surjectivity.

We prove the low-energy estimate for compactly supported words, subtracting the full zero-energy projection $Q_\xi = \mathsf{E}_\xi(\{\mathbf{p}^4 = 0\})$. Let $A^\dagger$ be its word adjoint, reversing order and applying label and test conjugation, and put $\alpha_z A = \mathsf{T}_\xi(z) A \mathsf{T}_\xi(z)^*$. Bound all support times and spatial norms by $r$. Set $R = 4r + 1$, $\mu = \mu_{A\Omega_\xi}$ and $\lambda = \mu_{A^\dagger\Omega_\xi}$. The common distribution bounds give $\mu(\mathbb{R}^4) + \lambda(\mathbb{R}^4) \le M$, uniformly in $\xi$. Covariance and locality imply

$$a(y,t) = \widehat{\mu}_+(y,t) - \widehat{\lambda}_+(-y,-t) = \langle \Omega_\xi, [A^\dagger, \alpha_{(y,t)} A]\Omega_\xi \rangle = 0 \quad (|y| > R,\ |t| \le 1)\,.$$

Here $\widehat{\mu}_+(z) = \int e^{ip\cdot z}\, d\mu(p)$, and likewise for $\lambda$. Write $E = p^4$ for energy. The time average $\bar{a}(y) = \int_0^1 [a(y,t) - a(y,-t)]/(2i)\, dt$ is bounded by $M$ and supported in $\overline{B}_R$. It is the Fourier transform of two positive pushforwards: $\pi(E)d\mu$ under $p \mapsto \vec{p}$ and $\pi(E)d\lambda$ under $p \mapsto -\vec{p}$, where $\pi(E) = (1 - \cos E)/E$, $\pi(0) = 0$. On $0 \le E \le 1$, $\pi(E) \ge E/\pi$.

Choose a smooth momentum bump $b_u$, equal to one on $B_u$, supported in $B_{2u}$ and between zero and one. Its transform $\widehat{b_{u+}}(-y)$ has absolute value at most $8|B_1|u^3$. Fourier inversion and positivity therefore give

$$(2\pi)^3 \frac{u}{2\pi}\, \mu(\{u/2 < E \le u\}) \le \left| \int \bar{a}(y) \widehat{b_{u+}}(-y)\, dy \right| \le 8|B_1||B_R| M u^3\,.$$

Here $|\vec{p}| \le E$ on the spectral cone makes $b_u = 1$ on the indicated shell. Summing shells $u2^{-j}$ gives

$$\mu(\{0 < E \le u\}) \le C R^3 M u^2 \qquad (0 < u \le 1)\,. \tag{17.22}$$

For $u > 1$ use the total mass bound. Since $e^{-sE} = \int_E^\infty s e^{-st}\, dt$, Tonelli gives

$$0 \le \langle A\Omega_\xi, (e^{-sH_\xi} - Q_\xi) A\Omega_\xi \rangle = \int_{E>0} e^{-sE}\, d\mu \le C_A s \int_0^\infty t^2 e^{-st}\, dt = 2C_A s^{-2}\,. \tag{17.23}$$

The heat pairing is continuous in $\xi$: (17.20) gives continuous Schwartz frequency cutoffs, with error $Me^{-sN}$ outside radius $2N$ on $\overline{V}_+$. Subtract $|\langle \Omega_\xi, A\Omega_\xi \rangle|^2$. Almost-everywhere uniqueness and full support extend (17.23) to this continuous pairing in every phase. Let $s \to \infty$ to obtain

$$\|Q_\xi A\Omega_\xi\|^2 = |\langle \Omega_\xi, A\Omega_\xi \rangle|^2\,.$$

Since $Q_\xi \Omega_\xi = \Omega_\xi$, density of compact word vectors gives $Q_\xi = P_{0,\xi}$. Surjectivity and heat intertwining give the OS vacuum line, completing W0.

For irreducibility, let $B$ satisfy the weak relation in Lemma 17.7. The field adjoint law and iteration move words across matrix entries without requiring $B$ to preserve the domain. For a word $A$, put $u = A\Omega_\xi$, $v = A^\dagger \Omega_\xi$ and, in the upper half-plane,

$$p(z) = \langle u, e^{izH_\xi} B\Omega_\xi \rangle, \qquad q(z) = \langle B^*\Omega_\xi, e^{izH_\xi} v \rangle\,.$$

These functions are bounded and continuous to the boundary, where covariance and the weak relation give $p(t) = q(-t)$. Morera and Liouville make their bounded pasted continuation constant. Hence $\langle u, e^{-sH_\xi} B\Omega_\xi\rangle = \langle u, B\Omega_\xi\rangle$. Heat convergence gives $B\Omega_\xi = b\Omega_\xi$. Moving words again gives $\langle A'\Omega_\xi, BA\Omega_\xi\rangle = b\langle A'\Omega_\xi, A\Omega_\xi\rangle$, so $B = b1$ by density. A scalar satisfies the relation by the adjoint law. This proves the equivalence in Lemma 17.7. □

*Proof of Theorem 6.15.* Lemmas 17.6 and 17.7 prove the faithful W0–W3 theories, Definitions 6.13–6.14, the onto comparison, vacuum lines and scalar weak commutant required by Theorem 6.15. Lemmas 17.8 and 17.9 supply the uniform rates, phase averages and continued stress, retaining $\xi_*$. □

# 18 The trace law and non-free comparisons

The stress tensor constructed in Section 16 has a trace whose leading free comparison vanishes. To determine its first nonzero term one must retain the next coefficient. We center the trace in the phase fixed in Section 6, identify its positive spectral measure, and prove Theorem 7.2 by a fourth-order comparison with the free field. The coefficient combines the one-loop trace coefficient and the free curvature-square correlation. The spectral law distinguishes this field from the finite comparisons of Proposition 7.4 and the polynomial and photon comparisons in Theorem B.

Conservation determines the scalar curvature-square coefficient from the divergence of a lattice stress insertion. A two-dimensional linear equation removes the other tensor components; the Ward identity fixes the remaining normalization. An estimate on the divergence must be more accurate than the slowly growing coefficients of this equation. Adjustable stopping levels give a subsequence of every prescribed null sequence; positivity then removes lattice rounding and gives the full-radius limit.

Finite mass-shell products have expansions with no negative logarithmic powers, which contradict the trace law. In photon Fock space, relative locality forces polynomial identities under momentum doubling. The trace law permits only sectors with at most three particles; their spectral growth is then too small.

Throughout, retain the fields, inclusions and units of Definition 4.9 and Proposition 4.11, the tensor of Theorem 5.4, and the selected reconstruction of Theorem 6.15. The symbols $\tau, e_{\mathrm{tr}}, \Theta, G_\Theta$ and $A_G$ have the meanings of Definition 7.1. Write $\Phi_\Theta(f) = \Phi(\iota_4\Theta, f)$, and use the word domain of Definition 6.6.

## 18.1 Trace and spectrum

Centering must hold on the common operator domain, as well as in separated correlations. The proof below identifies the subtraction after continuation.

**Lemma 18.1** (Centering as an operator identity)**.** *For every $f \in \mathcal{S}$, on the word domain,*

$$\Phi_\Theta(f) = \Phi(\iota_4\tau, f) - e_{\mathrm{tr}}\left(\int f\right)\mathrm{Id}_{\mathcal{H}}, \qquad \langle \Omega, \Phi_\Theta(f)\Omega\rangle = 0\,.$$

*Proof.* Use unit deletion in Definition 4.4, the quotient calculus of Lemma 15.4, and the continuation and field realization of Lemma 17.6. Deleting the unit insertion leaves a Euclidean correlation independent of its position. On each ordered-time region its continuation is therefore constant in that variable. Uniqueness in Lemma 17.6 identifies it with the correlation containing the unit. Boundary values and smearing give $\langle u, \Phi(\iota_4\mathbf{1}_4, f)v\rangle = (\int f)\langle u, v\rangle$ for word vectors $u, v$. Density

and linearity give the identity. The zero Euclidean one-point function continues to zero and has zero boundary distribution, proving the centering assertion. □

We use the Fourier convention of Section 7:

$$\widehat{f}_+(\mathbf{p}) = \int e^{i\mathbf{p}\cdot x} f(x)\,dx, \qquad (\mathcal{W}_{\mathrm{rel}}\psi)(\mathbf{p}) = \int e^{i\mathbf{p}\cdot(y-x)}\psi(x,y)\,dx\,dy\,.$$

In particular, $\mathcal{W}_{\mathrm{rel}}(\bar f \otimes h) = \overline{\widehat{f}_+}\widehat{h}_+$. Write $W_2^{\xi*}(\Theta,\Theta;\bar f\otimes h) = \langle \Phi_\Theta(f)\Omega, \Phi_\Theta(h)\Omega\rangle$ for the joint tempered boundary distribution of Lemma 17.6. The next lemma constructs its positive measure.

**Lemma 18.2** (Unique trace spectral measure)**.** *There is a unique positive tempered-growth Borel measure $\mu_\Theta$, supported in $\overline{V}_+$, such that*

$$d\mu_{\Phi_\Theta(f)\Omega} = |\widehat{f}_+|^2\,d\mu_\Theta \qquad (f\in\mathcal{S})\,.$$

*It is Lorentz invariant in momentum variables, and*

$$W_2^{\xi*}(\Theta,\Theta;\psi) = \int \mathcal{W}_{\mathrm{rel}}\psi\,d\mu_\Theta \qquad (\psi\in\mathcal{S}_2),$$
$$G_\Theta(r) = \int e^{-r\mathbf{p}^4}\,d\mu_\Theta \qquad (r>0)\,.$$

*Proof.* Use the positive boundary distributions of Lemma 17.6. For $v_f = \Phi_\Theta(f)\Omega$, translation covariance gives $\widehat{h}_+(P)v_f = \widehat{f}_+(P)v_h$. Here $\widehat{h}_+(P) = \int \widehat{h}_+(\mathbf{p})\,d\mathsf{E}(\mathbf{p})$, with $\mathsf{E}$ the joint translation spectral resolution of Theorem 6.15. Both sides smear the field with $f*h$. Choose tests whose transforms do not vanish on successively larger balls. On such a ball define $d\mu_\Theta = |\widehat{f}_+|^{-2}d\mu_{v_f}$. The displayed identity makes these definitions agree on overlaps. Polarization gives $\langle v_f, v_h\rangle = \int \overline{\widehat{f}_+}\widehat{h}_+\,d\mu_\Theta$. Product tests determine the joint distribution.

The spectrum condition gives cone support. Schwartz continuity, applied to inverse transforms of cutoffs equal to one on $\{|\mathbf{p}|\le R\}$, gives $\mu_\Theta(\{|\mathbf{p}|\le R\}) \le C(1+R)^k$. Lemma 17.9 continues $T$ to a symmetric Hermitian Lorentz tensor. Contraction and invariance of the unit (Lemma 16.10) make $\iota_4\Theta$ a Hermitian Lorentz scalar. Covariance of $W_2$ and uniqueness of the measure give Lorentz invariance.

Tempered growth and $|\mathbf{p}| \le \sqrt{2}\mathbf{p}^4$ on the cone make the Fourier–Laplace integral holomorphic at positive imaginary time, with boundary $W_2$. To identify it with the Euclidean continuation of Lemma 17.6, take enough primitives to obtain continuous boundary values, using its polynomial boundary bound. Their zero difference extends by Schwarz reflection [26, Theorem 11.14]; holomorphic uniqueness and differentiation give equality and the Laplace formula. In particular $G_\Theta$ is real, nonnegative and nonincreasing. □

**Comparison at prescribed radii.** Use the scaling frames of Definition 5.1: $U_i(d) : M_d^{\mathrm{Max}} \to F_d$. Insert $U_i(d)A$ at $r_i x$, and read tensors by $r_i^4 U_i(4)^{-1}$. Write $x^{(i)}$ for nearest points to $x$ in $r_i^{-1}a_{n_i}\mathbb{Z}^4$. Jets use positional derivatives of smooth interpolants and their normalized lattice differences, including derivatives of the inserted fields. A shell flux is the integral against $-\partial_a\chi$ in Definition 5.3. Define $\beta_i = \lim_{(N,p)}\beta_{N,p,k_i}$ along the fixed coefficient refinement; the radius level $n_i$ precedes the insertion level $k_i$.

The next lemma extends Lemma 16.6 to prescribed radii. Its proof also transfers the derivative argument of Lemma 16.12 to these frames.

**Lemma 18.3** (Prescribed radii)**.** *Every positive null sequence has a subsequence $r_i$ with frames preserving reality, units, inclusions and $H_4$, and $n_i, k_i \to \infty$, such that*

$$\beta_i \to \infty, \qquad \frac{\log(1/r_i)}{\beta_i} \to \kappa_G^{-1}\,. \tag{18.1}$$

*For degrees $d_j \le D$, $A_j^{(i)} \to A_j$ and compact $K \subset \mathrm{Gen}_n$,*

$$\sup_{x\in K}\left|S_n(U_i(\boldsymbol{d})\boldsymbol{A}^{(i)}; r_i x^{(i)}) - S_n^{\mathrm{Max}}(\boldsymbol{A}; x)\right| \to 0\,.$$

*The same holds for each fixed jet and translation-shell flux. For $n = 2$ without derivatives it holds uniformly over the phases.*

*Proof.* Use Definition 3.9, Lemmas 14.8–14.25 and the finite-dimensional construction of Lemma 16.6. Put $M_n = M_n^{\mathrm{cut}}$. At radius level $n$ allow

$$r(n,u) = a_n M_n^u, \qquad u \in [\alpha_0, \alpha_0+1], \qquad k(n) = \min\{m > n : \log(a_n/a_m) \ge (\log\widehat{\beta}_n)^{48}\}\,. \tag{18.2}$$

The separation in $k(n)$ leaves room for all fixed-degree marked transports; its logarithmic size is negligible compared with $\widehat{\beta}_n$. For degree $D$ and arity $n_{\mathrm{ar}}$ choose

$$c < 0, \quad \alpha_1 + 6 < \alpha_0, \quad \alpha_2 + 5 < \alpha_0 + c, \quad (\alpha_0-\alpha_1)Dn_{\mathrm{ar}} + c < 0, \quad (\alpha_0-\alpha_2)Dn_{\mathrm{ar}} - \alpha_0 - c < 0\,.$$

Choose the differences $\alpha_0 - \alpha_1 > 6$, then $-c > (\alpha_0-\alpha_1)Dn_{\mathrm{ar}}$, then $\alpha_0 - \alpha_2 > 5 - c$, and finally increase all three $\alpha_j$ together. At scaled mesh $h_n = M_n^{-u}$, choose an intermediate displacement $\delta_n = M_n^c$. The marked moment bounds of Lemma 14.8, transported from exponent $\alpha_1 + u - \alpha_0$, cost at most $CM_n^{(\alpha_0-\alpha_1)D}$ per insertion. Lemmas 14.12–14.25 give the same bound on reflected products. Transport from exponent $\alpha_2 + u - \alpha_0$ bounds one positional difference, with cost $CM_n^{(\alpha_0-\alpha_2)D}$. Telescoping the $n_{\mathrm{ar}}$ insertions therefore bounds the two comparison errors by

$$CM_n^{(\alpha_0-\alpha_1)Dn_{\mathrm{ar}}}\delta_n, \qquad CM_n^{(\alpha_0-\alpha_2)Dn_{\mathrm{ar}}}h_n/\delta_n\,.$$

These are moment transport across $\delta_n$ and lattice interpolation across one mesh cell; both vanish uniformly in $u \in [\alpha_0, \alpha_0+1]$. The remaining reflected-product errors have bound $\varepsilon_n \to 0$ from Lemma 14.12. Applied to every repeated reflected product, these bounds control the norm of the corresponding zero-energy multiplication operator. Evaluation at any character of the phase algebra of Theorem 6.2 is bounded by that norm. This proves phase-uniformity for the two-point errors, including moving coefficients by finite-dimensional multilinearity.

We impose on the entry bounds of row 6 in Section 9 that both adjacent stage ratios obey $\ell_m \le \log\widehat{\beta}_m$. Thus $\ell_n \le M_{n+1}$ eventually. Monotonicity of $M_n$ gives $r(n,\alpha_0) \le r(n+1,\alpha_0+1)$, while $r(n,\alpha_0) \to 0$. First crossing of any prescribed small radius, followed by continuity in $u$, puts that radius in one of these intervals. Diagonal extraction over degrees, arities, compact sets and jet orders gives $n_i, k_i = k(n_i)$ and frames, using Lemma 14.7 for symmetries and inclusions. Fix the exponents before each extraction, and require $(\alpha_0+1)\log M_{n_i} \le (\log\widehat{\beta}_{n_i})^{48}$.

For jets apply the corrected differences of Lemma 14.19 and the same estimates at the higher degree. The difference quotient converges to the positional derivative by smoothness and the integral formula for a difference; this is the argument of Lemma 16.12 on the present frames. Integration on the fixed compact shell gives the flux limit. Minimality of $k(n)$ gives $\log(a_n/a_{k(n)}) = O((\log\widehat{\beta}_n)^{48}) = o(\widehat{\beta}_n)$. The running law of Theorem 3.6, in (3.14), gives $\beta_i/\widehat{\beta}_{k_i} \to 1$. Since $\widehat{\beta}_n = \bar{\beta} + \kappa_G \log(1/a_n)$, the imposed bound on $u \log M_n$ proves (18.1). □

**The fourth-order Ward coefficient.** Set $L = T^{\mathrm{Max}}$, $D_{ab} = \delta_{ab}\mathcal{O}_{F^2}$, and $\Xi_{aa} = 0$, $\Xi_{ab} = \sum_{c,\alpha} : F^{\alpha}_{ac}F^{\alpha}_{bc} :$ for $a \neq b$. Simplicity excludes invariant linear terms and fixes the colour bilinear form. Reflections and permutations then leave exactly these three centered real symmetric $H_4$-tensorial tensors. Recall $\mathcal{O}_{F^2} = \frac{1}{4}\sum_{a,b,\alpha} : F^{\alpha}_{ab}F^{\alpha}_{ab} :$.

Before imposing the Maxwell equation, divergence has three coordinates, represented in component $b$ by

$$\sum_{c}\langle F_{bc}, \partial_b F_{bc}\rangle, \quad \sum_{a,c}\langle F_{ac}, \partial_b F_{ac}\rangle, \quad \sum_{a,c}\langle F_{bc}, \partial_a F_{ac}\rangle\,.$$

For diagonal entries $z_0\sum_c\langle F_{ac}, F_{ac}\rangle + z_1\sum_{b,c}\langle F_{bc}, F_{bc}\rangle$ and off-diagonal entries $z_2\sum_c\langle F_{ac}, F_{bc}\rangle$, the divergence coordinates are $(2(z_0 - z_2), 2z_1 + z_2/2, z_2)$. Indeed Bianchi gives $\sum_{a,c}\langle F_{ac}, \partial_a F_{bc}\rangle = \frac{1}{2}\sum_{a,c}\langle F_{ac}, \partial_b F_{ac}\rangle$. Thus the functional $\rho_{\mathrm{tr}}(x, y, z) = -2x - 8y$ satisfies

$$\rho_{\mathrm{tr}}(\mathrm{div}_{\mathrm{raw}}\, B)\mathcal{O}_{F^2} = -\operatorname{tr} B, \qquad \rho_{\mathrm{tr}}(\operatorname{div} L) = \rho_{\mathrm{tr}}(\operatorname{div}\Xi) = 0, \qquad \rho_{\mathrm{tr}}(\operatorname{div} D) = -4\,. \tag{18.3}$$

It annihilates the third coordinate, the Maxwell equation direction. The Maxwell quotient of this raw space is two dimensional.

We construct the lattice insertions from plaquette curvatures. At a site $x$ put $f_{ab}(x) = \mathrm{proj}_{\mathfrak{g}}(U_{\square(x;a,b)} - U^{-1}_{\square(x;a,b)})/2$ and

$$J_{ab} = \frac{\delta_{ab}}{4}\sum_{c,d}\langle f_{cd}, f_{cd}\rangle - \sum_{c}\langle f_{ac}, f_{bc}\rangle, \qquad (Y^{x}_{b})_{e} = \mathbf{1}_{e^{-}=x} f_{b,e^{\mathrm{dir}}}(x)\,.$$

These are the initial stress current and left displacement. Let $\mathsf{S} = \mathsf{S}_{N,m}$ be the leading analytic path action without its coupling, defined in Section 12. Set $\mathcal{E}_b = \sum_a \nabla_a J_{ab} - D\mathsf{S}[Y_b]$, where $\nabla_a = \sum_{j=1}^{4} c_j(T_a^j - T_a^{-j})$, $(c_1, c_2, c_3, c_4) = (4/5, -1/5, 4/105, -1/280)$ and $T_a q(x) = q(x + e_a)$. Initially $\operatorname{tr} J = 0$ and $J_{ab} - J_{ba} = 0$.

The averaging kernels must respect this particular difference. For odd block ratio $\ell$, define commuting finite sums

$$P_a = \ell^{-1}\sum_{j=0}^{\ell-1} T_a^j, \qquad B_a = \sum_{j=1}^{4} c_j \sum_{v=-j}^{j-1} T_a^v, \qquad N_a = B_a(T_a^{\ell})P_a, \qquad Q_0 = \prod_a P_a N_a\,.$$

Here $B_a(T_a^{\ell})$ replaces every shift by its $\ell$th power; $B_a(T_a - 1) = \nabla_a$. Write $F_a = \prod_{c\neq a} P_c N_c$ and put $Q_a = B_a P_a F_a$, $Q_a^{\mathrm{edge}} = N_a F_a$. Since $P_a(T_a - 1) = \ell^{-1}(T_a^{\ell} - 1)$,

$$Q_0\nabla_a = \ell^{-1}\nabla_a^{\mathrm{coarse}} Q_a, \qquad Q_0(T_a - 1) = \ell^{-1}(T_a^{\ell} - 1)Q_a^{\mathrm{edge}}, \qquad Q_a - Q_0 = (T_a^{\ell} - 1)Q_a^{\mathrm{tr}}\,.$$

The last quotient is a finite Laurent polynomial: $B_a - B_a(T_a^{\ell})P_a$ vanishes modulo $T_a^{\ell} - 1$ after multiplication by $P_a$, because $P_a$ averages all residue classes and both $B_a$ factors have mass one. Polynomial division defines $Q_a^{\mathrm{tr}}$. The support radius of $Q_0$ is at most $56\ell$, its mass is one and its absolute coefficient sum is at most $32^4$.

For Lemma 11.6's stationary background $U_{\mathrm{stat}}(V)$, stationary substitution is $\mathcal{I}F(V) = F(U_{\mathrm{stat}}(V))$. The stationary equation is $D_U(\mathsf{S} + \mathsf{S}_{\mathrm{block}}) = 0$ on the gauge-fixed fine variations. Let $R_f$ and $R_0$ be the fine-coordinate and root-frame differentials in Lemma 11.7. The induced coarse displacement is $\mathcal{I}_* Y = (R_f + d_V R_0)Y(U_{\mathrm{stat}}(V))$. With $\delta_a = T_a - 1$, transfer by

$$J'_{ab} = \ell^4 Q_a \mathcal{I} J_{ab}, \quad Y'_b = \ell^5 Q_0 \mathcal{I}_* Y_b, \quad \mathcal{E}'_b = \ell^5 Q_0 \mathcal{I}\mathcal{E}_b, \quad q'_a = \ell^3 Q_a^{\mathrm{edge}}\mathcal{I} q_a + \ell^4 Q_a^{\mathrm{tr}}\mathcal{I} J_{aa}\,.$$

Starting with $q_a = 0$, this preserves $\operatorname{tr} J = \sum_a \delta_a q_a$. The antisymmetric part, which need not remain zero, is likewise a sum of forward differences: its new correction in direction $a$ is $\ell^4 Q_a^{\mathrm{tr}} \mathcal{I} J_{ab}$ and in direction $b$ is $-\ell^4 Q_b^{\mathrm{tr}} \mathcal{I} J_{ba}$, in addition to the transferred old differences.

For an analytic local function $F$, denote its degree-$d$ polynomial read-out by $\mathcal{P}_d F$: on a radial jet $\mathsf{j}$ it is $\sum_{v=0}^{d} [\varepsilon^v] F(U(\mathrm{sc}_\varepsilon \mathsf{j}))$, in the basis of Definition 2.3. The weighted read-out is $\mathsf{D}^{\mathrm{fac}}_{t_m} \mathcal{P}_d F$, using Definition 3.9. Symmetrize $J$, average over $H_4$ and conjugation, apply the degree-four read-out and marked transport, and subtract the mixture mean. Call the resulting field $B_i^L$. The same operations on the local contractions defining $\Xi, D$ give $B_i^{\Xi}, B_i^D$.

**The regulated Ward coefficient.** We first calculate the scalar coefficient before marked transport. For a chart function $F$ define its Hessian Fourier kernel by

$$\widehat{F}_{c\alpha,c\alpha}(p,q) = \sum_{x,y} e^{i(p\cdot x + q\cdot y)} D^2 F(0)[e_{x,c,\alpha}, e_{y,c,\alpha}] \,.$$

Fix a colour $\alpha$. For a family $F_b$ write

$$\mathcal{L}(F) = \operatorname{Re} \sum_{a\neq c,b} i[z^2 t] \widehat{F_b}_{c\alpha,c\alpha}(z e_a, -z e_a + t e_b),$$
$$\mathcal{Q}(F) = |\Lambda|^{-1} \operatorname{Re} \sum_{a\neq c} [z^2] \widehat{F}_{c\alpha,c\alpha}(z e_a, -z e_a) \,.$$

Coefficients use $[z^r t^s] F = (r!s!)^{-1} \partial_z^r \partial_t^s F(0)$. Put $\mathcal{H}(F) = \mathcal{L}(F) - \frac{1}{2} \operatorname{Re} \sum_{a\neq c} [z^2] \widehat{F_c}_{c\alpha,c\alpha}(z e_a, -z e_a)$. The Haar coefficient is $\mathfrak{h}(Y) = \mathcal{H}(\operatorname{div}_{\mathrm{Haar}} Y)$; the subtraction corrects the lattice derivative. Define

$$\Gamma_\mu = \tfrac{1}{2} \log\det(H + dd^* + \mu^2 I) - \log\det(d^* d + \mu^2 I), \quad R_\Lambda(\mu) = \mathcal{Q}(\Gamma_\mu), \quad \mathcal{D}_\Lambda(\mu) = -\mu R_\Lambda'(\mu),$$

where $H$ is the left Hessian of $\mathsf{S}$ and $d$ is the infinitesimal gauge insertion. The next lemma computes this coefficient; its proof follows.

**Lemma 18.4** (Regulated Ward coefficient)**.** *At a fixed level, for the transferred currents above with uniformly bounded exponentially weighted derivatives, as their fourth-order current defects tend to zero and the period tends to infinity,*

$$\mathfrak{h}(Y) \longrightarrow -12\kappa_G \,.$$

*Proof.* Use Definition 2.1 and Lemmas 11.6, 11.10, 12.4, 12.6, 12.9. We keep the contacts in the determinant identity. Write $\mathcal{R}_\mu = (H + dd^* + \mu^2 I)^{-1}$, $\mathcal{G}_\mu = (d^* d + \mu^2 I)^{-1}$. For $\Psi_s(U) = e^{sY(U)} U$ let $\mathcal{Y}_U(z) = \partial_s|_0 \log(\Psi_s(e^z U)\Psi_s(U)^{-1})$ and $\mathcal{V} = D\mathcal{Y}_U(0)$. If $\mathcal{J}_U$ is the mixed derivative of $\log(\mathrm{gauge}_{e^v}(e^z U) U^{-1})$ in $v, z$ at zero, gauge invariance gives $Hd = -\mathcal{T}$, $\mathcal{T} = D\mathsf{S}[\mathcal{J}_U]$, hence

$$\mathcal{R}_\mu d = d\mathcal{G}_\mu + \mathcal{R}_\mu \mathcal{T} \mathcal{G}_\mu \,.$$

The product rule for $D^2(D\mathsf{S}[Y])$ and cyclicity of trace now give

$$\tfrac{1}{2} \operatorname{tr}(\mathcal{R}_\mu D^2(D\mathsf{S}[Y])) - \operatorname{div}_{\mathrm{Haar}} Y = D\Gamma_\mu[Y] - \mu^2 \operatorname{tr}(\mathcal{R}_\mu \mathcal{V}) + D\mathsf{S}[Z], \tag{18.4}$$

with $Z = \frac{1}{2} \operatorname{contr}(\mathcal{R}_\mu, D^2 \mathcal{Y}_U(0)) - \operatorname{contr}(\mathcal{G}_\mu d^* (\mathcal{V} + \mathcal{V}^*) \mathcal{R}_\mu, \mathcal{J}_U)$. Contractions sum the indicated matrix indices. Thus the off-shell contact is retained.

Here is the cancellation under $\mathcal{L}$. The leading quadratic kernel of an invariant scalar action at the required momentum order is a colour multiple of $|p|^2 I - p \otimes p$. Write $X_{acb}$ for the coefficient

of momentum $e_a$ in the diagonal polarization-$c$ entry of $DY_b$, with its colour indices contracted against the action kernel. Gauge covariance makes the mixed polarization term $-X_{cac}$ and gives $X_{aaa} = 0$. In the tested variation the three terms, after summing the intermediate polarization, are

$$-2X_{aca} - X_{cac} + \sum_b X_{bcb} \qquad (a \neq c).$$

Summing over $a \neq c$ makes the first and second sums $-3\sum_{b,c} X_{bcb}$, cancelling the third sum. Higher momentum orders do not contribute. This applies to $D\Gamma_\mu[Y]$, $D\mathsf{S}[Z]$ and the further action variation below, uniformly on each fixed $\mu \in [\delta, 1]$. For the first term on the left of (18.4), substitute $D\mathsf{S}[Y_b] \equiv_4 \sum_a \nabla_a J_{ab} - \mathcal{E}_b$. The coefficient linear in the total momentum $te_b$ selects $a = b$; summing $b$ gives $\operatorname{tr} J = \sum_a \delta_a q_a$. Its additional total-momentum factor has zero coefficient linear in $t$. The defect $\mathcal{E}_b$ tends to zero in the weighted seminorms, so its contracted coefficient does too. Here $\equiv_4$ denotes equality of field derivatives through order four. The correction in $\mathfrak{h}$ tends to zero by the same identity.

We compute the surviving regulator term explicitly. For $\chi(x) = \mathbf{1}_{x=0}$ define $M_b h(x) = \chi(x) h_b(x)$, $(D_b h)_a(x) = \delta_a \chi(x) h_b(x)$ and $(T_b^{(r)} h)(x) = \delta_b \chi(x) h(x)$ on $r$-forms, $r = 0, 1$. Set

$$W_b^{(1)} = \operatorname{sym}(\mathcal{V}_b - dM_b - D_b) + \tfrac{1}{2} T_b^{(1)}, \qquad W_b^{(0)} = \operatorname{sym}(M_b d) - \tfrac{1}{2} T_b^{(0)},$$

where $\operatorname{sym} A = (A + A^*)/2$. The inverse identity above yields

$$\begin{aligned} \operatorname{tr}(\mathcal{R}_\mu \mathcal{V}_b) = {} & \operatorname{tr}\big(\mathcal{R}_\mu (D_b - \tfrac{1}{2} T_b^{(1)})\big) + \tfrac{1}{2} \operatorname{tr}(\mathcal{G}_\mu T_b^{(0)}) + D\mathsf{S}[Z_b'] + \mathcal{W}_b, \\ \mathcal{W}_b = {} & \operatorname{tr}(\mathcal{R}_\mu W_b^{(1)}) + \operatorname{tr}(\mathcal{G}_\mu W_b^{(0)}), \qquad Z_b' = \operatorname{contr}(\mathcal{G}_\mu M_b \mathcal{R}_\mu, \mathcal{J}_U). \end{aligned}$$

The tested $D_b$ term tends to $\mathcal{Q}(\operatorname{tr} \mathcal{R}_\mu)$; each of the four $T_b^{(r)}$ terms tends to $\mathcal{Q}(\operatorname{tr} \mathcal{R}_\mu)$ or $\mathcal{Q}(\operatorname{tr} \mathcal{G}_\mu)$ respectively. This follows by summing $\delta_b \chi$ against the two Fourier waves: its symbol is $e^{-it} - 1 = -it + O(t^2)$, paired with the factor $i$ in $\mathcal{L}$. Translation invariance identifies the local anchor sum with the volume-normalized trace in $\mathcal{Q}$. Consequently their sum is $-\mathcal{Q}(\operatorname{tr} \mathcal{R}_\mu) + 2\mathcal{Q}(\operatorname{tr} \mathcal{G}_\mu)$. Differentiation of $\Gamma_\mu$ in $\mu$ gives the same expression multiplied by $\mu^2$, namely $\mathcal{D}_\Lambda(\mu)$. We have proved

$$\mathfrak{h}(Y) = \mathcal{D}_\Lambda(\mu) + \mu^2 \mathcal{L}(\mathcal{W}) + o(1), \tag{18.5}$$

with error uniform on every fixed $[\delta, 1]$.

For the remainder use the inverse bounds in the proof of Lemma 12.9: $\|\partial^r R_\mu(q)\| \leq C(|q| + \mu)^{-2-r}$. Differentiating $R_\mu A_\mu = I$ once more gives the same bound for $r = 3$. The stationary displacement bounds of Lemmas 11.6, 11.10, give bounded weighted derivatives of $W^{(r)}$. Their zero-background symbols and first momentum derivatives tend to zero, as does the zero-momentum first field insertion: the free curvature displacement cancels $dM_b + D_b - \frac{1}{2} T_b^{(1)}$ in the displayed formula. Put $z = |q| + \mu$. Taylor expansion gives factors $z^2, z, 1$ for the zero-, one- and two-field insertions of these symbols. In the product rule they pair with inverse bounds $Cz^{-4}, Cz^{-3}, Cz^{-2}$, respectively. Thus the scaled derivative bound through order three is $C(z^2 z^{-4} + 2z z^{-3} + z^{-2}) \leq C' z^{-2}$. Unscaling the quadratic coefficient gives $Cz^{-4} \leq C' z^{-5}$. In four dimensions the integral of this bound is at most $C/\mu$: the ball $|q| \leq \mu$ costs $C/\mu$, and the annulus $2^j \mu < |q| \leq 2^{j+1}\mu$ costs $C\mu^{-1} 2^{-j}$. At fixed $\mu > 0$ the finite sums converge; hence $|\mu^2 \mathcal{L}(\mathcal{W})| \leq C\mu + o(1)$ with $C$ independent of $\mu \leq 1$.

The bound following (12.3) in Lemma 12.9's proof gives $|R(\mu) - 12\kappa_G \log \mu| \leq C$ for $0 < \mu \leq 1$. Indeed $R/12 = \alpha_\alpha(Y_\mu)$: the Hessian is twice the quadratic-action kernel, exactly the factor 2 in that coefficient. At fixed positive $\mu$, Riemann sums give $R_\Lambda(\mu) \to R(\mu)$. Choose $u < 0$, then a length

$L_* > 0$. Once the period controls the endpoints, the mean value theorem on $[u - L_*, u]$ selects $\mu_\Lambda \in [e^{u-L_*}, e^u]$ with $|\mathcal{D}_\Lambda(\mu_\Lambda) + 12\kappa_G| \le (2C + o(1))/L_*$. Apply (18.5) on this fixed interval, then let $u \to -\infty$ and $L_* \to \infty$. This proves the limit without differentiating an asymptotic error. □

**Transfer to the continuum.** Put $b_i = \widehat{\beta}_{k_i}$, $E_i = \exp((\log\log b_i)^2/5)$, $R_i = \exp((\log\log b_i)^2/6)$, and $\mathsf{A}_i = \beta_i r_i^4 U_i(4)^{-1}$, $K_i = \beta_i r_i^5 U_i(5)^{-1}\,\mathrm{div}$. The image of divergence on centered real symmetric $H_4$ tensors is an $i$-dependent plane $W_i \subset (M_5^{\mathrm{Max}})^4$: its kernel is the conserved line of Lemma 16.7. Functionals below act on the ambient real tensorial vector space, with any fixed norm.

This estimate transfers the scalar coefficient to that plane. Its proof follows.

**Lemma 18.5** (Fourth-order Ward transfer)**.** *There are uniformly bounded real functionals $\lambda_i$ and $C < \infty$, depending only on the chosen construction, such that*

$$\begin{aligned} &\|\beta_i K_i B_i^L\| \le C R_i, && \lambda_i(\beta_i K_i B_i^L) \to -\kappa_G, \\ &K_i B_i^B \to \mathrm{div}^{\mathrm{Max}}\, B, && E_i[\lambda_i(K_i B_i^B) - \rho_{\mathrm{tr}}(\mathrm{div}_{\mathrm{raw}}\, B)] \to 0 \quad (B = \Xi, D). \end{aligned} \tag{18.6}$$

*Moreover*

$$\mathsf{A}_i B_i^B \to B \ (B = L, \Xi, D), \qquad b_i^p\, \mathrm{tr}(\mathsf{A}_i B_i^B) \to 0 \ (B = L, \Xi;\ p \in \mathbb{N}), \qquad \beta_i/b_i \to 1\,. \tag{18.7}$$

*Proof.* Use Definition 3.9, Lemmas 11.6–11.8, 12.4, 12.6–12.9, 14.8, 14.9, 14.16–14.19, and Lemmas 18.3, 18.4.

*Current estimates.* Differentiating the stationary equation eliminates its fine variation. Fourth-order matching in Lemma 12.6 preserves $\sum_a \nabla_a J_{ab} - \mathcal{E}_b \equiv_4 D\mathsf{S}[Y_b]$. The kernel identities above preserve the trace and antisymmetric parts as differences. For a chart function define the weighted order-$r$ seminorm by the sum of absolute coefficients of $D^r F(0)$, with the exponential distance and polynomial weights of Lemma 11.6; for a displacement use its link components. Divide the order-$r$ current defect by $(\log \widehat{\beta}_m)^r$. At orders $r = 3, 4$, the linearized stationary fields contribute $\ell^{-2}$ each, so transfer costs $C_j \ell^{5-2r}$. At order two the matched low-momentum terms must first be removed. The affine and quadratic parts of a stationary test field have weighted sizes $C_j(1 + |x|)/\ell^2$ and $C_j(1 + |x|)^2/\ell^3$; the remainder is at most $C_j(1 + |x|)^3/\ell^4$, by Lemmas 11.6, 11.9. Subtracting the affine–affine and affine–quadratic products leaves quadratic–quadratic and affine–remainder products, each with factor $\ell^{-6}$. Multiplication by $\ell^5$ gives $C_j/\ell$. A bounded matched contribution gives the additional $C_j/(\log \widehat{\beta}_m)^2$. More precisely the order-two factor is bounded by

$$C_j \left\{ \frac{1}{(\log \widehat{\beta}_m)^2} + \frac{(\log \widehat{\beta}_{m+1})^2 + 1}{\ell_m (\log \widehat{\beta}_m)^2} \right\}.$$

Nonlinear chain-rule partitions involve fewer than $r$ derivatives and gain inverse logarithmic powers. Choose each ratio large, then its entry coupling large; the resulting triangular map contracts by $1/2$. After $n_{\mathrm{step}}$ transfers the normalized defects are $O(2^{-n_{\mathrm{step}}})$.

*The finite coefficient.* At level $k$ set $h_k = (\log\log \widehat{\beta}_k)^2$ and use $M_k = M_k^{\mathrm{cut}}$. Define $\mathcal{P}_{5,k} = t_k^{-2} \mathsf{D}_{t_k}^{\mathrm{fac}} \mathcal{P}_5$, followed by real $H_4$ averaging. Haar integration by parts in the density gives the degree-five coefficient in the form

$$\Sigma_k = \mathcal{P}_{5,k}(\mathrm{div}_{\mathrm{Haar}}\, Y - D\mathcal{L}_{N,k}[Y]) + \varepsilon_{\mathrm{stat}} + \varepsilon_{\mathrm{tail}} + \varepsilon_{\mathrm{cl}} + \varepsilon_{\mathrm{mark}}\,. \tag{18.8}$$

Here $\mathcal{L}_{N,k}$ is the full quadratic loop action of Definition 12.2. The errors are stationary substitution, omitted loop paths, the classical defect and the marked remainder. Each has norm at most $e^{-h_k}$:

truncate stationary kernels and loop paths at $C_j(\log \widehat{\beta}_k)^2$ using Lemmas 11.6, 12.3; the tails then decay exponentially. Lemmas 14.8, 14.9 bound the other terms by $C_j \widehat{\beta}_k^{-\eta_j} M_k^{C_j} + C_j \widehat{\beta}_k^{C_j} e^{-M_k^{1/5}}$, with $\eta_j > 0$, after paying the finite projection losses. Both are $o(e^{-h_k})$; delayed entry makes the stated bounds simultaneous.

The loop term in (18.8) has zero $\rho_{\mathrm{tr}}$ coefficient. If $H_{\mathrm{loop}}$ is its quadratic link kernel, differentiation of $D\mathcal{L}[Y]$ pairs $H_{\mathrm{loop}}$ with $DY$. The cubic momentum test is a sum of pairings of constant and quadratic polynomial links and of two affine polynomial links. Constant links are pure gauge; $H_{\mathrm{loop}}$ also kills affine links by both moment cancellations of Lemma 12.4. Thus every pairing vanishes. The lattice correction in $\mathfrak{h}$ is the corresponding correction to this coefficient read-out. Choose the transfer depth and period at this fixed $k$ so that $\mathfrak{h}(Y)$ differs from $-12\kappa_G$ by at most $\widehat{\beta}_k^{-100}$, using Lemma 18.4. Choose the momentum-Taylor approximation to its finite local read-out within $\widehat{\beta}_k^{-30}$. The exact read-out identity gives the factor $1/12$ for the twelve ordered transverse planes. Delayed entry bounds the retained loops $2M_k^{\theta J_k}$ and stage losses by $M_k^{1/6}$. Therefore

$$\|\Sigma_k\| \le C M_k^{1/6}, \qquad |\rho_{\mathrm{tr}}(\Sigma_k) - \mathfrak{h}(Y)/12| \le 4\|\rho_{\mathrm{tr}}\| e^{-h_k} + \widehat{\beta}_k^{-30}/12\,.$$

Passing marked coefficients to the fixed refinement adds $O(e^{-h_k/4})$ by Lemma 14.9. Denote the limiting real averaged coefficient again by $\Sigma_k$. Thus $\rho_{\mathrm{tr}}(\Sigma_k) \to -\kappa_G$ and $\|\Sigma_{k_i}\| \le C R_i$.

*The equation direction.* Let $s_i$ be degree-five marked transport, followed by $\beta_i r_i^5 U_i(5)^{-1}$, on the three-dimensional raw vector space. Let $s$ be the free Maxwell quotient map; $v_{\mathrm{eom}} = (0,0,1)$, $\ker s = \mathbb{R} v_{\mathrm{eom}}$ and $\rho_{\mathrm{tr}}(v_{\mathrm{eom}}) = 0$. Let $j_i$ be the raw coefficient of the transferred current's divergence after the same threshold normalization. Taylor matching and marked comparison give

$$j_i \to -v_{\mathrm{eom}}, \quad |\rho_{\mathrm{tr}}(j_i)| \le C e^{-h_{k_i}/4}, \quad v_i^{\mathrm{eq}} = \beta_i^{-1}\Sigma_{k_i} - j_i \to v_{\mathrm{eom}}\,.$$

In particular $E_i \rho_{\mathrm{tr}}(v_i^{\mathrm{eq}}) \to 0$, since $E_i/\beta_i \to 0$ and $\rho_{\mathrm{tr}}(\Sigma_{k_i})$ is bounded. For the other rate, localize $\Sigma_{k_i} - \beta_i j_i$ before transport, with a fixed degree-four spectator. Its exterior moment is zero by Haar integration by parts, since the varied link is outside the spectator stencil. The marked errors of Lemmas 14.8, 14.9 decay across the separating interval. Write $H_n = (\log \widehat{\beta}_n)^{48}$. Their two-point bound at radius level $n_i$, before the degree-five scaling, is $e^{-H_{n_i}/100}$. We may require $u_i \le \log\log \widehat{\beta}_{n_i}$ in the diagonal extraction. The extra scaling factor is then at most $\exp(5(\log\log \widehat{\beta}_{n_i})^3)$, while $b_i/\widehat{\beta}_{n_i} \to 1$. Thus each rescaled evaluation after multiplication by $b_i^p$ is bounded by

$$C b_i^p \exp\{5(\log\log \widehat{\beta}_{n_i})^3 - H_{n_i}/100\} \longrightarrow 0\,.$$

Finite separating free two-point tests give $b_i^p s_i(v_i^{\mathrm{eq}}) \to 0$.

Construct right inverses $t_i$ of $s_i$ from the frames and corrected degree-five projections: lift a free vector by the common frame, undo its threshold normalization and marked transport, and use the frame inverse to check $s_i t_i = I$. Their coefficient lifts converge on a finite basis by Lemma 18.3, so $t_i \to t$ with $st = I$. These identities give rank two; hence $\ker s_i$ is a line. Its generator is $v_i^{\mathrm{eq}} - t_i s_i v_i^{\mathrm{eq}}$. The bases formed by $t_i$ and this kernel generator converge to the free basis, giving $s_i \to s$. Define $\lambda_i = \rho_{\mathrm{tr}} t_i$. Decomposing along this line and $\operatorname{im} t_i$ gives

$$\|\lambda_i s_i - \rho_{\mathrm{tr}}\| \le C\big(|\rho_{\mathrm{tr}}(v_i^{\mathrm{eq}})| + \|s_i v_i^{\mathrm{eq}}\|\big) = o(E_i^{-1}).$$

The actual divergence satisfies $\beta_i K_i B_i^L - s_i \Sigma_{k_i} \to 0$ by the same equation estimate; for $B = \Xi, D$, $K_i B_i^B - s_i \operatorname{div}_{\mathrm{raw}} B = o(b_i^{-p})$. Since $R_i/E_i \to 0$, these estimates prove (18.6).

*Tensor traces.* Taylor matching gives $\mathsf{A}_i B_i^B \to B$. The trace of $\Xi$ is zero; the transferred trace of $L$ is a forward difference. Its degree-three projection is a unit, since simplicity leaves no nonconstant

invariant below degree four. Average the degree-four difference over $m$ sites: it telescopes to two endpoint errors divided by $ma_k$. Choose $m \to \infty$ with $ma_k \to 0$ and endpoint errors $o(ma_k)$ using Lemma 14.9; smoothness gives the same separating correlation in the limit. A small inverse power of $a_k$ for $m$ gives $O(a_k^{\eta})$ for some $\eta > 0$. Finite separating tests and subexponential inverse transport preserve $o(b_i^{-p})$ for every $p$. This proves the trace assertion in (18.7); its coupling assertion is Lemma 18.3. □

Conservation converts these divergence estimates into the scalar coefficient. Lemmas 18.3 and 18.5 give

$$\beta_i r_i^4 U_i(4)^{-1}\Theta \longrightarrow -\kappa_G \mathcal{O}_{F^2}\,. \tag{18.9}$$

*Derivation of* (18.9). Use the field calculus of Lemma 15.4, the free tensor calculation of Lemma 16.1, the conserved line of Lemma 16.7, and the normalized shell laws of Lemma 16.11. The free divergences of $\Xi, D$ are independent: a kernel combination would be a multiple of $L$, impossible in these two tensor coordinates. Thus $f_i : \mathbb{R}^2 \to W_i$, $f_i(c) = c_0 K_i B_i^{\Xi} + c_1 K_i B_i^{D}$, are eventually invertible with bounded inverses by (18.6). Set $c_i = -f_i^{-1}(\beta_i K_i B_i^{L})$. Then $|c_i| \le C R_i$ and

$$X_i = \beta_i B_i^{L} + c_{i,0} B_i^{\Xi} + c_{i,1} B_i^{D}$$

is conserved. Applying $\lambda_i$ to its zero divergence gives the exact identity

$$4c_{i,1} = \lambda_i(\beta_i K_i B_i^{L}) + c_{i,0}\lambda_i(K_i B_i^{\Xi}) + c_{i,1}\big[\lambda_i(K_i B_i^{D}) + 4\big]\,.$$

Since $R_i/E_i \to 0$, the last two terms vanish and $c_{i,1} \to -\kappa_G/4 = d_2$.

Equation (18.7) and $R_i/\beta_i \to 0$ give $r_i^4 U_i^{-1} X_i \to L$. Its weighted trace $\beta_i \operatorname{tr}(r_i^4 U_i(4)^{-1} X_i)$ tends to $4d_2\mathcal{O}_{F^2}$, since the other traces vanish faster than every inverse power. Lemma 16.7 now gives $X_i = \nu_i T$ with $\nu_i \in \mathbb{R}$, since these tensors and the fixed $T$ are centered in the mixture. By Lemma 16.1, choose $A \in M_4^{\mathrm{Max}}$, a separating spectator functional $o$, and an axis $b$ with $y = o(\partial_b^{\mathrm{Max}} A) \neq 0$. For these fixed tests the flux of $r_i^4 U_i^{-1} X_i$ tends to the flux of $L$, namely $y$, by Lemma 18.3. The flux of $T$, multiplied by $r_i$, tends to $y$ by the normalized Ward identity of Lemma 16.11 and the jet limit of Lemma 18.3. The first flux is $\nu_i$ times the second. Dividing by the latter proves $\nu_i \to 1$. This gives the weighted trace limit for $T$.

Finally $\Theta = \sum_a T_{aa} - e_{\mathrm{tr}}\mathbf{1}_4$ and $U_i^{-1}\mathbf{1}_4 = \mathbf{1}_4$. The extra term has size $|e_{\mathrm{tr}}|\beta_i r_i^4 \to 0$ by (18.1). □

*Proof of Theorem 7.2.* The phase reflection identity of Lemma 17.5 makes $e_{\mathrm{tr}}$ real. Lemmas 16.10, 16.11 and 17.9 show that $\Theta$ is real and $O(4)$ invariant and that its continued label is Hermitian and Lorentz scalar. Lemma 18.1 gives vacuum centering. For the asymptotic, take any prescribed positive null sequence and the subsequence in (18.1). By (18.9), the phase comparison, and the free normalization in Lemma 16.1, every generic pair $x = (x_0, x_1)$ with $|x| < 100$ and $\mathrm{sep}_{\infty}(x) > 1/100$, with $x^{(i)} \in r_i^{-1} a_{n_i}\mathbb{Z}^4$ nearest to $x$, satisfies

$$r_i^8 \log^2(1/r_i) G_{\Theta}(r_i|x_0^{(i)} - x_1^{(i)}|) \longrightarrow \frac{A_G}{|x_0 - x_1|^8}\,.$$

Translation and $O(4)$ invariance (Lemma 16.11) give $S_2^{\xi_*}(\Theta, \Theta; rx_0, rx_1) = G_{\Theta}(r|x_0 - x_1|)$. Here $\kappa_G^{-2}(4d_2)^2 = 1$.

Take $x_0 = 0$ and $x_1 = (t/2, t/2, t/2, t/2)$, with $1/2 < t < 2$. This pair is generic and has distance $t$. Monotonicity from Lemma 18.2 bounds the lower limit at radius $r_i$ by $A_G/t^8$ for $t > 1$, and the upper limit by that expression for $t < 1$. Let $t \to 1$. The limit along this subsequence is $A_G$. The subsequence criterion gives the full-radius limit. Here $A_G = 3d_{\mathfrak{g}}/\pi^4 > 0$ since $d_{\mathfrak{g}} \ge 1$; identifying it uses $d_2 = -\kappa_G/4 \neq 0$. Thus $\Theta \neq 0$. □

The following high-energy law follows from TR and Lemma 18.2, as proved below.

**Lemma 18.6** (High-energy trace limit)**.** *For every $\varphi \in C_c(\mathbb{R}^4;\mathbb{C})$ and every $\varphi \in \mathcal{S}$,*

$$\lim_{R_{\mathrm{E}}\to\infty} R_{\mathrm{E}}^{-8}(\log R_{\mathrm{E}})^2 \int \varphi(\mathbf{p}/R_{\mathrm{E}})\, d\mu_\Theta(\mathbf{p}) = \frac{A_G}{1536\pi} \int_{\overline{V}_+} \mathsf{m}(\mathbf{p})^2 \varphi(\mathbf{p})\, d^4\mathbf{p}\,.$$

*Proof.* Use Definition 7.1, Lemma 18.2 and the full-radius conclusion of TR. Put $\nu_{R_{\mathrm{E}}} = R_{\mathrm{E}}^{-8}\log^2 R_{\mathrm{E}}\, \mu_\Theta(R_{\mathrm{E}}\,\cdot)$. For every future timelike Laplace covector $u$, Lorentz invariance reduces its transform to the time-axis transform at $s = \sqrt{\mathsf{m}(u)}$. Hence

$$\int e^{-u\cdot\mathbf{p}}\, d\nu_{R_{\mathrm{E}}} = R_{\mathrm{E}}^{-8}\log^2 R_{\mathrm{E}}\, G_\Theta(s/R_{\mathrm{E}}) \longrightarrow A_G s^{-8}\,.$$

To identify the constant, write $t = \mathbf{p}^4$ and integrate the spatial radius $q$:

$$\int_{\overline{V}_+} e^{-st}\mathsf{m}(\mathbf{p})^2\, d^4\mathbf{p} = 4\pi \int_0^\infty e^{-st} \int_0^t q^2(t^2-q^2)^2\, dq\, dt = 1536\pi s^{-8}\,.$$

The $s = 1$ transform bounds compact masses. Diagonal extraction and Riesz representation give vague subsequential limits [26, Theorem 2.14], applied to positive functionals on $C_c(\mathbb{R}^4)$. The $u/2$ transform controls the $u$-damped tail, giving the displayed transform for every limit. Uniqueness follows by holomorphic uniqueness in each coordinate for timelike real part, then Fourier inversion on product Schwartz tests against the finite damped measures [26, Theorems 10.18 and 9.11]; finiteness justifies Fubini. Thus every extracted limit is the cone density in the statement.

For Schwartz tests positivity gives

$$\nu_{R_{\mathrm{E}}}\{\mathbf{p}^4 \le R\} \le e\, R_{\mathrm{E}}^{-8}\log^2 R_{\mathrm{E}}\, G_\Theta((R_{\mathrm{E}}R)^{-1}) \le CR^8 \qquad (R \ge 1,\ R_{\mathrm{E}} \ge R_{\mathrm{E},0})\,.$$

Use the trace law at $1/(R_{\mathrm{E}}R)$ and $\log R_{\mathrm{E}} \le \log(R_{\mathrm{E}}R)$. A Schwartz bound of order greater than eight sums the dyadic energy tails uniformly, since $|\mathbf{p}| \le \sqrt{2}\mathbf{p}^4$ on the cone. The $C_c$ limit then gives the Schwartz limit. □

For $\psi_{R_{\mathrm{E}}}(x,y) = R_{\mathrm{E}}^8\psi(R_{\mathrm{E}}x, R_{\mathrm{E}}y)$, $\mathcal{W}_{\mathrm{rel}}\psi_{R_{\mathrm{E}}}(\mathbf{p}) = \mathcal{W}_{\mathrm{rel}}\psi(\mathbf{p}/R_{\mathrm{E}})$. For $\psi \in \mathcal{S}_2$, $\mathcal{W}_{\mathrm{rel}}\psi \in \mathcal{S}$. Substitution in Lemma 18.6 gives the position-space consequence

$$R_{\mathrm{E}}^{-8}\log^2 R_{\mathrm{E}}\, W_2^{\xi*}(\Theta,\Theta;\psi_{R_{\mathrm{E}}}) \longrightarrow \frac{A_G}{1536\pi}\int_{\overline{V}_+} \mathsf{m}(\mathbf{p})^2 \mathcal{W}_{\mathrm{rel}}\psi(\mathbf{p})\, d^4\mathbf{p}\,.$$

## 18.2 Precise non-free comparisons

The radial integrals of Definition 7.3 imply the following exclusion; its proof follows.

**Lemma 18.7** (Finite Wick obstruction)**.** *For every finite Wick profile $W_{\mathrm{Wick}}$ and every $\delta > 0$, there is $r \in (0,\delta)$ such that*

$$G_\Theta(r) \ne W_{\mathrm{Wick}}(r)\,.$$

*Proof.* Use Definition 7.3 and the positive limit in TR. For one factor, polar coordinates and $\omega_{m_{\mathrm{sh}}} = \sqrt{m_{\mathrm{sh}}^2 + q^2}$ give

$$K_{m_{\mathrm{sh}},k_E,h_{\mathrm{mom}}}(r) = \frac{1}{2}\left(\int_{S^2} h_{\mathrm{mom}}\right)\int_{m_{\mathrm{sh}}}^\infty \omega_{m_{\mathrm{sh}}}^{k_E}(\omega_{m_{\mathrm{sh}}}^2 - m_{\mathrm{sh}}^2)^{(j_h+1)/2} e^{-r\omega_{m_{\mathrm{sh}}}}\, d\omega_{m_{\mathrm{sh}}}\,.$$

For $m_{\mathrm{sh}} = 0$ this is proportional to $r^{-j_h-k_E-2}$. Otherwise split at $2m_{\mathrm{sh}}$; the bounded interval is analytic. On the tail expand $(1 - m_{\mathrm{sh}}^2/\omega_{m_{\mathrm{sh}}}^2)^{(j_h+1)/2}$. Taking enough terms makes the remainder integrable with any prescribed number of energy moments, hence gives a Taylor remainder of the required order. The retained terms are $I_a(r) = \int_{2m_{\mathrm{sh}}}^{\infty} v^a e^{-rv}\, dv$, $a \in \mathbb{Z}$. Integrate explicitly for $a \geq 0$; $I'_{-1} = -e^{-2m_{\mathrm{sh}}r}/r$ gives the logarithm, and $I'_a = -I_{a+1}$ gives all lower powers. Only nonnegative logarithmic powers occur. For products, choose each remainder order above the other factors' pole orders.

Thus $\operatorname{Re} W_{\mathrm{Wick}} = \sum_{a,k} c_{a,k} r^a \log^k r + O(1)$ with finitely many terms. After multiplying by $r^8 \log^2(1/r)$, the terms with $a > -8$ vanish. Among the others, the least $a$, then largest $k$, gives a nonzero divergent leading term. A finite limit is therefore zero. Lemma 18.2 makes $G_\Theta$ real, so equality with $W_{\mathrm{Wick}}$ implies equality with its real part. The positive trace limit contradicts equality on $(0, \delta)$, proving also Proposition 7.4. □

To describe polynomial mass-shell Wick contractions, put $q_m(\mathbf{k}) = (\mathbf{k}, \sqrt{m^2 + |\mathbf{k}|^2})$, $m \geq 0$. Fix a finite set of elementary field indices $\mathsf{x}, \mathsf{y}$; words $u, u'$ list its elements. For words of length $k$, $\sigma$ ranges over permutations of $\{1, \dots, k\}$. Comparison data specify finite mass–polynomial lists $\mathcal{M}_{\mathrm{pair}}(\mathsf{x}, \mathsf{y})$, with $P \in \mathbb{C}[\mathbf{p}^1, \dots, \mathbf{p}^4]$, and complex contraction coefficients $c(u, u', \sigma)$.

**Definition 18.8** (Polynomial mass-shell pairings)**.** Set

$$v_{m,P}(\mathbf{k}) = \frac{P(q_m(\mathbf{k}))}{2\omega_m(\mathbf{k})}, \qquad L_{m,P}(\varphi) = \int v_{m,P}(\mathbf{k}) \varphi(q_m(\mathbf{k}))\, d^3\mathbf{k}\,.$$

For words $u = (\mathsf{x}_i)_{i=1}^k$, $u' = (\mathsf{y}_i)_{i=1}^k$, define the convolutions of the functionals $L_{m,P}$ by

$$\Pi(u, u'; \varphi) = \sum_\sigma c(u, u', \sigma) \sum_{(m_i, P_i) \in \prod_i \mathcal{M}_{\mathrm{pair}}(\mathsf{x}_i, \mathsf{y}_{\sigma(i)})} \int \prod_i v_{m_i,P_i}(\mathbf{k}_i)\, \varphi\left(\sum_i q_{m_i}(\mathbf{k}_i)\right) \prod_i d^3\mathbf{k}_i\,.$$

For unequal lengths put $\Pi = 0$. A zero denominator at zero momentum gives value zero.

For finite polynomial fields $Q = \sum_i \alpha_i u_i$, including empty words, the bilinear pairing is $W^{\mathrm{pol}}(Q, Q'; \varphi) = \sum_{i,j} \alpha_i \alpha'_j \Pi(u_i, u'_j; \varphi)$. Empty contractions evaluate $\varphi$ at zero. The shell measure has no $(2\pi)^{-3}$ factor.

The next lemma separates each comparison by one product test; its proof follows.

**Lemma 18.9** (Polynomial countertest)**.** *For every comparison data and every $Q, Q'$, there are $f, h \in \mathcal{S}$ with compactly supported $\mathcal{W}_{\mathrm{rel}}(\bar f \otimes h)$ such that*

$$\langle \Phi_\Theta(f)\Omega, \Phi_\Theta(h)\Omega \rangle \neq W^{\mathrm{pol}}(Q, Q'; \mathcal{W}_{\mathrm{rel}}(\bar f \otimes h))\,.$$

*Proof.* Use Definitions 7.3, 18.8 and Lemmas 18.2, 18.7. Equality for all product tests implies equality for every $\varphi \in C_c^\infty$: choose $\widehat{h}_+ = \varphi$ and real $\widehat{f}_+ = 1$ on its support. Approximate $e^{-r\mathbf{p}^4}$ by compact cutoffs. Dominated convergence applies: Lemma 18.2 controls $\mu_\Theta$; exponential damping controls comparison polynomials at infinity, and $d^3\mathbf{k}/|\mathbf{k}|$ is integrable at zero. Expanding the numerators into monomials factors the integrals into the kernels of Definition 7.3, with the empty words giving constants. Hence $G_\Theta(r) = W_{\mathrm{Wick}}(r)$ for all $r > 0$, contrary to Lemma 18.7. □

For the photon comparison use Hilbert tensor products.

**Definition 18.10** (Maxwell Fock space)**.** For $n \in \mathbb{N}$ colours put

$$d\sigma(\mathbf{k}) = \frac{d^3\mathbf{k}}{2|\mathbf{k}|}, \qquad \Pi_{ab}(\mathbf{k}) = \delta_{ab} - \frac{\mathbf{k}_a \mathbf{k}_b}{|\mathbf{k}|^2},$$
$$\mathcal{K}_n = L^2(d\sigma; \mathbb{C}^n \otimes \mathbb{C}^3), \qquad \mathcal{F}_n = \bigoplus_{k \geq 0} \mathcal{K}_n^{\otimes k}.$$

The density and projector are zero at zero momentum. Physical vectors are symmetric and transverse in each slot. Translations multiply sector $k$ by $\exp\{iz \cdot \sum_j (\mathbf{k}_j, |\mathbf{k}_j|)\}$. The sector-zero unit is $\Omega_0$.

**Definition 18.11** (Coherent comparison vectors)**.** For Schwartz tests $h = (h_{\alpha,a})$, put

$$v_h(\alpha, a; \mathbf{k}) = |\mathbf{k}| \sum_b \Pi_{ab}(\mathbf{k}) \widehat{h}_{\alpha,b,+}(\mathbf{k}, |\mathbf{k}|), \qquad \mathcal{E}(h) = e^{-\|v_h\|^2/2} \bigoplus_{k \geq 0} \frac{v_h^{\otimes k}}{\sqrt{k!}}.$$

A comparison $U : \mathcal{H} \to \mathcal{F}_n$ is a complex linear isometry with physical image, $U\Omega = \Omega_0$, intertwining translations. Set $\psi_U(f) = U\Phi_\Theta(f)\Omega$.

Real tests have real components. Causal separation means $\mathsf{m}(x - y) < 0$ for every pair of points in their closed supports, where $\mathsf{m}(x) = (x^4)^2 - \sum_{a=1}^3 (x^a)^2$.

Relative locality would impose the equality excluded below. The proof uses spectral growth without restricting particle amplitudes.

**Lemma 18.12** (Fock countertest)**.** *For every $n \geq 0$ and every comparison $U$, for every ball about zero there are Schwartz $f$ compactly supported in that ball and real compactly supported Schwartz $h_{\alpha,a}$, each causally separated from $f$, such that*

$$\langle \mathcal{E}(-h), \psi_U(f) \rangle \neq \langle \psi_U(\bar{f}), \mathcal{E}(h) \rangle .$$

*Proof.* Use Definitions 18.10–18.11, Lemmas 18.1–18.6 and the translation covariance of Lemma 17.6. Fix a ball and suppose equality always holds.

*Sector amplitudes.* Set $p_j = (\mathbf{k}_j, |\mathbf{k}_j|)$ and $p_{\mathrm{tot}} = \sum_j p_j$. The convolution argument of Lemma 18.2 gives sector amplitudes $Y_{k,s}$, with $s$ listing colour and spatial indices:

$$\psi_U(f)_{k,s} = \widehat{f}_+(p_{\mathrm{tot}}) Y_{k,s}, \qquad \mu_\Theta = (p_{\mathrm{tot}})_* \sum_{k,s} |Y_{k,s}|^2 d\sigma^{\otimes k}. \tag{18.10}$$

Box masses are finite. Cauchy–Schwarz and polynomial spectral growth give integrability against sufficiently high inverse polynomials in each momentum.

The common sector unitary $i^k$ preserves the equality, physical vectors and translations. Write $\mathsf{y} = i^k Y_{k,s}$.

*Locality and doubling.* Expand the coherent equality in a real parameter multiplying $h$, justified by Cauchy–Schwarz and the exponential series. Polarization and complexification give independent complex slot tests. Transversality removes the projector in contractions with physical waves. Shrink nonnegative unit-integral $f$ to zero. Then $|\widehat{f}_+| \leq 1$ and $\widehat{f}_+ \to 1$; dominated convergence gives, for $g_j$ supported spacelike to zero,

$$\int \mathsf{y}(\boldsymbol{k}) \prod_j \left[ |\mathbf{k}_j| \widehat{g}_{j,-}(p_j) \right] d\sigma^{\otimes k} = \int \overline{\mathsf{y}(\boldsymbol{k})} \prod_j \left[ |\mathbf{k}_j| \widehat{g}_{j,+}(p_j) \right] d\sigma^{\otimes k}. \tag{18.11}$$

Here $\widehat{g}_-(p) = \widehat{g}_+(-p)$; the sector factors $i^k$ give exactly these signs.

Fix other slots and use $g(x,t) = a(x)\eta_\epsilon^{(b)}(t)$, with $\eta_\epsilon$ approximating unit mass and $b = 0, 1$. If $0 \notin \operatorname{supp} a$, these are spacelike for small $\epsilon$. Thus both Cauchy data of (18.11) are supported at zero. Weighted integrability bounds their order by $J$, independently of the other tests.

Each datum depends only on the $J$-jet: multiplication by a cutoff in $B(0,\epsilon)$ leaves the distribution unchanged. For a zero-$J$-jet test, Taylor's formula bounds derivatives through order $J$ by $O(\epsilon)$; let $\epsilon \downarrow 0$.

The massless wave equation separates the frequency branches from these two Cauchy data away from zero momentum. After smearing other slots, each branch is integrable, bounded and holomorphic in its timelike tube, with continuous real boundary. Vanishing persists under small translations of the other spacelike tests. Schwarz reflection and holomorphic uniqueness [26, Theorems 11.14 and 10.18] make the zero set invariant under adding $\mathbb{R}u$ for every future timelike $u$. These vectors span $\mathbb{R}^4$, giving vanishing at every real translation. Successive Fourier inversion [26, Theorem 9.11] of the smeared integrable amplitudes gives zero almost everywhere. Countably many shrinking bumps remove the other tests: their transforms are eventually nonzero at each momentum.

Let $D_j\mathsf{y}(\boldsymbol{k}) = \mathsf{y}(\mathbf{k}_1, \dots, 2\mathbf{k}_j, \dots, \mathbf{k}_k)$. In the time-order-$b$ Cauchy datum ($b = 0, 1$), this substitution gives $2^{-3-b}$ times the datum at $a(x) \mapsto 8a(2x)$, from $d\sigma(2\mathbf{k}) = 4d\sigma(\mathbf{k})$ and $|\mathbf{k}|^{1+b}$. Spatial jets of order $l$ scale by $2^{3+l}$, giving eigenvalues $2^{l-b}$. The finite jets and branch argument therefore give, for some $J'$,

$$\prod_{a=0}^{J'-1} (D_j - 2^{a-1})\mathsf{y} = 0 \qquad \text{in every slot } j\,. \tag{18.12}$$

*Growth.* For $R = 2^l$ let $M_\mathsf{y}(R) = \int_{|\mathbf{k}_j| \le R} |\mathsf{y}|^2 d\sigma^{\otimes k}$. By (18.10) and the trace law,

$$M_\mathsf{y}(2^l) \le \mu_\Theta\{\mathbf{p}^4 \le (k+1)2^l\}, \qquad 2^{-8l} M_\mathsf{y}(2^l) \longrightarrow 0\,. \tag{18.13}$$

For the limit, bound the energy mass by $e\, G_\Theta(((k+1)2^l)^{-1})$.

Lagrange interpolation in the commuting $D_j$ splits $\mathsf{y}$ into joint eigencomponents $D_j\mathsf{y}_{\boldsymbol{a}} = 2^{a_j - 1}\mathsf{y}_{\boldsymbol{a}}$. Each is a finite combination of doubled copies, so retains (18.13). Moreover $M_{\mathsf{y}_{\boldsymbol{a}}}(2^l) = 2^{2l\sum_j a_j} M_{\mathsf{y}_{\boldsymbol{a}}}(1)$, since $d\sigma(2\mathbf{k}) = 4d\sigma(\mathbf{k})$. If $a_j = 0$, contraction preserves box mass, contradicting local square integrability and absence of an atom at zero unless the component vanishes. Equation (18.13) excludes $\sum_j a_j \ge 4$. Thus $a_j \ge 1$, $\sum_j a_j \le 3$, and $k \le 3$, with box mass $\le CR^6$ for dyadic $R \ge 1$. The vacuum sector vanishes by Lemma 18.1 and $U\Omega = \Omega_0$.

Summing (18.10) gives $\mu_\Theta\{\mathbf{p}^4 \le R\} \le CR^6$ at dyadic $R \ge 1$. Multiplication by $R^{-8}\log^2 R$ gives zero in the limit, contradicting Lemma 18.6 on a nonnegative compact test in $\{0 < \mathbf{p}^4 < 1,\ \mathsf{m}(\mathbf{p}) > 0\}$. □

For a probability space $(Z, \lambda)$, let each $P_z : \mathcal{F}_n \to \mathcal{F}_n$ preserve physical inner products and fix $U\mathcal{H}$ pointwise, without assuming linearity. Nonintegrable scalar functions have integral zero (Definition 6.13).

**Lemma 18.13** (Localized averaged countertest)**.** *For every such $n, U, (Z,\lambda), (P_z)$ and $r > 0$, there are open $A, B$, with $\overline{A} \subset B(0,r)$ and $\mathsf{m}(x-y) < 0$ on $\overline{A} \times \overline{B}$, and tests $f, h$ as in Lemma 18.12, supported respectively in $A, B$, such that*

$$\int_Z \langle P_z\mathcal{E}(-h), \psi_U(f)\rangle\, d\lambda(z) \ne \int_Z \langle \psi_U(\bar{f}), P_z\mathcal{E}(h)\rangle\, d\lambda(z)\,.$$

*Proof.* For physical $v$ and $w \in U\mathcal{H}$ (Definition 18.11), $\langle P_z v, w\rangle = \langle P_z v, P_z w\rangle = \langle v, w\rangle$. Thus averaging preserves both integrands.

Apply Lemma 18.12 in $B(0, r/2)$. Compactness and strict spacelike separation give support neighbourhoods $A, B$ with spacelike separated closures and $\overline{A} \subset B(0, r)$. □

# 19 Products and contact subtractions

A cluster of separated insertions can be replaced, to any prescribed degree, by a finite sum of fields at one point. The coefficients must be chosen before the exterior insertions and the phase. We construct them from finitely many correlations of the mixed table, impose the rotation and reality laws by averaging, and remove the lower degrees successively. Theorem 7.6 follows from this construction. Lemmas 19.11–19.16 give its integrated and nested forms, used in Theorem B.

The lattice estimate uses the signed local and graded quotient bounds of Lemmas 19.1 and 19.2. Blocking loses marker weights when insertion points approach one another. Grouping nearby markers bounds this loss by a power of the ratio of insertion and fusion scales. Taylor subtraction and coefficient transport produce one field and a small remainder. Reflection positivity converts its chessboard root into a norm bound in the commutative compression algebra, valid in every phase. Derivative induction removes the grid restriction; tree induction allows several clusters to shrink independently.

For contacts, a pointwise bound away from collisions is insufficient. We subtract normal Taylor jets of the test function, with a different order for each retained field degree. Partial collisions require the same operation inside each smaller cluster. The resulting finite recursion gives the distribution in Theorem 7.7; its absolute convergence follows by summing normal shells. The contact projections have their own probes and clearances, while using the same original fields and correlation table as the product expansion.

Throughout, the lattice insertions and quotient fields are those of Definitions 3.9 and 4.9; inclusions and derivatives are those of Lemma 15.4. We use the rotations of Lemma 16.11, the centered bounds of Lemma 14.12, and the compression algebra $\mathfrak{A}_0$ and phase tables of Lemmas 17.4–17.5. Write $|\boldsymbol{A}|_{\deg} = \sum_i \deg A_i$ and $\zeta \in \mathsf{X} \sqcup \{\mathrm{mix}\}$, as in Definition 7.5.

## Inputs from the marked expansion and the quotient

We first establish two estimates needed when the field to be extracted varies with the cutoff. Throughout this subsection, a local marked expression has the form of Definition 14.4 and the regularity, zero radial jets and forced supports of Lemma 14.9. Write

$$\mu(x) = \sum_{\alpha} |x_\alpha|, \qquad x^\circ = x - x_0\mathbf{1}, \qquad \mathfrak{r}_i = \mathcal{E}_i + \widetilde{\epsilon}_i^{-1/2}\mathcal{K}_i\,.$$

The last quantity is the weighted remainder mass of Lemma 14.8. Fix an output degree $t$. The diagonal rescaling $(\mathsf{d}_i x)_\alpha = \ell_i^{t-\deg\alpha} x_\alpha$ accounts for the engineering degrees inside $V_t$. Put $C_i = C_{J_i}$ in this subsection. The normalized columns of Lemma 14.6 are denoted by $\mathcal{H}_i(b)$, their coefficient maps by $\mathsf{X}_i$, and their remainder bounds are $C_i\mu((\mathsf{X}_i b)^\circ)$ by Lemma 14.8.

Let $K_j \geq 1$ be the nondecreasing stage bound of item 1 in Lemma 14.1. The degree rescaling above is the map $\mathsf{S}_L$ in the proof of Lemma 14.5. For an input coefficient $v$ and remainder mass $q$, its averaged marked estimate gives an output coefficient $u$ and remainder mass $r$ with

$$\begin{aligned} \mu(u - M_i\mathsf{d}_i v) &\leq \epsilon_i[\mu(v^\circ) + q], \\ r &\leq K_{J_{i+1}}\mu((\mathsf{d}_i v)^\circ) + \epsilon_i q, \qquad \epsilon_i = \ell_i^{-3/8}. \end{aligned} \tag{19.1}$$

Here $M_i$ is the leading polynomial map, fixes the unit, preserves the nonconstant subspace, and satisfies $\|M_i^{\pm 1}\| \le K_{J_{i+1}}$. These estimates follow from (14.14) and the symmetry averaging in that proof.

Write $C_l = C_{J_l}$. In the threshold order of Section 9, the construction chooses

$$64K_{J_l+1}^2 \le C_l, \quad 10^5 K_{J_{l+1}+1}^6 (1+C_{l+1})^2 \le \ell_l^{1/8}, \quad 1+2C_l \le \ell_l^{1/40}. \tag{19.2}$$

The column choice imposes the first inequality; the ratio choice imposes the other two. Stage growth gives $J_{l+1} \le J_l + 1$.

The following estimate retains the signed difference produced by subtracting a column. For a continuation on $[k_*, m]$, let $x_l, r_l$ be its coefficients and remainder bounds. Write $q_{\mathrm{fin}}$ for the combined coefficient and remainder bound of the original expression at $m$, before the affine column subtraction.

**Lemma 19.1** (Signed local continuation)**.** *At sufficiently late levels, normalized columns plus a zero-coefficient remainder admit an affine coarse continuation satisfying* (19.1) *and*

$$H_i = \ell_i^t \mathcal{B}^{(i)}_{\beta_{N,p,i+1}} H_{i+1}\,.$$

*If $x_{k_*} = 0$, then, for $k_* \le l \le m$,*

$$r_l + \mu(x_l) \le 4(1+C_m) q_{\mathrm{fin}} (a_m/a_l)^{1/8}, \qquad r_m \le 2(1+C_m) q_{\mathrm{fin}}\,.$$

The proof derives the interval bound from the signed one-step inequalities.

*Proof.* We use the marked integration of Lemma 14.5 and (14.14), obtained from Lemmas 13.4, 13.8 and 13.12, the local format of Lemma 14.9, and the normalized columns of Lemmas 14.6–14.8, with the insertion normalization of Definition 3.9.

Let $R$ have coefficient zero and remainder bound $q_0$. Its regular terms have zero radial jets through $t$; its forced terms retain their eligibility events. In the marked integration of Lemma 14.5, the former gain an inverse block factor before the connected sums, and the latter keep half the forced suppression. The stage constants precede the ratio and its entry threshold, so both resulting bounds $C_j/\ell_i + o_{\widehat{\beta}_i}(1)$ are at most $\eta_i = \ell_i^{-2/5}$. Apply (14.14) with zero primary coefficient. Sum the regular masses $g_a$ and forced masses $k_b/\widetilde{\epsilon}_{i+1}^{1/2}$ before taking the output norm. Their sum is at most $q_0$, so the output coefficient and remainder are each at most $\eta_i q_0$. Summability in Lemma 13.4 justifies termwise integration and retains the exact density identity. The last inequality of (19.2) gives $(1+2C_i)\eta_i \le \epsilon_i$. If $Z$ denotes the resulting local expression and $z$ its coefficient, then

$$H(Z) = \ell_i^t \mathcal{B}^{(i)}_{\beta_{N,p,i+1}} H(R), \qquad \mu(z) \le \eta_i q_0, \qquad \mathfrak{r}_i(Z) \le \eta_i q_0\,.$$

Write the fine affine family as $\mathcal{H}_{i+1}(a+b) + R$, with $b$ the free parameter. Set $c = \mathsf{X}_i^{-1} z$ and replace it at the coarse level by

$$\mathcal{H}_i(a+c+b) + R', \qquad R' = Z - \mathcal{H}_i(c)\,.$$

The coefficient of $R'$ is zero, and $\mathfrak{r}_i(R') \le (1+C_i)\eta_i q_0$. Linearity of the column density gives $H(\mathcal{H}_i(a+c+b)) + H(R') = H(\mathcal{H}_i(a+b)) + H(Z)$. Together with the exact column identity of Lemma 14.6, this proves the asserted density identity for every $b$, including all forced terms. The supports remain fixed in $b$ by Lemma 14.8.

For the column alone use (19.1), with fine remainder bound $q_{\mathrm{col}}$. Adding $Z$ changes the coefficient by at most $\eta_i q_0$. For the remainder, recentering costs at most $C_i\eta_i q_0$, and shifting the column parameter costs at most another $C_i\eta_i q_0$. The total extra cost is $(1+2C_i)\eta_i q_0 \le \epsilon_i q_0$. With $q = q_{\mathrm{col}} + q_0$, these are the asserted bounds. Repeating this construction from the fine end to the chosen coarse end produces the affine family on the whole interval.

For later use put $U = \mu(u)$, $Y = \mu(\mathsf{d}_i v)$, $K = K_{J_{i+1}}$ and retain $r, q$ for the remainder bounds. Applying $M_i^{-1}$ and using $\mu(v^\circ) \le Y$ gives

$$Y \le K[U + \epsilon_i(Y+q)], \qquad r \le KY + \epsilon_i q\,. \tag{19.3}$$

For an interval $[k_*, m]$, choose the free affine parameter to cancel its coefficient at $k_*$; this uses the invertibility of $\mathsf{X}_{k_*}$. The signed residual then starts with coefficient zero. In step $l$ put $K = K_{J_{l+1}}$ and, in (19.3), put $A_l = 32(1+C_l)^2$ and induct from $k_*$ toward $m$. Equation (19.2) implies

$$A_l \ge 4K^2, \quad 4K\epsilon_l \le 1, \quad 4KA_{l+1}\epsilon_l \le 1, \quad (4K^2+1)\ell_l^{1/8}\epsilon_l \le 1.$$

From $A_l U \le r \le KY + \epsilon_l q$ and $A_l \ge 4K^2$ obtain $4KU \le Y + \epsilon_l q$. Substitution into (19.3) gives $Y \le 4K\epsilon_l q$. Thus $A_{l+1}\mu(x_{l+1}) \le q$, since $\mu(x_{l+1}) \le Y$, and $\ell_l^{1/8} r \le q$. Iteration gives the discount $(a_m/a_{k_*})^{1/8}$. At the finest level the affine subtraction has $r_m \le (1+C_m)q_{\mathrm{fin}} + C_m\mu(x_m)$. The cone gives $2C_m\mu(x_m) \le r_m$, hence $r_m \le 2(1+C_m)q_{\mathrm{fin}}$ and $r_m + \mu(x_m) \le 4(1+C_m)q_{\mathrm{fin}}$. Applying the iterated discount between any $l$ and $m$ proves the interval bound. □

The next estimate recovers the output-degree power after passing to the quotient. Fix any norm on $F_t$. Put $r_t = m_{\mathrm{ref}}(t)$ and $\|v\|_m = \mu(x_m(v))$, where $x_m(v) = x_{N,p,m;t}(v)$ is the transferred coefficient of Definition 3.9. A refinement below is a nontrivial refinement of the ordered cutoff limits (14.22); it may depend on the cluster geometry.

**Lemma 19.2** (Graded quotient bound)**.** *For $0 \le i \le t$ there are positive $C_{t,i}(m)$, independent of the refinement and label family, with $\log C_{t,i}(m)/s_m \to 0$. If labels $v_{N,p} \in V_t$ converge in $V_t$ to $v$ on such a refinement and eventually $\|v_{N,p}\|_m \le \epsilon_{\mathrm{bd}}$, then*

$$\mathrm{dist}_{F_t}([v], \jmath_{i-1,t}F_{i-1}) \le C_{t,i}(m)a_m^{-(t-i)}\epsilon_{\mathrm{bd}}\,. \tag{19.4}$$

*Here $[v] \in F_t$ is the class of $v$, $F_{-1} = 0$, and $m$ is sufficiently large.*

*Proof.* We use the triangular comparison in Lemma 14.5 and the marked bounds of Lemma 14.8, and the physical quotient of Definition 4.9 with the corrected inclusions of Lemma 15.4. The leading polynomial matching reads the prescribed engineering jets of the local sections of Lemma 10.2. A basis section has no lower engineering jet, and its jet in its own degree is its specified basis polynomial. Consequently, for the leading map,

$$(M_l)_{\alpha\beta} = 0 \quad (\alpha \ne \beta,\ \deg\alpha \le \deg\beta).$$

The diagonal entries are the nonzero fine-to-coarse normalization factors. The full coefficient map differs from this leading map by the marked error. Write $P$ for coordinate projection onto engineering degrees at most $i-1$, first assuming $i > 0$. The leading map $M_l$ and its inverse preserve $\ker P$: triangularity gives preservation by $M_l$, and injectivity on this finite-dimensional subspace gives preservation by its inverse. Consequently $\mu(Px) \le K_{J_{l+1}}\mu(PM_l x)$. The degree rescaling expands $Px$ by at least $\ell_l^{t-i+1}$, whereas it expands $x - Px$ by at most $\ell_l^{t-i}$. The marked error is bounded by $\epsilon_l\mu(x^\circ)$.

The two resulting estimates are

$$\mu(x_{l+1}) \le C_{l+1}\mu(Px_{l+1}) \implies \begin{cases} \mu(x_l) \le C_l \mu(Px_l), \\ \ell_l^{t-i+1}\mu(Px_{l+1}) \le C_{l+1}\mu(Px_l), \end{cases}$$
$$Px_{l+1} = 0 \implies \mu(x_l) \le C_{l+1}\ell_l^{t-i}\mu(x_{l+1})\,.$$

To obtain the first line, put $Y = \mu(P\mathsf{d}_l x_{l+1})$. The low-cone hypothesis and $C_{l+1} \le \ell_l$ give $\mu(\mathsf{d}_l x_{l+1}) \le 2Y$. Applying the inverse leading map to the low coordinates and using $2K_{J_{l+1}}\epsilon_l(1 + C_{l+1}) \le 1$ gives $Y \le 2K_{J_{l+1}}\mu(Px_l)$. The full coefficient is at most $3K_{J_{l+1}}Y$, so $64K_{J_{l+1}}^2 \le C_l, C_{l+1}$ gives both conclusions. In the second line use $\mu(\mathsf{d}_l x_{l+1}) \le \ell_l^{t-i}\mu(x_{l+1})$ and $K_{J_{l+1}} + 1 \le C_{l+1}$. These numerical inequalities follow from (19.2).

At the bare level take the plane $PV_t$. Its transported image remains in the low cone by the first line. The map from this plane to its low coordinates at any level is injective by its expansion bound, hence onto $PV_t$. Subtracting a transported vector with the same low coordinates at level $m$ therefore leaves zero low coordinates and increases the full size by at most $1 + C_m$. Repeat the subtraction at every next coarser level, applying the second line between subtractions. This gives a vector $y_{N,p}$ in the transported bare low plane with

$$\|v_{N,p} - y_{N,p}\| \le G_m(1 + C_m)(a_{r_t}/a_m)^{t-i}\epsilon_{\mathrm{bd}}, \qquad G_m = \prod_{l<m}(1 + C_l)C_{l+1}\,.$$

Allowing a fixed norm-equivalence factor, put $\mathcal{J}_m = \prod_{l<m} C_{l+1}$. The first-line expansion, iterated from the bare level, shows that bounded vectors $y_{N,p}$ in this plane have bare coefficients bounded by $a_N^{t-i+1}e^{o(s_N)}$. Multiplying the same polynomial by $a_N^{i-1}$ instead of $a_N^t$ therefore gives a degree-$(i-1)$ local observable with subpower healthy derivative and all-configuration supremum bounds. Its stencil is unchanged. To match this observable to the selected lower-degree insertions, apply Lemma 14.14: it gives admissible degree-$(i-1)$ coefficients $c_{N,p}$ whose insertions differ from these observables by a small error. Lemma 15.3 removes that error in every separated spectator test.

To pass to a physical lower field, let $\Pi_{N,p}$ be the equation projections of Lemma 14.16. The normalized equation remainder

$$\frac{c_{N,p} - \Pi_{N,p}c_{N,p}}{1 + \|\Pi_{N,p}c_{N,p}\|}$$

remains admissible: its first term is a bounded scalar multiple of $c_{N,p}$ and its second is bounded. Lemma 14.17 and Lemma 15.3 make every separated spectator test of it tend to zero. Take a simultaneous cluster point $(\lambda, z)$ of $((1 + \|\Pi c\|)^{-1}, \Pi c/(1 + \|\Pi c\|))$ on a further refinement. The projection convergence (15.8) gives $\Pi_{i-1}z = z$ and $\lambda + \|z\| = 1$. If $y$ is a limit of the bounded labels $y_{N,p}$, every spectator functional has limit $L(z) = \lambda L(y)$. If $\lambda = 0$, the all-spectator identity and (15.11) put $z$ in $\ker \Pi_{i-1}$, contradicting $\|z\| = 1$. Thus $z/\lambda \in V_{i-1}$ has exactly the limiting correlations of $y$. The corrected inclusions of Lemma 15.4 and the all-spectator definition of the null space now give $[y] \in \jmath_{i-1,t}F_{i-1}$. Boundedness of $y_{N,p}$ follows from the displayed distance bound and convergence of $v_{N,p}$. If these bounds hold only on the chosen refinement, set $y_{N,p} = 0$ off the set where they hold; this extends the bounded plane family to the original tails without changing its limit. Passing to this limit proves the distance estimate.

For $i = 0$ no plane is removed. Iterating the reverse coefficient estimate of Lemma 14.8 gives $\|v\| \le \mathcal{J}_m(a_{r_t}/a_m)^t\epsilon_{\mathrm{bd}}$ and hence the same bound in $F_t$. Finally $\log \mathcal{J}_m = o(s_m)$ by row 5 of Section 9 and Lemma 14.1. Since $C_l \ge 2$, $G_m C_m \le (\mathcal{J}_m)^3 C_0$. Thus $\log G_m = o(s_m)$ as well. Fixed reference-spacing and quotient-norm factors are absorbed in $C_{t,i}(m)$. No estimate depends on the refinement used to take the limits. □

## 19.1 Products and nested clusters

A probe word is a fixed finite list of fields at distinct nonzero points. Correlating a field at the origin against such a word gives a linear functional on $F_j$. The intersection of the kernels of all these functionals is zero, by the definition of $\mathcal{N}_j$. Finite dimensionality therefore permits a finite separating collection.

Choose the probe functionals recursively. Given biorthogonal fields $u_\alpha$ and probe functionals $\ell_\alpha$ below degree $j$, continue each $\ell_\alpha$ to $F_j$ by the same correlation formula; inclusions preserve that formula by Lemma 15.4. The retraction

$$\mathsf{t}_j A = \sum_{\deg\alpha<j} \ell_\alpha(A)\jmath_{\deg\alpha,j-1}u_\alpha, \qquad \mathsf{t}_j\jmath_{j-1,j} = 1$$

allows a section $s$ of the new quotient to be replaced by $s-\jmath\mathsf{t}_j s$. Lift a quotient basis with this section and realize its dual by finite combinations of probe functionals, which span $F_j^*$. The enlarged system is biorthogonal. In particular $\ell_\alpha$ vanishes on $\jmath F_{\deg\alpha-1}$. For $j=0$ take $u_0=\mathbf{1}$. Let $\mathcal{P}_j$ be the union of the finite probe sets used through degree $j$, with index $\beta$. Write $\ell_\alpha(A)=\sum_\beta m_{\alpha\beta}S(A(0),B_{j,\beta})$ in this finite collection.

**Definition 19.3** (Finite correlation projection)**.** Define the decoder on arbitrary data by

$$D_j(y)=\sum_{\deg\alpha\le j}\left(\sum_{\beta\in\mathcal{P}_j} m_{\alpha\beta}y_\beta\right)\jmath_{\deg\alpha,j}u_\alpha\,.$$

It satisfies $D_j((S(A(0),B_{j,\beta}))_\beta)=A$ for $A\in F_j$. For a word $w$ separated from these probes, set

$$P_j^{\mathrm{raw}}w=D_j\big((S(w,B_{j,\beta}))_{\beta\in\mathcal{P}_j}\big)\,.$$

This extracts a field from finitely many correlations using the fixed biorthogonal coordinates.

Here and below a word is a finite product of labelled insertions; sums of words are read linearly. Rotations act on every position and label, and conjugation acts on labels and scalar coefficients. Translation of a word by $u$ is denoted by $\tau_u$.

**Definition 19.4** (Symmetric projection)**.** When every rotation of $w$ is separated from every probe in $\mathcal{P}_j$, set

$$A_jw=\int_{O(4)} R_j(h)P_j^{\mathrm{raw}}(h^{-1}w)\,dh, \qquad B_jw=\tfrac12\big(A_jw+\overline{A_j\bar w}\big)\,.$$

The Haar measure has mass one. Outside this domain set both maps to zero. At center $o$, apply these formulas to $\tau_{-o}w$. These averages impose rotation covariance and reality.

Let $I_qA=A(0)$ for $A\in F_q$. The inclusions $\jmath_{i,j}$ identify the same field in different degrees.

**Definition 19.5** (OPE coefficients)**.** Put $L_{i,q}=B_iI_q$ and define

$$Q_0=1, \qquad Q_j=1-\sum_{i<j}\jmath_{i,j}Q_iL_{i,j}, \qquad E_j=\bigcap_{i<j}\ker(Q_iL_{i,j})\,.$$

For the chosen real basis $(O_{j,\lambda})_{\lambda\in\mathrm{Ch}_j}$ of $E_j$, let $\omega_{j,\lambda}:E_j\to\mathbb{C}$ be its coordinate functionals and set

$$c_{j,\lambda}(\boldsymbol{A};x,o)=\omega_{j,\lambda}\left(Q_jB_j\left[\prod_i A_i(x_i-o)\right]\right)\,.$$

The recursion removes lower degrees before reading coordinates and is independent of arity and truncation.

The following splitting makes the coefficients a single choice for all later estimates. The proof uses the biorthogonal choices just made.

**Lemma 19.6** (One channel prescription)**.** *The real fields $O_{j,\lambda}$ form complex bases of conjugation-invariant complements:*

$$F_j = \jmath_{j-1,j} F_{j-1} \oplus E_j, \qquad E_j = \operatorname{span}_{\mathbb{C}}\{O_{j,\lambda} : \lambda \in \mathrm{Ch}_j\}, \qquad F_{-1} = 0\,.$$

*The degree-zero vector spans the unit line and need not equal the unit. All coefficients are fixed before spectators, states and truncations.*

*Proof.* Use the field calculus of Lemma 15.4 and the preceding biorthogonal construction. Its reconstruction formula fixes every field of $F_j$. Continuing an old functional means using the same finite correlations, since $\iota_j \jmath_{q,j} = \iota_q$; no extension of a functional is chosen arbitrarily.

For $i \geq q$, this reconstruction and the rotation and reality laws give $L_{i,q} = \jmath_{q,i}$. Also $L_{i,j}\jmath_{q,j} = L_{i,q}$ for $i < q \leq j$: both sides evaluate the same single-field word before averaging. For all $q \leq j$, induction in $j$ now gives

$$Q_j L_{j,q} = 0 \ (q < j), \qquad \sum_{i \leq j} \jmath_{i,j} Q_i L_{i,q} = \jmath_{q,j}\,. \tag{19.5}$$

For $q = j$ the second identity is the recursion; for $q < j$, compose that recursion with $\jmath_{q,j}$ and use the compatibility just proved and the induction hypothesis. In the same induction assume $Q_k F_k \subset E_k$ and $Q_k = 1$ on $E_k$ for $k < j$. Applying $Q_i L_{i,j}$ to $Q_j A$ then cancels its matching $i$ term; terms with $k < i$ vanish because the $i$ cut kills lower fields, and terms with $i < k < j$ vanish because $Q_k A \in E_k$. Thus $Q_j F_j \subset E_j$. On $E_j$, the defining sum for $Q_j$ is zero, so $Q_j = 1$. Its kernel is exactly $\jmath F_{j-1}$ by (19.5). Haar invariance and simultaneous input and output conjugation show that every cut intertwines rotations and conjugation. Thus $E_j$ is invariant under both. Its fixed real subspace spans it, since $v = (v + \bar{v})/2 + i(v - \bar{v})/(2i)$; a real basis is also a complex basis. This proves the stated splitting and the simultaneous choice. □

For $T \to \infty$ write $f \preccurlyeq e^{bT}$ when, for every $\tau > 0$, there is $T_0$ such that for every $T \geq T_0$ an ordered cutoff tail gives $\|f\| \leq e^{(b+\tau)T}$, uniformly in the cutoff and volume on that tail. Products add exponents and finite sums take their maximum. Factors of exponent zero are called subpower. A level $k$ has height $b$ when $s_k/T \to b$; thus $a_k = e^{-bT+o(T)}$. Lemma 14.1 supplies these levels by taking the first with $s_k \geq bT$, or $s_k \geq \sqrt{T}$ if $b = 0$.

Introduce one square-zero variable for each insertion in each reflected cell. A live marker is one distinct insertion group; its positive weight multiplies every coefficient containing it. If $F_{X,A}$ is a coefficient with support $X$ and marker set $A$, its weighted majorant is

$$\sum_{X,A} e^{\sigma \operatorname{cov}(X)} \|F_{X,A}\| \prod_{\iota \in A} w_\iota\,.$$

Here $\|F_{X,A}\|$ is the regular derivative or forced majorant of Definitions 3.3–3.4; at a support ball use its anchored sum. A grouped coefficient carries the product of its live marker weights, denoted by $\omega$.

For the marked expansion put

$$\delta_l^{\mathrm{mark}} = \widehat{\beta}_{l+1}^{-2}, \qquad \phi_l = \widehat{\beta}_{l+1}^{-1}, \qquad \Xi_l = \exp\sqrt{\widehat{\beta}_{l+1}}, \qquad z_{t,l} = \widehat{\beta}_l^{-\nu_t}.$$

The fixed integer $\nu_t$ bounds the local degree-$t$ polynomial losses, and $z_{t,l}C_l \le 1$ at entry. The four factors pay respectively for support enlargement, regular transport, Taylor extraction and insertion normalization. Their logarithms and $\log C_l$ are $o(s_l)$. For an initial weight $\varpi > 0$ at $k_*$ define its later weights by

$$\varpi_{k_*} = \varpi, \qquad \varpi_{l+1} = \phi_l \varpi_l \quad (l \ge k_*).$$

The marked entry choices require $\phi_l \ell_l^{t+1} z_{t,l} \le z_{t,l+1}$ for $l \ge r_t$. This follows the zero-jet regular remainder through the integration.

Write $\mathfrak{b}_l$ for the support radius $B_l$ of Section 9. Fix a reflected cell at level $k$ and choose

$$k_* = \min\{l \ge k : \mathfrak{b}_l^2 \le a_k/a_l\}, \qquad \mathcal{S}_k = \exp\{(a_k/a_{k_*})^4 \mathsf{v}_{k_*} + 5\},$$

where $\mathsf{v}_l = q_l$ is the per-site logarithmic bound (14.1). These quantities ensure that the marked stencils fit in the cell and bound its density normalization.

For a grouped coefficient at level $m+1$ of weight $\omega > 0$, put

$$q_m = 4\Xi_m a_{m+1}^t/\omega, \qquad \mathfrak{E}_m = 8(1+C_m)q_m\,.$$

The terminal weight $\varpi$ must satisfy

$$\begin{aligned} \varpi(a_m/a_{k_*})^{t+1} z_{t,m}/z_{t,k_*} &\le \delta_m^{\mathrm{mark}}\omega/(16\Xi_m), \\ 32(1+C_m)\varpi\, a_{k_*}^{-t}\mathfrak{E}_m(a_m/a_{k_*})^{1/8} &\le z_{t,k_*}/2. \end{aligned} \tag{19.6}$$

These conditions pay for extraction at $m$ and for the continued residual, respectively.

**Lemma 19.7** (Marked extraction and insertion costs)**.** *A grouped coefficient of weight $\omega$ has a degree-$t$ local extraction with coefficient and remainder each at most $q_m$. With $k_*$ as above, sufficiently late, and whenever (19.6) holds, the matched field has coefficient size at most $2\mathfrak{E}_m$. Its interior reflected remainder in the level-$k$ cell has root at most $2\mathcal{S}_k/\varpi$.*

*Proof.* We use the norms and insertions of Definitions 3.3–3.4 and 3.9, the marked integration of Lemma 14.5, its convergent majorants from Lemmas 13.4 and 13.11, and the local format of Lemma 14.9. Divide a grouped coefficient by its available weight $\omega$. Extract its radial polynomial through $t$ and write the regular complement by the integral Taylor formula. The finite derivative sums, coefficient conversion and remainder constants are at most $\Xi_m$ after the entry levels have been chosen: their logarithms are bounded by stage constants and powers of $\log \widehat{\beta}_{m+1}$, whereas $\log \Xi_m = \sqrt{\widehat{\beta}_{m+1}}$. Multiplication by $a_{m+1}^t$ gives $q_m$. The factor four bounds the polynomial and complement separately. Forced terms keep their original majorants and the forced weight of Definition 14.4.

The combined cost is at most $2q_m$. Subtract the column matching the continuation at $k_*$. Lemma 19.1 gives, for $k_* \le l \le m$,

$$\mathfrak{r}_l + \mu(x_l) \le \mathfrak{E}_m(a_m/a_l)^{1/8}, \qquad \mu(x_m^{\mathrm{field}}) \le 2\mathfrak{E}_m.$$

The factor $4(1+C_m)$ multiplying $2q_m$ includes the finest residual and coefficient together. The entry requirement $\phi_l \ell_l^{t+1} z_{t,l} \le z_{t,l+1}$ gives the regular zero-jet gain: iterating $\varpi_{l+1} = \phi_l \varpi_l$ yields $\varpi_m \le \varpi(a_m/a_{k_*})^{t+1} z_{t,m}/z_{t,k_*}$. The first condition of (19.6) permits extraction at $m$. For the second, the level-$l$ ratio of required to permitted residual weight is bounded by its level-$k_*$ value times $(a_l/a_{k_*})^{7/8} \le 1$: use $\varpi_l \le \varpi(a_l/a_{k_*})^{t+1} z_{t,l}/z_{t,k_*}$, the residual bound $\mathfrak{E}_m(a_m/a_l)^{1/8}$ and divide

by $z_{t,l}$. Thus the condition at $k_*$ permits the signed residual and compensating columns throughout the interval. Their density identities are exact by Lemma 19.1.

In the square-zero variables defined above, a marked polynomial is measured by the sum of the absolute coefficients times the product of their assigned weights. Consequently the coefficient with one residual per cell is at most $\varpi^{-1}$ per cell times that sum. Summing the unmarked majorants and dividing by the two positive density normalizations bounds the integral per cell by $\mathcal{S}_k$. The singleton, forced-profile, connected regular and eligible forced sums contribute at most $2, 1, 1, 1$ per cell, giving the additive constant five. Apply this to the original local difference and the correction representing its column, and use the root triangle inequality, obtained by expanding the reflected product and applying (15.4) to each term. This gives $2\mathcal{S}_k/\varpi$.

To verify that $k_*$ has the required height, let $\Phi_l = \max(1, \mathsf{v}_l)$ bound the normalization cost. The entry choices satisfy

$$\mathfrak{b}_{l+1}^2 \le 2\mathfrak{b}_l^2, \qquad (\ell_l \mathfrak{b}_l^2)^4 \Phi_{l+1} = o(s_{l+1}). \tag{19.7}$$

The level $k_*$ exists since $a_l \mathfrak{b}_l^2$ decreases by at least $2/3$ at each step. If $k_* = k$, the spacing ratio is one. Otherwise $k_* = l + 1$, and minimality gives $a_k/a_{k_*} \le \ell_l \mathfrak{b}_l^2$. Therefore $(a_k/a_{k_*})^4 \mathsf{v}_{k_*} = o(s_{k_*})$. Also $0 \le s_{k_*} - s_k = \log(a_k/a_{k_*}) \le (a_k/a_{k_*})^4 \Phi_{k_*} = o(s_{k_*})$. This proves $s_{k_*}/s_k \to 1$ and $\log \mathcal{S}_k = o(s_{k_*})$. At height zero it still gives $\log \mathcal{S}_k = o(T)$. Finally $\log \Xi_m = o(s_m)$ follows from $\widehat{\beta}_m = \bar{\beta} + \kappa_G s_m$ and $s_{m+1}/s_m \to 1$. The other displayed functions are fixed powers of $\widehat{\beta}$ or the subpower column cost of Lemma 14.1. □

Fix nonempty lattice labels $v_\iota \in V_{d_\iota}$, let $n_{\mathrm{cl}}$ be their number and put $W = \sum_\iota d_\iota$. Choose levels $k, m, j$ of heights $0 \le b < a < c$. The input points lie on the spacing-$a_j$ grid, in the centered level-$(m+1)$ block $[-a_{m+1}/2, a_{m+1}/2)^4$, and their separation is at least $\mathfrak{b}_j^2 a_j$, with the support radius $\mathfrak{b}_j$ defined above. The configurations belong to $\mathrm{Gen}_{n_{\mathrm{cl}}}$, and all stencils are interior to the level-$k$ cell. These conditions hold eventually for clusters of extent $e^{-xT+o(T)}$ and reciprocal separation at most $e^{xT+o(T)}$, where $a < x < c$. The chosen $k_*$ has height $b$ by (19.7).

**Lemma 19.8** (Lattice cluster replacement)**.** *For every output degree $t$, the preceding configurations admit labels $v_{N,p}^{\mathrm{out}} \in V_t$ on an ordered cutoff tail such that, with $\mathcal{Z} = Wc + 4n_{\mathrm{cl}}(c-a)$,*

$$\|v_{N,p}^{\mathrm{out}}\|_m \preccurlyeq e^{(\mathcal{Z}-ta)T},$$
$$\left\| \prod_\iota O_{N,p}(v_\iota, x_\iota) - O_{N,p}(v_{N,p}^{\mathrm{out}}, 0) \right\|_{\mathcal{T}_k(0)} \preccurlyeq e^{[\mathcal{Z}-(t+1/16)(a-b)]T}.$$

*The difference is interior. The bounds are uniform in cutoff and volume on that tail.*

*Proof.* Use Definition 14.4, Lemmas 14.8–14.9, and Lemmas 19.1 and 19.7. In the square-zero variables defined above, the coefficient of their product in $\operatorname{dens} \rho_N \prod_\iota (1 + z_\iota O_{N,p}(v_\iota, x_\iota))$ is the product insertion times the density. Until level $j$ the stencils are disjoint, so their marked integrations are separate.

Moving a marker to its representative through at most $h$ cover balls gives

$$\operatorname{cov}(X \cup \{y_\iota : \iota \in A\}) \le \operatorname{cov}(X) + h|A|. \tag{19.8}$$

If at most $\mathfrak{n}$ representatives occupy a ball, shrinking each affected weight by $\eta e^{-\sigma h}$ gives the original term times $\eta^{|A|}$. In the anchored sum, terms incident on the old ball use its original majorant; any new incidence can be assigned to an old marker on the joining path. There are at most $n_{\mathrm{cl}}$ old markers for each new representative and $\mathfrak{n}$ representatives. Taking $\eta \le [(1+\mathfrak{n})n_{\mathrm{cl}}]^{-1}$ pays for these

assignments. Unmarked terms keep their supports. Hence the enlarged weighted series is bounded by the original majorant series of Lemma 13.11.

Keep a maximal $\mathfrak{b}_l^2$-separated set of representatives at each level, retaining the distinguished one at the origin, and attach each discarded point to a retained neighbor. A fusion decreases the number of groups, so there are at most $n_{\mathrm{cl}} - 1$ fusions. The joining-cover bound can be taken as $h_l = 16[1 + \ell_l(\mathfrak{b}_l + \mathfrak{b}_{l+1} + 1)^2]^4$. The entry choices give $(\sigma_{J_{l+1}} + 1)h_l = o(s_l)$ and subpower occupancy. Choose one $\alpha_j \le 1$ below all the resulting enlargement allowances for $m \le l < j$. It is subpower: the logarithm of the running minimum is bounded by the maximum logarithmic cost; split this maximum into a fixed initial segment and a tail bounded by $\tau s_j$, then send $\tau$ to zero. This uses the entrance requirement on $h_l$ in rows 6–7 of Section 9 and Lemma 14.1. The factor $[(1 + \mathfrak{n})n_{\mathrm{cl}}]^{-1}$ above is included in this allowance $\alpha_j/8$.

Between fusions the one-mark integration costs at most $6\ell_l \le \ell_l^4$ in each weight. This is a bound on the marked regular and forced masses: for an incoming majorant $\mathcal{L}$, the rooted integration and local subtraction of Lemmas 13.4 and 13.12 give $2(p_l + \ell_l\mathcal{L}) + f_l$, where the stage choices give $2p_l \le \mathcal{L}$ and $2f_l \le \mathcal{L}$. Thus the output is at most $(2\ell_l + 3/2)\mathcal{L} \le 6\ell_l\mathcal{L}$. With $g$ groups remaining, their product weight at level $l$ is consequently at least

$$(a_j/a_l)^{4n_{\mathrm{cl}}}(\alpha_j/8)^{n_{\mathrm{cl}}(n_{\mathrm{cl}}-g)} \prod_{\iota} w_\iota. \tag{19.9}$$

Indeed a separated step multiplies the weight by $\ell_l^{-4g} \ge \ell_l^{-4n_{\mathrm{cl}}}$. A fusion to $g' < g$ uses at most $n_{\mathrm{cl}}$ factors $\alpha_j/8$, whereas $n_{\mathrm{cl}}(g - g') \ge n_{\mathrm{cl}}$.

At $m + 1$ all representatives have the same ancestor. The initial normalized-field weights can be chosen with $\prod_\iota w_\iota = e^{-WcT+o(T)}$ by Lemma 14.8. Thus (19.9) gives $\omega = e^{-ZT+o(T)}$ and $\mathfrak{E}_m = e^{(\mathcal{Z}-ta)T+o(T)}$. Choose

$$\varpi = \exp\{[(t + 1/16)(a - b) - \mathcal{Z}]T\}. \tag{19.10}$$

The second condition of (19.6) has exponent $-(a - b)/16 < 0$; the first has exponent $-\mathcal{Z} - 15(a - b)/16$, strictly below the exponent $-\mathcal{Z}$ of its right side. Both therefore hold, including every subpower factor. Lemma 19.7 gives the coefficient bound $2\mathfrak{E}_m$ and root bound $2\mathcal{S}_k/\varpi$ asserted here. All enlargements fit the prescribed cell margin. Reflection and conjugation preserve the majorants, so the same calculation applies to all cells and all cutoffs on the common tail. □

We next state the bounds when the exterior distances also shrink. Write $w = \prod_i A_i(x_i)$, $W = |\boldsymbol{A}|_{\deg}$ and $V = |\boldsymbol{B}|_{\deg}$. As $T \to \infty$, suppose the cluster lies within $e^{-\alpha T+o(T)}$ of $o$, its pairwise distances are at least $e^{-\alpha T-o(T)}$, and $\alpha > 0$. Exterior points are at distance at least $e^{-\beta T-o(T)}$ from every cluster point and $o$, and mutually at least $e^{-\upsilon T-o(T)}$ apart, where $0 \le \beta < \alpha$ and $\upsilon \ge \beta$. The field lists are fixed, or range over fixed compact sets of coefficients. For the raw decoder, let $\lambda_{i,\kappa}^{\mathrm{raw}}$ be its biorthogonal coordinates and $A_h$ its truncation through degree $h$. Write $\mathcal{R}_D^{\mathrm{raw}}$ for the resulting tested remainder in $\mathfrak{A}_0$.

The following estimates retain the exterior distance powers needed for nested clusters and contacts.

**Lemma 19.9** (Power-separated estimates)**.** *For every $i, D \geq 0$ and $j \leq h$, the stated geometry paths satisfy*

$$\lambda_{i,\kappa}^{\mathrm{raw}} \preccurlyeq e^{(W-i)\alpha T}, \qquad \mathcal{R}_{D}^{\mathrm{raw}} \preccurlyeq e^{[W\alpha-(D+1)(\alpha-\beta)+V\upsilon]T}, \tag{19.11}$$

$$Q_j B_j w - Q_j L_{j,h} A_h \preccurlyeq e^{(W-h-1)\alpha T}. \tag{19.12}$$

*The final coefficients and tested remainder of Definition 19.5 obey the same respective bounds, in norm before state evaluation.*

The proof converts lattice roots to algebra norms, removes the grid restriction by derivative descent, and then averages the decoder.

*Proof.* We use Lemma 19.8, the derivative calculus of Lemma 15.4, the root bounds of Lemmas 14.12 and 15.2, and the compression limit of Lemma 17.4. Write $W = |\boldsymbol{A}|_{\deg}$, $V = |\boldsymbol{B}|_{\deg}$ and $n_{\mathrm{cl}} = |I_{\mathrm{cl}}|$.

First convert roots into algebra norms. If a lattice difference has centered root at most $h$, and exterior insertions have roots at most $h_l$, apply (15.4) and Lemma 15.2 to $2n$ widely translated copies, alternately conjugated. Put the difference in a coarse cell and the exterior insertions in finer disjoint cells. Their stencils fit by Lemma 15.2; the bound is $(h\prod_l h_l)^{2n}$. For the positive time-axis OS Hamiltonian, let $E$ be its spectral resolution. Then

$$\|(e^{-tH_4^{\mathrm{OS}}} - P_4^0)\psi\|^2 = \int_{(0,\infty)} e^{-2t\lambda}\, d\langle E_\lambda \psi, \psi\rangle \longrightarrow 0\,.$$

Inserting this strong limit between bounded slabs in the compression construction of Lemma 17.4 identifies the limit of these translated copies with the moment of $(U^*U)^n$. The ordered limits and then infinite translation separation give

$$\int_{\mathsf{X}} |\xi(U)|^{2n}\, d\nu(\xi) \leq (h\prod_l h_l)^{2n}\,. \tag{19.13}$$

Here $U$ is the corresponding difference in $\mathfrak{A}_0$. If $|\xi(U)| > h\prod_l h_l$ anywhere, continuity and the full support of $\nu$ in Theorem 6.2 give an open set of positive measure where it exceeds this bound by a fixed factor. Letting $n \to \infty$ contradicts (19.13). Thus $\|U\| \leq h\prod_l h_l$. The same argument bounds a word by its insertion roots, giving crude exponent $W\alpha + V\upsilon$ on the stated geometries.

We give the sharper calculation first for the generic grid configurations defined above in a cluster of height $\alpha > 0$. Choose $0 < b < a < \alpha < c$, with exterior distance exponent $\beta < b$, and set $\mathcal{Z} = Wc + 4n_{\mathrm{cl}}(c-a)$. Apply Lemma 19.8 with an auxiliary degree $t \geq D$. For each fixed cluster geometry the output coefficients are bounded at fixed $m$; the reverse transfer inequality of Lemma 14.8 bounds them also at the reference level. Along a refinement of the ordered cutoff tails they converge to a field $A^{\mathrm{out}} \in F_t$. The preceding norm argument bounds the replacement error by exponent

$$\mathcal{Z} - (t + 1/16)(a-b) + V\upsilon\,.$$

Testing the same difference against the fixed probes bounds the raw coordinate errors. Reference-level coefficient convergence and Lemma 19.2 apply on this geometry-dependent refinement. The degree-$i$ raw coordinate annihilates $F_{i-1}$ by the biorthogonality before Definition 19.3, and is a fixed bounded functional on the quotient. With $\epsilon_{\mathrm{bd}} \preccurlyeq e^{(\mathcal{Z}-ta)T}$, (19.4) therefore bounds that coordinate by exponent $\mathcal{Z} - ia$.

Choose $t$ large enough that $(t+1/16)(a-b) \geq (D+1)a$ and also large enough for every coefficient being tested. Exact reconstruction of $A^{\mathrm{out}}$ by the prescription separates the degrees $i \leq D$ from $D < i \leq t$. An omitted degree contributes at most

$$\mathcal{Z} - ia + i\beta + V\upsilon \leq \mathcal{Z} - (D+1)(a-\beta) + V\upsilon\,.$$

The low-degree coordinate errors satisfy the same bound, by the chosen auxiliary horizon. We have therefore proved grid exponents $\mathcal{Z} - ia$ for raw degree-$i$ coordinates and $\mathcal{Z} - (D+1)(a-\beta) + V\upsilon$ for the raw remainder.

Run the crude, grid and derivative bounds uniformly over all characters $\xi \in \mathsf{X}$. The algebra $\mathfrak{A}_0$ is commutative, so $\|U\| = \sup_\xi |\xi(U)|$. To remove the grid restriction, consider these character evaluations for every input label list and every positional derivative. A derivative replaces its input label by a field of degree one higher, by Lemma 15.4. For suitable constants $c_0, d_0$ independent of the input degrees, the crude bound has exponent $\alpha W + c_0$ and the grid bound exponent $cW + d_0$. At all $W$ with $(c-\alpha)W \geq c_0 - d_0$, the crude bound already proves the grid exponent at arbitrary points. Descend in $W$. Move the configuration by $O(a_j)$ to generic level-$j$ grid points; the intervening segment remains separated at exponent $\alpha$. The induction hypothesis bounds its derivative by $c(W+1) + d_0$, and multiplication by $a_j = e^{-cT+o(T)}$ gives $cW + d_0$ for the change. This proves the descent for every slot and axis.

Now let $a \uparrow \alpha$ and $c \downarrow \alpha$, keeping $\beta < b < a$. Translation moves the origin to $o$. This proves (19.11) for arbitrary configurations. The remainder estimate is in $\mathfrak{A}_0$, and empty clusters give zero remainder.

Finally transfer these estimates to Definition 19.5. Let $A_h \in F_h$ be the raw truncation at an auxiliary degree $h$. In every rotated probe correlation, the difference between the cluster and $A_h(0)$ has exponent $(W-h-1)\alpha$. The marked norm bounds and multilinearity make (19.11) uniform on compact coefficient sets, hence on the rotation orbit. Haar averaging, conjugation and the fixed maps $Q_j$ preserve the exponent. This proves (19.12). With $h = j$, all lower raw modes vanish under $Q_j L_{j,j}$ by (19.5); the remaining modes have exponent $(W-j)\alpha$. With $h = D$, use (19.5) to reconstruct $A_D$, and telescope the raw remainder with the errors (19.12) for $j \leq D$. Testing an error in $F_j$ costs $j\beta + V\upsilon$, so their largest exponent is $W\alpha - (D+1)(\alpha-\beta) + V\upsilon$. □

The lattice replacement gives two small-radius estimates with different fixed data. For the coefficient fix $\boldsymbol{A}, M_{\mathrm{geo}}, \eta, \varepsilon, j, \lambda$; for the remainder fix $\boldsymbol{A}, \boldsymbol{B}, M_{\mathrm{geo}}, \eta, \delta_{\mathrm{sp}}, \varepsilon, D$. The parameters $M_{\mathrm{geo}}$ are nonnegative and the separation and error parameters are positive, as in Definition 7.5.

For the algebra-valued remainder put

$$\mathcal{R}_D = \prod_i \widehat{A_i(x_i)} \prod_l B_l(y_l) - \sum_{j \leq D,\, \lambda \in \mathrm{Ch}_j} c_{j,\lambda}\, \widehat{O_{j,\lambda}(o)} \prod_l B_l(y_l).$$

Here $\widehat{w} = M^0(w)$ is the compressed separated word of Definition 6.1 and Lemma 17.4, and its characters and vacuum state give the phase and mixed remainders of Definition 7.5.

**Lemma 19.10** (Separated estimates)**.** *There are thresholds $r_0$ for the coefficient bound and $r_1$ for the remainder bound, preceding positions and states, such that*

$$|c_{j,\lambda}(\boldsymbol{A}; x, o)| \leq r^{j - |\boldsymbol{A}|_{\deg} - \varepsilon}, \qquad |\mathcal{R}_D^{\zeta}| \leq r^{D+1-|\boldsymbol{A}|_{\deg}-\varepsilon}$$

*on the respective admissible configurations for $0 < r < r_0$ or $0 < r < r_1$. The second bound also holds in $\mathfrak{A}_0$ before state evaluation.*

*Proof.* Apply Lemma 19.9 to Definition 7.5. Set $T = -\log r$, $\alpha = 1$ and $\beta = \upsilon = 0$. Every admissible geometry has these bounds on extent and reciprocal distances. If either claimed threshold failed, choose violating configurations at radii tending to zero. The estimates just proved along that sequence, with loss $\varepsilon/2$, eventually contradict the violation with loss $\varepsilon$. Constants and subpower factors are absorbed by $r^{\varepsilon/2}$. Taking $M_{\mathrm{geo}} r < \delta_{\mathrm{sp}}/2$ also ensures separation. Character evaluation and the mixed vacuum are contractions on $\mathfrak{A}_0$. This proves both estimates with coefficient one and the stated separate dependencies. □

*Proof of Theorem 7.6.* Lemma 19.6 gives the fields and coefficients of Definition 7.5 before spectators, state indices or truncation. Lemma 19.10 gives its coefficient bound and its remainder bound with their respective thresholds. These are exactly the two assertions of Theorem 7.6. □

Integration requires bounds for the full and truncated expressions individually. Fix the two field lists, $M_{\mathrm{geo}} \geq 0$, $\eta, \delta_{\mathrm{sp}}, \varepsilon > 0$, $D \in \mathbb{N}$, a measurable space $Z$ and $B_0 \geq 0$. Write $\vartheta_Z$ for the complex integration weight.

**Lemma 19.11** (Integrated separated estimate)**.** *Some $r_0 > 0$ works for all smaller radii, measures $\lambda_Z$, integrable weights with $\int |\vartheta_Z|\, d\lambda_Z \leq B_0$, and almost-everywhere measurable geometry and phase maps admissible on $\{\vartheta_Z \neq 0\}$:*

$$\left| \int_Z \vartheta_Z \mathcal{R}_D^{\xi(z)}\, d\lambda_Z \right| \leq B_0 r^{D+1-|\boldsymbol{A}|_{\deg}-\varepsilon}.$$

*The weighted full and truncated expressions are integrable. The same conclusion holds for the mixed state.*

*Proof.* Use Definition 7.5, Lemma 19.10, and the phase continuity of Lemma 17.5. The root argument in Lemma 19.9 also bounds the full correlation uniformly over these configurations and states for each fixed $r$. The finite truncated sum is the full expression minus the uniformly bounded remainder, so it too is uniformly bounded. Separate continuity in geometry and phase gives joint measurability by rounding the geometry in its open domain to a countable dense set and passing to the pointwise limit. The coefficients are smooth where their rotated probes are clear; reducing $r_0$ puts all admissible geometries there. Their compositions are measurable, and multiplication by $\vartheta_Z$ gives integrable majorants. Integrate the remainder bound against $|\vartheta_Z|$. □

A nested configuration retains leaf positions and internal centers, so it also describes partially replaced words. Call an internal vertex other than the root proper, and fix its truncation $D_v \in \mathbb{N}$.

**Definition 19.12** (Nested geometry and rates)**.** Give each vertex $v$ of a finite rooted tree a position $z_v$, each leaf a field $A_i$, and each internal vertex a scale $\varrho_v > 0$. Its root $o_{\mathrm{root}}$ is internal. Require $\varrho_{o_{\mathrm{root}}} = r$, $\varrho_v \leq \varrho_{\mathrm{par}(v)}$, $|z_u - z_v| \leq M_{\mathrm{geo}} \varrho_v$ for every descendant $u$, and $|z_u - z_{u'}| \geq \eta \varrho_v$ for positions in different child subtrees. Put

$$W_v = \sum_{i \text{ below } v} \deg A_i, \qquad q_v = \varrho_v / \varrho_{\mathrm{par}(v)},$$

$$\mathcal{Q}_{\varepsilon,q}(\varrho) = r^{q - W_{o_{\mathrm{root}}} - \varepsilon} \prod_{v \neq o_{\mathrm{root}} \text{ internal}} q_v^{-W_v - \varepsilon}, \qquad \mathsf{e}_{\boldsymbol{D}}(\varrho) = \sum_{v \neq o_{\mathrm{root}} \text{ internal}} q_v^{D_v + 1}.$$

Spectators are $\delta_{\mathrm{sp}}$-clear of every tree position and each other.
The recursion retains every leaf, including those above a neighboring truncation.

**Definition 19.13** (Iterated coefficient)**.** A leaf contributes its field and coefficient one. At a proper internal vertex $v$, replace its children by

$$\sum_{j\le D_v}\sum_{\lambda\in \mathrm{Ch}_j} c_{j,\lambda}(\text{child fields};\text{child centers}, z_v)O_{j,\lambda},$$

multiplying their coefficients and summing. At the root retain only $(q,\lambda)$, giving $C^{\mathrm{it},\boldsymbol{D}}_{q,\lambda}$. The direct coefficient is $C^{\mathrm{dir}}_{q,\lambda}=c_{q,\lambda}(\boldsymbol{A};z_{\mathrm{leaves}},z_{o_{\mathrm{root}}})$. This compares successive and direct expansion.

Fix the tree, leaf list, output $(q,\lambda)$, proper-vertex truncations, $M_{\mathrm{geo}}\ge 0$ and $\eta,\varepsilon>0$. No lower bound on the ratios $q_v$ is imposed.

**Lemma 19.14** (Nested coefficient estimate)**.** *There is $r_0>0$ such that all geometries of Definition 19.12 with $0<r<r_0$ satisfy*

$$|C^{\mathrm{dir}}_{q,\lambda}|\le \mathcal{Q}_{\varepsilon,q},\qquad |C^{\mathrm{dir}}_{q,\lambda}-C^{\mathrm{it},\boldsymbol{D}}_{q,\lambda}|\le \mathcal{Q}_{\varepsilon,q}\mathsf{e}_{\boldsymbol{D}}\,.$$

*Both expressions are smooth in neighborhoods of these geometries.*

*Proof.* Use Definitions 19.5, 19.12 and 19.13 and Lemma 19.9. Along any sequence of admissible radii and scales tending to the root collision, put $T=\sum_{v\ \mathrm{internal}}\log(1/\varrho_v)$. Eventually every summand is positive. After a subsequence, $\log(1/\varrho_v)/T\to a_v\ge 0$, with $\sum_v a_v=1$ and $a_v\ge a_{\mathrm{par}(v)}$. Set

$$\mathcal{B}=W_{o_{\mathrm{root}}}a_{o_{\mathrm{root}}}+\sum_{v\ne o_{\mathrm{root}}\ \mathrm{internal}} W_v(a_v-a_{\mathrm{par}(v)})\,.$$

We prove by induction on the tree that a direct output of degree $q$ has exponent $\mathcal{B}-qa_{o_{\mathrm{root}}}$, and its error against the prescribed finite iteration has that exponent minus $\gamma=\min_{v\ne o_{\mathrm{root}}\ \mathrm{internal}}(D_v+1)(a_v-a_{\mathrm{par}(v)})$. With no proper internal vertices the two coefficients coincide exactly.

First flatten the edges $u\to v$ with $a_u=a_v$: combine their grouping before making any field replacement. The word at $v$ then consists of original leaves in this equal-height component and outputs from strictly finer children; its distinct positions are separated at exponent $a_v$ by the tree geometry. For a child with $a_u>a_v$, expand to an auxiliary horizon $H_u$. The remainder exponent has the improvement $(H_u+1)(a_u-a_v)$ by (19.11). Choose the children in a fixed order, choosing $H_u$ only after the outputs already used as exterior fields are fixed. Each improvement tends to infinity with $H_u$, whereas the exterior degree in that step is fixed. It can therefore exceed any prescribed error exponent. The identity

$$\prod_{i=1}^{n}X_i-\prod_{i=1}^{n}Y_i=\sum_{i=1}^{n}\left(\prod_{h<i}Y_h\right)(X_i-Y_i)\left(\prod_{h>i}X_h\right) \tag{19.14}$$

justifies these successive replacements in every correlation and in every probe defining a coefficient. It is an identity of separated expressions, with their scalar coefficient factors retained.

For a fixed collection of child outputs $j_u$, the child exponents include $-j_ua_u$, and the parent coefficient contributes $j_ua_v$. Their sum contains $-j_u(a_u-a_v)$; the remaining leaf weights telescope to $\mathcal{B}-qa_{o_{\mathrm{root}}}$. Nonnegative output degrees therefore never increase this exponent. If the root height is zero, apply the crude estimate to this root word of child outputs; its finitely many field degrees cost exponent zero, while their coefficients already carry the child contributions to $\mathcal{B}$. This proves the direct bound, by the preceding arbitrary-accuracy replacements and the induction hypothesis. Remove auxiliary outputs above $D_u$. Each discarded term has at least one $j_u\ge D_u+1$ and thus

gains at least $\gamma$. Replacing the remaining direct child coefficients by their iterated coefficients uses (19.14) again and gains $\gamma$ from one child error. If $\gamma = 0$, the direct and iterated bounds themselves give the error exponent. If $\gamma > 0$, every proper internal scale increase is positive and the auxiliary horizons can be chosen to make the preceding replacement errors smaller than this exponent. This proves the induction for the difference, with one improvement rather than a product of improvements.

The logarithmic exponent of $\mathcal{Q}_{\varepsilon,q}$ is

$$\mathcal{B} - q a_{o_{\mathrm{root}}} + \varepsilon A_*, \qquad A_* = a_{o_{\mathrm{root}}} + \sum_{v \neq o_{\mathrm{root}} \text{ internal}} (a_v - a_{\mathrm{par}(v)}) > 0 \,.$$

Positivity follows because $\sum_v a_v = 1$: if every displayed nonnegative term vanished, every $a_v$ would vanish. The exponent of $\mathtt{e}_{\boldsymbol{D}}$ is $-\gamma$. Thus the estimates with loss $\varepsilon/2$ imply the displayed bounds with loss $\varepsilon$ and coefficient one along this sequence. A sequence of violations would have the subsequence just treated, a contradiction. This proves a threshold uniform in the scale vector, even when some scales decrease faster than every power of $r$. All positions stay separated and, for small $r$, clear of every finite rotated probe family used in the recursion. The averaging integrands are smooth on an open neighborhood with that clearance; their derivatives are bounded on compact neighborhoods times $O(4)$. Differentiating the finite sums and Haar integrals proves smoothness. □

For tested products, retain the original leaves at the root; only the degree of its output is truncated. Fix additionally the root truncation $m \in \mathbb{N}$, a spectator list $\boldsymbol{B}$ and $\delta_{\mathrm{sp}} > 0$.

**Lemma 19.15** (Nested tested estimate)**.** *For some $r_0 > 0$, every smaller radius, admissible nested geometry, admissible spectator configuration and state index $\zeta$ satisfy*

$$|\mathcal{R}^{\zeta}_{m}(\boldsymbol{A}, z_{\mathrm{leaves}}; \boldsymbol{B}, y; z_{o_{\mathrm{root}}})| \leq \mathcal{Q}_{\varepsilon,m+1}(\varrho) \,.$$

*The same bound holds in the norm of $\mathfrak{A}_0$ before evaluation in a character or the mixed vacuum.*

*Proof.* Use Definition 7.5 and the tree induction in Lemma 19.14. Keep the spectators in every replacement. Their reciprocal distances from every tree position and from one another have exponent zero. They therefore add no term to $\mathcal{B}$. After the arbitrary-accuracy child replacements, the root remainder estimate of Lemma 19.9, has exponent $\mathcal{B} - (m+1) a_{o_{\mathrm{root}}}$. When $a_{o_{\mathrm{root}}} = 0$, apply the crude bound to the root word of child outputs after these replacements. The child coefficient powers already account for $\mathcal{B}$; the finitely many root output fields and the spectators cost exponent zero. The positive quantity $\varepsilon A_*$ used above again absorbs all finite sums and subpower factors. The contradiction argument is unchanged. The norm estimate follows from (19.13); applying a character or the mixed vacuum proves the scalar bounds with the same threshold. □

Fix now the data of Lemma 19.14 or 19.15, a measurable parameter space and $B_0 \geq 0$. A scale vector $\varrho$ in the following statement is fixed throughout each integral.

**Lemma 19.16** (Integrated nested estimates)**.** *A threshold preceding the measure, weight, scale vector and measurable geometry maps gives the respective integral bounds*

$$B_0 \mathcal{Q}_{\varepsilon,q}, \qquad B_0 \mathcal{Q}_{\varepsilon,q} \mathtt{e}_{\boldsymbol{D}}, \qquad B_0 \mathcal{Q}_{\varepsilon,m+1}$$

*for integrable weights of absolute integral at most $B_0$ and geometries admissible almost everywhere where the weight is nonzero. The weighted compared expressions are integrable. The tested bound permits a measurable phase map or the mixed state.*

*Proof.* For fixed $\varrho$, Lemma 19.14 bounds the direct coefficient and its difference from the iterated coefficient, hence also the latter. Lemma 19.15 bounds the tested difference; the root argument (19.13) bounds the full word for these fixed positive separations and hence its truncated expression. Smoothness and the joint measurability argument of Lemma 19.11 give measurability. Multiply these constant bounds by the absolute value of the weight and integrate, as in Lemma 19.11. The thresholds are independent of $\varrho$, although the majorants depend on this fixed vector. □

## 19.2 Contact subtractions

We work with finite sums of field words, identifying $A(x)$ with $(\jmath_{d,e}A)(x)$ and imposing linearity in $A$. Their product is concatenation. The correlation of a word is read only at distinct positions; elsewhere give it value zero. This convention is used inside the algebraic formulas below, before any integration. The underlying prescription and its fields are those fixed in Lemma 19.6; the finite probes chosen next are independent of its OPE probes.

**Definition 19.17** (Contact detector projection)**.** Choose contact probes and decoders as in Definition 19.3. Choose $a_q^{\mathrm{ct}} > 0$ with every probe point at distance at least $2a_q^{\mathrm{ct}}$ from zero. For a word $w$, set

$$K_q w = \mathbf{1}_{\{\text{all positions of } w \text{ lie in } \overline{B}(0,a_q^{\mathrm{ct}})\}} w\,.$$

Apply the averaging and reality formula of Definition 19.4 to this clipped word with the contact probes; extend linearly to sums, obtaining $B_q^{\mathrm{ct}}$. This extracts contact fields on a fixed region clear of its probes.

For contact expressions $I_q A = A(0)$ still denotes a single field. Unlike the domain restriction in Definition 19.4, clipping here acts on each monomial before summing it.

**Definition 19.18** (Coherent contact projections)**.** With $j_q = \jmath_{q,q+1}$, set

$$P_0 = B_0^{\mathrm{ct}}, \qquad P_{q+1} = j_q P_q + B_{q+1}^{\mathrm{ct}} - j_q P_q I_{q+1} B_{q+1}^{\mathrm{ct}}, \qquad P_{\le q}^{\mathrm{ct}}(o) w = P_q(\tau_{-o} w)\,.$$

The last term removes the component already measured at lower degrees. In particular, $P_{\le q}^{\mathrm{ct}}(o)[A(o)] = A$ for $A \in F_q$.

**Definition 19.19** (Contact grades and clearance)**.** The exact-degree components and common clearance are

$$\begin{aligned} P_{=0}^{\mathrm{ct}}(o) w &= B_0^{\mathrm{ct}}(\tau_{-o} w), \\ P_{=q}^{\mathrm{ct}}(o) w &= (1 - \jmath_{q-1,q} P_{q-1} I_q) B_q^{\mathrm{ct}}(\tau_{-o} w) \quad (q > 0), \\ \delta_q^{\mathrm{pres}} &= \min_{0 \le d \le q} a_d^{\mathrm{ct}}\,. \end{aligned}$$

These give the graded contact subtractions and their common range of separated evaluation.

The projections are defined before any test function or state. Their strata, rather than the OPE strata, provide the real bases used in the contact construction.

**Lemma 19.20** (Real bases and scales)**.** *The contact projections are covariant under translations, $O(4)$ and conjugation. Their filtered strata have real bases, with the unit in degree zero. There are positive subtraction scales $\delta_j^{\mathrm{ct}}$ such that*

$$\delta_j^{\mathrm{ct}} \to 0, \qquad 4\delta_{j+1}^{\mathrm{ct}} < \delta_j^{\mathrm{ct}}, \qquad 2(j+1)\delta_j^{\mathrm{ct}} \le \delta_j^{\mathrm{pres}}\,.$$

*Proof.* Use the inclusions, rotations and conjugation of Part I, as in Lemma 19.6. A ball centered at zero is rotation invariant. Haar invariance proves rotation covariance of $B_q^{\mathrm{ct}}$, and its reality formula proves conjugation covariance. These maps fix $A(0)$: before averaging the decoder returns $R_q(h^{-1})A$, and after multiplication by $R_q(h)$ it returns $A$. Define $T_q = P_q I_{q+1} : F_{q+1} \to F_q$. The recursion gives

$$P_q I_q = 1, \qquad T_q j_q = 1, \qquad T_q P_{q+1} = P_q, \qquad P_{\le q}^{\mathrm{ct}}(o) = \sum_{d \le q} \jmath_{d,q} P_{=d}^{\mathrm{ct}}(o)\,. \tag{19.15}$$

For the third identity, apply $T_q$ to the recursion and cancel $T_q B_{q+1}^{\mathrm{ct}}$; the fourth follows by iteration. Thus the degree-$q$ stratum is $\ker T_{q-1}$ for $q > 0$, and $F_0$ for $q = 0$. Each is invariant under conjugation and rotations. Lemma 19.6 gives real bases; choose the unit in $F_0$. Covariance passes through the recursion, with translations following from $\tau_{-(o+u)}\tau_u = \tau_{-o}$.

Put $g_j = \min(2^{-j}, \delta_j^{\mathrm{pres}}/[2(j+1)])$, and choose

$$\delta_0^{\mathrm{ct}} = g_0, \qquad \delta_{j+1}^{\mathrm{ct}} = \min(\delta_j^{\mathrm{ct}}/8, g_{j+1})\,.$$

Positivity and the three inequalities follow directly. These scales can also be bounded by any prescribed positive sequence by including it in $g_j$. Lemma 19.22 uses only the single length $s_*$ below; the scales $\delta_j^{\mathrm{ct}}$ are recorded for Theorem B. □

We give the finite subtraction formula before proving its convergence. Fix $\boldsymbol{A} \in \mathbb{F}^n$, put $d_i = \deg A_i$, $W = \sum_i d_i$, and choose

$$B_* = \min(1, \delta_W^{\mathrm{pres}}), \qquad s_* = B_*/4\,.$$

Expand each input in the contact stratum bases using (19.15), retaining its original degree tag $d_i$ in every subtraction order. For $h \subset \{1, \dots, n\}$, $|h| \ge 2$, let $\Delta_h$ be its collision diagonal, $x_h^{\perp}$ the orthogonal component normal to $\Delta_h$, and $\bar{x}_h = |h|^{-1} \sum_{i \in h} x_i$. Thus $x_h^{\perp}$ has entries $x_i - \bar{x}_h$ on $h$ and zero elsewhere. Write

$$d_h = \sum_{i \in h} d_i, \qquad q_h = 4(|h| - 1), \qquad L_h = \max(0, d_h + 1 - q_h)\,.$$

Choose a smooth radial cutoff $\chi$ equal to one on the unit ball and zero outside the ball of radius two. The normal Taylor operator is

$$(\mathsf{T}_{h,k}^{\perp} f)(x) = \chi(x_h^{\perp}/s_*) \sum_{l=0}^{k} \frac{1}{l!} D^l f(x - x_h^{\perp})[x_h^{\perp}, \dots, x_h^{\perp}]\,. \tag{19.16}$$

It subtracts only normal variables, leaving the centroid and all exterior variables as parameters.

An expression $F \otimes U$ means the field expression $F(x)$ multiplied by $(Uf)(x)$ when tested against $f$. Products compose the test operators. On a monomial write $F = F_h F_{h^c}$ for its factors in and outside $h$. Let $\mathsf{R}_{h,w}$ replace the subword of $F(x)$ on $h$ by the field $P_{=w}^{\mathrm{ct}}(\bar{x}_h) F_h(x)$ at $\bar{x}_h$, leaving its test operator unchanged. Define recursively

$$\mathcal{C}_{\{i\}}^{\mathrm{ct}} = A_i(x_i) \otimes 1, \qquad \Pi_h^{\mathrm{punct}} = \sum_{\substack{\pi \text{ partition of } h \\ |\pi| \ge 2}} \prod_{b \in \pi} \mathcal{C}_b^{\mathrm{ct}}, \tag{19.17}$$

$$\mathcal{C}_h^{\mathrm{ct}} = -\sum_{w < L_h} \mathsf{R}_{h,w} \Pi_h^{\mathrm{punct}} \mathsf{T}_{h, L_h - 1 - w}^{\perp}, \qquad \Pi_h^{\mathrm{full}} = \Pi_h^{\mathrm{punct}} + \mathcal{C}_h^{\mathrm{ct}} \quad (|h| \ge 2)\,.$$

Singleton and empty full expressions are respectively $A_i(x_i)\otimes 1$ and the unit. In a nested term, the parent Taylor operator acts first on the test and the child operators act afterwards. Operators on disjoint slot sets commute. The recursion terminates because every child has fewer slots; expanding it gives forests of disjoint or nested collision subsets.

Before summing shells we need estimates for complete signed expressions after child subtractions. Normalize $T \to \infty$ so that all separation heights lie in $[0,1]$, and fix a finite set $\mathcal{P}$ of position paths. A finite separated sum of cost $c$ at height $s$ has terms

$$a(T)\prod_{\iota} A_\iota(x_\iota(T)), \qquad |a| \preccurlyeq e^{(c-s\sum_\iota \deg A_\iota)T}, \qquad |x_\iota - x_{\iota'}| \geq e^{-sT-o(T)} \quad (\iota \neq \iota'),$$

with $x_\iota \in \mathcal{P}$. This definition permits different terms to use overlapping subsets of $\mathcal{P}$.

Write $\mathcal{U} \in \mathcal{A}_s(c)$ if it has a finite height-one representation of some finite cost and satisfies the following approximation property. For every $L > 0$, there is a finite separated sum $g$ of cost $c$ at height $s$, using $\mathcal{P}$, such that for every $d \in \mathbb{R}$ and every exterior finite separated sum $f$ of cost $d$ at height 1,

$$\big|S^\zeta((\mathcal{U}-g)f)\big| \preccurlyeq e^{(-L+d)T}. \tag{19.18}$$

Every exterior path must be at distance at least $e^{-sT-o(T)}$ from every path in $\mathcal{P}$. The bound permits any varying character $\zeta(T)$ and the mixed state. Thus exterior internal separations are measured at height one, while its distance from $\mathcal{P}$ is measured at height $s$. Scalar factors of rate $b$ change the cost to $c+b$.

Suppose the paths in $\mathcal{P}$ lie within $e^{-sT+o(T)}$ and within $2s_*$ of a center $o \in \mathcal{P}$. For contact projections assume $2s_* \leq a_w^{\mathrm{ct}}$. Write $I_q^o A = A(o)$ for insertion at this center. These fixed-distance requirements ensure that the contact probes are clear; the exponential distances determine the rates.

**Lemma 19.21** (Signed layer transfer)**.** *For $0 \leq u \leq s \leq 1$ and $\mathcal{U} \in \mathcal{A}_s(c)$, every clear contact degree $w$ satisfies*

$$P^{\mathrm{ct}}_{=w}(o)\mathcal{U} \preccurlyeq e^{(c-ws)T}, \qquad \mathcal{U} - I_w^o P^{\mathrm{ct}}_{\leq w}(o)\mathcal{U} \in \mathcal{A}_u\big(c-(w+1)(s-u)\big).$$

*Separated products add their costs; finite signed sums take their maximum. Coarsening from $s$ to $u$ without subtracting retains cost $c$.*

The proof orders the approximation precisions before applying the projections.

*Proof.* For the fields of Definition 4.9, use Lemma 19.9, including its uniformity on compact label sets, the telescope (19.14), and the exact contact identities (19.15).

First check products, which explains the strength of the approximation property. Let $\mathcal{U},\mathcal{V}$ have costs $c,d$ and a separation at height $s$. Suppose a height-one representation of $\mathcal{V}$ has cost $b$. Choose an approximant $g$ to $\mathcal{U}$ of precision $L+b$. Its own height-one representation has some finite cost $b'$. Next choose an approximant $g'$ to $\mathcal{V}$ of precision $L+b'$. Then

$$\mathcal{U}\mathcal{V} - gg\prime = (\mathcal{U}-g)\mathcal{V} + g(\mathcal{V}-g\prime)$$

has precision $L$ against every permitted exterior sum. The product $gg\prime$ has cost $c+d$, since its word positions are separated at height $s$ and its scalar coefficient rates add. Iterate in a fixed order for more factors. Addition and subtraction follow by taking the maximum cost and making every approximation as accurate as required.

For $u < s$, expand a finite separated approximant to $\mathcal{U}$ at $o$ to an auxiliary degree $H$. Applied to each term, (19.11) gives output fields $A_d(o)$ whose coefficient rates are at most $c - ds$, and an error at exterior height $u$ of rate at most $c - (H+1)(s-u)$. Increasing $H$ gives any specified precision. The original approximation error permits these farther exterior tests. This gives an arbitrarily accurate single-point sum $g$ at $o$, of cost at most $c$ at height $u$. If $u = s$, retain the original approximant. This proves coarsening.

For $s > 0$, approximate with exterior height zero. A contact probe of total degree $p$ costs at most $p$ at height one; raise the precision by $p$. Compactness of $O(4)$ makes this uniform over rotated and conjugated probes, so averaging preserves arbitrary precision. On a degree-$d$ field the degree-$w$ contact row is zero for $d < w$ and is a fixed linear map for $d \geq w$. Its rate is at most $c - ds \leq c - ws$, proving the row bound. At $s = 0$ test directly against the fixed clear probes: their positions have height zero and add no exponent. The same conclusion follows.

For the residual, use $u < s$ and the single-point sum $g$ just constructed. Every field of degree $d \leq w$ cancels exactly under $1 - I_w^o P_{\leq w}^{\mathrm{ct}}$. For $d \geq w+1$ its coefficient rate at the outgoing height is $c - ds$. Viewed at height $u$, its cost is at most $c - d(s-u) \leq c - (w+1)(s-u)$. The lower fields produced by its contact projection obey the same inequality, since their degrees are at most $d$. Thus $g - I_w^o P_{\leq w}^{\mathrm{ct}} g$ has the required cost. The exact error is $(\mathcal{U} - g) - I_w^o P_{\leq w}^{\mathrm{ct}}(\mathcal{U} - g)$. The first term has arbitrary precision by choice of $H$; the finite probes defining the second cost only finitely many additional exponents. Choose the precision to pay for them and for the requested exterior test. This proves the residual's approximation property. Its finite height-one representation follows from the initial one and the finite probe formulas. At $u = s$ the row bound makes the projected finite sum have cost $c$ at height $s$, so subtraction retains that cost directly. □

**Lemma 19.22** (Contact extension)**.** *For every state $\zeta$, arity $n$ and field list $\boldsymbol{A}$, the normal subtractions* (19.17) *with the contact projections, bases and scales fixed above define a complex linear continuous functional $\mathfrak{D}_{\boldsymbol{A}}^{\zeta}$ on $C_c^\infty(\mathbb{R}^{4n})$. For every $f \in C_c^\infty(\mathrm{Conf}_n)$,*

$$\mathfrak{D}_{\boldsymbol{A}}^{\zeta}(f) = \int f(x) S_n^{\zeta}(\boldsymbol{A}; x)\, dx\,.$$

*Proof.* Use the original fields of Definition 4.9, the contact projections of Definitions 19.17–19.19, and the lattice-derived estimates of Lemma 19.9. We prove absolute integrability of the complete subtracted expression on every compact set, with a finite test seminorm.

At a group $h$, reorganize the full expression as

$$\left(1 - \sum_{w < L_h} \mathsf{R}_{h,w}\right) \Pi_h^{\mathrm{punct}} f + \sum_{w < L_h} \mathsf{R}_{h,w} \Pi_h^{\mathrm{punct}} (1 - \mathsf{T}_{h,L_h-1-w}^{\perp}) f\,. \tag{19.19}$$

These are identities of field expressions with test operators; individual unsubtracted summands need not be integrable. Every nonzero Taylor term lies within $2s_* \leq B_*$ of its diagonal, so all its field projections have their required contact clearance. If $L_h = 0$, no projection is subtracted and $q_h - d_h \geq 1$ already gives integrability. For $L_h > 0$, (19.15) removes every degree below $L_h$ from the first arm. We apply Lemma 19.21 to the punctured expression, including the child subtractions. Here is the induction invariant: after fixing the child arms and their test operators, the field expression belongs to $\mathcal{A}_s(c)$ at its outgoing height; a projected arm of degree $w$ has coefficient rate $c - ws$, and a residual arm through $w$ has cost $c - (w+1)(s-u)$ at a parent height $u$. Singleton fields start this invariant. Products over separated partition blocks preserve it by the product clause of Lemma 19.21; its projection and residual clauses give the next arms, and its coarsening

clause handles equal heights. Keep every child Taylor operator attached to its arm throughout these operations. Each derivative expansion has finitely many scalar test factors of finite rate. The scalar-factor and product clauses of Lemma 19.21 include these in the approximation cost and retain the invariant for each differentiated arm.

Rescale the patch so all scales are at most one. Fix its finite collision tree, write $\rho_b$ for its scales and $R = \rho_{\mathrm{par}(h)}$, and set

$$T = \log(1/\rho_{\min}), \quad s_b = \log(1/\rho_b)/T, \quad u = \log(1/R)/T, \quad \rho_{\min} = \min_b \rho_b \,.$$

After a subsequence these heights converge in $[0,1]$, with $s_h \geq u$. Retain the child size factor $\mathcal{H}_h = \prod_{b \text{ proper below } h} (\rho_b/\rho_{\mathrm{par}(b)})^{-d_b}$. Before child gains, the incoming cost is $c = d_h s_h + \sum_{b \text{ proper below } h} d_b(s_b - s_{\mathrm{par}(b)})$. Lemma 19.21 gives a degree-$w$ row of rate $c - w s_h$, and a residual through degree $L_h - 1$ of cost $c - L_h(s_h - u)$. Thus, after retaining $\mathcal{H}_h$ and the outer field costs, their physical size bounds are respectively

$$C_\tau \mathcal{H}_h \rho_h^{w-d_h} \rho_{\min}^{-\tau}, \qquad C_\tau \mathcal{H}_h \rho_h^{-d_h} (\rho_h/R)^{L_h} \rho_{\min}^{-\tau}. \tag{19.20}$$

With one unresolved scale, $\rho_{\min} = \rho_h = \rho$, giving the row bound $C_\tau \rho^{w-d_h-\tau}$ and residual gain $(\rho/R)^{L_h}$. With several scales, retain the loss at $\rho_{\min}$ and the child gains; the induction below distributes the loss between ancestors and descendants. If these estimates failed uniformly in a sector, choose a violating sequence and take the convergent height vector just described. The signed invariant, including equal heights, gives the rates with loss $\tau/2$, contradicting the violation with loss $\tau$. This proves (19.20) uniformly over the sector.

Taylor's integral remainder gives, on a shell $|x_h^\perp| \asymp \rho$ within the subtraction range,

$$(1 - \mathsf{T}^\perp_{h,L_h-1-w}) f = O\Big( \rho^{L_h - w} \max_{|\alpha| \leq L_h - w} \|\partial^\alpha f\|_\infty \Big) \,.$$

Including normal volume, every term of (19.19) therefore gains $\rho^{q_h + L_h - d_h - \tau}$, whose exponent is at least $1 - \tau$.

Partial collisions on a normal shell require a further induction. Induct on $|h| - |\pi|$, where $\pi$ is the currently separated partition. The case of singleton blocks is a separated correlation against a smooth compactly supported test, bounded by Lemma 15.5 and (19.13). For a nonsingleton block, cut its normal coordinate into dyadic shells and rescale to $1/2 \leq |x_h^\perp| \leq 2$. At any point of this compact annulus its equality classes form a proper partition. The centroids of nonempty subgroups contained in different classes are distinct there. There are finitely many such subgroups; continuity gives a neighborhood where their distances have a common positive lower bound. A finite smooth partition of unity on the annulus thus gives sectors with all subsequent subgroup centroids separated. Within a sector apply the induction hypothesis to this strictly finer partition. On overlaps the resulting expressions agree: a group crossing two sectors' separated blocks has a Taylor jet on a diagonal disjoint from the localized test support, and contributes zero. The subdivision therefore preserves the full expression and its ordered test operators.

For the shell estimate, let $a > 0$ be the shell radius divided by its parent scale. The rescaled Taylor remainder through order $k$ obeys

$$\|(1 - \mathsf{T}^\perp_{h,k}) f\|_{C^l \text{ on the rescaled shell}} \leq C \|f\|_{C^{l+k+1}} \begin{cases} a^{k+1}, & a \leq 1, \\ a^k, & a > 1. \end{cases} \tag{19.21}$$

For $a \leq 1$, differentiate the integral Taylor remainder, including the cutoff; for $a > 1$, use the bounded test and the degree-$k$ polynomial. The cutoff derivatives have fixed bounds at the parent

scale. For a projected component $w$, use Lemma 19.21 to transfer the signed field expression and put $k = L_h - 1 - w$. Its volume and coefficient power $a^{q_h+w-d_h}$ times the Taylor factor is at most $a$ when $a \le 1$, and at most 1 when $a > 1$. Indeed $q_h + w - d_h + k = 0$ whenever $w < L_h$. For the unprojected remainder the power is $q_h + L_h - d_h \ge 1$; its support restricts $a$ above by a fixed geometric constant. Thus, for every $0 < \theta \le 1$, the combined estimate is at most $Ca^\theta$.

Choose $\gamma > 0$ with $(3^{|h|-|\pi|} + 1)\gamma \le 1$. At a finer partition with $d'$ remaining splits, the just-proved signed invariant remains valid for every projected and residual arm. Allocate loss $3^{d'}\gamma$ to its coefficient and test estimates. One split strictly decreases the remaining number of splits, and $3^{d'+1}\gamma \ge 2 \cdot 3^{d'}\gamma + \gamma$ leaves a factor $a^\gamma$ after the preceding estimate. To see that this is uniform in the parent scale $R$, combine the child loss $\rho^{-\tau}$ with its ratio improvement by

$$(\rho/R)^{\gamma+\tau}\rho^{-\tau} = \rho^\gamma R^{-(\gamma+\tau)} .$$

The last factor is charged to the parent's allocated loss. Finite sector sums change only the constant. For shells $\rho \asymp 2^{-j}$, the bound is consequently $C2^{-j\gamma}$ times a fixed test seminorm. Summing this geometric series completes the induction, and gives absolute convergence also for iterated choices of disjoint shells.

Finally cover a given compact set by neighborhoods of the equality classes at each of its points. The same separation argument gives finitely many patches, including patches outside the total-collision subtraction range. On these patches the recursion reduces to the product of the full expressions in the separated classes: any crossing Taylor term has zero jet. A smooth partition of unity and the Leibniz rule combine their bounds. Each descent costs at most $W + 2$ derivatives by (19.21); there are at most $n$ descents. We obtain

$$\int \left| \left\langle \Pi^{\mathrm{full}}_{\{1,\dots,n\}} f \right\rangle_\zeta (x) \right| dx \le C_K \max_{|\alpha| \le (W+2)n} \|\partial^\alpha f\|_\infty \quad (\operatorname{supp} f \subset K) . \tag{19.22}$$

Define $\mathfrak{D}^\zeta_{\boldsymbol{A}}(f)$ by this integral. Absolute convergence preserves complex linearity; (19.22) proves distribution continuity.

For separated $f$, every normal jet on a collision diagonal vanishes. Each nontrivial term of (19.17) has a largest Taylor operator acting first; disjoint operators commute. Only $f(x)\prod_i A_i(x_i)$ remains, proving the restriction also for empty and singleton lists. □

The compact estimate gives the local distribution order.

**Lemma 19.23** (Local distribution order)**.** *For every fixed $\zeta, \boldsymbol{A}$ and compact $K \subset \mathbb{R}^{4n}$, there are $q_K \in \mathbb{N}$ and $C_K < \infty$ such that*

$$|\mathfrak{D}^\zeta_{\boldsymbol{A}}(f)| \le C_K \max_{|\alpha| \le q_K} \|\partial^\alpha f\|_\infty \qquad (\operatorname{supp} f \subset K) .$$

*Proof.* By (19.22) and the integral triangle inequality, take $q_K = (2 + |\boldsymbol{A}|_{\deg})n$. □

*Proof of Theorem 7.7.* Definitions 19.17–19.19 fix the contact projections on the prescription of Lemma 19.6. Lemma 19.20 gives the bases and scales; Lemma 19.22 gives the extension, and Lemma 19.23 its local order. □

The conclusions of Theorem 7.7 assert neither uniqueness, positivity of the extension nor contact Ward identities. Estimates along general collision subspaces are not stated here and are not used in Theorem B.

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